Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
And
KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of

Information Technology & Digital Panel

Case 23-G-0225
Case 23-G-0226

Date: September 22, 2023

Rebuttal Testimony of the Information Technology & Digital Panel

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I. Introduction

Q. Please introduce the members of the Information Technology & Digital Panel ("IT&D") Panel.

A. The Panel consists of Erik Barthel, Chris Murphy, and Christophe Chirol.

Q. Is this the same Panel that previously submitted direct and corrections and updates testimony in these proceedings?

A. Erik Barthel and Chris Murphy previously provided testimony for the IT&D Panel as part of the Companies' April 28, 2023 filings and the Companies' June 30, 2023 corrections and updates filings. Christophe Chirol has been added to the IT&D Panel replacing Jeffrey Knighton for the Company's rebuttal testimony. The defined terms in the Panel's direct testimony and corrections and updates testimony have the same meanings in this rebuttal testimony.

Q. Mr. Chirol, please state your full name and business address.

A. My name is Christophe Chirol. My business address is 2 Hanson Place, Brooklyn, New York 11217.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company as a Lead Analyst of Revenue Requirements for New York, supporting National Grid's New York operating companies, including KEDNY and KEDLI, and am responsible for providing input and management of the preparation of various exhibits and analyses associated with the Company's cost of service and revenue requirement filings in this case.

Q. Please describe your educational background and professional experience.

A. I received a Bachelor of Science in Business Administration with a concentration in Finance and Marketing from Marist College in 2008. I joined National Grid in December of 2008. Prior to my current position I worked in Finance under the Management Reporting and Planning team as a Decision Support Analyst for various areas such as Human Resources, Finance, the Transaction Delivery Center, and Property Services (2008-2014). My responsibilities included assisting in the development, consolidation, and reporting of the five-year business plans, monthly reporting, quarterly forecasts, and annual operating and capital budgets. I also assisted in various audits and the collection and documentation of data and responses for companywide regulatory rate case filings. Since

2014, I have worked on the New York Revenue Requirements team as a Regulatory Analyst with increasing responsibility in support of the New York operating companies' (KEDNY, KEDLI, and Niagara Mohawk Power Corporation d/b/a National Grid ("Niagara Mohawk")) rate case activities.

II. Purpose of Testimony

Q. What is the purpose of the Panel's rebuttal testimony?

A. The purpose of the Panel's rebuttal testimony is to respond to certain recommendations in the testimony of the Department of Public Service Staff's ("Staff") Information Technology Panel ("SITP"). Specifically, the Panel will address the following topics:

(i) the SITP's proposed IT&D forecasting adjustment that would reduce the Rate Year Service Company IT&D capital expenditures ("CAPEX") forecast by 62.60 percent from $307.922 million to $115.115 million;

(ii) the SITP's corresponding downward adjustments to the Company's Rate Year operating expenditures ("OPEX") and run-the-business ("RTB") costs;

(iii) the adequacy of the support provided by the Company for the projected IT&D CAPEX, OPEX, and RTB costs;

(iv) the SITP's proposal to add an additional reporting requirement that would obligate the Company to notify Staff as physical security or cybersecurity projects become partially or fully sanctioned, are merged with other projects, or are discontinued; and

(v) the SITP's proposals that the Company develop enhanced standards for information required in individual IT&D projects to be considered for inclusion in approved capital budgets and revenue requirements, and that the Company be directed to provide an initial proposal concerning such enhancements within 60 days after issuance of a rate order by the Commission in these proceedings.

Q. Does the Panel sponsor any exhibits as part of its rebuttal testimony?

A. Yes. The Panel sponsors the following exhibits that were prepared and /or compiled under our direction and supervision:

(i) Exhibit ___ (ITDP-1R) is the "Report on Information Technology Reporting Format" dated December 22, 2021 and submitted in the 2019 KEDNY and KEDLI Rate Cases;

(ii) Exhibit ___ (ITDP-2R) consists of the Companies' redacted quarterly Information Technology Capital Investment Reports for the period October 2021 through June 30, 2023;

(iii) Exhibit ___ (ITDP-3R) contains the Companies' Information Technology Capital Investment Plans for fiscal year ("FY") 2023 and FY 2024;

(iv) Exhibit ___ (ITDP-4R) is an excerpt from the Companies' Implementation Plan Update dated October 31, 2022 and submitted in the Companies' Management and Operations Audit proceeding (Case 18-M-0185); and

(v) Exhibit ___ (ITDP-5R) responses to information requests ("IR") provided in these proceedings and cited in this rebuttal testimony.

Q. Do other witness panels address recommendations by Staff concerning the Companies' IT&D costs?

A. Yes. The following witness panels address Staff's IT&D-related recommendations:

(i) The Companies' Customer Panel addresses the Staff Efficiency Panel's recommendations concerning the proposed Green Button Connect and Clean Energy Platform projects;

(ii) The Companies' Gas Infrastructure and Operations Panel ("GIOP") addresses the Staff Policy Panel and the Staff Pipeline Safety Panel testimony regarding Advanced Metering Infrastructure ("AMI") investments; and

(iii) The Companies' Capital Structure Panel and witness Joshua C. Nowak address the SITP's recommendations concerning the Service Company's cost of capital.

III. The SITP's CAPEX, OPEX, and RTB Adjustments

Q. Please describe the SITP's proposed adjustment to the Companies' CAPEX budget.

A. The SITP recommends a forecasting adjustment that includes the removal of 30 projects from the Companies' CAPEX budget. After adjusting for the proposed removal from the Rate Year IT&D CAPEX forecast, the Service Company's IT&D CAPEX forecast would be reduced by 62.60 percent, from $307.922 million to $115.115 million.

Q. Please describe the SITP's proposed adjustment to the Companies' OPEX and RTB levels.

A. As a result of the SITP's proposed removal of 30 projects from the Companies' CAPEX budget, the Service Company OPEX and RTB levels would be reduced by the SITP by $43.031 million and $14.713 million, respectively, in FY 2025.

Q. Did the SITP explain why it believes that the Companies could meet their legal obligations with a CAPEX budget of $115.115 million?

A. No. The SITP never explains why it believes that the Companies could continue to operate effectively with an annual CAPEX budget of only $115.115 million when IT&D actual CAPEX has exceeded that in each of the last five fiscal years.

Q. What was the IT&D CAPEX amount reflected in the Companies' rates during the FY 2021 to FY 2023 period?

A. The Companies' rate allowance for IT&D CAPEX was $225 million in FY 2021 and $150 million in each of FY 2022 and FY 2023.

Q. What is the significance of this information?

A. The fact that the Companies invested $261.379 million and $307.906 million in FY 2022 and FY 2023, respectively, compared to a rate allowance of $150 million in each of these years demonstrates that a much larger level of IT&D investment was deemed necessary for the Companies to fulfill their service obligations. The Companies' need to invest between $111.379 million and $157.906 million in IT&D systems that were not compensated in rates provides compelling evidence that the Companies

cannot possibly manage their systems under the $115.115 million IT&D CAPEX budget recommended by the SITP.

Q. What would the likely consequences be if the Commission were to set rates reflecting the SITP's recommended level of IT-related CAPEX?

A. Setting rates that reflected the SITP's proposed amount of IT&D CAPEX in the Rate Year would severely compromise the Companies' ability to continue to provide safe and reliable service. The SITP's recommended budget would prevent the Companies' from advancing strategic IT&D projects as it would require them to solely focus capital spending on maintaining existing core systems. Of particular concern, removing the C2C Enhancements project, the Physical Security and Cybersecurity projects (consisting of Identity and Access Management, Network Security Program, Operational Technical Security, Security Orchestration Automation and Response ("SOAR"), and Security Vulnerability Management Program), the Digital Platforms and Infrastructure Technology ("DPIT") Networks investments, and the Backoffice Refresh project create undue risks to the Companies' ability to provide safe, reliable service. The SITP's removal of these high-value investments will restrict the Companies' ability to make necessary enhancements to its customer information system after completing KEDNY's conversion to

CSS and compromise the Companies' ability to address security risks to its physical infrastructure and information systems with investments that are a continuation of existing in-flight initiatives.

IT&D investments provide the critical backbone to every other function in the Companies, whether it be customer-facing services, critical infrastructure assets, or day-to-day operations. These projects are essential to the Companies' efforts to strengthen physical and digital assets, enhance customer information systems, and maintain the IT&D systems that enable all other systems to operate. The maintenance of the Companies' IT&D systems is absolutely critical to the Companies' ability to safely and reliably serve customers now and in the future. Adoption of SITP's IT&D CAPEX budget for ratemaking purposes would not provide sufficient funds to permit the Companies to successfully manage their IT&D functions.

Q. The SITP (at 63-64) states that it is concerned about the Service Company's ability to meet its budgets and states that if the Commission rejects the SITP's project-by-project cuts then it should consider applying an adjustment to reflect the Companies' historic underspending. Do you have any comments concerning this recommendation?

A. Yes. As shown on the SITP's Exhibit __ (SITP-5), over the period of the current rate plans, FY 2021 through FY 2023, the combined KEDLI and KEDNY average of CAPEX spending for core IT projects was $267.179 million, only 0.6 percent below the average CAPEX budget of $268.675 million. The five-year average of IT&D CAPEX core IT spending during the period FY 2019 to FY 2023 was $214.559 million, only 1.3 percent below the average CAPEX budget of $217.333 million. These modest budget variances completely refute the justification for the SITP's recommended disallowance of nearly 62 percent of the Companies' Rate Year CAPEX budget.

Due to the nature of IT&D projects, it is not unusual for projects to be delayed or accelerated to meet the needs of rapidly changing businesses. The Companies face considerable challenges every year in meeting and managing their IT&D budgets. The Joint Proposal adopted in the 2019

KEDNY and KEDLI Rate Cases recognizes this challenge by requiring the Companies to reconcile their IT&D spending in the aggregate, not at the individual project level. Despite the challenges involved in managing an IT&D budget, the Companies' budget versus actual CAPEX spending over the term of the current rate plans reflects an underspending of only 0.6 percent. Given this performance, it is difficult for the Panel to understand why the SITP believes it is reasonable to reduce the Companies' forecast Rate Year CAPEX budget by nearly 62 percent, or over $190 million. It is inconceivable that the Companies will have a budget shortfall of anything approaching that amount. From the Companies' perspective, the Companies' recent performance in managing their IT&D CAPEX budget would not support any adjustment greater than the 1.3 percent budget variance experienced over the last 5 years.

Q. Regarding OPEX/RTB costs, the SITP (at 63) also expresses concerns about the Companies' ability to meet its OPEX and RTB budgets. Do you have any comments concerning the SITP's stated concerns?

A. Yes. Once again, over the term of the Companies' current rate plan, FY 2021 to FY 2023, the Companies' actual IT&D-related OPEX & RTB costs have totaled approximately $127.4 million in comparison to a budget of approximately $120.1 million, a variance of approximately 6.0 percent.

As the Panel discussed above, these types of budget variances occur because of changes in the timing of projects in a rapidly evolving IT environment. More importantly, a difference of 6.0 percent in IT&D-related OPEX spending cannot be said to justify the SITP's proposed reduction of $66.8 million of the Companies' budgeted OPEX/RTB Rate Year costs of $93.7 million, a reduction of over 71 percent of the Companies' total budget. The SITP's OPEB/RTB expense recommendation would leave the Companies woefully short of the funds needed to manage their IT&D systems effectively.

Q. Do the Companies agree with the SITP (at 24) that the risk of overspending a budget identified in an Investment Estimation Tool ("IET") is a reasonable basis for the SITP to remove a project from the IT&D CAPEX forecast?

A. No. While early-stage cost estimates included in the IET generally allow for a greater variance than at the time of sanctioning, the Companies are subject to a downward-only reconciliation mechanism for Service Company Rents, Net Utility Plant, and Depreciation that shields customers from any overspending above the IT&D revenue requirement approved by the Commission. The reconciliation mechanism also prevents the Companies from recovering more than the revenue requirement when the

Companies spend less than what was initially budgeted. As a result, the risk associated with overspending is wholly borne by the Company.

Q. Do the Companies and Staff agree with continuing the downward-only reconciliation mechanism proposed by Staff (at 67)?

A. Yes, the Companies and Staff agree that the downward-only reconciliation mechanism should remain in effect.

Q. The SITP asserts (at 15) that IETs do not provide enough information to substantiate moving forward with multi-million dollar IT&D investments. Do you agree?

A. No. In the Companies' responses to IR Nos. DPS-457 and DPS-457 Supplemental, which Staff included in Exhibit ___ (SITP-1), IETs were provided for all projects not-yet-sanctioned. The IETs provide key attributes of proposed future investments, including investment descriptions, cost estimates, scope, timing, and the expected benefiting operating company, among other details. While prepared when a project is in conceptual development, a range of criteria are evaluated including risks to achieve, CAPEX, OPEX, and RTB cost considerations, alternatives, and benefits to ensure that IT&D projects that reach the sanctioning stage have been vetted for overall success with the

information available to the project team at the outset. IETs are developed with input from project manager experience, historical cost trends, and labor, hardware, software, and other anticipated expenses. Further, IT&D projects that continue to advance after an IET is prepared are still required to complete both partial and full sanctioning before progressing to the start-up phase.

Q. Are the IETs provided in this case consistent with the information that is utilized by National Grid to develop its IT&D CAPEX budgets each year?

A. Yes. Due to the nature of IT&D projects and the Companies' rapidly changing business and information technology environments, National Grid's IT&D CAPEX budget for any given year will include projects that are in the start-up phase and for which only IETs have been developed, projects that have been partially sanctioned, and projects that have been fully sanctioned. To manage its business efficiently and effectively, National Grid must develop a budget that includes the IT&D CAPEX projects that need to be placed in service in a given year, as well as multi-year investments still under development. Because the IT&D budget serves as the foundation for the Companies' rate filings (which, in New York, utilize a fully forecast rate year), it is unavoidable that a significant

number of projects included in the Rate Year forecast will be in the IET phase, or only partially sanctioned for prior fiscal years. Additionally, the IT&D strategic business plan is developed annually, with a ten-year horizon, and is informed by and inclusive of forecast cost estimates contained in project specific IETs. Clearly, IETs are a necessary and proper part of a successfully functioning budget process and the costs estimates reflected in IETs should be reflected in budgets that form the basis of reasonable utility rates.

Q. Do the Companies currently provide information that allows Staff to review the Companies' IT&D budgets and compare actual IT&D CAPEX to those budgets?

A. Yes. In the 2019 KEDNY and KEDLI Rate Cases, the Commission adopted a Joint Proposal that required the Companies to collaborate with Staff to develop modified and improved IT&D annual and quarterly reporting requirements. The revised process was intended to provide greater transparency into the Companies' IT&D investments, sanctioning, budgeting, and approval process. The new process also seeks to ensure that Staff is provided with executive-level information regarding the status of IT&D-related projects enabling easier monitoring of project status and performance in terms of schedule, scope, and budget for each of the

program categories and individual projects within each category, as well as major projects. The modified quarterly reporting is intended to identify project issues and risks, identify project milestones, and provide details on project governance for each of the individual projects. In furtherance of the revised process to review IT&D investments, the Companies conduct a semi-annual meeting with Staff to discuss overall performance to budget, project progress, and project-specific challenges and benefits.

As a result of the collaboration between the Companies and Staff, the Companies filed a "Report on Information Technology Reporting Format" dated December 22, 2021. A copy of the report is attached hereto as Exhibit ___ (ITDP-1R). Likewise, in accordance with the revised reporting format, the Companies have since filed their annual IT&D capital plans, as well as quarterly reports, which provide Staff with comprehensive information concerning the Companies' IT&D plans, budgets, and results. Copies of the redacted quarterly reports for the period 2021 to 2023 are set forth in Exhibit ___ (ITDP-2R) and copies of the Companies' redacted annual capital plans for FY 2023 and FY 2024 are set forth in Exhibit ___ (ITDP-3R).

Q. Is it the Panel's understanding that the Companies' IT&D reporting satisfies the goals and requirements of the Joint Proposal adopted in the 2019 KEDNY and KEDLI Rate Cases?

A. Yes. Compliance with the IT&D reporting requirements was noted in the Companies' Implementation Plan Update dated October 31, 2022 in the Companies' most recent management and operations audit in Case 18-M-0195 ("2018 Management Audit"). Staff did not challenge the Companies' assertions that they were in compliance with the requirements of the Joint Proposal adopted in the 2019 KEDNY and KEDLI Rate Cases or the 2018 Management Audit. Thus, it is reasonable to conclude that the Companies' reports provide Staff with information sufficient to enable effective monitoring of the Companies' IT&D planning, budgeting, and spending.

Q. Does the SITP recommend that the Companies continue their current reporting?

A. Yes. While the SITP (at 64-67) recommends certain additional reporting requirements that the Panel will address later in this rebuttal testimony, the SITP states (at 65) that the Companies should continue their current reporting.

Q. What is the significance of the extensive IT-related reporting that the Companies have provided to Staff over the term of the current rate plan?

A. The extensive reporting provided to Staff, coupled with the Companies' performance in managing their IT&D CAPEX budget over the term of the rate plan should have provided Staff with a level of confidence that the Companies are able to manage their IT&D CAPEX portfolio closely to the budget. From the Companies' perspective, it is simply not reasonable for the SITP to conclude that because many projects in the Companies' IT&D CAPEX forecast are based on IET estimates they should be subject to complete disallowance.

Q. Do the Companies agree with the SITP's proposed (at 22) removal of Rate Year IT&D projects for lack of a sanction paper?

A. No. The IT&D budgeting and sanctioning process, outlined in the Companies' response to IR Nos. DPS-457, DPS-457 Supplemental, and DPS-943, which Staff included in Exhibit ___ (SITP-1), is a multi-phased process that works across various functions within the IT&D organization, as well as with executive leadership, to develop annual budgets that consider project criticality and costs before sanctioning a project and beginning development; IETs are a foundational step in both the budgeting

and sanctioning processes, but not the only evaluation of a project before it begins. Before an IT&D project is completed and placed in service it must pass through National Grid's extensive review process including partial and full sanctioning. The IT&D reporting requirements agreed to by the Companies and Staff afford Staff the ability to review IT&D projects as they move through the development process.

Q. Does the removal of projects recommended by the SITP include projects that are already in flight?

A. Yes. As discussed earlier, and because the Companies use a fully forecast Rate Year in developing the revenue requirement, the rigorous annual IT&D budgeting exercise is relied on to develop the Companies' Rate Year forecast. Given the complexity and time needed to develop various ongoing IT&D initiatives and projects, many require a multi-year investment before being placed in service. As a result, several projects included in the Companies' Rate Year forecast have been through the annual IT&D budgeting review process, have been sanctioned through FY 2024, and are well underway in development. The SITP's recommendation to remove those projects already in flight would result in stranded, undeveloped projects without funding to continue. The result would have a widespread and detrimental affect on a many IT&D projects

that aim to maintain the Companies' network infrastructure, improve security systems, enhance customer engagement, and ensure safe and reliable service in an ever-changing IT environment.

A. C2C Enhancements

Q. Will removal of the C2C Enhancements project proposed by the SITP (at 31) hinder the Companies' development of a more robust and effective customer information system?

A. Yes. The C2C Enhancements project addresses broader CSS functionality beyond the CRIS-to-CSS conversion underway in KEDNY. This project provides the underlying capability needed to support new, complex billing regulations, energy efficiency programs, net metering, security requirements, and data management, among other functions. Consistent with the IT&D sanctioning procedure, the development team leading the project has been tasked with refining and validating all potential development concepts and to provide business cases for each proposal. To date, those efforts have supported a conclusion that the C2C Enhancements project should continue to pursue security and resiliency enhancements, data management, and batch optimization, among other concepts. Given the significance of this project, it is clearly unreasonable for SITP to disallow the entire project costs because they

may be refined over time. Such refinements are consistent with the normal IT&D budgeting process.

Q. Do the Companies agree with the SITP's assertion that it has not adequately addressed the C2C Enhancements project's dependency on the CRIS-to-CSS conversion (at 32-33)?

A. No. To mitigate the project's dependency on KEDNY's conversion to CSS, the Companies are taking a combined and staggered resource approach between KEDNY's CSS transition and C2C Enhancements projects. The development team, furthermore, has built a delivery timeline that allows for multiple simulated launches of CSS before its placed in service, and before beginning the C2C Enhancements project development. A portion of developer resources tasked with the underlying CRIS-to-CSS conversion effort will transition to the C2C Enhancements project as the conversion nears completion, and in doing so, subject-matter experts will pivot from the initial phase of CSS implementation work to C2C Enhancement efforts in-step with the launch of CSS in KEDNY. Development teams tasked with the CSS conversion and the subsequent C2C Enhancements project are relying on past CSS conversions across four other National Grid operating companies to adopt best practices and lessons learned to ensure timely

delivery. Conversions to CSS by Niagara Mohawk and KEDLI have benefitted over 2.3 million customers and the Companies are leveraging that experience to develop delivery timelines based on real-world experience and to mitigate potential delays before they occur.

B. DPIT Network Infrastructure Project

Q. Please describe the SITP's basis for proposing to remove the Companies' DPIT Network Infrastructure project.

A. The SITP indicated that deficiencies in the project's IET formed the basis of the recommendation to remove the DPIT Network Infrastructure project from the Companies' Rate Year forecast. The SITP (at 48) points to a lack of a "basic level of project conceptualization" in the IET as the basis for its removal.

Q. Do the Companies agree with the SITP's analysis and rationale for removing the DPIT Network Infrastructure project?

A. No. The Companies provided robust responses to IR Nos. DPS-457, which Staff included in Exhibit ___ (SITP-1) and IR Nos. DPS-459, and DPS-850, included in Exhibit ___ (ITDP-5R), specific to the DPIT Network Infrastructure project. In sum, the responses provide the project's anticipated risk margin, cost consideration, long-term

maintenance and staffing plans, an overview of historic spend associated with the DPIT Networks investment in prior fiscal years, the budgeting process deployed to forecast costs in the Rate Year, and, most importantly, DPIT Network Infrastructure project sanction papers with detailed information for FY 2023 and FY 2024 as accompaniments to the IET for the forecast Rate Year investment for this multi-year, ongoing effort.

As budgeted and sanctioned investments, and in line with IT&D budgeting and sanctioning procedures, CAPEX for DPIT Networks Infrastructure investments was $20.688 million in FY 2023 and $17.837 million in FY 2024. Cancelling these investments in the Rate Year would have far-reaching impacts on the Companies' ability to maintain its IT&D network infrastructure used across nearly all business units, chief among them customer, operations, and cybersecurity systems. The DPIT Networks Infrastructure investments enable wireless connections across the Companies' field and operations teams, provide updated telephony systems, and provide the Companies with a secure network that protects its assets, facilities, and operations from vulnerabilities. As important, DPIT Network Infrastructure investments enable collaboration and service in an increasingly digital environment that is in constant

change, and require a constant investment to maintain its resiliency and effectiveness.

C. Physical and Cybersecurity

Q. Please elaborate as to how the reduction in the IT&D CAPEX budget recommended by the SITP would have a detrimental impact on the Companies' ability to maintain physical security and cybersecurity.

A. The Companies' response to IR Nos. DPS-879 and DPS-942, provided here in Exhibit ___ (ITDP-5R), explained the process of evaluating alternatives before concluding that the proposed physical security investments should progress based on their overall significance. In deciding to proceed with these projects, the Companies reviewed annual vulnerability assessments, dedicated threat intelligence team evaluations, and information from multiple external information groups and, based on this information, designed these projects to specifically address identified physical and cybersecurity vulnerabilities.

Removal of the projects presents an undue risk to the Companies' facilities, and ultimately, their ability to safely serve customers. A reduction in the IT&D CAPEX budget, as advocated by the SITP, will prolong the Companies' exposure to identified risks and reduce their

ability to effectively implement cybersecurity controls that improve identification, protection, detection, and response capabilities through continuous improvement to keep pace with emerging threats, and increased threat actor activity, and would hinder the Companies' ability to respond to and mitigate the impact of potential cyber events.

D. Backoffice Refresh

Q. Please explain the basis for the SITP's removal of the Backoffice Refresh project from the Companies' Rate Year forecast.

A. The SITP (at 27) explained that the information the Companies provided in the Backoffice Refresh IET pointed to a lack of experienced internal and external resources capable of delivering the project and a lack of overall project detail sufficient to support its inclusion in the Companies' revenue requirements.

Q. Do the Companies agree with the SITP's basis for removing the Backoffice Refresh project?

A. No. Resourcing project teams with experienced personnel is always a risk that is addressed at the conceptual phase of development. The Backoffice Refresh team engaged Pricewaterhouse Coopers ("PwC") to mitigate resourcing risk and to help develop cost forecasts, and in so

doing, leveraged a capable partner at the project's conceptual phase to develop a strategic roadmap that addresses those risks. Over the past four years, with PwC's support, the project team has prepared multiple business cases with benefits, costs, duration, resourcing need, governance models, and team structure. The current structure of the total Backoffice Refresh project utilizes a compartmentalized approach of discreet initiatives, but similar to the initial, sanctioned, iteration of the project that was paused because of the COVID-19 pandemic, the current project will advance beyond the IET phase and will be sanctioned in accordance with IT&D sanctioning and budgeting processes discussed throughout this testimony.

Q. Is it the Companies' position that removal of the Backoffice Refresh project presents system-wide risk to finance, payroll, human resourcing, and supply chain functions?

A. Yes. As discussed in the IT&D Panel's initial testimony in these proceedings, the Companies currently utilize an enterprise resource planning ("ERP") solution that was implemented over ten years ago, in 2012. The SAP ECC 6.0 ERP is confronting the end of its useful life and SAP has indicated that it will no longer support this platform after 2025. Allowing critical back-office platforms to go off standard support from

SAP risks exposure to cybersecurity vulnerability and evolving regulatory and tax obligations to the detriment of customer and employee data protection, as well as software incompatibility across other systems used by the Companies. Such risks far outweigh those raised by the SITP.

IV. IT&D Reporting

Q. Do the Companies agree with the SITP's proposal (at 66) that KEDNY and KEDLI should notify Staff as physical or cybersecurity projects become partially or fully sanctioned, are merged with other projects, or are discontinued?

A. No. As discussed previously, following Commission approval of the Joint Proposal in the 2019 KEDNY and KEDLI Rate Cases, the Companies collaborated with Staff to update their reporting process to improve, simplify, and provide a clearer view of the Company's entire IT&D investment portfolio. As a result of that collaboration, the Companies provide quarterly capital investment reports to Staff that include an update on IT&D investment plans, descriptions of program investment categories, and a comparison of budgeted to actual spending. Each quarterly report includes a detailed breakdown of individual project spending by portfolio compared against the budgeted spending for each of those investments.

The updated reporting format is both effective and comprehensive. Requiring the Company to provide information on physical and cybersecurity projects in shorter reporting cycles and in near real-time would be an unduly burdensome duplication of efforts that are already undertaken in a manner that permits Staff to provide the Companies with meaningful input concerning their ongoing IT&D projects and programs, including those that involve cybersecurity and physical security.

Q. Do the Companies agree with the SITP recommendation (at 66-67) that the Companies be required to engage with Staff to improve and enhance the IT&D rate case reporting process prior to its next rate filing?

A. As described above, the Companies' current framework for providing IT&D capital investment reports was a collaborative effort developed with input from Staff. The comprehensive annual budgeting process ensures that the Companies only include projects in their rate filings that have been properly vetted to support the delivery of safe and reliable service to customers. As part of these rate proceedings, in addition to their testimony describing individual IT&D projects, the Companies have provided responses to over ninety requests for information related to its IT&D projects and proposals, including explanations describing the

budgeting, sanctioning, and organizational structure it relies on to support projects included in its revenue requirements. The Companies believe the current process developed through collaboration with Staff provides a comprehensive view of the Companies' IT&D decision making and should remain unchanged.

V. Conclusion

Q. Does this conclude the IT&D Panel's rebuttal testimony?

A. Yes.

List of Exhibits

Exhibit ___ (ITDP-1R)	Report on Information Technology Reporting Format
Exhibit ___ (ITDP-2R)	Redacted quarterly Information Technology Capital Investment Reports for the period October 2021 through June 30, 2023
Exhibit ___ (ITDP-3R)	Information Technology Capital Investment Plans for fiscal year ("FY") 2023 and FY 2024
Exhibit ___ (ITDP-4R)	Excerpt from the Companies' Implementation Plan Update
Exhibit ___ (ITDP-5R)	Responses to information requests ("IR") provided in these proceedings and cited in this rebuttal testimony

Exhibit ___ (ITDP-1R)

Report on Information Technology Reporting Format

The Brooklyn Union Gas Company d/b/a National Grid NY
KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 and 19-G-0310

Report on
Information Technology Reporting Format

SUBMITTED TO:
NEW YORK STATE PUBLIC SERVICE COMMISSION

December 22, 2021

nationalgrid

Report on
National Grid's Information Technology Reporting Format

I. Background

Pursuant to Section IV.6.5 of the Joint Proposal approved by the New York State Public Service Commission ("Commission") on August 12, 2021 in Case 19-G-0309 *et al.*,[1] The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (together "National Grid" or "the Companies") submit this Report on National Grid's Quarterly and Annual Information Technology Reporting Format ("Report"). The Joint Proposal provides that the Companies are to collaborate with Department of Public Service Staff ("Staff") to modify and improve upon the format of quarterly and annual information technology ("IT") reporting Staff currently receives from the Companies' Upstate New York affiliate.[2] The Joint Proposal further requires that the Companies file a report with the Commission Secretary no later than 90 days prior to the start of Rate Year Three of the rate plan describing the reporting requirements agreed to through the collaborative process between Staff and the Companies.[3] This Report describes the Companies' planned reporting format in accordance with the requirements in Section IV.6.5 of the Joint Proposal.

II. Joint Proposal Requirements

The Joint Proposal sets forth general requirements for quarterly and annual IT investment reporting by the Companies. Among the goals for a modified reporting format are to enable easier monitoring and understanding by Staff of individual IT project status and progress, as well as providing a clear view of the entire IT investment portfolio. This would include greater transparency of the Companies' IT investments, sanctioning, budgeting and approval process, while "avoiding placing an undue burden on both the Companies to compile the information and on Staff to review the information."[4] The Joint Proposal provides that the modified reporting format should streamline quarterly reporting to present strategic, executive-level information on IT project status to make it easier to monitor project status and progress in terms of schedule, scope, and budget for each of the investment categories (as well as for individual projects within each category) and for major IT projects (e.g., transformation investments, such as Gas Business

[1] Case 19-G-0309, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service*, Case 19-G-0310, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service*, and Case 18-M-0270 *Petition for Approval Pursuant to Public Service Law Section 113(2), of a Proposed Allocation of Certain Tax Refunds Between KeySpan Gas East Corporation d/b/a National Grid and Ratepayers*, Order Approving Joint Proposal as Modified, and Imposing Additional Requirements (issued and effective August 12, 2021). Hereinafter, the Joint Proposal approved in Cases 19-G-0309 and 19-G-0310 is referred to as the "Joint Proposal."

[2] Staff currently receives quarterly and annual IT reports from Niagara Mohawk Power Corporation d/b/a National Grid ("Niagara Mohawk") pursuant a requirement from Niagara Mohawk's most recently approved rate case. *See* Cases 17-E-0238 and 17-G-0239, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service*, Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans (issued and effective March 15, 2018), Niagara Mohawk Joint Proposal, Section IV.8.6.

[3] Joint Proposal, Section IV.6.5. Rate Year Three begins April 1, 2022.

[4] Joint Proposal, Section IV.6.5.

Enablement, Advanced Metering Infrastructure, Customer Information System, etc.). For quarterly IT reports, the modified format should, at a minimum:

- Identify project issues and risks
- Identify project milestones
- Provide details on project governance for each of the individual projects.

For the annual IT reports, in addition to the requirements of the quarterly reports, the Companies should provide more detailed descriptions of the status of projects, expanding as necessary to address any issues identified.

Because the IT portfolio is managed at the National Grid USA Service Company level for the benefit of all the US operating companies, IT reporting is to be consistent across all New York jurisdictions and show the full IT portfolio. In addition, variance reporting (including root cause determination of variances and corrective actions), allocations to each company, and status of savings associated with retirement of legacy systems being replaced by new IT investments would continue to be included.

III. Current Reporting Format

The current IT quarterly report provides an explanation of the top ten projects by annual budget in the following investment categories:

(1) Regulatory, Legal and Compliance Mandates.
(2) Business Programs and Projects.
(3) Technology/Infrastructure Foundation and Reliability.

For any of the top ten projects in the foregoing categories with a budget-versus-actual variance greater than 10 percent, the report provides an explanation of the reason for the variance. The quarterly report also provides a listing of all projects delivered in that quarter, all projects discontinued, all new demand, and a budget exception report.

The final report of the year is the annual report, In addition to the information included in the quarterly reports, the annual report format currently includes a listing of all projects delivered in the fiscal year, and also is accompanied by the sanction papers for all approved projects over $1 million.

IV. Collaboration with Staff and Planned Reporting Format

On October 13, 2021, the Companies met with Staff to review the status of the Companies' IT investment program, and to discuss a proposed modified format of the periodic IT investment reports. At that meeting, Staff provided feedback on the Companies' proposed quarterly and annual IT reporting formats. Based on that feedback, the Companies plan to format future IT investment reports as follows.

First, the Companies intend to provide their Annual IT Investment Plan at the beginning of each fiscal year as is currently the practice with Niagara Mohawk. This Annual IT Investment Plan provides a list of proposed IT investments with associated budgets and planned in service dates for the forthcoming year.

As discussed at the October 13 meeting, the current quarterly and annual reporting format is very segmented and detailed but does not provide Staff much insight as to how National Grid is delivering the IT Strategy. The Companies therefore plan to modify the report format to include information in the format used in National Grid's internal IT management reporting to provide Staff with the same level of insight provided to senior IT management. The proposed reporting would be at the portfolio level spanning six separate areas of IT:

1) Security
2) Information and Operations
3) Customer
4) Group Functions
5) Gas
6) Electric

Each report would include the following information for each portfolio:

- Financial Section – Financial Summary, Details of Opex & Capex Spend for the portfolio as a whole, a financial forecast.
- Project Delivery Section – A commentary with Key Challenges, a Mobilization Summary as to totals of completed, inflight and pipeline projects, a Delivery Methodology summary.
- Priority Programs Summary – Status of the programs, Red/Amber/Green status, and Budget.

Included in Attachment 1 is the planned format of the quarterly IT report. The modified format is intended to address concerned voiced by Staff regarding the level of detail being provided in the IT portfolio reports. The Power Business Intelligence report for each IT portfolio comparing actual-versus-budget for the fiscal year to date, together with a year-end forecast-versus-budget, is intended to provide the necessary detail for Staff to effectively evaluate the status of National Grid's IT investment program. Information from the portfolio report will provide the strategic insight, and the individual project detail from the Power Business Intelligence report will enable a complete picture of project performance.

The Companies will continue to provide a list of all projects placed into service by fiscal year end. However, the Companies are proposing not to send the large volume of sanction papers that have been sent in the past. Rather, the Companies think it would be more efficient to send only those papers specifically requested by Staff. At this stage, the Companies also are not proposing to include a Budget Exception Report, or a listing of projects placed in service at each quarter end.

V. Next Steps

The Companies plan to file their first quarterly IT report using the modified format within sixty days of the end of the third quarter of FY 2022 (December 31, 2021). The Companies expect to continue to coordinate with, and take feedback from, Staff in the future to further evolve the report format to achieve a product that is easy for Staff to use to better understand the Companies' IT capital investments and that is not unduly burdensome for National Grid to produce.

National Grid

IT & Digital Reporting Format Modifications

December 2021

nationalgrid

Proposed Format

National Grid will leverage the Portfolio reviews to provide a quarterly summary to DPS Staff.

National Grid will report for each of Gas, Electric, Central Functions, Customer, Security and Information & Operations (I&O), and provide a Power BI Dashboard showing budget vs. actual performance to date and year-end forecast vs. budget for the six areas.

The Portfolio Dashboard gives an overview of project delivery and costs.

Portfolio Dashboard

Financial Section	Priority Programs
Summary	Status
Opex & Capex Spend	Red/Amber/Green status
Forecast	Budget
Delivery	
Commentary – Key Challenges	
Mobilization Summary	
Delivery Methodology	

Samples from the four IT business areas are shown on the next pages.

The Companies would add I&O and Cyber to these.

These are illustrative and show the type of information that would be provided.

Note these reports are not in final form and the format may evolve further.

Executive Summary - Group Functions

Group Functions continues to be focused on the Rhode Island Divestiture and delivering the FY22 portfolio. All IT Foundations projects in the US are active and all US Core Q2 planned start-ups are active. Significant progress has been made in financial training for PMs in accruals and forecasting to improve financial record keeping. A major area of risk continues to be contention with RI in both resources and systems.

Financial Summary

Totex

Budget ($M)
69.2

Forecast ($M)
60.9

Act. YTD ($M)
13.2

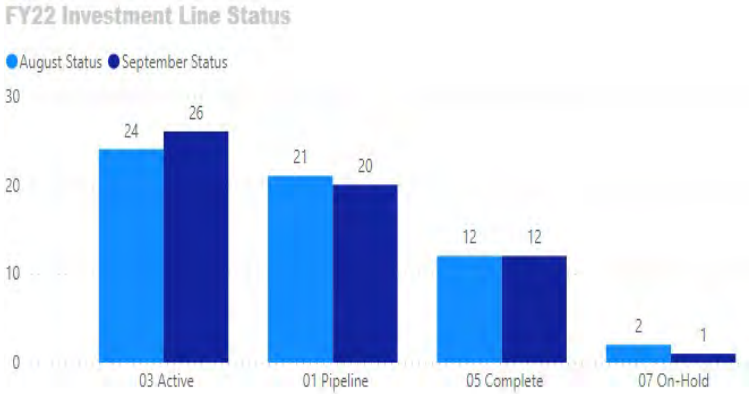
Priority Programs

Program	Status	Overall RAG	Totex Budget ($M)	Funding Type
HR & Payroll Service Pack	Active D&T		1.4	Regulated
401(k) Admin Changes	Active D&T		1.0	Regulated
FTOM: SRM to Ariba	Active Start-Up		5.0	Regulated
FTOM: Closing Simplification	Active Start-Up		1.7	Regulated
SAP Portals to Fiori	Start-Up		0.6	Regulated

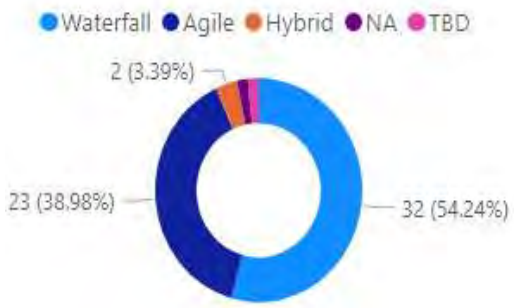
Key Challenges

Challenge	Escalation Point
RI Divestiture – Resource allocation from US GF requires backfill. Multiple dependencies across the portfolio, Delayed Day 1 mitigates some issues.	Joel Semel
Accrual & Forecasting accuracy has negatively impacted financials. Training has been developed, and multiple sessions delivered, to minimize errors in financial reporting going forward.	Joel Semel

Mobilization Summary



Delivery Methodology



*please see financial overview slide for further information

**actuals based on April through August 21 data

Executive Summary - Customer

To achieve Customer's North Star vision of improving customer engagement and reducing the cost to serve we are taking a "Digital first" approach. This means (i) improving the customer-facing systems to drive up self-service (ii) reengineering back office processes to generate internal efficiencies (iii) Addressing the legacy technology that limit our speed to value (iv) and building a data layer that provides real-time information to both customers and customer service representatives.

Financial Summary

Totex

Budget ($)

51.3m

Forecast ($)

51.9m

Actuals *YTD* ($)*

21.3m

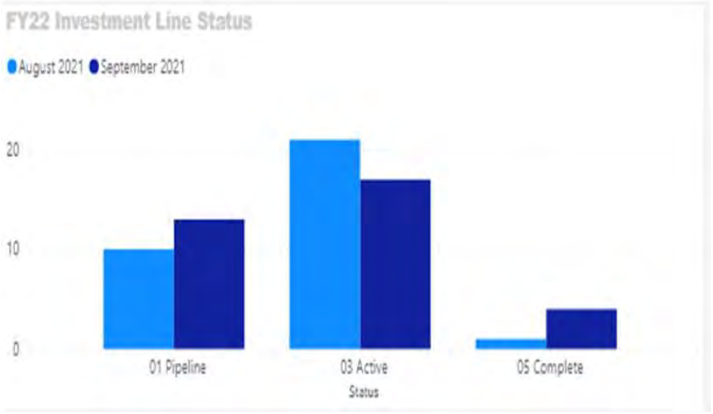
Priority Programs

Program	Status	RAG	Budget ($m)	Funding Type
Unified Web Portal	*Inflight*		*11.6m*	*IT funded*
CIS: CRIS Migration	*Pipeline*		*NA*	*TBD*
Customer Data Platform	*Sprint 0*	*?*	*1.3m*	*IT Funded*
AMI	*Sprint 0*	*?*	*NA*	*Business Funded*
Zero-Based Redesign	*Pipeline*	*?*	*NA*	*Business Funded*
RI Transition	*Start-up*		*NA*	*Not Funded in Customer*
Customer Ecosystem Stabilization	*Inflight*		*7.6m*	*IT Funded*
Clean Energy	*Validation*		*NA*	*Business Funded*

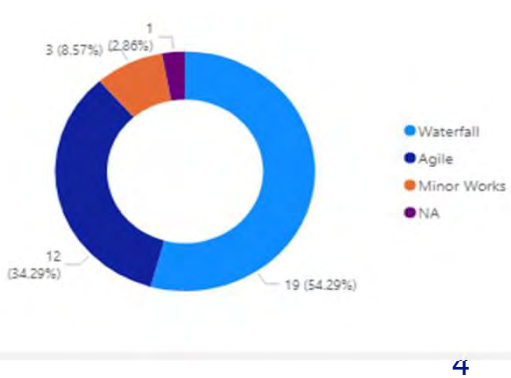
Key Challenges

Challenge	Escalation Point
Product Model Resources and Training	*Anil/Tony*

Mobilization Summary



Delivery Methodology



Executive Summary - GBU

The IT US Gas and Capital Delivery portfolio is aligned to the strategic focus areas of the Gas Business through thematic strategic programs like Gas Business Enablement to support the transformation of the future gas business and execution of tactical initiatives that are tied to shorter term business goals. The business outcomes for these programs and projects focus on: Implementing capabilities not previously available to Gas, Improving E2E capital efficiency while improving safety and reliability, Delivery of field performance in efficiency, safety, compliance, and customer satisfaction, Implementing standardized processes, tools & technologies, Standardize work practices and processes in the field using enterprise-wide systems and technologies.

FY22 Budget was approved for 14 projects out of 17 proposed. 2 of the approved projects will be delayed/differed to later in FY22 or FY23 due to business changes. Budget allotted to those projects is being reallocated to PIF, NSRMS(New Storms) and DR(Demand Response) projects which will provide value to the busines(See financial overview)

Financial Summary

Totex

Budget ($)

17.4m

Forecast ($)

18.5m

Actuals YTD ($)

3.1m

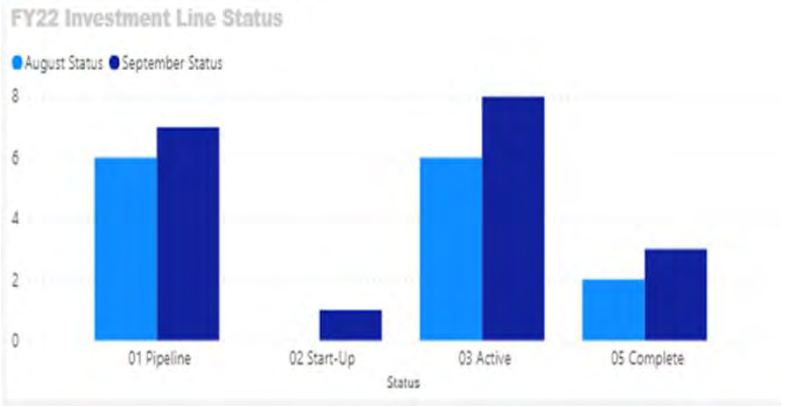
Priority Programs

Program	Status	Overall RAG	Budget ($m)	Funding Types
GBE	Inflight		-	Business Funded
LNG Asset & Work Mgmt.	Inflight		3.003	
NRG GIS Replacement	Inflight		4.175	
Primavera P6 Upgrade	Closure		.484	
NSRMS	Start-Up		2.026	
Urbint – Safety and Risk AI for field	Pilot			
Gas Storm Hardening Phase 2	Inflight		.743	
Gas App Rationalization	Inflight		.681	

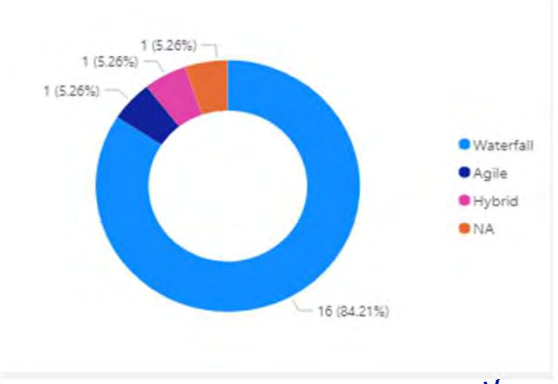
Key Challenges

Challenge	Escalation Point
Consistent challenges with sanction paper approvals due to RTB validation with service owners, and I&O approver declining papers, causing a restart and delay in project starts.	Helen Smith/ PMO
Delay in receiving final invoices extends project closure by at least 3 months	Strategy Office/ PMO

Mobilization Summary



Delivery Methodology



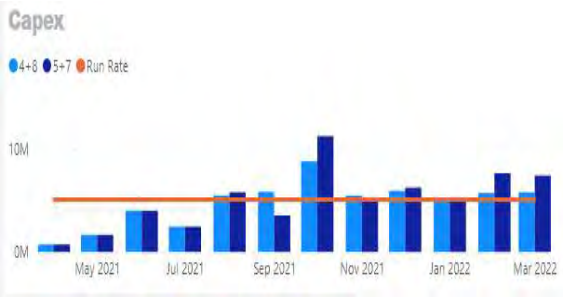
*please see financial overview slide for further information

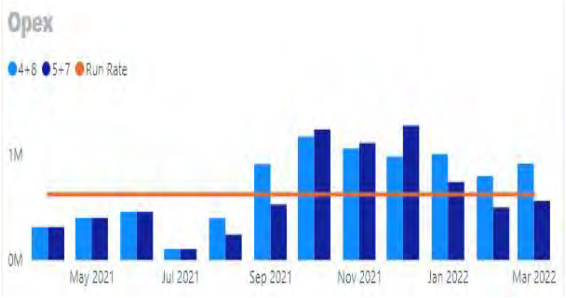
**actuals based on April & May 21 data

Executive Summary - EBU

- RI Divestiture: The January 1st Day 1 planning date has been updated based upon the agreed upon procedural schedule. The planned date for regulatory approval is February 25, 2022, which would lead to a Day 1 cutover in early March. All streams currently on track for Day 1 following closure of key Finance, HR and Supply Chain decisions.
- GIS Phase 1 Go-Live: Successfully completed on September 6th , which enables new types of equipment such as Electric Vehicle Charging Stations, Energy Storage, Inverters and Substation Equipment to be available for use in GIS, and interface downstream systems/processes
- Active Resource Integration (ARI): Product received approval from electric leadership at the August 30th Digital Change Board Meeting to move forward; *validation work expected to start on 11/1*.
- Discussions underway with Business around Digital initiatives (ICU Out and GridMD) slated for FY22 yet are still pending approval from the Digital Change Board.
- Network & Telecom IT/OT Operating Model, Transformation Roadmap and Savings Opportunities Analysis on target to be delivered in October

Totex
Budget ($)*
64.2m
Forecast ($)
68.5m
Actuals YTD ($)
16.2m

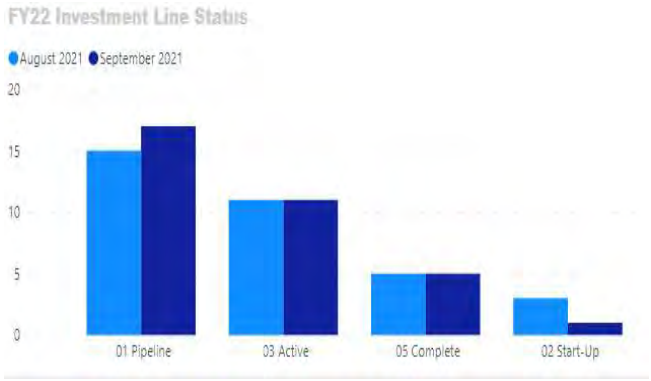




Key Challenges

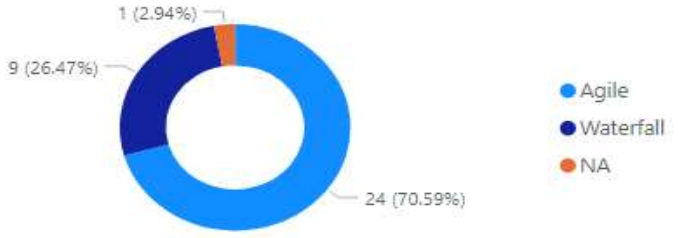
Challenge	Escalation Point
MA ROE OPEX Challenge	Senior Leadership (NELT)
Digital ways of working require ramp up with resources, processes and technology.	CIO
Significant change across the organization and number of initiatives pulling on the same business and IT resources.	Senior Leadership

Mobilization Summary



Priority Programs

Program	Status	RAG	Budget Totex (FY22)	Funding Types
GridMod	Inflight		35.5	Business/ Regulatory
ADMS	Inflight		30.1	Business/ Regulatory
EBU Digital Programs	Inflight		42.6	Self-Funding/ Regulatory
Electric MV90 Upgrade	Inflight		1.8	Regulatory
FCS Upgrade	Inflight		1.8	Regulatory
Energy Accounting Migration	Inflight		2.5	Regulatory
Transmission Outage	Delivered	Actuals>10 %	0.8	Regulatory
Price Transmission	Initiation		1.9	Regulatory
MTS Retail Settlement	Initiation		1.7	Regulatory
Corporate PI	Initiation		1.3	Regulatory
CYME Gateway	Initiation		1.1	Regulatory



*please see financial overview slide for further information

Power BI Reports

The following are samples of Power BI reports National Grid has developed. The next four pages illustrate the type of information the Companies can provide from a total portfolio perspective down to individual project performance.

The report would provide an overall portfolio review of year-to-date performance to budget and year-end forecast to budget for each portfolio similar to what appears on page 8. A portfolio view with information similar to that shown on page 9 would also be provided.

Budget versus actual view for each project would be provided as shown on page 10.

Finally, the report would include each project's actual and forecasted spend profile similar to that shown on page 11.

Although there would be numerous pages the Companies can readily automate production and provide a consistent report. The Power BI reports would provide a comprehensive and transparent presentation of the IT program from the portfolio level down to individual project level and enable review down to an appropriate level of detail to be able to understand overall performance and identify areas of concern.

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US IT Investments
FY22 4 + 8 2021

MTD | YTD | FY

Rollup: USIT

Fcst / QRF: FY22 4 + 8 20 / FY22 3 + 9

CIO / BP: All / All

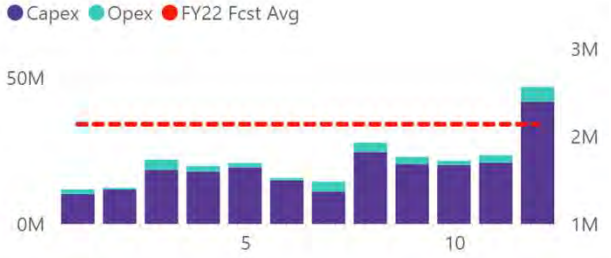
Program: All

Portfolio | Project

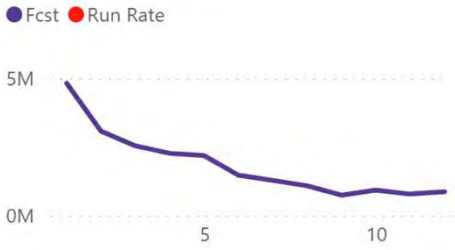
Portfolio	Capex BUD	Capex QRF	Capex Fcst	Capex Var	Opex BUD	Opex QRF	Opex Fcst	Opex Var	Totex BUD	Totex QRF	Totex Fcst	Totex Var
Customer Operations	31.8M	32.1M	2.1M	29.7M	4.5M	6.8M	0.4M	4.1M	**36.3M**	**38.8M**	**2.5M**	**33.8M**
Infrastructure	60.1M	61.6M	1.9M	58.2M	12.1M	16.3M	0.1M	11.9M	**72.2M**	**77.9M**	**2.0M**	**70.2M**
IT Projects(CTA)	12.0M	10.5M		12.0M	10.9M	1.2M		10.9M	**22.9M**	**11.7M**		**22.9M**
NE & US Electric	18.3M	18.4M	2.2M	16.1M	3.0M	4.7M	0.9M	2.1M	**21.3M**	**23.2M**	**3.1M**	**18.1M**
NY & US Gas	13.6M	13.4M	5.4M	8.2M	3.0M	3.7M	0.8M	2.2M	**16.6M**	**17.0M**	**6.2M**	**10.4M**
Security	30.4M	30.5M	7.2M	23.2M	6.0M	6.1M	0.0M	6.0M	**36.4M**	**36.6M**	**7.2M**	**29.2M**
US Group Functions	53.9M	50.1M	3.1M	50.8M	15.3M	13.4M	1.0M	14.2M	**69.2M**	**63.5M**	**4.1M**	**65.1M**
Total	**220.2M**	**216.5M**	**22.0M**	**198.2M**	**54.6M**	**52.2M**	**3.2M**	**51.4M**	**274.8M**	**268.7M**	**25.2M**	**249.6M**
Customer Digital	12.3M	12.1M		12.3M	2.9M	0.7M	0.5M	2.4M	**15.2M**	**12.8M**	**0.5M**	**14.7M**
EBU Digital	38.8M	38.6M		38.8M	4.1M	3.4M		4.1M	**42.9M**	**42.0M**		**42.9M**
Total Digital	**51.1M**	**50.7M**		**51.1M**	**7.0M**	**4.1M**	**0.5M**	**6.6M**	**58.1M**	**54.9M**	**0.5M**	**57.7M**
All Other		-2.5M			-4.6M	0.7M		-4.6M	**-4.6M**	**-1.7M**		**-4.6M**
Total	**271.3M**	**264.8M**	**22.0M**	**249.3M**	**57.0M**	**57.0M**	**3.7M**	**53.3M**	**328.3M**	**321.8M**	**25.7M**	**302.7M**

FY22 Fcst Chart

FY21 Actuals Chart



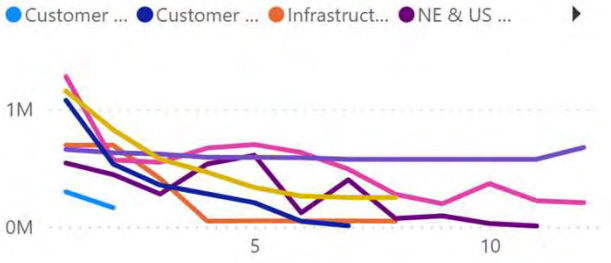
Capex MTD Chart



Opex MTD Chart

Totex YTD Chart

Totex MTD Chart



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US IT Investments
FY22 4 + 8 2021

Show BP

Project Type: Capex

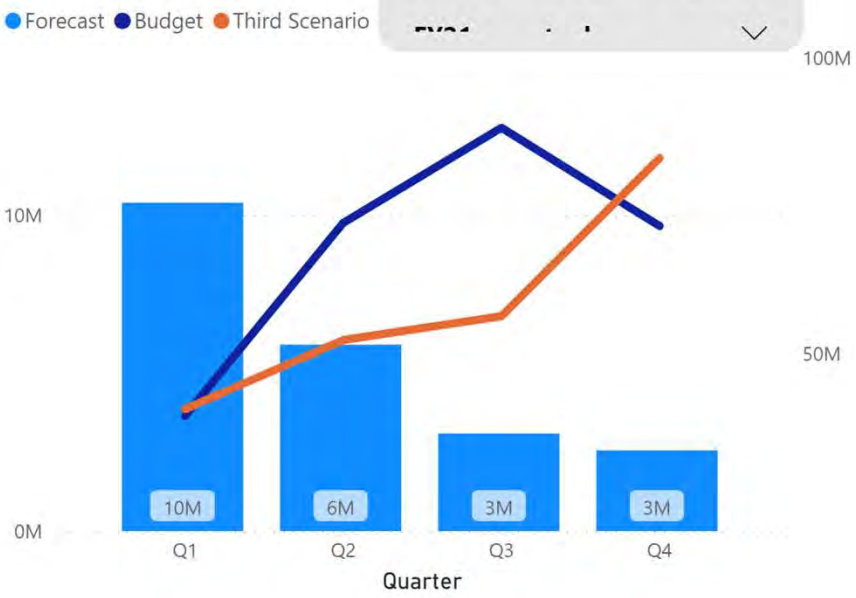
Rollup: USIT

Scenario: FY22 4 + 8 2021

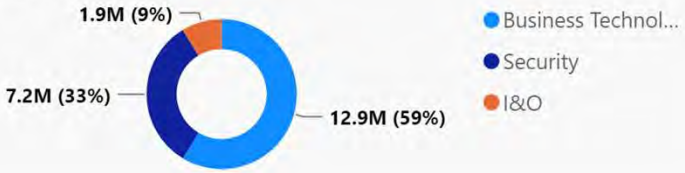
Portfolio: All

CIO	Q1	Q2	Q3	Q4	Total	BUD	Funding	RRATE
US Group Fun...	2.1M	0.7M	0.4M		**3.1M**	53.9M	57.4M	
Infrastructure	1.7M	0.1M	0.1M		**1.9M**	60.1M	67.2M	
NE & US Electric	1.0M	1.1M	0.1M	0.0M	**2.2M**	57.1M	57.1M	
Security	1.9M	1.7M	1.7M	1.8M	**7.2M**	30.4M	30.4M	
Customer	1.7M	0.5M			**2.1M**	44.1M	44.1M	
IT Projects(CTA)						12.0M	1.9M	
NY & US Gas	2.1M	1.8M	0.8M	0.7M	**5.4M**	13.6M	13.7M	
All Other								
Grid Mod								
RI PPL								
Total	**10.4M**	**5.9M**	**3.1M**	**2.6M**	**22.0M**	**271.3M**	**271.9M**	

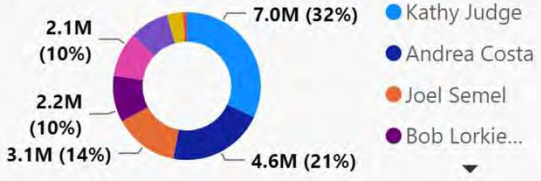
Select third scenario for graph



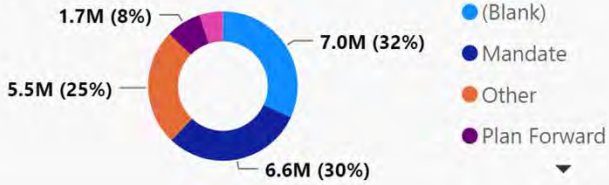
by Portfolio



by BP



by Focus Area



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US IT Investments
FY22 4 + 8 2021

Project Type	Rollup	Scenario 1 / Scenario 2	Period	CIO / BP	Inv ID
Capex	USIT	FY22 4 + 8 20 FY22 3 + 9	7	All All	All

MTD | YTD | FY

CIO	Scen 1	Scen 2	Var	Bud	Var
Total	**1.3M**	**33.3M**	**32.0M**	**33.1M**	**31.9M**
US Group Functions	0.2M	5.5M	5.3M	5.8M	5.7M
Infrastructure	0.0M	9.3M	9.3M	9.9M	9.9M
NE & US Electric	0.1M	9.5M	9.4M	8.0M	7.9M
Security	0.6M	3.5M	3.0M	3.4M	2.8M
Customer		3.3M	3.3M	3.0M	3.0M
IT Projects(CTA)		1.4M	1.4M	1.9M	1.9M
NY & US Gas	0.4M	1.2M	0.8M	1.1M	0.7M
All Other		-0.4M	-0.4M		

FY / Scenario	Q1	Q2	Q3	Q4	Total
FY22 4 + 8	10.4M	5.9M	3.1M	2.6M	**22.0M**
FY22 3 + 9	39.3M	75.4M	83.2M	66.8M	**264.8M**
budget	39.5M	72.1M	88.2M	71.6M	**271.3M**

INV ID	Project Description	Scen 1	Scen 2	Var	Bud	Var
Total		**1.3M**	**33.3M**	**32.0M**	**33.1M**	**31.9M**
3617	Physical Security	0.6M		-0.6M		-0.6M
3617B	DNY Capitalized Asset Replacement & Installation		0.1M	0.1M		
3617D	UNY Capitalized Asset Replacement & Installation		0.3M	0.3M		
3617H	Physical Security - KEDLI - Equip Replacement		0.1M	0.1M		
3617I	Physical Security - MASS Electric - Equip Replacement		0.1M	0.1M		
3617J	Physical Security - Boston Gas		0.1M	0.1M		
3617K	Physical Security - Colonial Gas		0.0M	0.0M		
3617L	Physical Security - RI Electric		0.0M	0.0M		
3617M	Physical Security - RI Gas		0.0M	0.0M		
3617N	Physical Security - NIMO EGT		0.0M	0.0M		
3683USN	SOAR - Security Orchestration Automation and Response		0.7M	0.7M		
3870A	F&A and Implementation - Vision FM Replacement		0.2M	0.2M	0.0M	0.0M
3932	Customer Contact Center / SDC Technology Upgrade Implement Solution				0.0M	0.0M
4481	US MDS-Energy Accounting System (EAS) migration to Wholesale Settlement Application (WSA)				0.0M	0.0M
4483	US Control-Transmission Outage Application (iTOA) Upgrade to HTML 5 Web				0.0M	0.0M
4562	US SAP: Business Warehouse (BW) Consolidation to HANA Enterprise Cloud (HEC)		0.1M	0.1M	0.0M	0.0M
4704R	NY REV Street Lighting Delivery				0.0M	0.0M
4750	Customer Exp. Transformation technology program				0.4M	0.4M
4769	Gas Storm Hardening Using Remote Service Valves (RSV's)				0.0M	0.0M
4808	Asset Management System for LNG	0.2M	0.2M	0.0M	0.2M	0.1M

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Filters

Showing top 25 projects, click here to show all

CIO	Project Type	Rollup	Project Manager
All	Capex	All	All
	Business Partner	Portfolio	SDM
	All	All	All

INV ID	Project Description	Portfolio	1	2	3	4	5	6	7	8	9	10	11	12	Total
Total			**4.8M**	**3.1M**	**2.5M**	**2.3M**	**2.2M**	**1.5M**	**1.3M**	**1.1M**	**0.7M**	**0.9M**	**0.8M**	**0.9M**	**22.0M**
3617	Physical Security	Security	0.6M	0.6M	0.6M	0.6M	0.6M	0.6M	0.6M	0.6M	0.6M	0.6M	0.6M	0.7M	**7.0M**
4898	Asset Management System for LNG	Business Technology	0.8M	0.3M	0.3M	0.4M	0.2M	0.4M	0.3M	0.2M	0.2M	0.3M	0.2M	0.2M	**3.8M**
5497	Lease Accounting Updates and CCM - Phase 2	Business Technology	0.3M	0.3M	0.3M	0.3M	0.2M	0.2M	0.2M	0.2M					**1.8M**
4481	US MDS-Energy Accounting System (EAS) migr...	Business Technology	0.1M	0.3M	0.1M	0.3M	0.5M								**1.3M**
5380A	Blackline: US Implementation	Business Technology	0.4M	0.3M	0.2M										**0.9M**
5160	NY Stack VDER and Portal Integration	Business Technology	0.2M	0.2M	0.2M	0.2M	0.1M								**0.9M**
4769	Gas Storm Hardening Using Remote Service Val...	Business Technology	0.1M	0.1M	0.1M	0.1M	0.3M	0.1M							**0.8M**
5274A	New Storms Respose Management System Imp...	Business Technology	0.2M	0.1M	0.1M	0.1M	0.1M	0.1M	0.1M	0.0M					**0.8M**
6046	SD WAN FY21 Expansion	I&O	0.3M	0.3M	0.1M										**0.7M**
4987	SOE (Windows) Upgrade and Device Refresh	I&O	0.2M	0.2M	0.2M										**0.6M**
5750	Directory Services - US	I&O	0.1M	0.1M	0.1M	0.0M	0.0M	0.0M	0.0M	0.0M					**0.6M**
5877A	Intelligent Substation Design	Business Technology	0.0M	0.0M	0.0M	0.2M	0.1M	0.1M	0.1M	0.0M	0.0M	0.0M			**0.6M**
5795F	CDG Net Metering	Business Technology	0.1M	0.1M	0.1M	0.1M	0.0M	0.0M							**0.4M**
6066	FCP Upgrade for CSS - Plan Forward	Business Technology	0.4M												**0.4M**
5987	Smart City Phase II	Business Technology	0.3M	0.0M											**0.3M**
5889A	Control Room Event Recording	Business Technology	0.1M	0.1M	0.1M										**0.3M**
5842A	US SAP_ GRC Upgrade	Business Technology	0.2M												**0.2M**
5345B	Bill Print and Bill Composition Implementation	Business Technology	0.1M	0.0M	0.0M										**0.2M**
3683USH	Network Security Improvements	Security	0.1M	0.1M	0.0M										**0.2M**
5841A	Pension Administrator Replacement	Business Technology	0.0M	0.0M	0.0M	0.0M									**0.1M**
5486	AVLS Modem Reconfiguration	Business Technology	0.1M												**0.1M**
5843A	SAP Platform Upgrade: OS and Databases	Business Technology	0.1M												**0.1M**
5785C	Stabilization of Critical Gas Legacy Applications ...	Business Technology	0.0M	0.0M											**0.1M**
5489A	IBM SNOW Automation	I&O	0.0M	0.0M											**0.0M**
5489B	WiPro SNOW Automation	I&O	0.0M	0.0M											**0.0M**

Exhibit ___ (ITDP-2R)

Redacted quarterly Information Technology Capital Investment Reports for the period October 2021 through June 30, 2023

Niagara Mohawk Power Corporation d/b/a National Grid

The Brooklyn Union Gas Company d/b/a National Grid NY

KeySpan Gas East Corporation d/b/a National Grid

Cases 20-E-0380, 20-G-0381, 19-G-0309 and 19-G-0310

INFORMATION TECHNOLOGY CAPITAL INVESTMENT REPORT FOR QUARTER ENDED DECEMBER 31, 2021

SUBMITTED TO:
NEW YORK STATE DEPARTMENT OF PUBLIC SERVICE STAFF

March 1, 2022

nationalgrid

Information Technology Capital Investment Report for Quarter Ended December 31, 2021

I. Introduction

Niagara Mohawk Power Corporation (Niagara Mohawk), The Brooklyn Union Gas Company (KEDNY), and KeySpan Gas East Corporation (KEDLI) (together "National Grid" or the "Companies") submit this Information Technology ("IT") Capital Investment Quarterly Report ("IT Quarterly Report") for the third quarter of fiscal year 2022 ("FY22").[1] Pursuant to the Public Service Commission's ("Commission") January 20, 2022 Order in Cases 20-E-0380 20-G-0381, and August 12, 2021 Order in Cases 19-G-0309 and 19-G-0310, the Company is to provide quarterly variance reports to Department of Public Service Staff ("Staff") regarding IT capital projects and investments.[2] The IT Quarterly Report provides an explanation of the Company's overall IT investment plan approach; descriptions of program investment categories; and a comparison of budgeted to actual spending.

IT capital projects and investments that are shared investments across National Grid's operating companies are implemented and owned by National Grid USA Service Company, Inc. ("Service Company") and allocated to the benefiting US operating companies, including Niagara Mohawk, KEDNY and KEDLI in the form of Service Company rent expense once the investments are in-service. The Service Company rent expense includes a return on the capital investment (net of deferred taxes) plus booked depreciation expense.

Through Q3 FY22, National Grid spent $183.0 million compared to a year-to-date Service Company budget of $228.1 million, representing a $45.1 million underspend. Attachment 1 includes detail on budgets and actual spending for each of the portfolios and commentary regarding performance which has been presented to each of the Chief Information Officers responsible for the respective portfolios.

Attachment 2 includes a detailed breakdown of individual project spend by portfolio compared to the budgeted spend for each of those projects.

II. Program Categories/Spending Rationales

National Grid has re-organized the US Information Technology organization into six areas:

[1] The third quarter of FY22 is October 1, 2021 through December 31, 2021 ("Q3 FY22").
[2] *See* Cases 20-E-0380 and 20-G-0381, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service*, "Order Adopting Terms of Joint Proposal, Establishing Rate Plans and Reporting Requirements" (issued and effective January 20, 2022), Joint Proposal, Section IV.10.5; and Cases 19-G-0309 and 19-G-0310, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Energy East Corporation d/b/a National Grid for Gas Service*, "Order Approving Joint Proposal, as Modified, Imposing Additional Requirements" (issued and effective August 12, 2021), Joint Proposal, Section IV.6.5.

(1) Group Functions
(2) United States Electric
(3) United States Gas
(4) Customer
(5) Infrastructure and Operations
(6) Security

IT capital investment also is tracked based on these area designations. Descriptions of each portfolio category/spending rationale are provided below.

Group Functions
Group Functions executes National Grid's digital roadmaps, leveraging technology and reimagining processes in support of Finance, Enterprise Business Services, Human Resources, Legal, and Safety Health & Environmental, continuing to upgrade SOX related systems, and retiring outdated systems.

United States Electric
Provides IT Support to our US Electric Distribution and Transmission businesses. Modernizing our key legacy operational systems while building capabilities that will effectively enable the energy transition and transformation to a Distribution System Operator (DSO). Through digital delivery, the Electric CIO organization provides greater efficiency and data driven insights into our workforce operations:

- More automated and reliable grid controls systems
- Capital programs and projects that support the proliferation of Distributed Energy Resources (DERs)
- A growing market of prosumer electric customers

United States Gas
Provides IT Support to our US Gas Distribution and Transmission businesses developing digital products based on the foundation of gas business enablement, that deliver continuous improvement of our asset lifecycle management capabilities through:

- Risk based asset investment planning
- Performance based maintenance
- AI based work optimization
- Mobile field products

Customer
Provides IT Support to our US Customer business, delivering best-in-class digital products that increase engagement and reduce cost to serve:

- Modernizing our legacy "back-end" IT estate to support more digital products
- Liberating data and creating an abstracted and real-time customer data platform to assist with achieving energy efficiency ambitions

Infrastructure and Operations ("I&O")
Supports our internal IT Architecture covering Cloud and Hosting, Critical Network Infrastructure, Networks, End User Experience and Operations. I&O is focused on reducing National Grid's

technical debt through cost effective, modernized IT infrastructure enabling cloud capabilities that meet our risk profile. The organization enables a connected, seamless enterprise that is flexible, secure, reliable and scalable while maintaining acceptable levels of technical debt and risk.

Security

The Security function protects our people, premises, and digital systems with the objective of maintaining trust in National Grid services, focusing on:

- Reduction in likelihood and impact of security incidents
- Compliance with relevant laws, standards and regulations (e.g., NERC, PCI DSS, GDPR, NIS-R)
- Improved user experience and standard of security practices across National Grid

The Security function aims to reduce the likelihood and impact of physical and cyber security incidents affecting National Grid. It does this by championing security best practices, technology and promoting a security-awareness culture.

The summaries below identify the subjects presented in the attachments to this Q3 FY22 report.

A. Attachment 1

1. Summary (pages 2 – 3) – Overview of US IT as a whole.
    a. High-level executive summary, including work and challenges faced across the portfolios and a brief explanation around the forecast (p. 2).
    b. Actual expenditure versus budget by portfolio year to date, and full fiscal year forecast versus annual budget by portfolio (p. 2).
    c. Actual and forecasted spend by month for FY22 compared to budget, prior year and FY22. Each bar is color-coded to give an overview of the comparative spend per portfolio (p. 3).

2. Portfolio Sections – The remainder of Attachment 1 includes six sections cover; one for each of the portfolios: Group Functions; US Electric, US Gas, Customer, I&O and Security. Each portfolio consists of 3 pages:
    a. The first page provides a narrative financial and business overview. The three charts on this page show: operating expenses, capital expenses, and total expenses; FY22 actuals and forecast by period versus budget, with trend lines showing prior year expenditure and run rate. This page also shows the full year total expenditure budget and forecast and the Q3 actual total expenditure compared to the Q3 budget.
    b. The second page of the portfolio sections has two charts on the top left of the page: the top chart is actual expenditure versus budget by program year to date. The second chart compares full fiscal year forecast versus annual budget by program. A priority investments list for the portfolio and their status appears at the top right hand of the page. Blue means complete, green means on track, amber means some risk, and red means focused intervention required. On the bottom half of this page is a chart showing actual and forecasted spend for FY22

compared to budget, prior year and FY22 for the portfolio. Each bar is color-coded to give an overview of the comparative spend per program.

c. The third page of each portfolio section provides a status update on the various projects that are active, complete, and in the pipeline, as well as the methodology by which we are delivering the projects. There is also a key challenges section and the escalation point for that challenge.

Attachment 1 is based on National Grid's monthly Portfolio Delivery Group (PDG) reviews and reporting cadence to senior management of each portfolio and ultimately the IT leadership team. Security and Infrastructure and Operations do not currently participate in the PDG reviews but provide their senior management with monthly updates.

B. Attachment 2

INVP Level Variances - This attachment provides a detailed listing of individual projects in each of the six portfolios. Actual spend to date is provided, together with the budgeted allowance and the year-to-date variance. A monthly forecast walk also is provided to give the year-end forecast, with a comparison to year-end budget and corresponding variance.

In addition to detailing each of the six portfolios, Attachment 2 also includes a section for "Other" costs. The "Other" category has three elements:

- The IT4IT sub-category. This is essentially a funding source for net present value positive projects to enable efficiency savings in National Grid's IT run-the-business operations.
- The "other" sub-category is the allocation of overheads such as Enterprise Architecture and Solution Delivery operating expenses to a centralized account. The capital charges for this overhead are made directly to individual projects.
- Finally, the burdens are the net removal of operating expense burdens to ensure IT reporting agrees to Corporate Controllable Cost reporting.

National Grid

Quarterly Report – Q3 FY22
Information Technology

1st March 2022

nationalgrid

Executive Summary – US IT

US IT has faced significant challenges this year which have caused us to fall behind with our investment plan. Efforts are underway to close gaps and utilize additional resources from our application development/application maintenance (ADAM) partners and work closely with our vendors to overcome supply chain issues. Notwithstanding the challenges, significant progress has been made with reducing technical debt and delivering capabilities to benefit both our internal and external customers.

Financial Summary

Portfolio View - Year to Date $M

Portfolio	Capex			Opex			Totex		
	Actuals YTD	Budget YTD	Variance	Actuals YTD	Budget YTD	Variance	Actuals YTD	Budget YTD	Variance
US Electric	31.2	37.8	6.6	3.3	5.0	1.8	34.4	42.8	8.4
US Gas	5.9	9.4	3.5	0.9	2.0	1.1	6.8	11.4	4.6
Customer	38.8	31.6	(7.2)	2.8	5.2	2.4	41.5	36.8	(4.8)
Group Functions	27.7	34.2	6.6	4.4	10.2	5.7	32.1	44.4	12.3
I&O	43.2	49.9	6.7	6.7	9.6	2.9	49.9	59.5	9.6
Security	16.2	19.6	3.4	3.8	4.5	0.6	20.0	24.1	4.1
Other	0.0	5.9	5.9	(1.8)	3.3	5.1	(1.8)	9.2	11.0
Total	**162.9**	**188.4**	**25.5**	**20.1**	**39.7**	**19.6**	**183.0**	**228.1**	**45.1**

Portfolio View - Full Year $M

Portfolio	Capex			Opex			Totex		
	FY Forecast	FY Budget	Variance	FY Forecast	FY Budget	Variance	FY Forecast	FY Budget	Variance
US Electric	45.0	57.1	12.1	5.0	7.1	2.1	50.0	64.2	14.2
US Gas	15.3	13.6	(1.7)	2.3	3.0	0.7	17.6	16.6	(1.0)
Customer	53.0	44.1	(8.9)	4.5	7.4	2.9	57.5	51.5	(6.0)
Group Functions	53.7	53.9	0.3	10.0	15.3	5.2	63.7	69.2	5.5
I&O	67.0	60.1	(6.9)	14.0	12.1	(1.9)	81.0	72.2	(8.8)
Security	30.4	30.4	0.0	6.0	6.0	(0.0)	36.4	36.4	(0.0)
Other	0.9	12.0	11.1	0.1	6.3	6.1	1.0	18.3	17.3
Total	**265.2**	**271.3**	**6.1**	**41.9**	**57.0**	**15.1**	**307.2**	**328.3**	**21.1**

Many of the challenges the business has faced this year are also impacting the wider economy.

Capacity and Resources
1.Availability of IT and business resources to support delivery
2.Recruiting & retaining high quality permanent resources is proving challenging
3.Significant change across the organization
4.Covid related issues with third-party contractor resource pools (earlier in year)

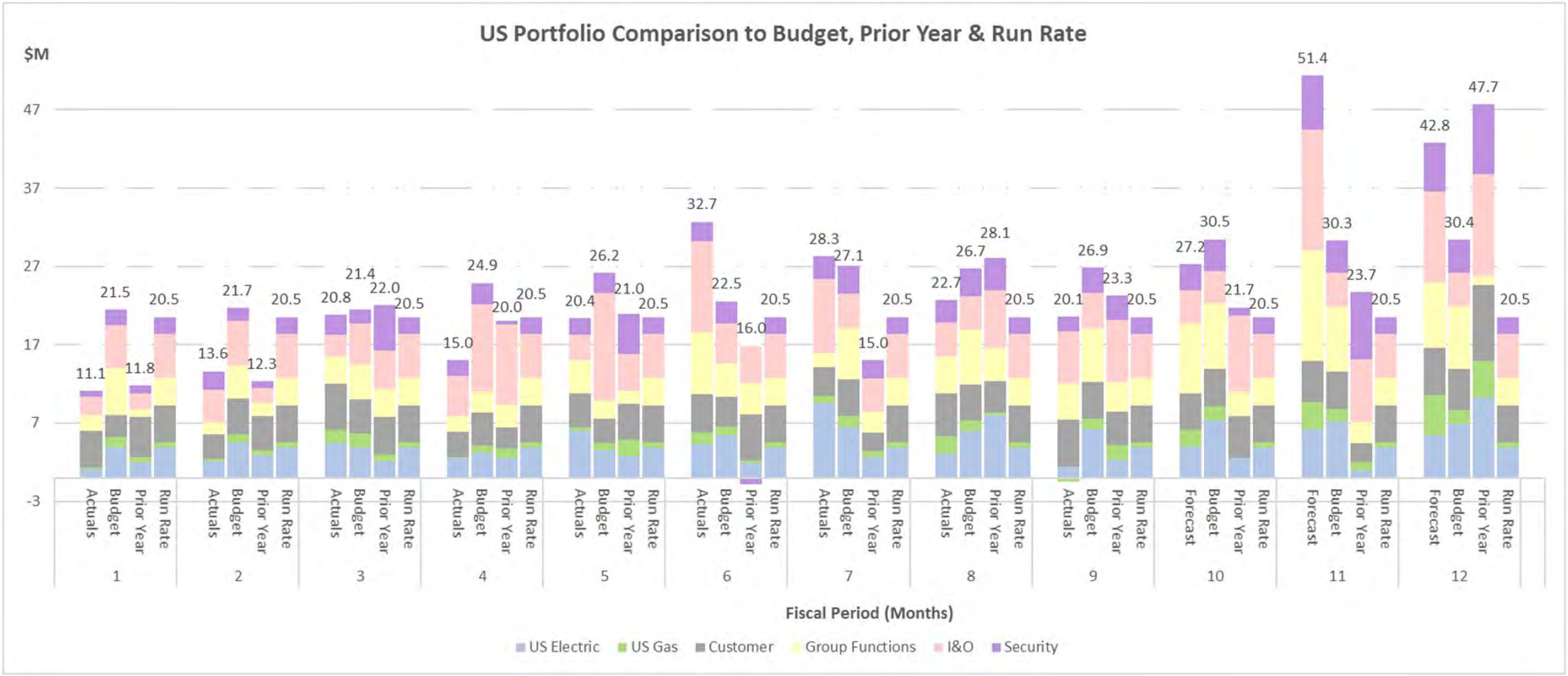
Supply Chain Issues
Commercial and Contracting Challenges & Chip Shortages

Divestures
Portfolios are impacted with separation, leading to potential delays & stoppage of work.

The gap is expected to be closed towards the end of the year with planned items above current project run rates as follows:

$15M Hardware planned delivery in February – Infrastructure
$7M Hardware / Software purchases on orders planned receipt before fiscal year-end – Security
$7M Pathfinder Leak Survey – Gas portfolio

Portfolio Comparison



US Group Functions
Q3 FY22 Report

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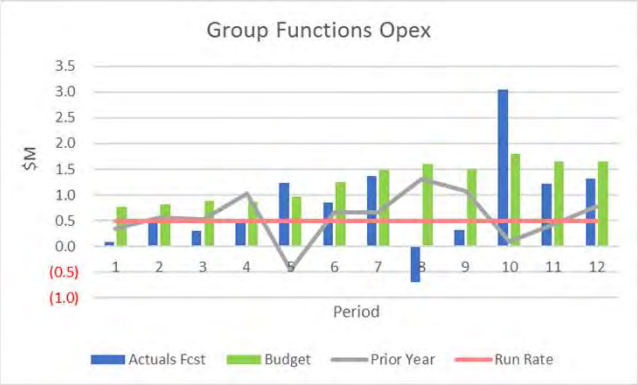
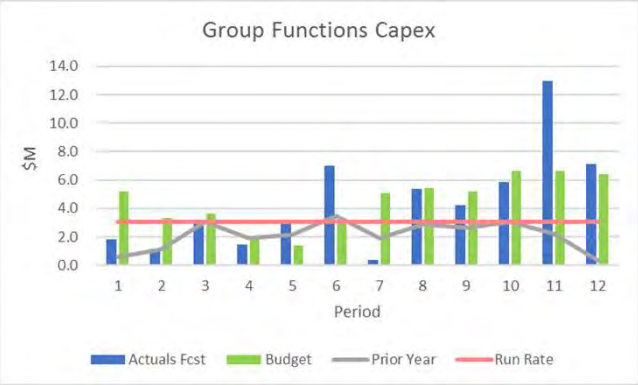
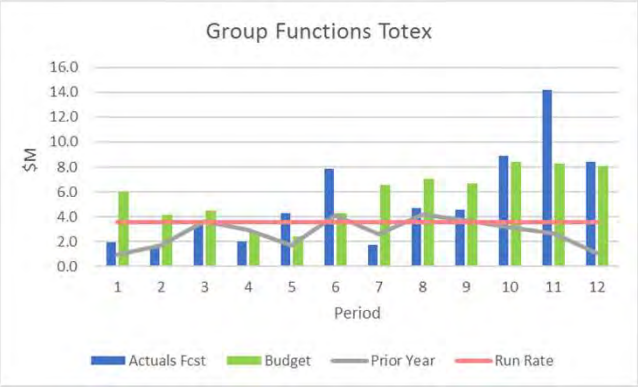
Executive Summary - Group Functions

Group Functions continues to be focused on the Rhode Island divestiture and delivering the FY22 portfolio. All original IT Foundations projects in the US are active and all US Core Q3 planned start-ups are active. Progress has been made with substituting new projects to offset the underspend of Capex. A major area of risk continues to be contention with RI in both resources and systems.

Financial Summary







Totex

Budget ($M)	Forecast ($M)
69.2	63.7
YTD Budget($M)	**YTD Actual ($M)**
44.4	32.1

Portfolio

- Substitutions helping close the gap on YTD Capex spend
- Continuing to mitigate resource constraints due to RI divestiture through ADAM partners and filling vacancies

Core Group Functions

- Intentional Capex overspend to offset Global project underspend; 6426 Investment Recovery, 5843D InfoSteward Upgrade, 6289D Blackline: Task Management
- Extended Palantir project scope will increase spend by additional $1.2M (not in December forecast)

IT Foundations

- GridStack infrastructure investment progressing through SGA and sanctioning to deliver in Q4.

Group Functions Program View of Financials

Group Functions Program View Year to Date $M

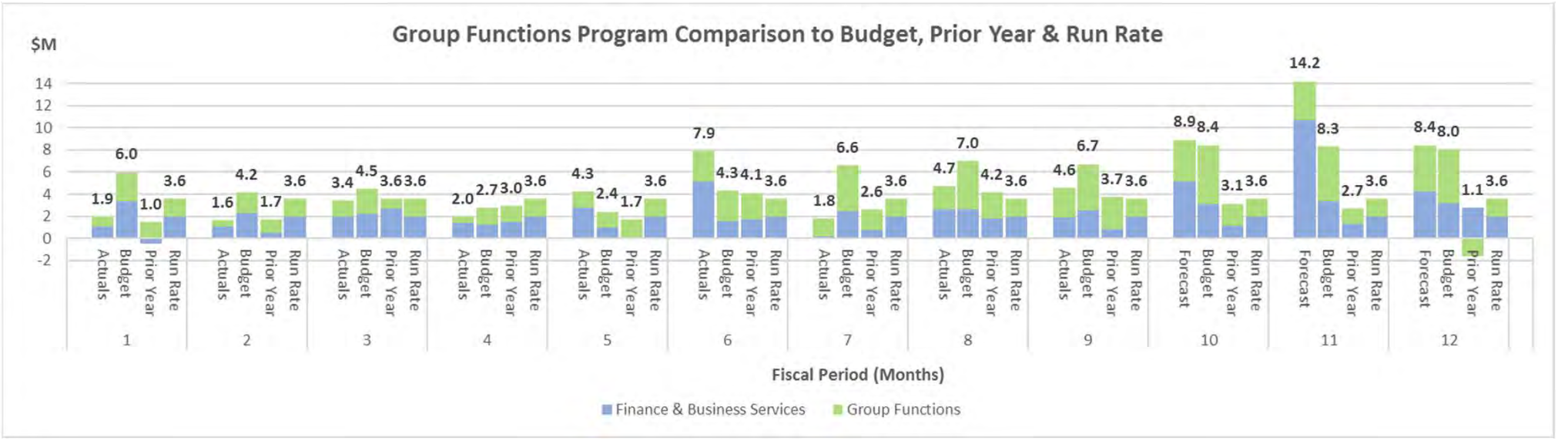
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Group Functions	Finance & Business Services	15.3	14.0	(1.3)	2.6	5.2	2.6	17.9	19.3	1.3
	Group Functions	12.3	20.2	7.9	1.9	4.9	3.1	14.2	25.1	11.0
	Totals	**27.7**	**34.2**	**6.6**	**4.4**	**10.2**	**5.7**	**32.1**	**44.4**	**12.3**

Group Functions Program View Full Year $M

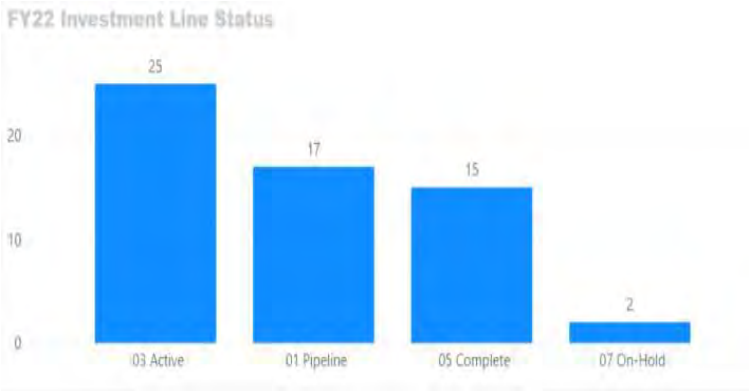
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Group Functions	Finance & Business Services	32.1	21.3	(10.9)	5.8	7.6	1.8	38.0	28.9	(9.1)
	Group Functions	21.5	32.7	11.1	4.2	7.6	3.4	25.7	40.3	14.6
	Totals	**53.7**	**53.9**	**0.3**	**10.0**	**15.3**	**5.2**	**63.7**	**69.2**	**5.5**

Priority Investments

Programme	Status	Overall RAG	Totex Budget ($M)	Funding Type
HR & Payroll Service Pack	Complete		1.4	Regulated
401(k) Admin Changes	Active D&T		1.0	Regulated
FTOM: SRM to Ariba	Active R&D		5.3	Regulated
FTOM: Closing Simplification	Active Implementation		1.7	Regulated
SAP Portals to Fiori	Active R&D		0.6	Regulated



Mobilisation Summary



No. of Initiatives Sanctioned

56

No. of Initiatives to be Sanctioned

3

No. of initiatives to be Sanctioned within the next month

1

Totex Sanctioned ($M)

$48.9M

Totex to be Sanctioned ($M)

$5.3M

Status

- Based on 8+4 forecast three projects left to sanctioned for FY22

Resources

- Three new Project Managers have been hired, onboarded and are leading FY22 projects

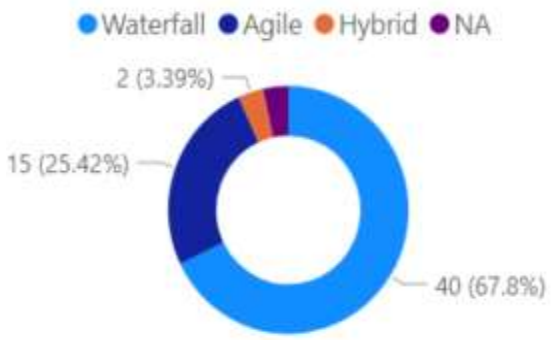
Scope

- Remaining pipeline projects need to complete scope assessment as part of business case analysis and sanctioning process.

Key Challenges

Challenge	Escalation Point
RI Divestiture – Resource allocation from US GF requires backfill. Multiple dependencies across the portfolio.	Group Functions CIO
Resource constraint in getting FY22 GridStack investment through SGA and sanctioning process in time to spend Capex	Group Functions CIO

Delivery Methodology



US Gas
Q3 FY22 Report

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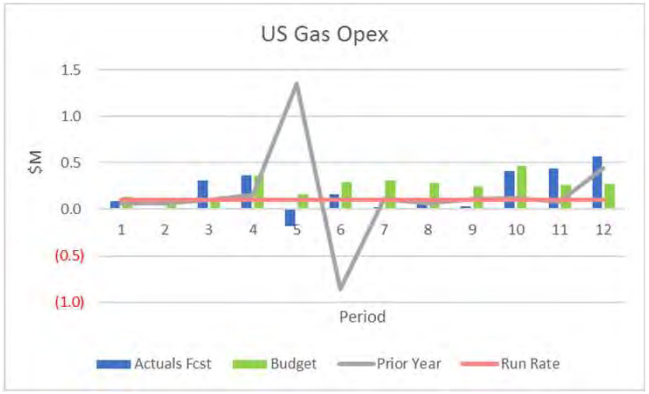
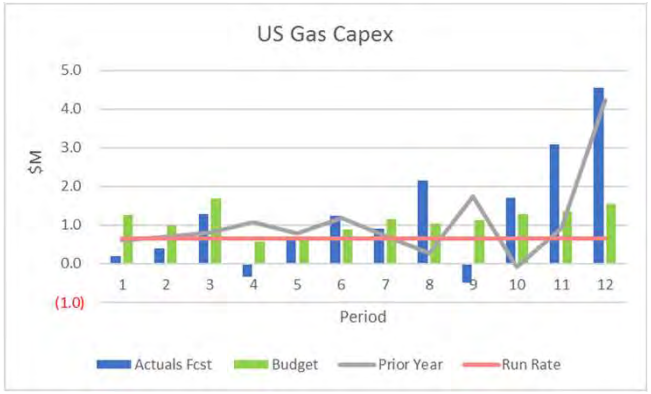
Executive Summary – US Gas

Gas Business Enablement: The November release successfully added 1200+ users of Long Island on to the GBE solution and work has switched to the MA & NYC (May 2022) Release: Detailed planning is underway for post-Nov against May scenarios optimizing data and training management.

The Unified Web Portal was successfully launched for NYC GAS customer on 11/21

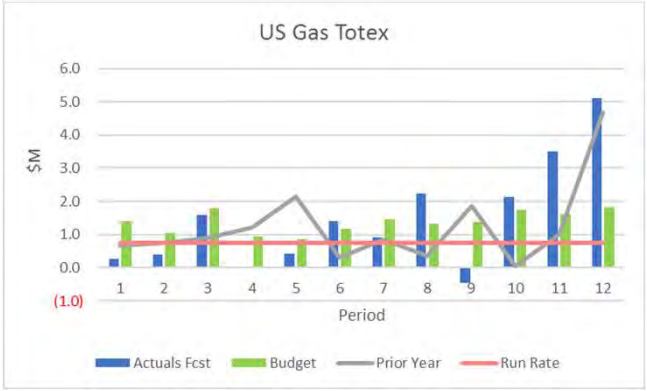
Financial Summary





Totex

Budget ($M)	Forecast ($M)
16.7	17.6
YTD Budget($M)	YTD Actual ($M)
11.4	6.8



App rationalization budget in Gas portfolio transferred back Opex $0.682m

The February forecast includes three new projects to be added to the portfolio(adjusted within portfolio)

LNG IT Support for OT Systems- $0.2 Opex, $0.600 Capex

I&R and Meter Ops Field Device Improvements- $0.200 Capex

Pathfinder for Leak Survey for MA with an estimated forecast of $7M capex.

US Gas Program View of Financials

Gas Program View Year to Date $M

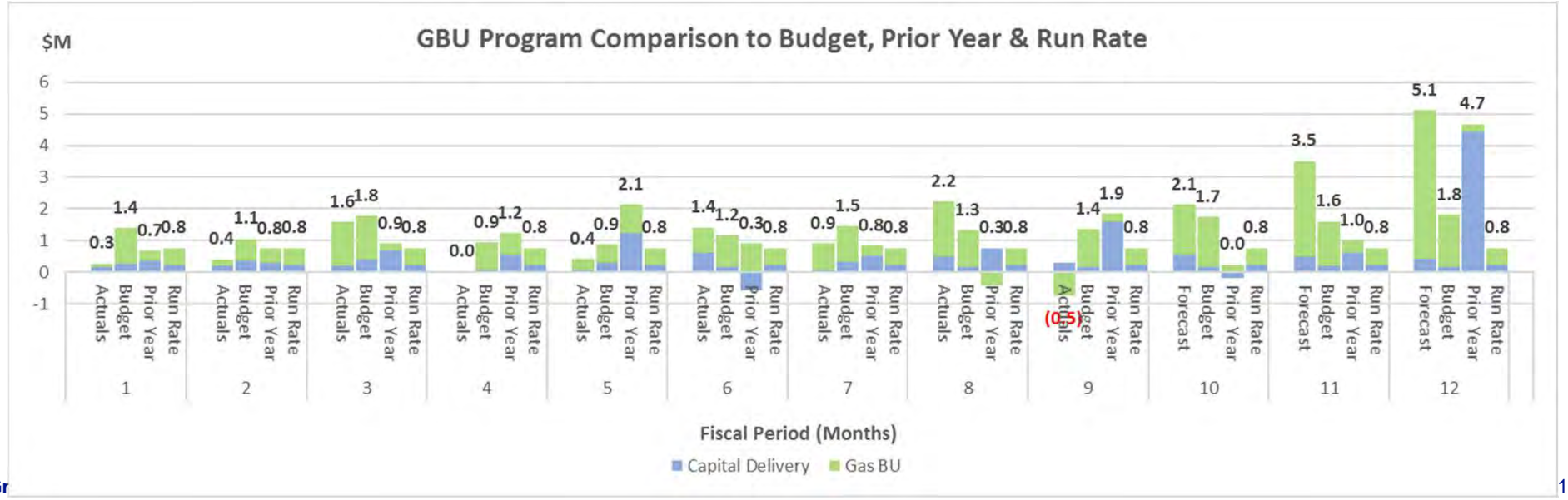
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
US Gas	Capital Delivery	1.9	2.0	0.2	0.2	0.2	(0.0)	2.1	2.2	0.2
	Gas BU	4.1	7.4	3.3	0.7	1.8	1.1	4.7	9.2	4.4
		5.9	**9.4**	**3.5**	**0.9**	**2.0**	**1.1**	**6.8**	**11.4**	**4.6**

Gas Program View Full Year $M

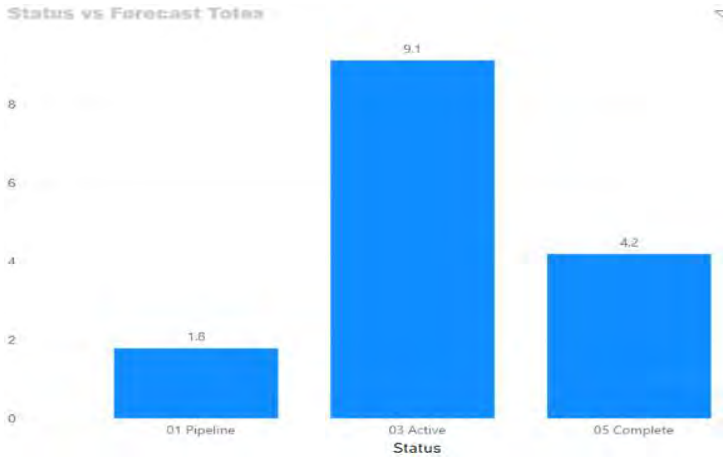
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
US Gas	Capital Delivery	3.3	2.5	(0.7)	0.3	0.2	(0.0)	3.5	2.8	(0.8)
	Gas BU	12.0	11.1	(0.9)	2.0	2.7	0.7	14.0	13.8	(0.2)
		15.3	**13.6**	**(1.7)**	**2.3**	**3.0**	**0.7**	**17.6**	**16.6**	**(1.0)**

Priority Investments

Program	Status	Overall RAG	Budget ($m)	Funding Types
GBE	Inflight		-	Business Funded
LNG Asset & Work Mgmt.	Inflight		3.003	IT Core
NRG GIS Replacement	PIS		4.175	IT Core
Continuous Improvement for Gas Applications	Start-up		2.6	IT Core
NSRMS	Inflight		2.026	IT Core
ER&S (Enterprise Scalability and Resilience	Start-up		2.50	IT Core
Gas App Rationalization	Inflight		.681	CTA
GTIS	Inflight		1.18	IT Core



Mobilisation Summary



As of January, 2022, all FY22 projects, except FCS (pending EBU project dependencies) are now either pending sanction (LNG IT support for OT systems, I&R and Meter Ops field devices improvements, Pathfinder for leak survey for MA) or have been sanctioned and are underway

No. of Initiatives Sanctioned

16

No. of Initiatives to be Sanctioned

3

No. of initiatives to be Sanctioned within the next month

3

Totex Sanctioned ($M)

$13.7M

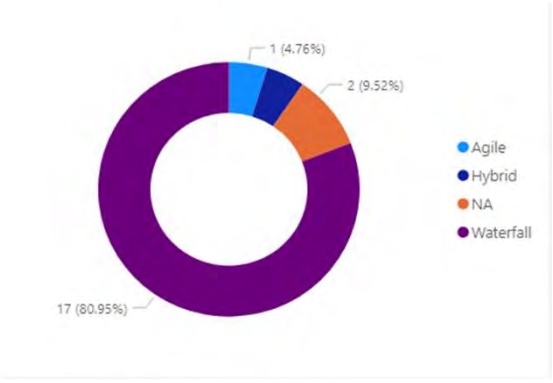
Totex to be Sanctioned ($M)

$1.4M

Key Challenges

Challenge	Escalation Point
• Delay in receiving final invoices extends project closure by at least 3 months	Strategy Office/ PMO
• Ability to force close accounting for projects(ex-4990)	Finance/ PMO
• COH changed from 5.7% to 6.9%, a net increase of 1.2%- Creates accurate forecast challenges	Finance/ PMO
• Resource challenges specially PM's	BP team

Delivery Methodology



US Electric
Q3 FY22 Report

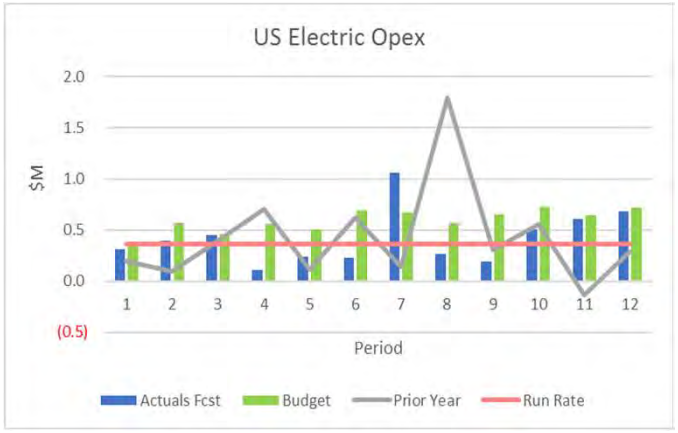
nationalgrid

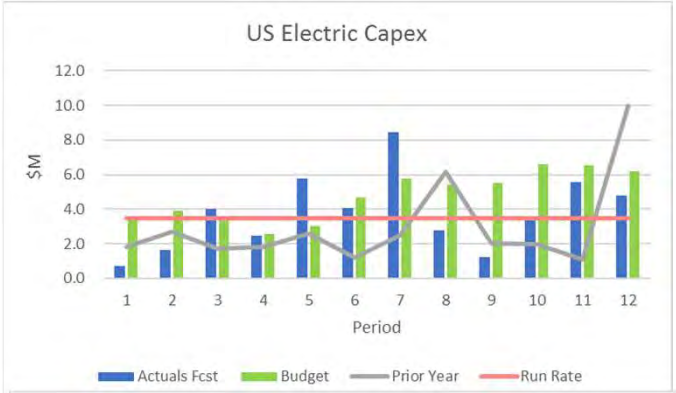
Executive Summary – US Electric

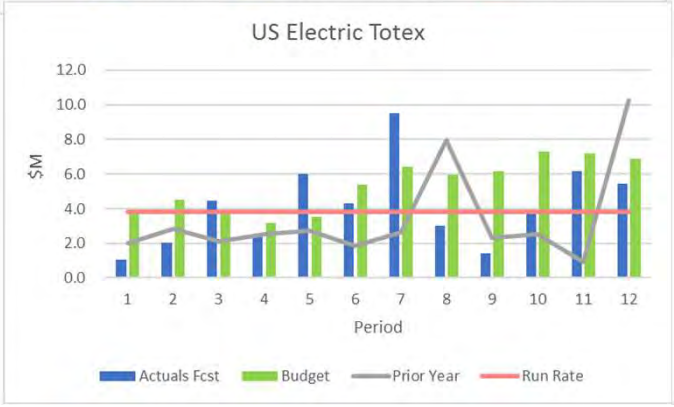
In FY22, Electric IT focused on -

1) reducing operational risk by upgrading / replacing end of life systems, e.g., Energy accounting system migration to Wholesale Settlement application, US Control-Transmission Outage Application (iTOA) application upgrade, MV90 (in-flight – application captures Electric Meter Reads, Commercial/ Industrial/Wholesale supports 40% of Electric Revenue), Field Collection System upgrade (in-flight)
2) reimagining ways of working for people in field, for investment planners etc.

- Vegetation Management Optimization: implemented MVP allowing for optimizing cycle prune and hazard tree removal plans. VMO MVP is expected to drive ~$2M in Type 1 Opex Contractor Costs in FY22.,
- OnMyWay - digitization of Confirming Work Request (CWR) forms for people in field; crews in NY and NE completed a total of 50,000+ jobs using the OMW app in the first 10 months of deployment, Benefits in FY22 are expected to be: Type 1: ~$3M Capex / ~$0.5M Opex ; Type 2: ~$1.5M Capex / ~$0.5M Opex
- FutureNow: This product will leverage more accurate, data driven value modeling and integrated TOTEX planning through its simulation/optimization capabilities to drive workplan outcomes at a significantly reduced cost. FutureNow MVP Roll Out on 2/14/22, which will help drive the following savings in FY23 through its workplan optimization capabilities: Type 1: ~$10M Capex / ~$1.5M Opex







Totex

Budget ($M)	Forecast ($M)
64.2	44.4*
YTD Budget($M)	**YTD Actual ($M)**
42.8	34.4

*Reforecasting FY22 OPEX down from $5.0M to $4.2M; this is a $790K decrease for remainder of year OPEX forecast.

*Reforecasting FY22 CAPEX down from $45.0m to $40.2m; this is a $4.77m decrease for remainder of year CAPEX forecast.

Meeting scheduled with Commercial team to review and discuss current Vendor negotiation delays, and need for timely execution of contracts that have licensing spend forecasted at the end of FY22.

Portfolio deep dives will continue through the end of the FY, with an increased focus on forecasts around product resourcing, and our ability to fill those positions in a timely fashion (internal and external).

Regular meetings with the business to align on forecast, backlog of new demand, delivery challenges and FY22 Digital products.

US Electric Program View of Financials

Electric Program View Year to Date $M

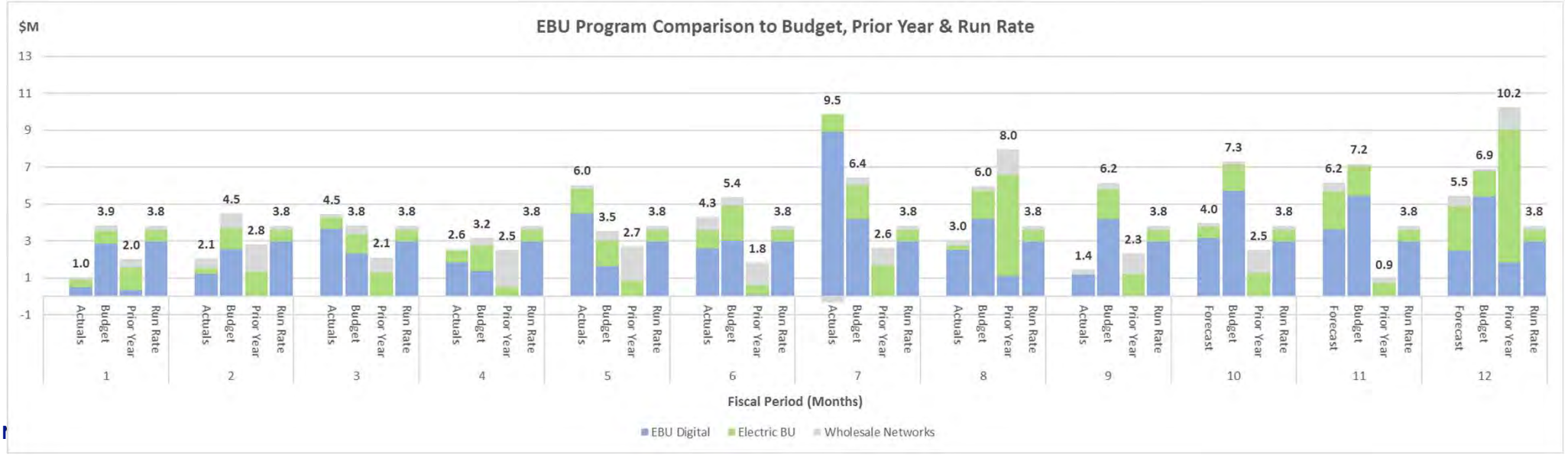
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
US Electric	EBU Digital	25.5	23.5	(1.9)	1.5	2.8	1.4	26.9	26.3	(0.6)
	Electric BU	4.1	10.6	6.5	1.4	1.9	0.5	5.5	12.5	7.0
	Wholesale Networks	1.6	3.7	2.1	0.4	0.3	(0.1)	2.1	4.0	2.0
	Totals	**31.2**	**37.8**	**6.6**	**3.3**	**5.0**	**1.8**	**34.4**	**42.8**	**8.4**

Electric Program View Full Year $M

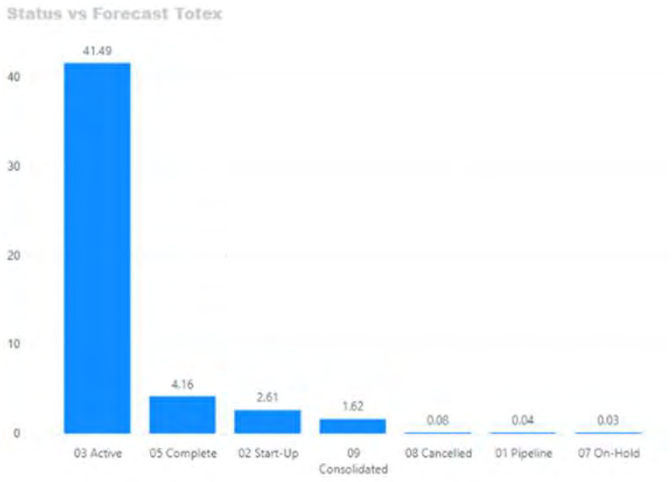
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
US Electric	EBU Digital	33.8	38.8	5.0	2.4	4.1	1.8	36.2	42.9	6.8
	Electric BU	8.3	14.4	6.0	2.2	2.5	0.3	10.5	16.9	6.3
	Wholesale Networks	2.8	3.9	1.1	0.5	0.5	0.0	3.3	4.4	1.1
	Totals	**45.0**	**57.1**	**12.1**	**5.0**	**7.1**	**2.1**	**50.0**	**64.2**	**14.2**

Priority Investments

Programme	Status	RAG	9+3 Totex Forecast	Budget Totex (FY22)	Funding Types
GridMod	*Inflight*			*35.5*	*Business/ Reg*
ADMS	*Inflight*			*30.1*	*Business/ Reg*
EBU Digital Programs	**Inflight**		**36.2**	**42.9**	**Regulatory**
Electric MV90 Upgrade	**Inflight**		**1.4**	**1.8**	**Regulatory**
FCS Upgrade	**Inflight**		**1.3**	**2.1**	**Regulatory**
Transmission Outage	**Delivered**		**1.6**	**0.7**	**Regulatory**
Wholesale Settlement	**Delivered**		**2.3**	**2.2**	**Regulatory**
Price Transmission	**Inflight**		**0.5**	**1.6**	**Regulatory**
MTS Retail Settlement	**Inflight**		**0.6**	**1.9**	**Regulatory**
Corporate PI	**Inflight**		**0.4**	**0.6**	**Regulatory**
NYISO PMU – Large Generators	**Inflight**		**0.4**	**0.5**	**Regulatory**



Mobilisation Summary



- FY22: *Completed Initiatives*: 6
  Active Initiatives: 23
  Pipeline Initiatives: 2
- Two (2) Additional Pipeline projects planned to begin before end of FY22 --- *Control Center Operation, & Power Quality Upgrade*
- Two (2) Go-lives planned for remainder of FY22 --- *Fluke / WECO, & Troy Energy Storage System IS Network*
- Products driving large Amber RAG status for Schedule portion of portfolio are FutureNow, OMW, & Generation Asset Management
- CYME Gateway has been put On-Hold, pending the completion of GIS Phase 2; this is now not expected to start until Q4FY23 / Q1FY24.

No. of Initiatives Sanctioned

21

No. of Initiatives to be Sanctioned

2

No. of initiatives to be Sanctioned within the next month

2

Totex Sanctioned ($M)

$41.2M

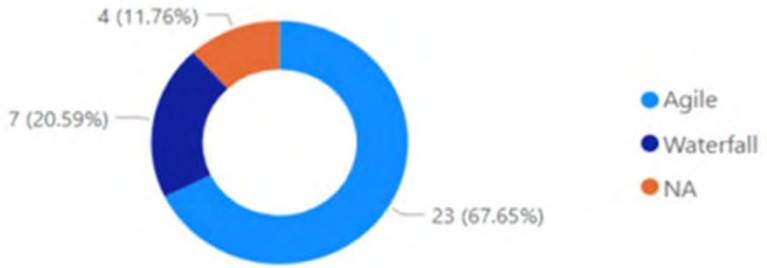
Totex to be Sanctioned ($M)

$2.9M

Key Challenges

Challenge	Escalation Point
1. Digital ways of working require ramp up with resources, processes and technology.	CIO / HR / Commercial *(ADAM Partners)*
2. Negotiations with Vendors regarding Licensing and general terms and conditions	Commercial
3. Significant change across the organization and number of initiatives pulling on the same business and IT resources.	Senior Leadership

Delivery Methodology



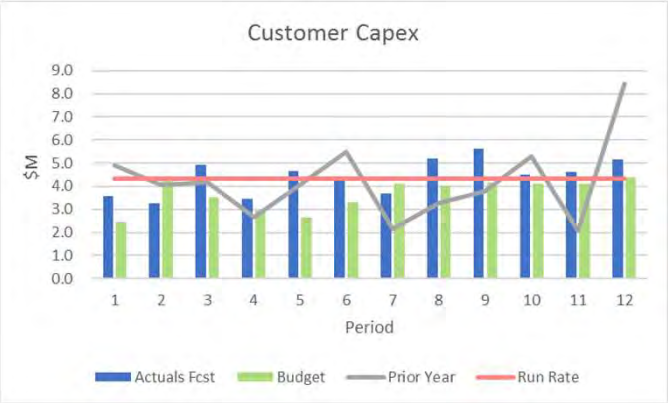
Customer
Q3 FY22 Report

nationalgrid

Executive Summary – Customer

Customer is making significant strides in digital enablement and improving the customer experience. Two advancements of note are the delivery of a unified web portal to NYC Gas customers in November integration of contact center third party vendors onto the AIMS (IVR suite) to provide a seamless call experience to customers. The User Experience scores have increased significantly (over 20 points) for NYC Gas customers since the migration to the new web platform, however due to stability problems we have put on hold deploying to UNY and LI.

Financial Summary



Totex

Budget ($M)	Forecast ($M)
51.5	57.5

YTD Budget($M)	YTD Actual ($M)
36.8	41.5

The portfolio is on target to utilize the full budget for delivery.

In Q4 we will see the run rate of Customer Data Platform increase as the team matures and additional resources come on board. We also have two new mandate and compliance efforts kicking off in Q4.

The largest variance is in the Unified Web Portal (UWP) effort. Leading up to the UWP launch in November and since, we have had persistent stability issues due to multiple root causes including bad data; missing or incorrect user profiles; software errors; and poorly tuned servers. There is an active war room to address the issues We have made progress with daily incidents reducing from 103 to 37. The additional resources required to stabilize the experience is the primary driver of additional costs. The UWP project will continue to run over budget into early FY23 as the stability problems are addressed.

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Customer Program View of Financials

Customer Program View Year to Date $M

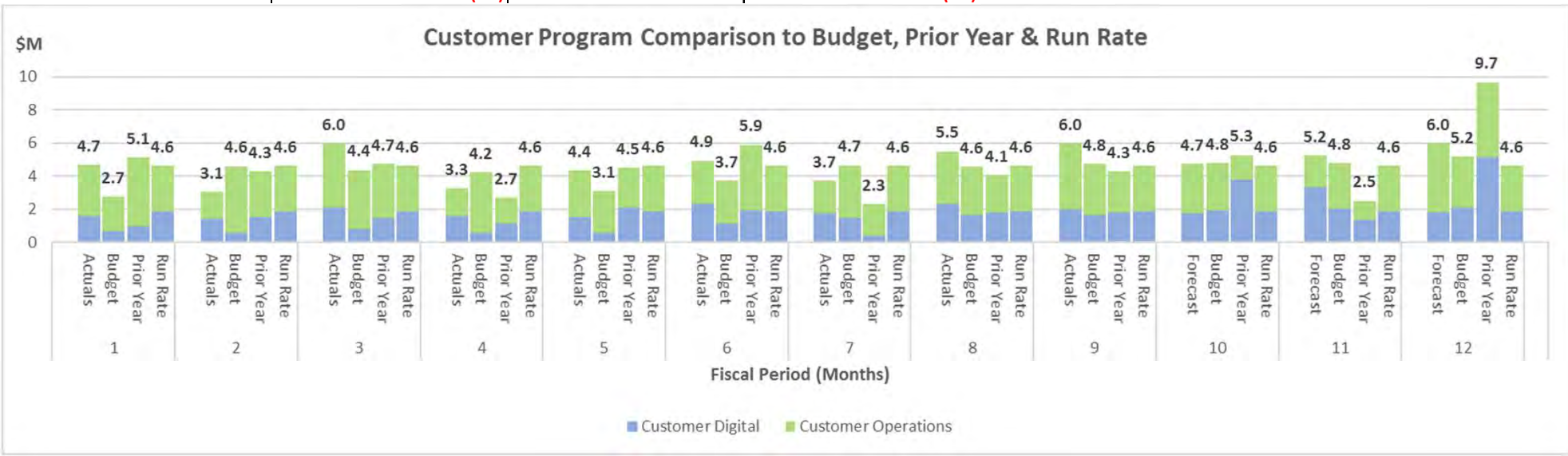
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Customer	Customer Digital	16.5	7.4	(9.1)	0.1	1.7	1.6	16.6	9.1	(7.5)
	Customer Operations	22.3	24.2	1.9	2.6	3.5	0.9	24.9	27.7	2.8
	Totals	**38.8**	**31.6**	**(7.2)**	**2.8**	**5.2**	**2.4**	**41.5**	**36.8**	**(4.8)**

Customer Program View Full Year $M

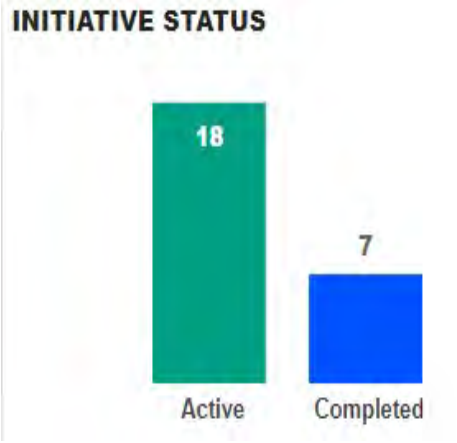
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Customer	Customer Digital	22.8	12.3	(10.5)	0.7	2.9	2.2	23.5	15.2	(8.3)
	Customer Operations	30.2	31.8	1.7	3.7	4.5	0.7	33.9	36.3	2.4
	Totals	**53.0**	**44.1**	**(8.9)**	**4.5**	**7.4**	**2.9**	**57.5**	**51.5**	**(6.0)**

Priority Investments

Program	Status	RAG	Budget ($m)	Funding Type
Unified Web Portal	Inflight		19.0m	IT funded
CIS: CRIS Migration	Inflight		10.2m	Business Funded
Customer Data Platform	Inflight		5.0m	IT Funded
AMI	Inflight		7.7m	Business Funded
Zero-Based Redesign	Start up		6.1m	Business Funded
RI Transition	Inflight		3.0m	Not Funded in Customer
Customer Ecosystem Stabilization	Inflight		5.0m	IT Funded
Clean Energy	Validation		NA	Business Funded



Mobilisation Summary



No. of Initiatives Sanctioned

30

No. of Initiatives to be Sanctioned

6

No. of initiatives to be Sanctioned within the next month

4

Totex Sanctioned ($M)

$58.1M

Totex to be Sanctioned ($M)

$0.29M

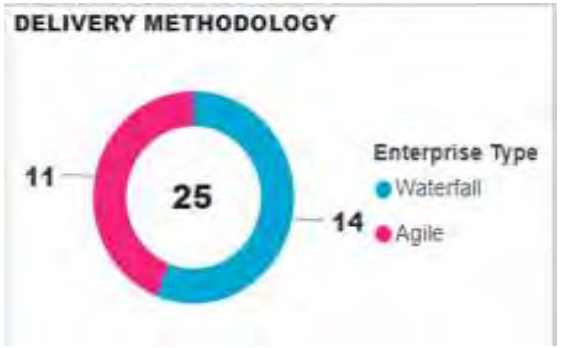
There are three pipeline additional efforts to kick-off in FY22:

- PCI Remediation is another project in startup. Resources have been identified and a solution is in final review..
- There is a second audit item remediation required for NACHA. There is forecast for an F&A/Validation phase. Resources are being identified to work this item.
- Preferred Pronoun is a third compliance effort kicking off this quarter.

Key Challenges

Challenge	Escalation Point
Product Model Resources and Training	*Anil/Tony*

Delivery Methodology



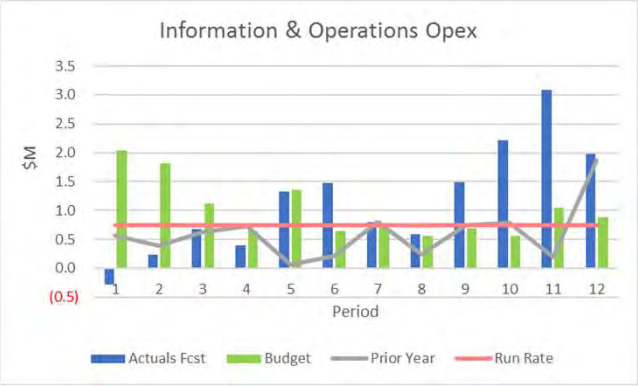
Infrastructure & Operations
Q3 FY22 Report

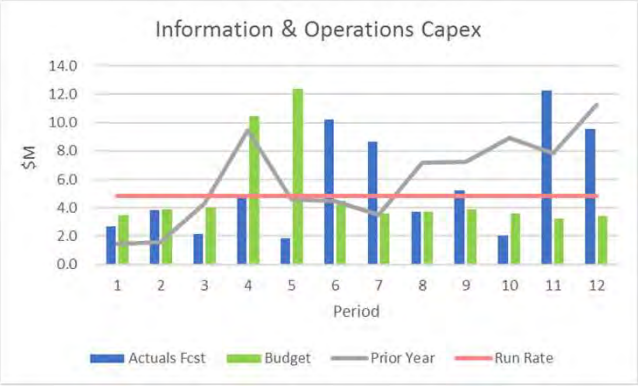
nationalgrid

Executive Summary – I&O

I&O has improved the level of tech debt, increased network performance, resiliency and reliability and worked to improve end-user experiences. Challenges around available skilled resources and hardware delivery delays have created some challenges although we are expecting the mainframe purchase early in 2022.

Financial Summary

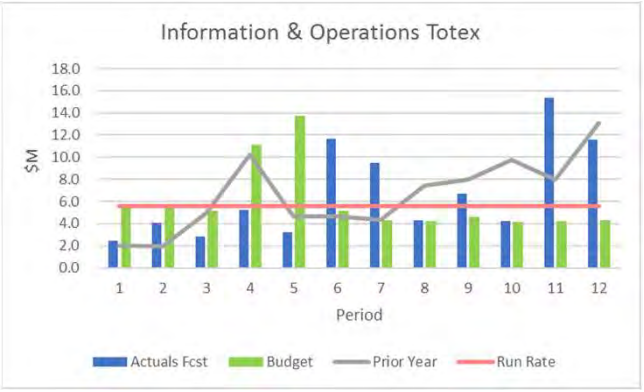




Totex

Budget ($M)	Forecast ($M)
72.2	81.0

YTD Budget($M)	YTD Actual ($M)
59.5	49.9



Data Center Transformation: Mainframe purchase decided, equipment ordered, expected receipt in Jan/Feb – enables migration from existing DXC data centers ($15m)

Cloud & Hosting:

Migrations from legacy infrastructure. Reduce technical debt, reduce risk of unsupported hardware and enabler for DCT.

Progress year to date: US - HCI 60%, DPS 58%, Block Storage 73%

Networks: US – WiFi 40/57 Sites (319/~800 WAPs Removed) // EOL – Replaced 442/~1291 Devices

Operations:

Completed prod rollout of Dynatrace OneAgent to 7 Apps (inc. Settlements, GBE, MBA & UWP)

Synthetic monitoring POC completed (which enables automated availability monitoring)

End User:

VDI SoftLayer to Azure migration 90% complete for US

146 Printers removed (100% of target)

Global: 118 MTR rooms commissioned (121 target)

.

I&O Program View of Financials

I&O Program View Year to Date $M

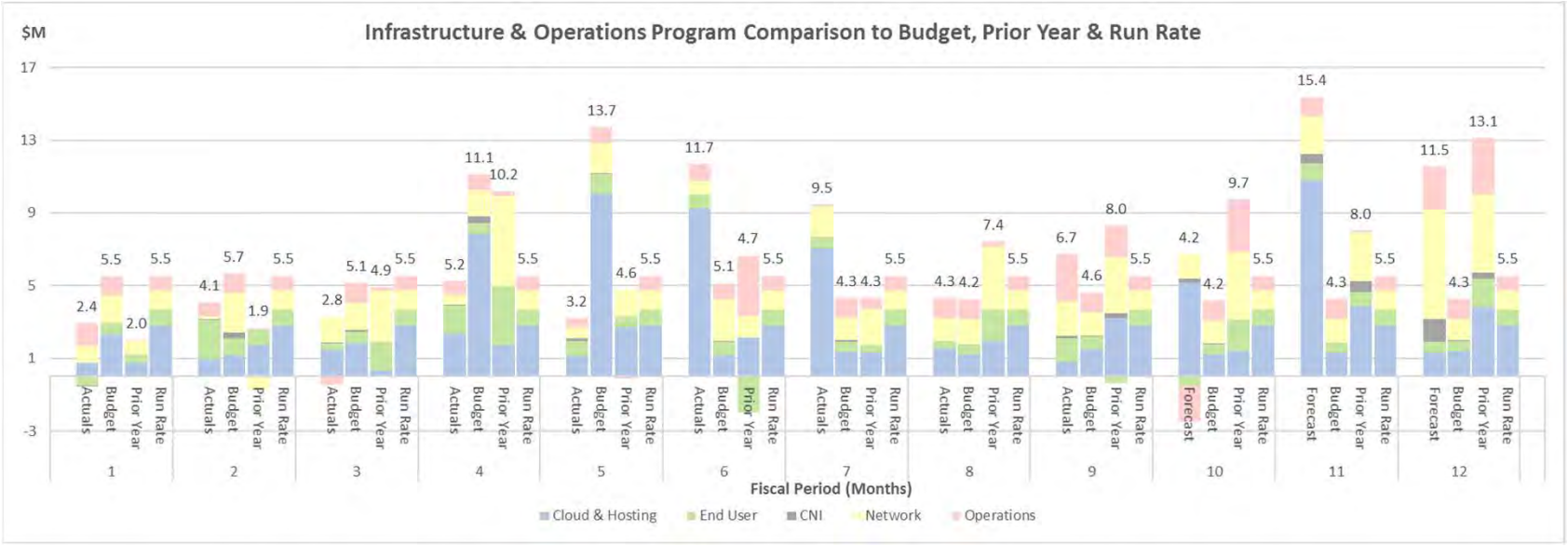
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
I&O	Cloud & Hosting	19.5	21.3	1.8	5.8	7.2	1.4	25.3	28.5	3.2
	End User	7.4	5.2	(2.2)	0.1	1.1	1.0	7.5	6.3	(1.2)
	CNI	0.3	1.0	0.7	0.1	0.0	(0.1)	0.4	1.0	0.6
	Network	8.9	13.7	4.9	0.3	0.7	0.4	9.2	14.5	5.3
	Operations	7.1	8.7	1.6	0.3	0.5	0.1	7.4	9.1	1.7
	Totals	**43.2**	**49.9**	**6.7**	**6.7**	**9.6**	**2.9**	**49.9**	**59.5**	**9.6**

I&O Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
I&O	Cloud & Hosting	32.5	23.0	(9.5)	10.1	9.4	(0.7)	42.6	32.4	(10.2)
	End User	7.7	6.9	(0.7)	0.9	1.1	0.2	8.6	8.1	(0.5)
	CNI	2.0	1.1	(0.9)	0.4	0.0	(0.4)	2.4	1.1	(1.3)
	Network	17.7	17.4	(0.3)	0.9	0.8	(0.1)	18.6	18.2	(0.4)
	Operations	7.1	11.7	4.6	1.7	0.7	(1.0)	8.8	12.5	3.6
	Totals	**67.0**	**60.1**	**(6.9)**	**14.0**	**12.1**	**(1.9)**	**81.0**	**72.2**	**(8.8)**

Priority Investments

Programme	Status	RAG	Forecast Totex (FY22)	Funding Types
Data Center Transformation	Inflight		31.2	IT
Networks	Inflight		20.2	IT
Cloud & Hosting	Inflight		11.5	IT
Operations	Inflight		7.7	IT
End User Experience	Inflight		5.6	IT
Critical Network Infrastructure (CNI)	Inflight		1.7	IT

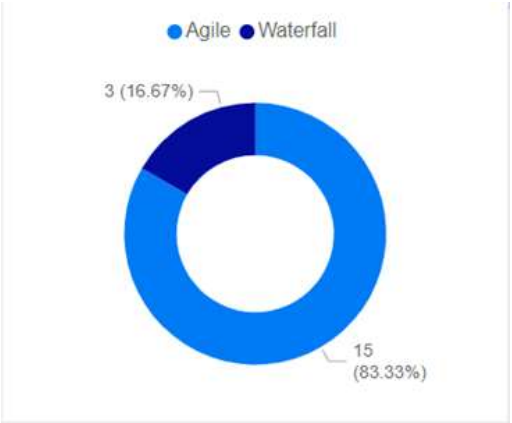


Mobilisation Summary

Key Challenges

Challenge	Escalation Point
End User: Fibre delivery lead times for UK mobility in-building solution circuits, impact's ability to migrate office-based users to O2 until solution in place to address network coverage constraints, option around wi-fi calling being explored (ng-p)	Vendor Lead Times

Delivery Methodology



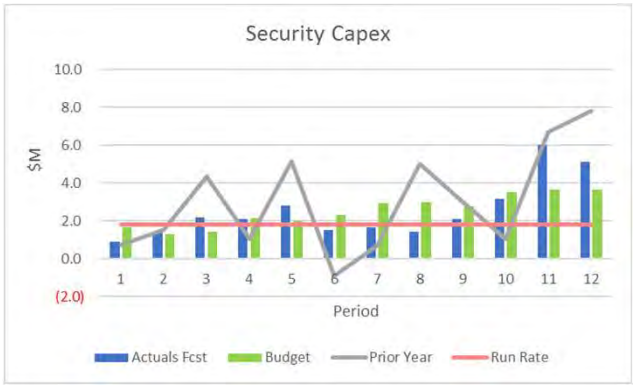
US Security
Q3 FY22 Report

nationalgrid

Executive Summary – Security

Security has prioritized delivery of key risk reducing Security Capabilities into Operational Technology environments, while sustaining capabilities established in the Enterprise and Critical National Infrastructure environments with consideration of the changing threat landscape. The US Operational Technology Engineering Team has on-boarded the necessary resources to establish Security Point of Presence at identified Operational Sites. The Identity and Access Management Program is set to close in the first quarter of the upcoming fiscal year, ahead of schedule.

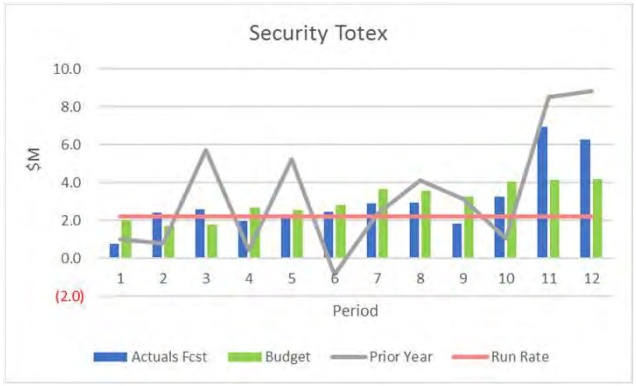
Financial Summary



Totex

Budget ($M)	Forecast ($M)
36.4	36.4

YTD Budget($M)	YTD Actual ($M)
24.1	20.0



Security Portfolio –

Operational Technology Intrusion Detection Project sanctioned in FY22 Q3. Intrusion Detection Hardware arrival anticipated in FY22 Q4, approx. $1.4m.

Identity and Access Management Program has been accelerated and is planned to be completed ahead of schedule. Approx. $2.5m anticipated in FY22Q4.

Platform Security Hardware expected to arrive in FY22 Q4, approx. $3.6m.

Security Program View of Financials

Security Program View Year to Date $M

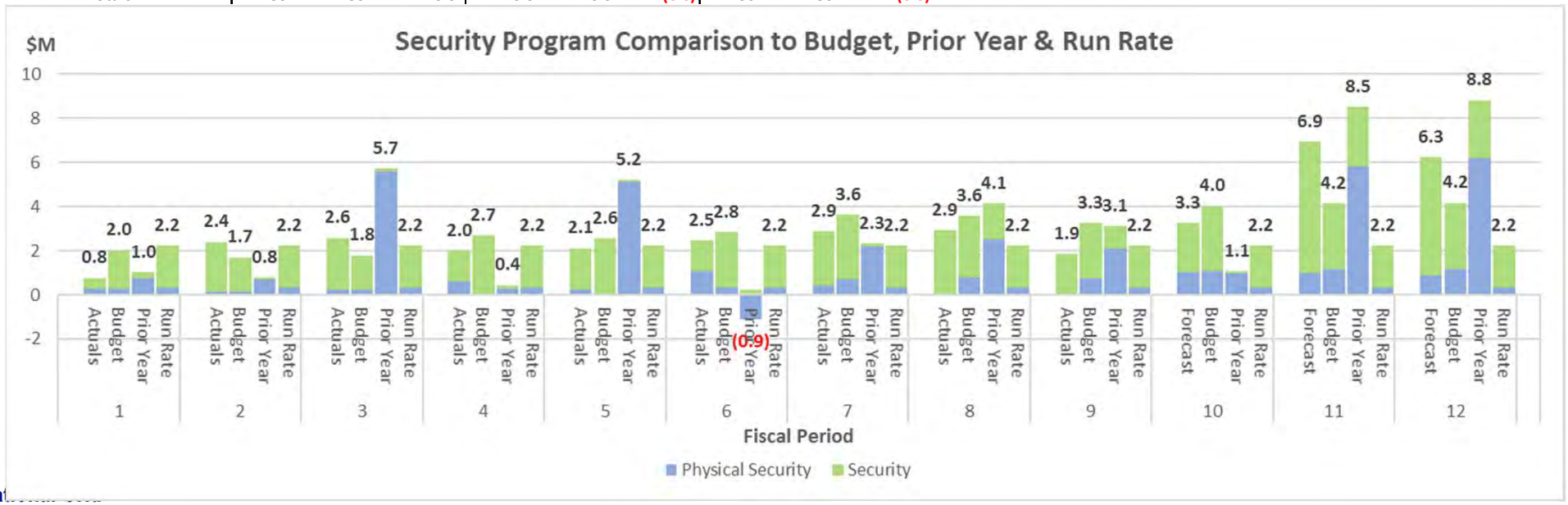
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Security	Physical Security	2.9	3.3	0.4	0.1	0.0	(0.1)	3.0	3.3	0.3
	Security	13.3	16.3	3.0	3.7	4.5	0.8	17.0	20.8	3.8
	Totals	**16.2**	**19.6**	**3.4**	**3.8**	**4.5**	**0.6**	**20.0**	**24.1**	**4.1**

Security Program View Full Year $M

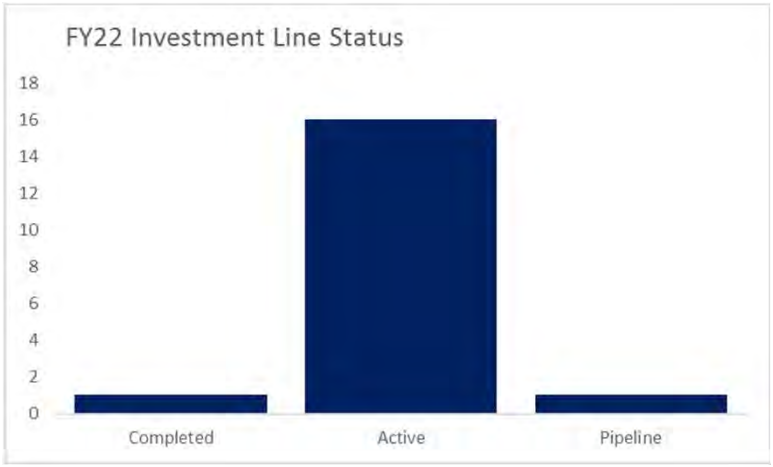
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Security	Physical Security	5.8	6.7	0.9	0.1	0.0	(0.1)	5.9	6.7	0.8
	Security	24.7	23.7	(0.9)	5.9	6.0	0.1	30.5	29.7	(0.8)
	Totals	**30.4**	**30.4**	**0.0**	**6.0**	**6.0**	**(0.0)**	**36.4**	**36.4**	**(0.0)**

Priority Investments

Programme	Status	RAG	Budget Totex (FY22)	Funding Types
Operational Technology	Inflight		5.9	IT
Platform Security	Inflight		6.9	IT
Network Security	Inflight		4.3	IT
Identity and Access Management	Inflight		9.2	IT
Vulnerability Management	Inflight		1.3	IT
Security Orchestration Automation and Response	Inflight		2.1	IT
Physical Security	Inflight		6.7	IT



Mobilisation Summary



No. of Initiatives Sanctioned

17

No. of Initiatives to be Sanctioned

1

No. of initiatives to be Sanctioned within the next month

1

Totex Sanctioned ($M)

$36.7M

Totex to be Sanctioned ($M)

$0.14M

Status

- 1 Project (Vulnerability Mgt) remains to be sanctioned for FY22.
- 1 Project (OT Intrusion Detection) will be re-sanctioned to increase scope and sustain deployment schedule.
- 12 Projects will complete and close by FY22 end.
- 3 Projects will complete and close in FY23Q1.

Resources

- Operational Technology Security Engineering Team has on-boarded resources to delivery against plan.

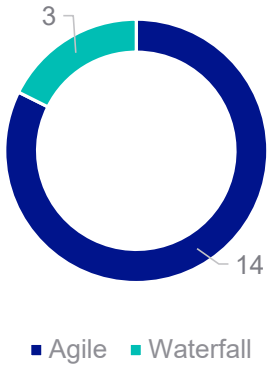
Scope

- Scope development has been completed for FY23.

Key Challenges

Challenge	Escalation Point
Supply Chain constraints i.e. chip shortage may impact delivery timelines. Extended lead-times have been factored into project timelines, as reflected in FY22Q4 forecasts.	Commercial/ Procurement
Changes to the threat landscape and regulatory mandates may result in need to pivot to address emerging threats, risk areas and compliance.	CISO

Delivery Methodology



Project View - Group Functions

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Variance	Forecast 10	Forecast 11	Forecast 12	Year End Forecast	Year End Budget	Variance
Group Fun	Finance & Business Services	3870A	F&A and Implementation - Vision FM Replacement	1,763,663	1,730,000	(33,663)	152,609	159,383	154,826	2,230,481	1,730,000	(500,481)
		4562	US SAP: Business Warehouse (BW) Consolidation to HANA Enterprise Cloud (HEC)	620,019	480,000	(140,019)	16,433	38,965		675,417	480,000	(195,417)
		4998	Allegro Upgrade to Horizon	9,803	0	(9,803)				9,803		(9,803)
		5360	Lease Accounting Updates and Contract Management	24,909	0	(24,909)				24,909		(24,909)
		5375	SAP/PowerPlan/Front Office Maintenance of Business (MOB) - FY20	(17,945)	0	17,945				(17,945)		17,945
		5378	FSSC&HR Systems (Non-SAP) Operational Requirements & Upgrades FY20	(93,379)	0	93,379				(93,379)		93,379
		5380A	Blackline: US Implementation	133,598	795,000	661,402				133,598	795,000	661,402
		5466	AMAG HW and SW Upgrade	1,205,452	563,000	(642,452)	52,380	204,934	180,329	1,643,095	563,000	(1,080,095)
		5497	Lease Accounting Updates and CCM - Phase 2	50,820	769,000	718,180	2,417	178,983	147,158	379,378	769,000	389,622
		5507	US Legal: Document Management	2,746	0	(2,746)				2,746		(2,746)
		5551	Fleet improvements	0	44,717	44,717					75,000	75,000
		5605	US SAP: Infrastructure Landscape FY21	(1,919)	0	1,919				(1,919)		1,919
		5606	US SAP: Infrastructure Landscape FY22	2,818,186	0	(2,818,186)	304,993	335,611	337,172	3,795,962		(3,795,962)
		5609	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY21	825	0	(825)				825		(825)
		5610	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY22	1,012,666	1,346,721	334,055	153,988			1,166,654	1,424,000	257,346
		5631	Implementation Costs for Booking Mgmt Project	155,463	123,000	(32,463)				155,463	123,000	(32,463)
		5777	US IMS Upgrade	851	0	(851)				851		(851)
		5841A	Pension Administrator Replacement	157,507	249,000	91,493				157,507	249,000	91,493
		5841C	401(k) Administration Changes	483,772	283,431	(200,341)	134,143	108,415	118,154	844,484	546,000	(298,484)
		5841D	Credit Card MSP Changes	0	89,433	89,433					150,000	150,000
		5842A	US SAP_ GRC Upgrade	982,980	941,000	(41,980)				982,980	941,000	(41,980)
		5842B	Mandatory FERC Form Changes	653,068	421,706	(231,362)				653,068	825,000	171,932
		5843A	SAP Platform Upgrade: OS and Databases	556,831	450,000	(106,831)				556,831	450,000	(106,831)
		5843B	US SAP Platform Upgrade: Minor Releases	408,621	373,712	(34,909)				408,621	485,000	76,379
		5843C	US SAP Healthcheck: Roadmap and assessment of critical components	0	67,373	67,373					113,000	113,000
		5843D	US SAP Healthcheck: Upgrade Critical Components - FY22	600,069	858,088	258,018	61,904	232,591	232,935	1,127,499	1,706,000	578,501
		5884	Operational excellence - IT solutions for Property Services	0	44,717	44,717					75,000	75,000
		5895A	Investment Recovery IT Strategy & Roadmap	0	38,000	38,000					38,000	38,000
		5895B	Everbridge / Critical Event Management	357,424	290,000	(67,424)				357,424	290,000	(67,424)
		5896A	US Tax: SABRIX Upgrade	168,936	120,000	(48,936)				168,936	120,000	(48,936)
		5896B	Warehouse SAP Scanner Replacement (Barcoding)	218,583	510,000	291,417	3,421	3,421	3,935	229,360	510,000	280,640
		5896C	Wellness: RedBrick to Virgin Pulse Migration	(55,647)	4,000	59,647				(55,647)	4,000	59,647
		5896D	WEM Waste Environmental Management	346,208	129,000	(217,209)	5,635	9,632	2,640	364,115	129,000	(235,116)
		5902	Applications Health Check (Non-SAP) Program	0	180,179	180,179					358,000	358,000
		5918	Non-SAP Minor Enhancements	(59,351)	0	59,351				(59,351)		59,351
		5919	US SAP Minor Enhancements	(14,611)	0	14,611				(14,611)		14,611
		5964	Safety Line Inspection	(17)	0	17				(17)		17
		5979	Federal PAC Reporting Payroll Interface	0	22,656	22,656					38,000	38,000
		5983	DB Landscape Optimization	3,426	0	(3,426)	38,148	193,144	271,082	505,800		(505,800)
		6137	Enterprise PMO, Jira Align release 2	0	0	0		254,292	129,233	383,525		(383,525)
		6219	US F&BS Minor Enhancements (Non-SAP)	217,695	143,093	(74,602)				217,695	240,000	22,305
		6220	US F&BS Minor Enhancements (SAP)	668,833	355,665	(313,167)	109,618	41,181	41,302	860,934	675,000	(185,934)
		6221	Tech Debt: RiskMaster Replacement	2,556	377,495	374,938	165,334	93,076	76,385	337,351	743,000	405,649
		6231	US Treasury: Allegro Upgrade Phase 2	141,435	190,834	49,399	95,151	95,268	120,346	452,200	375,000	(77,200)
		6232	Cybergrant Replacement	0	210,601	210,601	45,000	45,000	45,000	135,000	416,000	281,000
		6233	Legal: Digitization	0	200,282	200,282	45,288	5,095	38,350	88,733	383,000	294,267
		6234	HR Benefit Vendor Integrations	0	178,131	178,131					338,000	338,000
		6235	Tech Debt: RFS / SIR / Remedy	0	121,995	121,995	22,400	29,500	29,500	81,400	236,000	154,600
		6236	Tech Debt: Carousel	0	133,414	133,414					263,000	263,000
		6237	US Group Functions Operational Enhancements FY22	0	44,717	44,717					75,000	75,000
		6289	Financial Transformation (FTOM) - Record To Report (US)	0	4,335,798	4,335,798		637,788	637,788	1,275,576	8,280,000	7,004,424
		6289B	FTOM: Record to Report	1,873,875	0	(1,873,875)	28,003	28,003	5,095	1,934,976		(1,934,976)
		6289D	FTOM: Record to Report - Task Management with Blackline	0	0	0	442,894	273,043	273,487	989,424		(989,424)
		6290	Financial Transformation: Source to Pay?	0	1,868,599	1,868,599					3,570,000	3,570,000
		6290A	Source To Pay	548,411	0	(548,411)	538,002	690,507	731,411	2,508,331		(2,508,331)
		6377	US Group Function Operational Requests	480,918	0	(480,918)	72,360	72,325	72,039	697,642		(697,642)
		6421	P2P Data Analysis Tool	1,502,991	0	(1,502,991)	1,280,689	13,492		2,797,172		(2,797,172)
		6426	Investment Recover Digitalization	6,943	0	(6,943)	117,863	145,162	113,033	383,001		(383,001)
		6458US	QES IT Tool Sunset	571	0	(571)	5,555	5,555	7,315	18,996		(18,996)
		6461USA	Business Planning F&A (FP&A Enablement & Modelling Tool)	0	0	0	130,000	214,000	220,000	564,000		(564,000)
		6468	GridStack - Infrastructure	0	0	0	1,104,329	6,547,474	279,413	7,931,216		(7,931,216)
		FIN22	Finance + Placeholder	0	167,919	167,919					320,000	320,000
	Finance & Business Services Total			**17,937,787**	**19,252,277**	**1,314,490**	**5,128,557**	**10,655,840**	**4,267,928**	**37,990,112**	**28,900,000**	**(9,090,112)**
	Group Functions	4418US	RSA Archer Upgrade v6.8 and enhancements	(47,221)	0	47,221				(47,221)		47,221
		4418USA	Archer Optimisation	0	0	0	21,710			21,710		(21,710)
		4418USB	RSA Archer Upgrade & Refresh	223,978	0	(223,978)	119,612	61,915	179,748	585,253		(585,253)
		5158	MyHub Compliance & Enhancements	453,358	6,536,700	6,083,342	324,899	324,899	349,615	1,452,771	7,263,000	5,810,229

	5158A	myHub System Stabilization (30/60/90 Initiative)	0	179,843	179,843				340,000	340,000
	5317	Employee Digital Workplace	(46)	0	46				(46)	46
	5472	GRC Archer - Risk and Migration	436	0	(436)				436	(436)
	5506	Integration Services	107,559	0	(107,559)				107,559	(107,559)
	5555	Fieldglass Replacement	0	917,251	917,251				1,800,000	1,800,000
	5899A	Cloud API Migrations Phase 2	2,795,529	2,228,995	(566,534)	118,441	341,918	371,541	3,627,429	2,270,000	(1,357,430)
	5900A	ServiceNow Upgrade to version Orlando	4,318	8,000	3,683				4,318	8,000	3,683
	5901B	Application Lifecycle Management Toolset Phase 2	192,753	8,000	(184,753)				192,753	8,000	(184,753)
	5901C	MS Project Online Enhancements	10,577	8,000	(2,577)				10,577	8,000	(2,577)
	5901D	EA Strategic Portfolio Planning & Delivery (SPPD)	973,948	868,000	(105,948)	77,662			1,051,610	868,000	(183,610)
	5917	IT4IT: Data Management	32,608	0	(32,608)				32,608		(32,608)
	5917A	Data Mgmt_ Tooling	466,959	0	(466,959)	(90,136)			376,823		(376,823)
	5917B	Tableau Rationalization & Power BI Enhancements	0	255,235	255,235					494,000	494,000
	5917C	Business Objects / DataWarehouse Replacement	0	422,120	422,120					810,000	810,000
	5917E	Master Data Management	0	245,000	245,000					245,000	245,000
	5935	Solution Development Lab	8,427	11,000	2,573				8,427	11,000	2,573
	6182US	Global Career Site	0	0	0	37,914	37,914	34,425	110,253		(110,253)
	6183US	HR Reporting Analytics	7,177	0	(7,177)	272,000	31,700	31,700	342,577		(342,577)
	6185US	Onboarding 2.0	0	0	0	22,900	39,200	39,200	101,300		(101,300)
	6238US	Stakeholder and Internal Applications (SIA)	0	344,933	344,933					657,000	657,000
	6239US	Gridhome US Enhancements	0	379,149	379,149					683,000	683,000
	6241US	RSA Archer Upgrade & Refresh	0	294,138	294,138					580,000	580,000
	6242US	ARIBA System Enhancements	0	264,755	264,755	125,575	68,165	68,165	261,905	522,000	260,095
	6243US	BoardVantage	0	73,932	73,932					124,000	124,000
	6252US	Data Mgmt Tools and Initiatives	1,553,871	643,313	(910,557)	340,466	282,738	471,064	2,648,139	1,283,000	(1,365,139)
	6253US	Cloud API Migration - Legacy Integration Tools	1,323,817	266,232	(1,057,585)	367,561	375,662	498,055	2,565,095	525,000	(2,040,095)
	6254US	Platform as a Service (PaaS) on Azure	555,006	443,444	(111,561)	99,677	98,923	171,855	925,461	885,000	(40,461)
	6255US	Digital Foundation Tooling and DevSecOps	498,678	443,444	(55,234)	106,653	98,775	166,984	871,090	885,000	13,910
	6256US	Platform Upgrades: Salesforce/Maximo/Mulesoft	537,993	381,760	(156,233)	61,912	99,079	130,021	829,005	758,000	(71,005)
	6257US	GSD Engineering Lab/Data Synchronization	0	939,983	939,983					1,873,000	1,873,000
	6258US	Data Archival	0	731,919	731,919					1,463,000	1,463,000
	6258USA	Data Archival Platform Implementation	1,515,370	0	(1,515,370)	39,525	67,763		1,622,658		(1,622,658)
	6258USB	Data Archival: US Group Function Portfolio	224,147	0	(224,147)	245,073	204,500	193,862	867,582		(867,582)
	6259US	EA Foundation: Tech Debt/Resiliency/Rationalization	0	843,413	843,413					1,650,000	1,650,000
	6260US	EA Foundation: Increased Capability	854,086	1,918,600	1,064,513	137,748	34,940	24,940	1,051,714	3,755,000	2,703,285
	6261US	ServiceNow Upgrade: Stay Current FY22	447,203	128,644	(318,559)	93,926	128,054	106,901	776,084	255,000	(521,084)
	6262US	Additional ServiceNow (SNOW) Capability	0	128,644	128,644					255,000	255,000
	6263US	IT Enterprise Project Mgmt Office (PMO)	0	1,065,433	1,065,433					2,093,000	2,093,000
	6264US	Apptio Development	273,979	695,972	421,993	25,000			298,979	1,398,000	1,099,021
	6265	HR Policy Compliance	0	98,973	98,973		21,000	21,000	42,000	166,000	124,000
	6266US	EA Strategic Portfolio Planning & Delivery / Additional Capability	510,233	376,990	(133,243)	371,874	337,997	462,142	1,682,246	750,000	(932,246)
	6270US	Digitizing Learning	0	177,488	177,488		112,000	56,000	168,000	350,000	182,000
	6273	Workforce Data Domain	0	0	0	168,000	258,000	395,000	821,000		(821,000)
	6273US	Workforce Data Domain HR data governance Tools Processes EDM	151,454	848,675	697,221	132,810	200,760	88,300	573,324	1,610,000	1,036,676
	6346USA	Strategic External Channel Governance - Persona Based Websites	0	0	0	151,178	58,465	86,324	295,967		(295,967)
	6346USB	CA:Website Analytics	0	0	0	5,000	10,000	5,000	20,000		(20,000)
	6346USC	Domain Management Refresh	0	0	0	77,887	61,815	2,210	141,912		(141,912)
	6346USE	Brand Central Design and Implementation	0	0	0	71,370	37,114	37,114	145,598		(145,598)
	6346USF	Brand Central Design and Implementation	0	0	0	82,259	4,427	3,990	90,676		(90,676)
	6346USG	External Emailing Capability	0	0	0	15,612	19,329	1,783	36,724		(36,724)
	6346USI	Video Strategic Management and Archiving (Enterprise)	0	0	0	34,936	24,533	896	60,365		(60,365)
	6360US	IT Planning: Enterprise PMO	502,836	0	(502,836)	134,792	115,109	163,972	916,709		(916,709)
	HR22	HR + Placeholder	0	1,966,828	1,966,828					3,631,000	3,631,000
Group Functions Total			**14,179,033**	**25,144,808**	**10,965,774**	**3,793,836**	**3,556,694**	**4,161,807**	**25,691,370**	**40,276,000**	**14,584,630**
Group Functions Total			**32,116,820**	**44,397,084**	**12,280,264**	**8,922,393**	**14,212,534**	**8,429,735**	**63,681,482**	**69,176,000**	**5,494,518**

Project View - US Electric

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Variance	Forecast 10	Forecast 11	Forecast 12	Year End Forecast	Year End Budget	Variance
US Electric	**EBU Digital**	5054	On My Way (EBU Digital Solution) Phase 2	3,812,478	7,608,730	3,796,252	779,755	1,038,505	852,483	6,483,221	9,400,000	2,916,779
		5912	On My Way (EBU Digital Solution)	(86,635)	0	86,635				(86,635)		86,635
		6073	VMO Validation Phase	4,731,578	5,043,521	311,943	766,690	452,682	454,487	6,405,437	6,400,000	(5,437)
		6076	FutureNow Validation Phase	18,022,919	0	(18,022,919)	930,279	1,582,501	687,046	21,222,745		(21,222,745)
		6291	ICU-Out (Electric Digita)	31,095	3,015,918	2,984,823	43,841	43,841	223,735	342,512	6,000,000	5,657,488
		6294	GridMD	0	2,759,393	2,759,393					5,500,000	5,500,000
		6425	DER Active Resource Integration NY	350,068	0	(350,068)	440,122	487,506	213,739	1,491,435		(1,491,435)
		6437	GridMD	54,844	0	(54,844)	206,010	31,010	31,010	322,874		(322,874)
		EBU22	EBU Digital Placeholder	0	327,401	327,401					584,000	584,000
		FN22	Future Now - Next Phase	0	4,184,180	4,184,180					8,499,999	8,499,999
		PD22	Predictament	0	3,385,427	3,385,427					6,550,000	6,550,000
	EBU Digital Total			**26,916,346**	**26,324,570**	**(591,776)**	**3,166,697**	**3,636,045**	**2,462,500**	**36,181,588**	**42,933,998**	**6,752,410**
	Electric BU	4481	US MDS-Energy Accounting System (EAS) migration to Wholesale Settlement Application (WSA)	2,166,362	2,127,000	(39,362)	52,132	47,688	45,865	2,312,047	2,127,000	(185,047)
		4483	US Control-Transmission Outage Application (iTOA) Upgrade to HTML 5 Web	1,505,378	743,000	(762,378)	82,483			1,587,861	743,000	(844,861)
		4704R	NY REV Street Lighting Delivery	98,146	137,000	38,854				98,146	137,000	38,854
		5241	East Pulaski Energy Storage System IS Network	13,472	3,000	(10,472)	2,200	2,780	46,000	64,452	3,000	(61,452)
		5242	Buffalo Energy Storage System IS Network	64,220	51,000	(13,220)	2,465	2,476	17,500	86,661	51,000	(35,661)
		5471Y	NY Clean Innovation	0	192,580	192,580					323,000	323,000
		5486	AVLS Modem Reconfiguration	45,471	17,000	(28,471)				45,471	17,000	(28,471)
		5570	Electric Field Crew iPad Time Entry and Basic Capability -NE	(37,279)	0	37,279				(37,279)		37,279
		5702	GridStack	(108,028)	0	108,028				(108,028)		108,028
		5773B	Visitor Logging for NERC-CIP Reliability Compliance	14,221	0	(14,221)	23,871	93,566	157,194	288,852		(288,852)
		5871	NY Energy Storage Systems (ESS)	0	167,539	167,539					281,000	281,000
		5871A	NY Battery Energy Storage Systems	0	0	0		10,000	30,000	40,000		(40,000)
		5871D	Clean Energy Initiatives	0	0	0		25,000	50,000	75,000		(75,000)
		5887A	ARCOS - Reporting Replacement	19,352	578,467	559,114	19,392	104,551	106,220	249,515	586,000	336,485
		5888B	Power Quality Upgrade	0	0	0	37,500	45,000	45,000	127,500		(127,500)
		5889A	Control Room Event Recording	53,816	263,000	209,184				53,816	263,000	209,184
		5889B	Operational Excellence, Operational Support (Power Quality Upgrade) *B	0	66,960	66,960					128,000	128,000
		5889C	Electric Regulatory Enhancements	31,413	0	(31,413)	10,000	10,000	25,000	76,413		(76,413)
		5890A	Corporate PI	27,181	435,000	407,819	75,910	138,132	151,714	392,937	630,000	237,063
		5890B	CYME Gateway	24,878	494,000	469,122				24,878	754,000	729,122
		5890C	Computapole Replacement F&A	0	0	0		25,000	45,000	70,000		(70,000)
		5891A	Electric MV90 Upgrade	697,131	1,710,240	1,013,109	80,020	343,495	257,788	1,378,434	1,808,000	429,566
		5891B	Field Collection System (FSC) Upgrade POC	(2,609)	0	2,609				(2,609)		2,609
		5891C	FCS Upgrade	370,999	1,397,456	1,026,458	64,358	541,513	362,825	1,339,695	1,800,000	460,305
		5891E	MTS Retail Settlement	212,597	840,000	627,403	36,931	61,749	279,985	591,262	1,890,000	1,298,738
		5891I	Price Transmission (Retail Price) System Replacement	141,393	0	(141,393)	50,652	147,371	179,450	518,866		(518,866)
		5891X	Operational Excellence - Meter Data Services (Price Transmission) *X	0	798,696	798,696					1,575,000	1,575,000
		5892	Operations Excellense - Control Center Operation	0	432,530	432,530					851,000	851,000
		5892A	Control Room EOL Server and Tool Replacement	0	0	0	20,000	325,000	501,250	846,250		(846,250)
		5892C	IDS-DVD Dig Safe DVD Replacement	17,717	476,471	458,753	21,279	103,609	110,054	252,659	675,000	422,341
		5893A	Mobility Mwork Replacement for Trouble Workers	219,172	1,530,000	1,310,828	25,923	22,410	19,502	287,007	2,190,000	1,902,993
		5915USIT	INVP 5915 Grid Mod Cyber Security	(6,172)	0	6,172				(6,172)		6,172
		5976	Clifton Park Innovative Pricing Rate Design Project	0	13,117	13,117					22,000	22,000
		6043	Plan Forward	24,622	0	(24,622)				24,622		(24,622)
		6051	STORMS Extended Performance	(170,306)	0	170,306				(170,306)		170,306
		6320	Telecom & Network Deep Dive	35,416	0	(35,416)				35,416		(35,416)
	Electric BU Total			**5,458,563**	**12,474,054**	**7,015,491**	**605,116**	**2,049,340**	**2,430,347**	**10,543,366**	**16,854,000**	**6,310,634**
	Wholesale Networks	5756	Digital Substation Transformation	(463)	0	463				(463)		463
		5773D	NYISO PMU Requirement for Large Generators	86,368	239,907	153,539	41,891	84,580	151,189	364,028	473,000	108,972
		5822C	Generation Asset Management Upgrade	1,449,803	2,482,788	1,032,985	105,248	211,543	302,924	2,069,518	2,548,000	478,482
		5973	FY21 Wholesale Networks and US Capital Delivery Minor Works	(16,708)	0	16,708				(16,708)		16,708
		6043G1	Content Management for Engineering	554,692	1,303,457	748,765	38,267	176,923	104,280	874,162	1,394,000	519,838
	Wholesale Networks Total			**2,073,693**	**4,026,153**	**1,952,460**	**185,406**	**473,046**	**558,393**	**3,290,538**	**4,415,000**	**1,124,462**
US Electric Total				**34,448,602**	**42,824,777**	**8,376,175**	**3,957,219**	**6,158,431**	**5,451,240**	**50,015,492**	**64,202,998**	**14,187,506**

Project View - US Gas

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Variance	Forecast 10	Forecast 11	Forecast 12	Year End Forecast	Year End Budget	Variance
US Gas	Capital Delivery	4990	Primavera Upgrade/Stabilization	438,095	484,000	45,905				438,095	484,000	45,905
		5274A	New Storms Respose Management System Implementaition	1,040,210	1,229,186	188,976	297,977	297,574	319,315	1,955,076	1,236,000	(719,076)
		5830H	Capital Delivery - Safety and Risk Artificial Intelligence in the Field	0	520,722	520,722					1,046,000	1,046,000
		6222	PIF Replacement	605,326	0	(605,326)	249,289	179,351	120,260	1,154,226		(1,154,226)
	Capital Delivery Total			**2,083,631**	**2,233,909**	**150,278**	**547,266**	**476,925**	**439,575**	**3,547,397**	**2,766,000**	**(781,397)**
	Gas BU	4769	Gas Storm Hardening Using Remote Service Valves (RSV's)	611,723	743,000	131,277	92,320	93,660	56,212	853,915	743,000	(110,915)
		4898	Asset Management System for LNG	355,621	3,431,518	3,075,897	303,627	480,963	1,061,791	2,202,002	4,548,000	2,345,998
		5785C	Stabilization of Critical Gas Legacy Applications (ePermits)	84,714	0	(84,714)				84,714		(84,714)
		5785I	NRG GIS Replacement Project	3,553,838	1,030,000	(2,523,838)	103,364	3,048		3,660,250	1,030,000	(2,630,250)
		5785J	Security & Resilience - SalesForce	(0)	1,409,083	1,409,083				(0)	2,740,000	2,740,000
		5785K	Security & Resilience - Maximo	10,300	1,133,189	1,122,888				10,300	2,260,000	2,249,700
		5785L	Security & Resilience - Mulesoft	(420,916)	639,946	1,060,862				(420,916)	1,243,000	1,663,916
		5785O	Itron 100G to 500/550G ERT Transition	0	0	0			8,000	8,000		(8,000)
		5785P	Storm Hardening - Itron Intelis Meter Implementation	0	0	0		16,000	168,000	184,000		(184,000)
		5785Q	Field Collection Systems (FCS) Upgrade (Gas Only)	0	0	0	150,000	258,000	318,000	726,000		(726,000)
		5787A	Gas Control Center Simulator	3,463	0	(3,463)				3,463		(3,463)
		5787D	GTIS Functional Upgrades	37,784	218,850	181,066	63,149	63,149	63,149	227,231	377,000	149,769
		5787F	Gas Transportation Information System (GTIS) Phase 3 (F&A)	0	75,124	75,124					126,000	126,000
		5787G	F&A Study for Gas Demand Response (DR)	67,866	0	(67,866)	39,003	29,446	27,202	163,517		(163,517)
		5788A	FY21 Gas Regulatory and Operational Enhancements (Minor Works)	(3,678)	0	3,678				(3,678)		3,678
		5788D	Interface MyHub To OnBoard LMS OQ External Site	122,001	155,000	32,999	8,000			130,001	155,000	24,999
		5788G	FY22 Gas Regulatory and Operational Enhancements (Minor Works)	68,047	149,056	81,009	51,900	73,331	56,769	250,047	250,000	(47)
		5789B	Stabilization of Critical Gas Legacy Applications (Portal for ArcGIS)	2,684	15,000	12,316				2,684	15,000	12,316
		5789C	Stabilization of Critical Gas Legacy Applications (CWQ)	(2,194)	0	2,194				(2,194)		2,194
		5789E	Summons and Corrective Actions Response (CAR) Work Tracker	25,889	157,000	131,111	67,955	68,292	53,089	215,225	305,000	89,775
		6043B6	Mobile devices rollout for Plan Forward	21,288	0	(21,288)				21,288		(21,288)
		6353A	FY22 Gas Application Rationalization	56,697	0	(56,697)				56,697		(56,697)
		6392	Contractor Ipad for Gas Business	136,768	0	(136,768)		253,000	260,000	649,768		(649,768)
		6441	Scalability & Resilience for Enterprise Gas Platforms	9,938	0	(9,938)	700,000	900,000	800,000	2,409,938		(2,409,938)
		AC	Andrea Costa Efficiency Adj	0	0	0		800,000	1,800,000	2,600,000		(2,600,000)
	Gas BU Total			**4,741,832**	**9,156,764**	**4,414,932**	**1,579,318**	**3,038,889**	**4,672,212**	**14,032,251**	**13,792,000**	**(240,251)**
US Gas Total				**6,825,463**	**11,390,673**	**4,565,210**	**2,126,584**	**3,515,814**	**5,111,787**	**17,579,648**	**16,558,000**	**(1,021,648)**

Project View - Customer

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Variance	Forecast 10	Forecast 11	Forecast 12	Year End Forecast	Year End Budget	Variance
Customer	Customer Digital	4750	Customer Exp. Transformation technology program	0	1,874,263	1,874,263					3,719,000	3,719,000
		4750J	Web Self Service	11,435,005	0	(11,435,005)	1,033,833	1,254,411	85,938	13,809,187		(13,809,187)
		4750JX	Web Self Service Decomm	0	0	0	50,000	150,000	88,000	288,000		(288,000)
		4750K	Personalization	(508,171)	0	508,171				(508,171)		508,171
		4750M	ODS	2,620,089	0	(2,620,089)	309,295	326,971	12,077	3,268,432		(3,268,432)
		4750N	Automated Customer Services	(407,137)	0	407,137				(407,137)		407,137
		5673	My Business Account (Nucleus Phase 2)	3,257,967	4,145,221	887,254	382,158	361,559	377,481	4,379,165	5,600,000	1,220,835
		5859A2	Customer and Agent Experience - IVR Baseline	0	2,068,408	2,068,408					3,940,000	3,940,000
		5859B2	Customer and Agent Experience - IVR Minor Enhancements	0	1,017,081	1,017,081					1,950,000	1,950,000
		5905	Nucleus Project	192,914	0	(192,914)				192,914		(192,914)
		6206	Customer Smart Target	23,362	0	(23,362)		1,250,000	1,250,000	2,523,362		(2,523,362)
	Customer Digital Total			**16,614,030**	**9,104,974**	**(7,509,057)**	**1,775,286**	**3,342,941**	**1,813,496**	**23,545,753**	**15,209,000**	**(8,336,753)**
	Customer Operations	3932	Customer Contact Center / SDC Technology Upgrade Implement Solution	4,790,096	2,965,000	(1,825,096)	33,168	20,161		4,843,425	2,965,000	(1,878,425)
		4904	Upgrade Servers for CCAE Reporting	190,942	94,000	(96,942)				190,942	94,000	(96,942)
		5069	RightFax Upgrade/Remediation	0	129,837	129,837					257,000	257,000
		5125	Customer Connection Program	2,490,553	1,850,225	(640,328)	287,128	288,417	291,175	3,357,273	2,558,000	(799,273)
		5146	Smart Cities Schenectady	17	0	(17)				17		(17)
		5160	NY Stack VDER and Portal Integration	1,340,797	1,557,412	216,614	234,884	302,252	303,657	2,181,590	2,111,000	(70,591)
		5345B	Bill Print and Bill Composition Implementation	896,690	1,174,369	277,678	93,305	74,689	150,382	1,215,066	1,680,000	464,934
		5430	NY Green Button Connect	1,443,582	890,000	(553,582)		148,045		1,591,627	890,000	(701,627)
		5625	Customer Billing Minor Enhancements FY21	104,678	0	(104,678)				104,678		(104,678)
		5627B	Experian Platform Upgrade (Tallyman & PMP)	0	66,953	66,953					132,000	132,000
		5627C	Outreach Optimization	0	133,414	133,414					263,000	263,000
		5627D	Post COVID Collections Restart	0	155,474	155,474					300,000	300,000
		5722	KEDLI Rate Case (CRIS)	322,552	58,000	(264,552)	59,424	37,721	10,142	429,839	58,000	(371,839)
		5723	KEDNY Rate Case	609,877	313,000	(296,877)	131,414	126,946	129,849	998,086	313,000	(685,086)
		5778	Customer Billing Improvement Program - Phase 1	1,922,727	8,173,295	6,250,568	641,923	613,833	326,236	3,504,719	11,250,000	7,745,281
		5795E	MA Electric Vehicle Rebate Program	13,791	0	(13,791)				13,791		(13,791)
		5795F	CDG Net Metering	606,612	902,361	295,749	53,261	55,333	119,158	834,364	992,000	157,636
		5795G	Telephone Consumer Protection Act (TCPA)	58,111	0	(58,111)				58,111		(58,111)
		5836	Lockbox Payments and Credit Payments Implementation	631,279	518,000	(113,279)	(4,435)	624		627,468	518,000	(109,468)
		5859A	Customer and Agent Experience - IVR Baseline	0	359,100	359,100			1,033,038	1,033,038	513,000	(520,038)
		5859B	Customer and Agent Experience - IVR Minor Enhancements	0	777,000	777,000					1,110,000	1,110,000
		5863	Fix the Foundation	0	1,753,867	1,753,867		100,000	150,000	250,000	3,450,000	3,200,000
		5863D	AIMS Stability	740,227	0	(740,227)				740,227		(740,227)
		5903	Customer Operational Enhancements FY21	1,037	0	(1,037)				1,037		(1,037)
		5926A	NiMO Rate Case	1,198,831	663,750	(535,081)	20,854	54,215	46,161	1,320,061	975,000	(345,061)
		5926B	MA Gas Rate Case	0	326,600	326,600					488,000	488,000
		5987	Smart City Phase II	130,627	265,573	134,946	21,199	41,448	3,633	196,907	290,000	93,093
		5996	Simon Baker's Efficiency Target	0	0	0		(2,000,000)		(2,000,000)		2,000,000
		6007	Inside Agent	876,649	1,166,369	289,721	60,509	90,508	42,865	1,070,531	1,187,000	116,469
		6032	Customer Billing COVID Transition	443,052	244,908	(198,144)	25,394	25,040	24,684	518,170	450,000	(68,170)
		6042	Customer US Retail Web Infrastructure Upgrade	10,268	0	(10,268)				10,268		(10,268)
		6066	FCP Upgrade for CSS - Plan Forward	1,477,043	1,922,920	445,877	192,606	105,036	127,544	1,902,229	2,080,000	177,771
		6112	WFH Hardware - Plan Forward	231,377	0	(231,377)				231,377		(231,377)
		6115	Customer COVID-19 Payment Plan for IVR and Web	226,603	0	(226,603)				226,603		(226,603)
		6214	Customer Operational Enhancements FY22	84,220	112,500	28,280	16,629	14,679	24,850	140,378	150,000	9,622
		6217	Customer Billing Minor Works FY22	311,724	214,640	(97,084)	33,289	33,289	33,289	411,591	360,000	(51,591)
		6250	EV End-End Solution	0	863,000	863,000			50,000	50,000	863,000	813,000
		6302	2022 AIMS Platform	67,353	0	(67,353)	5,000	84,000	84,000	240,353		(240,353)
		6302X	2022 AIMS Platform	0	0	0		75,000	25,000	100,000		(100,000)
		6349	Customer Data Platform	2,467,483	0	(2,467,483)	1,051,266	1,178,288	1,008,311	5,705,348		(5,705,348)
		6393	Technology Validation for CRIS to CSS Migration	372,772	0	(372,772)	2,516			375,288		(375,288)
		6427	RI Purchase of Receivables (POR) Program	810,356	0	(810,356)			66,163	876,519		(876,519)
		6434	CRIS to CSS Migration	26,727	0	(26,727)				26,727		(26,727)
		6435	Customer PCI Remediation	0	0	0	10,000	418,700	114,000	542,700		(542,700)
	Customer Operations Total			**24,898,653**	**27,651,566**	**2,752,914**	**2,969,334**	**1,888,224**	**4,164,137**	**33,920,348**	**36,297,000**	**2,376,652**
Customer Total				**41,512,683**	**36,756,540**	**(4,756,143)**	**4,744,620**	**5,231,165**	**5,977,633**	**57,466,101**	**51,506,000**	**(5,960,101)**

Project View - I&O

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Variance	Forecast 10	Forecast 11	Forecast 12	Year End Forecast	Year End Budget	Variance
I&O	Cloud & Hosting	4489	Active Directory Improvements	1,005	0	(1,005)				1,005		(1,005)
		5750	Directory Services - US	16,114	0	(16,114)				16,114		(16,114)
		5837	Apllication Virtualization	8,566	0	(8,566)				8,566		(8,566)
		5874	Cloud Center of Excellence	(930,350)	0	930,350	1,929,700			999,350		(999,350)
		5878	Dell Discovery	69,336	0	(69,336)				69,336		(69,336)
		5942	Legacy AD Remediation	65,767	0	(65,767)				65,767		(65,767)
		6018	Security Operations Services (SOS)	583,092	0	(583,092)				583,092		(583,092)
		6022	US Logical Database Transition	562	0	(562)				562		(562)
		6049	Hosting Infrastructure Refresh to HCI	(207,372)	0	207,372				(207,372)		207,372
		6050	Migration to IBM P9	44,287	0	(44,287)				44,287		(44,287)
		6052	Migration of Current Storage to Strategic Deployment ??? US	(32,502)	0	32,502				(32,502)		32,502
		6088	6088 - IBM P8 Digital Asset Build	36,041	0	(36,041)				36,041		(36,041)
		6106	Data Protection Suite (DPS)	46,701	0	(46,701)				46,701		(46,701)
		6117	HCI Lite Design & Deployment	(90,974)	0	90,974				(90,974)		90,974
		6200	I&O US Cloud and Hosting FY22	9,776,879	8,285,448	(1,491,431)	925,310	1,177,617	891,906	12,771,712	10,995,000	(1,776,712)
		6223	I&O US Data Center Transformation FY22	15,945,496	20,230,305	4,284,809	2,310,203	9,615,994	401,969	28,273,662	21,420,000	(6,853,662)
	Cloud & Hosting Total			**25,332,649**	**28,515,754**	**3,183,105**	**5,165,213**	**10,793,611**	**1,293,875**	**42,585,348**	**32,415,000**	**(10,170,348)**
	CNI	5867	Network Security Infrastructure Program	(51,513)	0	51,513				(51,513)		51,513
		5867A	OMS Router Replacement & IT Service Management	30,891	0	(30,891)				30,891		(30,891)
		5867E	Substation Access Points	59,191	0	(59,191)				59,191		(59,191)
		6199	I&O US CNI FY22	365,175	1,010,441	645,266	199,870	519,013	1,251,667	2,335,725	1,080,000	(1,255,725)
	CNI Total			**403,743**	**1,010,441**	**606,698**	**199,870**	**519,013**	**1,251,667**	**2,374,293**	**1,080,000**	**(1,294,293)**
	End User	4987	SOE (Windows) Upgrade and Device Refresh	4,608,572	0	(4,608,572)	(475,423)			4,133,149		(4,133,149)
		5487	Collaboration & Unified Communications	3	0	(3)				3		(3)
		5487E	ICE Public Folders - Migration/Decom	91,113	0	(91,113)				91,113		(91,113)
		5487G	Azure Cloud Mail Routing & Exchange 2010 Decom	(111,764)	0	111,764		29,000		(82,764)		82,764
		6091	Return to Office Network Changes	14,205	0	(14,205)				14,205		(14,205)
		6097	Cloud Video Interoperability	159,382	0	(159,382)				159,382		(159,382)
		6108	SharePoint to SharePoint Online	74,281	0	(74,281)				74,281		(74,281)
		6136	System Center Configuration Manager (SCCM)	(2,211)	0	2,211				(2,211)		2,211
		6198	I&O US End User Experience FY22	2,690,092	6,335,844	3,645,752	(32,188)	897,164	638,550	4,193,618	8,055,000	3,861,382
	End User Total			**7,523,674**	**6,335,844**	**(1,187,830)**	**(507,611)**	**926,164**	**638,550**	**8,580,777**	**8,055,000**	**(525,777)**
	Network	5521	LAN Infrastructure Program	394	0	(394)				394		(394)
		5521A	Verizon - Aruba Wireless LAN (Waltham)	(68,300)	0	68,300				(68,300)		68,300
		5522A	End User VPN Replacement	(89,799)	0	89,799				(89,799)		89,799
		5523	Voice Infrastructure Program	(101)	0	101				(101)		101
		5523A	Northborough and Avaya PBX Replacement/Upgrade (NE Control Room sites)	(563,441)	0	563,441				(563,441)		563,441
		5523B	Upstate NY PBX Replacement	16,838	0	(16,838)				16,838		(16,838)
		5523F	Dedicated CNI Voice Network Buildout	219	0	(219)				219		(219)
		5523G	Downstate NY PBX Replacement Upgrade	22,219	0	(22,219)				22,219		(22,219)
		5524A	WAN Diversity at 18 Regional Storm Sites	24,432	0	(24,432)				24,432		(24,432)
		5754	Network Equipment End of Life Replacement - LAN	93,809	0	(93,809)				93,809		(93,809)
		5759	UPS Infrastructure	(33,517)	0	33,517				(33,517)		33,517
		5774	WiFi Phase 2	(813)	0	813				(813)		813
		5838	Managed Networks	526,898	0	(526,898)				526,898		(526,898)
		5846	Substation Network Connectivity & Comms	86,368	0	(86,368)				86,368		(86,368)
		6046	SD WAN FY21 Expansion	(1,035,699)	0	1,035,699				(1,035,699)		1,035,699
		6197	I&O US Networks FY22	10,239,837	14,483,709	4,243,873	1,349,958	2,076,035	5,993,319	19,659,149	18,195,001	(1,464,148)
	Network Total			**9,219,342**	**14,483,709**	**5,264,367**	**1,349,958**	**2,076,035**	**5,993,319**	**18,638,654**	**18,195,001**	**(443,653)**
	Operations	5161	Device Refresh	447,637	0	(447,637)				447,637		(447,637)
		5489	Emerging Technology	(34,229)	0	34,229				(34,229)		34,229
		5489A	IBM SNOW Automation	(256,487)	0	256,487				(256,487)		256,487
		5489B	WiPro SNOW Automation	(206,704)	0	206,704				(206,704)		206,704
		5580	Managed Workspace Services Transition & Tx	879	0	(879)				879		(879)
		5876	Pure Technical Debt Remediation	405,676	0	(405,676)	66,985			472,661		(472,661)
		5997	Caitlin Davidson's Efficiency Target	0	0	0		500,000	500,000	1,000,000		(1,000,000)
		6114	Managed Workspace Services Transformation	(10,766)	0	10,766	(160,443)			(171,209)		171,209
		6196	I&O US Operations FY22	7,061,237	9,140,563	2,079,326	(1,888,874)	537,873	1,863,335	7,573,571	12,450,002	4,876,431
	Operations Total			**7,407,244**	**9,140,563**	**1,733,319**	**(1,982,332)**	**1,037,873**	**2,363,335**	**8,826,120**	**12,450,002**	**3,623,882**
I&O Total				**49,886,653**	**59,486,311**	**9,599,658**	**4,225,098**	**15,352,696**	**11,540,746**	**81,005,193**	**72,195,003**	**(8,810,190)**

Project View - Security

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Variance	Forecast 10	Forecast 11	Forecast 12	Year End Forecast	Year End Budget	Variance
Security	**Physical Security**	3617A	Physical Security - G20	89,889	0	(89,889)	150,000	50,000	100,000	389,889		(389,889)
		3617F	PS - NIMOE	1,668,660	0	(1,668,660)	300,000	150,000	100,000	2,218,660		(2,218,660)
		3617G	Physical Security - KEDNY - Equip Replacement	43,855	0	(43,855)	200,000	400,000	300,000	943,855		(943,855)
		3617H	Physical Security - KEDLI - Equip Replacement	483,408	0	(483,408)	100,000	150,000	50,000	783,408		(783,408)
		3617I	Physical Security - MASS Electric - Equip Replacement	387,987	0	(387,987)	100,000	50,000	50,000	587,987		(587,987)
		3617J	Physical Security - Boston Gas	100,894	0	(100,894)	80,000	100,000	188,000	468,894		(468,894)
		3617K	Physical Security - Colonial Gas	34,696	0	(34,696)	30,000	50,000	50,000	164,696		(164,696)
		3617L	Physical Security - RI Electric	76,189	0	(76,189)	30,000	10,000	10,000	126,189		(126,189)
		3617M	Physical Security - RI Gas	18,815	0	(18,815)	30,000	10,000	20,000	78,815		(78,815)
		3617N	Physical Security - NIMO EGT	(11)	0	11				(11)		11
		5748	Security Control Center Back-Up	124,185	0	(124,185)				124,185		(124,185)
		6244	Physical Security	0	3,291,719	3,291,719					6,687,001	6,687,001
	Physical Security Total			**3,028,567**	**3,291,719**	**263,152**	**1,020,000**	**970,000**	**868,000**	**5,886,567**	**6,687,001**	**800,434**
	Security	3683USD	Develop Robust Incident Response	0	0	0				0		(0)
		3683USF	Enterprise Centralized Patch Management	43,392	0	(43,392)				43,392		(43,392)
		3683USH	Network Security Improvements	512,162	0	(512,162)	(283,798)			228,364		(228,364)
		3683USN	SOAR - Security Orchestration Automation and Response	474,859	0	(474,859)		734,000		1,208,859		(1,208,859)
		3683USX	Application Security As a Service	(1,703)	0	1,703				(1,703)		1,703
		5702USA	Malware Defense - Endpoint Protection	(627,148)	0	627,148				(627,148)		627,148
		5702USC	Malware Defense - System Hardening	(39,656)	0	39,656				(39,656)		39,656
		5820	Data Protection	7,679	0	(7,679)				7,679		(7,679)
		5828	Secure Web Gateway	(790)	0	790				(790)		790
		5848	Vulnerability Management	183,852	1,007,038	823,186	103,200	103,200	327,884	718,136	1,281,000	562,864
		5850	Network Security	(61,473)	0	61,473				(61,473)		61,473
		5850US	Network Security US	667,473	0	(667,473)	(148,216)	(145,193)		374,064		(374,064)
		5850USA	SSN Optimization US	169,924	0	(169,924)				169,924		(169,924)
		5850USB	vSTIG Firewall Upgrade US	(31,368)	0	31,368				(31,368)		31,368
		5850USC	Intrusion Detection System and Intrusion Prevention System US	(314)	0	314				(314)		314
		5850USD	Intrusion Detection System and Intrusion Prevention System US	675,390	0	(675,390)	(131,953)		30	543,467		(543,467)
		5870	Identity and Access Management	10,492,169	7,162,020	(3,330,149)	1,847,935	1,160,461	675,430	14,175,995	9,239,000	(4,936,995)
		5933	US Strategic Risk Remediation	149,002	3,170,083	3,021,082	450,450	1,315,187	1,102,513	3,017,152	5,856,000	2,838,848
		5940	Global VM Enterprise	19,943	0	(19,943)	(60,000)			(40,057)		40,057
		5963US	Platform Projects Annual Program FY21	(448,217)	0	448,217	22,333			(425,884)		425,884
		5998	Elaine Wilson's Efficiency Target	0	0	0	(1,156,528)	(1,156,528)	(1,156,528)	(3,469,584)		3,469,584
		6210US	FY22 Platform Tower	3,167,737	4,859,073	1,691,335	(44,369)	1,552,082	2,248,630	6,924,080	6,948,000	23,920
		6211US	FY22 Network Tower	1,254,173	2,988,107	1,733,935	579,170	953,671	660,574	3,447,588	4,299,000	851,412
		6245	Security Orchestration Automation & Response	0	1,583,250	1,583,250					2,111,000	2,111,000
		6245US	Security Orchestration Automation & Response	35,411	0	(35,411)	57,630	599,951	804,517	1,497,509		(1,497,509)
		6416	Operational Technology Privileged Access Workstation	320,737	0	(320,737)	957,641	434,699	545,899	2,258,976		(2,258,976)
		6417US	Vulnerability Management T2 FY22	0	0	0	51,933	401,440	176,280	629,653		(629,653)
	Security Total			**16,963,235**	**20,769,571**	**3,806,336**	**2,245,428**	**5,952,970**	**5,385,229**	**30,546,862**	**29,733,999**	**(812,863)**
Security Total				**19,991,802**	**24,061,290**	**4,069,488**	**3,265,428**	**6,922,970**	**6,253,229**	**36,433,429**	**36,421,000**	**(12,429)**

Project View - Other

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Variance	Forecast 10	Forecast 11	Forecast 12	Year End Forecast	Year End Budget	Variance
Other	**IT4IT**	6297	IT Project Evolution (FY22 Cost Savings & CTA)	0	12,405,902	12,405,902	200,000	200,000	265,000	665,000	22,900,000	22,235,000
		6297A	FY22 CTA - App Maintenance	0	0	0	200,000	200,000	250,000	650,000		(650,000)
		6297B	FY22 CTA - DC & Cloud	0	0	0	200,000	200,000	200,000	600,000		(600,000)
		6297C	FY22 CTA - App Licenses	0	0	0	200,000	200,000	200,000	600,000		(600,000)
		6297F	FY22 CTA - NOC Insource / RFP	45,578	0	(45,578)				45,578		(45,578)
		6297J	FY22 CTA - End User	0	0	0	100,000	100,000	100,000	300,000		(300,000)
		6297K	FY22 CTA - Help Desk	0	0	0	286,000	286,000	286,000	858,000		(858,000)
		6368USL	Mid Term Initiatives Opex CTA US Networks RFP	0	0	0				0		0
	IT4IT Total			**45,578**	**12,405,902**	**12,360,324**	**1,186,000**	**1,186,000**	**1,301,000**	**3,718,578**	**22,900,000**	**19,181,422**
	Other	#	COH reconcile to 4403	(3,495,922)	0	3,495,922				(3,495,922)		3,495,922
		5598	Application Rationalization	(167)	0	167				(167)		167
		6082	FY22 Planning Initiative	7,066	0	(7,066)				7,066		(7,066)
		COCU	5110S-IS2608 ServCo AJE COH-OpEx AR	157,783	0	(157,783)				157,783		(157,783)
		COEA	5110S-IS2706 ServCo AJE COH-OpEx AC	365,681	0	(365,681)				365,681		(365,681)
		COEL	5110S-IS2605 ServCo AJE COH-OpEx AE	46,563	0	(46,563)				46,563		(46,563)
		COGA	Capital OH for Gas Organization	80,095	0	(80,095)				80,095		(80,095)
		COPM	5110S-IS2612 ServCo AJE COH-OpEx AC	662,799	0	(662,799)				662,799		(662,799)
		COSC	5110S-IS2679 ServCo AJE COH-OpEx AC	395,090	0	(395,090)				395,090		(395,090)
		COSE	5110S-IS2735 ServCo AJE COH-OpEx AC	498,539	0	(498,539)				498,539		(498,539)
		COTD	5110S-IS2714 ServCo AJE COH-OpEx AC	2,005,404	0	(2,005,404)				2,005,404		(2,005,404)
	Other Total			**722,932**	**0**	**(722,932)**	**0**	**0**	**0**	**722,932**	**0**	**(722,932)**
	Other Business Tech	Burden	Burden	(2,561,487)	(3,233,455)	(671,968)	(300,000)	(300,000)	(300,000)	(3,461,487)	(4,648,000)	(1,186,513)
	Other Business Tech Total			**(2,561,487)**	**(3,233,455)**	**(671,968)**	**(300,000)**	**(300,000)**	**(300,000)**	**(3,461,487)**	**(4,648,000)**	**(1,186,513)**
Other Total				**(1,792,978)**	**9,172,447**	**10,965,424**	**886,000**	**886,000**	**1,001,000**	**980,022**	**18,252,000**	**17,271,977**

Niagara Mohawk Power Corporation d/b/a National Grid

The Brooklyn Union Gas Company d/b/a National Grid NY

KeySpan Gas East Corporation d/b/a National Grid

Cases 20-E-0380, 20-G-0381, 19-G-0309 and 19-G-0310

INFORMATION TECHNOLOGY CAPITAL INVESTMENT REPORT FOR QUARTER ENDED March 31, 2022

SUBMITTED TO:
NEW YORK STATE DEPARTMENT OF PUBLIC SERVICE STAFF

May 27, 2022

nationalgrid

Information Technology Capital Investment Report for Quarter Ended March 31, 2022

I. Introduction

Niagara Mohawk Power Corporation (Niagara Mohawk), The Brooklyn Union Gas Company (KEDNY), and KeySpan Gas East Corporation (KEDLI) (together "National Grid" or the "Companies") submit this Information Technology ("IT") Capital Investment Quarterly Report ("IT Quarterly Report") for the fourth quarter of fiscal year 2022 ("FY22").[1] Pursuant to the Public Service Commission's ("Commission") January 20, 2022 Order in Cases 20-E-0380 and 20-G-0381, and August 12, 2021 Order in Cases 19-G-0309 and 19-G-0310, the Company is to provide quarterly variance reports to Department of Public Service Staff ("Staff") regarding IT capital projects and investments.[2] The IT Quarterly Report provides an explanation of the Company's overall IT investment plan approach; descriptions of program investment categories; and a comparison of budgeted to actual spending.

IT capital projects and investments that are shared investments across National Grid's operating companies are implemented and owned by National Grid USA Service Company, Inc. ("Service Company") and allocated to the benefiting US operating companies, including Niagara Mohawk, KEDNY and KEDLI, in the form of Service Company rent expense once the investments are in-service. The Service Company rent expense includes a return on the capital investment (net of deferred taxes) plus booked depreciation expense.

Through Q4 FY22, National Grid spent $299.1 million compared to a year-to-date Service Company budget of $328.3 million, representing a $29.2 million underspend. Attachment 1 includes detail on budgets and actual spending for each of the portfolios and commentary regarding performance which has been presented to each of the Chief Information Officers responsible for the respective portfolios.

Attachment 2 includes a detailed breakdown of individual project spend by portfolio compared to the budgeted spend for each of those projects.

II. Program Categories/Spending Rationales

National Grid has re-organized the US Information Technology organization into six areas:

[1] The fourth quarter of FY22 is January 1, 2022 through March 31, 2022 ("Q4 FY22").

[2] *See* Cases 20-E-0380 and 20-G-0381, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service*, "Order Adopting Terms of Joint Proposal, Establishing Rate Plans and Reporting Requirements" (issued and effective January 20, 2022), Joint Proposal, Section IV.10.5; and Cases 19-G-0309 and 19-G-0310, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Energy East Corporation d/b/a National Grid for Gas Service*, "Order Approving Joint Proposal, as Modified, Imposing Additional Requirements" (issued and effective August 12, 2021), Joint Proposal, Section IV.6.5.

(1) Group Functions
(2) United States Electric
(3) United States Gas
(4) Customer
(5) Digital Platforms and Infrastructure Technology (Formerly Infrastructure and Operations)
(6) Security

IT capital investment is tracked based on these area designations. Descriptions of each portfolio category/spending rationale are provided below.

Group Functions

Group Functions executes National Grid's digital roadmaps, leveraging technology and reimagining processes in support of Finance, Enterprise Business Services, Human Resources, Legal, and Safety Health & Environmental, continuing to upgrade SOX related systems, and retiring outdated systems.

United States Electric

Provides IT Support to our US Electric Distribution and Transmission businesses. Modernizing our key legacy operational systems while building capabilities that will enable the energy transition and transformation to a Distribution System Operator (DSO). Through digital delivery, the Electric CIO organization provides greater efficiency and data driven insights into our workforce operations that provide:

- More automated and reliable grid controls systems
- Capital programs and projects that support the proliferation of Distributed Energy Resources (DERs)
- A growing market of prosumer electric customers

United States Gas

Provides IT Support to our US Gas Distribution and Transmission businesses by developing digital products based on the foundation of gas business enablement that deliver continuous improvement of our asset lifecycle management capabilities through:

- Risk-based asset investment planning
- Performance-based maintenance
- AI-based work optimization
- Mobile field products

Customer

Provides IT Support to our US Customer business, delivering best-in-class digital products that increase engagement and reduce costs to serve by:

- Modernizing our legacy "back-end" IT estate to support more digital products
- Liberating data and creating an abstracted and real-time customer data platform to assist with achieving energy efficiency targets

Digital Platforms and Infrastructure Technology (DPIT) (Formerly Infrastructure and Operations)
Supports our internal IT Architecture covering Cloud and Hosting, Critical Network Infrastructure, Networks, End User Experience and Operations. DPIT is focused on reducing National Grid's technical debt through cost effective, modernized IT infrastructure, enabling cloud capabilities that meet our risk profile. The organization enables a connected, seamless enterprise that is flexible, secure, reliable and scalable while maintaining acceptable levels of technical debt and risk.

Security
The Security function protects our people, premises, and digital systems with the objective of maintaining trust in National Grid services, focusing on:

- Reduction in likelihood and impact of security incidents
- Compliance with relevant laws, standards and regulations (*e.g.*, NERC, PCI DSS, GDPR, NIS-R)
- Improved user experience and standard of security practices across National Grid

The Security function aims to reduce the likelihood and impact of physical and cyber security incidents affecting National Grid. It does this by championing security best practices, technology, and promoting a security-awareness culture.

Other Category

For accounting purposes there is an additional category listed in Attachment 2 to this report. The "Other" category has 3 elements;

- The "IT4IT" sub-category is essentially a funding bucket for NPV-positive projects to enable efficiency savings in our IT-run the business operations.
- The "Other" sub-category is the allocation of operating expense overheads such as Enterprise Architecture and Solution Delivery charges to a centralized account. The capital charges for this overhead are made directly to individual projects.
- Finally, "Burdens" are the net removal of operating expense burdens to ensure our reporting agrees to our Corporate Controllable Cost reporting. Operating expense burdens are consolidated and reported separately under non-controllable costs in our corporate reporting packs.

The summaries below explain the subjects presented in the attachments to the Q4 FY22 report.

A. Attachment 1

1. Summary (pages 2 – 3) – Overview of US IT as a whole.
    a. High-level executive summary, including work and challenges faced across the portfolios and a brief explanation around the forecast (p. 2).
    b. Actual expenditure versus budget by portfolio for the full fiscal year, actual versus annual budget by portfolio (p. 2).
    c. Actual spend by month for FY22 compared to budget, prior year and run rate. Each bar is color-coded to give an overview of the comparative spend per portfolio (p. 3).

2. Portfolio Sections – The remainder of Attachment 1 includes six sections, one for each of the portfolios: Group Functions; US Gas; US Electric; Customer; DPIT; and Security. Each portfolio consists of 3 pages:
    a. The first page provides a narrative financial and business overview. The three charts on the first page show: (1) operating expenses; (2) capital expenses; and, (3) total expenses. Each chart shows FY22 actuals by period versus budget, with trend lines showing prior year expenditure and run rate. This page also shows the full year Q4 actual total expenditure compared to the Q4 budget.
    b. The second page of the portfolio sections has one chart on the top left of the page. This chart compares full fiscal year actual expenditure versus annual budget, by program. A priority investments list for the portfolio and their status appears at the top right hand of the page. Blue means complete, green means on track, amber means some risk, and red means focused intervention required. On the bottom half of this page is a chart showing actual spend for FY22 compared to budget, prior year and run rate for the portfolio. Each bar is color-coded to give an overview of the comparative spend per program.
    c. The third page of each portfolio section provides a status update on the various projects that are active, complete, and in the pipeline, as well as the methodology by which we are delivering the projects. There is also a key challenges section and the escalation point for that challenge.

    Attachment 1 is based on National Grid's monthly Strategic Portfolio Reviews (SPR) and reporting cadence to senior management of each portfolio, and ultimately to the IT leadership team. Security and DPIT do not currently participate in the SPR reviews but provide their respective senior management with monthly updates.

B. Attachment 2

INVP Level Variances - This attachment provides a detailed listing of individual projects in each of the six portfolios explaining spend in FY22.

(1) Total actual spend compared to budget and variance.
(2) A variance reason for all capital investments which are outside the 10% tolerance to budget.

C. Attachment 3

FY 22 Closure Summary - This attachment provides a detailed listing of individual projects in each of the six portfolios which were closed during the year. The attachment shows:

(1) The list of closure papers written and approved in FY22
(2) Total costs incurred in the closure papers
(3) Explanation of the benefits derived from the closure papers
(4) Summary of important/relevant lessons learned

In total, 105 projects were closed during FY22 with a total cost of $259.693 million.

National Grid

Quarterly Report – Q4 FY22
Information Technology

27th May 2022

nationalgrid

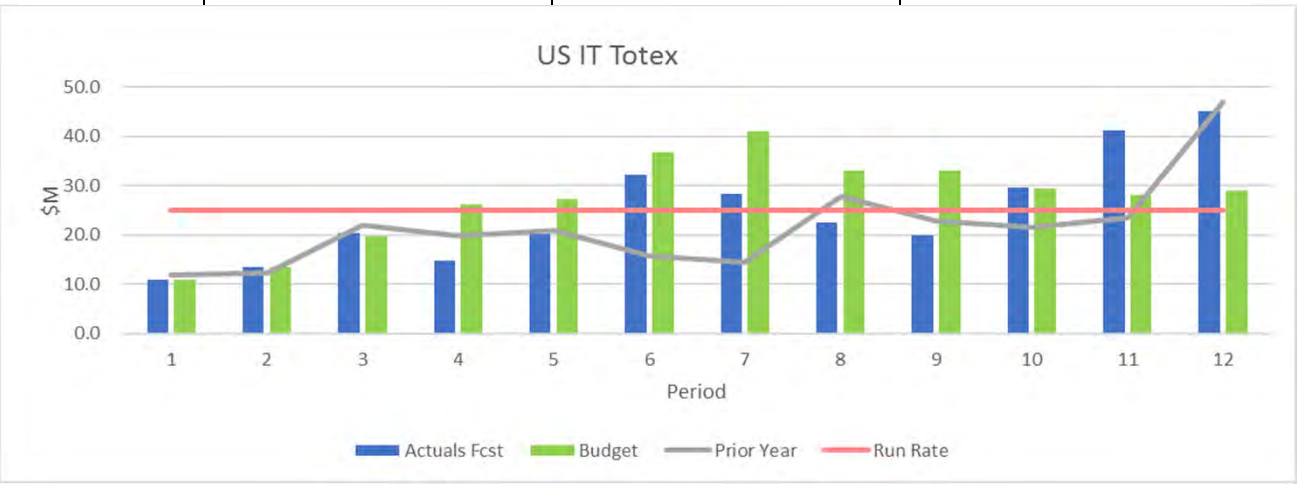
Executive Summary – US IT

US IT faced significant headwinds this year that caused delays across our investment plans. Staffing shortages, SME availability, supply chain issues, and divesture challenges all impacted progress. We were able to offset underspend in portfolios by pulling in out year investments and accelerating work where appropriate and advantageous. Notwithstanding the challenges, significant progress has been made with reducing technical debt (23.9%) and delivering capabilities to benefit both our internal and external customers. Key achievements were delivering the Unified Web Portal in Downstate New York, Future Now (data driven investment planning tool), Vegetation Management Optimization and Transmission Outage for our Electric Business. We continue to invest heavily in cyber security to mitigate threats to our assets.

Financial Summary

Portfolio View - Full Year $M

Portfolio	Capex			Opex			Totex		
	FY Actual	FY Budget	Variance	FY Actual	FY Budget	Variance	FY Actual	FY Budget	Variance
US Electric	43.2	57.1	13.9	5.3	7.1	1.8	48.5	64.2	15.7
US Gas	20.6	13.6	(7.0)	1.6	3.0	1.4	22.1	16.6	(5.6)
Customer	48.3	44.1	(4.2)	3.7	7.4	3.7	52.0	51.5	(0.5)
Group Functions	52.2	53.9	1.7	9.5	15.3	5.8	61.7	69.2	7.4
DPIT	68.9	60.1	(8.8)	13.8	12.1	(1.7)	82.6	72.2	(10.4)
Security	28.2	30.4	2.2	5.9	6.0	0.1	34.0	36.4	2.4
Other	0.0	12.0	12.0	(1.9)	6.3	8.2	(1.9)	18.3	20.2
Total	**261.4**	**271.3**	**9.9**	**37.7**	**57.0**	**19.3**	**299.1**	**328.3**	**29.2**



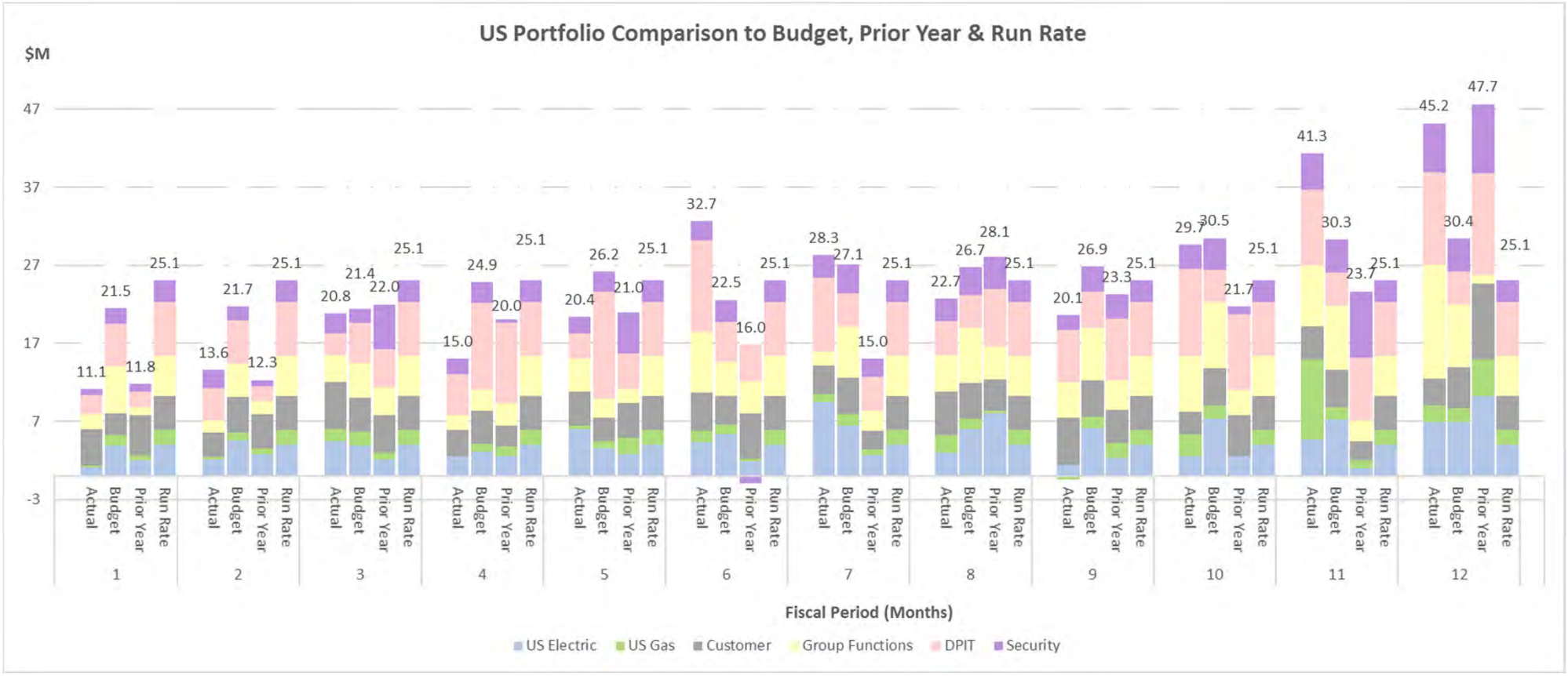
Many of the challenges the business faced this year impacted the wider economy.

Capacity and Resources
1. Availability of IT and business resources to support delivery
2. Recruiting and retaining high quality permanent resources is proving challenging
3. Significant change across the organization
4. Covid related issues with third-party contractor resource pools (earlier in year)

Supply Chain Issues
Commercial and Contracting Challenges, Supply Chain issues and chip shortages

Divestures
Portfolios have been impacted with separation leading to potential delays and stoppage of work to ensure oversight of investments for the benefit of our remaining customers.

Portfolio Comparison



Grid Q4 FY22 IT

National

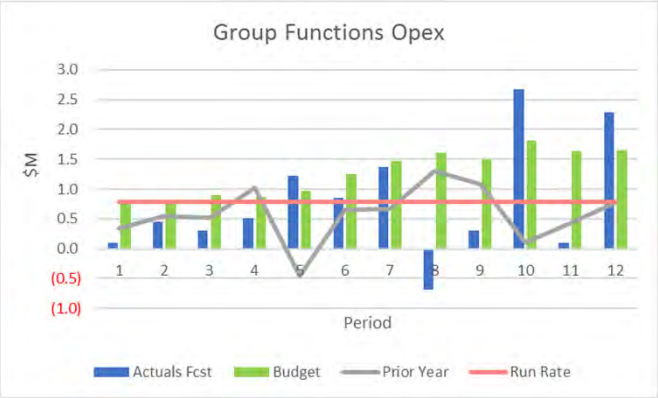
US Group Functions
Q4 FY22 Report

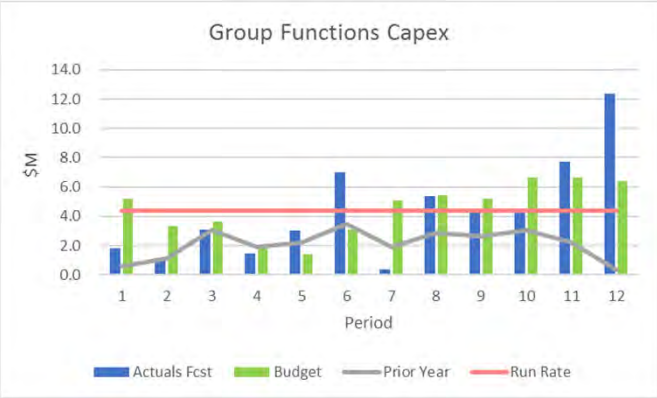
nationalgrid

Executive Summary - Group Functions

We executed our FY22 portfolio plan to meet budget expectations and delivering 97% of Capex to Budget. Key delivered capabilities include: **(1) Blackline** - delivering labor efficiencies through automation, communications, and enhanced accuracy and accountability; **(2) FERC Forms** - federal mandate upgrading form submission system resulting in automated regulatory filings and improved business information processing; **(3) HR Service Pack** - deliver needed updates to the SAP application portfolio in order to comply with federal, state, and local government requirements; **(4) BW Consolidation** - consolidating two separate reporting platforms into one, reducing cost of IT support and hardware; **(5) Pension Admin. Replacement** – delivers reduced administrative costs and provides a better user experience for both plan sponsor and participants; **(6) 401(k) Enhancements** - merging two 401(k) plans into one resulting in participant savings of 30% year over year and adding a Roth 401(k) to the plan, meeting the needs of our multi-generational workforce.

Financial Summary



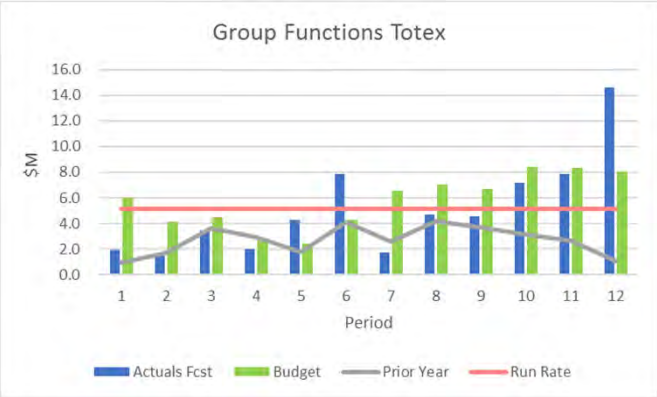


Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
69.2	61.7	63.7

Capex

Budget ($M)	Actual ($M)
53.9	52.2



Summary

- OPEX variance of $5.8m was the main driver of the overall variance of $7.5m to TOTEX budget, due in part to the general industry challenges experienced across IT that delayed the OPEX intensive start and closure phases of certain investments

Significant Project Updates

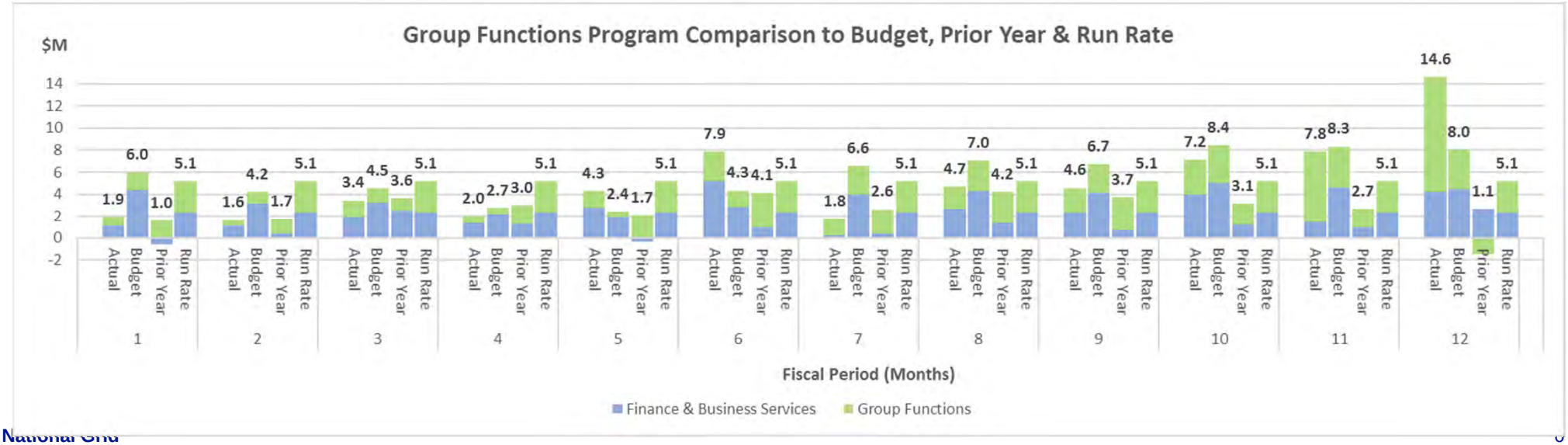
- Palantir-mandated walk-in added spend of $2.9m not previously budgeted
- SAP landscape added spend of $3.6m spend to set up SAP project environments
- MyHub $6.3m and Fieldglass $1.8m underspend due to business constraints leading spend to shift from FY22 to FY23
- Source to Pay Transformation spend of $3M shifted from FY22 to FY23 due to contract license negotiations with SAP
- GridStack infrastructure investment progressed through SGA and sanctioning to deliver in Q4 of FY22 with additional $10.2m spend
- Cloud Migration required additional $2.0m for increase in licences and usage

Group Functions Program View of Financials

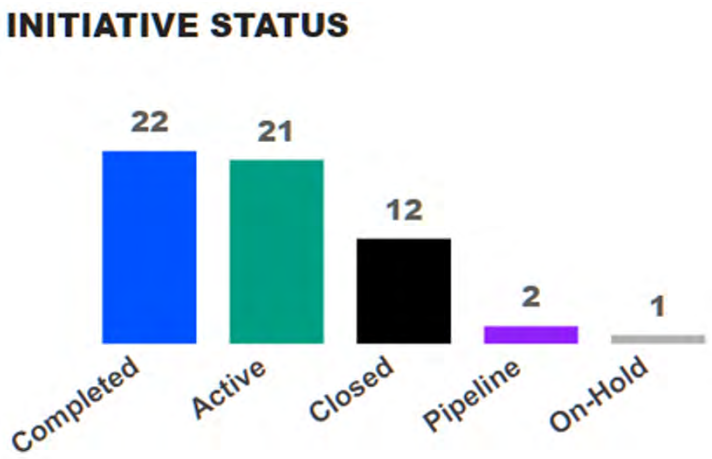
Portfolio	Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
Group Functions	Finance & Business Services	23.2	32.8	9.6	4.9	11.3	6.3	28.1	44.1	15.9
	Group Functions	29.1	21.2	(7.9)	4.6	4.0	(0.6)	33.6	25.1	(8.5)
	Totals	**52.2**	**53.9**	**1.7**	**9.5**	**15.3**	**5.8**	**61.7**	**69.2**	**7.4**

Priority Investments

Program	Status	Overall RAG	Totex Budget ($M)	Funding Type
HR & Payroll Service Pack	Complete	B	1.4	Regulated
401(k) Admin Changes	Active D&T	G	1.0	Regulated
FTOM: SRM to Ariba	Active Sprint Delivery	G	5.3	Regulated
FTOM: Closing Simplification	Complete	B	1.7	Regulated
SAP Portals to Fiori	Active D&T	G	0.6	Regulated



Mobilisation Summary



Status

- FY22 delivered projects total $40.9M Totex
- FY23 projects started up in FY22:
  - S2P Transformation Release 2
  - Investment Recovery
  - Blackline Release 3 - Task Management
  - HR Service Pack Summer 2022

Resources

- Group Functions is actively recruiting and on-boarding resources to deliver on our plan.

Scope

- Project scope development has been completed for FY23.

Key Challenges

Challenge	Escalation Point
Resource challenges (*e.g.*, Project Managers) across the portfolio.	Group Functions CIO

Delivery Methodology

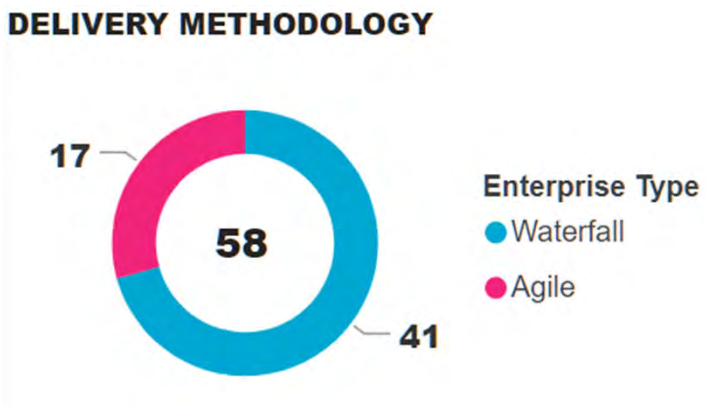


*Chart data as of May 13

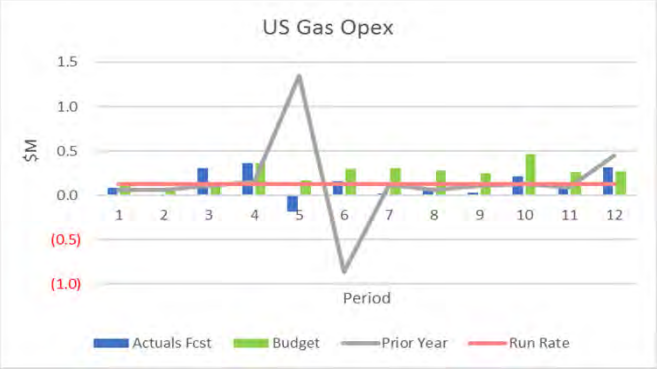
US Gas
Q4 FY22 Report

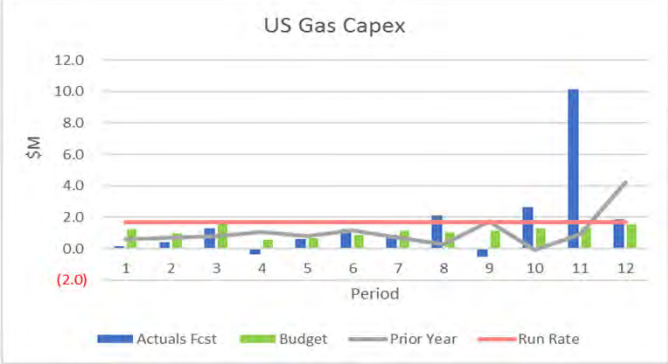
nationalgrid

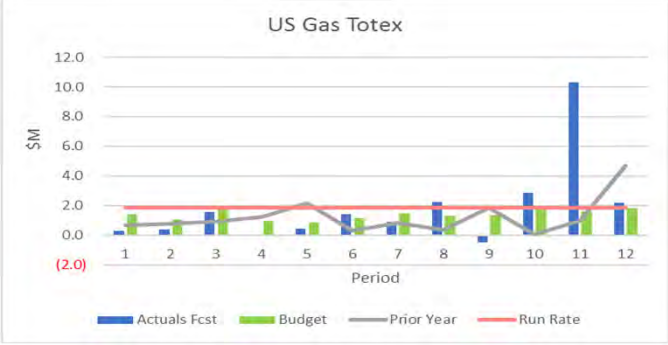
Executive Summary – US Gas

The November release of Gas Business Enablement (GBE) successfully added 1,200+ users of Long Island to the GBE solution and work has switched to the MA and NYC (May 2022) Release: Detailed planning is underway for post-Nov against May scenarios optimizing data and training management.

The US Gas business delivered: **(1) NRG GIS replacement** - convert and move data from an unstable GIS platform to an enterprise strategic format and stable platform that helps the business with less outages and business continuity plans; **(2) Interfaced MyHub To Onboard LMS OQ External Site** - built an employee information interface between SAP/MyHub and Gas Operational Qualifications (OQ) certification provider (ITS) to eliminate errors due to manual processing and reduce fines and re-work performed by field technicians; **(3) Primavera Upgrade/Stabilization** - provides a significant expansion in functionality for complex capital gas and electric project planning, scheduling, and forecasting activities, including risk management and new web-enabled / mobile capabilities that will help better decision making for complex capital projects risks, costs and forecast accuracy; **(4) PIF replacement** - platform upgrade that reduces security risk and complexity for maintenance and support; **(5) Minimum Viable Product release for LNG Asset & Work management** - implemented asset and work management framework to track and proactively maintain assets of LNG plants throughout their lifecyle on an enterprise platform.







Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
16.6	22.1	17.6

Capex

Budget ($M)	Actual ($M)
13.6	20.6

The general industry challenges impacted the Gas portfolio but we were able to manage delays by substitutions and pulling work forward.

- Stabilization of Critical Gas Legacy Applications - $1.6m incremental spend to address the security risks and platform stability of enterprise platforms (Salesforce, Maximo and MuleSoft) that would reduce the numbers of P1/P2 incidents and reallocation of resources from support to other priorities. However, much of the planned work has been delayed until FY23 ($6.6m).
- Continuous improvement for Gas applications $3m - Schedule and resourcing enhancement work for Gas Business.
- Pathfinder for Leak Survey which only impacts the MA jurisdiction $7m. The effort was to replace the Palm pilot standalone application with existing integration to access database.
- Summons for Corrective Actions and LNG IT for OT support were postponed to FY23 due to resource challenges and technical issues $3.4m.
- Gas I&R, Meter Ops, and field device replacement was moved to FY23 due to late schedule of laptops from the supplier $2.5m.
- Capital Delivery - Safety and Risk Artificial Intelligence in the Field – Proof of concept was run by the business before a decision to proceed was made $1m.

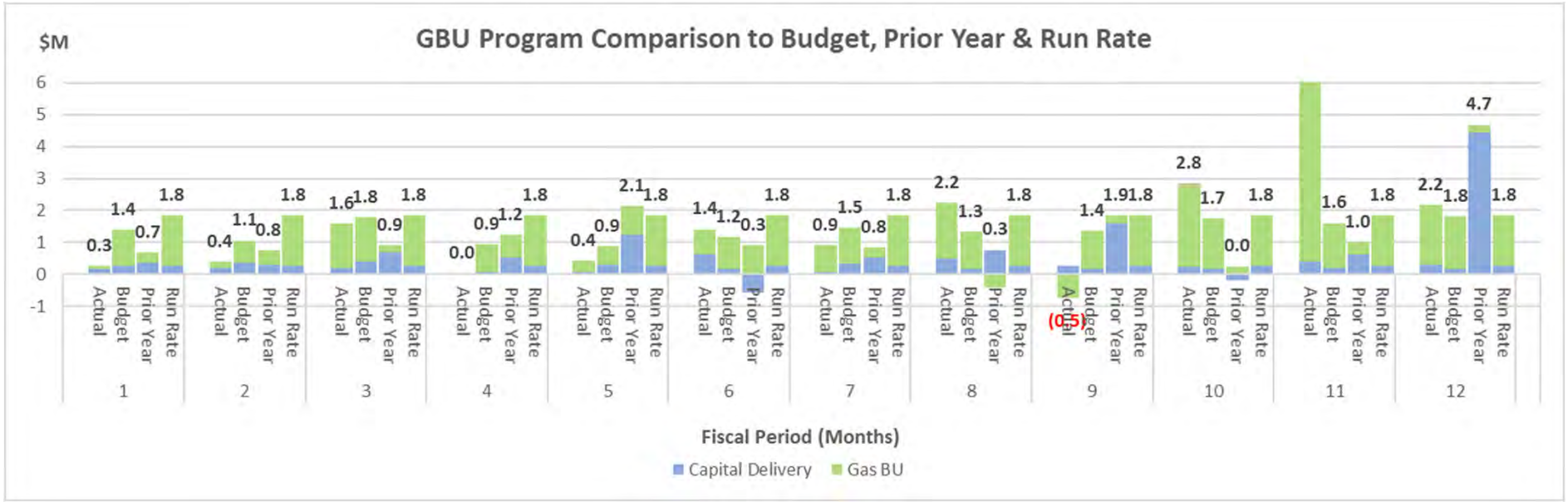
US Gas Program View of Financials

Gas Program View Full Year $M

Portfolio	Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
US Gas	Capital Delivery	2.8	2.5	(0.2)	0.2	0.2	0.0	3.0	2.8	(0.2)
	Gas BU	17.8	11.1	(6.7)	1.3	2.7	1.4	19.1	13.8	(5.3)
		20.6	**13.6**	**(7.0)**	**1.6**	**3.0**	**1.4**	**22.1**	**16.6**	**(5.6)**

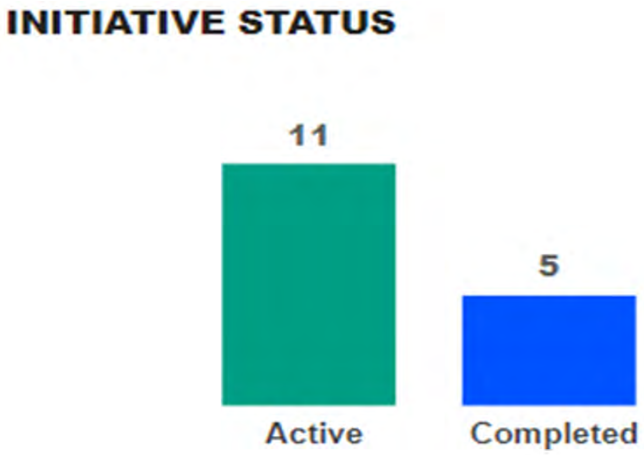
Priority Investments

Program	Status	Overall RAG	Budget ($m)	Funding Types
GBE	Inflight		-	Business Funded
LNG Asset & Work Mgmt.	Inflight		3.00	IT Core
NRG GIS Replacement	PIS		4.18	IT Core
Continuous Improvement for Gas Applications	Start-up		2.60	IT Core
NSRMS	Inflight		2.03	IT Core
ER&S (Enterprise Scalability and Resilience	Start-up		2.50	IT Core
Gas App Rationalization	Inflight		0.68	CTA
GTIS	Inflight		1.18	IT Core



Grid Q4 FY22 IT
National

Mobilisation Summary



As of March 2022, all FY22 projects, except FCS (pending EBU project dependencies, LNG IT support for OT systems) have been sanctioned and are underway. These remaining projects would be moved to FY23

No. of Initiatives Sanctioned

12

No. of Initiatives to be Sanctioned

1

No. of initiatives to be Sanctioned within the next month

1

Totex Sanctioned ($M)

$27.9M

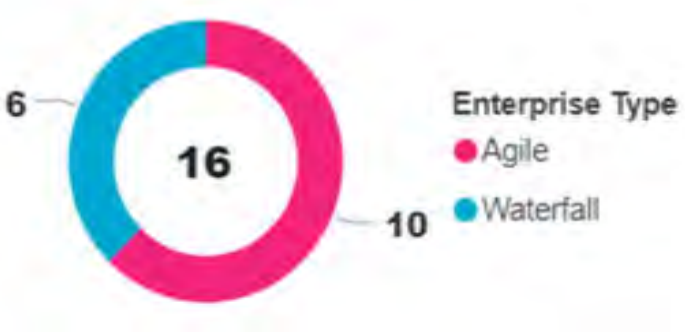
Totex to be Sanctioned ($M)

$0.800

Key Challenges

Challenge	Escalation Point
• Delay in receiving final invoices extends project closure by at least 3 months	Strategy Office/ PMO
• Resource challenges specially PM's	BP team

Delivery Methodology



US Electric
Q4 FY22 Report

nationalgrid

Executive Summary – US Electric

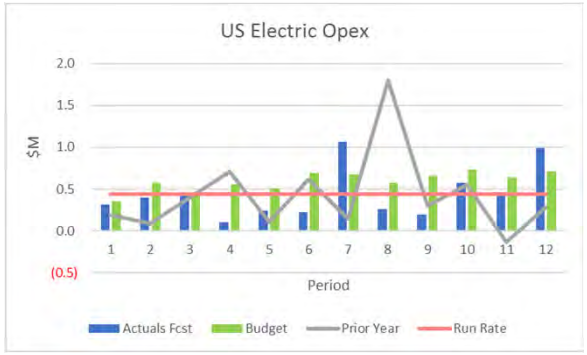
In FY22, Electric IT Focused On –

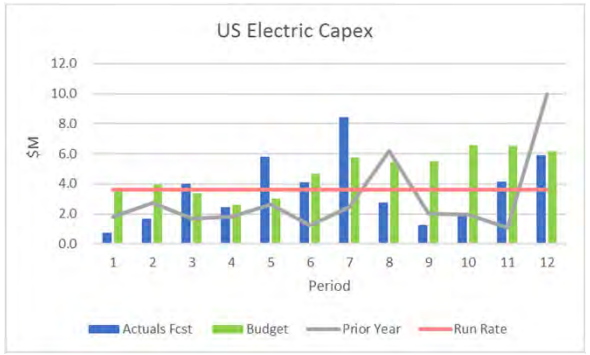
Grid Q4 FY22 IT

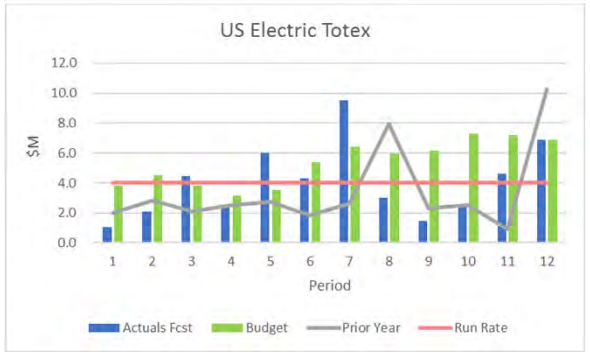
1) Reducing operational risk by upgrading / replacing end of life systems, *e.g., Energy Accounting System Migration to Wholesale Settlement application, US Control-Transmission Outage Application (iTOA) application upgrade, MV90 (in-flight – application captures Electric Meter Read for Commercial/ Industrial/Wholesale Customers which supports 40% of Electric Revenue), and Field Collection System Upgrade (in-flight)*

National

2) Re-imagining ways of working for Field Employees, providing them with new Business Capabilities to help drive operational efficiencies through the releases of new and transformational Digital products:
   - **Vegetation Management Optimization (VMO):** Minimum viable product (MVP) was launched on Nov 15th, providing capabilities to leverage and drive a reduction in contractor costs by optimizing our cycle prune and hazard tree removal plans. This product drove $2.04M in Type 1 Contractor Savings in FY22 and is expected to drive $4.47M in FY23.
   - **OnMyWay:** Enables electric field crews to self-assign short cycle jobs while in the field. Through the end of the year, Field Workers were self-assigning work at an average of 74% for all short cycle jobs, with some barns reporting 99-100% self-assignment. These self-assignments are done based on experience or guidance from supervisors, which enables our field force to be more efficient. Through the end of our FY, the team/product achieved $3.2M in actual benefits since initiation.
   - **FutureNow**: Launched MVP on February 14th, providing the business with a more accurate, data driven, TOTEX planning tool. Through its simulation and optimization capabilities, this tool will enable the business to significantly reduce the costs of future Distribution and Transmission workplan; *FY23 benefits are expected to be $10M CAPEX and $1.5M OPEX.*







Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
64.2	48.5	44.4*

Capex

Budget ($M)	Actual ($M)
57.1	43.2

National Grid *Forecast updated subsequent to input into financial systems

The main driver behind the variance to budget was within the digital portfolio which has shifted spend into FY23 due to general industry delays.

In FY23, Electric IT expects to invest ~$86M, which will be used to modernize our existing aged systems and infrastructure, implement transformational products that will support the 'Future of Electricity' while enabling the interconnected utility model, and propel our business to be at the heart of a clean, fair, reliable and affordable energy future, with a focus and investments in the following areas:

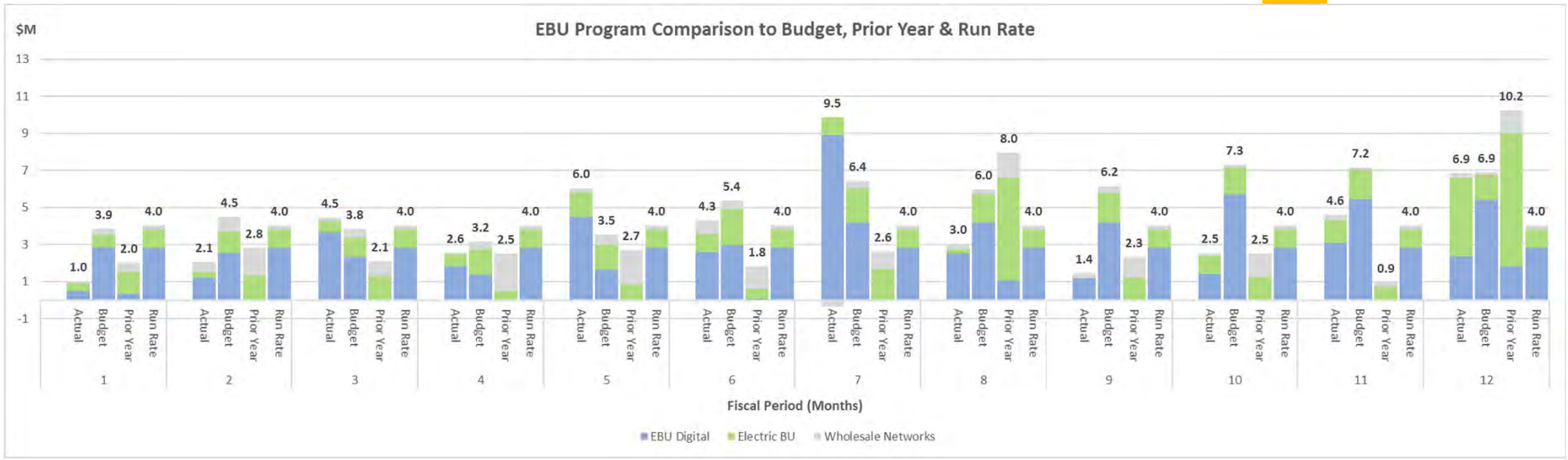
- Meter Data Services
- Field and Control Center Operations
- Data/Asset Management and Modernization
- Regulatory Compliance
- Digital Lighthouse Products
- GridMod / ADMS

US Electric Program View of Financials

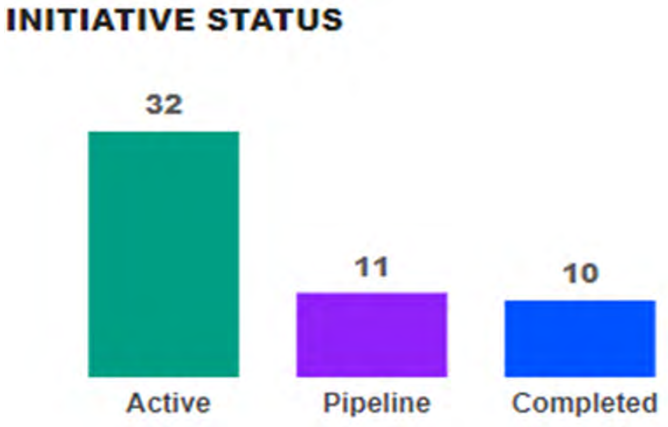
Portfolio	Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
US Electric	EBU Digital	31.5	38.8	7.3	2.3	4.1	1.8	33.8	42.9	9.1
	Electric BU	9.3	14.4	5.1	2.5	2.5	(0.0)	11.8	16.9	5.0
	Wholesale Networks	2.4	3.9	1.6	0.4	0.5	0.1	2.8	4.4	1.6
	Totals	**43.2**	**57.1**	**13.9**	**5.3**	**7.1**	**1.8**	**48.5**	**64.2**	**15.7**

Priority Investments

Programme	Status	RAG	Budget Totex (FY22)	Funding Types
GridMod	*Inflight*		*35.5*	*Business/ Reg*
ADMS	*Inflight*		*30.1*	*Business/ Reg*
EBU Digital Programs	**Inflight**		**42.9**	**Regulatory**
Electric MV90 Upgrade	**Inflight**		**1.8**	**Regulatory**
FCS Upgrade	**Inflight**		**2.1**	**Regulatory**
Transmission Outage	**Delivered**		**0.7**	**Regulatory**
Wholesale Settlement	**Delivered**		**2.2**	**Regulatory**
Price Transmission	**Inflight**		**1.6**	**Regulatory**
MTS Retail Settlement	**Inflight**		**1.9**	**Regulatory**
Corporate PI	**Inflight**		**0.6**	**Regulatory**
NYISO PMU – Large Generators	**Inflight**		**0.5**	**Regulatory**



Mobilisation Summary



- **FY22**: ***Completed Initiatives***: 10
  Active Initiatives: 32
- Five (5) Products with Amber RAG Status' are due to projects having to shift to the right for Commercial and/or Resourcing issues
- Due to a schedule risk, Generation Asset Management Upgrade (INVP 5822C) is currently carrying a red RAG status. This risk is due to the necessary integrated (testing) environments, that are required to complete development and testing, not currently being available; *Solution Delivery is actively driving this towards resolution*

No. of Initiatives Sanctioned

38

No. of Initiatives to be Sanctioned

15

No. of initiatives to be Sanctioned within the next month

1

Totex Sanctioned ($M)

$175.1M

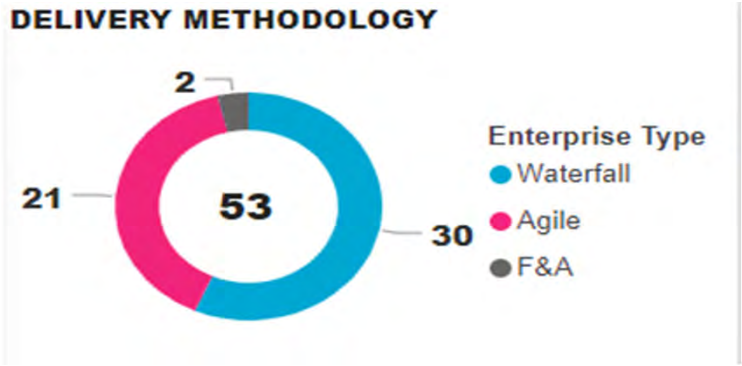
Totex to be Sanctioned ($M)

$29.7M

Key Challenges

Challenge	Escalation Point
1. Digital ways of working require ramp up with resources, processes and technology.	CIO / HR / Commercial *(ADAM Partners)*
2. Negotiations with Vendors regarding Licensing and general terms and conditions	Commercial
3. Significant change across the organization and number of initiatives pulling on the same business and IT resources.	Senior Leadership

Delivery Methodology



Customer
Q4 FY22 Report

nationalgrid

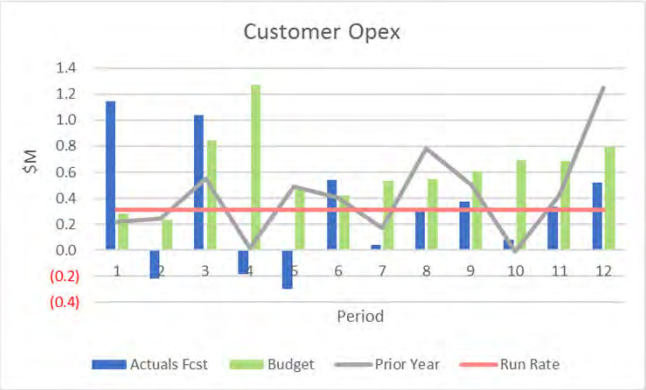
Executive Summary – Customer

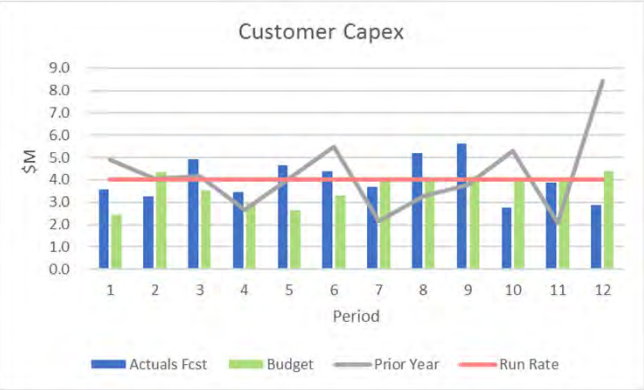
Grid Q4 FY22 IT

Customer is making significant strides in digital enablement and improving the customer experience. Two advancements of note in Q4 include a major AMI milestone and increased adoption of our Business Customer on the My Business Account platform. The AMI team successfully connected live meters to AMI applications in lab and generating bills. Momentum continues in Q1 as end-to-end testing progresses well. My Business Account adoption reached 20.05% by the end of FY22 - this provides our business customers the ability to pay and manage multiple accounts, provides them insights into their energy use, and allows them to better manage their solar credits.

National

Financial Summary





Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
51.5	52.0	57.5

Capex

Budget ($M)	Actual ($M)
44.1	48.3

The Customer portfolio utilized the full budget for the year with a small variance to target.

The largest variance is in the Unified Web Portal (UWP – our new transaction web portal) project. After the UWP launch in November to NYC Gas Customers, we have had stability issues due to multiple root causes including bad data; missing or incorrect user profiles; software errors; and poorly tuned servers. From a customer perspective, this would have resulted in poor transaction response times, or even, in very rare occasions, transaction time-outs. The additional resources required to stabilize the experience is the primary driver of additional costs. The UWP project will continue to run over budget into Q1 FY23 for launch readiness..

In Q4 we decided to delay the re-start of the new Smart Target tool in order to focus on higher priority efforts. That is the primary driver of the Q3 forecast miss. We intend to re-start Smart Target in the second half of FY23.

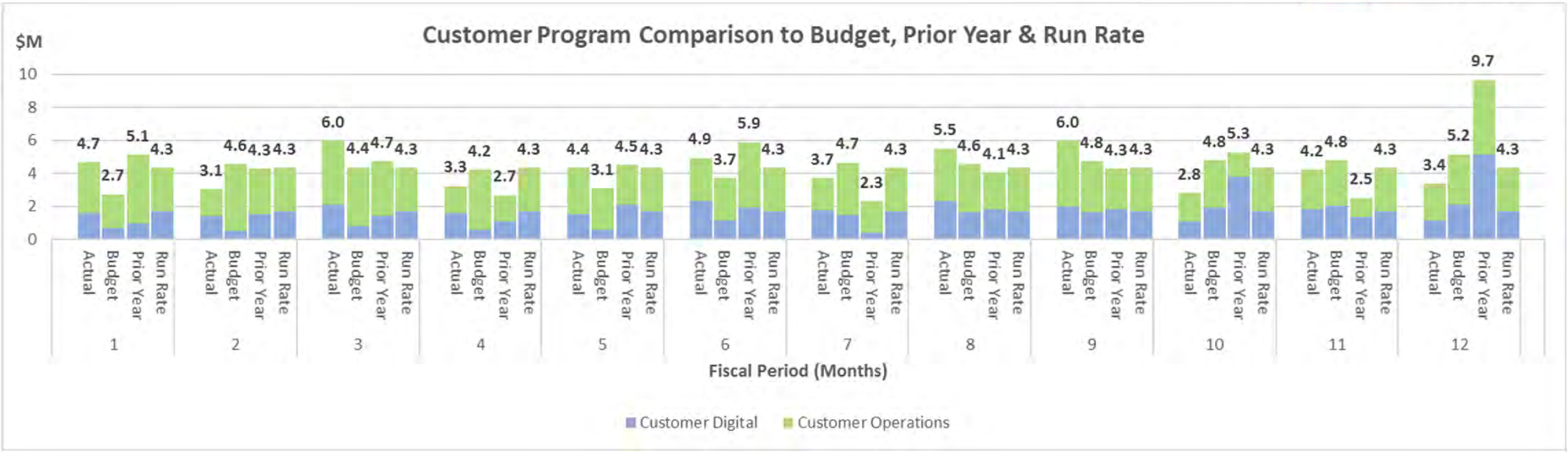
Customer Program View of Financials

Customer Program View Full Year $M

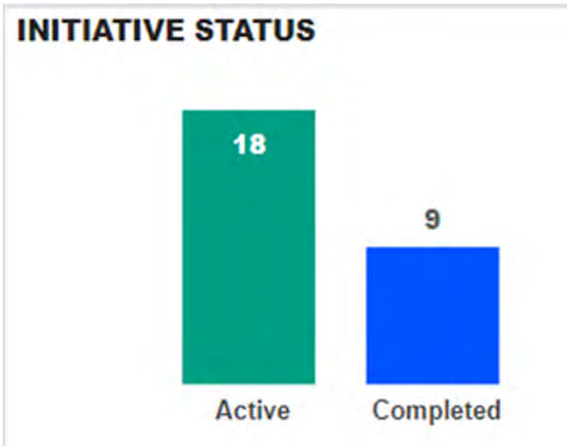
Portfolio / Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
Customer Customer Digital	20.3	12.3	(8.0)	0.3	2.9	2.6	20.6	15.2	(5.4)
Customer Operations	28.0	31.8	3.8	3.4	4.5	1.1	31.3	36.3	5.0
Totals	**48.3**	**44.1**	**(4.2)**	**3.7**	**7.4**	**3.7**	**52.0**	**51.5**	**(0.5)**

Priority Investments

Program	Status	RAG	Budget ($m)	Funding Type
Unified Web Portal	*Inflight*		*19.0m*	*IT funded*
CIS: CRIS Migration	*Inflight*		*10.2m*	*Business Funded*
Customer Data Platform	*Inflight*		*5.0m*	*IT Funded*
AMI	*Inflight*		*7.7m*	*Business Funded*
Zero-Based Redesign	*Start up*		*6.1m*	*Business Funded*
RI Transition	*Inflight*		*3.0m*	*Not Funded in Customer*
Customer Ecosystem Stabilization	*Inflight*		*5.0m*	*IT Funded*
Clean Energy	*Validation*		*NA*	*Business Funded*



Mobilisation Summary



No. of Initiatives Sanctioned

31

No. of Initiatives to be Sanctioned

2

No. of initiatives to be Sanctioned within the next month

1

Totex Sanctioned ($M)

$58.1

Totex to be Sanctioned ($M)

$0.3M

Overall, teams are mobilized to deliver FY22 and FY23 Budget.

Product Area

- As FY22 ends, we have five Products supported by permanent squads that will have continuous delivery in FY23.

Project Area

- In the area that is starting off the year in the project model, we have a full slate of active projects that will continue into FY23.
- These include three recent projects that we have kicked-off in Q4: (1) PCI Compliance; (2) NACHA Compliance; and (3) Preferred Pronoun.

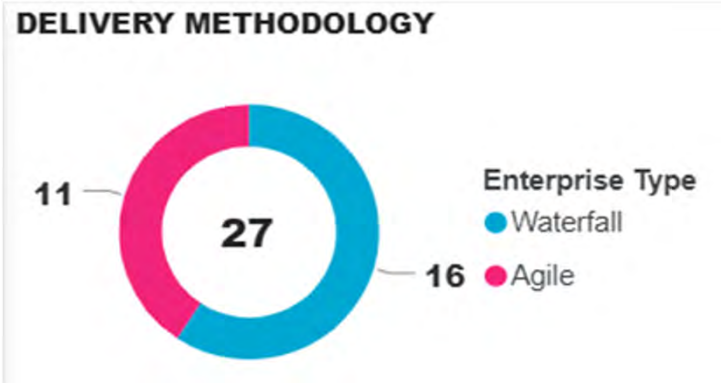
Starts Planned for Q2-Q4 (FY23)

- New Mandate Bucket – funding in future quarters to deliver new mandates.

Key Challenges

Challenge	Escalation Point
***Business Resources**: Business resources have been prioritized to Transformation efforts including ZBR and CRIS to CSS. As a result, Smart Target development has been delayed to FY23.*	*Simon Baker*
***UWP Launch**: Stability problems are close to being resolved. The current blocker is the dependency on RI.*	*Simon Baker*

Delivery Methodology



Digital Platforms and Infrastructure Technology (DPIT) (formerly I&O)

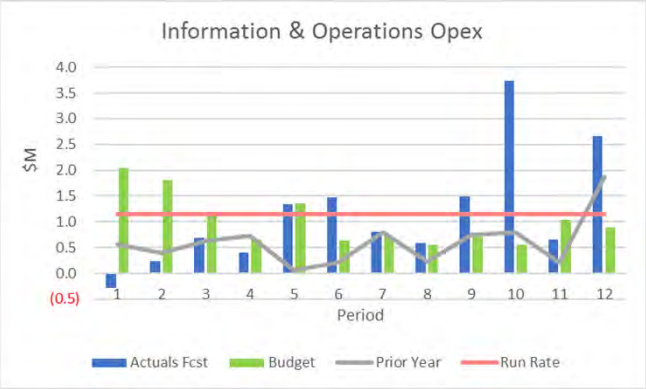
Q4 FY22 Report

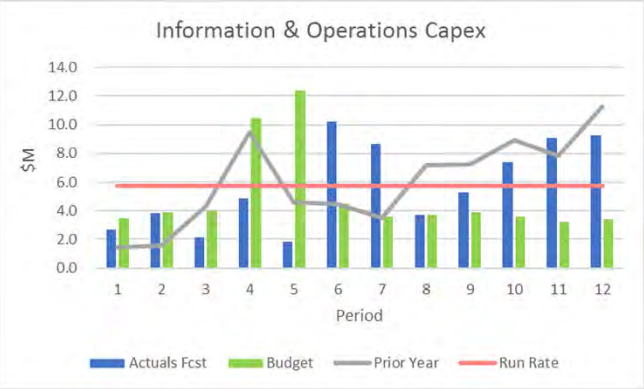
nationalgrid

Executive Summary – DPIT

DPIT has improved the level of tech debt, now at 23.9%, increased network performance, resiliency and reliability, and worked to improve end-user experiences. Challenges around available skilled resources and hardware delivery delays have had an impact. However, we delivered all FY22 Objectives per our roadmap.

Financial Summary



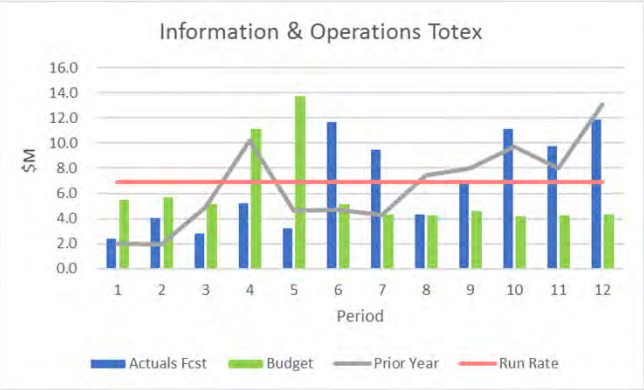


Totex

*Budget ($M)	Actual ($M)	Q3 Forecast ($M)
72.2	82.4	81.0

Capex

*Budget ($M)	Actual ($M)
60.1	68.9



Data Center Transformation: Completed delivery and physical installation of core network, compute, and data storage infrastructure across 2 new US data center sites. $4M variance to budget driven by OPEX underspend (CTA funded - outside of DPIT)

Cloud & Hosting: Migrations from legacy infrastructure - Reduce technical debt, reduce risk of unsupported hardware and enabler for DCT. $1M overspend variance to budget, driven by construction overhead costs and purchase of Cloud Health subscription SW to support work in FY23

Progress FY22: HCI 95%, DPS 86%, Block Storage 77%

Networks: US 30% reduction in Technical Debt – EOL – Replaced 442/~1291 Devices // WiFi 45/57 Sites (291 Access Points replaced), 45 Switches across15 Sites // SDWAN 36 Sites transformed. $1.4M overspend variance due to purchase of additional HW needed to deliver scope

Operations: Delivered single view of infrastructure health, Reduction of Technical Debt with 130 servers decommissioned. $0.4M overspend variance to budget due to construction overhead charges

End User: VDI SoftLayer to Azure migration complete for US, 50% Technical Debt reduction, Windows 10 program complete, enabling launch of Windows-as-a-Service for continuous application & computer updates. $1.5M underspend variance due to overestimated cost for delivering scope

146 Printers removed (100% of target)
34 video conferencing updated with MS Teams for collaboration

* Budget was filed before priorities and cost estimates went through a final approval. The final plan was increased by 15% with a variance to actual spend of 3.2% (0.2% CAPEX)

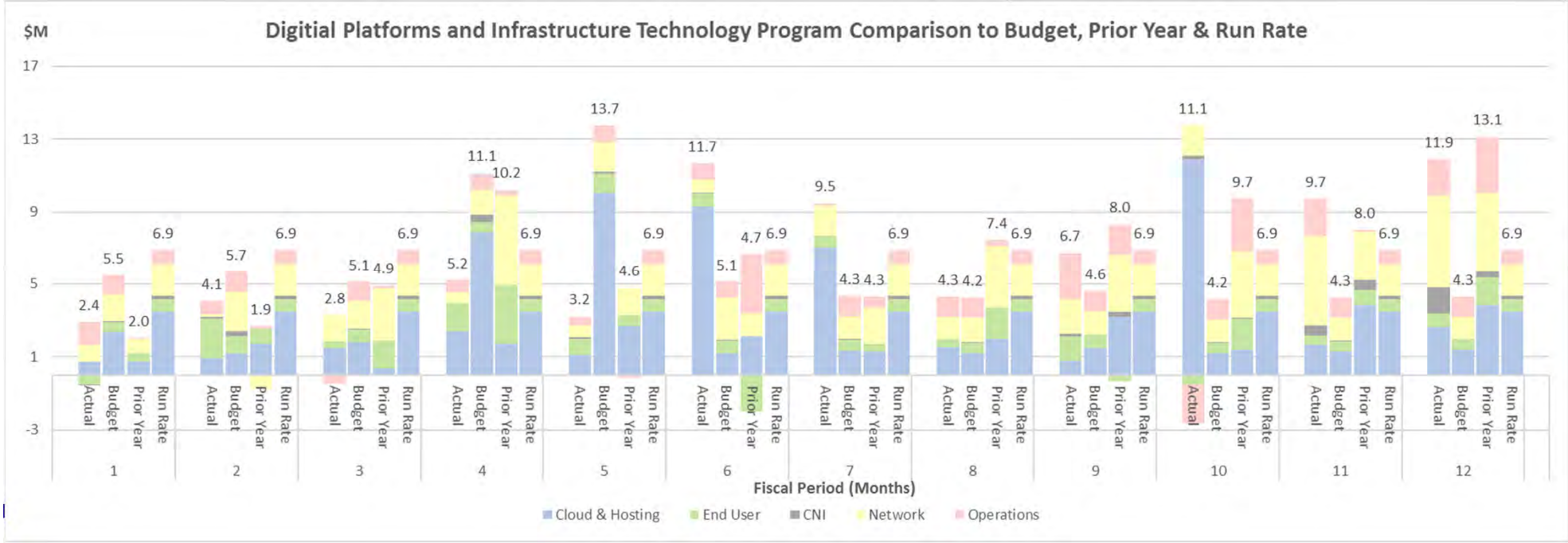
DPIT Program View of Financials

DPIT Program View Full Year $M

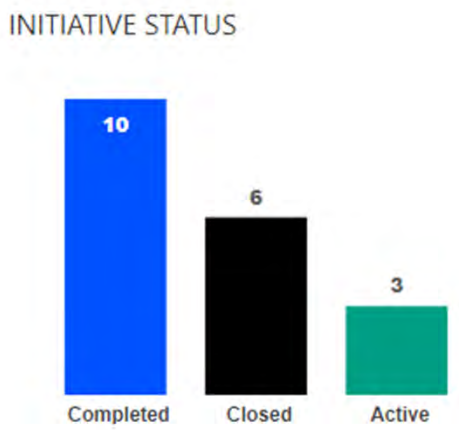
Portfolio	Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
I&O	Cloud & Hosting	31.7	23.0	(8.6)	9.9	9.4	(0.5)	41.5	32.4	(9.1)
	End User	7.2	6.9	(0.3)	1.0	1.1	0.1	8.2	8.1	(0.2)
	CNI	2.1	1.1	(1.1)	0.5	0.0	(0.5)	2.6	1.1	(1.6)
	Network	19.7	17.4	(2.3)	1.2	0.8	(0.4)	20.9	18.2	(2.7)
	Operations	8.1	11.7	3.6	1.2	0.7	(0.4)	9.3	12.5	3.1
	Totals	**68.9**	**60.1**	**(8.8)**	**13.8**	**12.1**	**(1.7)**	**82.6**	**72.2**	**(10.4)**

Priority Investments

Programme	Status	RAG	Actual Totex (FY22)	Funding Types
Data Center Transformation	Inflight		27.1	IT
Networks	Inflight		20.9	IT
Cloud & Hosting	Inflight		14.1	IT
Operations	Inflight		9.3	IT
End User Experience	Inflight		8.2	IT
Critical Network Infrastructure (CNI)	Inflight		2.6	IT



Mobilisation Summary



No. of Initiatives Sanctioned for FY22

6

Totex Sanctioned for FY22 ($M)

$73M

No. of initiatives Sanctioned for FY23

5

Totex Sanctioned for FY23 ($M)

$66.9M

Status

- FY22 Sanctioned Initiatives In Services as of April 6. All other carryover initiatives/projects in FY22 are completed and closed.

Resources

- All teams resourced appropriately

Scope

- Delivered all FY22 Objectives per roadmap.

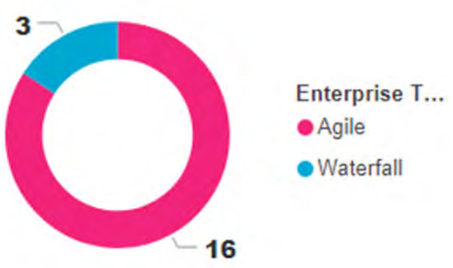
Key Challenges

Grid Q4 FY22 IT

Challenge	National	Escalation Point

No challenges for FY22

Delivery Methodology



US Security
Q4 FY22 Report

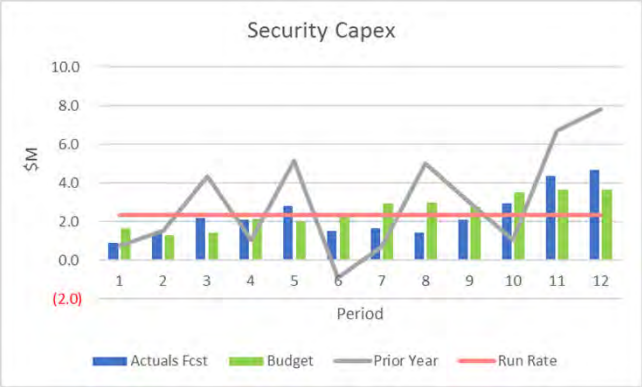
nationalgrid

Executive Summary – Security

[redacted]

Grid Q4 FY22 IT

Financial Summary



Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
36.4	34.0	36.4

Capex

Budget ($M)	Actual ($M)
30.4	28.2

Security Portfolio

Security experienced similar challenges to the rest of the industry in terms of supply chain and resourcing issues.

The threats to our networks are constantly evolving and require monitoring to ensure threats are mitigated. As a result, funds were reprioritized across the portfolio to relieve these pressures.

[redacted]

Total spend was in line with the budget allocation across the cyber investments.

Physical security was underspent by $2m due to break/fix actuals being less than original estimates.

Security Program View of Financials

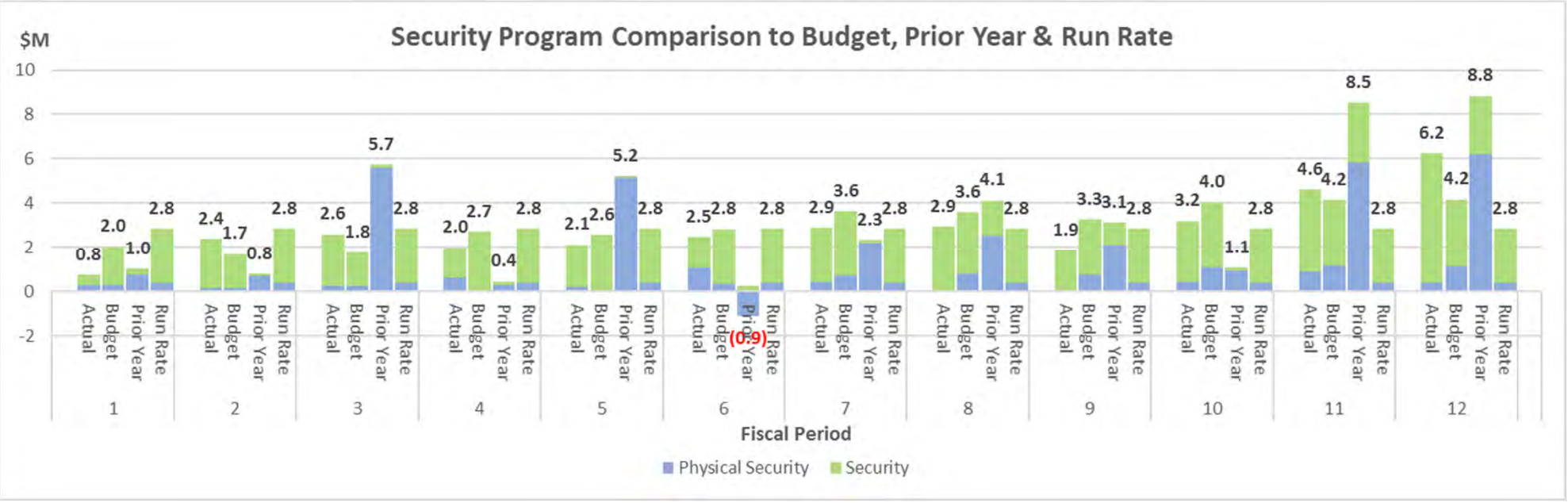
Security Program View Full Year $M

Portfolio	Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
Security	Physical Security	4.7	6.7	1.9	(0.0)	0.0	0.0	4.7	6.7	1.9
	Security	23.4	23.7	0.3	5.9	6.0	0.1	29.3	29.7	0.4
	Totals	**28.2**	**30.4**	**2.2**	**5.9**	**6.0**	**0.1**	**34.0**	**36.4**	**2.4**

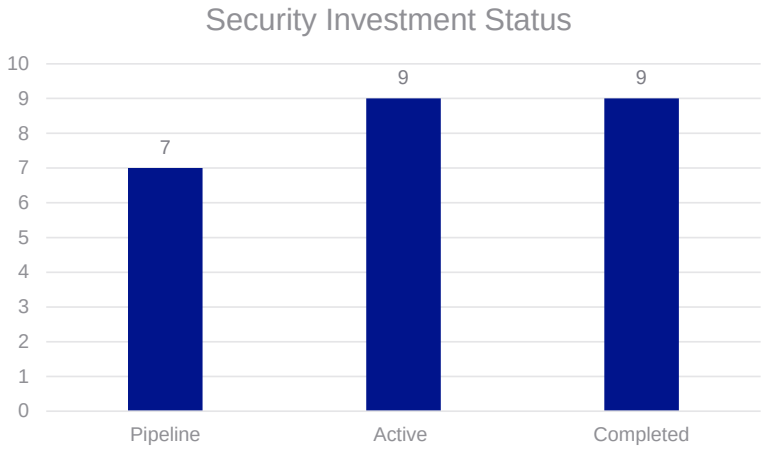
Priority Investments

Grid Q4 FY22 IT

Programme	Status	RAG	Budget Totex (FY22)	Funding Types
Operational Technology	Inflight	National	5.9	IT
Platform Security	Inflight		6.9	IT
Network Security	Inflight		4.3	IT
Identity and Access Management	Inflight		9.2	IT
Vulnerability Management	Inflight		1.3	IT
Security Orchestration Automation and Response	Inflight		2.1	IT
Physical Security	Inflight		6.7	IT



Mobilisation Summary



No. of Initiatives Sanctioned

18

No. of Initiatives to be Sanctioned

7

No. of initiatives to be Sanctioned within the next month

2

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]
- [redacted]
- [redacted]

Totex Sanctioned ($M)

$36.8M

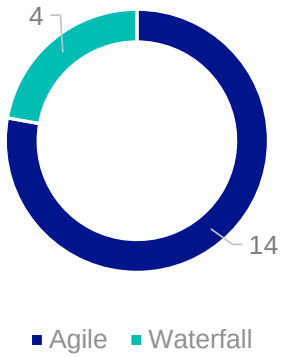
Totex to be Sanctioned ($M)

$23.4M

Key Challenges

Challenge	Escalation Point
Supply Chain constraints (*i.e.*, chip shortages) may impact delivery timelines. Extended lead-times have been factored into project timelines, as reflected in FY22Q4 forecasts.	Commercial/ Procurement
Changes to the threat landscape and regulatory mandates may result in need to pivot to address emerging threats, risk areas and compliance.	CISO

Delivery Methodology



Portfolio	Program	Project INVP	Project Name	Year End Actual	Year End Budget	Variance	Reason
Customer	Customer Digital	4750	Customer Exp. Transformation technology program	0	3,719,000	3,719,000	Funding Bucket
		4750J	Web Self Service	14,387,807	0	(14,387,807)	Complexity
		4750K	Personalization	(508,171)	0	508,171	Other
		4750M	ODS	3,438,103	0	(3,438,103)	Complexity
		4750N	Automated Customer Services	(407,137)	0	407,137	Other
		5673	My Business Account (Nucleus Phase 2)	3,544,978	5,600,000	2,055,022	Imprecise Estimates
		5859A	Customer and Agent Experience - IVR Baseline	0	3,940,000	3,940,000	Funding Change
		5859B	Customer and Agent Experience - IVR Minor Enhancements	0	1,950,000	1,950,000	Funding Change
		5905	Nucleus Project	167,669	0	(167,669)	Accounting
		6206	Customer Smart Target	23,362	0	(23,362)	Accounting
	Customer Digital Total			**20,646,611**	**15,209,000**	**(5,437,611)**	
	Customer Operations	3932	Customer Contact Center / SDC Technology Upgrade Implement Solution	4,937,001	2,965,000	(1,972,001)	Complexity
		4904	Upgrade Servers for CCAE Reporting	191,010	94,000	(97,010)	Complexity
		5069	RightFax Upgrade/Remediation	0	257,000	257,000	Funding Change
		5125	Customer Connection Program	3,285,070	2,558,000	(727,070)	Funding Change
		5146	Smart Cities Schenectady	17	0	(17)	Immaterial
		5160	NY Stack VDER and Portal Integration	1,950,831	2,111,000	160,169	Within Target
		5345B	Bill Print and Bill Composition Implementation	1,128,985	1,680,000	251,015	Scope changes
						300,000	Funding Change
		5430	NY Green Button Connect	1,596,546	890,000	(123,546)	Timing
						(100,000)	Funding Change
						(450,000)	Imprecise Estimates
						(33,000)	Other
		5625	Customer Billing Minor Enhancements FY21	104,678	0	(104,678)	Opex Investment
		5627B	Experian Platform Upgrade (Tallyman & PMP)	0	132,000	132,000	Funding Change
		5627C	Outreach Optimization	0	263,000	263,000	Funding Change
		5627D	Post COVID Collections Restart	0	300,000	300,000	Funding Change
		5722	KEDLI Rate Case (CRIS)	381,520	58,000	(323,520)	Funding Change
		5723	KEDNY Rate Case	846,926	313,000	(533,926)	Scope changes
		5778	Customer Billing Improvement Program - Phase 1	3,007,573	11,250,000	8,142,427	Funding Change
						100,000	Timing
		5795E	MA Electric Vehicle Rebate Program	13,791	0	(13,791)	Accounting
		5795F	CDG Net Metering	760,508	992,000	231,492	Other
		5795G	Telephone Consumer Protection Act (TCPA)	58,111	0	(58,111)	Accounting
		5836	Lockbox Payments and Credit Payments Implementation	612,988	518,000	(94,988)	Imprecise Estimates
		5859A	Customer and Agent Experience - IVR Baseline	121,515	513,000	391,485	Funding Change
		5859B	Customer and Agent Experience - IVR Minor Enhancements	0	1,110,000	1,110,000	Funding Change
		5863	Fix the Foundation	0	3,450,000	3,450,000	Funding Change
		5863D	AIMS Stability	741,200	0	(741,200)	Scope changes
		5903	Customer Operational Enhancements FY21	1,037	0	(1,037)	Opex Investment
		5926A	NiMO Rate Case	1,318,974	975,000	(343,974)	Imprecise Estimates
		5926B	MA Gas Rate Case	0	488,000	488,000	Funding Change
		5987	Smart City Phase II	136,560	290,000	153,440	Efficiencies
		6007	Inside Agent	1,041,901	1,187,000	145,099	Timing
		6032	Customer Billing COVID Transition	477,253	450,000	(27,253)	Within Target
		6042	Customer US Retail Web Infrastructure Upgrade	10,268	0	(10,268)	Opex Investment
		6066	FCP Upgrade for CSS - Plan Forward	1,772,466	2,080,000	307,534	Other
		6112	WFH Hardware - Plan Forward	232,192	0	(232,192)	Complexity
		6115	Customer COVID-19 Payment Plan for IVR and Web	226,686	0	(226,686)	Timing
		6214	Customer Operational Enhancements FY22	83,096	150,000	66,904	Opex Investment
		6217	Customer Billing Minor Works FY22	536,694	360,000	(176,694)	Opex Investment
		6250	EV End-End Solution	0	863,000	863,000	Opex Investment
		6302	2022 AIMS Platform	282,383	0	(282,383)	Timing
		6349	Customer Data Platform	4,116,739	0	(4,116,739)	Funding Change
		6393	Technology Validation for CRIS to CSS Migration	372,688	0	(372,688)	Opex Investment
		6427	RI Purchase of Receivables (POR) Program	768,081	0	(768,081)	Funding Change
		6434	CRIS to CSS Migration	28,426	0	(28,426)	Opex Investment
		6435	Customer PCI Remediation	187,334	0	(187,334)	Funding Change
		6492	NACHA Compliance- Security Encryption Effort	7,986	0	(7,986)	Opex Investment
		6496	NY Gender Pronoun and Name Mandate	1,063	0	(1,063)	Opex Investment
	Customer Operations Total			**31,340,095**	**36,297,000**	**4,956,905**	
Customer Total				**51,986,706**	**51,506,000**	**(480,706)**	

Portfolio	Program	Project INVP	Project Name	Year End Actual	Year End Budget	Variance	Reason
US Gas	Capital Delivery	4990	Primavera Upgrade/Stabilization	435,110	484,000	48,890	Imprecise Estimates
		5274A	New Storms Respose Management System Implementaition	1,646,421	1,236,000	(410,421)	Funding Change
		5830H	Capital Delivery - Safety and Risk Artificial Intelligence in the Field	0	1,046,000	1,046,000	Funding Change
		6222	PIF Replacement	928,885		(928,885)	Funding Change
	Capital Delivery Total			**3,010,416**	**2,766,000**	**(244,416)**	
	Gas BU	4769	Gas Storm Hardening Using Remote Service Valves (RSV's)	677,533	743,000	65,467	Within Target
		4898	Asset Management System for LNG	1,189,466	4,548,000	3,358,534	Scope changes
		5785C	Stabilization of Critical Gas Legacy Applications (ePermits)	84,714	0	(84,714)	Accounting
		5785I	NRG GIS Replacement Project	3,491,540	1,030,000	(2,461,540)	Funding Change
		5785J	Security & Resilience - SalesForce	0	2,740,000	2,740,000	Funding Change
		5785K	Security & Resilience - Maximo	10,300	2,260,000	2,249,700	Funding Change
		5785L	Security & Resilience - Mulesoft	(420,916)	1,243,000	1,663,916	Accounting
		5787A	Gas Control Center Simulator	3,463	0	(3,463)	Opex Investment
		5787D	GTIS Functional Upgrades	771,007	377,000	(394,007)	Scope changes
		5787F	Gas Transportation Information System (GTIS) Phase 3 (F&A)	0	126,000	126,000	Opex Investment
		5787G	F&A Study for Gas Demand Response (DR)	127,549	0	(127,549)	Opex Investment
		5788A	FY21 Gas Regulatory and Operational Enhancements (Minor Works)	(3,678)	0	3,678	Opex Investment
		5788D	Interface MyHub To OnBoard LMS OQ External Site	86,499	155,000	68,501	Scope changes
		5788G	FY22 Gas Regulatory and Operational Enhancements (Minor Works)	143,279	250,000	106,721	Opex Investment
		5789B	Stabilization of Critical Gas Legacy Applications (Portal for ArcGIS)	2,684	15,000	12,316	Efficiencies
		5789C	Stabilization of Critical Gas Legacy Applications (CWQ)	(2,194)	0	2,194	Imprecise Estimates
		5789E	Summons and Corrective Actions Response (CAR) Work Tracker	174,917	305,000	130,083	Other
		6015	Continuous Improvement for Gas Applications	3,060,035	0	(3,060,035)	Funding Change
		6043B6	Mobile devices rollout for Plan Forward	19,255	0	(19,255)	Opex Investment
		6353A	FY22 Gas Application Rationalization	114,585	0	(114,585)	Opex Investment
		6392	Contractor Ipad for Gas Business	383,359	0	(383,359)	Other
		6441	Scalability & Resilience for Enterprise Gas Platforms	1,612,813	0	(1,612,813)	Funding Change
		6479	LNG IT Support for OT Systems	4,164	0	(4,164)	Opex Investment
		6481	Pathfinder for Leak Survey for MA	7,589,804	0	(7,589,804)	Funding Change
	Gas BU Total			**19,120,179**	**13,792,000**	**(5,328,179)**	
US Gas Total				**22,130,595**	**16,558,000**	**(5,572,595)**	

Exhibit ___ (ITDP-2R)
Attachment 2
Page 3 of 9

Portfolio	Program	Project INVP	Project Name	Year End Actual	Year End Budget	Variance	Reason
US Electric	EBU Digital	5054	On My Way (EBU Digital Solution) Phase 2	5,292,485	9,400,000	1,807,515	Funding Change
						2,300,000	Imprecise Estimates
		5912	On My Way (EBU Digital Solution)	(68,303)	0	68,303	Timing
		6073	VMO Validation Phase	5,919,341	6,400,000	480,659	Within Target
		6076	FutureNow Validation Phase	21,723,243	0	(21,723,243)	Imprecise Estimates
		6291	ICU-Out (Electric Digita)	162,341	6,000,000	5,837,659	Funding Change
		6294	GridMD	0	5,500,000	5,500,000	Timing
		6425	DER Active Resource Integration NY	511,917	0	(511,917)	Funding Change
		6437	GridMD	295,404	0	(295,404)	Opex Investment
		EBU22	EBU Digital Placeholder	0	584,000	584,000	Funding Change
		FN22	Future Now - Next Phase	0	8,499,999	8,499,999	Timing
		PD22	Predicament	0	6,550,000	6,550,000	Timing
	EBU Digital Total			**33,836,427**	**42,933,998**	**9,097,571**	
	Electric BU	4481	US MDS-Energy Accounting System (EAS) migration to Wholesale Settleme	2,321,006	2,127,000	(194,006)	Within Target
		4483	US Control-Transmission Outage Application (iTOA) Upgrade to HTML 5 W	1,578,072	743,000	(835,072)	Complexity
		4704R	NY REV Street Lighting Delivery	111,020	137,000	25,980	Accounting
		5241	East Pulaski Energy Storage System IS Network	17,112	3,000	(14,112)	Scope changes
		5242	Buffalo Energy Storage System IS Network	65,973	51,000	(14,973)	Complexity
		5471XB	DER Aggregation Management - Registration, Operation and Settlement	0	0	0	Opex Investment
		5471Y	NY Clean Innovation	0	323,000	323,000	Opex Investment
		5486	AVLS Modem Reconfiguration	45,471	17,000	(28,471)	Accounting
		5570	Electric Field Crew iPad Time Entry and Basic Capability -NE	(37,279)	0	37,279	Timing
		5702	GridStack	2,917,222	0	(2,917,222)	Funding Change
		5773B	Visitor Logging for NERC-CIP Reliability Compliance	59,209	0	(59,209)	Timing
		5773E	NERC-CIP Reliability Compliance	102,072	0	(102,072)	Accounting
		5871	NY Energy Storage Systems (ESS)	0	281,000	281,000	Opex Investment
		5871E	Power Marketer Pilot	52,708	0	(52,708)	Opex Investment
		5887A	ARCOS - Reporting Replacement	62,514	586,000	523,486	Timing
		5889A	Control Room Event Recording	53,818	263,000	209,182	Accounting
		5889B	Operational Excellence, Operational Support (Power Quality Upgrade) *B	0	128,000	128,000	Timing
		5889C	Electric Regulatory Enhancements	70,868	0	(70,868)	Opex Investment
		5890A	Corporate PI	247,996	630,000	382,004	Other
		5890B	CYME Gateway	25,137	754,000	728,863	Timing
		5890C	Computapole Replacement F&A	5,541	0	(5,541)	Opex Investment
		5891A	Electric MV90 Upgrade	1,404,792	1,808,000	403,208	Timing
		5891B	Field Collection System (FSC) Upgrade POC	(2,609)	0	2,609	Opex Investment
		5891C	FCS Upgrade	1,267,859	1,800,000	532,141	Timing
		5891E	MTS Retail Settlement	532,622	1,890,000	1,357,378	Timing
		5891I	Price Transmission (Retail Price) System Replacement	346,038	0	(346,038)	Timing
		5891X	Operational Excellence - Meter Data Services (Price Transmission) *X	0	1,575,000	1,575,000	Funding Change
		5892	Operations Excellence - Control Center Operation	0	851,000	851,000	Funding Bucket
		5892A	Control Room EOL Server and Tool Replacement	276,891	0	(276,891)	Funding Change
		5892C	IDS-DVD Dig Safe DVD Replacement	42,104	675,000	632,896	Timing
		5893A	Mobility Mwork Replacement for Trouble Workers	317,741	2,190,000	1,872,259	Timing
		5915USIT	INVP 5915 Grid Mod Cyber Security	(6,172)	0	6,172	Accounting
		5976	Clifton Park Innovative Pricing Rate Design Project	0	22,000	22,000	Opex Investment
		6043	Plan Forward	29,940	0	(29,940)	Opex Investment
		6051	STORMS Extended Performance	(170,262)	0	170,262	Efficiencies
		6320	Telecom & Network Deep Dive	104,751	0	(104,751)	Opex Investment
	Electric BU Total			**11,842,154**	**16,854,000**	**5,011,846**	
	Wholesale Networks	5756	Digital Substation Transformation	(463)	0	463	Opex Investment
		5773D	NYISO PMU Requirement for Large Generators	148,898	473,000	324,102	Timing
		5822C	Generation Asset Management Upgrade	1,938,552	2,548,000	609,448	Timing
		5973	FY21 Wholesale Networks and US Capital Delivery Minor Works	(16,708)	0	16,708	Opex Investment
		6043G1	Content Management for Engineering	717,140	1,394,000	676,860	Complexity
	Wholesale Networks Total			**2,787,420**	**4,415,000**	**1,627,580**	
US Electric Total				**48,466,002**	**64,202,998**	**15,736,997**	

Portfolio	Program	Project INVP	Project Name	Year End Actual	Year End Budget	Variance	Reason
Group Funct	Finance & Business Services	3870A	F&A and Implementation - Vision FM Replacement	2,205,145	1,730,000	(475,145)	Scope changes
		4562	US SAP: Business Warehouse (BW) Consolidation to HANA Enterprise Clou	800,971	480,000	(320,971)	Scope changes
		4998	Allegro Upgrade to Horizon	9,803	0	(9,803)	Opex Investment
		5158	MyHub Compliance & Enhancements	980,489	7,263,000	6,282,511	Timing
		5158A	myHub System Stabilization (30/60/90 Initiative)	0	340,000	340,000	Timing
		5317	Employee Digital Workplace	(46)	0	46	Immaterial
		5360	Lease Accounting Updates and Contract Management	24,909	0	(24,909)	Timing
		5375	SAP/PowerPlan/Front Office Maintenance of Business (MOB) - FY20	(17,945)	0	17,945	Opex Investment
		5378	FSSC&HR Systems (Non-SAP) Operational Requirements & Upgrades FY20	(93,379)	0	93,379	Opex Investment
		5380A	Blackline: US Implementation	133,598	795,000	661,402	Imprecise Estimates
		5466	AMAG HW and SW Upgrade	1,405,321	563,000	(842,321)	Scope changes
		5497	Lease Accounting Updates and CCM - Phase 2	373,752	769,000	395,248	Timing
		5507	US Legal: Document Management	2,747	0	(2,747)	Timing
		5551	Fleet improvements	0	75,000	75,000	Opex Investment
		5555	Fieldglass Replacement (Global Workforce Tracker)	0	1,800,000	1,800,000	Timing
		5605	US SAP: Infrastructure Landscape FY21	46,133	0	(46,133)	Timing
		5606	US SAP: Infrastructure Landscape FY22	3,592,906	0	(3,592,906)	Funding Change
		5609	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY21	825	0	(825)	Immaterial
		5610	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY22	1,022,923	1,424,000	401,077	Scope changes
		5631	Implementation Costs for Booking Mgmt Project	65,602	123,000	57,398	Opex Investment
		5777	US IMS Upgrade	851	0	(851)	Immaterial
		5841A	Pension Administrator Replacement	157,507	249,000	91,493	Imprecise Estimates
		5841C	401(k) Administration Changes	792,483	546,000	(246,483)	Scope changes
		5841D	Credit Card MSP Changes	0	150,000	150,000	Opex Investment
		5842A	US SAP_ GRC Upgrade	982,980	941,000	(41,980)	Within Target
		5842B	Mandatory FERC Form Changes	638,192	825,000	186,808	Scope changes
		5843A	SAP Platform Upgrade: OS and Databases	560,406	450,000	(110,406)	Timing
		5843B	US SAP Platform Upgrade: Minor Releases	395,845	485,000	89,155	Timing
		5843C	US SAP Health check: Roadmap and assessment of critical components	0	113,000	113,000	Opex Investment
		5843D	US SAP Health check: Upgrade Critical Components - FY22	1,048,896	1,706,000	657,104	Timing
		5884	Operational excellence - IT solutions for Property Services	0	75,000	75,000	Opex Investment
		5895A	Investment Recovery IT Strategy & Roadmap	0	38,000	38,000	Opex Investment
		5895B	Everbridge / Critical Event Management	359,039	290,000	(69,039)	Scope changes
		5896A	US Tax: SABRIX Upgrade	151,234	120,000	(31,234)	Complexity
		5896B	Warehouse SAP Scanner Replacement (Barcoding)	235,056	510,000	274,944	Complexity
		5896C	Wellness: RedBrick to Virgin Pulse Migration	(55,647)	4,000	59,647	Opex Investment
		5896D	WEM Waste Environmental Management	347,852	129,000	(218,852)	Imprecise Estimates
		5902	Applications Health Check (Non-SAP) Program	0	358,000	358,000	Efficiencies
		5918	Non-SAP Minor Enhancements	(58,913)	0	58,913	Opex Investment
		5919	US SAP Minor Enhancements	(50,111)	0	50,111	Timing
		5964	Safety Line Inspection	(17)	0	17	Opex Investment
		5979	Federal PAC Reporting Payroll Interface	0	38,000	38,000	Opex Investment
		6137US	Enterprise PMO, Jira Align release 2	382,306	0	(382,306)	Funding Change
		6182US	Global Career Site	76,974	0	(76,974)	Opex Investment
		6183US	HR Reporting Analytics	488,588	0	(488,588)	Funding Change
		6185US	Onboarding 2.0	13,974	0	(13,974)	Funding Change
		6219	US F&BS Minor Enhancements (Non-SAP)	142,785	240,000	97,215	Opex Investment
		6220	US F&BS Minor Enhancements (SAP)	566,637	675,000	108,363	Timing
		6221	Tech Debt: RiskMaster Replacement	67,090	743,000	675,910	Timing
		6231	US Treasury: Allegro Upgrade Phase 2	182,993	375,000	192,007	Timing
		6232	Cybergrant Replacement	0	416,000	416,000	Efficiencies
		6233	Legal: Digitization	0	383,000	383,000	Efficiencies
		6234	HR Benefit Vendor Integrations	0	338,000	338,000	Efficiencies
		6235	Tech Debt: RFS / SIR / Remedy	0	236,000	236,000	Efficiencies
		6236	Tech Debt: Carousel	0	263,000	263,000	Efficiencies
		6237	US Group Functions Operational Enhancements FY22	0	75,000	75,000	Opex Investment
		6265	HR Policy Compliance	8,637	166,000	157,363	Opex Investment
		6270US	Digitizing Learning	17,095	350,000	332,905	Timing
		6273US	Workforce Data Domain HR data governance Tools Processes EDM	961,402	1,610,000	648,598	Funding Change
		6289	Financial Transformation (FTOM) - Record To Report (US)	0	8,280,000	8,280,000	Funding Bucket
		6289B	FTOM: Record to Report	1,898,729	0	(1,898,729)	Funding Change
		6289D	FTOM: Record to Report - Task Management with Blackline	527,301	0	(527,301)	Funding Change
		6289E	FERC Form Integration with wDesk	15,163	0	(15,163)	Opex Investment
		6290	Financial Transformation: Source to Pay?	0	3,570,000	3,570,000	Funding Bucket
		6290A	Source To Pay	1,114,584	0	(1,114,584)	Funding Change
		6290B	Source To Pay: Release 2 - Supplier Onboarding	1,307	0	(1,307)	Opex Investment
		6377	US Group Function Operational Requests	1,597,682	0	(1,597,682)	Efficiencies
		6421	P2P Data Analysis Tool	2,865,782	0	(2,865,782)	Funding Change
		6426	Investment Recover Digitalization	362,443	0	(362,443)	Funding Change
		6450x	Accelerated Telecom Networks Audit (POTS and Circuits)	156,739	0	(156,739)	Opex Investment
		6461USA	Business Planning F&A (FP&A Enablement & Modelling Tool)	621,199	0	(621,199)	Opex Investment
		FIN22	Finance + Placeholder	0	320,000	320,000	Funding Bucket
		HR22	HR + Placeholder	0	3,631,000	3,631,000	Funding Bucket
	Finance & Business Services Total			**28,128,817**	**44,060,000**	**15,931,183**	

Group Functions	4418US	RSA Archer Upgrade v6.8 and enhancements	(26,543)	0	26,543	Opex Investment
	4418USB	RSA Archer Upgrade & Refresh	352,536	0	(352,536)	Efficiencies
	5472	GRC Archer - Risk and Migration	436	0	(436)	Immaterial
	5506	Integration Services	107,559	0	(107,559)	Timing
	5899A	Cloud API Migrations Phase 2	3,766,212	2,270,000	(1,496,212)	Scope changes
	5900A	ServiceNow Upgrade to version Orlando	4,523	8,000	3,477	Immaterial
	5901B	Application Lifecycle Management Toolset Phase 2	194,575	8,000	(186,575)	Timing
	5901C	MS Project Online Enhancements	10,577	8,000	(2,577)	Immaterial
	5901D	EA Strategic Portfolio Planning & Delivery (SPPD)	1,025,906	868,000	(157,906)	Scope changes
	5917	IT4IT: Data Management	32,840	0	(32,840)	Opex Investment
	5917A	Data Mgmt_ Tooling	337,270	0	(337,270)	Funding Change
	5917B	Tableau Rationalization & Power BI Enhancements	0	494,000	494,000	Efficiencies
	5917C	Business Objects / DataWarehouse Replacement	0	810,000	810,000	Efficiencies
	5917E	Master Data Management	0	245,000	245,000	Efficiencies
	5935	Solution Development Lab	8,427	11,000	2,573	Imprecise Estimates
	5983	DB Landscape Optimization	116,976	0	(116,976)	Timing
	6238US	Stakeholder and Internal Applications (SIA)	0	657,000	657,000	Funding Change
	6239US	Gridhome US Enhancements	0	683,000	683,000	Funding Change
	6241US	RSA Archer Upgrade & Refresh	0	580,000	580,000	Funding Bucket
	6242US	ARIBA System Enhancements	238,412	522,000	283,588	Imprecise Estimates
	6243US	BoardVantage	0	124,000	124,000	Opex Investment
	6252US	Data Mgmt Tools and Initiatives	3,047,584	1,283,000	(1,764,584)	Funding Change
	6253US	Cloud API Migration - Legacy Integration Tools	2,596,101	525,000	(2,071,101)	Scope changes
	6254US	Platform as a Service (PaaS) on Azure	874,635	885,000	10,365	Within Target
	6255US	Digital Foundation Tooling and DevSecOps	738,577	885,000	146,423	Funding Change
	6256US	Platform Upgrades: Salesforce/Maximo/MuleSoft	853,428	758,000	(95,428)	Funding Change
	6257US	GSD Engineering Lab/Data Synchronization	0	1,873,000	1,873,000	Efficiencies
	6258US	Data Archival	0	1,463,000	1,463,000	Funding Bucket
	6258USA	Data Archival Platform Implementation	1,625,496	0	(1,625,496)	Funding Change
	6258USB	Data Archival: US Group Function Portfolio	842,662	0	(842,662)	Timing
	6259US	EA Foundation: Tech Debt/Resiliency/Rationalization	0	1,650,000	1,650,000	Efficiencies
	6260US	EA Foundation: Increased Capability	1,455,450	3,755,000	2,299,550	Funding Change
	6261US	ServiceNow Upgrade: Stay Current FY22	810,598	255,000	(555,598)	Scope changes
	6262US	Additional ServiceNow (SNOW) Capability	0	255,000	255,000	Efficiencies
	6263US	IT Enterprise Project Mgmt Office (PMO)	0	2,093,000	2,093,000	Efficiencies
	6264US	Apptio Development	886,127	1,398,000	511,873	Timing
	6266US	EA Strategic Portfolio Planning & Delivery / Additional Capability	1,185,629	750,000	(435,629)	Funding Change
	6313US	GDPR Purge Compliance	115,676	0	(115,676)	Opex Investment
	6346US	CA: Group We Digital Evolution Phase 1 - FY22	89,411	0	(89,411)	Funding Change
	6346USC	Domain Management Refresh	1,017,021	0	(1,017,021)	Opex Investment
	6346USE	Brand Central Design and Implementation	50,564	0	(50,564)	Opex Investment
	6346USF	Brand Central Design and Implementation	130,033	0	(130,033)	Opex Investment
	6346USI	Video Strategic Management and Archiving (Enterprise)	103,352	0	(103,352)	Opex Investment
	6360US	IT Planning: Enterprise PMO	861,492	0	(861,492)	Funding Change
	6458US	QES IT Tool Sunset	571	0	(571)	Opex Investment
	6468	GridStack - Infrastructure	10,164,995	0	(10,164,995)	Funding Change
Group Functions Total			**33,619,107**	**25,116,000**	**(8,503,107)**	
Group Functions Total			**61,747,925**	**69,176,000**	**7,428,075**	

Portfolio	Program	Project INVP	Project Name	Year End Actual	Year End Budget	Variance	Reason
DPIT	Cloud & Hosting	4489	Active Directory Improvements	1,005	0	(1,005)	Immaterial
		5750	Directory Services - US	16,114	0	(16,114)	Imprecise Estimates
		5837	Apllication Virtualization	8,566	0	(8,566)	Imprecise Estimates
		5874	Cloud Center of Excellence	982,604	0	(982,604)	Imprecise Estimates
		5878	Dell Discovery	69,336	0	(69,336)	Accounting
		5942	Legacy AD Remediation	65,767	0	(65,767)	Imprecise Estimates
		6018	Security Operations Services (SOS)	583,114	0	(602,445)	Funding Change
						19,331	Imprecise Estimates
		6022	US Logical Database Transition	271,027	0	(271,027)	Opex Investment
		6049	Hosting Infrastructure Refresh to HCI	(207,372)	0	207,372	Other
		6050	Migration to IBM P9	44,287	0	(44,287)	Complexity
		6052	Migration of Current Storage to Strategic Deployment ??? US	(32,502)	0	32,502	Accounting
		6088	6088 - IBM P8 Digital Asset Build	36,041	0	(36,041)	Other
		6106	Data Protection Suite (DPS)	46,701	0	(46,701)	Accounting
		6117	HCI Lite Design & Deployment	(90,974)	0	90,974	Accounting
		6200	I&O US Cloud and Hosting FY22	12,591,052	10,995,000	(1,596,052)	Imprecise Estimates
		6223	I&O US Data Center Transformation FY22	27,142,188	21,420,000	(5,880,561)	Funding Change
						158,373	Imprecise Estimates
	Cloud & Hosting Total			**41,526,953**	**32,415,000**	**(9,111,954)**	
	CNI	5867	Network Security Infrastructure Program	(51,513)	0	51,513	Imprecise Estimates
		5867A	OMS Router Replacement & IT Service Management	30,891	0	(30,891)	Imprecise Estimates
		5867E	Substation Access Points	59,191	0	(59,191)	Funding Change
		6199	I&O US CNI FY22	2,596,494	1,080,000	(1,026,086)	Funding Change
						(490,408)	Imprecise Estimates
	CNI Total			**2,635,062**	**1,080,000**	**(1,555,062)**	
	End User	4987	SOE (Windows) Upgrade and Device Refresh	4,108,583	0	(4,108,583)	Funding Change
		5487	Collaboration & Unified Communications	3	0	(3)	Opex Investment
		5487E	ICE Public Folders - Migration/Decom	91,113	0	(91,113)	Opex Investment
		5487G	Azure Cloud Mail Routing & Exchange 2010 Decom	(111,764)	0	111,764	Opex Investment
		6091	Return to Office Network Changes	14,205	0	(14,205)	Opex Investment
		6097	Cloud Video Interoperability	159,382	0	(159,382)	Imprecise Estimates
		6108	SharePoint to SharePoint Online	74,281	0	(74,281)	Imprecise Estimates
		6136	System Center Configuration Manager (SCCM)	(2,211)	0	2,211	Imprecise Estimates
		6198	I&O US End User Experience FY22	3,896,858	8,055,000	2,871,479	Funding Change
						1,286,663	Imprecise Estimates
	End User Total			**8,230,451**	**8,055,000**	**(175,451)**	
	Network	5521	LAN Infrastructure Program	394	0	(394)	Opex Investment
		5521A	Verizon - Aruba Wireless LAN (Waltham)	(68,300)	0	68,300	Accounting
		5522A	End User VPN Replacement	(89,799)	0	89,799	Accounting
		5523	Voice Infrastructure Program	(101)	0	101	Opex Investment
		5523A	Northborough and Avaya PBX Replacement/Upgrade (NE Control Room sit	(563,320)	0	563,320	Accounting
		5523B	Upstate NY PBX Replacement	16,838	0	(16,838)	Imprecise Estimates
		5523F	Dedicated CNI Voice Network Buildout	219	0	(219)	Opex Investment
		5523G	Downstate NY PBX Replacement Upgrade	22,291	0	(22,291)	Imprecise Estimates
		5524A	WAN Diversity at 18 Regional Storm Sites	25,420	0	(25,420)	Imprecise Estimates
		5754	Network Equipment End of Life Replacement - LAN	93,809	0	(93,809)	Imprecise Estimates
		5759	UPS Infrastructure	(33,517)	0	33,517	Accounting
		5774	WiFi Phase 2	(813)	0	813	Immaterial
		5838	Managed Networks	527,946	0	(527,946)	Imprecise Estimates
		5846	Substation Network Connectivity & Comms	86,368	0	(86,368)	Imprecise Estimates
		6046	SD WAN FY21 Expansion	(1,035,568)	0	1,035,568	Accounting
		6197	I&O US Networks FY22	21,949,940	18,195,001	(3,754,939)	Scope
	Network Total			**20,931,804**	**18,195,001**	**(2,736,803)**	
	Operations	5161	Device Refresh	447,637	0	(447,637)	Funding Change
		5489	Emerging Technology	(34,229)	0	34,229	Accounting
		5489A	IBM SNOW Automation	(256,487)	0	256,487	Accounting
		5489B	WiPro SNOW Automation	(206,704)	0	206,704	Accounting
		5580	Managed Workspace Services Transition & Tx	879	0	(879)	Opex Investment
		5876	Pure Technical Debt Remediation	538,019	0	(538,019)	Imprecise Estimates
		6114	Managed Workspace Services Transformation	(134,978)	0	134,978	Imprecise Estimates
		6196	I&O US Operations FY22	8,949,381	12,450,002	3,500,621	Funding Change
	Operations Total			**9,303,518**	**12,450,002**	**3,146,483**	
I&O Total				**82,627,789**	**72,195,003**	**(10,432,787)**	

Portfolio	Program	Project INVP	Project Name	Year End Actual	Year End Budget	Variance	Reason
Security	Physical Security	3617A	Physical Security - G20	298,710	0	(298,710)	Funding Change
		3617F	PS - NIMOE	2,250,577	0	(2,250,577)	Funding Change
		3617G	Physical Security - KEDNY - Equip Replacement	1,048,667	0	(1,048,667)	Funding Change
		3617H	Physical Security - KEDLI - Equip Replacement	406,502	0	(406,502)	Funding Change
		3617I	Physical Security - MASS Electric - Equip Replacement	480,407	0	(480,407)	Funding Change
		3617J	Physical Security - Boston Gas	109,549	0	(109,549)	Funding Change
		3617K	Physical Security - Colonial Gas	34,696	0	(34,696)	Funding Change
		3617L	Physical Security - RI Electric	94,064	0	(94,064)	Funding Change
		3617M	Physical Security - RI Gas	24,365	0	(24,365)	Funding Change
		3617N	Physical Security - NIMO EGT	(2)	0	2	Immaterial
		5748	Security Control Center Back-Up	(8,627)	0	8,627	Opex Investment
		6244	Physical Security	0	6,687,001	6,687,001	Funding Bucket
	Physical Security Total			**4,738,906**	**6,687,001**	**1,948,094**	
	Security	3683USD	Develop Robust Incident Response	[redacted]			
		3683USF	Enterprise Centralized Patch Management				
		3683USH	Network Security Improvements				
		3683USN	SOAR - Security Orchestration Automation and Response				
		3683USX	Application Security As a Service				
		5702USA	Malware Defense - Endpoint Protection				
		5702USC	Malware Defense - System Hardening				
		5820	Data Protection				
		5828	Secure Web Gateway				
		5848	Vulnerability Management				
		5850	Network Security				
		5850US	Network Security US				
		5850USA	SSN Optimization US				
		5850USB	vSTIG Firewall Upgrade US				
		5850USC	Intrusion Detection System and Intrusion Prevention System US				
		5850USD	Intrusion Detection System and Intrusion Prevention System US				
		5870	Identity and Access Management				
		5933	Operational Technology Intrusion Detection				
		5940	Global VM Enterprise				
		5963US	Platform Projects Annual Program FY21				
		6210US	FY22 Platform Tower				
		6211US	FY22 Network Tower				
		6245	Security Orchestration Automation & Response				
		6245US	Security Orchestration Automation & Response				
		6416	Operational Technology Privileged Access Workstation				
	Security Total			**29,305,557**	**29,733,999**	**428,442**	
Security Total				**34,044,463**	**36,421,000**	**2,376,537**	

Portfolio	Program	Project INVP	Project Name	Year End Actual	Year End Budget	Variance	Reason
Other	IT4IT	6297	IT Project Evolution (FY22 Cost Savings & CTA)	0	22,900,000	22,900,000	Funding Bucket
		6297F	FY22 CTA - NOC Insource / RFP	(2,983,720)	0	2,983,720	Imprecise Estimates
		6368USL	Mid Term Initiatives Opex CTA US Networks RFP	0	0	0	Opex Investment
	IT4IT Total			**(2,983,720)**	**22,900,000**	**25,883,719**	
	Other	#	COH reconcile to 4403	(8,902,217)	0	8,902,217	Accounting
		5598	Application Rationalization	(167)	0	167	Immaterial
		6082	FY22 Planning Initiative	7,066	0	(7,066)	Funding Change
		6470X	Digital Performance Dashboard	310,402	0	(310,402)	Funding Change
		6494	Digital Performance Dashboard Ph.2	1,077,613	0	(1,077,613)	Funding Change
		6495	Brightening Digital Lighthouses	2,169,735	0	(2,169,735)	Funding Change
		COCU	5110S-IS2608 ServCo AJE COH-OpEx AR	344,983	0	(344,983)	Accounting
		COEA	5110S-IS2706 ServCo AJE COH-OpEx AC	339,154	0	(339,154)	Accounting
		COEL	5110S-IS2605 ServCo AJE COH-OpEx AE	94,642	0	(94,642)	Accounting
		COGA	Capital OH for Gas Organization	585,179	0	(585,179)	Accounting
		COPM	5110S-IS2612 ServCo AJE COH-OpEx AC	1,595,702	0	(1,595,702)	Accounting
		COSC	5110S-IS2679 ServCo AJE COH-OpEx AC	1,079,784	0	(1,079,784)	Accounting
		COSE	5110S-IS2735 ServCo AJE COH-OpEx AC	3,896,652	0	(3,896,652)	Accounting
		COTD	5110S-IS2714 ServCo AJE COH-OpEx AC	2,789,466	0	(2,789,466)	Accounting
	Other Total			**5,387,995**	**0**	**(5,387,995)**	
	Other Business Tech	Burden	Burden	(4,332,165)	(4,648,000)	(315,835)	Accounting
	Other Business Tech Total			**(4,332,165)**	**(4,648,000)**	**(315,835)**	
Other Total				**(1,927,889)**	**18,252,000**	**20,179,889**	
Grand Total				**299,075,591**	**328,311,000**	**29,235,409**	

Category	Type of Variance	Explanation
Project Delivery - Scope change	Scope changes	Spend impacted by both reduced or increased scope
Project Delivery - Initial scope delivered	Imprecise Estimates	Cost and Quantity of resources were incorrectly assessed when creating the budget. (eg. Price Increases)
Project Delivery - Initial scope delivered	Complexity	Unforeseen challenges in delivering the solution drove costs up. No change in scope but amount of work to deliver increased or potentially decreased
Project Delivery - Initial scope delivered	Efficiencies	More cost effective implementation options - Found a better way to deliver the solution which reduced costs
Strategic Direction	Funding Bucket	Source of undefined funding where funds will be moved to other projects
Strategic Direction	Funding Change	Move $ to or from another project
Annual Accounting	Accounting	Over or under accruals which reverse at year end leaving a balance on the INVP
Annual Accounting	Timing	Project slipped and intended scope was not delivered this fiscal year
Other	Other	

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
4377A	Melville Data Centre Clearance (DXC DC Capacity Increase)	$858,609	$449,191	$1,307,800	Supplier DXC needs constant management to stay on task.	This project purchased the necessary hardware to increase capacity at the DXC Data Centers to support the migration of equipment from National Grid's Melville data center.
4193	RI Street Light Metering Pilot	$0	$58,000	$58,000	The workshop held that including all business and application owners to cover the end-to-end process of meter lifecycle proved highly valuable in identifying changes to existing technology, gaps in [illegible] in existing business processes to accommodate the new types of meters.	The Rhode Island Public Utilities Commission (RI-PUC) ordered National Grid to assess the accuracy of new metering technology for street light applications including the impact to information systems and costs associated for automation of billing operations. National Grid performed a feasibility study to assess the impact to metering and billing applications required to support the new technology, options for deployment, costs, and timeline for deployment. There was no technology deployments associated with this investment.
3870	F&A and Implementation - Vision FM Replacement	$0	$180,000	$180,000	1. Not understanding who was the appropriate contact for ServiceNow (demand manager vs Enterprise Platform Architect) caused delay and confusion in identifying the right ServiceNow vendor to be engaged with. 2. Not enough submissions from the business SMEs on weighted requirements and demo assessments, skewed the overall score results for the vendors.	This project conducted a feasibility study to replace 3 antiquated and unsupported applications used by Property services namely 1) Facilities work order management system called "Vision FM" 2)PAM (Property Asset management) & 3) PRIA (Property Real Information Application) used to manage Property rights & Easements. National Grid's enterprise platforms IBM's Maximo & Service Now with Nuvvolo have been analyzed in depth as replacement platform for the 3 applications stated above. IBM's Maximo solution has been concluded to be best suited for the project requirements and recommended for replacing Vision FM, PAM & PRIA.
5487C	Box & Huddle to SharePoint Online	$456,000	$25,000	$481,000	Engagement: Customer engagement earlier on in the project would have made planning for Implementation easier Daily stand up meetings with the vendor worked well for the team once in Implementation stage Ensure full understanding of security requirements as agreed upon by IT Security Knowledge - Training: Post Implementation Support (PIS) team would benefit from further O365 training for faster turnaround on issue resolution	Migrated Box and Huddle to either SharePoint Online (SPO) or OneDrive for Business (ODB) Performed analysis to identify migration to SPO or ODB Decommissioned Huddle Reduced number of Box licenses to only essential personnel
5487B	SharePoint to SharePoint Online	$0	$0	$0	The scope of the discovery phase of the project was set without full consideration for the future use of the information and project delivery approach. This resulted in later rework and project delay to interpret the information and plan for the next phase. Ensure adequate time is allowed in the startup phase to develop the project delivery approach and to reach agreement with all stakeholders. The size & complexity of the project was not articulated consistently to all parties & reflected in the UK & US pipeline planning. Improved governance in pipeline planning and reporting. The approach to sanction a group of similar projects into a program results in complex management of program finances and sanctions. This is especially true if the projects are of differing durations and global. Re-sanction at program level is often required only for one of the projects in the program, resulting in frequent sanction steps. Either progress sanction at the project level or allow wider flexibility for program sanctions to cover longer time periods & stages and be governed by PMB or Product Owner.	Migrated SharePoint 2007, 2010 and 2013 to either SharePoint Online (SPO) or Microsoft OneDrive. Discovery on current SharePoint environments Operated in SP Online or SP 2016 Group membership, access rights, and site owners follows to SPO Unused sites decommissioned
5981	Microsoft Teams	$1,451,000	$217,000	$1,668,000	What went well - The MS Teams project was launched as COVID-19 Emergency project to increase IT capability for staff to work from home. The tight timeframe for delivery required a deviation from the normal protocols for project delivery. The necessary measures were given the full support of management and the project teams adapted admiralty to the challenges ahead. This resulted in MS Teams being enabled for circa 25,000 users within 6 weeks. What went well - The urgency to deliver a swift solution meant that the usual Governance and Service Transition approach would not naturally fit to the aggressive implementation timeline. Governance and Service Transition continue to show process flexibility to accommodate urgent project delivery. It's noted that the Governance and Service Transition processes have recently been adapted to accommodate the new Agile Delivery approach in I&O. What didn't go so well - Native MS Teams does allow end users to create their own Teams however this solution did not fulfill security requirements. The operational model approach, i.e. raise a service now ticket had a Service Level Agreement of 5 days to create a Team. This wasn't acceptable to management. During the project an interim solution was implemented with a request form in SharePoint and a dedicated member of the Project Team who created Teams. The turnaround time for this process was 1-4 hours which set an expectation for the end user community. Recommendation - Earlier review of the solutions being sought and thorough impact analysis against the risk that the desired solution may be delayed. Revisit alternative solutions, previously discounted to see if they can be adapted to meet the need in a shorter timeframe.	National Grid is a global organization that builds teams that deliver business goals. There is no expectation that those teams are co-located and team members are frequently based in different offices or even different regions. A greater choice of tools will improve the opportunities for teams to select the right tools for their needs and allow them to become productive in a shorter space of time. The scope of this project included all US staff who use the Office 365 suite of products and have sent an email in the last six (6) months. This included office and field based staff who may access Office 365 using either a laptop or a mobile device.
4570	US CNI Tech Services-Network Equipment Lifecycle Replacements	$10,105,000	$1,000	$10,106,000	Refreshed network infrastructure, appliance software and hardware, renewed vendor support services delivered improved system resiliency, stability, latency and connectivity to control rooms and substation in the New York and New England territories. As technology advancements are more rapid today - planning advanced procurement of hardware and software must be made with discretion.	This investment addressed the potential failure of aged resources that were no longer supported by the network hardware and software vendors that would have led to unacceptable disruption and\or loss of monitoring and system control. INVP 4570 was a Policy-driven investment which procured networking assets needed to replace out of warranty equipment and support infrastructure in the CNI Data Centers, Communications rooms and Operations Centers across National Grid service territories in New York and New England. Schedule

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
5271	F&A for CIS Request for Proposal	$0	$418,000	$418,000	1. Based on the detailed process workshop results, to deliver this scope, data and information flows in the to-be solution need to be solidified further. 2. The resulting implementation considerations and options will be used in later deliverables.	This project conducted a feasibility study to analyze National Grid's current Customer Information Systems landscape and assess high level requirements for transitioning from two legacy systems to one platform. During this roadmap and technical discovery, the National Grid team, with support from TMG, built the foundation and path forward. In addition, this project will work towards defining a solution scope, solution confirmation, and risk mitigation reviews.
4377	Data Centre Migration & Capacity Increase	$261,000	$1,767,000	$2,028,000	All DC migration projects should have network, storage and disaster recovery (DR) architects as an integral part of the project team. Thorough understanding of legacy equipment is needed before migration is attempted. Complete and accurate inventory of all equipment in the data center should be available at the beginning of the project.	The Melville Data Centre Clearance project migrated corporate applications to the DXC data centers and subsequently removed National Grid equipment (hardware/software) from the Melville Data Center.
4975USG	Gateway Upgrades	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
4975USB	CNI Intrusion Detection System (IDS) refresh Phase1	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
5461	Total Reward Statement	$450,000	$154,000	$604,000	In case of business managed vendors, clearly communicate list of controls, security evidences, certifications required by vendors at the beginning. Ensure services like Single Sign On, being provided as guidelines by EA and Security teams to connect to third party apps/websites, are actually ready to be implemented for entire population. Understand exception management process for any controls that are not in place or cannot be enforced due to various limitations (e.g. vendor is contractually not obligated to provide performance test results or physical security controls at their locations).	National Grid offers comprehensive benefits & compensation packages to the workforce. National Grid's Benefits team implemented the Digital Total Rewards solution to give employees real-time access to the true value of their compensation and benefits. With data aggregated from a number of internal and external sources the solution provides an up-to-date view of an employee's total rewards, demonstrating the value of the rewards package and enabling employees to use their benefits to their full potential.
5545	Learning Interfaces integration with SAP SuccessFactors LearningLink	$531,000	$96,000	$627,000	Identify 3rd party project management support at start-up. This project had particular issues as Myhub support was being transitioned from Accenture to Wipro. Obtain development and test environments with filtered PII data to comply with data privacy guidelines upfront during the start-up phase Approval processes around securing right Myhub environments, access allowed to consultants, and exception procedures need to be better agreed upon by all stakeholders. Get third party vendor agreement on expectations with their product early. Look for business stakeholders to help take the conversations forward if the relationship is managed by business.	This project integrated SAP SuccessFactors' (MyHub) with the RollCall mobile application and Sterling's Background check system. This integration allowed National Grid to improve the quality, efficiency, and data integrity by removing manual data entry or file exchanges with vendors.
3614B7	CNI Secure Communication	$8,784,600	$2,461,000	$11,245,600	Lessons Learned - Limited technical experts available in US CNI environments and spread too thinly. Advice to Future projects - Bring in additional resources sooner to ensure knowledge acquisition doesn't slow timescales. Lessons Learned - Significant increase in the scope, cost and timeline of 3614B7 US CNI Network Security. Project Implementation and go-live dates had to be readjusted. Advice to Future projects - Scope items like Security Services Network (SSN) extensions for US CNI environments which have significant impact on cost and time should be avoided where possible. A detailed impact assessment should be undertaken at the earliest to avoid significant impact on project's parameters of cost and time.	•Security Risk mitigation: The project implemented an appropriate encryption solution for securing the SCADA communication and hence, mitigate the risk reference 826 as quoted in National Grid risk register and security risk review by Deloitte, "Unencrypted, broadcast traffic may be eavesdropped, modified, or replayed. An alteration to this communication may result in an undesired action by the SCADA system or RTU, which could result in unstable gas/electricity transmission and distribution or a safety incident." •Continued Compliance to Security policies: Ensured continued compliance to National Grid's security policies by securing the SCADA communication with an appropriate encryption solution, thereby preventing an operational incident and ensuring the safe operation of the control systems •Reliability and Availability: This investment leads to the protection of the communication channels utilized by critical cyber assets supporting the CNI operations. It ensures reliability of SCADA communications with remote sites, thereby contributing to availability and reliable operation of National Grid's physical networks
4802	DocuSign	$979,000	$98,000	$1,077,000	Better communication with the vendor in addressing and resolving defects is needed. Disagreements and resolution of defects may require better project engagement with IT commercial for added influence. Additionally, better communication with IT security is needed to avoid new unexpected requirements to be added late in the project as occurred in this case. The project team should be actively engaged with the security team to avoid this.	This mandated project addressed National Grid's obligation in New York State to implement Electronic DPAs (Deferred Payment Agreements) per the NY Public Service Commission and established a platform for digital workflow and signature capabilities.
5609	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY21	$1,354,000	$276,000	$1,630,000	Using the phase delivery approach proves to be effective. We are planning to use the phased delivery approach going forward.	This project provided a funding base and governance structure that allowed the Information Technology (IT) organization to effectively deliver the needed updates to the US SAP application portfolio in order to comply with federal, state, and local government requirements.

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
4398	Storms-IScheduler Upgrade Stabilization	$14,611,000	$1,045,000	$15,656,000	1. Lack of technical understanding of applications and integration within National Grid (NG) Enterprise. The Project team needs to gain a full understanding of integrations within NG and improve requirements gathering early in the project lifecycle. 2. Business Process Support (BPS) team should have engaged with business to validate and approve business requirements; Although business readiness meetings were conducted made for a challenging UAT validation based on specific business teams solely owned their specific functionality to perform their work. 3. Lack of pro-active anticipation and planning for post go-live and Post Implementation Support (PIS). More of a reaction, just-in-time approach to command center. Priority 2 issue responsiveness was similar - no pro-active root-cause performed, largely due to critical response required. An updated Responsible, Accountable, Consulted, Informed (RACI) Matrix, inclusive of Support Team for post go-live and PIS, should be a requirement for Business As Usual (BAU) hand-off on all projects.	As the primary Work Management System (STORMS) and Work Management Scheduling tool (iScheduler) for the legacy National Grid service territories, these systems were critical applications in support of both Electric and Gas Operations, and became increasingly unstable, experiencing multiple outages over several years, and were no longer supported by the vendor. This project upgraded STORMS to the latest version of technology including: server hardware, system software and database software, along with bringing both standard and custom application code to the latest version of the technology. Additionally, the aged middleware components were replaced with new supported components, and iScheduler was replaced with the vendor's latest scheduling tool and integrated with the STORMS product.
4761	US Foundation Hosting Renewal	$10,359,674	$4,416,727	$14,776,401	Perform thorough tests in the non-production environments prior to any upgrades being performed to ensure a sound working environment prior to migrating to production systems. Implement a process in place to identify certificates before expiration to avoid connectivity issues. Need to have a documented approach to Post-Implementation Support: How it will operate; How long it will run; What the success criteria is for existing to the Closure stage; and, Who are the stakeholders that sign-off on both the criteria and its fulfillment at the conclusion of Post-Implementation Support. Clearly document key areas/descriptions of and how they will transition with the go-live. Perhaps some type of RACI chart to help illustrate that. Engage Service Transition Team by providing a 30-day notice of a go-live event, which explains the impact and gives insight into what types of communication will occur leading up to and throughout the event	To address its growing business environment, National Grid enhanced its SAP and High Performance Analytic Appliance (HANA) application support and hosting services. SYNTAX (formerly known as FIT) supplied Platform as a Service (PaaS) for SAP and HANA applications, and ancillary applications including PowerPlan, Open Text, uPerform and SABRIX. National Grid IS worked with SYNTAX to move the SAP application portfolio to a new datacenter. By moving to the new platform, National Grid eliminated the need to renegotiate contract extensions with current hosting providers SAP, T-Systems and Wipro as well as having to conduct costly upgrades of the SAP infrastructure hosted by T-Systems.
6013	Azure Cloud Foundation	$1,502,000	$17,000	$1,519,000	The list of products and capabilities needs to be defined in the beginning of the project to avoid delays in Testing and Closing the project. A Daily Development Forum call was utilized nicely for Team Collaboration. It opened up room for reverse shadowing. NG technical resources actively participated which helped MS to close out tasks and the transition demos.	This project created a Microsoft Azure Cloud hosting solution environment. The new Azure Cloud services provided by this project supports future business initiatives, improved flexibility and agility. This enables faster deployment cycles and scalability of resources to meet increasing business demands.
5777	US IMS Upgrade	$485,000	$75,000	$560,000	New security requirements identified late in the project cycle added additional scope to the project and impacted the overall project timeline / Baseline security requirements must be defined and locked no later than design phase. Existing contract with vendor had gaps with NG's standard App support requirements / An amendment was introduced into the existing Contract to include the standard operating procedure(SOP).	The Incident Management System (IMS) is used by the entire Company to report and manage safety, occupational health, environmental and security information. As part of aligning processes and metrics between the US and UK SHE organizations, the US version of IMS was upgraded to match the functionality of the UK IMS system along with a Global Metrics Reporting module developed to allow for shared views of data across National Grid US and UK. Additionally, the application was moved from the legacy hosting environment DXC to NG's Azure Cloud.
4975USA	Perimeter enhancements	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
5886B	Device Refresh & Break/Fix	$4,539,000	$0	$4,539,000	1) It should be ensured that all the devices are following the business as usual (BAU) procedure of being added to the Configuration Management Database (CMDB). 2) The engagement of a Service Owner from the onset of the project and through a weekly cadence aided in a consistent understanding of progress.	This project was part of a program of work to implement a refreshed and continuously updated work environment for employees using modern tools and management techniques. The project focused on updating the workforce with new computers enabling them to be more secure, efficient, and productive.

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
5837	Application Virtualization	$898,000	$355,000	$1,253,000	1) The technical requirements provided by the vendor in the original SOW were not complete. For example, Horizon Cloud on Azure not being compatible with Legacy Active Directory was not identified. National Grid needs to have a full understanding of what the vendor technical requirements are prior to having the vendor draft a proposal. 2) Legacy applications are not compatible with the modern OS version used by Horizon Cloud on Azure. A plan for how to proceed with legacy application transition from Citrix needs to be identified.	This project performed analysis of existing virtualized applications hosted in the legacy Citrix environment. The project identified actively used applications to determine which applications to migrate to a hosted cloud environment. The project supported enablement of migration of selected applications and users to the VMware Horizon Cloud on Azure.
5878	Dell Discovery	$1,040,000	$230,000	$1,270,000	National Grid project manager should ensure that suppliers working on projects are familiar with non-functional requirements (stage gates, change control, service level agreements, governance). Make sure suppliers understand the complexity and duration for deploying system credentials to the environment as National Grid has unique cyber security, compliance and regulatory oversight, with many levels of review and approval being necessary.	National Grid engaged Dell EMC Services to provide a design of the future state infrastructure. The project delivered two primary outcomes, which ran as parallel work-streams: A comprehensive analysis of the current technical architecture which delivered an “as-is” and "to-be" artifact for the future design of the new computing environment. Data Center Asset Cloud Maturity Assessment and design/architectural plan to support a new computing environment.
5906	MA Distributed Solar Charge	$674,000	$51,000	$725,000	Innovative development & testing approach resulted in project completed ahead of time & under-budget Engaging an experienced billing business person as part of project from beginning resulted in no scope creep, and misinterpreted requirements. Service Transition was well received There were some hiccups while changing meter data service application, which was not changed for long time. Such changes can be addressed in the early stages.	Net metered customers that over-generate are not billed the distributed solar charge. Their total kilowatt hour (kWh) for a billing period is negative when they over-generate. The charge was only billed on usage greater than 0 kWh. Effective October 1, 2020, the Department of Public Utilities (DPU) ordered Massachusetts (MA) utilities to ensure the Distributed Solar Charge is applied to customer that over-generate. This project automated the changes for billing in the Customer Services System (CSS) accordingly.
6081	Post COVID - Data Access Development	$0	$208,000	$208,000	N/A	This is an agile project to deliver a self service data environment to enable COVID list creation and segmentation of customer data. This is an interim solution in support of COVID related requirements until the Marketing Cloud solution is fully deployed.
5146	Smart Cities Schenectady	$467,715	$536,562	$1,004,277	1: Project re-sanctioning was done to include Network and device testing. Include Network and Security teams early on of the evaluation phase of the project to determine network and device requirements and include in the initial sanctioning process. 2: The vendor supporting network penetration testing did not have a SOC2 report available. Confirm the vendor conducting penetration testing is SOC2 compliant to avoid additional connectivity solutions/cost and timeline delays. 3: Unexpected device testing delays due to resource availability caused by NY travel limitations and quarantine requirement. Although somewhat unavoidable based on travel restrictions, for future testing develop a contingency plan to leverage secondary resources and an alternate solution to device testing.	This mandated project provided Information Technology (IT) consulting support to the City of Schenectady Smart City REV Demonstration phase one project that included installation of light-emitting diode (LED) street lights, network lighting control (NLC) nodes, and a primary Internet of Things (IoT) network utilizing a low-power wireless personal area network in two zones of the city. The technology platform provided by the vendors was evaluated from a Solution Architect, Enterprise Architect, Digital Risk and Security (DR&S) and Network perspective and evaluation reports and recommendations were provided for expanding this LED pilot. As per security team's recommendation, penetration testing was planned to be completed. The project was re-sanctioned for Network & Device testing. The vendor supporting network penetration testing did not have a SOC2 report available and thus a new VPN had to be set up, resulting in additional cost. Also, there were unexpected delays to device testing, due to resource availability caused by NY travel limitations and quarantine requirements. Testing was put on hold and restarted when travel restrictions were relaxed. But, this increased the cost of testing.
6136	System Center Configuration Manager (SCCM)	$605,000	$7,000	$612,000	Public Key Infrastructure (PKI) certificates for new Active Directory (AD) implemented 3 months late due to vendor resources not being available Require vendor to have committed resources to meet delivery schedule Atos estimate of cost to deliver were too high resulting in over budget allocation for NG National Grid requires a full understanding of what the estimating costs are from the vendors technical requirements prior to having the vendor draft a proposal	This project implemented System Center Configuration Manager (SCCM) Solution and integrated it with the New Global Active Directory (AD) infrastructure. SCCM allows for rollout of application phase deployments, security patches and critical software updates that will be required for Windows 10.
4975USF	Internal Public Key Infrastructure (PKI)	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
5891B	Field Collection System (FSC) Upgrade POC	$0	$176,000	$176,000	1) It would have been timelier and more efficient to include the requirements for connectivity ports with external hardware when submitting the initial build request for the test server. Asking for the ports to be added at a later date contributed to the timeline delay. 2) The project would have benefitted from having an assigned resource from CMS, as that party could have facilitated the testing with the mobile collection unit. 3) The project's Security Architect reminded the team of the following points: a) POCs can only reside in a development or test environment, b) POCs can't use production data, c) POCs can't communicate with production systems, devices or appliances, and d) POCs can't allow for connectivity with external third parties	This Feasibility and Analysis investment was a Proof of Concept (POC) that the current Field Collection System (FCS) can be upgraded to the latest version of the application/hardware and continue to support the current legacy process and tools, as well as support the new Itron Mobile Cloud solution. National Grid needed to ensure the current version of the Itron Field Collection System v4.4 is compatible with both the Itron legacy mobile collection software (MCS), which is currently in-use at National Grid, as well as the current Itron Mobile Application. The investment was the basis for a road map for a larger investment to perform the actual upgrade of the production FCS system and install the new Mobility Platform.

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
5896C	Wellness: RedBrick to Virgin Pulse Migration	$0	$179,000	$179,000	NG BP&I team recommends using SSO for all internal and external applications. Current logistics issues with Union employees (not having devices, not using their AD ID actively) does not allow for a successful implementation of SSO/MFA . This has been a major hurdle in many HR projects for over 2 years. Projects have to go for exception process as underlying service is not fully rolled out due to logistics issue(SSO/MFA using MS Azure) . It is important to resolve these before projects are requested to use SSO/MFA The project had to go-live on 1st January (year end freeze). Project should have planned in advance to ensure all parties are informed and aligned for a project launch during the freeze period to avoid last minute urgent meetings.	After a competitive bid, Virgin Pulse was selected as National Grid's Health and Wellness platform beginning Jan 1, 2021. A data feed from National Grid to Empyrean was modified according to need of the new wellness platform (Virgin Pulse). IT worked with Employee Services, HR, SHEA and Virgin Pulse and Empyrean to ensure the platform receives required data, Virgin Pulse meets baseline security requirements and platform is accessible from web and mobile devices from within or outside of the National Grid network.
6037	US vSTIG Bandwidth Upgrade (10G)	$1,672,000	$0	$1,672,000	Pre-Installation site survey - assessments of legacy environments. Adequate site preparation by the supplier (Verizon) prior to scheduling change window. Cabling issues at both US locations caused delays and rescheduling of changes. Pre-configuration of routers and switches to ensure that equipment is ready prior to change window.	This project included upgrades in the (US) region vSTIGs (Verizon Secure Telecommunications Internet Gateway) located in Ashburn, VA, and Billerica, MA. The scope of this project included critical internet capacity expansions, to support the extended periods of working remotely due to the current COVID-19 pandemic, “the new normal".
6106	Data Protection Suite (DPS)	$3,258,831	$52,000	$3,310,831	Constant collaboration with suppliers (meetings more than twice a week) greatly helped the delivery of the project. Creating multiple status project calls for the project duration and multiple consistent status meetings is the lesson learned There was an issue with maintenance and support incorrectly allocated to the capex accounting code. In order to prevent this issue in the future ensure the Purchase Order is created with the correct allocation between opex and capex codes.	This project procured, installed and configured a new back-up storage solution that replaced the current Tivoli Storage Manager (TSM) in National Grid’s DXC hosted data centers.
5487D	Intelligent Communications	$0	$301,000	$301,000	IT Solutions Delivery team determined that Zoom as a video conferencing solution is not fit for purpose for National Grid and therefore will not deliver the best possible end user experience. As a result, the Zoom pilot project has been closed. MS Teams is our enterprise communications and collaboration tool.	Zoom was investigated to replace WebEx as the corporate teleconferencing tool to provide a more efficient and cost effective solution to meetings throughout the National Grid Estate.
6108	Sharepoint to Sharepoint Online	$636,000	$69,000	$705,000	The scope of the discovery phase of the project was set without full consideration for the future use of the information and project delivery approach. This resulted in later rework and project delay to interpret the information and plan for the next phase. Ensure adequate time is allowed in the startup phase to develop the project delivery approach and to reach agreement with all stakeholders. The size & complexity of the project was not articulated consistently to all parties & reflected in the UK & US pipeline planning. Improved governance in pipeline planning and reporting. The approach to sanction a group of similar projects into a program results in complex management of program finances and sanctions. This is especially true if the projects are of differing durations and global. Re-sanction at program level is often required only for one of the projects in the program, resulting in frequent sanction steps. Either progress sanction at the project level or allow wider flexibility for program sanctions to cover longer time periods & stages and be governed by PMB or Product Owner.	National Grid utilizes multiple instances of SharePoint (2007, 2010, 2013) leading to inconsistencies in control, support and site storage. To lower IT operational spend and ensure only systems within support are used, National Grid has consolidated the numerous SharePoint instances into SharePoint Online.
5522A	End User VPN Replacement	$1,571,879	$6,149	$1,578,028	Stage impacted: Design Area impacted: Quality Root cause: Technology Lessons Learnt +/-: Negative Lesson Learnt: Ensure product is aligned to IT Strategic Road map and Reference Architecture Recommended Action: 1. Procurement to complete Due diligence -does product fully meet requirements 2. Ensure NG/Supplier meet the standard required Infrastructure Stage impacted: Requirements Area impacted: Customer Root cause: Planning Lessons Learnt +/-: Negative Lesson Learnt: No discovery activity in order to identify various business critical Apps/Areas/functions as project was Agile Recommended Action: 1.Agile activity should have been broken into sprints 2. Update CMD/SCCM to ensure data is current 3. HR data by area made available 4. CSMs to provide app utilize by area Stage impacted: Implementation Area impacted: Quality Root cause: Technology Lessons Learnt +/-: Positive Lesson Learnt: Fix forward approach Recommended Action: Worked well, better/advance knowledge of EUC Apps	The Juniper infrastructure on which this solution was hosted was no longer under vendor support and consequently any infrastructure failure at best needed to be resolved under a best endeavors basis. The impact of any infrastructure failure would have resulted in extended service unavailability, resulting in remote users being unable to access the National Grid network for an extended period. Note, a Juniper infrastructure failure occurred in the US earlier in 2019 resulting in a loss of VPN service. As the Juniper infrastructure was no longer under vendor support, the issue was resolved through adhoc support processes. Additionally, since the original sanction the COVID-19 response has required all employees to work from home where possible. This significantly increased the numbers of users connecting from home simultaneously and required the expedited deployment of a new solution to avoid failure of the existing due to the increased load.

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
5522	Network Security Infrastructure Program	$1,571,879	$61,617	$1,633,496	Lessons Learned from INVP 5522A: Need to ensure product is aligned to IT Strategic Roadmap and Reference Architecture. Procurement to complete due diligence. Does product fully meet requirements? Ensure NG/Supplier meets the standard required infrastructure There was no discovery phase to identify various business critical apps/areas/functions as project was agile. Agile activity should have been broken into sprints Update CMD/SCCM to ensure data is current HR data by area should be made available CSMs/SD's to provide app utilization by area The project used a fix forward approach which worked well and provided a better/advanced knowledge of EUC apps	The Network Security Infrastructure Program was initiated for a series of projects intended to refresh the network infrastructure and provide new capabilities to improve network performance, mobility and user experience. To ensure that the services continued to be secure, reliable and efficient it was important that services were maintained within support and new capabilities were added as technology advanced. In addition, the changing threat landscape and expanding use of cloud based services required a change from premise based security appliances towards more agile cloud and virtual security platforms. The program prioritized work and delivered the most critical first, considering current and future needs of the business. The purpose of this program paper was initially to request funds for the following initiatives listed under the Program INVP 5522 - Network Security Infrastructure Program: 5522A: Verizon – End User Virtual Private Network (VPN) Replacement 5522C: Verizon – Verizon Strategic Internet Gateway (vSTIG) Device Refresh 5522D: Verizon – Verizon - Tufin Upgrade 5522F: Verizon – vSTIG re-architecture - New Core Firewall The only project completed under this program was: INVP 5522A: Verizon - End User Virtual Private Network (VPN) Replacement The remaining projects were de-scoped and defunded from this program: 5522C - vSTIG refresh was completed under a separate initiative in FY21, INVP 6037 for the upgrade and hardware refresh is part of FY22 Network initiative sanction 5522D - not addressed under any other initiatives 5522F - this work was to be owned by the IT Security Group
3956	WiFi for Fleet Services Diagnostic Laptops	$521,000	$3,000	$524,000	Require the vendor to ensure equipment needed for the project is readily available seek alternate solutions sooner.	This project introduced Wi-Fi into each of the Fleet Service garages to allow for its use during vehicle diagnostic testing. The majority of the Fleet garages did not have Wi-Fi making it difficult and hazardous to perform this testing. In some cases the vehicles needed to be sent out to other facilities due to the lack of connectivity in the Fleet Service garage to perform the diagnostics.
5787A	Gas Control Center Simulator F&A	$0	$253,082	$253,082	Identify scope and validate if other NG IT teams are needed for the project to ensure no cost variance in the end of the project. RFP timelines depend on the vendor engagement, hence get commitment for vendors on timely responses and include risk margin to accommodate and delays to stay on top of schedule slips.	This is a Feasibility & Analysis study. The purpose of this phase of the project was to select a SCADA Gas Control Simulator software vendor that can provide software to allow extensive "almost on-the-job training" for Gas Control operators, without adversely impacting the gas distribution and transmission network. The idea is that Gas Controllers in the Control Center will be able to simulate various situations that occur on the production SCADA system and train in a disconnected environment to handle such situations, without actually causing any harm, if something were to not go as planned.
5789C	Stabilization of Critical Gas Legacy Applications (CWQ)	$268,269	$39,345	$307,614	After considerable analysis, the CWQ project team determined that is was impossible to upgrade the planned components without upgrading the entire application. Consequently, we obtained approval from the CWQ business sponsor (Caroline Hon) to repurpose CWQ's QA Environment as a DR Fail-over site, and conclude the CWQ project by handing off to Production Support the DR Fail-over Runbook and configuration definition. Therefore, the knowledge transfer session with the Production Support team was completed on 22 Feb 2021. During this session, details of the DR Fail-over Runbook and Hardware / Software Configuration Matrix were delivered, thereby illustrating that assets were delivered through the suite of DR Fail-over documentation provided to Production Support Team.	The original focus of the CWQ project was to upgrade the Citrix servers (or replace Citrix with an alternate application virtualization technology), thereby stabilizing the application and improving its performance. Following considerable evaluation / research, the CWQ team reached the conclusion that It is not possible to upgrade the planned components without upgrading the entire application. Since many of the application's components are no longer offered by their vendor, this would require customization of a replacement component. Upgrading any of these legacy components would result in the need to upgrade not just the entire CWQ application, but all MQ instances within interfacing applications. This is well beyond the original scope anticipated for this project. Consequently, we have discussed this project conclusion alternative with the CWQ business sponsor, and she agreed with repurposing of CWQ's QA Environment as DR Fail-over site and concluding the CWQ project. By taking this minimum cost alternative, availability of the Production environment will be elevated considerably in the event of a major outage. Acknowledging that the CWQ application must continue to function in Production until its replacement by GBE MAXIMO initiative (18 – 24 months).
4489	Active Directory Improvements	$2,249,000	$272,000	$2,521,000	Regular governance reporting provided the right level of detail & confidence of control to project leadership (including hours / financial tracking). Agreement on RAG status of risk & issues clear definition was required. Lesson Learned is to create and agree on a joint definition of Risks & Issues. Strong communication plans with key stakeholders were imperative to deliver hardware and to gain acceptance of delays in shipping during the COVID-19 pandemic. Working closely with facilities management to fully understand the site closure schedules and site restrictions during the Covid-19 pandemic helped the project team develop a strong plan and accurate schedule for hardware installations.	Active Directory (AD) is a key service that supports core authentication for all National Grid computers and servers logging onto the corporate network in both the United States (US) and United Kingdom (UK). Therefore, AD provides access to all National Grid Information Systems (IS). This project implemented a refreshed global AD infrastructure and support services.
6117	HCI Lite Design & Deployment	$3,238,000	$119,000	$3,357,000	1. Project was completed in 17 weeks using the lessons learned from Hosting Infrastructure Refresh to HCI - US. Using previous lessons learned helped to accelerate delivery. 2. Primary Resource for completion of the SRM portion fell ill. Plans were made to mitigate risk. Re-planned for enough backfill and used to deliver project on time.	The Virtual Computing Environment (VCE) that hosted National Grid (NG) critical applications reached end of service in March 2018. This created a performance and security risk to National Grid. This project procured new Hyper Converged Infrastructure (HCI), installed it in DXC data centers and configured it to make it operationally ready to accept workloads.
3683USO	Network Segregation	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
6091	Return to Office Network Changes	$0	$171,000	$171,000	1) Due to COVID restrictions, it was difficult for the team to test the work completed at specific sites where we could not get a resource on site to complete testing Ensure there is a viable plan for testing at remote sites in advance of implementation 2) Team was very efficient at deploying the bulk of the upgrades early on in the project by completing the work in batches Work with vendor to identify locations where bulk upgrades can be completed at the same time 3) Team was successful at completing the upgrades at priority sites with high traffic early on in the project followed by low priority sites Work with vendor to prioritize upgrades at priority sites with the most traffic first as they will have the greatest impact	This solution implemented quality of service (QoS) changes to the Verizon managed network and network security devices which optimized Microsoft Teams traffic. This ensured end users returning to the office experience the best possible video and audio performance when using Microsoft Teams and safeguards National Grid's security and network architecture policies and strategies.
5935	Solution Development Lab	$8,000	$102,000	$110,000	Vendor, Amazon Web Services (AWS), could have been engaged earlier in the project to better aid in design/environment set-up. Understand exception management process for any controls that are not in place or cannot be enforced due to various limitations (e.g. vendor is contractually not obligated to provide performance test results or physical security controls at their locations). Stronger awareness by Project Manager around active Solution Development Framework (SDF) Refresh activities could have avoided extra documentation build-out by the project team.	Solution Development Lab enabled virtual access to mobile devices (iPad, iPhones, tablets, etc.) for end-to-end solution testing by the Quality Engineering Services team. Testing is able to mimic the total customer experience received by National Grid external customers who access our websites on their personal devices.
3683USP	IAM: Role Based Access control	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
5580	Managed Workspace Services Transition & Tx	$4,696,000	$23,000	$4,719,000	Improve scope definition for future End User Experience (EUX) transitions (split transition and transformation more clearly) Take more time planning the program timescales Improve communication with key stakeholders to avoid incorrect charges being recorded against INVP	National Grid reached the lease termination date and was required to purchase the PC's on lease or return them. Laptops under four years old (DXC contract terms) have a Net Present Value (NPV) which required payment by National Grid.
3431A	Portfolio/Demand Management Replacement	$0	$65,000	$65,000	Clear roles and responsibilities identified at the outset with commitment from Sponsors and Project Team to fulfil their roles.	Demand/Portfolio Planning & Management (D/PPM) is the ability to manage National Grid's investment portfolio to achieve strategic objectives. UK and US are at the center of a transforming energy landscape with record level investment across both regions, with a potential UK RIIO T2 spend of £2bn and US strategic business plan of $3.6bn in the next five years. Progress has been made in each region in establishing and agreeing to the transformation programs at an exec level. To achieve our strategic objectives, there are significant risks across both regions to ensure the portfolio is optimized to confirm strategic objectives are not compromised. This investment completed a Feasibility and Analysis Study to identify the right product for a D/PPM tool with findings presented to the Sponsorship Group.
6052	Design and Deployment of New Storage Data	$0	$158,000	$158,000	Interviewing the project teams from recent migration efforts at National Grid helped increase the quality of the migration planning for this project. Staffing this project with people who were closely involved with recent migration efforts at National Grid helped increase the quality of the migration plan delivered.	This project created a migration plan to utilize new storage infrastructure that was recently acquired to mitigate risk of outages due to failures on end of life infrastructure. The project is being closed and the remaining project scope has moved into the Cloud & Hosting Agile Release Train.
5789B	Stabilization of Critical Gas Legacy Applications (Portal for ArcGIS)	$0	$240,000	$240,000	Project delays were related to DXC server availability. Confirm and get commitment from DXC for server and resources availability before project kick off. Engaging the right person for the job was not clear which contributed for the delay. Get a detail level escalation process from DXS. Engage DXC management assistance to engage right team at DXC and provide guidance for problem solving.	The project stabilized the Portal for ArcGIS application. This application had frequent unscheduled downtime that caused business inefficiencies. This has been remedied by upgrading the outdated infrastructure, core database/application software, and by increasing the capacity of the Disaster Recovery (DR) environment

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
3683USAO	High Value Asset Discovery and Implementation	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
5380B	Forecasting and Long Term Planning Solution (Metapraxis)	$2,305,757	$82,500	$2,388,257	1. Positive Early Engagement of IT Security and Tech Risk Team resulted into timely completion of Penetration Testing 2. Negative Delay in background check process of vendor consultants has resulted into delay in on-boarding which caused the delay in setting up servers 3. Negative Ensure that request for new hardware is made much in advance (possibly in requirement phase) to avoid the delay of server build	This project integrated the three-year forecasting and long term strategic planning process to improve efficiencies and business ownership of the data using the solution provided by Metapraxis.
4914	EMS Lifecycle Hardware and Software Upgrade	$17,179,000	$3,067,000	$20,246,000	EMS upgrade presented several improvement opportunities noticeably captured; Working through IT commercial process discovered to be lengthy, tying out POs Invoices and charges to project closure presented project closure delays. Vendor Statement of Work (SOW) need concise verbiage detailing work to be performed, detail of work breakdown of work vendor resources to perform, breakdown of cost structure by resource\hours. Reduce iterative vendor translation of SOWs and outline expectations at project start-up.	INVID 4914 Energy Management System - The Project delivered new and updated EMS server hardware, operating system version and application software, new network hardware and security appliances in the New York and New England Regional Control Centers. INVID 4914 Energy Management System hardware and software investment was delivered from supporting INVP's: 4568-EMS Lifecycle Hardware and Software Upgrade and INVP 4570-Tech Services - Network Equipment Lifecycle replacements.
6050	IBM P9 Digital Asset Build	$432,000	$483,000	$915,000	Verification of application and database information for the in scope servers was needed to confirm the accuracy of the CMDB (Configuration Management Database) data. Vendor to ensure that the use of the Live Partition Mobility (LPM) utility is compatible with the in scope server configurations. Staging and OS patching activities can be performed by the vendor ahead of time to expedite migration activities.	This project enabled the recently purchased and deployed IBM Power 9 (P9) servers to perform the business functionality and applications for which they were purchased. Migrated workloads from existing P7 servers to the new P9 servers in the Norwich Data Centers. This allowed us to decommission P7 to reduce technical debt.
3683USAN	DNS Resolution & Protection	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
5867A	OMS Router Replacement & IT Service Management	$468,000	$31,000	$499,000	Project was dependent on resources that were not dedicated to the project or the project's direct control leading to work that was not completed in a timely manner as CNI business teams had high priority work to complete for production and other projects in the New York control rooms. In future projects, we will confirm resource dedications or define the scope as best efforts to complete work based on available resources. During the F&A study portion of the project, we identified that a redundant F&A study had been planned by the business area. As a result, we were able to collaborate and avoid redundant work while ensuring that IT and Business roadmaps are in alignment.	This project replaced four end of service life (EOSL) CNI (Critical National Infrastructure) OMS (Outage Management System) routers for the New England OMS control room circuits at Henry Clay Boulevard Data Center (2) and Northborough Data Center (2). The project also implemented the four routers for the New York OMS control room circuit in a powered and burn-in pre-production status and confirmed that final implementation will be completed by CNI staff in FY22 (INVP 6199 I&O US CNI). (In total, eight routers should be considered as used and useful.) This project also completed a Feasibility and Analysis (F&A) study to see which IT Service Management (ITSM) and Configuration Management Database (CMDB) tool will be able to support CNI applications and to be used by all functional areas in US CNI.
6049	Hosting Infrastructure Refresh to HCI	$2,170,000	$480,000	$2,650,000	Work closer with the National Grid Active Directory team in the future to make sure that the project team properly adopts the latest active directory standards. Reusing the National Grid security teams tools streamlined the onboarding of the penetration test team	The Virtual Computing Environment (VCE) that hosts National Grid (NG) applications reached end of service in March 2018. This created a performance and security risk to National Grid. This project procured new Hyper Converged Infrastructure (HCI), installed and configured for new workload use in order to mitigate the risk above.
5788A	FY21 Gas Regulatory and Operational Enhancements (Minor Works)	$0	$69,000	$69,000	The anticipated demand for minor works in the Gas portfolio did not come through for various reasons. (Org changes, divestiture, etc.) We have learned this lesson and have streamlined the intake process via Gas BP to address this issue for FY 2022 minor works.	The requests approved under this Minor Works project will each require less than $100K (typically, substantially less) and will represent a mix of mandatory, operational, strategic, and tactical initiatives. A Minor Works project request at or below the $30K threshold is prioritized and governed via discussions between the Business and the IT Business Partner team. Minor Works requests exceeding $30K will be required to follow the IT project governance path (i.e., they will require their own investment proposal and associated approvals). Each minor works request will be tracked and periodically reviewed with the business. Any associated issues related to benefits or Run-The-Business (RTB) cost implications will be addressed with those organizations at that time.

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
5942	Legacy AD Remediation	$555,000	$35,000	$590,000	This project has migrated the legacy Active Directory domain controllers to new virtual servers on new Active Directory infrastructure. The operating system on legacy Active Directory domain controllers were upgraded to operate on supported Windows Operating Systems and will be regularly patched to ensure they maintain a strong security posture. The upgraded servers now benefit from the latest and future Microsoft security patches and will remain secure against threats. This project also preformed a series of hardening actives to the current Active Directory estate by enhancing the security settings, removing unused user accounts, updating administrative policies, and completing penetration testing. The project is being closed and the remaining project scope has moved into the Cloud & Hosting Agile Release Train.	Worked with vendors to fully understand technology tools before utilizing proved to be very important during the project. Team met with vendor to understand the Connection Logger tool multiple times which prevented issues from occurring during implementation. Detailed runbook for each change made communications and implementation seamless.
5489A	IBM SNOW Automation	$188,000	$12,000	$200,000	National Grid was under contractual obligations to implement IBM's automation solution made up of robotic processing automation (RPA) and cognitive AI solutions to run system health checks, monitor systems, and report on incidents data in the SNOW ticketing system. By not implementing this solution, National Grid would have run the risk of incurring penalties for not meeting the agreed terms of the contract.	Security and Enterprise Architecture needs to be involved in the project from the onset to understand the requirements of the project and how to plan and budget for those requirements. Having the Service Owner participate as an active member of the project team was a huge benefit. Future projects should request Service Owner engage as an active member of project team throughout the lifecycle of the project
5317	Employee Digital Workplace	$1,528,000	$1,966,000	$3,494,000	The sanctioning process for the group program was complex with differing standards between US and UK taking 6 months to complete. Steps to optimize this have been taken. Defining a North Star (Primary metric) as part of the business case helped with program direction and day to day decision making leading to a positive outcome. Guiding principles defined at the start of the program, such as making data driven decisions, reduced subjectivity and erroneous decisions. Diversity of experience from different functions (content, employee data, product, user experience, change management) across the Working Group led to a more thought-out program and execution Communicating a clear and concise view of impediments allowed everyone to support the resolutions and helped the program succeed.	National Grid's intranet wasn't fit for purpose, it was difficult to find the latest documents, old content wasn't always removed and it wasn't an effective communication channel being limited to laptop users in our offices. This program established the primary portal to access all others as part of a modern digital workplace. It has transformed employee productivity, experience and engagement by connecting people, knowledge and tasks. It is accessible via mobile, tablets and desktops.
3683USA	US Endpoint Security	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
5846	Network Connectivity Refresh	$13,174,000	$0	$13,174,000	1. Collaboration with the facilities teams was critical to scheduling on-site work in order to provide the survey teams with site access. A detailed day-by-day schedule was developed and shared with the facilities teams in advance to get approval to access the site. 2. Access to many sites was restricted due to COVID-19, HSOA's, and operational activities When site access was not available and the team was put on-hold, it was critical to develop a detailed ramp-up plan to quickly engage the team when restricted were lifted. 3. The on-site visits for surveying created a great deal of documentation, including sites surveys, WiFi surveys, structured cabling surveys and photographs. This documentation was required for use by other projects. Service level agreements were established to identify agreed upon timelines for posting documentation so that other teams could plan their work accordingly.	This project delivered Verizon services that designed and proposed to build recommendations for updated National Grid corporate network connectivity and communications aligned with the National Grid network strategy. The recommendation that was delivered proposed recommendations based on audit findings and provided a roadmap for updating the existing Verizon services that delivered and implemented the operation and maintenance of services.

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
5521	LAN/Wi-Fi Infrastructure Program	$3,307,000	$338,000	$3,645,000	1. Power Survey did not account for additional power supply requirements for a new switch. Ensure that future surveys correctly estimate the required number of power supplies (4) for the new equipment 2. Anchor Desks and Executive Offices not included in switch port design; this was a change based on the location of the desks and offices which impacted design. Include requirements for Anchor Desks and Executive Offices in future phases; ensure the business is aware of the impact of any changes mid-stream. 3. Test equipment was not readily available / credentialed properly. Include test equipment and configuration user stories/requirements for future projects as needed.	The LAN (Local-Area Network)/Wi-Fi Infrastructure Program identified a series of projects that refreshed the network infrastructure which improved network performance, mobility, and user experience. The following initiatives are listed under the Program INVP 5521 – LAN/Wi-Fi Infrastructure Program: 5521A: Verizon – Aruba LAN and Wireless LAN (Waltham) 5521B: Verizon – US Wi-Fi Refresh (De-scoped)
5841A	Pension Administrator Replacement	$680,000	$114,232	$794,232	1. On-Boarding new technologies. Introduction of MuleSoft, as new Managed File Transfer platform, impacted the cost. The dependency on new platform could have been avoided by using existing services. 2. Improve stakeholder alignment. In a program of work where the IT team has a supporting role, the engagement with the stakeholders should be well defined. This would avoid impact on the delivery timeline and allow time to submit exceptions (for example, for usage of Live Data) as well as better understanding of the requirements (Single Sign On design).	National Grid's contractual agreement with Pension Administrator, Willis Towers Watson (WTW), was set to expire in December 2021 and, as a result, National Grid made the decision to move to a new vendor, Alight. This project supported the IT components of transitioning to the new vendor.
3614	DRS Program - DR&S 1	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
5785K	Security & Resilience - Maximo	$810,000	$0	$810,000	Follow PMO assistance before sanction to ensure processes were not changed or altered. Request a dedicated resource to manage Coreflex charges against projects.	The current GBE Maximo 7.6.0.8 instance is implemented to support only a single organization, single site. Current implementation does not follow Maximo Best Practices, thus Future National Grid Organizations like LNG, FM will automatically inherit the business logic and configurations that were enabled in GBE design unless refactored.
5785L	Security & Resilience - Mulesoft	$611,000	$0	$611,000	1. Follow PMO assistance before sanction to ensure processes were not changed or altered. 2. Request a dedicated resource to manage Coreflex charges against projects.	The Resilience, Scalability, and Security initiative for the GBE Mulesoft platform aims stabilize the platform and enable scalability for future growth, resulting in an application which will be easier to support and modify. Improvements to the Software and Data Architecture of the platform were focused on aligning to industry best practices, implementing more robust and standardized design patterns, introducing application isolation mechanisms, improving error handling, and improving User Experience.
5486	AVLS Modem Reconfiguration	$1,220,000	$48,000	$1,268,000	Enabled National Grids Electric fleet continued support on Verizon's 4G network upgrading Grids vehicle modems and antenna's as Verizon's 3G network reached end of support. Additionally the annual Code Blue testing was performed on all vehicles having completed 100% success rate. Align with yard Supervisors, their Directors and Executive Sponsors in advance of installs ensuring vehicles are available at the agreed upon location in advance of the installer arriving per plan.	Safety: Real-time vehicle tracking functionality is crucial for the safety of National Grid crews in the field. The project will upgrade modems, mitigating the risk due to discontinuation of Verizon 3G support. Efficiency / Productivity : Increased reliability gained by the availability of newer, faster, and more reliable devices. Also, once in the AVLS system, specialty vehicles such as Backhoes, Diggers, Dump truck, Excavators can be quickly located for deployment or recovered from previous job sites. Speed / Connectivity : Faster devices with better connectivity through the use of faster cellular connectivity cards. Ensure Security : The new modems will be designated as the only devices allowed to connect to the NG network in the vehicles. By utilizing the latest wireless technology, security enhancements that are now built into the protocol, and the incorporation of VPN to create a secure and encrypted tunnel, the data being sent wirelessly by these devices will be protected from eavesdropping or other cyber attacks.

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
5754	Network Equipment End of Life Replacement - LAN	$4,585,000	$216,000	$4,801,000	Collaboration with the Faclities team was critical when scheduling on-site work in order to provide the survey team with site access. 1. As a collaboration with the facilities team was critical when scheduling on-site work in order to provide the survey team with site access. We learned how to and delivered a detailed schedule for on-site work while contacting the facilities team in advance to get approval to access the site. 2. During the Covid-19 Pandemic, HSOA's and operational activities access to many sites were restricted. When site access was not available and the team was put on hold, we discovered the criticality of developing a detailed plan to quickly get the team ramped up when restrictions were lifted. 3. Verizon provided field technicians to complete on-site work that was properly badged and credentialed to enter the property. The project continued to use the resources because it meant that facilities did not need to provide additional resources for access and escorting of the premises. This sped up delivery and coordination between the teams and facility staff members.	1. Mitigated the risk of critical systems failure from existing unsupported switches. 2. Provided a centralized expandable environment for Wireless Access Point interfaces to be implemented. 3. Enabled the decommissioning of obsolete switches. 4. Provided functional benefits which will enable improvements in the efficiency of data and file transfer 5. Provided an audit ready technology refresh for non compliant switches. The new switches will have the upgraded software patches and will be in compliance for a cyber security audit.
5903	Customer Operational Enhancements FY21	$0	$178,000	$178,000	N/A	The requests worked under this project are expected to contribute to improved system reliability and business functionality, fulfill the organization's operating requirements, and comply with regulatory mandates.
5487E	ICE Public Folders - Migration/Decom	$0	$359,000	$359,000	- Much time was spent creating documentation for data that ended up not being used or created no value. For similar migrations there should be a focus to deliver slimmed down architectural artifacts. - When moving to Office 365 all data was migrated with the exception of Public Folders which was de-scoped leading to continuing IBM costs. For future exchange migrations, scope should be reviewed by multiple parties to ensure no gaps.	- ICE Exchange 2010 platform is out of support - Removed the RTB costs of maintaining the Public Folders on the legacy ICE Exchange 2010 platform - Enabled standardization of collaboration tools on O365
5785C	Stabilization of Critical Gas Legacy Applications (ePermits)	$369,000	$135,000	$504,000	The Project team needs to have a solid understanding about the network configuration changes required, prior to go-live. All tickets for Firewall Change and Reverse Proxy changes need to be submitted early enough to resolve any inconsistencies between Firewall and Reverse Proxy changes, and engage technicians from Verizon SOC knowledgeable of these changes to ensure a smooth deployment process. Moreover, the Azure support team was scheduled for availability during deployment, in the event that their help was required with Azure VM connectivity.	1. Key business benefits resulted from this project are: - Not having to resort to a paper-based permitting process (and consequently higher cost of a manual process) in case of unavailability of an e-Permitting interface in Maximo Gas Business Enablement (GBE) - Avoided cost of turning back to paper process due to device upgrades in Municipalities 2. Key IT benefits resulting from this project are: - Reduced cost of IT work resulting from minor incidents - Avoided cost of IT security risk on end-of-life servers
5785J	Security & Resilience - SalesForce	$0	$0	$0	1. Follow PMO assistance before sanction to ensure processes were not changed or altered. 2. Request a dedicated resource to manage Coreflex charges against projects and feasibility of reclassification before project kick off.	The Resilience, Scalability, and Security initiative for the GBE Salesforce platform aims stabilize the Salesforce platform and enable scalability for future growth, resulting in an application which will be easier to support and modify. Improvements to the Software and Data Architecture of the platform focused on aligning to industry best practices, implemented more robust and standardized design patterns, introduced application isolation mechanisms, improved error handling, and improved User Experience.
6042	Customer US Retail Web Infrastructure Upgrade	$0	$358,000	$358,000	1.No prior experience of migrating this system to the Azure environment existed. - Share experience with other team members on evaluating complexity as it relates to multiple complex system interfaces 2. Key individuals were not available and committed to the project. Ravi Kandasamy is spread too thin and he needed to take personal leave which severely impacted this project. - 3. Work should be done more in a product mode where resources are dedicated to the project. 4. Verizon had several missteps throughout the project implementation, requiring rework, extended timeline and cost overruns. - Greater knowledge and familiarity with the NG overall network and applications is required. Again, Kevin Enny has been the only trusted resource. 5. Include COH in the estimates.	The server upgrade provided Microsoft supported environment for National Grid customer web site. The upgrade supported compliance with security policies for accessing this environment.
3683USD	Develop Robust Incident Response	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
3683USX	Application Security As a Service	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
5360	Lease Accounting Updates and Contract Management	$8,671,000	$758,000	$9,429,000	1. Improvement - Technical Delivery: To address scope change the team did a Proof Of Concept. For the Fleet Journal Entry Settlement End-To-End process prior to finalizing the new design. In addition, extended test cycles should be planned to include thorough testing of the new Fleet Journal Entry Settlement process. 2. Improvement - Hosting: The practice of utilizing a program approach for SAP hosting charges and related projects has proven to be unnecessarily complex and ineffective. A new process has since been developed for projects starting in FY21 that allows more transparent management of hosting charges and the associated SAP projects. 3. Improvement - Governance: The Cognitive Contract Management (CCM) project was bundled with PowerPlan Lease and sanctioned together as a single Investment. The projects ran on 2 very different timelines and it often became difficult to manage CCM individually. PowerPlan Lease had higher visibility and CCM issues/risks were often less of a priority. A complex project like CCM should be sanctioned and managed as its own project to ensure it receives the appropriate focus that is required. 4. Improvement - Technical Delivery: IT security review of the application occurred too late in the process. Security challenged several design aspects that had to be adjusted late in the project timeline. IT Security engagement in the future should occur prior to product selection to allow for a review of conceptual and logical architecture prior to the start of implementation. At the time Security requested additional requirements, the project timeline was impacted. Early and ongoing conversations regarding technical security / non-functional requirements is needed to reach a common understanding of these types of requirements to proactively mitigate risk and delays. 5. Improvement-Technical Delivery: Another reason for the delay of this project was the shift to work-from-home related to the COVID-19 pandemic. The testing team on this project was required to focus all of its time on ensuring a success virtual year end close.	The main benefits of this project were: - Compliance with mandatory accounting standards (IFRS16 and ASC842). - Ensured that leases are correctly reflected on the Company's financial statements. - Ensured an effective control environment around the gathering and tracking of lease agreements and accounts. - Captured and maintaining of complete and accurate lease contract information. - Established an automated solution to review new agreements which should increase efficiencies and accuracy in data gathering.
5378	FSSC&HR Systems (Non-SAP) Operational Requirements & Upgrades FY20	$0	$198,000	$198,000	N/A	The requests worked under this project are expected to contribute to improved system reliability and business functionality, fulfill the organization's operating requirements, and comply with regulatory mandates.
5392	US SAP: Infrastructure Landscape FY20	$830,000	$1,000	$831,000	The practice of utilizing a program approach for SAP hosting charges and related projects have proven to be unnecessarily complex and ineffective. A new process has been developed for FY21 that allows a more transparent management of hosting charges impacting or have a relationship to SAP projects. This closure represents the last investment completed under program approach.	This project is intended to support mandated projects by: - Reduced the lead time to start projects and initiatives within the portfolio - Increased accuracy of cost estimates - Alleviated project startup bottlenecks - Increased reliability for SAP related project delivery - Reduced one-time capex startup costs associated with standing up new environments for each project
5523B	Upstate NY PBX Replacement/Upgrade (Upstate NY Control Room Sites)	$1,058,000	$0	$1,058,000	1. The IPT (Internet Protocol Telephony) workbook was too flat for programming control room phones effectively as was difficult for ongoing operational support to the business. - The recommended actions for improvement are; Add a tab for: published lines, inventory of phones with MAC addresses, and telephone layouts for buttons with associated lines for each default phone. 2. The AS-IS documentation needs to be maintained by National Grid and not by third-party vendors. - There was no documentation for the control room phones within National Grid, the recommendation is to generate new documentation with published telephone lines listed for each control room. 3. Using the same model for control room phones allows for non-critical phones to be substituted if a problem arises on a critical phone. - This project discovered having the commonality of models allows the internal phone configurations to be the same, which increases the quality, ensures consistent delivery and support for telephone services that reduce operational costs.	1. Supports more efficient use of National Grid facilities 2. More effective collaboration 3. Increased end-user experience 4. Ease of the Maintenance 5. This program of work is aligned to the strategies and other efforts within National Grid. 6. IPT Systems will afford the greater potential to support the integration of voice data services and applications
5867E	Substation Access Points	$734,000	$10,000	$744,000	1. Projects need to be started earlier for proper resourcing. 2. Request Electric PTO Ops (Protection Telecommunication Operations) dedicated resources ahead of time. 3. Received estimates from another provider of comparable equipment.	1. Reduced technical debt within the NG CNI estate. 2. Ensured regulatory compliance. 3. Enhanced system security by swapping legacy devices out for new kit. 4. Increased operational availability & vendor support.
5795E	MA Electric Vehicle Rebate Program	$614,000	$54,000	$668,000	Lesson 1: Corporate Standards can change at any time and impact scope and design - example: Mulesoft for file transfers which vendor could not accommodate. Action 1: Be flexible and creative to look for new approaches, and get agreement from all sponsors. Lesson 2: Ensure transparency and clear communications with vendor. Action 2: Set communication expectations and escalate if not being met, before project schedule and quality are impacted. Lesson 3: Consider IT aspects of vendor before signing to evaluate level of quality they will provide to National Grid. Action 3: This vendor was very small and could not meet standards which drove more manual effort for business.	This project complies with DPU 18-150 and will enable customers to have easy access to view rebate credits via their National Grid bill rather than receiving them via Gift Card or PayPal.
5161	Device Refresh	$5,011,000	$0	$5,011,000	1. Device refresh must occur in alignment with the three-year refresh policy to ensure device failures are not greater than expected capacity. 2. A defined risk margin of additional hardware should be procured to account for unplanned device breakage or failure.	1. A trusted environment created by managed software/devices and associated security rules 2. Like for like replacement of end of life end user devices, inline with three-year refresh policy 3. Increased user morale created by an improved user experience

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
5795G	Telephone Consumer Protection Act (TCPA)	$809,000	$87,000	$896,000	The project team needs to do a deep dive or impact analysis with interfacing applications during the R&D phase. This would lead to better planning with regard to cost estimates and project schedule.	Completing this project assisted National Grid in complying with the TCPA. It improved customer experience and increased customer satisfaction because it assisted National Grid in identifying customers who have indicated that they do not wish to receive outbound collection calls placed by automated or pre-recorded voice messages.
5867	Network Security Infrastructure Program	$1,176,000	$68,000	$1,244,000	5867A - OMS Router Replacement EOSL Refresh / ITSM F&A 1. The project was dependent on resources that were not dedicated to the project or the project's direct control leading to work that was not completed in a timely manner as CNI business teams had high priority work to complete for production and other projects in the New York control rooms. 2. In future projects, we will confirm resource dedications or define the scope as best efforts to complete work based on available resources 3. During the F&A study portion of the project, we identified that a redundant F&A study had been planned by the business area. As a result, we were able to collaborate and avoid redundant work while ensuring that IT and Business roadmaps are in alignment. 5867E - Substation Access Points EOSL Refresh 1. Projects need to be started earlier for proper resourcing. 2. Request Electric PTO Ops (Protection Telecommunication Operations) dedicated resources ahead of time. 3. Received estimates from another provider of comparable equipment.	1. Reduction of assets that are end of service life within the National Grid CNI estate 2. Ensured regulatory compliance 3. Enhanced system security by swapping legacy devices out for new kit 4. Increased operational availability & vendor support
5973	FY21 Wholesale Networks and US Capital Delivery Minor Works	$0	$61,000	$61,000	N/A	The requests worked under this project contributed to improved system reliability and business functionality, fulfill the organization's operating requirements, and comply with regulatory mandates.
5489B	WiPro SNOW Automation	$0	$1,000	$1,000	N/A	1. Identification and End to End automation of incident 2. Provided predictive ticketing and identifies and remediates issues even before they are reported by user 3. Reduced effort by automating common tasks and allowing support staff to focus on critical problems. 4. Integrated with core Ticketing Tools like Remedy and ServiceNow 5. Enhanced end user experience 6. Executed Health checks on applications to identify problems
5489	Emerging Technology	$42,000	$0	$42,000	Security and Enterprise Architecture need to be involved in the project from the onset to understand the requirements of the project and how to plan and budget for those requirements. Have Security and Enterprise Architecture involved early on in the process and throughout planning Having the Service Owner participate as an active member of the project team was a huge benefit Request Service Owner engage as an active member of project team throughout the lifecycle of the project	1. Reduced operating costs as repeatable tasks can be moved to AI or Bots 2. The ability to fix devices in the field that may be breaking before the employee or customer is aware a problem exists 3. Limited 24/7 support services via chat services 4. Technology allowed IT to engage employees ahead of problems to repair issues 5. Ability to identify threat vectors ahead of notification from vendors 6. This program of work has aligned the strategies and other efforts within National Grid
5506	Integration Services	$3,697,000	$98,000	$3,795,000	N/A	1. Introduced modern integration capabilities utilizing current technology 2. Implementated additional capacity in the middleware suite 3. Improved resiliency of the integration platform 4. Risk mitigation of the current integration environment 5. A flexible environment that has grown as business needs increase 6. Established cloud ready integration platform 7. Simplified adoption by IT and Business organizations
5901B	DevOps Tools	$1,634,000	$117,000	$1,751,000	1. IT Commercial was able to negotiate better pricing on software licenses. 2. Building out the Azure Kubernetes cluster was complex and required additional time and collaboration of Security, I&O, and Solution Engineering teams. 3. More time must be planned for security vulnerability scans and CIS Benchmarking.	1. Automated Regression Efficiency – Reduced IT Service Costs 2. Performance Testing Efficiency – Reduced IT Service Costs 3. Improved Test Coverage – Cost Avoidance 4. Automated Deployment Efficiency – Reduced IT Service Costs 5. Build Automation Efficiency – Reduced IT Service Costs
5901C	MS Project Online Enhancements	$307,000	$42,000	$349,000	1. Avoid Go-Live between 15-Mar and 31-Mar as crucial time for Finance team. 2. Engage IT Commercial early in the software procurement process. 3. Follow-up on resource time entry on a weekly/monthly basis to reduce FY year-end adjustments.	The completed work to align these processes benefited the strategic implementation going forward and help to shape more informative detailed requirements. • Improved visibility into project performance. Allowed National Grid to make timely decisions and address issues as they arise, helping to prevent project delays and potentially reducing budget overruns • Increased project management performance. By streamlining reporting and reducing the amount of administrative workload on project managers, allowing them to be more productive
5507	US Legal: Document Management	$348,000	$184,000	$532,000	Separate Feasibility & Analysis (F&A) / Options Analysis should be performed prior to initiating implementation project. Include global roadmap as input in F&A / Options Analysis. From mid-March to mid-April, Commercial & Procurement Teams are busy with Fiscal Year End activities and there could be significant delays in their responses.	Move the application onto a supported version of Open Text that is compatible with Office 365 and supports direct integration with Office 365 products. Improved system performance. Additional functionality included document redaction capability. Document level encryption will be implemented as per Security recommendations.

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
5570	Electric Field Crew iPad Time Entry and Basic Capability -NE	$942,000	$0	$942,000	1. Project team learned how to successfully re-design the training effort to deliver remotely rather than in-person due to Covid restrictions. 2. The remote training method provided the benefit of flexibility for Operations, removing one or two field personnel rather than a classroom full, thus lessening the impact of the training. It also was flexible in that last minute changes could be made with individuals either reporting out sick or needed on an emergency job. 3. Since the remote training was conducted one on one, it was tailored to the individual, creating a better overall experience. 4. With remote training, the philosophy was changed to offer training slots for an entire region rather than one physical facility, increasing the efficiency in which the training could be completed. 5. We learned that the team had to set some days aside for building and distributing of the iPads since the remote trainings were consuming the entire day rather than just the mornings as in the past classroom setting.	This project drove the replacement of a manual paper process for the capture and entry of field crew time leading to increased accuracy, reduce delays and improve efficiencies for field crew time capture. A suite of tools was introduced which allowed the field to remotely access the latest standards and operating manuals reducing the need to print paper and carry paper copies in the trucks. Crews have remote access to Maps, facility records and safety information improving improved awareness and safety in the field Direct cost savings (Annual): - Cost savings will be from paper reduction in MA and RI - Indirect financial benefits: - Safety – reduces injury and risk to employees and customers - Technology platform – basis for future digital initiatives - Improved data quality – reduce time entry data errors
5896A	US Tax: SABRIX Upgrade	$299,000	$80,000	$379,000	1. Coordinate with other projects in advance so all are aligned at the overall program level. 2. The technology version compatibilities should be confirmed early in the project.	Tax Determination relies on monthly software updates from Thomson Reuters and must be kept in support to ensure tax compliance and avoid significant financial penalties due to incorrect tax calculations and filings. Operating System and Database upgrades allowed us to maintain consistent patching and to comply with IT Security vulnerability management standards.
5838	Managed Networks	$20,612,000	$77,000	$20,689,000	Agile Delivers Value to the customer faster 1. Embraced the one team model had lead to successful staff augmentation model, full transparency and constant communication, and full team accountability on delivering work 2. Able to onboard, train, and launch a full operational team remotely discovering that colocation is no required 3. Discovered success in agile delivery in infrastructure where teams are self-organized driving to success empowering the team in owning their work/outcomes and striving for relentless improvement 4. Role Clarity where team member should understand their role and how to be successful prior to the first jumpstart 5. Metrics and agreements on defining success measures	Summary of Benefits: The work completed during this project will enable future requirements for Grid Modernization as well as Corporate network modernization. Future capabilities include delivering a platform of solutions, services, devices and applications that enable us to operate our distribution system more reliably, effectively and efficiently to the benefit of our customers. Improvement of the end user experience by utilizing subject matter experts to reduce technical debt. Support of more device types in the environment which will increase effective collaboration and networking capabilities. This work was the driving force to ensure we are meeting our IT Focus Areas and other efforts within IT National Grid. Supported more efficient use of National Grid facilities by improving resilience of communications across sites. This resulted in Run the Business (RTB) Operations and Maintenance (O&M) savings of ~ $3,000,000.
5380A	Blackline: US Implementation	$1,473,000	$831,000	$2,304,000	1. Positive: High IT involvement from the project onset with the business team was a critical success factor. Ensure that NG US IT is engaged with business from the onset of the project with dedicated leadership for each function. 2. Positive: Availability of key resources with knowledge of the National Grid Solution Delivery Framework (SDF) allowed for deliverables to be planned and completed successfully. Ensure that IT resources available during the planning phase have working knowledge of the National Grid SDF framework. 3. Negative: Document review sessions for SDF milestone deliverables were not immediately followed up with clear requests for sign off. Follow up document review sessions with sign off request emails to quickly close off documentation. 4. Positive: Stakeholder engagement through Yammer posts and change champion meetings established good relationships with end users and built momentum / excitement for the project. Engage end users throughout the project to aid in solution adoption post deployment. 5. Positive: Strong engagement from the Steering Committee allowed for effective communication and approval of desired process and technology changes. Ensure frequent engagement with the Steering Committee in all phases of the project. 6. Positive: All relevant parties (National Grid and vendor) were consulted during requirements gathering, ensuring that the requirements captured covered IT, Operational Finance and the Controllership. Involve several end users from different departments when gathering requirements. 7. Negative: Reporting requirements were not completely developed during the design phase. Build in more design sessions specifically targeted at reporting requirements. 8. Positive: The extensive documentation of configuration requirements provided the guidance needed for environment build. Document and sign off on all build decisions. 9. Negative: Complexity in National Grid general ledger account structure and volume of data increased solution complexity. Address data challenges at the root cause or source vs. creating solutions to manage data challenges in end state application. 10. Positive: Good mix of self taught and instructor-led sessions. Provide access to self taught courses prior to instructor led trainings.	- Enhanced automation within key close processes, reducing manual intervention and associated risk of human error - Standardization of activities to unlock further automation opportunities - Opportunity to automate and strengthen key SOX controls - Opportunity to move from transaction processing to analysis activities earlier in the close process - Increased integration with the SAP platform - Enhanced visibility and transparency into the close process - Offers a path to longer term efficiencies

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
6097	Cloud Video Interoperability	$1,171,000	$1,000	$1,172,000	Cloud Video Interop (CVI) was implemented on circa 170 devices globally. Given the impact comprehensive testing would have on time, resource effort and cost it was decided to conduct a full test over a weekend across a selected number of video conference rooms (15 in total) and all other rooms were tested against a minimum viable product set of test cases. - Consider a similar approach for future work where there is a requirement for repetitive testing. COVID site access restrictions and geographical distances between sites required the support of team members outside of the Project Team. The Project Team were thankful for support from the US Exec Support Teams who were often on site or closer to site and were not subject to the tighter site access restrictions implemented due to COVID. - When faced with on site testing at a variety of locations consider the option of utilizing local resource. The CVI extended feature One Button to Join (OBTJ) was not included in the original project scope. - Consider the benefits of including the complete service in the original scope versus the option of delivering the MVP solution and then retrofitting additional features.	- Enabled Video Conference Devices to facilitate Microsoft Teams meetings - Reduced the number of WebEx licenses required to facilitate meetings using video conference devices
6018	Security Operations Services (SOS)	$1,332,000	$71,000	$1,403,000	1. Lesson: The Security Baseline Requirements deliverable ran well over the scheduled milestone timeline although the BP&I team was engaged early on. Action: When introducing new technology, we need to ensure that the team is engaged early on. 2. Lesson: The Microsoft Defender for Identity scope was increased dramatically after the initial project Sanctioning. Action: All integrating business units need to participate in the scoping process and project scheduling.	- Enables National Grid to gain visibility and control over security resources, workflows, and policies. - Increased security posture within National Grid data centers - Ability to integrate Azure security alerts into one Security Information Event Management (SIEM) solution - These services will assist in protecting against advanced threats
5759	UPS Infrastructure	$268,000	$0	$268,000	Extensive pre-installation site readiness survey. Identify any additional electrical requirements needed and evaluate and identify any additional space requirements (such as cabinets and unit rack space needed). Post implementation photos to confirm site is left in acceptable operational conditions to eliminate additional site visits to resolve any issues such as open cabinets.	- Ensures business operations continuity at sites (during normal and extreme weather conditions) - Protects equipment from electrical spikes and dips, and Protects from shutdown - Increases reliability - Management / Monitoring of the operating system - Management/Monitoring of installed applications - Backups of the virtual machine, OS or Application
5523G	Downstate NY PBX Replacement Upgrade	$381,000	$80,000	$461,000	1. The AS-IS documentation needs to be maintained by National Grid and not by third-party vendors. There was no documentation for the control room phones within National Grid. It is recommended that new documentation should have all published listed for each control room. 2. Existing control rooms with Cisco phones must be treated like new control rooms which led to performing development and full user acceptance testing at locations. 3. Resiliency plans for control rooms using a call from home can have issues if jabber fails and the cell phone takes too long to ring in some areas. The recommended action is; for smaller control rooms we should implement Normal Call Flow and emergency call flow, Emergency Call Flow rings all jabber phones, desk phones, and cell phones immediately, and larger control rooms may need to incorporate cell phones into a normal call flow.	The key business benefits are to have: - The new voice recording system will comply with Section 215 of the Federal Power Act (FPA), which requires control room calls to be retained for three years. - The legacy voice recording system is at a critical risk of unrecoverable failure. The system is no longer vendor-supported and has had failures that have resulted in the loss of recording capability. - The recording system is on Windows 2000 operating system, which is no longer vendor supported. - Modern, supported, and reliable telephony infrastructure to record incoming and outgoing calls into Gas Dispatch. - Ease of the Maintenance. - The ability to playback recorded calls reliably as and when needed, in order to meet business, regulatory, and audit needs. - A better Quality Control process to streamline how we receive calls, retrieve information, and communicate that information to internal and external customers. - Enhanced end-user experience. - This project is aligned to the strategies of National Grid.
5523A	Northborough and Avaya PBX Replacement/Upgrade (NE Control Room sites)	$0	$0	$0	1. The AS-IS documentation needs to be maintained by National Grid and not by third-party vendors. There was no documentation for the control room phones within National Grid. It is recommended that new documentation should have all published listed for each control room. 2. Existing control rooms with Cisco phones must be treated like new control rooms which led to performing development and full user acceptance testing at locations. 3. Resiliency plans for control rooms using a call from home can have issues if jabber fails and the cell phone takes too long to ring in some areas. The recommended action is; for smaller control rooms we should implement Normal Call Flow and emergency call flow, Emergency Call Flow rings all jabber phones, desk phones, and cell phones immediately, and larger control rooms may need to incorporate cell phones into a normal call flow.	- Supports more efficient use of National Grid facilities - More effective collaboration - Increased end-user experience - Ease of the Maintenance reduced cost - This program of work is aligned to the strategies and other efforts within National Grid. - IPT Systems will afford the greater potential to support the integration of voice data services and applications

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned	Benefits
6393	Technology Validation for CRIS to CSS Migration	$0	$373,000	$373,000	We spent some time getting access to legacy documentation - rather than get access to legacy servers. We should have moved the necessary files to a current server. There were some complexities with access and timeliness.	The result of this F&A would be a an understanding whether this approach is feasible from a technology standpoint. In addition, this approach will help assure that all of the "big rocks" have a move forward option.
5842A	US SAP: GRC Upgrade	$1,921,000	$234,000	$2,155,000	1. Runtime errors were received while executing the jobs which was caused by Deadlock due to multiple processes are trying to update the same entry in table. Need to ensure that Parallel jobs are not running trying to update same table entries. 2. File Folder access permissions were not set up prior to cut over. Syntax will update folder permission if needed. 3. Monitor Project activities or higher and lower level activity times against the project planned activities. Adjust activities as need for project resources.	This project will to contribute to improved GRC environments reliability and business functionality, fulfill the organization's operating requirements, and support compliance with regulatory mandates. In addition, this upgrade project will remediate audit recommendation (SAP APD.11). In addition, this project will benefit National Grid in various ways: - Avoid higher out of support R&M costs through the upgrade to a supported software version - Fewer hours spent resolving issues by avoiding spending time researching old software versions - Reduction in time spent connecting to new platforms through broader support for integration with newer versions of SAP systems - Improved synchronization jobs take less time/resources to complete and cause fewer errors requiring support resolution - Mass maintenance features reduce time spent updating firefighter controller and owner master data
4990	Primavera Upgrade/Stabilization	$7,504,000	$684,000	$8,188,000	1. The need for a Mulesoft integration platform development resource was not planned for well enough in advance. Recommendation: As soon as new technology is identified, assess whether National Grid has resources that can be allocated to the project. 2. The project experienced significant technical resource turnover during the Requirements and Design phase. This had a negative impact as it took time to bring new resources up to speed. Recommendation: Retain a strong core IT team throughout the project. 3. The security requirements changed midway into the project. This created a significant amount of -re-work in the existing production system. Recommendation: As National Grid gets more and more security conscious, these requirements will get tighter and, therefore, prior to security rules/requirements being updated for National Grid systems, the full impact of such changes to the in-flight projects needs to be evaluated. 4. The Solution that was incorporated significant SAP changes was much more complex than originally anticipated and design documentation was not available. Recommendation: For every new integration implemented – the detailed design document should be created for future dereference.	This investment facilitated US gas and electric business units' ability to achieve enabling of the 20% improvement in National Gird Capex performance by providing the following benefits: - The upgraded Primavera P6 version will provide a significant expansion in functionality for complex capital project planning, scheduling, and forecasting activities including risk management and new web-enabled / mobile capabilities, and therefore, it will increase visibility on and enable better decision making of complex capital project risks, associated costs, and forecast accuracy, so projects can be managed more effectively. - The upgraded Primavera P6 version capabilities will enable the business to answer regulatory entities queries in a timely manner, with more accurate complex capital project cost and schedule data, and therefore: - - Reduce negative impacts to customers - - Avoid penalties for under performance - - Increase credibility with regulators - - Improve regulatory filing outcomes based on an improved, on time, on budget delivery of complex capital delivery projects.
	Total	**$222,910,045**	**$36,783,152**	**$259,693,197**		

Niagara Mohawk Power Corporation d/b/a National Grid

The Brooklyn Union Gas Company d/b/a National Grid NY

KeySpan Gas East Corporation d/b/a National Grid

Cases 20-E-0380, 20-G-0381, 19-G-0309 and 19-G-0310

INFORMATION TECHNOLOGY CAPITAL INVESTMENT REPORT FOR QUARTER ENDED June 30, 2022

SUBMITTED TO:
NEW YORK STATE DEPARTMENT OF PUBLIC SERVICE STAFF

August 29, 2022

nationalgrid

Information Technology Capital Investment Report for Quarter Ended June 30, 2022

I. Introduction

Niagara Mohawk Power Corporation (Niagara Mohawk), The Brooklyn Union Gas Company (KEDNY), and KeySpan Gas East Corporation (KEDLI) (together "National Grid" or the "Companies") submit this Information Technology ("IT") Capital Investment Quarterly Report ("IT Quarterly Report") for the first quarter of fiscal year 2023 ("FY23").[1] Pursuant to the Public Service Commission's ("Commission") January 20, 2022 Order in Cases 20-E-0380 and 20-G-0381, and August 12, 2021 Order in Cases 19-G-0309 and 19-G-0310, the Company is to provide quarterly variance reports to Department of Public Service Staff ("Staff") regarding IT capital projects and investments.[2] The IT Quarterly Report provides an explanation of the Company's overall IT investment plan approach; descriptions of program investment categories; and a comparison of budgeted to actual spending.

IT capital projects and investments that are shared investments across National Grid's operating companies are implemented and owned by National Grid USA Service Company, Inc. ("Service Company") and allocated to the benefiting US operating companies, including Niagara Mohawk, KEDNY and KEDLI, in the form of Service Company rent expense once the investments are in-service. The Service Company rent expense includes a return on the capital investment (net of deferred taxes) plus booked depreciation expense.

Through Q1 FY23, National Grid spent $59.3 million compared to a year-to-date Service Company budget of $65.0 million, representing a $5.7 million underspend. Attachment 1 includes detail on budgets and actual spending for each of the portfolios and commentary regarding performance which has been presented to each of the Chief Information Officers responsible for the respective portfolios.

Attachment 2 includes a detailed breakdown of individual project spend by portfolio compared to the budgeted spend for each of those investments.

II. Program Categories/Spending Rationales

National Grid has re-organized the US Information Technology organization into six areas:

[1] The first quarter of FY23 is April 1, 2022 through June 30, 2022 ("Q1 FY23").

[2] *See* Cases 20-E-0380 and 20-G-0381, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service*, "Order Adopting Terms of Joint Proposal, Establishing Rate Plans and Reporting Requirements" (issued and effective January 20, 2022), Joint Proposal, Section IV.10.5; and Cases 19-G-0309 and 19-G-0310, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Energy East Corporation d/b/a National Grid for Gas Service*, "Order Approving Joint Proposal, as Modified, Imposing Additional Requirements" (issued and effective August 12, 2021), Joint Proposal, Section IV.6.5.

(1) Group Functions
(2) United States Electric
(3) United States Gas
(4) Customer
(5) Digital Platforms and Infrastructure Technology (Formerly Infrastructure and Operations)
(6) Security

IT capital investment is tracked based on these area designations. Descriptions of each portfolio category/spending rationale are provided below.

Group Functions

Group Functions executes National Grid's digital roadmaps, leveraging technology and reimagining processes in support of Finance, Enterprise Business Services, Human Resources, Legal, and Safety Health & Environmental, continuing to upgrade SOX related systems, and retiring outdated systems.

United States Electric

Provides IT Support to our US Electric Distribution and Transmission businesses. Modernizing our key legacy operational systems while building capabilities that will enable the energy transition and transformation to a Distribution System Operator (DSO). Through digital delivery, the Electric CIO organization provides greater efficiency and data driven insights into our workforce operations that provide:

- More automated and reliable grid controls systems
- Capital programs and projects that support the proliferation of Distributed Energy Resources (DERs)
- A growing market of prosumer electric customers

United States Gas

Provides IT Support to our US Gas Distribution and Transmission businesses by developing digital products based on the foundation of gas business enablement that deliver continuous improvement of our asset lifecycle management capabilities through:

- Risk-based asset investment planning
- Performance-based maintenance
- AI-based work optimization
- Mobile field products

Customer

Provides IT Support to our US Customer business, delivering best-in-class digital products that increase engagement and reduce costs to serve by:

- Modernizing our legacy "back-end" IT estate to support more digital products
- Liberating data and creating an abstracted and real-time customer data platform to assist with achieving energy efficiency targets

Digital Platforms and Infrastructure Technology (DPIT) (Formerly Infrastructure and Operations)
Supports our internal IT Architecture covering Cloud and Hosting, Critical Network Infrastructure, Networks, End User Experience and Operations. DPIT is focused on reducing National Grid's technical debt through cost effective, modernized IT infrastructure, enabling cloud capabilities that meet our risk profile. The organization enables a connected, seamless enterprise that is flexible, secure, reliable and scalable while maintaining acceptable levels of technical debt and risk.

Security
The Security function protects our people, premises, and digital systems with the objective of maintaining trust in National Grid services, focusing on:

- Reduction in likelihood and impact of security incidents
- Compliance with relevant laws, standards and regulations (*e.g.*, NERC, PCI DSS, GDPR, NIS-R)
- Improved user experience and standard of security practices across National Grid

The Security function aims to reduce the likelihood and impact of physical and cyber security incidents affecting National Grid. It does this by championing security best practices, technology, and promoting a security-awareness culture.

Other Category

For accounting purposes there is an additional category listed in Attachment 2 to this report. The "Other" category has 3 elements:

- The "IT4IT" sub-category acts as a funding bucket for NPV-positive projects to enable efficiency savings in our IT-run business operations.
- The "Other" sub-category is the allocation of operating expense overheads such as Enterprise Architecture and Solution Delivery charges to a centralized account. The capital charges for this overhead are made directly to individual projects.
- Finally, "Burdens" are the net removal of operating expense burdens to ensure our reporting agrees with our Corporate Controllable Cost reporting. Operating expense burdens are consolidated and reported separately under non-controllable costs in our corporate reporting packs.

The summaries below explain the subjects presented in the attachments to the Q1 FY23 report.

A. Attachment 1

1. Summary (pages 2 – 3) – Overview of US IT as a whole.
    a. High-level executive summary, including work and challenges faced across the portfolios and a brief explanation around the forecast (p. 2).
    b. Actual expenditure versus budget by portfolio year to date, and full fiscal year forecast versus annual budget by portfolio (p. 2).
    c. Actual and forecasted spend by month for FY23 compared to budget, prior year and run rate. Each bar is color-coded to give an overview of the comparative spend per portfolio (p. 3).

2. Portfolio Sections – The remainder of Attachment 1 includes six sections, one for each of the portfolios: Group Functions; US Gas; US Electric; Customer; DPIT; and Security. Each portfolio consists of 3 pages:
    a. The first page provides a narrative financial and business overview. The three charts on the first page show: (1) operating expenses; (2) capital expenses; and, (3) total expenses. Each chart shows FY23 actuals and forecast by period versus budget, with trend lines showing prior year expenditure and run rate. This page also shows the full year forecast and year to date total and capital expenditure compared to the budget.
    b. The second page of the portfolio sections has two charts on the top left of the page: the top chart is actual expenditure versus budget by program year to date. The second chart compares full fiscal year forecast versus annual budget by program. There is also a key challenges section and the escalation point for that challenge. On the bottom half of this page is a chart showing actual and forecasted spend for FY23 compared to budget, prior year and run rate for the portfolio. Each bar is color-coded to give an overview of the comparative spend per program.
    c. The third page of each portfolio section provides a status update on the various projects. A priority investments list for the portfolio and their status appears at the top right hand of the page. Blue means complete, green means on track, amber means some risk, and red means focused intervention required.

    Attachment 1 is based on National Grid's monthly Strategic Portfolio Reviews (SPR) and reporting cadence to senior management of each portfolio, and ultimately to the IT leadership team.

B. Attachment 2

INVP Level Variances - This attachment provides a detailed listing of individual projects in each of the six portfolios explaining spend in FY23.

(1) Total actual spend compared to budget and variance.
(2) A variance reason for all capital investments which are outside the 10% tolerance to budget.

National Grid

Quarterly Report – Q1 FY23
Information Technology

28th August 2022

nationalgrid

US IT

Financial Performance

Portfolio View - Year to Date $M

Portfolio	Capex			Opex			Totex		
	Actuals YTD	Budget YTD	Variance	Actuals YTD	Budget YTD	Variance	Actuals YTD	Budget YTD	Variance
US Electric	8.2	13.6	5.4	1.0	1.3	0.4	9.2	14.9	5.8
US Gas	4.0	3.5	(0.5)	0.4	0.9	0.5	4.4	4.4	(0.0)
Customer	10.2	10.3	0.1	3.6	1.5	(2.1)	13.8	11.8	(2.0)
Group Functions	5.7	7.9	2.2	1.3	1.9	0.7	7.0	9.9	2.9
DPIT	14.3	23.7	9.4	2.2	2.9	0.7	16.5	26.6	10.1
Security	6.2	5.3	(0.9)	1.9	1.3	(0.6)	8.1	6.6	(1.5)
Other	(0.0)	(6.8)	(6.8)	0.4	(2.3)	(2.7)	0.4	(9.1)	(9.5)
Total	**48.6**	**57.5**	**8.9**	**10.7**	**7.5**	**(3.2)**	**59.3**	**65.0**	**5.7**

Portfolio View - Full Year $M

Portfolio	Capex			Opex			Totex		
	FY Forecast	FY Budget	Variance	FY Forecast	FY Budget	Variance	FY Forecast	FY Budget	Variance
US Electric	60.6	59.5	(1.1)	8.8	8.8	0.0	69.4	68.3	(1.1)
US Gas	28.0	28.0	0.0	4.3	4.5	0.2	32.3	32.5	0.2
Customer	39.5	33.5	(6.0)	4.0	4.0	(0.0)	43.5	37.5	(6.0)
Group Functions	38.5	38.5	(0.0)	6.2	7.4	1.2	44.7	45.8	1.1
DPIT	96.0	92.7	(3.3)	8.1	13.3	5.2	104.1	106.1	2.0
Security	36.0	42.1	6.1	7.1	7.1	0.0	43.1	49.2	6.1
Other	0.6	4.9	4.3	3.2	(3.4)	(6.6)	3.9	1.5	(2.4)
Total	**299.3**	**299.3**	**0.0**	**41.7**	**41.7**	**(0.0)**	**341.0**	**341.0**	**(0.0)**

IT Commentary

Delivery

Q1 got off to a much better start than the corresponding period last year with investments sanctioned prior to the new fiscal year following improvements in the planning process.

Investments delivered include Myhub Manager Self Service, Blackline task management and FIORI. There has also been a focus on digital products to improve customer experience and security across the rest of the IT portfolio.

Challenges

Commercial and contracting challenges & chip shortages are still being experienced in the post-COVID era.

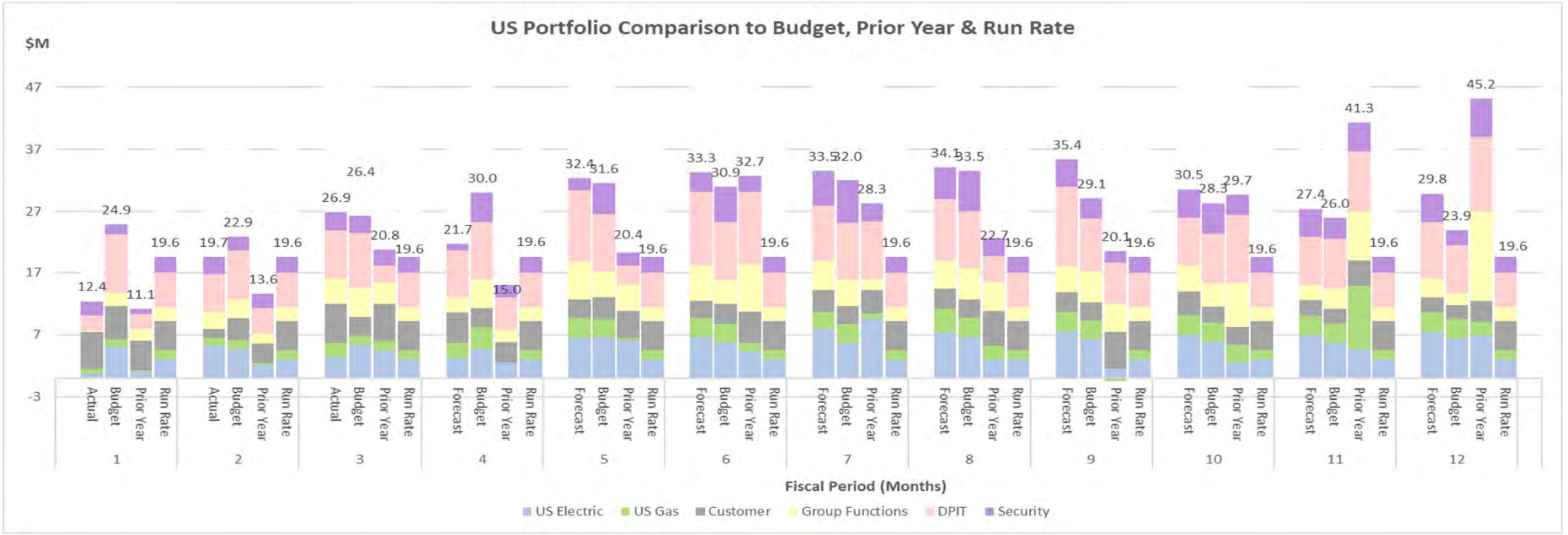
Specialist IT resources are proving a challenge to recruit and business resources difficult to secure given the overall scope and complexity of implementation. However, management is working through these issues to ensure capabilities are delivered as scheduled.

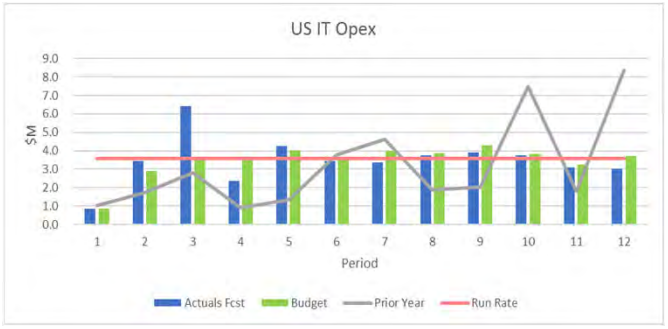
Financial Performance

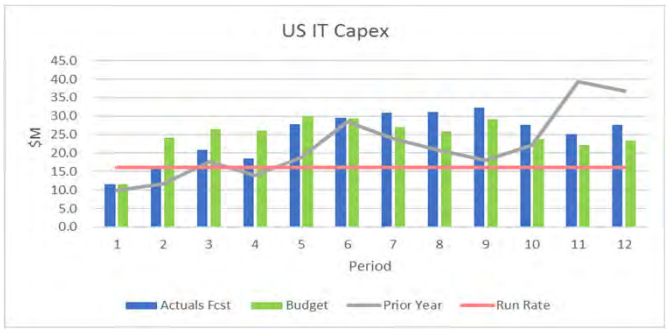
Overall financial performance was slightly under budget reflecting managing through some of the challenges. However, the largest variance was due primarily to budget phasing which has been corrected in the latest forecasts.

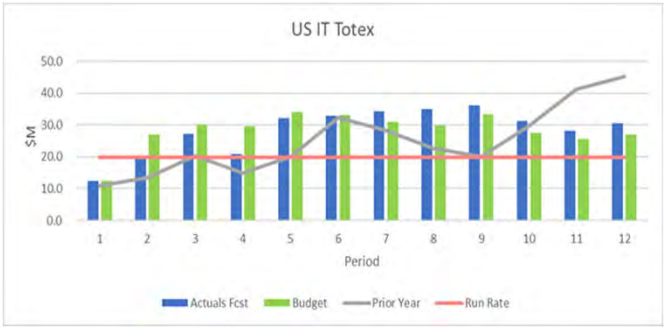
There is upward pressure on the year-end forecast particularly in the Customer portfolio and US Gas to provide desired customer and user enhancements.

Portfolio Comparison & Cost Charts







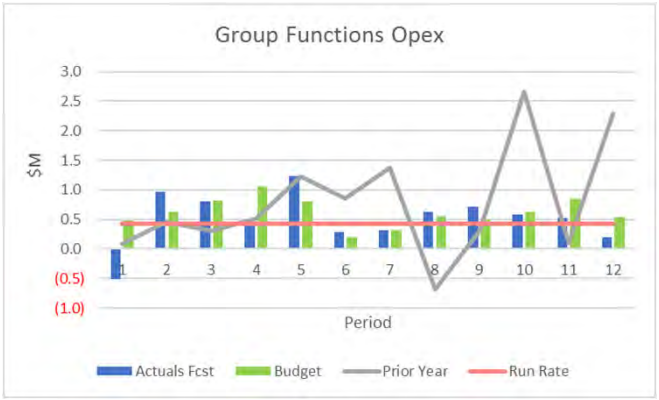


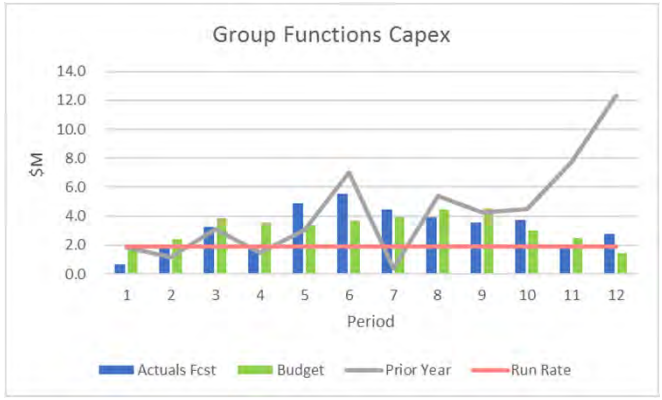
US Group Functions
Q1 FY23 Report

nationalgrid

Group Functions

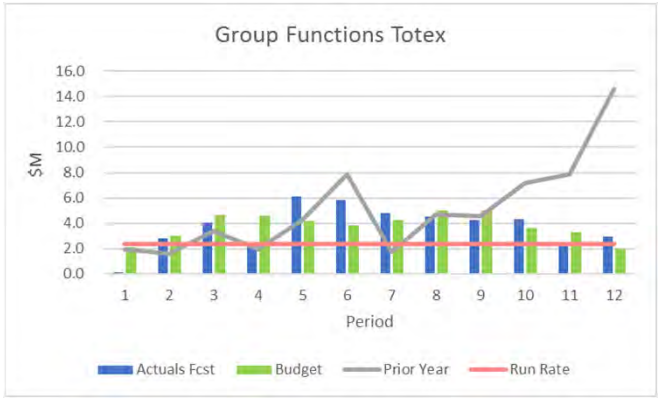
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	45.8	44.7
	YTD Budget($M)	YTD Actual ($M)
	9.9	7.0

Capex	Budget ($M)	Forecast ($M)
	38.5	38.5
	YTD Budget($M)	YTD Actual ($M)
	7.9	5.7



Portfolio Commentary

- MyHub 2.0 delivered the capability for managers self-serve with org structure changes and hiring initiations without having to interact with HR or business services. Delivers a potential reduction in rework activities by 40% (~350 hrs/mth) for the Business Services team.
- Successful release of Blackline Task Management which will enable >48% auto reconciliation of accounts in Q2, >55% by the end of FY23.
- Successful pilot launch to replace homegrown and customized US SAP end-user portals (home page, HR, BI, GRC and NUB) with Fiori apps to improve the overall employee experience, reduce support costs by $200K annually, and enable use on mobile devices.
- Reported capex spend of $5.7M, compared to budget of $7.9M two biggest contributors were:
  - Gridstack: -$1.2M due to BCG scope reduction,
  - SAP Landscape: -$0.9M due to RI Divesture transfer
- Despite resource challenges, the company is exploring ways to ensure we deliver commitments and achieve financial targets.

Group Functions Program View of Financials

Group Functions Program View Year to Date $M

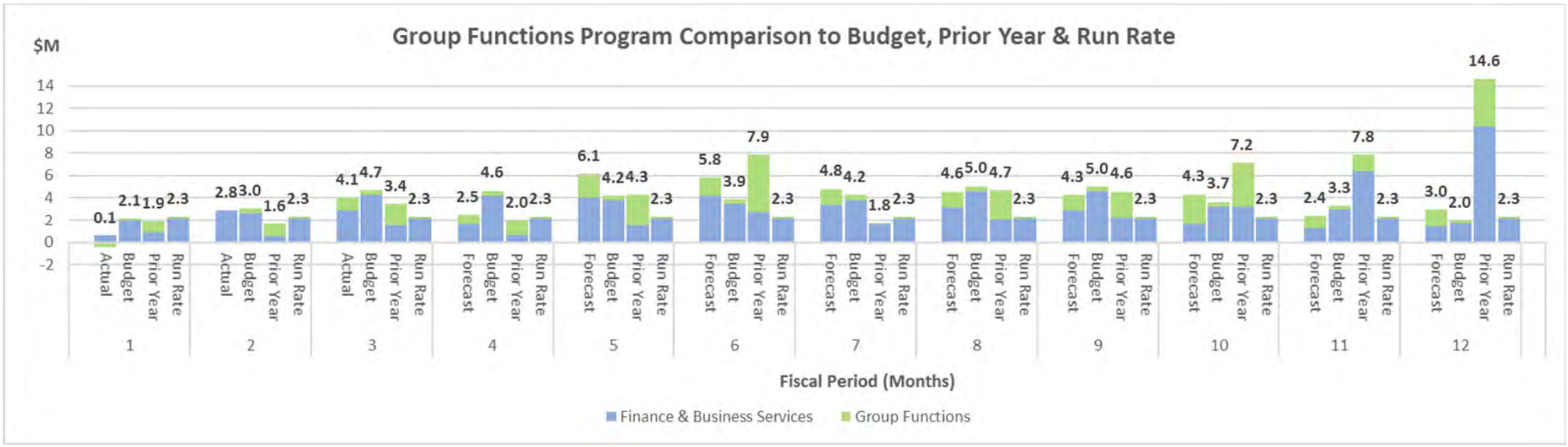
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Group Functions	Finance & Business Services	5.5	7.2	1.7	0.9	1.8	0.9	6.4	9.0	2.6
	Group Functions	0.3	0.7	0.5	0.4	0.1	(0.2)	0.6	0.9	0.2
	Totals	**5.7**	**7.9**	**2.2**	**1.3**	**1.9**	**0.7**	**7.0**	**9.9**	**2.9**

Group Functions Program View Full Year $M

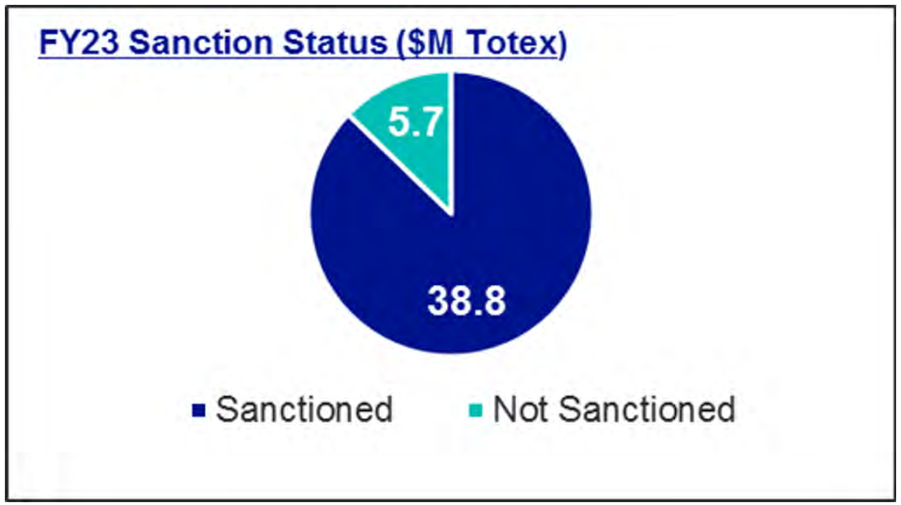
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Group Functions	Finance & Business Services	27.5	34.7	7.2	2.4	6.8	4.3	29.9	41.4	11.5
	Group Functions	11.0	3.8	(7.2)	3.8	0.6	(3.2)	14.8	4.4	(10.4)
	Totals	**38.5**	**38.5**	**(0.0)**	**6.2**	**7.4**	**1.2**	**44.7**	**45.8**	**1.1**

Key Challenges

Challenge	Escalation Point
• Resource constraints due to competing priorities and programs for both IT and Business resources	• Working with ADAM partners to backfill key roles • Recruitment ongoing to supplement with contractors
• Shortage of skilled/trained resources with Agile experience to help with transition to agile ways of working	• Upskill key resources in both IT and the business • Recruit and hire to fill open requisitions
• Risk – Capex moving to Opex due to new Finance policy on SaaS could impact ability to deliver financial commitments	• Working with finance to mitigate risk



Mobilisation Summary



Status

- Based on 3+9 forecast, 7 projects left to be sanctioned for FY23

Resources

- 3 new Project Managers have been hired, onboarded and are leading FY23 projects

Scope

- Remaining pipeline projects need to complete scope assessment as part of business case analysis and sanctioning process.

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
Blackline: Task Management	Complete	1.0		
Ariba Guided Buying	Inflight	1.4	Cost, Resources, & Schedule	
Core PowerPlan Upgrade and Lease Consolidation	Inflight	2.3	Schedule & Resources	
Warehouse Investment Recovery Digitization	Inflight	1.0	Resources & schedule	
R2R: Robotic Process Automation	Inflight	1.8		
SAP Portals to Fiori	Inflight	0.7	Cost & Technology	

US Gas
Q1 FY23 Report

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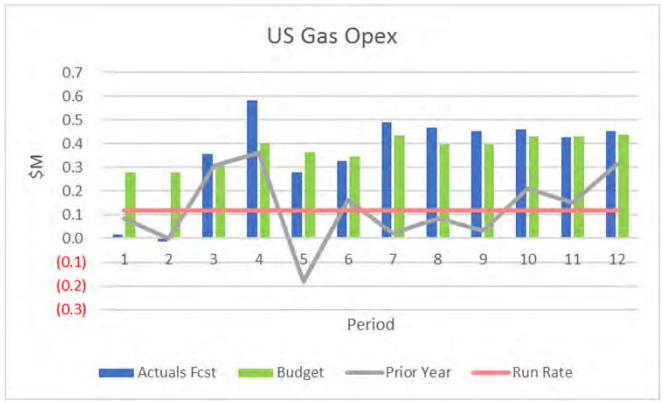
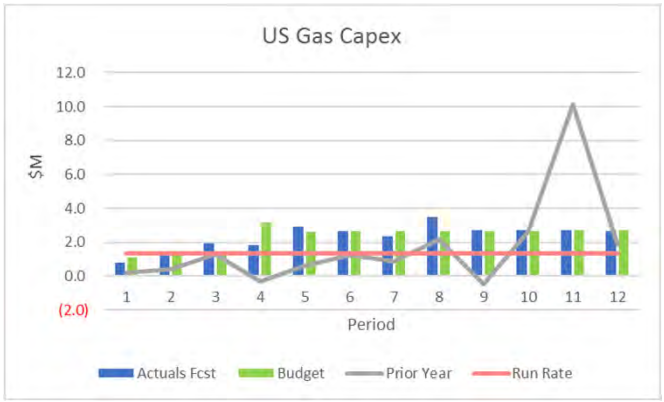
US Gas

Financial Performance



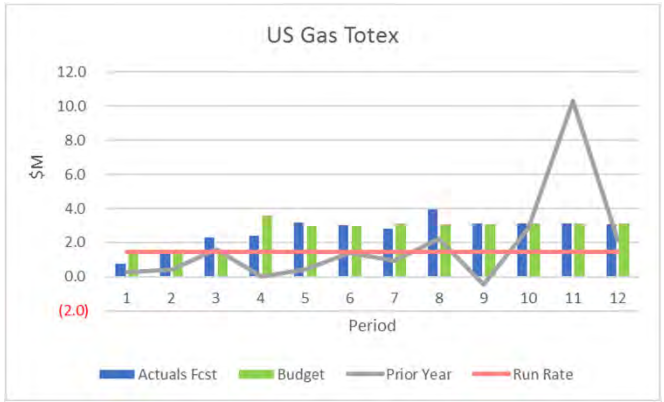


Totex

Budget ($M)	Forecast ($M)
32.5	32.3
YTD Budget($M)	YTD Actual ($M)
4.4	4.4

Capex

Budget ($M)	Forecast ($M)
28.0	27.9
YTD Budget($M)	YTD Actual ($M)
3.5	4.0



Portfolio Commentary

- The New Storms Response Management System, Go Live, has been postponed due to performance issues. The project went live in August.
- Gas Storm Hardening has been delayed due to required network equipment from a supplier being postponed because of sourcing issues.
- GTIS received additional funding (~$2.756M) to fast track the project in FY23 to mitigate any security risks due to legacy systems
- Overall, the Gas portfolio is on budget with some variation across the investments. Two key variances are as follows:
  - LNG Asset management June release postponed due to issues. Team planning Aug. release(~935K Capex Over)
  - New Gas demand DR is estimated to underspend Opex by about $700K as POC's would not be needed
- Aside from supplier delays and general resource challenges, the Gas portfolio is expected to deliver planned outcomes on budget for the fiscal year with some upward pressure due to user demands.

US Gas Program View of Financials

Gas Program View Year to Date $M

Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
US Gas	Capital Delivery	0.1	0.3	0.2	(0.0)	0.0	0.0	0.1	0.3	0.2
	Gas BU	3.9	3.2	(0.7)	0.4	0.9	0.5	4.3	4.1	(0.2)
		4.0	**3.5**	**(0.5)**	**0.4**	**0.9**	**0.5**	**4.4**	**4.4**	**(0.0)**

Gas Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
US Gas	Capital Delivery	0.1	0.3	0.1	0.0	0.0	0.0	0.1	0.3	0.1
	Gas BU	27.9	27.8	(0.1)	4.3	4.5	0.2	32.2	32.2	0.1
		28.0	**28.0**	**0.0**	**4.3**	**4.5**	**0.2**	**32.3**	**32.5**	**0.2**

Key Challenges

Challenge	Escalation Point
FCS(Gas Only) delayed to the dependency on completion of in Flight FCS Software upgrade by IT Electric	None
New Storms Response delivery delayed due to performance issues and readiness of contractors	New Code developed and deployed by CGI and additional mock tests being scheduled
LNG IT for OT support start is delayed due to contract negotiation issues.	Considering collaboration with a new vendor, NST
Master data interface issue in Maximo, SAP, PP affecting LNG	Changes required in Maximo & MuleSoft API in July/ August

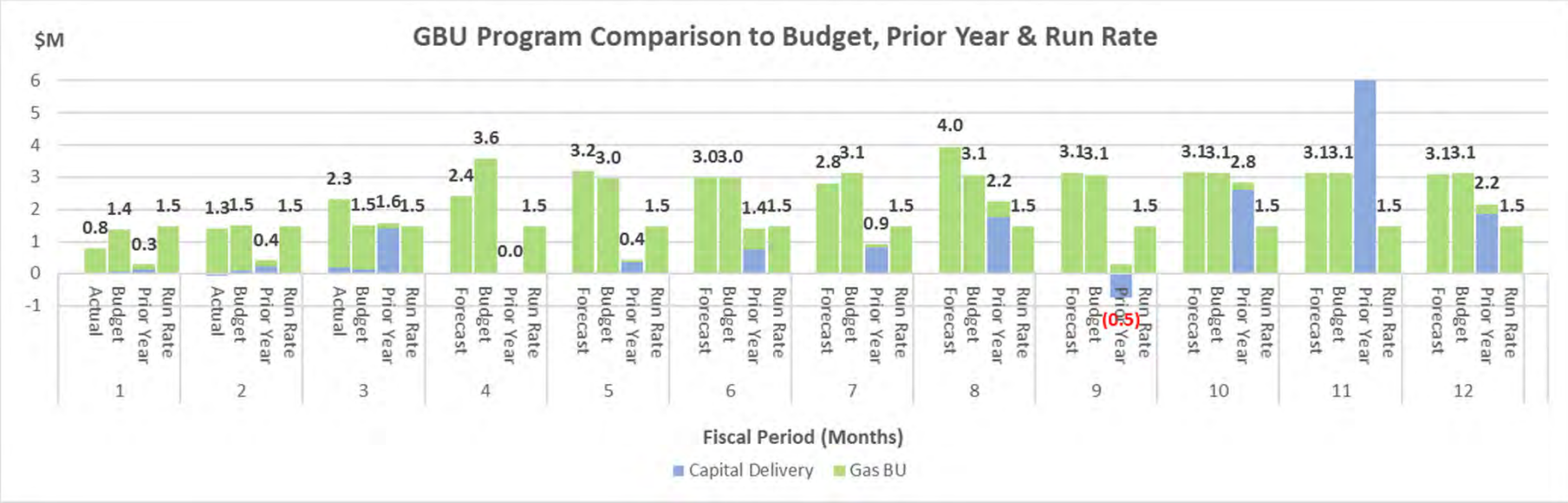
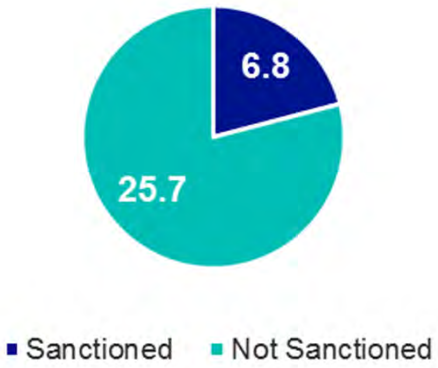


Exhibit ___ (11DP-2R)
Attachment 1
Page 1 of 27

Mobilisation Summary

FY23 Sanction Status ($M Totex)



Status

- Based on 3+9 forecast, 6 projects left to sanctioned for FY23

Resources

- 3 new Project Managers have been hired, onboarded and are leading FY23 projects

Scope

- Remaining pipeline projects need to complete scope assessment as part of business case analysis and sanctioning process.

Priority Investments

Program	Status	FY 23 Budget ($m)	RAG This Month	RAG Last Month
Gas Storm Hardening	Inflight	0.4		
LNG Asset & Work Mgmt.	Inflight	1.6		
Summons for Corrective Actions	PIS	0.1		
Enterprise Scalability and Resilience	Inflight	1.2		
Continuous Improvement for Gas Applications	Inflight	1.2		
PIF Replacement	PIS	0.0		
LNG IT support for OT	Start-up	0.9		
Gas I&R, Meter Ops, and Field Device Replacement	Inflight	0.1		
Enterprise Platform for Gas Retail Choice (GTIS)	Inflight	2.0		
Field Collection Systems (FCS) Upgrade	Prestart up	0.8		
Gas Management System (SCADA) Upgrade	Start-up	2.1		
New Gas DR Solution Evaluation	Inflight	0.9		
New Storms Response Management System Implementation	Inflight	0.3		
MOC Library & Action Item Automation	Inflight	0.2		
Continued Scaling & Evolution of Gas IT Transformation Program	Prestart up	20.1		
SOP Third-Party User Access for PE Review	Cancelled	0.1		
Enable Gas Curtailment Map in ESRI	Cancelled	0.04		
New System Operating Procedure Solution	Deferred/ FY24	0.1		
PIPES/Enterprise Maximo Integration	June decision	0.1		
Digital Asset & Work Management for CNG and I&R	Deferred/ FY25	0.3		
Enable Converted LI Gas Distribution Asset Data in ESRI	Deferred/ FY24	0.1		

US Electric
Q1 FY23 Report

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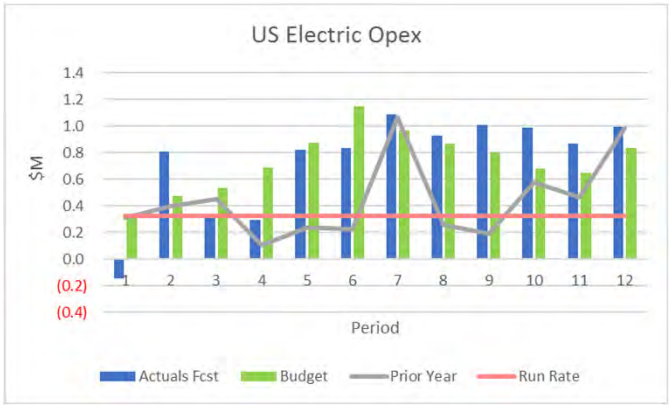
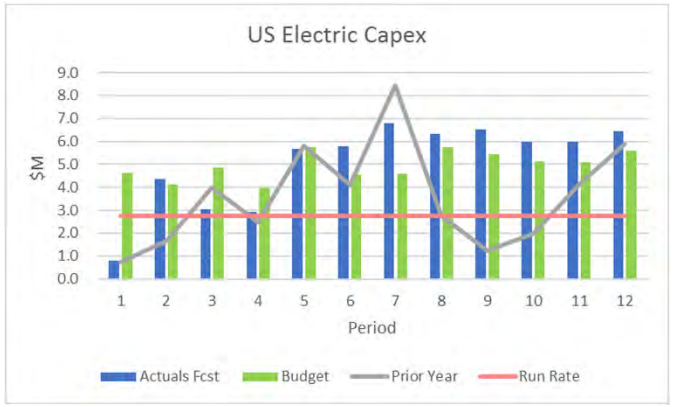
US Electric

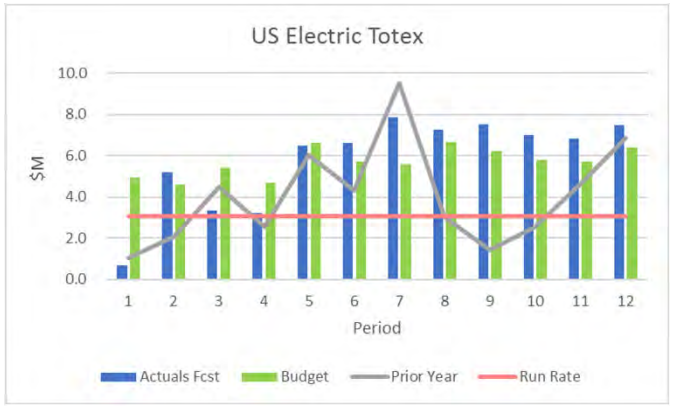
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	68.3	69.4
	YTD Budget($M)	YTD Actual ($M)
	14.9	9.2

Capex	Budget ($M)	Forecast ($M)
	59.5	60.6
	YTD Budget($M)	YTD Actual ($M)
	13.6	8.2



Portfolio Commentary

- SBP22 efforts are wrapping up and have been a major focus area for the team, specifically to ensure the Business and Financial Leadership are aligned with the products within our final SBP22 Portfolio/Submission, the benefits we expect to deliver, and the consequences associated the SBP22 products that have been removed or delayed (i.e. VOLT Funding, DSO, etc.)
- There are some delays attributed to technical challenges and global supply chain issues (i.e. MV90, NYISO-PMU), but those risks are actively being managed and do not require escalation at this time. All Core Products that were expected to be delivered in FY23 remain on track to do so.
- Q1 Total spend was $5.7M less than Budget; $2.7m was in digital products with the remainder across the portfolio largely due to mobilization efforts and ramping up of resources.
- FY23 Forecast is currently running ~$1.1m under target with the potential for these funds to be used for other products and Business needs (i.e. CNI/EMS Enhancements, Computapole F&A, etc.)

US Electric Program View of Financials

Electric Program View Year to Date $M

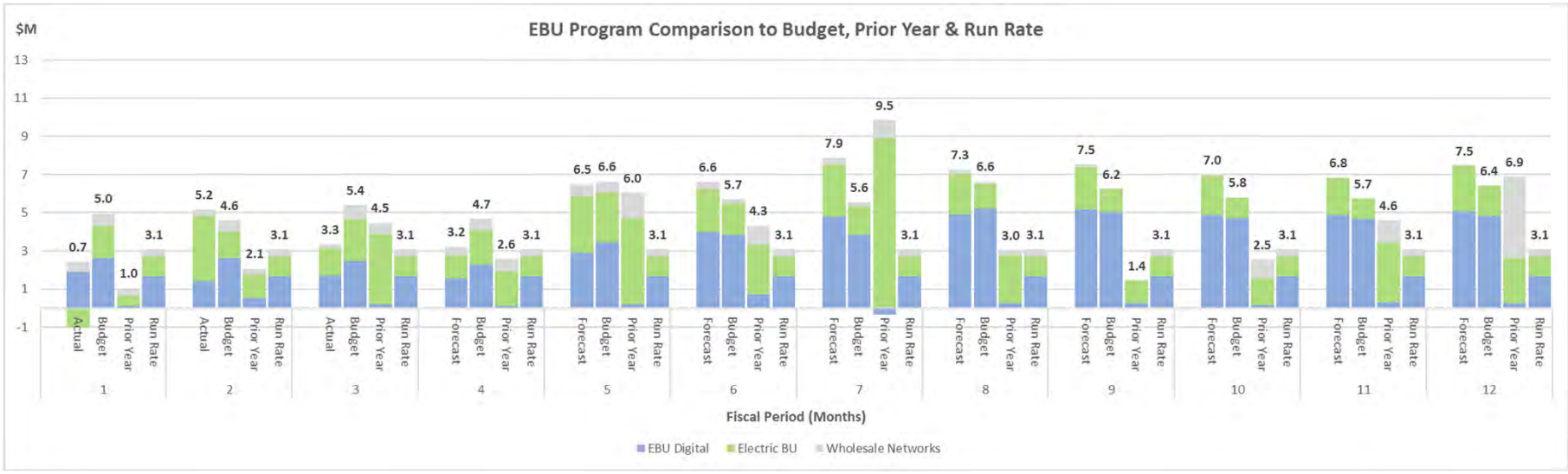
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
US Electric	EBU Digital	4.4	6.8	2.4	0.6	0.9	0.3	5.0	7.7	2.7
	Electric BU	2.8	4.8	2.0	0.3	0.4	0.1	3.1	5.2	2.1
	Wholesale Networks	1.0	2.1	1.0	0.1	0.0	(0.1)	1.1	2.1	1.0
	Totals	**8.2**	**13.6**	**5.4**	**1.0**	**1.3**	**0.4**	**9.2**	**14.9**	**5.8**

Electric Program View Full Year $M

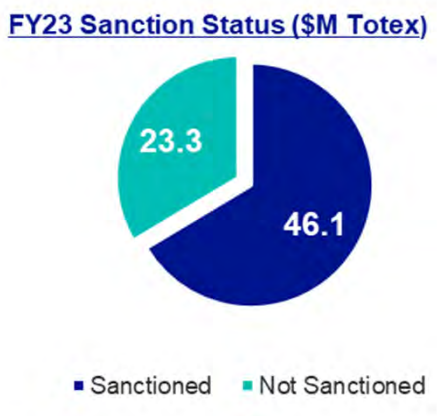
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
US Electric	EBU Digital	38.7	40.2	1.6	4.5	5.4	0.9	43.1	45.6	2.5
	Electric BU	18.9	15.6	(3.3)	4.1	3.3	(0.8)	23.0	18.9	(4.1)
	Wholesale Networks	3.0	3.6	0.6	0.2	0.1	(0.1)	3.3	3.8	0.5
	Totals	**60.6**	**59.5**	**(1.1)**	**8.8**	**8.8**	**0.0**	**69.4**	**68.3**	**(1.1)**

Key Challenges

Challenge	Escalation Point
1. Resource challenges still remain with WiPro in the Core Portfolio; *still a significant challenge within the Digital Lighthouse Portfolio.*	CIO / HR / Commercial *(ADAM Partners)*
2. Global Supply Chain delays with critical hardware needed to deploy products (i.e ADMS, NYSIO-PMU)	Commercial, ITLT



Mobilisation Summary



Status

- Based on 3+9 forecast, 13 projects left to sanctioned for FY23

Resources

- New Project Managers have been hired, onboarded and are leading FY23 projects

Scope

- Remaining pipeline projects need to complete scope assessment as part of business case analysis and sanctioning process.

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
GridMod	Inflight	45.0		
ADMS	Inflight	24.2		
FutureNow	Inflight	2.8		
Electric MV90 Upgrade	Inflight	0.7		
FCS Upgrade	Inflight	0.30		
Price Transmission	Inflight	1.0		
Corporate PI	Inflight	1.6		
NYISO PMU – Large Generators	Inflight	0.7		
Content Management	Inflight	1.5		
Generation Asset Management Upgrade	Inflight	1.6		
Control Room EOL Server and Tool Replacement	Inflight	0		

Customer
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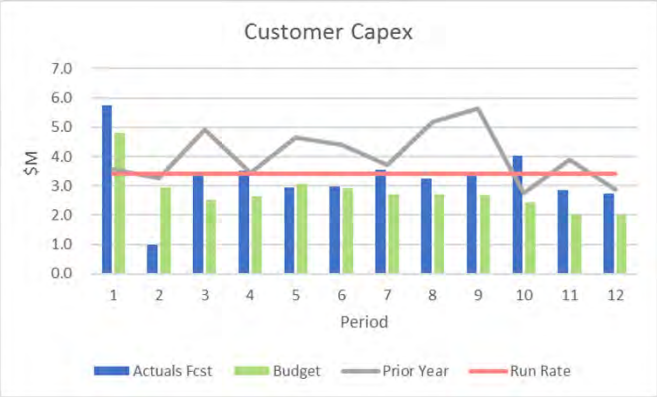
Customer

Financial Performance



Totex	Budget ($M)	Forecast ($M)
	37.5	43.5
	YTD Budget($M)	YTD Actual ($M)
	11.8	13.8

Capex	Budget ($M)	Forecast ($M)
	33.5	39.5
	YTD Budget($M)	YTD Actual ($M)
	10.3	10.2

Portfolio Commentary

- All Q1 budgeted efforts have active delivery teams with no lag in delivery at the start of the year.
- Q1 spend exceeded budget due primarily to Unified Wed Portal digital re-architecture and optimization. The current forecast for the year is to overspend by $6m, however there is upward pressure to spend $10M more than budget to deliver UWP 2.0 and IVR Re-platform. We are reviewing opportunities to re-allocate funding within the US IT portfolio to cover this demand.
- In Q1 we had continuous delivery on products to improve customer experience, improve efficiency and meet regulatory mandates. Of note were eight updates to the IVR call flow to improve service through that channel and saving $200k/month with higher containment, significant enhancements on recurring pay and notifications for our C&I customers in the My Business Account product and automating the EAP billing functionality for the KEDLI rate case.
- Looking ahead, we have had major releases in Q2 to-date including extending the Unified Web Portal to the final territories, LI and UNY, AMI MVP delivered with 5 live meters producing bills, significant improvements to the stability of core systems with the FCP upgrade and completed remediation to ensure PCI compliance.

Customer Program View of Financials

Customer Program View Year to Date $M

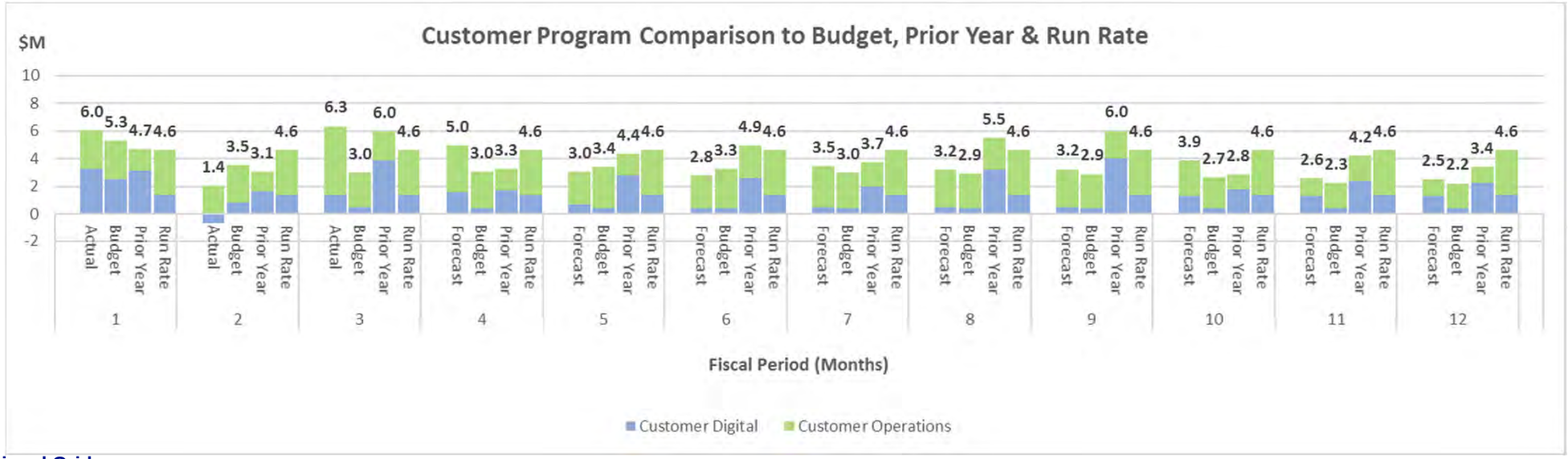
Portfolio / Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Customer Customer Digital	3.8	3.2	(0.6)	0.2	0.7	0.4	4.1	3.9	(0.2)
Customer Operations	6.4	7.1	0.7	3.3	0.8	(2.5)	9.7	8.0	(1.8)
Totals	**10.2**	**10.3**	**0.1**	**3.6**	**1.5**	**(2.1)**	**13.8**	**11.8**	**(2.0)**

Customer Program View Full Year $M

Portfolio / Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Customer Customer Digital	10.8	6.8	(3.9)	1.3	1.2	(0.1)	12.0	8.0	(4.0)
Customer Operations	28.7	26.7	(2.1)	2.7	2.8	0.1	31.5	29.5	(2.0)
Totals	**39.5**	**33.5**	**(6.0)**	**4.0**	**4.0**	**(0.0)**	**43.5**	**37.5**	**(6.0)**

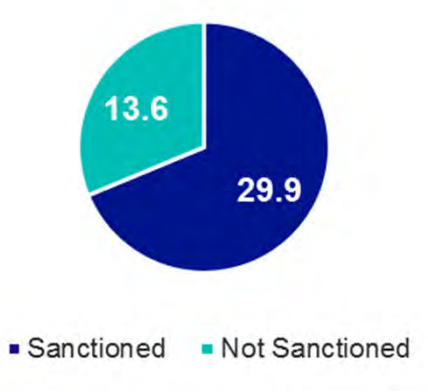
Key Challenges

Challenge	Escalation Point
***Business Resources**: Business resources have been prioritized to Transformation efforts including ZBR and CRIS to CSS. Working resource plan with Customer*	Simon Baker
***Data Center Transition Dependency**: CRIS to CSS has a significant dependency on the Data Center migration. Delays in Data Center impact the C2C schedule.*	Simon Baker and Shannon Soland
***Virtual Assistant Timeline** – The business case timeline for VA is at risk due to technical dependencies. (Part of the Zero Based Review initiative)*	Simon Baker



Mobilisation Summary

FY23 Sanction Status ($M Totex)



Status

- Based on 3+9 forecast, 3 projects left to sanctioned for FY23

Resources

- All planned Q1 projects are active and resourced

Scope

- The significant project to be sanctioned is the UWP 2.0 project which is a re-architecture and design of the web platform for sustainability and scalability of customer experience.

Priority Investments

Program	Status	Budget ($m)	RAG	RAG Last Month
Zero Based Redesign	Inflight	21.0		
CRIS to CSS	Inflight	36.2		
AMI	Inflight	22.3		
Clean Energy	Start up	9.3		
UWP Launch	Inflight	3.0		
Fix The Technology Foundations: UWP2.0	Start up	10		
Customer Data Platform	Inflight	7.0		

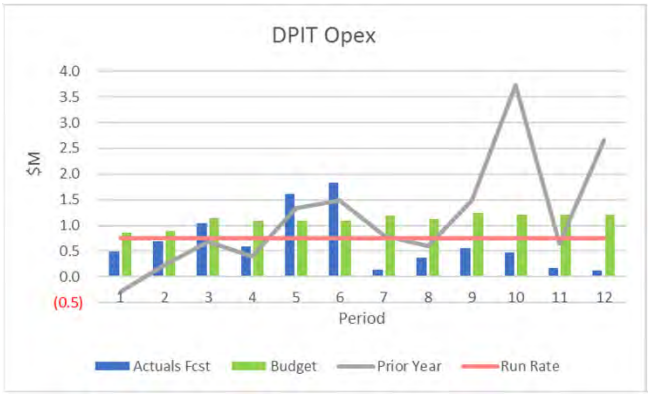
Note: UWP Launch has since gone live in all territories.

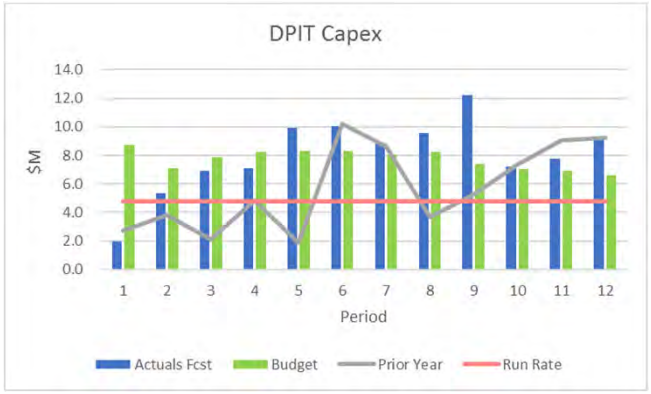
Infrastructure & Operations
Q1 FY23 Report

nationalgrid

DPIT

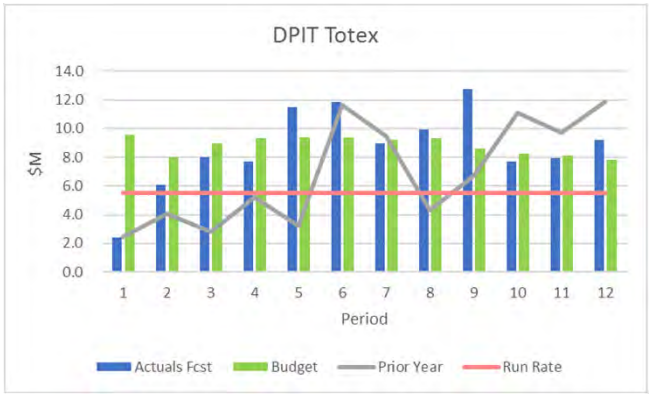
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	106.1	104.1
	YTD Budget($M)	YTD Actual ($M)
	26.6	16.5

Capex	Budget ($M)	Forecast ($M)
	92.7	96.0
	YTD Budget($M)	YTD Actual ($M)
	23.7	14.3



Portfolio Commentary

- All Product portfolios 100% sanctioned and inflight, mostly on-track to Outcomes and Key Results and financial forecast
- Data Center Transformation (DCT) is current exception – chip shortage, global supply chain issues, and network/infrastructure build delays are pushing schedule to FY24 and increased costs ~($5.1M).
  - DCT planning to shift delivery methodology to Agile Release Train
- Operations: Unified Technology Operations Centre (UTOC) will be a single operations center providing overwatch of Enterprise, CNI and Operational technology to maximize security and health of National Grid assets.
  - Minimum Viable Product (MVP) on track; 80% complete staffing for MVP; Construction delay due to Safety and HASP paperwork being completed by our construction vendor.
- Financials lagging slightly behind forecast in Q1 but this will be recovered in Q2/3,
  - Primary drivers are Digital Workplace Services (DWS) hardware and DCT Mainframe.

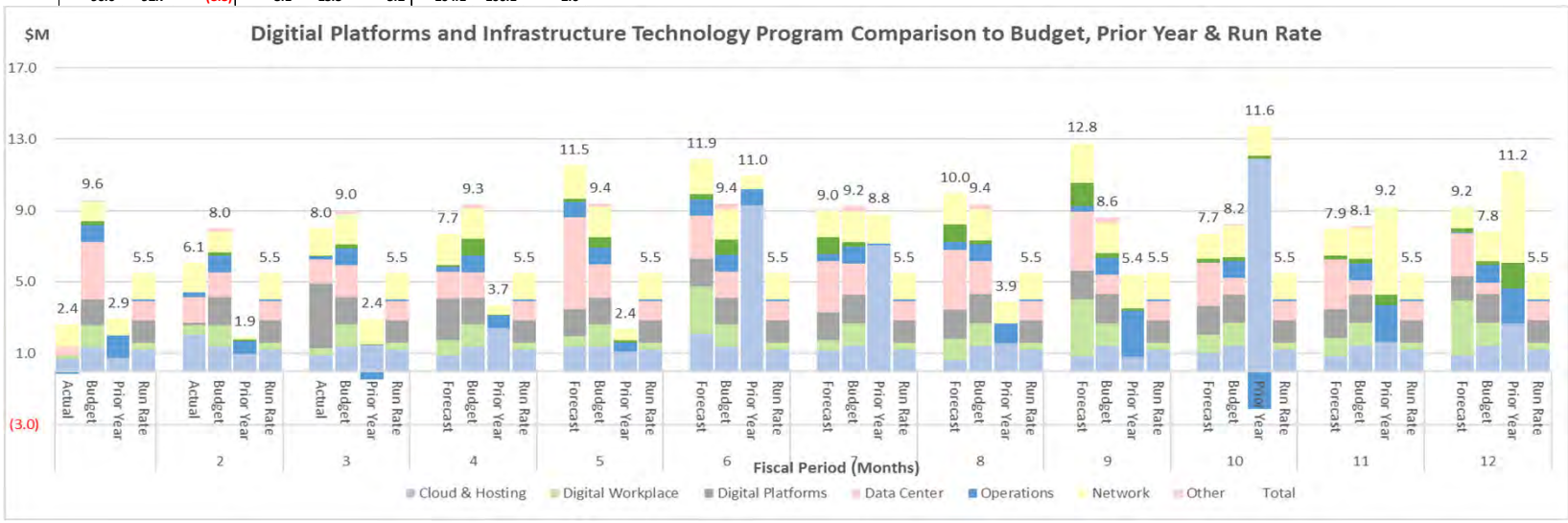
I&O Program View of Financials

DPIT Program View Year to Date $M

Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
DPIT	Cloud & Hosting	3.4	3.7	0.3	0.2	0.4	0.2	3.6	4.0	0.5
	Digital Workplace	1.0	3.5	2.6	0.2	0.2	(0.0)	1.2	3.7	2.6
	Digital Platforms	3.4	4.0	0.6	0.3	0.5	0.2	3.8	4.5	0.8
	Data Center	1.9	5.0	3.1	1.3	1.4	0.1	3.2	6.4	3.2
	Operations	0.2	2.7	2.4	0.0	0.1	0.1	0.3	2.8	2.5
	CNI	0.0	0.6	0.6	0.0	0.0	(0.0)	0.0	0.6	0.6
	Network	4.3	3.6	(0.7)	0.1	0.3	0.2	4.4	3.9	(0.5)
	Other	0.0	0.5	0.5	0.0	0.0	0.0	0.0	0.5	0.5
	Totals	**14.3**	**23.7**	**9.4**	**2.2**	**2.9**	**0.7**	**16.5**	**26.6**	**10.1**

DPIT Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
DPIT	Cloud & Hosting	11.8	15.0	3.2	1.5	1.6	0.1	13.3	16.6	3.4
	Digital Workplace	14.5	14.3	(0.2)	0.8	0.8	0.0	15.3	15.2	(0.1)
	Digital Platforms	16.4	16.4	0.0	2.2	2.2	(0.0)	18.6	18.6	(0.0)
	Data Center	16.6	11.5	(5.1)	12.8	6.8	(6.0)	29.4	18.3	(11.1)
	Operations	14.2	10.9	(3.3)	(10.6)	0.4	11.0	3.6	11.3	7.7
	CNI	4.2	4.3	0.1	0.2	0.2	0.0	4.4	4.5	0.1
	Network	18.3	18.0	(0.3)	1.2	1.3	0.1	19.5	19.3	(0.2)
	Other	0.0	2.3	2.3	0.0	0.0	0.0	0.0	2.3	2.3
	Totals	**96.0**	**92.7**	**(3.3)**	**8.1**	**13.3**	**5.2**	**104.1**	**106.1**	**2.0**

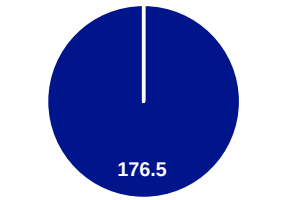


Key Challenges

Challenge	Escalation Point
Delays in US/UK migration, US mainframe, and server and application details are causing cost increases (under review) and schedule extension for US into FY24	• CTO aware of current status • Cost increases for Data Centre Transformation will be absorbed which may limit the ability to deliver full FY23 objectives
Re-prioritization of work will delay delivery teams and impact cost	• Q2: Launching Lean Portfolio Management to govern and prioritize epics

Mobilisation Summary

FY23 Sanction Status ($M Totex)



Global Sanction total

Status

- No projects left to sanctioned for FY23

Resources

- All Project Managers have been hired, onboarded and are leading FY23 projects

Scope

- FY23 projects have completed scope assessment and business case analysis for the sanctioning process.

Priority Investments

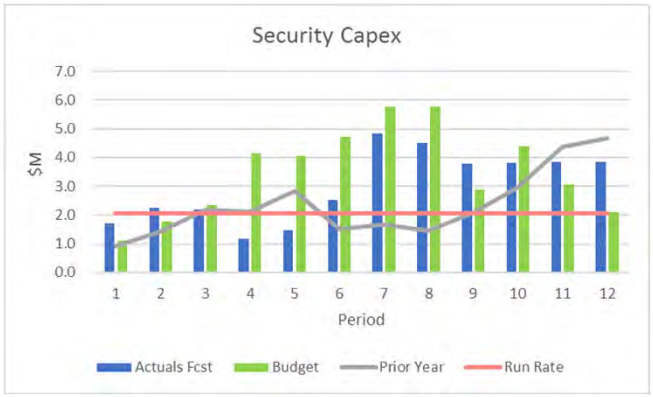
Program	Status	Budget ($m)	RAG This Month	RAG Last Month
Data Center Transformation (DCT) - *Modernization of Infrastructure & Networks*	Inflight	18.3	Cost & Schedule	
Cloud & Hosting - *Modernization of Infrastructure & Networks*	Inflight	16.6		
Digital Workplace Services - *Employee Satisfaction & Digital Product Delivery*	Inflight	15.2		
Networks - *Reduce Technical Debt*	Inflight	19.3		
Global Operations - *Employee Satisfaction & Digital Product Delivery*	Inflight	11.3		
Critical National Infrastructure (CNI) - *Reduce Technical Debt*	Inflight	4.5		
Digital Platforms - *Employee Satisfaction & Digital Product Delivery*	Inflight	18.6		

US Security
Q1 FY23 Report

nationalgrid

Security

Financial Performance



Totex

Budget ($M)	Forecast ($M)
49.2	43.1
YTD Budget($M)	YTD Actual ($M)
6.6	8.1

Capex

Budget ($M)	Forecast ($M)
42.1	36.0
YTD Budget($M)	YTD Actual ($M)
5.3	6.2

Portfolio Commentary

- [redacted]

Security Program View of Financials

Security Program View Year to Date $M

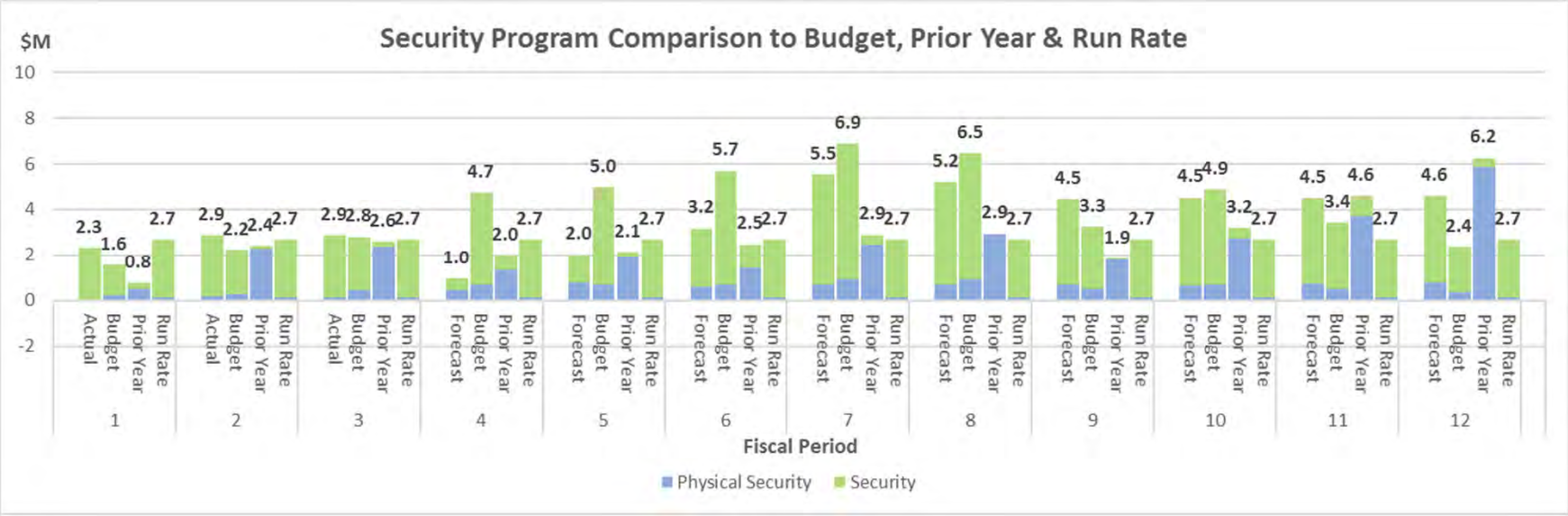
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Security	Physical Security	0.4	1.0	0.6	0.0	0.0	0.0	0.4	1.0	0.6
	Security	5.8	4.3	(1.5)	1.9	1.3	(0.6)	7.7	5.6	(2.1)
	Totals	**6.2**	**5.3**	**(0.9)**	**1.9**	**1.3**	**(0.6)**	**8.1**	**6.6**	**(1.5)**

Security Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Security	Physical Security	6.4	7.0	0.6	0.0	0.0	0.0	6.4	7.0	0.6
	Security	29.6	35.2	5.6	7.1	7.1	0.0	36.7	42.3	5.6
	Totals	**36.0**	**42.1**	**6.1**	**7.1**	**7.1**	**0.0**	**43.1**	**49.2**	**6.1**

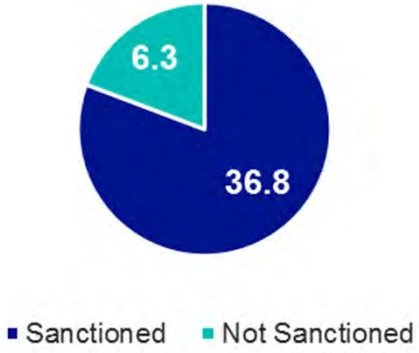
Key Challenges

Challenge	Escalation Point
[redacted]	[redacted]
[redacted]	[redacted]



Mobilisation Summary

FY23 Sanction Status ($M Totex)



Status

[redacted]

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
Cyber Operational Technology	In-Flight	[redacted]		
Identity and Access Management	In-Flight Add'l Sanction Aug	[redacted]		
Platform Security	In-Flight	[redacted]		
Network Security	In-Flight	[redacted]		
Physical Security	In-Flight Sanction for NIMO & BGC planned	[redacted]		
Security Orchestration Automation and Response	In-Flight	[redacted]		
Vulnerability Management	In-Flight	[redacted]		

Project View - Group Functions

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Variance	Non_Permanant Reason	Permanent Reason	Year End Forecast	Year End Budget	Variance	Permanent Reason
				Year to Date					Full Year Forecast			
Group Functions	**Finance & Business Services**	3870A	F&A and Implementation - Vision FM Replacement	395,169	-	(395,169)	Mischarges		19,952	-	(19,952)	Mischarges
		4418USB	RSA Archer Upgrade & Refresh	226,282	-	(226,282)		Accounting	226,282	-	(226,282)	Accruals
		4562	US SAP: Business Warehouse (BW) Consolidation to HANA Enterprise Cloud (HEC)	(54,604)	-	54,604		Accounting	(54,604)	-	54,604	Other
		5158	MyHub Compliance & Enhancements	(2,086)	700,000	2,086	Mischarges		2,897,914	2,900,000	2,086	
		5158	MyHub Compliance & Enhancements	See Above	See Above	700,000	Accruals					
		5466	AMAG HW and SW Upgrade	442,354	500,000	57,646	Delays		1,642,902	800,000	(842,902)	Imprecise Estimates
		5497	Lease Accounting Updates and CCM - Phase 2	551,703	500,000	(51,703)		Imprecise Estimates	1,569,405	2,300,000	730,595	Imprecise Estimates
		5555	Fieldglass Replacement	0	300,000	300,000		Timing	1,522,000	1,800,000	278,000	Timing
		5605	US SAP: Infrastructure Landscape FY21	18,813	-	(18,813)		Accounting	18,813	-	(18,813)	Accounting
		5606	US SAP: Infrastructure Landscape FY22	(932,492)	-	932,492		Funding Bucket	(932,492)	-	932,492	Funding Bucket
		5610	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY22	2,731	-	(2,731)		Timing	2,731	-	(2,731)	Timing
		5611	Annual HR & Payroll Mandatory Service Pack (HRSP) FY23	340,816	400,000	59,184		Imprecise Estimates	1,231,512	1,500,000	268,488	Imprecise Estimates
		5841C	401(k) Administration Changes	128,777	-	(128,777)		Timing	132,752	-	(132,752)	Timing
		5842B	Mandatory FERC Form Changes	(21,528)	-	21,528		Accounting	(21,528)	-	21,528	Accounting
		5843B	US SAP Platform Upgrade: Minor Releases	(162,731)	-	162,731		Scope Changes	(162,731)	-	162,731	Scope Changes
		5843D	US SAP Healthcheck: Upgrade Critical Components - FY22	813,159	700,000	(113,159)		Imprecise Estimates	846,279	700,000	(146,279)	Imprecise Estimates
		5895B	Everbridge / Critical Event Management	(37,893)	-	37,893		Complexity	135,480	-	(135,480)	Complexity
		5896B	Warehouse SAP Scanner Replacement (Barcoding)	(385)	-	385		Imprecise Estimates	(385)	-	385	Imprecise Estimates
		6002	Other's Efficiency Target	0	-	0			(9,774,000)	-	9,774,000	Accounting
		6137US	Enterprise PMO, Jira Align release 2	150,482	-	(150,482)		Funding Change	150,482	-	(150,482)	Funding Change
		6182US	Global Career Site	(49,000)	-	49,000		Accounting	(49,000)	-	49,000	Accounting
		6183US	HR Reporting Analytics	37,086	-	(37,086)		Accounting	37,086	1,300,000	36,914.3	Accounting
		6183US	HR Reporting Analytics								1,226,000	Funding Change
		6185US	Onboarding 2.0	95,998	-	(95,998)		Funding Change	368,554	-	(368,554)	Funding Change
		6219	US F&BS Minor Enhancements (Non-SAP)	28,316	-	(28,316)		Accounting	28,316	-	(28,316)	Accounting
		6220	US F&BS Minor Enhancements (SAP)	33,832	-	(33,832)		Accounting	33,832	-	(33,832)	Accounting
		6221	Tech Debt: RiskMaster Replacement	210,306	-	(210,306)		Funding Change	541,553	-	(541,553)	Funding Change
		6231	US Treasury: Allegro Upgrade Phase 2	104,631	-	(104,631)		Timing	107,341	-	(107,341)	Timing
		6242US	ARIBA System Enhancements (Spend Cube)	220,075	-	(220,075)		Timing	250,075	-	(250,075)	Timing
		6265	HR Policy Compliance	20,260	-	(20,260)		Funding Change	164,260	-	(164,260)	Funding Change
		6270US	Digitizing Learning	135,841	-	(135,841)		Funding Change	540,191	-	(540,191)	Funding Change
		6273US	Workforce Data Domain HR data governance Tools Processes EDM	254,782	400,000	145,218	Delays		1,162,692	700,000	(462,692)	Imprecise Estimates
		6289B	FTOM: Record to Report	(5,182)	-	5,182		Accounting	(5,182)	-	5,182	Accounting
		6289D	FTOM: Record to Report - Task Management with Blackline	1,163,708	600,000	(563,708)	Other	Imprecise Estimates	1,163,708	1,000,000	(163,708)	Imprecise Estimates
		6289E	FERC Form Integration with wDesk	195,330	500,000	304,670		Imprecise Estimates	436,392	1,100,000	663,608	Imprecise Estimates
		6290A	Source To Pay	1,326,142	433,000	(893,142)		Timing differences	4,952,742	1,433,000	(3,519,742)	Timing
		6290B	Source To Pay: Release 2 - Supplier Onboarding	61,134	200,000	138,866	Delays		2,776,674	1,300,000	(1,476,674)	Imprecise Estimates
		6346US	CA: Group We Digital Evolution Phase 1 - FY22	(88,050)	-	88,050		Accounting	(88,050)	-	88,050	Accounting
		6346USC	Domain Management Refresh	(466,839)	-	466,839		Accounting	(466,839)	-	466,839	Accounting
		6346USE	Brand Central Design and Implementation	(10,016)	-	10,016		Accounting	(10,016)	-	10,016	Accounting
		6346USF	Brand Central Design and Implementation	(67,193)	-	67,193		Accounting	(67,193)	-	67,193	Accounting
		6346USI	Video Strategic Management and Archiving (Enterprise)	(53,406)	-	53,406		Accounting	(53,406)	-	53,406	Accounting
		6377	US Group Function Operational Requests	(166,070)	-	166,070		Accounting	(166,070)	-	166,070	Accounting
		6390	HR + Placeholder	0	-	0					0	
		6421	P2P Data Analysis Tool	(20,000)	200,000	220,000	Delays	Accounting	2,139,973	1,200,000	(939,973)	Funding Change
		6426	Investment Recover Digitalization	114,703	330,000	215,297	Delays		1,004,698	1,000,000	(4,698)	Imprecise Estimates
		6450x	Accelerated Telecom Networks Audit (POTS and Circuits)	225,221	-	(225,221)		Funding Change	225,221	-	(225,221)	Funding Change
		5607	US SAP: Infrastructure Landscape FY23	659,322	1,000,000	340,678		Funding Bucket	2,867,954	4,000,000	1,132,046	Funding Bucket
		5843E	SAP BW/BI Upgrade	0	200,000	200,000	Delays		600,000	700,000	100,000	Timing
		5843F	SAP PI NW 7.40 (PI -> PO upgrade) to NW 750	0	200,000	200,000	Delays		700,000	900,000	200,000	Scope Changes
		6529	US Group Function Minor Enhancements (SAP)	223,636	200,000	(23,636)	Other	Imprecise Estimates	974,793	1,100,000	125,207	Imprecise Estimates
		6530	US Group Function Operational Requests - FY23	16,146	300,000	283,854	Timing differences		1,116,014	1,300,000	183,986	Other
		4418USC	CRO FY23: GRC Archer	0	375,000	375,000	Delays		1,452,600	1,500,000	47,400	Imprecise Estimates
		6514US	NG.com Refresh	0	45,000	45,000	Misclassifications			1,900,000	1,900,000	Funding Bucket
		6533US	SHE FY23: Group SHE Digital Evolution	0	195,709	195,709	Delays			1,500,000	1,500,000	Timing
		6534US	Legal FY23: Group Legal Digital Evolution	0	108,539	108,539	Delays			708,800	708,800	Timing
		Corp23/										
		6596	US Placeholder - Corp Tech Debt Remediation	0	200,000	200,000		Funding Bucket		1,800,000	1,800,000	Funding Bucket
		6270	Digitising Learning	0	200,000	200,000		Funding Bucket		600,000	600,000	Funding Bucket
		HR23/										
		6390	Tactical HR	0	-	0		Funding Bucket	226,000	1,800,000	1,574,000	Funding Bucket
		OPS23/										
		6574	Placeholder (Tech Debt & App Rat)	0	-	0		Funding Bucket		680,000	680,000	Funding Bucket
		6225US	R2R Reporting Use Cases	0	-	0				1,300,000	1,300,000	Other
		6289C	FTOM Record to Report: Process Mining with Celonis	0	-	0		Other		-	0	Other
		6289G	FTOM R2R: Robotic Process Automation of Manual Journal Entries (reason 1)	3,961	200,000	196,039		Funding Change	884,336	1,800,000	149,664	Funding Change
		6289G	FTOM R2R: Robotic Process Automation of Manual Journal Entries (reason 2)								766,000	Efficiencies
		R2R23	Finance + placeholder (Regulatory Compliance)	0	-	0				800,000	800,000	Funding Bucket
		JS	Efficiency Target						46,470		(46,470)	Funding Change
		6467	Union Bidding	7,720	-	(7,720)		Funding Change	233,953	-	(233,953)	Funding Change
		6596	US Placeholder - Corp Tech Debt Remediation	0	-	0			1,100,004	-	(1,100,004)	Funding Change
		6597	Finance + placeholder (Regulatory Compliance)	0	-	0			799,998	-	(799,998)	Funding Change
		6566US	HR Policy Compliance FY23	1,032	-	(1,032)		Funding Change	186,675	-	(186,675)	Funding Change
		6180US	CV Talent Screening / Blind CVs	180	-	(180)		Funding Change	306,304	-	(306,304)	Funding Change
		6424US	Infonet decommissioning	66,000	-	(66,000)		Funding Change	66,000	-	(66,000)	Funding Change
		3870B	Vision FM replacement, Phase 2	91,556	-	(91,556)		Funding Change	1,120,509	-	(1,120,509)	Funding Change
		6574	US Group Function: Tech Debt and Application Rationalization	0	-	0			699,996	-	(699,996)	Funding Change
		6289F	US GAAP Statutory reporting with WDesk	139,420	-	(139,420)		Funding Change	1,226,709	-	(1,226,709)	Funding Change
		6575	US Group Function: R2R Regulatory and Compliance	0	-	0			799,998	-	(799,998)	Funding Change
	Finance & Business Services Total			**6,369,151**	**8,987,248**	**2,618,096**			**29,886,631**	**41,421,800**	**11,535,169**	
	Group Functions	5899A	Cloud API Migrations Phase 2	62,549	-	(62,549)	Misclassifications		11,243	-	(11,243)	Other
		5917A	Data Mgmt_ Tooling	(108)	-	108		Accounting	(108)	-	108	Accounting
		5983	DB Landscape Optimization	495,735	-	(495,735)		Funding Change	2,492,487	-	(2,492,487)	Funding Change
		6137	Enterprise PMO, Jira Align release 2	0	-	0			1,232	-	(1,232)	Funding Change
		6252US	Data Mgmt Tools and Initiatives	(220,574)	-	220,574		Accounting	(183,755)	-	183,755	Accounting
		6253US	Cloud API Migration - Legacy Integration Tools	464,345	-	(464,345)		Funding Change	683,229	-	(683,229)	Funding Change
		6254US	Platform as a Service (PaaS) on Azure	12,229	-	(12,229)		Accounting	12,229	-	(12,229)	Accounting
		6255US	Digital Foundation Tooling and DevSecOps	13,578	-	(13,578)		Accounting	13,578	-	(13,578)	Accounting
		6256US	Platform Upgrades: Salesforce/Maximo/Mulesoft	(42,881)	-	42,881		Accounting	(35,235)	-	35,235	Accounting
		6258USA	Data Archival Platform Implementation	(4,761)	-	4,761		Accounting	(4,761)	-	4,761	Accounting
		6258USB	Data Archival: US Group Function Portfolio	431,709	300,000	(131,709)		Imprecise Estimates	3,011,020	1,100,000	(1,911,020)	Imprecise Estimates
		6260US	EA Foundation: Increased Capability	(1,816)	-	1,816		Accounting	52,127	-	(52,127)	Accounting
		6261US	ServiceNow Upgrade: Stay Current FY22	(1,869)	-	1,869	Misclassifications		(1,869)	-	1,869	Other
		6264US	Apptio Development	47,756	-	(47,756)		Funding Change	362,756	-	(362,756)	Funding Change
		6266US	EA Strategic Portfolio Planning & Delivery / Additional Capability	477,594	-	(477,594)		Funding Change	460,916	-	(460,916)	Funding Change
		6313US	GDPR Purge Compliance	851	-	(851)	Accruals	Accounting	851	-	(851)	Accounting
		6360US	IT Planning: Enterprise PMO	93,217	-	(93,217)		Funding Change	93,217	-	(93,217)	Funding Change
		6458US	QES IT Tool Sunset	20,411	-	(20,411)		Funding Change	40,539	-	(40,539)	Funding Change
		6461USA	Business Planning F&A (FP&A Enablement & Modelling Tool)	(204,166)	494,545	698,711		Accounting	1,901,500	2,500,000	598,500	Accounting
		6468	GridStack - Infrastructure	(1,110,822)	-	1,110,822		Accounting	(1,110,822)	-	1,110,822	Accounting
		6513US	CA: Group Web Digital Evolution - FY23	0	70,000	70,000		Funding Change	222,300	800,000	577,700	Funding Change
		6587US	Strategic Portfolio Planning & Delivery SPPD, Capability Expansion	83,195	-	(83,195)		Funding Change	1,102,682	-	(1,102,682)	Funding Change
		6522US	SHE: Digitisation of the CMT Process	0	-	0			100,621	-	(100,621)	Funding Change
		6533BUS	SHE: Sustainability Digital Evolution FY23	0	-	0			294,840	-	(294,840)	Funding Change
		6549US	SHE: Resilience Digital Evolution	0	-	0			465,000	-	(465,000)	Funding Change
		6391US	SHE: Consolidated Incident Management and Reporting	0	-	0			562,557	-	(562,557)	Funding Change
		6402US	Legal Operations	0	-	0			569,000	-	(569,000)	Funding Change
		6225UK	R2R Reporting Use Cases	0	-	0			1,980,000	-	(1,980,000)	Funding Change
		6564US	CA:NG.com Implementation	0	-	0			1,600,000	-	(1,600,000)	Funding Change
		6563US	CA:NG.com Design	0	-	0			116,000	-	(116,000)	Funding Change
	Group Functions Total			**616,172**	**864,545**	**248,373**			**14,813,374**	**4,400,000**	**(10,413,374)**	
Group Functions Total				**6,985,324**	**9,851,793**	**2,866,470**			**44,700,006**	**45,821,800**	**1,121,794**	

Project View - US Gas

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Year to Date Variance	Non_Permanant Reason	Permanent Reason	Full Year Forecast Year End Forecast	Year End Budget	Variance	Permanent Reason
US Gas	**Capital Delivery**	5274A	New Storms Respose Management System Implementaition	265,533	267,032	1,499	Delays		292,971	277,032	(15,939)	
		6222	PIF Replacement	(174,368)	-	174,368		Accounting	(164,355)	-	164,355	Accounting
	Capital Delivery Total			**91,165**	**267,032**	**175,867**			**128,616**	**277,032**	**148,416**	
	Gas BU	4769	Gas Storm Hardening Using Remote Service Valves (RSV's)	66,716	267,812	201,096	Delays		475,570	382,812	(92,758)	Timing
		4898	Asset Management System for LNG	1,373,394	985,000	(388,394)		Complexity	2,337,397	1,607,000	(730,397)	Complexity
		5785I	NRG GIS Replacement Project	672	-	(672)	Mischarges		672	-	(672)	
		5787D	GTIS Functional Upgrades	(51,780)	495,000	546,780		Scope changes	3,710,470	1,980,000	(1,730,470)	Scope changes
		5787G	F&A Study for Gas Demand Response (DR)	19,965	-	(19,965)		Accounting	19,965	-	(19,965)	Accounting
		5788D	Interface MyHub To OnBoard LMS OQ External Site	898	-	(898)		Accounting	898	-	(898)	Accounting
		5788G	FY22 Gas Regulatory and Operational Enhancements (Minor Works)	(7,958)	-	7,958		Accounting	(7,958)	-	7,958	Accounting
		5789E	Summons and Corrective Actions Response (CAR) Work Tracker	84,032	-	(84,032)		Timing	98,120	-	(98,120)	Timing
		6015	Continuous Improvement for Gas Applications	1,151,540	302,500	(849,040)	Mischarges		90,540	1,210,000	1,119,460	Funding Change
		6353A	FY22 Gas Application Rationalization	13,901	-	(13,901)		Accounting	13,901	-	(13,901)	Accounting
		6392	Contractor Ipad for Gas Business	12,041	-	(12,041)	Mischarges		12,041	-	(12,041)	
		6441	Scalability & Resilience for Enterprise Gas Platforms	960,966	297,500	(663,465)	Timing ($607k)	Accounting ($56k)	1,333,453	1,190,000	(143,452)	Imprecise Estimates
		6479	LNG IT Support for OT Systems	1,303	800,000	798,697	Delays		851,303	850,000	(1,303)	
		6481	Pathfinder for Leak Survey for MA	71,950	-	(71,950)		Funding Change	71,950	-	(71,950)	Funding Change
		5785Q	Field Collection Systems (FCS) Upgrade (Gas Only)	23,409	210,000	186,591	Delays		840,409	840,000	(409)	
		6472	MOC Library & Action Item Automation?	38,426	149,000	110,574	Accruals		144,628	150,000	5,372	
		6509	Gas I&R, Gas Meter Ops, and Electric Complex Construction Field Device Replacement	413,965	30,000	(383,965)		Accounting	540,402	134,000	(406,402)	Accounting
		6539	Asset Prism?	0	-	0			20,070,000	20,070,000	0	
		6542	New Gas Demand Response Solution POC and Pilot?	34,657	215,750	181,093	Timing		64,685	863,083	798,399	Funding Change
		6550	Gas Management System (SCADA) Upgrade?	79,024	440,833	361,809	Timing		1,966,735	2,098,333	131,598	Funding Change
		6552	System Operating Procedure Third-Party User Access for PE Review	-	10,000	10,000				100,000	100,000	Funding Change
		6553	Enable Gas Curtailment Map in ESRI	0	0	0				40,000	40,000	Funding Change
		6554	New System Operating Procedure (SOP) Solution	0	0	0				100,000	100,000	Funding Change
		6555	PIPES/Enterprise Maximo Integration	-	16,667	16,667		Funding	50,000	50,000	0	
		6556	Digital Asset & Work Management for CNG and I&R?	0	0	0				290,000	290,000	Funding Change
		6557	Enable Converted LI Gas Distribution Asset Data in ESRI	0	0	0				140,000	140,000	Funding Change
		6559	FY23 Gas Regulatory and Operational Enhancements (Minor Works)?	-	37,500	37,500		Funding Change		150,000	150,000	Funding Change
		6582	Gas Application Continuous Improvement	8,653	-	(8,653)		Funding Change	624,474	-	(624,474)	Funding Change
		5990	Efficiency Adj						(1,269,330)		1,269,330	Funding Change
		6573	FY23 Gas Regulatory and Operational Enhancements (Minor Works)	22,877	-	(22,877)	Other	Funding Change	149,877	-	(149,877)	Funding Change
	Gas BU Total			**4,318,650**	**4,257,562**	**(61,088)**			**32,190,201**	**32,245,229**	**55,028**	
US Gas Total				**4,409,815**	**4,524,595**	**114,779**			**32,318,817**	**32,522,261**	**203,444**	

Project View - US Electric

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Year to Date Variance	Non_Permanant Reason	Permanent Reason	Full Year Forecast Year End Forecast	Year End Budget	Variance	Permanent Reason
US Electric	EBU Digital	5054	On My Way (EBU Digital Solution) Phase 2	1,992,164	943,162	(1,049,002)		Imprecise Estimates	4,961,018	3,761,961	(1,199,058)	Imprecise Estimates
		5912	On My Way (EBU Digital Solution)	20,382	-	(20,382)		Imprecise Estimates	20,382	-	(20,382)	Imprecise Estimates
		6073	VMO Validation Phase	664,178	1,832,278	1,168,100		Other	8,569,948	8,566,395	(3,552)	Other
		6076	FutureNow Validation Phase	1,886,372	2,809,346	922,974		Other	11,184,454	11,186,890	2,436	Other
		6291	ICU-Out (Electric Digita)	57,571	962,642	905,072		Other	5,649,984	5,647,785	(2,198)	Other
		6425	DER Active Resource Integration NY	223,555	-	(223,555)		Funding Change	562,163	-	(562,163)	Funding Change
		6437	GridMD	4,688	525,170	520,483		Funding Bucket	4,688	1,020,103	1,015,415	Funding Bucket
		6388	EBU Placeholder	0	0	0			5,663,107	12,800,000	7,136,893	Funding Bucket
		GSFund	GridStack Funding	-	664,500	664,500		Funding Change		2,657,999	2,657,999	Funding Change
		6437A	GridMD	151,436	-	(151,436)		Funding Bucket	6,499,966	-	(6,499,966)	Funding Bucket
	EBU Digital Total			**5,000,346**	**7,737,098**	**2,736,752**			**43,115,710**	**45,641,134**	**2,525,424**	
	Electric BU	4481	US MDS-Energy Accounting System (EAS) migration to Wholesale Settlement Application (WSA)	(56,731)	-	56,731		Accounting	(56,731)	-	56,731	Accounting
		4483	US Control-Transmission Outage Application (ITOA) Upgrade to HTML 5 Web	54,155	-	(54,155)		Accounting	54,155	-	(54,155)	Accounting
		5241	East Pulaski Energy Storage System IS Network	7,202	-	(7,202)		Accounting	50,495	-	(50,495)	Accounting
		5242	Buffalo Energy Storage System IS Network	(1,240)	-	1,240		Accounting	(1,041)	-	1,041	Accounting
		5702	GridStack	(86,720)	-	86,720		Accounting	(86,720)	-	86,720	Accounting
		5773B	Visitor Logging for NERC-CIP Reliability Compliance	161,008	342,475	181,467		Imprecise Estimates	474,831	439,714	(35,117)	Imprecise Estimates
		5773E	NERC-CIP Reliability Compliance	68,119	-	(68,119)		Funding Change	96,325	-	(96,325)	Funding Change
		5871A	NY Battery Energy Storage Systems	-	125,000	125,000		Funding Change	2,478,000	2,675,000	197,000	Funding Change
		5871D	Clean Energy Initiatives	-	15,000	15,000	Timing differences		625,000	625,000	0	Other
		5871E	Power Marketer Pilot	108,210	-	(108,210)		Funding Change	185,319	-	(185,319)	Funding Change
		5871F	Battery Storage Communication Upgrades	-	160,000	160,000		Funding Change	1,325,000	1,500,000	175,000	Funding Change
		5887A	ARCOS - Reporting Replacement	298,501	369,376	70,875	Timing differences		795,193	780,689	(14,504)	Other
		5888D	Fluke & WECO Calibration licenses, hardware updates and data migration	71,588	112,000	40,412		Imprecise Estimates	284,213	118,000	(166,213)	Imprecise Estimates
		5889C	Electric Regulatory Enhancements	(22,540)	35,000	57,540		Funding Change	(22,540)	225,000	247,540	Funding Change
		5890A	Corporate PI	432,620	870,279	437,659	Timing differences		1,767,688	1,588,789	(178,899)	Imprecise Estimates
		5890C	Computapole Replacement F&A	0	0	0		Funding Change	464,000	-	(464,000)	Funding Change
		5891A	Electric MV90 Upgrade	368,169	415,510	47,341	Delays		1,210,866	741,520	(469,346)	Complexity
		5891C	FCS Upgrade	317,860	287,803	(30,057)		Scope changes	653,525	294,558	(358,967)	Scope changes
		5891D	IEE Integration/Upgrades	-	110,000	110,000		Funding Change	-	1,150,000	1,150,000	Funding Change
		5891E	MTS Retail Settlement	822,695	1,059,857	237,162	Timing differences		3,448,048	3,530,991	82,944	
		5891F	MDS Consolidation Phase 1	-	25,000	25,000	Timing differences		1,650,000	1,725,000	75,000	
		5891I	Price Transmission (Retail Price) System Replacement	329,077	618,088	289,011	Timing differences		1,053,643	1,005,377	(48,266)	
		5892A	Control Room EOL Server and Tool Replacement	180,307	-	(180,307)		Funding Change	225,307	-	(225,307)	Funding Change
		5892B	Electric Control Center Tool Replacement	-	125,000	125,000		Funding Change	1,119,000	1,380,000	261,000	Funding Change
		5892C	IDS-DVD Dig Safe DVD Replacement	18,261	490,704	472,443		Efficiencies	556,334	1,127,890	571,556	Scope changes
		5893A	Mobility Mwork Replacement for Trouble Workers	7,107	-	(7,107)		Accounting	7,107	-	(7,107)	Imprecise Estimates
		6043	Plan Forward	888	-	(888)		Accounting	888	-	(888)	Imprecise Estimates
		6320	Telecom & Network Deep Dive	20,661	-	(20,661)		Accounting	20,661	-	(20,661)	Imprecise Estimates
		6571	GridAPI - Electric	0	0	0			2,500,000	-	(2,500,000)	Funding Change
		BL	Efficiency Adj						1,948,236		(1,948,236)	Funding Change
		5889D	Electric FY23 Regulatory Mandates and Minor Enhancements	0	0	0			190,000	-	(190,000)	Funding Change
	Electric BU Total			**3,099,196**	**5,161,092**	**2,061,895**			**23,016,801**	**18,907,529**	**(4,109,273)**	
	Wholesale Networks	5773D	NYISO PMU Requirement for Large Generators	250,913	514,642	263,729	Timing differences		1,312,761	736,119	(576,642)	Other
		5822C	Generation Asset Management Upgrade	518,575	891,386	372,811	Timing differences		1,103,205	1,547,623	444,417	Other
		6043G1	Content Management for Engineering	311,772	645,244	333,472		Imprecise Estimates	851,514	1,493,248	641,734	Imprecise Estimates
	Wholesale Networks Total			**1,081,260**	**2,051,272**	**970,012**			**3,267,480**	**3,776,990**	**509,510**	
US Electric Total				**9,180,802**	**14,949,462**	**5,768,660**			**69,399,991**	**68,325,652**	**(1,074,339)**	

Project View - Customer

				Year to Date					Full Year Forecast			
Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Variance	Non_Permanant Reason	Permanent Reason	Year End Forecast	Year End Budget	Variance	Permanent Reason
Customer	Customer Digital	4750J	Web Self Service	2,364,237	1,238,257	(1,125,980)		Complexity	3,561,191	1,238,257	(2,322,934)	Complexity
		4750JX	Web Self Service Decomm	0	0	0						
		4750M	ODS	475,493	1,238,257	762,764		Funding Change	745,651	1,238,257	492,606	Funding Change
		5673	My Business Account (Nucleus Phase 2)	1,219,671	1,381,750	162,079		Timing	5,222,292	5,527,000	304,708	
		6206	Customer Smart Target	0	0	0			2,500,000	-	(2,500,000)	Funding Change
	Customer Digital Total			**4,059,401**	**3,858,264**	**(201,137)**			**12,029,134**	**8,003,514**	**(4,025,620)**	
	Customer Operations	3932	Customer Contact Center / SDC Technology Upgrade	(8,212)	-	8,212		Accounting	(8,212)	-	8,212	Accounting
		4904	Upgrade Servers for CCAE Reporting	(4,357)	-	4,357		Accounting	(4,357)	-	4,357	Accounting
		5125	Customer Connection Program	850,423	860,781	10,358			4,337,053	3,763,025	(574,028)	Scope Change
		5160	NY Stack VDER and Portal Integration	828,661	721,133	(107,528)		Complexity	3,304,038	2,295,586	(1,008,452)	Scope Change
		5345B	Bill Print and Bill Composition Implementation	286,778	398,160	111,382	Prioritizations ($85k)	Funding Change ($25k)	2,624,237	1,055,130	(1,569,107)	Funding Change
		5345B	Bill Print and Bill Composition Implementation									
		5345BX	Bill Print and Bill Composition Decomm									
		5722	KEDLI Rate Case (CRIS)	(1,640)	-	1,640		Accounting	(1,640)	-	1,640	Accounting
		5723	KEDNY Rate Case	313,866	367,167	53,301		Efficiencies	326,994	380,331	53,337	Efficiencies
		5778	Customer Billing Improvement Program - Phase 1	673,934	879,654	205,721	Delays ($190k)		1,020,848	991,583	(29,265)	Complexity
							Prioritizations ($35k)					
		5795F	CDG Net Metering	166,764	215,330	48,566		Efficiencies	292,636	223,254	(69,382)	Funding Change
		5859A	Customer and Agent Experience - IVR Baseline	1,523,825	700,000	(823,825)		Scope changes	5,643,247	2,800,000	(2,843,247)	Funding Change
		5863D	AIMS Stability	(53,029)	-	53,029		Accounting	(53,029)	-	53,029	Accounting
		5926A	NiMO Rate Case	4,523	-	(4,523)		Accounting	4,523	-	(4,523)	Accounting
		5987	Smart City Phase II	4,331	-	(4,331)		Accounting	7,455	-	(7,455)	Accounting
		5996	Simon Baker's Efficiency Target	0	0	0		Accounting	(16,005,694)	-	16,005,694	Accounting
		6007	Inside Agent	86,747	-	(86,747)		Accounting	86,747	-	(86,747)	Accounting
		6032	Customer Billing COVID Transition	35,873	110,496	74,623		Scope changes	286,732	303,258	16,526	Scope changes
		6066	FCP Upgrade for CSS - Plan Forward	336,603	368,964	32,361	Delays		408,440	372,526	(35,914)	Complexity
		6112	WFH Hardware - Plan Forward	(7,443)	-	7,443		Accounting	(7,443)	-	7,443	Accounting
		6214	Customer Operational Enhancements FY22	8,278	-	(8,278)		Accounting	8,278	-	(8,278)	Accounting
		6217	Customer Billing Minor Works FY22	(90,556)	-	90,556		Accounting	(90,556)	-	90,556	Accounting
		6250	EV End-End Solution	-	50,000	50,000		Funding Change		3,000,000	3,000,000	Funding Change
		6302	2022 AIMS Platform	111,682	-	(111,682)		Timing	111,682	-	(111,682)	Timing
		6349	Customer Data Platform	1,043,918	1,252,646	208,728	Delays		5,620,468	5,010,585	(609,883)	Funding Change
		6427	RI Purchase of Receivables (POR) Program	6,256	84,474	78,218		Accounting	6,256	87,197	80,941	Accounting
		6435	Customer PCI Remediation	222,282	389,679	167,397	Delays		373,017	389,679	16,662	
		6492	NACHA Compliance- Security Encryption Effort	257,443	315,703	58,260	Prioritizations	Scope changes	1,865,553	474,850	(1,390,703)	Scope changes
		6496	NY Gender Pronoun and Name Mandate	306,491	858,085	551,594	Timing		1,696,190	1,960,056	263,866	Timing
		6531	AIMS Upgrades and Enhancements	180,157	161,250	(18,907)	Prioritizations		721,214	645,000	(76,214)	Scope changes
		6538	Billing & Collections Mandates & Enhancements	-	104,327	104,327			2,800,000	5,294,426	2,494,426	Scope changes
		6543	Customer Minor Works	104,071	112,500	8,429			441,571	450,000	8,429	
		6584	UWP 2.0 Digital Re Architecture and Optimization	2,507,790	-	(2,507,790)		Funding Change	14,994,238	-	(14,994,238)	Funding Change
		6451	One Step Auto Cashering	34,476	-	(34,476)		Funding Change	660,533	-	(660,533)	Funding Change
	Customer Operations Total			**9,729,934**	**7,950,350**	**(1,779,585)**			**31,471,018**	**29,496,486**	**(1,974,533)**	
Customer Total				**13,789,335**	**11,808,614**	**(1,980,721)**			**43,500,152**	**37,500,000**	**(6,000,152)**	

Project View - DPIT

				Year to Date					Full Year Forecast			
Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Variance	Non_Permanant Reason	Permanent Reason	Year End Forecast	Year End Budget	Variance	Permanent Reason
DPIT	Cloud & Hosting	6200	I&O US Cloud and Hosting FY22	(14,805)	-	14,805		Accounting	(14,805)	-	14,805	Accounting
		6489	I&O US Cloud & Hosting FY23	3,606,684	4,042,110	435,427	Timing differences		13,275,981	16,622,203	3,346,223	Imprecise Estimates
	Cloud & Hosting Total			**3,591,878**	**4,042,110**	**450,232**			**13,261,175**	**16,622,203**	**3,361,028**	
	CNI	6199	I&O US CNI FY22	(35,966)	-	35,966		Accounting	(35,966)	-	35,966	Accounting
		6488	I&O US CNI FY23	66,955	641,985	575,030	Timing differences		4,447,157	4,499,596	52,439	
	CNI Total			**30,989**	**641,985**	**610,996**			**4,411,191**	**4,499,596**	**88,405**	
	Network	5774	WiFi Phase 2	(1,760)	-	1,760		Accounting	(1,760)	-	1,760	Accounting
		6197	I&O US Networks FY22	387,460	-	(387,460)		Imprecise Estimates	255,460	-	(255,460)	Imprecise Estimates
		6486	I&O US Networks FY23	4,029,917	3,916,745	(113,173)			19,258,004	19,300,124	42,120	
	Network Total			**4,415,618**	**3,916,745**	**(498,873)**			**19,511,705**	**19,300,124**	**(211,581)**	
	Operations	5876	Pure Technical Debt Remediation	(35,794)	-	35,794		Accounting	(35,794)	-	35,794	Accounting
		5997	Caitlin Davidson's Efficiency Target						(7,261,109)		7,261,109	
		6114	Managed Workspace Services Transformation	3,095	-	(3,095)		Timing	3,095	-	(3,095)	Timing
		6196	I&O US Operations FY22	(179,557)	-	179,557		Accounting	(670,838)	-	670,838	Accounting
		6485	I&O US Digital Operations FY23	492,989	2,786,710	2,293,721	Timing differences		11,582,545	11,318,284	(264,261)	
	Operations Total			**280,733**	**2,786,710**	**2,505,977**			**3,617,899**	**11,318,284**	**7,700,385**	
	Other	6520US	JIRA	-	300,000	300,000	Timing differences		-	900,000	900,000	Timing
		Apptio	Apptio	0	0	0			-	600,000	600,000	Timing
		EHUB	E-Hub	-	200,000	200,000	Timing differences		-	800,000	800,000	Timing
	Other Total			-	**500,000**	**500,000**			-	**2,300,000**	**2,300,000**	
	Data Center	6223	I&O US Data Center Transformation FY22	3,248,117	6,428,188	3,180,071		Imprecise Estimates	29,409,603	18,268,518	(11,141,086)	Imprecise Estimates
	Data Center Total			**3,248,117**	**6,428,188**	**3,180,071**			**29,409,603**	**18,268,518**	**(11,141,086)**	
	Digital Platforms	6519	DP GridStack	3,764,202	4,385,563	621,361	Timing differences		15,407,106	15,943,364	536,258	Timing
		6565	DP Secure & Compliant	-	142,190	142,190			3,197,318	2,656,677	(540,641)	Imprecise Estimates
	Digital Platforms Total			**3,764,202**	**4,527,753**	**763,552**			**18,604,424**	**18,600,041**	**(4,383)**	
	Digital Workplace	4987	SOE (Windows) Upgrade and Device Refresh	723	-	(723)		Timing	723	-	(723)	Timing
		6198	I&O US End User Experience FY22	152,524	-	(152,524)	Accruals		135,426	-	(135,426)	Accruals
		6487	I&O US End User Experience FY23	1,003,198	3,731,452	2,728,253	Timing differences		15,149,997	15,150,061	64	
	Digital Workplace Total			**1,156,445**	**3,731,452**	**2,575,006**			**15,286,146**	**15,150,061**	**(136,086)**	
DPIT Total				**16,487,983**	**26,574,943**	**10,086,960**			**104,102,144**	**106,058,827**	**1,956,683**	

Project View - Security

Portfolio	Program	Project INVP	Project Name	Year to Date Actual	Year to Date Budget	Year to Date Variance	Non_Permanant Reason	Permanent Reason	Full Year Forecast Year End Forecast	Year End Budget	Year End Variance	Permanent Reason
Security	**Physical Security**	3617A	Physical Security - G20	43,590	-	(43,590)		Funding Change	503,590	-	(503,590)	Funding Change
		3617F	PS - NIMOE	68,445	-	(68,445)		Funding Change	2,318,445	-	(2,318,445)	Funding Change
		3617G	Physical Security - KEDNY - Equip Replacement	56,379	-	(56,379)		Funding Change	526,379	-	(526,379)	Funding Change
		3617H	Physical Security - KEDLI - Equip Replacement	27,533	-	(27,533)		Funding Change	497,533	-	(497,533)	Funding Change
		3617I	Physical Security - MASS Electric - Equip Replacement	14,270	-	(14,270)		Funding Change	464,270	-	(464,270)	Funding Change
		3617J	Physical Security - Boston Gas	59,444	-	(59,444)		Funding Change	1,009,444	-	(1,009,444)	Funding Change
		3617K	Physical Security - Colonial Gas	18,437	-	(18,437)		Funding Change	228,437	-	(228,437)	Funding Change
		3617L	Physical Security - RI Electric	0	0	0		Funding Change	410000	0	-410000	Funding Change
		3617M	Physical Security - RI Gas	86,990	-	(86,990)		Funding Change	436,990	-	(356,990)	Funding Change
		3617	Physical Security	-	980,000	980,000		Funding Bucket		6,976,667	6,976,667	Funding Bucket
	Physical Security Total			**375,088**	**980,000**	**604,912**			**6,395,088**	**6,976,667**	**661,579**	
	Security	3683USF	Enterprise Centralized Patch Management									
		3683USH	Network Security Improvements									
		3683USN	SOAR - Security Orchestration Automation and Response									
		5702USA	Malware Defense - Endpoint Protection									
		5820	Data Protection									
		5848	Vulnerability Management									
		5850	Network Security									
		5850US	Network Security US									
		5850USA	SSN Optimization US									
		5850USD	Intrusion Detection System and Intrusion Prevention System US									
		5870	Identity and Access Management									
		5933	Operational Technology Intrusion Detection									
		5940	Global VM Enterprise									
		5998	Efficiency Target									
		6210US	FY22 Platform Tower									
		6211US	FY22 Network Tower									
		6245US	Security Orchestration Automation & Response									
		6416	Operational Technology Privileged Access Workstation									
		6417US	Vulnerability Management T2 FY22									
		6480us	Platform Security									
		6526us	Network Security									
		6540us	Security Orchestration Automation and Response									
		6510	OT PAW FY23									
		6417	Vulnerability Management T2 FY22									
		6526	FY23 Network Program									
		6613US	FY23 Identity and Access Management									
	Security Total											
Security Total												

Impacts Year end Forecast

Category	Type of Variance	Explanation
Project Delivery - Scope change	Scope changes	Spend impacted by both reduced or increased scope
Project Delivery - Initial scope delivered	Imprecise Estimates	Cost and Quantity of resources were incorrectly assessed when creating the budget. (eg. Price Increases)
Project Delivery - Initial scope delivered	Complexity	Unforeseen challenges in delivering the solution drove costs up. No change in scope but amount of work to deliver increased or potentially decrease
Project Delivery - Initial scope delivered	Efficiencies	More cost effective implementation options - Found a better way to deliver the solution which reduced costs
Strategic Direction	Funding Bucket	Source of undefined funding where funds will be moved to other project
Strategic Direction	Funding Change	Move $ to or from another project (please specify INVP moving to or moving from
Annual Accounting	Accounting	Over or under accruals which reverse at year end leaving a balance on the INV
Annual Accounting	Timing	Project slipped and intended scope was not delivered this fiscal yea
Other	Other	Please explain

No impact to forecast

Category	Type of Variance	Explanation
Accounting Errors	Mischarges	Temporary due to unavailable accounting codes or charging errors which will be corrected
Accounting Errors	Accruals	which may have been missed / over accrue
Accounting Errors	Misclassifications	eg. Between opex and capex or expense types
Timing Changes	Timing differences	such as budget phasing, early / later agreed payments or delivery methodology impacting phasing of costs
Timing Changes	Prioritizations	Projects will begin earlier or later than originally planned - Management Determined
Timing Changes	Delays	Materials (chip shortage), Procurement issue (negotiations), resource issues (internal and external) - Beyond Management Contro
Other	Other	Please explain

Niagara Mohawk Power Corporation d/b/a National Grid

The Brooklyn Union Gas Company d/b/a National Grid NY

KeySpan Gas East Corporation d/b/a National Grid

Cases 20-E-0380, 20-G-0381, 19-G-0309 and 19-G-0310

INFORMATION TECHNOLOGY CAPITAL INVESTMENT REPORT FOR QUARTER ENDED SEPTEMBER 30, 2022

SUBMITTED TO:
NEW YORK STATE DEPARTMENT OF PUBLIC SERVICE STAFF

November 29, 2022

nationalgrid

Information Technology Capital Investment Report for Quarter Ended September 30, 2022

I. Introduction

Niagara Mohawk Power Corporation (Niagara Mohawk), The Brooklyn Union Gas Company (KEDNY), and KeySpan Gas East Corporation (KEDLI) (together "National Grid" or the "Companies") submit this Information Technology ("IT") Capital Investment Quarterly Report ("IT Quarterly Report") for the second quarter (Q2) of fiscal year 2023 ("FY23").[1] Pursuant to the Public Service Commission's ("Commission") January 20, 2022 Order in Cases 20-E-0380 and 20-G-0381, and August 12, 2021 Order in Cases 19-G-0309 and 19-G-0310, the Company is to provide quarterly variance reports to Department of Public Service Staff ("Staff") regarding IT capital projects and investments.[2] The IT Quarterly Report provides an explanation of the Company's overall IT investment plan approach, descriptions of program investment categories, and a comparison of budgeted to actual spending.

IT capital projects and investments that are shared investments across National Grid's operating companies are implemented and owned by National Grid USA Service Company, Inc. ("Service Company") and allocated to the benefiting US operating companies, including Niagara Mohawk, KEDNY, and KEDLI, in the form of Service Company rent expense once the investments are in-service. The Service Company rent expense includes a return on the capital investment (net of deferred taxes) plus booked depreciation expense.

Through Q2 FY23, National Grid spent $144.9 million compared to a year-to-date Service Company budget of $146.9 million, representing a $1.9 million underspend. Attachment 1 includes detail on budgets and actual spending for each of the portfolios and commentary regarding performance which has been presented to each of the Chief Information Officers responsible for the respective portfolios.

Attachment 2 includes a detailed breakdown of individual project spend by portfolio compared to the budgeted spend for each of those investments.

II. Program Categories/Spending Rationales

National Grid has organized the US Information Technology organization into six areas:

[1] The second quarter of FY23 is July 1, 2022 through September 30, 2022 ("Q2 FY23").

[2] *See* Cases 20-E-0380 and 20-G-0381, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service*, "Order Adopting Terms of Joint Proposal, Establishing Rate Plans and Reporting Requirements" (issued and effective January 20, 2022), Joint Proposal, Section IV.10.5; and Cases 19-G-0309 and 19-G-0310, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Energy East Corporation d/b/a National Grid for Gas Service*, "Order Approving Joint Proposal, as Modified, Imposing Additional Requirements" (issued and effective August 12, 2021), Joint Proposal, Section IV.6.5.

(1) Group Functions
(2) United States Electric
(3) United States Gas
(4) Customer
(5) Digital Platforms and Infrastructure Technology (Formerly Infrastructure and Operations)
(6) Security

IT capital investment is tracked based on these area designations. Descriptions of each portfolio category/spending rationale are provided below.

Group Functions

Group Functions executes National Grid's digital roadmaps, leveraging technology and reimagining processes in support of Finance, Enterprise Business Services, Human Resources, Legal, and Safety Health & Environmental, continuing to upgrade SOX related systems, and retiring outdated systems.

United States Electric

Provides IT Support to our US Electric Distribution and Transmission businesses. Modernizing our key legacy operational systems while building capabilities that will enable the energy transition and transformation to a Distribution System Operator (DSO). Through digital delivery, the Electric CIO organization provides greater efficiency and data driven insights into our workforce operations that provide:

- More automated and reliable grid controls systems
- Capital programs and projects that support the proliferation of Distributed Energy Resources (DERs)
- A growing market of prosumer electric customers

United States Gas

Provides IT Support to our US Gas Distribution and Transmission businesses by developing digital products based on the foundation of gas business enablement that deliver continuous improvement of our asset lifecycle management capabilities through:

- Risk-based asset investment planning
- Performance-based maintenance
- AI-based work optimization
- Mobile field products

Customer

Provides IT Support to our US Customer business, delivering best-in-class digital products that increase engagement and reduce costs to serve by:

- Modernizing our legacy "back-end" IT estate to support more digital products
- Liberating data and creating an abstracted and real-time customer data platform to assist with achieving energy efficiency targets

Digital Platforms and Infrastructure Technology (DPIT) (Formerly Infrastructure and Operations)
Supports our internal IT Architecture covering Cloud and Hosting, Critical Network Infrastructure, Networks, End User Experience and Operations. DPIT is focused on reducing National Grid's technical debt through cost effective, modernized IT infrastructure, enabling cloud capabilities that meet our risk profile. The organization enables a connected, seamless enterprise that is flexible, secure, reliable and scalable while maintaining acceptable levels of technical debt and risk.

Security
The Security function protects our people, premises, and digital systems with the objective of maintaining trust in National Grid services, focusing on:

- Reduction in likelihood and impact of security incidents
- Compliance with relevant laws, standards and regulations (*e.g.*, NERC, PCI DSS, GDPR, NIS-R)
- Improved user experience and standard of security practices across National Grid

The Security function aims to reduce the likelihood and impact of physical and cyber security incidents affecting National Grid. It does this by championing security best practices, technology, and promoting a security-awareness culture.

Other Category
For accounting purposes, there is an additional category listed in Attachment 2 to this report. The "Other" category has 3 elements:

- The "IT4IT" sub-category acts as a funding bucket for NPV-positive projects to enable efficiency savings in our IT-run business operations.
- The "Other" sub-category is the allocation of operating expense overheads such as Enterprise Architecture and Solution Delivery charges to a centralized account. The capital charges for this overhead are made directly to individual projects.
- Finally, "Burdens" are the net removal of operating expense burdens to ensure our reporting agrees with our Corporate Controllable Cost reporting. Operating expense burdens are consolidated and reported separately under non-controllable costs in our corporate reporting packs.

The summaries below explain the subjects presented in the attachments to the Q2 FY23 report.

A. Attachment 1

1. Summary (pages 2 – 3) – Overview of US IT as a whole.
    a. High-level executive summary, including work and challenges faced across the portfolios and a brief explanation around the forecast (p. 2).
    b. Actual expenditure versus budget by portfolio year to date, and full fiscal year forecast versus annual budget by portfolio (p. 2).
    c. Actual and forecasted spend by month for FY23 compared to budget, prior year and run rate. Each bar is color-coded to give an overview of the comparative spend per portfolio (p. 3).

2. Portfolio Sections – The remainder of Attachment 1 includes six sections, one for each of the portfolios: Group Functions; US Gas; US Electric; Customer; DPIT; and Security. Each portfolio consists of 3 pages:
    a. The first page provides a narrative financial and business overview. The three charts on the first page show: (1) operating expenses; (2) capital expenses; and, (3) total expenses. Each chart shows FY23 actuals and forecast by period versus budget, with trend lines showing prior year expenditure and run rate. This page also shows the full year forecast and year to date total and capital expenditure compared to the budget.
    b. The second page of the portfolio sections has two charts on the top left of the page: the top chart is actual expenditure versus budget by program year to date. The second chart compares full fiscal year forecast versus annual budget by program. There is also a key challenges section and the escalation point for that challenge. On the bottom half of this page is a chart showing actual and forecasted spend for FY23 compared to budget, prior year and run rate for the portfolio. Each bar is color-coded to give an overview of the comparative spend per program.
    c. The third page of each portfolio section provides a status update on the various projects. A priority investments list for the portfolio and their status appears at the top right hand of the page. Blue means complete, green means on track, amber means some risk, and red means focused intervention required.

    Attachment 1 is based on National Grid's monthly Strategic Portfolio Reviews (SPR) and reporting cadence to senior management of each portfolio, and ultimately to the IT leadership team.

B. Attachment 2

INVP Level Variances - This attachment provides a detailed listing of individual projects in each of the six portfolios explaining spend in FY23, including:

(1) Total actual spend compared to budget and variance;
(2) A variance reason for all capital investments which are outside the 10% tolerance to budget.

National Grid

Quarterly Report – Q2 FY23
Information Technology

29th November 2022

nationalgrid

US IT

Financial Performance

Portfolio View - Year to Date $M

Portfolio	Capex			Opex			Totex		
	Actuals YTD	Budget YTD	Variance	Actuals YTD	Budget YTD	Variance	Actuals YTD	Budget YTD	Variance
US Electric	17.5	27.9	10.4	1.6	4.0	2.5	19.1	32.0	12.9
US Gas	14.4	11.9	(2.5)	1.5	2.0	0.5	16.0	13.9	(2.1)
Customer	23.7	18.9	(4.7)	3.1	2.6	(0.6)	26.8	21.5	(5.3)
Group Functions	18.3	18.6	0.2	4.3	4.0	(0.3)	22.6	22.6	(0.1)
DPIT	34.3	48.5	14.2	2.3	6.1	3.9	36.5	54.7	18.1
Security	12.3	18.2	5.9	2.3	3.8	1.5	14.6	22.0	7.4
Other	0.0	(14.6)	(14.6)	9.3	(5.2)	(14.5)	9.3	(19.8)	(29.1)
Total	**120.5**	**129.5**	**9.0**	**24.4**	**17.4**	**(7.0)**	**144.9**	**146.9**	**1.9**

Portfolio View - Full Year $M

Portfolio	Capex			Opex			Totex		
	FY Forecast	FY Budget	Variance	FY Forecast	FY Budget	Variance	FY Forecast	FY Budget	Variance
US Electric	48.7	59.5	10.8	8.7	8.8	0.1	57.4	68.3	11.0
US Gas	40.4	28.0	(12.4)	3.5	4.5	1.0	43.8	32.5	(11.3)
Customer	44.9	33.5	(11.4)	5.4	4.0	(1.4)	50.3	37.5	(12.8)
Group Functions	38.7	38.5	(0.3)	6.1	7.4	1.2	44.9	45.8	0.9
DPIT	87.2	92.7	5.5	6.3	13.3	7.0	93.5	106.1	12.5
Security	34.0	42.1	8.1	4.5	7.1	2.6	38.5	49.2	10.7
Other	1.4	4.9	3.5	12.4	(3.4)	(15.8)	13.7	1.5	(12.2)
Total	**295.2**	**299.3**	**4.0**	**46.9**	**41.7**	**(5.2)**	**342.1**	**341.0**	**(1.2)**

IT Commentary

Delivery

Progress on delivery continues with overall spend in line with expectations. Significant releases this quarter include The Electric Price Transmission System, and Unified Web Portal in Upstate New York and Long Island.

Challenges

Commercial and contracting challenges and chip shortages still exist in the post-COVID era.

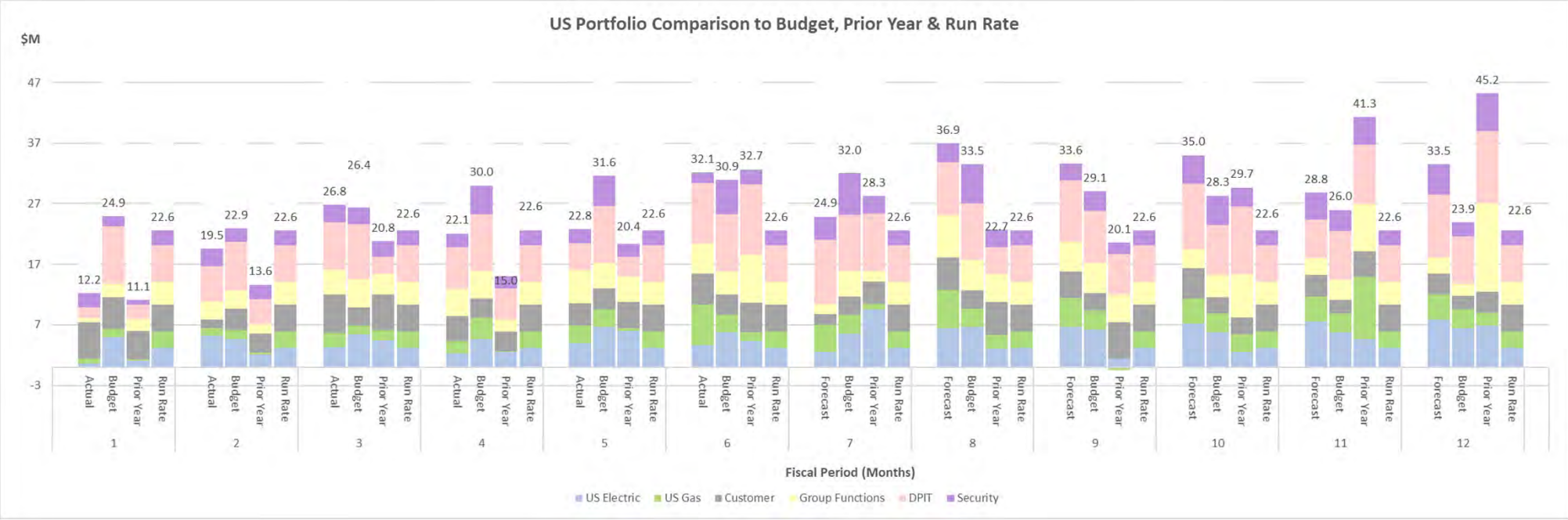
Specialist IT resources and business resources are proving difficult to secure. Management is working through these issues to ensure capabilities are delivered as scheduled. Although there are challenges in individual portfolios, across IT the situation is being managed.

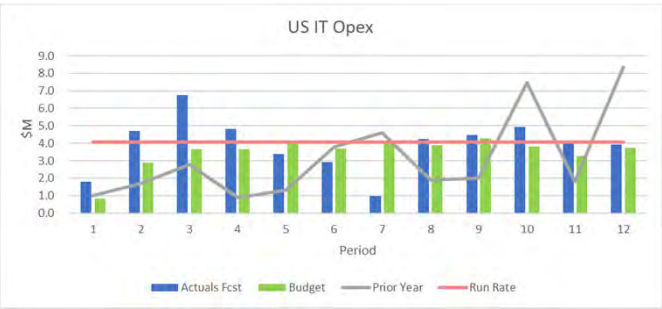
Financial Performance

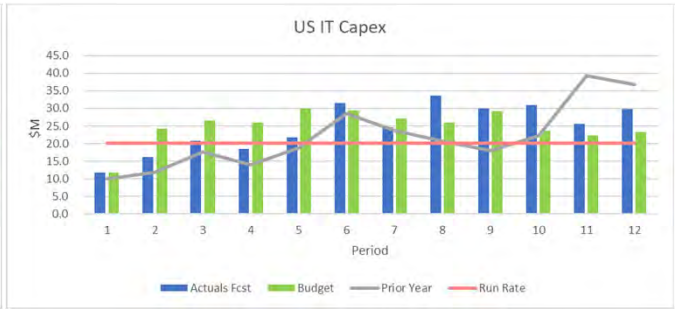
Financial performance is on budget. The 'Other' category is offsetting the underspend in the Electric and DPIT portfolios, with $20.5m budget phasing adjustment (following refinement of business plans).

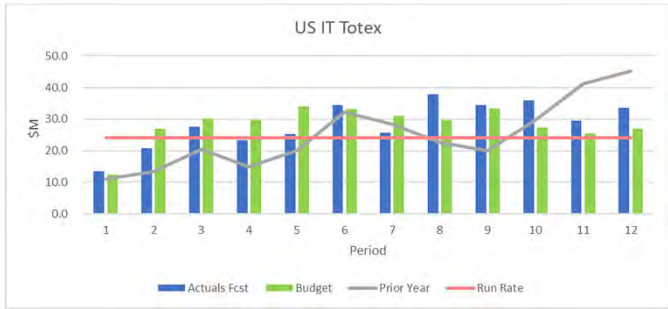
There is upward pressure on the year-end forecast, particularly in the Customer and US Gas portfolios, to provide desired customer and user enhancements. For example, Unified Web Portal 2.0 and Interactive Voice Response (IVR) are driving this demand.

Portfolio Comparison & Cost Charts







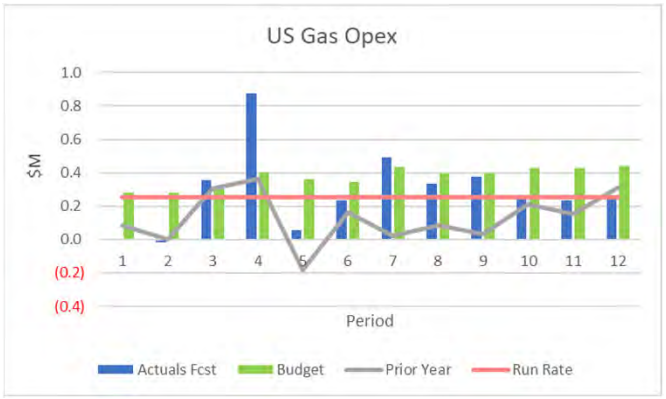


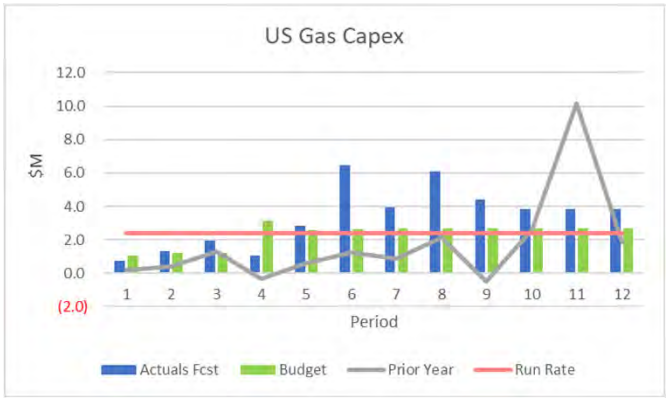
US Gas
Q2 FY23 Report

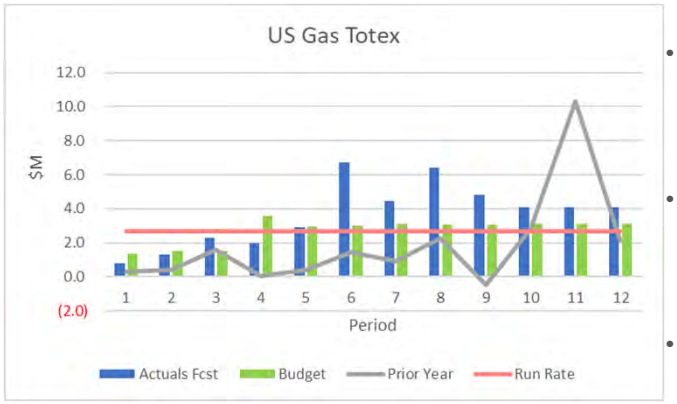
nationalgrid

US Gas

Financial Performance







Totex	Budget ($M)	Forecast ($M)
	32.5	43.8
	YTD Budget($M)	YTD Actual ($M)
	13.9	16.0

Capex	Budget ($M)	Forecast ($M)
	28.0	40.4
	YTD Budget($M)	YTD Actual ($M)
	11.9	14.4

Portfolio Commentary

- U.S. Gas variance is largely due to the CS&E program, which is driving the forecast from the original plan and year to date variance to budget, with overruns on vendor labor and misalignment on where NG labor costs would fall. Re-sanction for the program is almost complete and we're on forecast with the revised sanction budget.
- The Sanction process has been initiated for Gas Meter Management System Deployment and Integration (Tesco), FY23 - $3.182M, FY24 - $3.203M.
- Asset Management for LNG had multiple releases in Q2 that enabled integration with PowerPlan for unitization of asset data to improve asset tracking. It also provided compliance and audit reporting.
- PI replacement project in second phase; enabled interfaces with Primavera. It also enabled Service Now to access Primavera Unifier for capital project initiation with open authorization enabled.
- Gas I&R, Meter Operations, and field device replacement program replaced 124 aging tough books with Win. 10, rugged laptops and mitigated cybersecurity risks. Field teams now have personas defined for requesting and using laptops.
- New Storms Response Management System went live in Q2, implementing a new SaaS solution that is more reliable and automated, replacing an outdated and unsupported tool. This new system will digitize securing, mobilizing crews, and provides logistical support during storms and restoration events.

US Gas Program View of Financials

Gas Program View Year to Date $M

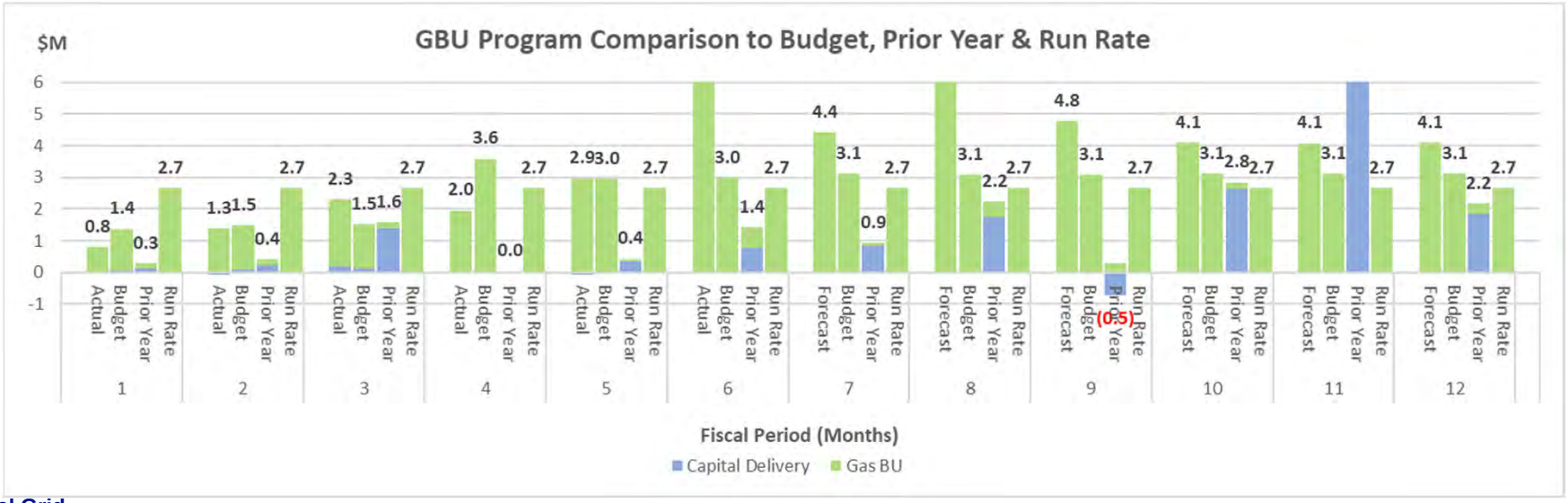
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
US Gas	Capital Delivery	0.0	0.3	0.2	(0.0)	0.0	0.0	0.0	0.3	0.2
	Gas BU	14.4	11.7	(2.7)	1.5	2.0	0.4	15.9	13.6	(2.3)
		14.4	**11.9**	**(2.5)**	**1.5**	**2.0**	**0.5**	**16.0**	**13.9**	**(2.1)**

Gas Program View Full Year $M

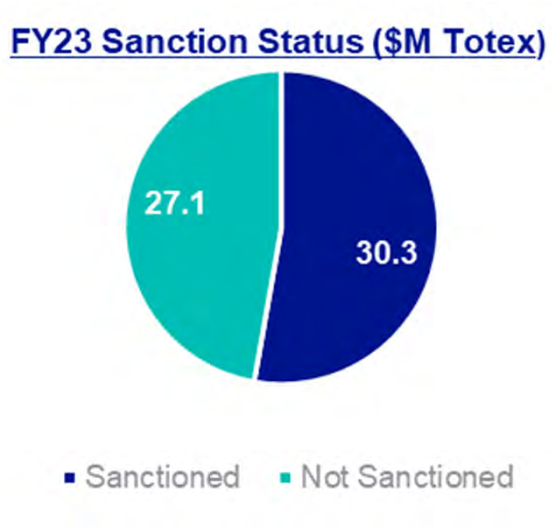
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
US Gas	Capital Delivery	0.0	0.3	0.2	0.0	0.0	0.0	0.1	0.3	0.2
	Gas BU	40.3	27.8	(12.6)	3.5	4.5	1.0	43.8	32.2	(11.6)
		40.4	**28.0**	**(12.4)**	**3.5**	**4.5**	**1.0**	**43.8**	**32.5**	**(11.3)**

Key Challenges

Challenge	Escalation Point
FCS(Gas Only) delayed to the dependency on completion of in Flight FCS Software upgrade by IT Electric	None



Mobilisation Summary



Status

- Based on 3+9 forecast, 6 projects left to sanctioned for FY23

Resources

- 3 new Project Managers have been hired, onboarded and are leading FY23 projects
- Overruns on vendor labor and misalignment on NG for CS&E Program

Scope

- Remaining pipeline projects need to complete scope assessment as part of business case analysis and sanctioning process.
- Additional Meter Shop project added to portfolio scope in FY23

Priority Investments

Program	Status	FY 23 Budget ($m)	RAG This Month	RAG Last Month
Gas Storm Hardening	Inflight	0.4		
LNG Asset & Work Mgmt.	Inflight	1.6		
Summons for Corrective Actions	PIS	0.1		
Enterprise Scalability and Resilience	Inflight	1.2		
Continuous Improvement for Gas Applications	Inflight	1.2		
PIF Replacement	PIS	0.0		
LNG IT support for OT	Start-up	0.9		
Gas I&R, Meter Ops, and Field Device Replacement	Inflight	0.1		
Enterprise Platform for Gas Retail Choice (GTIS)	Inflight	2.0		
Field Collection Systems (FCS) Upgrade	Prestart up	0.8		
Gas Management System (SCADA) Upgrade	Start-up	2.1		
New Gas DR Solution Evaluation	Inflight	0.9		
New Storms Response Management System Implementation	Inflight	0.3		
MOC Library & Action Item Automation	Inflight	0.2		
Continued Scaling & Evolution of Gas IT Transformation Program	Prestart up	20.1		
SOP Third-Party User Access for PE Review	Cancelled	0.1		
Enable Gas Curtailment Map in ESRI	Cancelled	0.04		
New System Operating Procedure Solution	Deferred/ FY24	0.1		
PIPES/Enterprise Maximo Integration	June decision	0.1		
Digital Asset & Work Management for CNG and I&R	Deferred/ FY25	0.3		
Enable Converted LI Gas Distribution Asset Data in ESRI	Deferred/ FY24	0.1		

US Electric
Q2 FY23 Report

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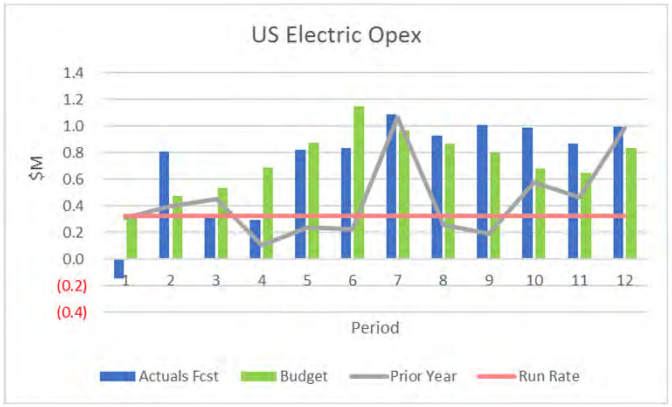
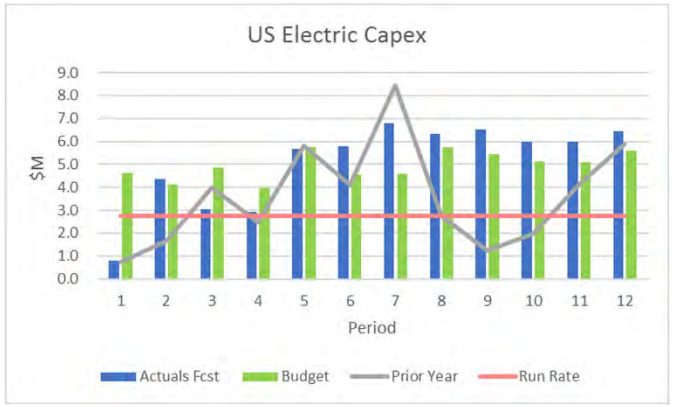
US Electric

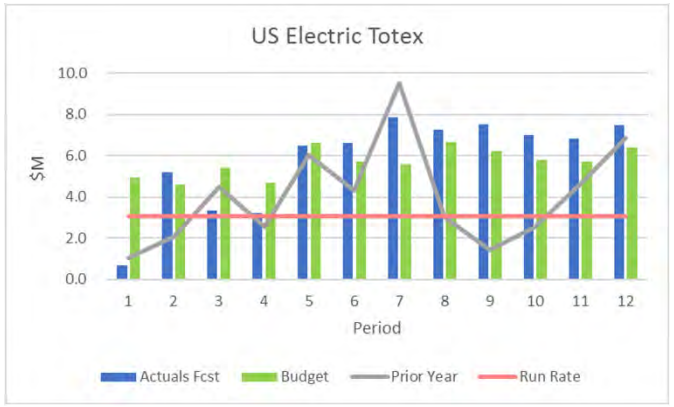
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	68.3	59.0
	YTD Budget($M)	YTD Actual ($M)
	32.0	19.1

Capex	Budget ($M)	Forecast ($M)
	59.5	50.2
	YTD Budget($M)	YTD Actual ($M)
	27.9	17.5



Portfolio Commentary

- Year to date under-spend of $10.4M CAPEX and $2.5M OPEX is being driven by supply chain delays, resource constraints, and various deliverability constraints that are delaying the start to planned products.
  - These delays are impacting the full year forecast and will reduce forecast v budget by $10.9M.
- **Price Transmission** successfully went live on October 14th. This enabled NGRID to remove a legacy vendor platform (tech debt) and build a Price Transmission system. This system receives hourly electricity prices a day ahead from the NYISO, based on zone. It then calculates our pricing based on the electric prices providing data to our Billing dept. to bill customers timely and accurately.
- **ADMS**: Through Q2, 1,040 of 2,800 (37%) US feeders are now ADMS ready, further extending our ability to visualize, monitor and simulate the Grid in real time; the program is on track to reach 50% FY23 Target.
- **UTOC (Unified Technology Operations Center) MVP** was successfully launched on September 1st at the Syracuse Office Complex (SOC). This will help improve response times to Network Outages, and provide additional data/network monitoring capabilities with our Remote Radio Devices (RRD) that support connectivity to:
  - Advanced Meter Infrastructure (AMI)
  - Volt/Var Optimization (VVO)
  - Fault Location, Isolation & Service Restoration (FLISR)

US Electric Program View of Financials

Electric Program View Year to Date $M

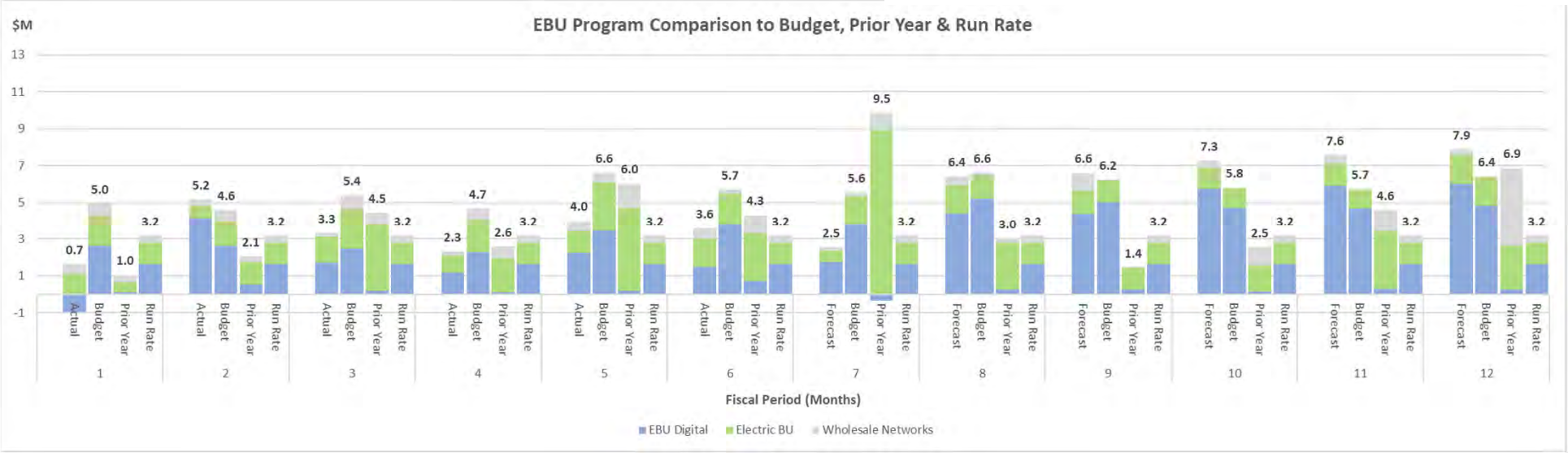
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
US Electric	EBU Digital	8.7	14.9	6.2	1.1	2.4	1.4	9.8	17.3	7.5
	Electric BU	6.4	9.7	3.3	0.4	1.5	1.1	6.8	11.2	4.4
	Wholesale Networks	2.4	3.3	1.0	0.1	0.1	(0.0)	2.5	3.4	0.9
	Totals	**17.5**	**27.9**	**10.4**	**1.6**	**4.0**	**2.5**	**19.1**	**32.0**	**12.9**

Electric Program View Full Year $M

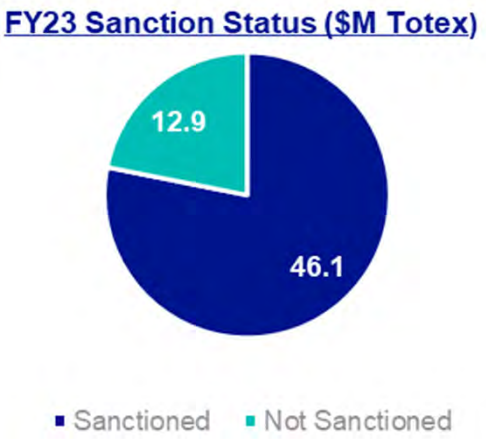
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
US Electric	EBU Digital	32.5	40.2	7.7	5.5	5.4	(0.1)	38.0	45.6	7.6
	Electric BU	11.0	15.6	4.6	3.1	3.3	0.2	14.1	18.9	4.8
	Wholesale Networks	5.1	3.6	(1.5)	0.1	0.1	0.0	5.3	3.8	(1.5)
	Totals	**48.7**	**59.5**	**10.8**	**8.7**	**8.8**	**0.1**	**57.4**	**68.3**	**11.0**

Key Challenges

Challenge	Escalation Point
Resource availability and resource quality challenges with ADAM Partners in the Core Portfolio; significant challenge within the Digital Lighthouse Portfolio.	CIO / HR / Commercial *(ADAM Partners)*
Global Supply Chain delays with critical hardware needed to deploy key programs/products (i.e. ADMS, resolved NYISO-PMU)	Commercial, ITLT
Generation Asset Management Upgrade is experiencing schedule delays due to reprioritization and availability of integrated environments that are necessary for development and testing; *multiple business units are engaging the same pool of resources/environments*	CIO, ITLT
Content Management team is manually converting data into the current solution during off hours (nights/weekends) to mitigate schedule risks/delays due to memory limitations with existing servers.	CIO, ITLT



Mobilisation Summary



Status

- Based on 6+6 forecast, ~78% of planned product spend has been Sanctioned.

Resources

- New Program Manager, overseeing Product Managers, has been successfully onboarded to ensure Products stay on track and to scope.
- US Electric CIO Product Operating Model on track to be rolled out by Fiscal Year End

Scope

- Remaining FY23 pipeline projects are beginning to complete scope assessment as part of business case analysis and sanctioning process.

Priority Investments

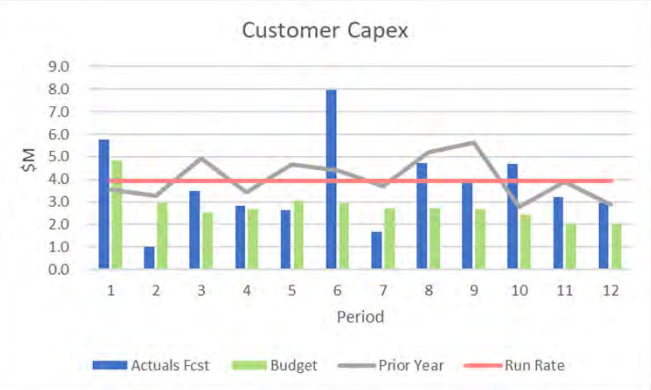
Program	Status	Budget ($m)	RAG This Month	RAG Last Month
GridMod	Inflight	45.0		
ADMS	Inflight	24.2		
FutureNow	Inflight	11.2		
VMO	Inflight	8.6		
OMW	Inflight	3.8		
Electric MV90 Upgrade	Inflight	0.7		
FCS Upgrade	Complete	0.3		
Price Transmission	Complete	1.0		
Corporate PI	Inflight	1.6		
NYISO PMU – Large Generators	Inflight	0.7		
Content Management	Inflight	1.5		
Generation Asset Management Upgrade	Inflight	1.6		
Control Room EOL Server and Tool Replacement	Inflight	1.4		
Retail Settlement	Inflight	3.5		

Customer
Q2 FY23 Report

nationalgrid

Customer

Financial Performance



Totex	Budget ($M)	Forecast ($M)
	37.5	50.3
	YTD Budget($M)	YTD Actual ($M)
	21.5	26.8

Capex	Budget ($M)	Forecast ($M)
	33.5	44.9
	YTD Budget($M)	YTD Actual ($M)
	18.9	23.7

Portfolio Commentary

- The Customer portfolio remains on track to meet additional demand from the business over the initial plan.
  - Demand for services has remained steady from Q1 to Q2; reallocation of funding identified to meet demand.
  - Unified Web Portal 2.0 – significant driver of the year to date increase over budgeted spend; will provide an improved user experience.
  - Zero Based Redesign (ZBR) – Continues to drive high value on Interactive Voice Response (IVR) (the other significant driver to year to date overspend); delivering improved user experience and increased number of self-service customer resolutions.
  - Clean Energy – Project has kicked off while finalizing business funding for NY portion of work. Will increase the level of customer engagement and third parties requesting clean energy products.
  - Technical Debt – Targeting technical debt remediations that support the Customer System and the Customer offices.
- Major releases in Q2 included;
  - Extending the Unified Web Portal to the final territories, LI and UNY.
  - Advanced Metering Infrastructure (AMI) Minimum Viable Product delivered with 5 live meters producing bills.
  - Significant improvements to the stability of core systems with the Foundation for Cooperative Processing (FCP) upgrade and completed remediation to ensure Payment Card Industry (PCI) compliance.
  - Implemented Community Distributed Generation (CDG) Net Metering – eliminates manual billing processes of customers participating in CDG billing program.

Customer Program View of Financials

Customer Program View Year to Date $M

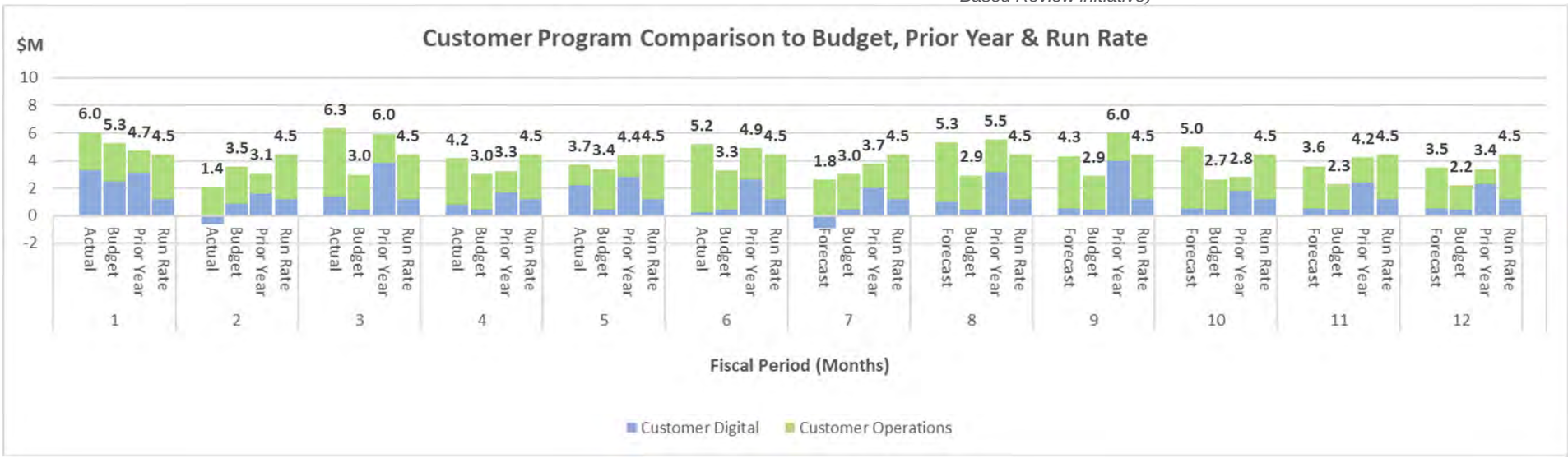
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Customer	Customer Digital	6.6	4.4	(2.2)	0.8	0.8	0.0	7.4	5.2	(2.1)
	Customer Operations	17.1	14.5	(2.6)	2.3	1.8	(0.6)	19.4	16.3	(3.2)
	Totals	**23.7**	**18.9**	**(4.7)**	**3.1**	**2.6**	**(0.6)**	**26.8**	**21.5**	**(5.3)**

Customer Program View Full Year $M

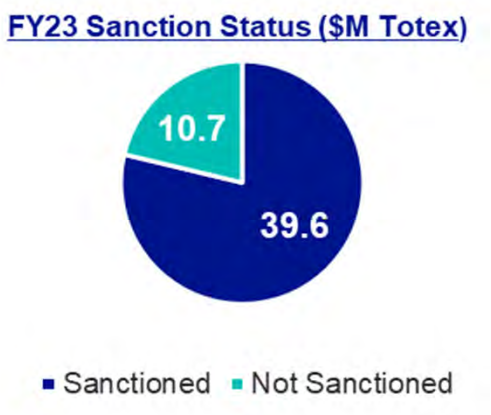
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Customer	Customer Digital	8.2	6.8	(1.3)	1.4	1.2	(0.3)	9.6	8.0	(1.6)
	Customer Operations	36.7	26.7	(10.1)	4.0	2.8	(1.1)	40.7	29.5	(11.2)
	Totals	**44.9**	**33.5**	**(11.4)**	**5.4**	**4.0**	**(1.4)**	**50.3**	**37.5**	**(12.8)**

Key Challenges

Challenge	Escalation Point
***Business Resources**: Business resources have been prioritized to transformation efforts including ZBR and CRIS to CSS. Working resource plan with Customer and HR*	Simon Baker
***Data Center Transition Dependency**: CRIS to CSS has a significant dependency on Data Center migration. Data Center migrations for Customer systems are planned in late Oct/early Nov*	Simon Baker and Shannon Soland
***Virtual Assistant Timeline** – The business case timeline for VA is at risk due to technical dependencies. An interim solution has been defined to target initial customer benefits while the longer term solution is developed. (Part of the Zero Based Review initiative)*	Simon Baker



Mobilisation Summary



Status

- A significant amount of the FY23 plan has been sanctioned

Resources

- All planned Q3 projects are active and resourced

Scope

- The significant project to be sanctioned is the UWP 2.0 project which is a re-architecture and design of the web platform for sustainability and scalability of customer experience.

Priority Investments

Program	Status	Budget ($m)	RAG	RAG Last Month
Zero Based Redesign	Inflight	21.0		
CRIS to CSS	Inflight	36.2	New Approved Date	
AMI	Inflight	22.3		
Clean Energy	Start up	9.3		
UWP Launch	Inflight	3.0		
Fix The Technology Foundations: UWP2.0	Start up	10		
Customer Data Platform	Inflight	7.0		

Notes:

UWP Launch has since gone live in all territories.

Zero Based Redesign has made great progress with IVR Self Service Resolutions and is rescoping plan for Digital Assistant implementation

Clean Energy resourcing plan finalized and project will be moving beyond start up

US Security
Q2 FY23 Report

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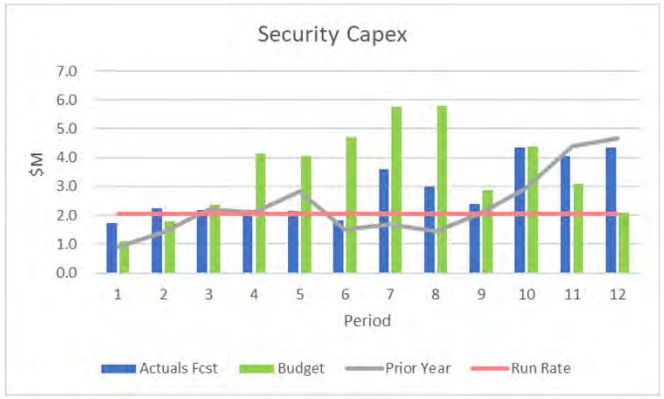
Security

Financial Performance



Totex	Budget ($M)	Forecast ($M)
	49.2	38.5
	YTD Budget($M)	YTD Actual ($M)
	22.0	14.6

Capex	Budget ($M)	Forecast ($M)
	42.1	34.0
	YTD Budget($M)	YTD Actual ($M)
	18.2	12.3

Portfolio Commentary

- All projects have been sanctioned.
- [redacted]

[redacted] Program View of Financials

Security Program View Year to Date $M

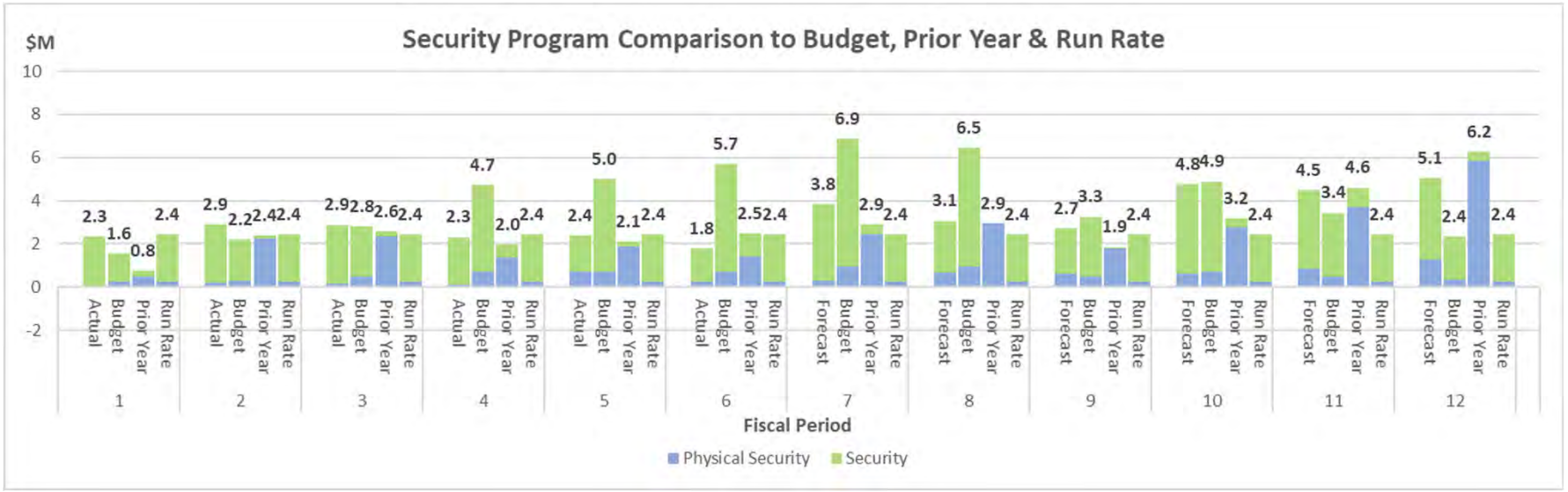
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Security	Physical Security	1.5	3.1	1.6	0.0	0.0	0.0	1.5	3.1	1.6
	Security	10.8	15.1	4.3	2.3	3.8	1.5	13.1	18.9	5.8
	Totals	**12.3**	**18.2**	**5.9**	**2.3**	**3.8**	**1.5**	**14.6**	**22.0**	**7.4**

Security Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Security	Physical Security	5.9	7.0	1.1	0.0	0.0	0.0	5.9	7.0	1.1
	Security	28.1	35.2	7.0	4.5	7.1	2.6	32.6	42.3	9.6
	Totals	**34.0**	**42.1**	**8.1**	**4.5**	**7.1**	**2.6**	**38.5**	**49.2**	**10.7**

Key Challenges

Challenge	Escalation Point
[redacted]	[redacted]
[redacted]	[redacted]
[redacted]	[redacted]



Mobilisation Summary

Status

- All projects have now been sanctioned.

Resources

[redacted]

Scope

[redacted]

- Commitments and scope continues to be refined and will be updated in Q4RF with assessment of impact of deliverables in FY24.

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
Cyber Operational Technology	In-Flight	[redacted]		
Identity and Access Management	In-Flight	[redacted]		
Platform Security	In-Flight	[redacted]		
Network Security	In-Flight	[redacted]		
Physical Security	In-Flight	[redacted]		
Security Orchestration Automation and Response	In-Flight	[redacted]		
Vulnerability Management	In-Flight	[redacted]		

US Group Functions
Q2 FY23 Report

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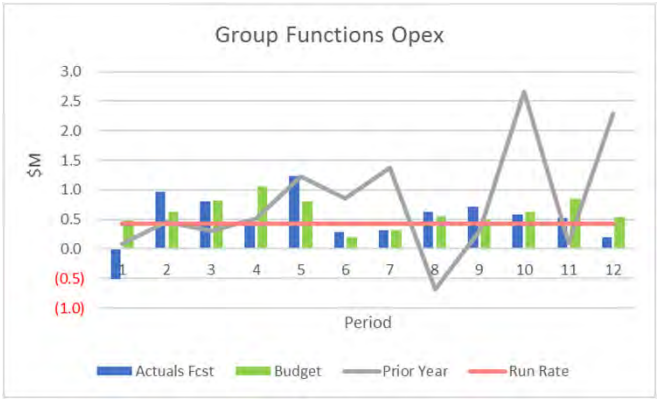
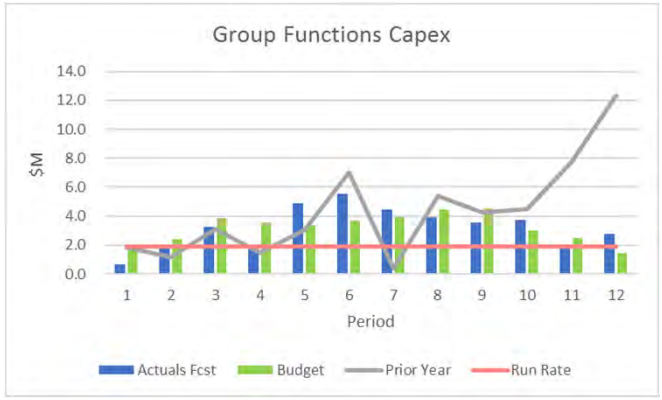
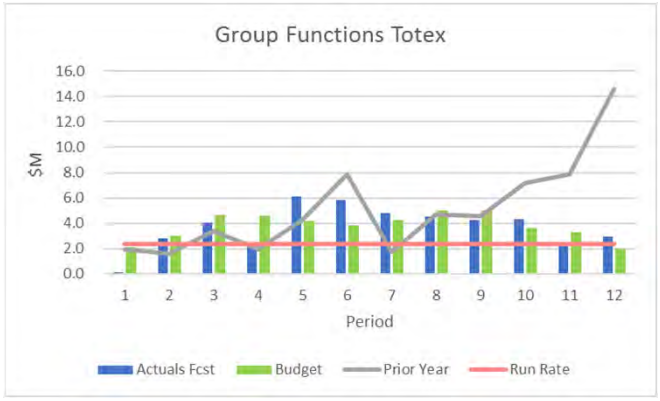
Group Functions

Financial Performance







Totex	Budget ($M)	Forecast ($M)
	45.8	44.9
	YTD Budget($M)	YTD Actual ($M)
	22.6	22.6

Capex	Budget ($M)	Forecast ($M)
	38.5	38.7
	YTD Budget($M)	YTD Actual ($M)
	18.6	18.3

Portfolio Commentary

Overall Group Functions is tracking to budget both year to date and full year forecast.

Key Accomplishments Delivered this Fiscal Year

Join to Leave (J2L) transformation through MyHub 2.0

- Delivered manager self-serve for org structure changes and hiring initiations for a potential reduction in rework activities by 40% (~350 hrs./mth) for the Business Services team.
- Digitizing learning - Supports the broader learning transformation enabling global learning support using a common platform which improves the user experience, the new platform provides more digital content which saves~9000 hours in face-to-face training annually.

Warehouse Investment Recovery- Completed release 1 which digitized the sale of scrap/obsolete materials from our warehouses leading to improved user experience and an opportunity for increased revenue generation with new efficiencies gained by use of a digital solution.

HR Support Pack Upgrade delivered in Q2 - These are mandatory annual changes requested by Federal and State agencies, such as the Internal Revenue Service (IRS) and various State Departments of Finance, as well as different municipalities. These changes must be applied to the SAP core solution in order to properly reflect employee wages, employee and Company withholdings, legal requirements and to comply with Federal and State regulatory reporting.

Group Functions Program View of Financials

Group Functions Program View Year to Date $M

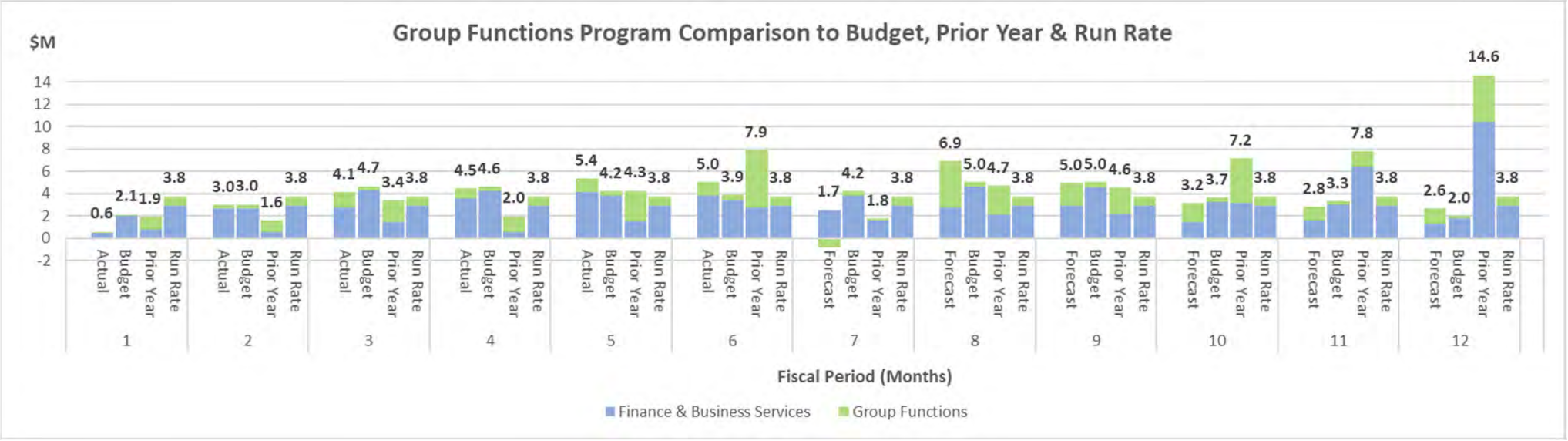
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Group Functions	Finance & Business Services	15.0	16.7	1.7	2.4	3.7	1.3	17.4	20.4	3.0
	Group Functions	3.3	1.9	(1.5)	1.9	0.3	(1.6)	5.2	2.1	(3.1)
	Totals	**18.3**	**18.6**	**0.2**	**4.3**	**4.0**	**(0.3)**	**22.6**	**22.6**	**(0.1)**

Group Functions Program View Full Year $M

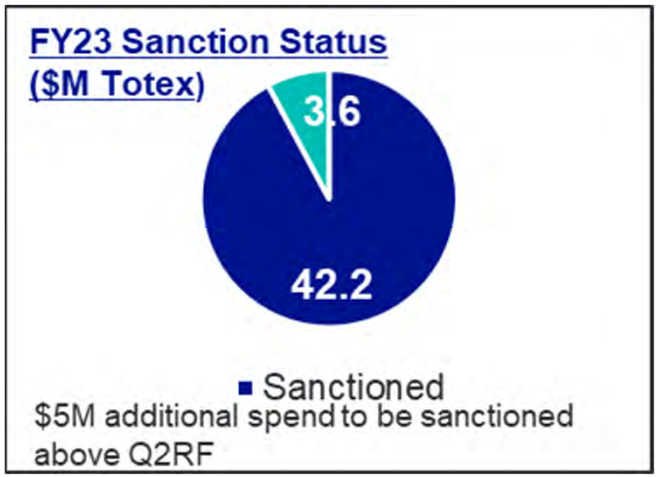
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Group Functions	Finance & Business Services	26.2	34.7	8.4	3.9	6.8	2.9	30.1	41.4	11.3
	Group Functions	12.5	3.8	(8.7)	2.3	0.6	(1.7)	14.8	4.4	(10.4)
	Totals	**38.7**	**38.5**	**(0.3)**	**6.1**	**7.4**	**1.2**	**44.9**	**45.8**	**0.9**

Key Challenges

Challenge	Escalation Point
• Resource constraints due to competing priorities and programs for both IT and Business resources	• Working with ADAM partners to backfill key roles • Recruitment ongoing to supplement with contractors
• Shortage of skilled/trained resources with Agile experience to help with transition to agile ways of working	• Upskill key resources in both IT and the business • Recruit and hire to fill open requistitions
• Risk – Capex moving to Opex due to new Finance policy on SaaS could impact ability to deliver financial commitments	• Working with finance to mitigate risk



Mobilisation Summary



Status

- We have had some recent walk-ins that will drive us above budget, working with Program Management Office to identify funding source for budget increase.
- The Ariba program needs additional funding over initial budget which will require a re-sanction, this also contributes to us exceeding our fiscal year budget. The funding is required for additional resources to cover increased scope and an extended timeline.

Resources

- Working with our vendor partners to staff for shortages so we can continue to deliver our business benefits planned for this fiscal year.

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
Blackline: Task Management	Complete	1.0		
Ariba Guided Buying	Inflight	1.4	Cost, Resources, & Schedule	
Core PowerPlan Upgrade and Lease Consolidation	Inflight	2.3	Schedule	
Joiners, Movers, Leavers	Start-up	3.2	Resources & schedule	
Warehouse Investment Recovery Digitization	Inflight	1.0		
R2R: Robotic Process Automation	Inflight	1.8		Cost & Technology
SAP Portals to Fiori	Complete	0.7		

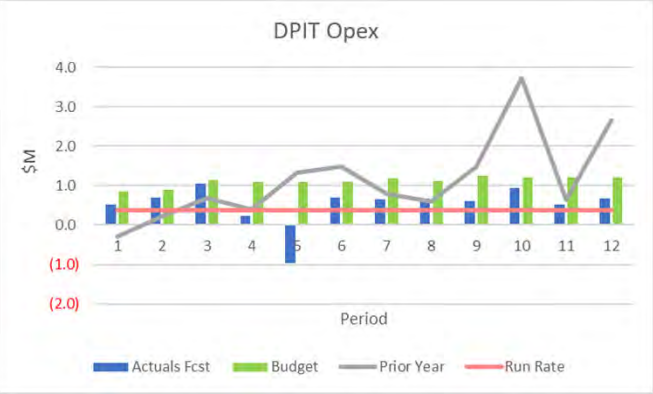
Digital Platforms Information Technology (DPIT)

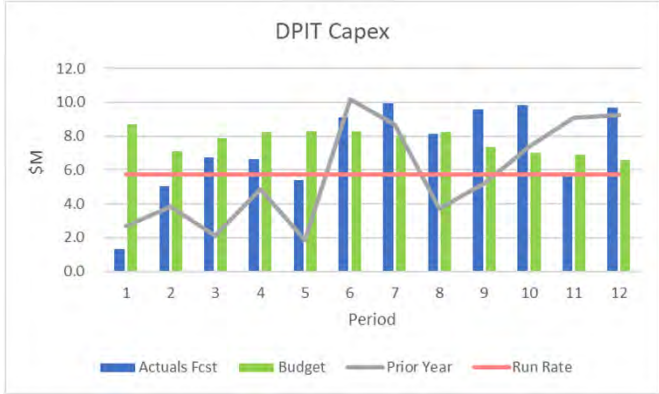
Q2 FY23 Report

nationalgrid

DPIT

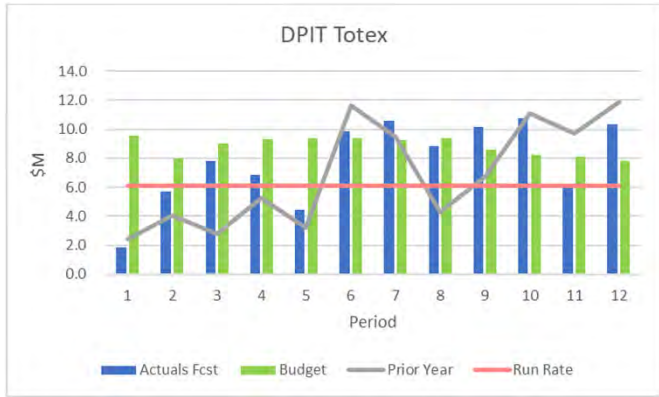
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	106.1	94.6
	YTD Budget($M)	YTD Actual ($M)
	54.7	36.5

Capex	Budget ($M)	Forecast ($M)
	92.7	87.2
	YTD Budget($M)	YTD Actual ($M)
	48.5	34.3



Portfolio Commentary

- **Financials:** Year to date ~$18M favorable to budget driven by budget phasing across the portfolio.
- The change in full year forecast since Q1 is due to OPEX cost shifting out of DPIT to Exceptional Funding. Otherwise DPIT is forecasting to spend to budget.
  - There will be a higher spend rate in the second half of the year driven by Ops Unified Technology Operations Centre (UTOC), Digital Workplace Services (HW, Atos), DCT (Mainframe SW, testing)
- **Product Model** enabling $29M incremental FY23 releases prior to year end:
  - Unified Technology Operations Center (UTOC) Minimum Viable Product
  - Network upgrades
  - Deployment of Windows as a Service updates to >23K devices
  - C&H improvements: to reduce compute resources required for applications, simplify application authentication, and ensure security stability
  - GridStack delivered centralized documentation, reusable code and automated data transformation for application development
- **Technical Debt:** Overall reduced to 19.8% (FY23 goal: 18%)
- **FY24 Investment Planning:** Product Managers are defining product line epics for FY24 for review in Nov/Dec
  - Product objectives will align with Infrastructure and Business Unit objectives

DPIT Program View of Financials

DPIT Program View Year to Date $M

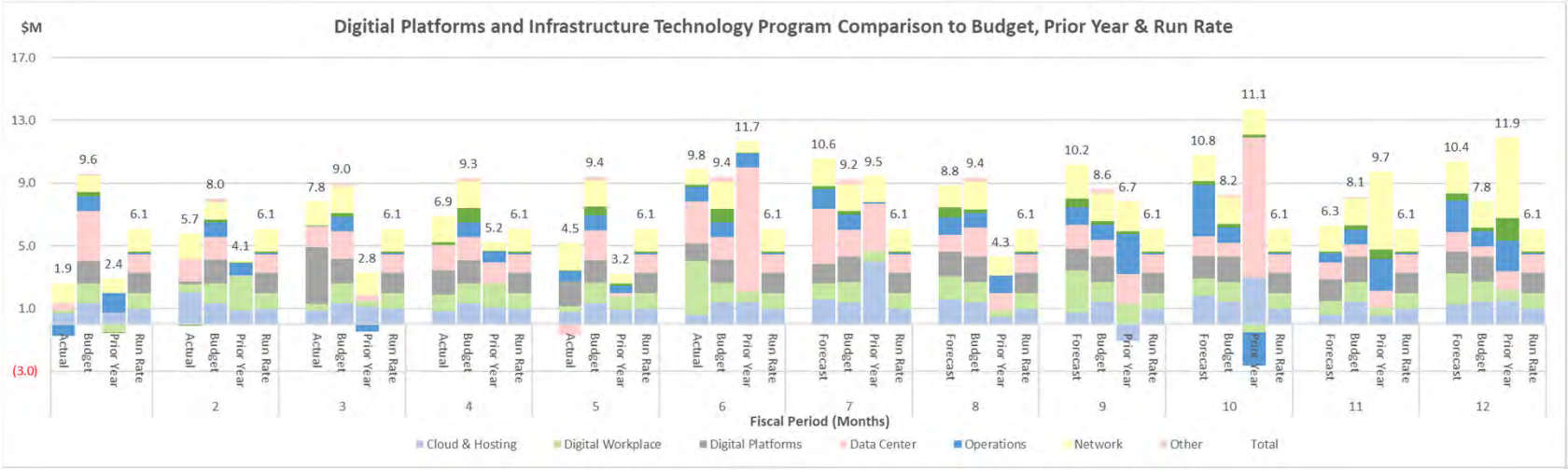
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
DPIT	Cloud & Hosting	5.5	7.4	1.9	0.4	0.7	0.3	5.9	8.1	2.3
	Digital Workplace	5.7	7.1	1.4	0.3	0.4	0.1	6.0	7.5	1.5
	Digital Platforms	7.5	7.8	0.3	0.5	1.1	0.6	8.0	8.9	0.9
	Data Center	6.4	8.2	1.9	0.4	3.0	2.6	6.8	11.3	4.4
	Operations	0.8	5.4	4.6	(0.1)	0.2	0.3	0.7	5.6	4.9
	CNI	0.4	2.9	2.6	0.0	0.1	0.0	0.4	3.0	2.6
	Network	8.1	8.4	0.2	0.7	0.7	(0.0)	8.8	9.0	0.2
	Other	0.0	1.2	1.2	0.0	0.0	0.0	0.0	1.2	1.2
	Totals	**34.3**	**48.5**	**14.2**	**2.3**	**6.1**	**3.9**	**36.5**	**54.7**	**18.1**

DPIT Program View Full Year $M

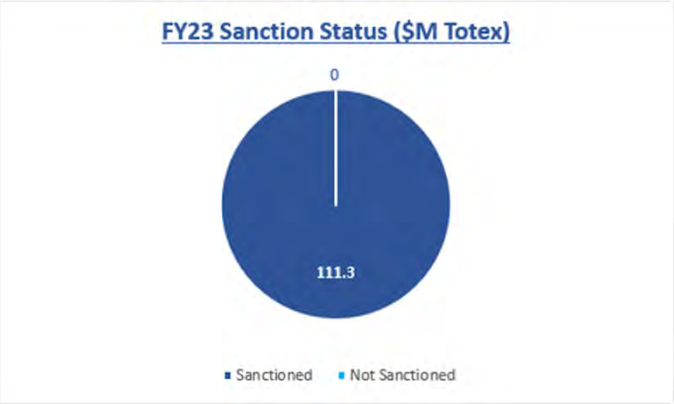
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
DPIT	Cloud & Hosting	11.7	15.0	3.3	1.7	1.6	(0.1)	13.4	16.6	3.3
	Digital Workplace	14.3	14.3	0.0	0.8	0.8	(0.0)	15.1	15.2	0.0
	Digital Platforms	15.1	16.4	1.3	1.4	2.2	0.8	16.5	18.6	2.1
	Data Center	16.1	11.5	(4.7)	0.4	6.8	6.5	16.5	18.3	1.8
	Operations	9.6	10.9	1.4	0.5	0.4	(0.1)	10.1	11.3	1.3
	CNI	2.5	4.3	1.8	0.1	0.2	0.1	2.6	4.5	1.9
	Network	17.9	18.0	0.1	1.4	1.3	(0.1)	19.4	19.3	(0.1)
	Other	0.0	2.3	2.3	0.0	0.0	0.0	0.0	2.3	2.3
	Totals	**87.2**	**92.7**	**5.5**	**6.3**	**13.3**	**7.0**	**93.5**	**106.1**	**12.5**

Key Challenges

Challenge	Escalation Point
Delays in US build, testing, and migration causing delays for US into FY24	• US Data Center is focus to complete in FY24
Spending increase in 2H FY23	• Multiple monthly cost and forecast reviews • US portfolio forecast details provide breakout and risks



Mobilisation Summary



*Total sanction includes exceptional costs as well as core portfolio costs

Status

- DPIT Product portfolios in the US are 100% sanctioned and inflight.

Resources

- No resource risks

Scope

- DPIT product portfolios on-track to both Objectives/Key Results (OKRs) and financial forecast of $94.6m, except:
  - Data Center Transformation: Amber - migrations started last weekend (Oct 21-24)

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
Data Center Transformation (DCT) - *Modernization of Infrastructure & Networks*	Inflight	18.3	Schedule	
Cloud & Hosting - *Modernization of Infrastructure & Networks*	Inflight	16.6		
Digital Workplace Services - *Employee Satisfaction & Digital Product Delivery*	Inflight	15.2		
Networks - *Reduce Technical Debt*	Inflight	19.3		
Global Operations - *Employee Satisfaction & Digital Product Delivery*	Inflight	11.3		
Critical National Infrastructure (CNI) - *Reduce Technical Debt*	Inflight	4.5		
Digital Platforms - *Employee Satisfaction & Digital Product Delivery*	Inflight	18.6		

					Year to Date Budget Variance $000's			Non_Permanant	Permanent	Full Year Forecast to Budget Variance $000's			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Explain	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	6 + 6 Total	Budget Total	YTD Variance	Reason
Customer	Customer Digital	4750	Customer Experience Program	No - Opex Only	16	0	(16)			16	0	(16)	
		4750J	Web Self Service	YTD & Forecast	4,465	1,238	(3,227)	Misclassifications	Complexity	3,229	1,238	(1,991)	Complexity
		4750M	ODS	YTD & Forecast	536	1,238	702		Funding Change	536	1,238	702	Funding Change
		5673	My Business Account (Nucleus Phase 2)	YTD Only	2,344	2,764	420	Accruals		5,159	5,527	368	
		6206	Customer Smart Target	Forecast Only	0	0	0			637	0	(637)	Complexity
	Customer Operations	3932	AIMS (TPV)	YTD & Forecast	(351)	0	351		Accounting	(351)	0	351	Efficiencies
		4904	Upgrade Servers for CCAE Reporting	No - Opex Only	(4)	0	4			(4)	0	4	
		5125	Customer Connection Program	YTD Only	2,223	1,819	(404)	Mischarges	Scope changes	3,975	3,763	(212)	
		5160	NY Stack VDER and Portal Integration	Forecast Only	1,630	1,516	(114)		Complexity	3,451	2,296	(1,155)	Complexity
		5345B	Bill Print and Bill Composition Implementation	YTD & Forecast	801	997	196		Other	2,572	1,055	(1,517)	Funding Change Funding Bucket
		5722	KEDLI Rate Case (Mandated)	No - Opex Only	(2)	0	2			(2)	0	2	
		5723	KEDNY Rate Case	YTD & Forecast	288	380	92		Efficiencies	288	380	92	Efficiencies
		5778	Customer Billing Improvement Program - Phase 1	YTD & Forecast	1	992	991		Funding Bucket	660	992	332	Funding Bucket
		5795F	Community Distributed Generation (CDG) Consolidated Billing	YTD & Forecast	295	223	(72)		Timing	295	223	(72)	Timing
		5859A	Customer and Agent Experience - IVR Baseline (UWP)	YTD & Forecast	1,867	1,400	(467)		Scope changes	1,867	2,800	933	Funding Change
		5863D	AIMS Stability	YTD & Forecast	(53)	0	53		Accounting	(53)	0	53	Accounting
		5926A	NIMO Rate Case	YTD & Forecast	(47)	0	47		Accounting	(47)	0	47	Accounting
		5987	Smart City Phase II	YTD & Forecast	26	0	(26)		Complexity	39	0	(39)	Complexity
		5996	Efficiency Target	Forecast Only	0	0	0			(1,500)	0	1,500	Other
		6007	Inside Agent	YTD & Forecast	77	0	(77)		Accounting	77	0	(77)	Accounting
		6032	Customer Billing COVID Transition	YTD & Forecast	90	211	121		Scope changes	169	303	134	Scope changes
		6066	FCP Upgrade for CSS - Plan Forward	YTD & Forecast	567	373	(194)		Complexity	573	173	(200)	Complexity
		6112	WFH Hardware - Plan Forward	No - Immaterial	(7)	0	7			(7)	0	7	
		6214	Customer Operational Enhancements FY22	No - Opex Only	8	0	(8)			8	0	(8)	
		6217	Customer Billing Minor Works FY22	No - Opex Only	(91)	0	91			(91)	0	91	
		6250	EV End-End Solution	YTD & Forecast	0	1,250	1,250		Funding Change	0	3,000	3,000	Funding Change
		6302	2022 AIMS Platform	YTD & Forecast	131	0	(131)		Timing	131	0	(131)	Timing
		6349	Customer Data Platform	YTD & Forecast	2,961	2,505	(456)		Complexity	6,852	5,011	(1,841)	Complexity
		6427	RI Purchase of Receivables (POR) Program	YTD & Forecast	55	87	32		Complexity	55	87	32	Complexity
		6435	Customer PCI Remediation	YTD & Forecast	470	390	(80)		Scope changes	431	390	(41)	Scope changes
		6492	NACHA Compliance- Security Encryption Effort	YTD & Forecast	563	475	(88)		Scope changes	1,560	475	(1,085)	Scope changes
		6496	NY Gender Pronoun and Name Mandate	YTD Only	853	1,940	1,087		Complexity	2,065	1,960	(105)	
		6531	AIMS Upgrades and Enhancements	YTD Only	368	323	(45)		Timing	673	645	(28)	
		6538	Billing & Collections Mandates & Enhancements	YTD & Forecast	0	1,187	1,187	Other		3,400	5,294	1,894	Other
		6543	Customer Minor Works	No - Opex Only	204	225	21			479	450	(29)	
		6584	UWP 2.0 Digital Re Architecture and Optimization	YTD & Forecast	5,400	0	(5,400)		Funding Change	11,424	0	(11,424)	Funding Change
		6451	One Step Auto Cashering	YTD & Forecast	44	0	(44)		Funding Change	646	0	(646)	Funding Change
		5778A	CRIS Infrast MGKEDNY	YTD & Forecast	89	0	(89)		Other	89	0	(89)	Other
		5778B	CSS Infrast NGLegv1	YTD & Forecast	612	0	(612)		Other	612	0	(612)	Other
		5778C	Encryption & DR AR	No - Immaterial	6	0	(6)			6	0	(6)	
		5778F	RightFax AR	YTD & Forecast	366	0	(366)		Other	366	0	(366)	Other
		5778D	Data Integ Svcs AR	No - Opex Only	6	0	(6)			6	0	(6)	
Customer Total					26,807	21,532	(5,275)			50,291	37,500	(12,791)	

					Year to Date Budget Variance $000's			Non_Permanant	Permanent	Full Year Forecast to Budget Variance $000's			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Explain	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	6 + 6 Total	Budget Total	YTD Variance	Reason
Group Functions	Finance & Business Services	3870A	Vision FM & Property Services Replacement	YTD & Forecast	73	0	(73)		Complexity	73	0	(73)	Complexity
		4418USB	RSA Archer Upgrade & Refresh	YTD & Forecast	136	0	(136)		Accounting	136	0	(136)	Accounting
		4562	US SAP: Business Warehouse (BW) Consolidation to HANA Enterprise Cloud (HEC)	YTD & Forecast	(82)	0	82		Accounting	(82)	0	82	Accounting
		5158	Myhub Compliance & Enhancements	YTD & Forecast	750	1,600	850		Other	750	2,900	2,150	Other
		5158A	myHub System Stabilization (30/60/90 Initiative)	Forecast Only	0	0	0			894	0	(894)	Other
		5466	AMAG HW and SW Upgrade	YTD & Forecast	913	800	(113)		Imprecise Estimates	1,706	800	(906)	Imprecise Estimates
		5497	Lease Accounting Updates and CCM - Phase 2	Forecast Only	1,124	1,100	(24)			1,654	2,300	646	Imprecise Estimates
		5555	Fieldglass Replacement	YTD & Forecast	0	1,000	1,000		Funding Change	0	1,800	1,800	Funding Change
		5605	US SAP: Infrastructure Landscape FY21	No - Immaterial	18	0	(18)			18	0	(18)	
		5606	US SAP: Infrastructure Landscape FY22	YTD & Forecast	(915)	0	915		Accounting	(915)	0	915	Accounting
		5610	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY22	No - Immaterial	3	0	(3)			3	0	(3)	
		5611	Annual HR & Payroll Mandatory Service Pack (HRSP) FY22	YTD & Forecast	579	800	221		Imprecise Estimates	914	1,500	586	Imprecise Estimates
		5841C	401(k) Administration Changes	YTD & Forecast	152	0	(152)		Timing	153	0	(153)	Timing
		5842B	Mandatory FERC Form Changes	YTD & Forecast	(21)	0	21		Accounting	(21)	0	21	Accounting
		5843B	US SAP Platform Upgrade: Minor Releases	YTD & Forecast	(162)	0	162		Accounting	(162)	0	162	Accounting
		5843D	US SAP Healthcheck: Upgrade Critical Components - FY22	YTD & Forecast	911	700	(211)		Imprecise Estimates	911	700	(211)	Imprecise Estimates
		5895B	Everbridge / Critical Event Management	YTD & Forecast	(40)	0	40		Complexity	125	0	(125)	Complexity
		5896B	Warehouse SAP Scanner Replacement (Barcoding)	No - Immaterial	(1)	0	1			(1)	0	1	
		5919	US SAP Minor Enhancements	No - Opex Only	0	0	0			0	0	0	
		6002	Other's Efficiency Target	Forecast Only	0	0	0			(4,998)	0	4,998	Other
		6182US	Global Career Site	YTD & Forecast	(49)	0	49		Accounting	(40)	0	40	Accounting
		6183US	HR Reporting Analytics	YTD & Forecast	12	500	488		Funding Change	12	1,300	1,288	Funding Change
		6185US	Onboarding 2.0	YTD & Forecast	266	0	(266)		Funding Change	344	0	(344)	Funding Change
		6219	US F&BS Minor Enhancements (Non-SAP)	No - Opex Only	14	0	(14)			14	0	(14)	
		6220	US F&BS Minor Enhancements (SAP)	YTD & Forecast	59	0	(59)		Imprecise Estimates	59	0	(59)	Imprecise Estimates
		6221	Tech Debt: RiskMaster Replacement	YTD & Forecast	477	0	(477)		Funding Change	521	0	(521)	Funding Change
		6231	US Treasury: Allegro Upgrade Phase 2	YTD & Forecast	90	0	(90)		Timing	90	0	(90)	Timing
		6242US	ARIBA System Enhancements	YTD & Forecast	166	0	(166)		Timing	166	0	(166)	Timing
		6265	HR Policy Compliance	No - Opex Only	34	0	(34)			130	0	(130)	
		6270US	Digitising Learning	YTD & Forecast	344	0	(344)		Funding Change	657	0	(657)	Funding Change
		6273US	Workforce Data Domain HR data governance Tools Processes EDM	Forecast Only	745	700	(45)			1,059	700	(359)	Imprecise Estimates
		6289	Financial Transformation (FTOM) - Record To Report (US)	No - Opex Only	0	0	0			0	0	0	
		6289B	FTOM: Record to Report	No - Immaterial	(4)	0	4			(4)	0	4	
		6289D	FTOM: Record to Report - Task Management with Blackline	YTD & Forecast	1,208	1,000	(208)		Imprecise Estimates	1,208	1,000	(208)	Imprecise Estimates
		6289E	FERC Form Integration with wDesk	YTD & Forecast	387	1,100	713		Imprecise Estimates	485	1,100	615	Imprecise Estimates
		6290A	Source To Pay: Release 1 – P2P Roll Out Program	YTD & Forecast	3,540	1,033	(2,507)		Imprecise Estimates	5,954	1,433	(4,521)	Imprecise Estimates
		6290B	Source To Pay: Release 2 - Supplier Onboarding	YTD & Forecast	102	800	698		Other	1,540	1,300	(240)	Other
		6346US	CA: Group We Digital Evolution Phase 1 - FY22	YTD & Forecast	(88)	0	88		Accounting	(88)	0	88	Accounting
		6346USC	Domain Management Refresh	No - Opex Only	(467)	0	467			(467)	0	467	
		6346USE	Brand Central Design and Implementation	No - Opex Only	(10)	0	10			(10)	0	10	
		6346USF	Brand Central Design and Implementation	No - Opex Only	(67)	0	67			(67)	0	67	
		6346USI	Video Strategic Management and Archiving (Enterprise)	No - Opex Only	(53)	0	53			(53)	0	53	
		6377	US Group Function Operational Requests	YTD & Forecast	(353)	0	353		Accounting	(353)	0	353	Accounting
		6421	P2P Data Analysis Tool	YTD & Forecast	1,542	500	(1,042)		Scope changes	1,925	1,200	(725)	Scope changes
		6426	Investment Recover Digitalization	YTD Only	561	670	109	Delays		1,052	1,000	(52)	
		6520US	Jira Align Phase 3	YTD & Forecast	957	0	(957)		Funding Change	1,958	0	(1,958)	Funding Change
		EHUB	E-Hub	No - Opex Only	0	0	0			0	0	0	
		5607	US SAP: Infrastructure Landscape FY23	YTD & Forecast	1,362	2,000	638		Funding Bucket	2,825	4,000	1,175	Funding Bucket
		5843E	SAP BW/BI Upgrade	YTD & Forecast	61	400	339		Timing	61	700	639	Timing
		5843F	SAP PI NW 7.40 (PI -> PO upgrade) to NW 750	YTD & Forecast	173	500	327	Delays		2,267	900	(1,367)	Scope changes
		6529	US Group Function Minor Enhancements (SAP)	No -Within Limits	519	500	(19)			998	1,100	102	
		6530	US Group Function Operational Requests - FY23	YTD Only	457	700	243	Timing differences		1,217	1,300	83	
		4418USC	CRO FY23: GRC Archer	YTD & Forecast	0	750	750	Prioritizations		1,263	1,500	237	Imprecise Estimates
		6514US	NG.com Refresh	YTD & Forecast	0	90	90		Funding Bucket	0	1,900	1,900	Funding Bucket
		6533US	SHE FY23: Group SHE Digital Evolution	YTD & Forecast	0	688	688		Timing	0	1,500	1,500	Timing
		6534US	Legal FY23: Group Legal Digital Evolution	YTD & Forecast	0	355	355		Timing	0	709	709	Timing
		Corp23	US Placeholder - Corp Tech Debt Remediation	YTD & Forecast	0	500	500		Funding Bucket	0	1,800	1,800	Funding Bucket
		6270	Digitising Learning	YTD & Forecast	0	300	300		Funding Bucket	0	600	600	Funding Bucket
		H823	Tactical HR	YTD & Forecast	0	300	300		Funding Bucket	0	1,800	1,800	Funding Bucket
		OPS23	Placeholder (Tech Debt & App Rat)	YTD & Forecast	0	280	280		Funding Bucket	0	680	680	Funding Bucket
		6225US	R2R Reporting Use Cases	YTD & Forecast	0	60	60		Funding Bucket	0	1,300	1,300	Funding Bucket
		6289C	FTOM Record to Report: Process Mining with Celonis	No - Opex Only	0	0	0			0	0	0	
		6289G	FTOM R2R: Robotic Process Automation of Manual Journal Entries	Forecast Only	697	700	3			1,327	1,800	473	Imprecise Estimates
		R2823	Finance + placeholder (Regulatory Compliance)	Forecast Only	0	0	0			0	800	800	Funding Bucket
		6467	Union Bidding	YTD & Forecast	149	0	(149)		Funding Change	407	0	(407)	Funding Change
		6596	US Placeholder - Corp Tech Debt Remediation	No - Opex Only	0	0	0			0	0	0	
		6597	Finance + placeholder (Regulatory Compliance)	No - Opex Only	0	0	0			0	0	0	
		6566US	HR Policy Compliance FY23	No - Opex Only	0	0	0			185	0	(185)	
		6180US	CV Talent Screening / Blind CVs	YTD & Forecast	7	0	(7)		Funding Change	315	0	(315)	Funding Change
		6424US	Infonet decommissioning	No - Opex Only	95	0	(95)			95	0	(95)	
		3870B	Vision FM replacement, Phase 2	YTD & Forecast	836	0	(836)		Funding Change	1,187	0	(1,187)	Funding Change
		6574	US Group Function: Tech Debt and Application Rationalization	No - Opex Only	0	0	0			200	0	(200)	
		6289F	US GAAP Statutory reporting with WDesk	YTD & Forecast	178	0	(178)		Funding Change	468	0	(468)	Funding Change
		6231US	US Treasury Allegro AR	No - Opex Only	26	0	(26)			26	0	(26)	
	Group Functions	5899A	Cloud API Migrations Phase 2	YTD & Forecast	66	0	(66)	Misclassifications		31	0	(31)	Misclassifications
		5917A	Data Mgmt _ Tooling	No - Opex Only	(1)	0	1			(1)	0	1	
		5983	DB Landscape Optimization	YTD & Forecast	953	0	(953)		Funding Change	1,788	0	(1,788)	Funding Change
		6137	Enterprise PMO	No - Opex Only	0	0	0			0	0	0	
		6137US	Enterprise PMO, Jira Align release 2	YTD & Forecast	83	0	(83)		Funding Change	83	0	(83)	Funding Change
		6252US	Data Mgmt Tools and Initiatives	YTD & Forecast	1,213	0	(1,213)		Other	(198)	0	198	Accounting
		6253US	Cloud API Migration - Legacy Integration Tools	YTD & Forecast	669	0	(669)		Funding Change	702	0	(702)	Funding Change
		6254US	Platform as a Service (PaaS) on Azure	No - Immaterial	17	0	(17)			17	0	(17)	
		6255US	Digital Foundation Tooling and DevSecOps	YTD & Forecast	(83)	0	83		Accounting	(83)	0	83	Accounting
		6256US	Platform Upgrades: Salesforce/Maximo/Mulesoft	YTD & Forecast	(32)	0	32		Accounting	(32)	0	32	Accounting
		6258USA	Data Archival Platform Implementation	No - Immaterial	16	0	(16)			16	0	(16)	
		6258USB	Data Archival: US Group Function Portfolio	YTD & Forecast	1,111	600	(511)		Scope changes	2,726	1,100	(1,626)	Scope changes
		6260US	EA Foundation: Increase Capability	YTD & Forecast	100	0	(100)		Accounting	54	0	(54)	Accounting
		6261US	ServiceNow Upgrade: Stay Current FY22	YTD & Forecast	31	0	(31)		Misclassifications	31	0	(31)	Misclassifications
		6264US	Apptio Development	YTD & Forecast	68	0	(68)		Funding Change	278	0	(278)	Funding Change
		6266US	EA Strategic Portfolio Planning & Delivery / Additional Capability	YTD & Forecast	521	0	(521)		Funding Change	451	0	(451)	Funding Change
		6360US	IT Planning: Enterprise PMO	YTD & Forecast	123	0	(123)		Funding Change	123	0	(123)	Funding Change
		6458US	QES IT Tool Sunset	No - Opex Only	0	0	0			1	0	(1)	
		6461USA	Business Planning F&A (FP&A Enablement & Modelling Tool)	YTD & Forecast	(153)	1,236	1,389	Delay		2,987	2,500	(487)	Scope changes
		6513US	CA: Group Web Digital Evolution - FY23	YTD & Forecast	70	307	237		Efficiencies	247	800	553	Efficiencies
		6587US	E-Hub	YTD & Forecast	313	0	(313)		Funding Change	1,655	0	(1,655)	Funding Change
		6522US	SHE: Digitisation of the CMT Process	Forecast Only	0	0	0			246	0	(246)	Funding Change
		6549US	SHE: Resilience Digital Evolution	Forecast Only	0	0	0			348	0	(348)	Funding Change
		6391US	SHE: Consolidated Incident Management and Reporting	No - Opex Only	0	0	0			245	0	(245)	
		6402US	Legal Operations	Forecast Only	0	0	0			169	0	(169)	Funding Change
		6563US	CA:NG.com Design	No - Opex Only	94	0	(94)			839	0	(839)	
		6621US	My Hub	YTD & Forecast	1	0	(1)		Funding Change	2,050	0	(2,050)	Funding Change
		6256	MuleSoft Dev Ops Inf	No - Opex Only	5	0	(5)			5	0	(5)	
		6313	GDPR Comp AC	No - Opex Only	1	0	(1)			1	0	(1)	
		6566	HR Policy FY23	No - Opex Only	16	0	(16)			16	0	(16)	
		6602	Microstrategy Mod	No - Opex Only	0	0	0			0	0	0	
Group Functions Total					22,645	22,569	(76)			44,886	45,822	936	

					Year to Date Budget Variance $000's			Non_Permanant	Permanent	Full Year Forecast to Budget Variance $000's			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Explain	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	6 + 6 Total	Budget Total	YTD Variance	Reason
Security	**Physical Security**	3617A	Physical Security - G20										
		3617D	Physical Security - NIMOE										
		3617F	PS - NIMOE										
		3617G	Physical Security - KEDNY - Equip Replacement										
		3617H	Physical Security - KEDLI - Equip Replacement										
		3617I	Physical Security - MASS Electric - Equip Replacement										
		3617J	Physical Security - Boston Gas										
		3617K	Physical Security - Colonial Gas										
		3617L	Physical Security - RI Electric										
		3617M	Physical Security - RI Gas										
		3617	Physical Security										
	Security	3683USF	Enterprise Centralized Patch Management										
		3683USH	Network Security Improvements										
		3683USN	SOAR - Security Orchestration Automation and Response										
		5702USA	Malware Defense - Endpoint Protection										
		5820	Data Protection										
		5848	Vulnerability Management										
		5850	Network Security										
		5850US	Network Security US										
		5850USA	SSN Optimization US										
		5850USB	vSTIG Firewall Upgrade US										
		5850USD	Intrusion Detection System and Intrusion Prevention System US										
		5870	Identity and Access Management										
		5933	Operational Technology Intrusion Detection										
		5940	Global VM Enterprise										
		5998	Security Efficiency Target										
		6210US	FY22 Platform Tower										
		6211US	FY22 Network Tower										
		6245US	Security Orchestration Automation & Response										
		6416	Operational Technology Privileged Access Workstation										
		6417US	Vulnerability Management T2 FY22										
		6480us	Platform Security										
		6526us	Network Security										
		6540us	Security Orchestration Automation and Response										
		6510	OT PAW FY23										
		6417	Vulnerability Management T2 FY22										
		6613US	FY23 Identity and Access Management										
Security Total													

					Year to Date Budget Variance $000's			Non_Permanant	Permanent	Full Year Forecast to Budget Variance $000's			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Explain	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	6 + 6 Total	Budget Total	YTD Variance	Reason
US Electric	EBU Digital	5054	On My Way (EBU Digital Solution) Phase 2	YTD & Forecast	2,741	1,867	(874)		Imprecise Estimates	5,397	3,762	(1,635)	Imprecise Estimates
		5702	GridStack	YTD & Forecast	27	0	(27)	Mischarges		27	0	(27)	Funding Bucket
		5912	On My Way (EBU Digital Solution)	YTD & Forecast	39	0	(39)		Funding Bucket	39	0	(39)	Funding Bucket
		6073	VMO Validation Phase	YTD & Forecast	1,657	4,455	2,798		Scope changes Imprecise Estimates	5,484	8,566	3,082	Scope changes Imprecise Estimates
		6076	FutureNow Validation Phase	YTD & Forecast	4,212	5,343	1,131		Efficiencies	8,188	11,187	2,999	Efficiencies
		6291	ICU-Out (Electric Digita)	YTD & Forecast	236	2,214	1,978		Timing	2,440	5,648	3,208	Timing
		6425	DER Active Resource Integration NY	YTD & Forecast	364	0	(364)		Funding Bucket	433	0	(433)	Funding Bucket
		6437	GridMD	YTD & Forecast	5	1,020	1,015		Timing	5	1,020	1,015	Timing
		6388	FY22 - EBU Digital Placeholder	YTD & Forecast	0	1,110	1,110		Timing	9,585	12,800	3,215	Timing
		GSFund	GridStack Funding	YTD & Forecast	0	1,329	1,329		Timing	0	2,658	2,658	Funding Change
		6437A	GridMod	YTD & Forecast	543	0	(543)		Funding Bucket	4,015	0	(4,015)	Funding Change
		6571	GridAPI Electric	Forecast Only	0	0	0			2,410	0	(2,410)	Funding Change
	Electric BU	4481	Energy Accounting System Migration to Wholesale Settlement Application - NY	YTD & Forecast	(94)	0	94		Accounting	(94)	0	94	Accounting
		4483	US Control-Transmission Outage Application (TOA) Upgrade to HTML 5 Web	YTD & Forecast	60	0	(60)		Timing	60	0	(60)	Timing
		5241	East Pulaski Energy Storage System IS Network	YTD & Forecast	20	0	(20)		Funding Change	50	0	(50)	Funding Change
		5242	Troy Energy Storage System IS Network	No - Immaterial	(1)	0	1			(1)	0	1	
		5773B	Visitor Logging for NERC-CIP Reliability Compliance	YTD & Forecast	239	440	201	Timing differences		274	440	166	Other
		5773E	NERC-CIP Reliability Compliance	YTD & Forecast	230	0	(230)		Other	231	0	(231)	Other
		5871A	NY Battery Energy Storage Systems	YTD & Forecast	0	975	975		Funding Bucket	0	2,675	2,675	Funding Bucket
		5871D	Clean Energy Initiatives	YTD & Forecast	0	145	145		Timing	332	625	293	Funding Change
		5871E	Power Marketer Pilot	No - Opex Only	213	0	(213)			199	0	(199)	
		5871F	Battery Storage Communication Upgrades	YTD & Forecast	0	590	590		Timing	930	1,500	570	Timing
		5887A	AICOS - Reporting Replacement	YTD & Forecast	472	699	227		Complexity	679	781	102	Complexity
		5888D	Fluke & WECO Calibration licenses, hardware updates and data migration	YTD & Forecast	255	118	(137)		Scope changes	366	118	(248)	Scope changes
		5889C	Electric Regulatory Enhancements	YTD & Forecast	(14)	90	104		Funding Bucket	(14)	225	239	Funding Bucket
		5890A	Corporate PI	YTD Only	755	1,342	587	Timing differences		1,638	1,589	(49)	Imprecise Estimates
		5890C	Computapole Replacement F&A	No - Opex Only	0	0	0			720	0	(720)	
		5891A	Electric MV90 Upgrade	YTD & Forecast	1,001	690	(311)		Timing	1,332	742	(590)	Timing
		5891C	Field Collection System (FCS) Upgrade	YTD & Forecast	810	295	(515)	Mischarges		415	295	(120)	Scope changes
		5891D	Meter Data Services IEE Upgrades	YTD & Forecast	0	660	660		Efficiencies	0	1,150	1,150	Efficiencies
		5891E	MTS Retail Settlement	YTD & Forecast	1,699	2,396	697	Delays		2,956	3,531	575	Timing
		5891F	MDS Consolidation Phase 1	YTD & Forecast	0	325	325		Timing	240	1,725	1,485	Timing
		5891I	Price Transmission (Retail Price) System Replacement	YTD & Forecast	891	990	99	Timing differences		1,259	1,005	(254)	Timing
		5892A	Control Room EOL Server and Tool Replacement	YTD & Forecast	224	0	(224)		Imprecise Estimates	224	0	(224)	Timing
		5892B	Electric Control Center Tool Replacement	YTD Only	0	540	540	Timing differences		1,375	1,380	5	
		5892C	IDS-DVD Dig Safe DVD Replacement	YTD & Forecast	34	927	893		Timing	412	1,128	716	Efficiencies
		5893A	Digitizing the Electric Business - Mobility (Mwork Replacement) *A	No - Opex Only	(35)	0	35			(35)	0	35	
		6043	Plan Forward	No - Opex Only	1	0	(1)			1	0	(1)	
		6320	Telecom & Network Deep Dive	No - Opex Only	12	0	(12)			12	0	(12)	
		5889D	Electric FY23 Regulatory Mandates and Minor Enhancements	No - Opex Only	24	0	(24)			174	0	(174)	
		5871DA	SD-WAN for Distributed Communications	Forecast Only	0	0	0			269	0	(269)	Funding Bucket
		5877A	Intelligent Substation Design (Bus.)	No - Opex Only	0	0	0			54	0	(54)	
		6043E1	Contant Mgmt Eng-T&D	No - Opex Only	1	0	(1)			1	0	(1)	
	Wholesale Networks	5274	New Storms Response Management System	No - Opex Only	4	0	(4)			4	0	(4)	
		5773D	NYISO PMU Requirement for Large Generators	YTD & Forecast	395	731	336	Timing differences		1,604	736	(868)	Complexity
		5822C	Generation Asset Management Upgrade	Forecast Only	1,584	1,513	(71)			2,734	1,548	(1,186)	Other
		6043G1	Content Management for Engineering	YTD & Forecast	473	1,159	686		Scope changes	927	1,493	566	Scope changes
US Electric Total					19,077	31,963	12,886			57,351	68,326	10,975	

					Year to Date Budget Variance $000's			Non_Permanant	Permanent	Full Year Forecast to Budget Variance $000's			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Explain	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	6 + 6 Total	Budget Total	YTD Variance	Reason
US Gas	Capital Delivery	5274A	New Storms Respose Management System Implementaition	No -Within Limits	294	277	(17)			303	277	(26)	
		6222	PIF Replacement	YTD & Forecast	(250)	0	250		Accounting	(250)	0	250	Accounting
		4769	Gas Storm Hardening Using Remote Service Valves (RSV's)	YTD & Forecast	151	323	172	Delays		443	383	(60)	Timing
		4898	Asset Management System for LNG	YTD & Forecast	2,364	1,567	(797)		Complexity	2,601	1,607	(994)	Complexity
		5785I	NRG GIS Replacement Project	No - Immaterial	1	0	(1)			1	0	(1)	
		5787D	GTIS Functional Upgrades	YTD & Forecast	551	990	439		Scope changes Complexity	4,168	1,980	(2,188)	Scope changes Complexity
		5787G	F&A Study for Gas Demand Response (DR)	No - Opex Only	14	0	(14)			14	0	(14)	
		5788D	Interface MyHub To OnBoard LMS OQ External Site	No - Immaterial	1	0	(1)			1	0	(1)	
		5788G	FY22 Gas Regulatory and Operational Enhancements (Minor Works)	No - Opex Only	(8)	0	8			(8)	0	8	
		5789E	Summons and Corrective Actions Response (CAR) Work Tracker	YTD & Forecast	99	0	(99)	Misclassifications		99	0	(99)	Accounting
		6015	Continuous Improvement for Gas Applications	YTD & Forecast	(225)	605	830		Funding Change	(225)	1,210	1,435	Funding Change
		6392	Contractor Ipad for Gas Business	YTD & Forecast	72	0	(72)	Misclassifications		72	0	(72)	Accounting
		6441	Scalability & Resilience for Enterprise Gas Platforms	YTD & Forecast	1,710	595	(1,115)		Other	1,324	1,190	(134)	Timing
		6479	LNG IT Support for OT Systems	YTD & Forecast	6	850	844	Delays		722	850	128	Timing
		6481	Pathfinder for Leak Survey for MA	YTD & Forecast	108	0	(108)	Prioritizations		108	0	(108)	Other
		5785Q	Field Collection Systems (FCS) Upgrade (Gas Only)	YTD & Forecast	30	420	390		Timing	499	840	341	Timing
		6472	MOC Library & Action Item Automation?	No - Opex Only	98	0	(98)			173	150	(23)	
		6509	Gas I&R, Gas Meter Ops, and Electric Complex Construction Field Device Replacement	YTD & Forecast	487	60	(427)		Other	496	134	(362)	Imprecise Estimates
		6539	Asset Prism?	YTD & Forecast	9,925	6,583	(3,342)		Scope changes	30,055	20,070	(9,985)	Scope changes
		6542	New Gas Demand Response Solution POC and Pilot?	No - Opex Only	76	432	356			76	863	787	
		6550	Gas Management System (SCADA) Upgrade?	YTD Only	292	993	701	Mischarges		2,100	2,098	(2)	
		6552	System Operating Procedure Third-Party User Access for PE Review	No - Opex Only	0	40	40			0	100	100	
		6553	Enable Gas Curtailment Map in ESRI	No - Opex Only	0	40	40			0	40	40	
		6554	New System Operating Procedure (SOP) Solution	No - Opex Only	0	0	0			0	100	100	
		6555	PIPES/Enterprise Maximo Integration	No - Opex Only	0	50	50			50	50	0	
		6556	Digital Asset & Work Management for CNG and I&R?	Forecast Only	0	0	0			0	290	290	Other
		6557	Enable Converted LI Gas Distribution Asset Data in ESRI	No - Opex Only	0	0	0			0	140	140	
		6559	FY23 Gas Regulatory and Operational Enhancements (Minor Works)?	No - Opex Only	0	75	75			0	150	150	
		6582	Gas Application Continuous Improvement	YTD & Forecast	92	0	(92)	Other		794	0	(794)	Funding Change
		6573	FY23 Gas Regulatory and Operational Enhancements (Minor Works)	No - Opex Only	79	0	(79)			233	0	(233)	
		6610	Gas App Rationalization	No - Opex Only	0	0	0			0	0	0	
US Gas Total					15,967	13,900	(2,067)			43,849	32,522	(11,327)	

CIO Group	it_program	inv_id	proj_level_descrip	Explain	Year to Date Budget Variance $000's			Non_Permanant Reason	Permanent Reason	Full Year Forecast to Budget Variance $000's			Permanent Reason
					Actual YTD	YTD Budget	YTD Variance			6 + 6 Total	Budget Total	YTD Variance	
DPIT	Cloud & Hosting	6200	I&O US Cloud and Hosting FY22	YTD Only	42	0	(42)			(67)	0	67	
		6200.01.01	INVP 6200.01.01 HCI / V-Block Migra	YTD Only	166	0	(166)			166	0	(166)	
		6200.01.02	INVP 6200.01.02 DPS Enablement (Mig	YTD Only	(87)	0	87			(87)	0	87	
		6200.02.01	INVP 6200.02.01 Storage Migration C	YTD Only	100	0	(100)			100	0	(100)	
		6200.02.02	INVP 6200.02.02 Data Management / A	YTD Only	(105)	0	105	Mischarges	Accounting	(105)	0	105	
		6200.03.01	INVP 6200.03.01 Public Cloud (Azure	YTD Only	185	0	(185)			185	0	(185)	
		6200.04.01	INVP 6200.04.01 Azure and GBE Palo	YTD Only	(146)	0	146			(146)	0	146	
		6200.04.02	INVP 6200.04.02 Horizon on Azure Ce	YTD Only	(47)	0	47			(47)	0	47	
		6200.05.01	INVP 6200.05.01 New Global Active D	YTD Only	(16)	0	16			(16)	0	16	
		6223	I&O US Data Center Transformation	YTD & Forecast	449	0	(449)			10,107	0	(10,107)	Funding Change
	Data Center	6223	I&O US Data Center Transformation FY22	YTD & Forecast	0	11,260	11,260			0	18,269	18,269	
		6223.01	INVP 6223.01 DCT Other Costs	YTD & Forecast	897	0	(897)			897	0	(897)	
		6223.02	INVP 6223.02 DCT Co-Location	YTD & Forecast	(692)	0	692	Prioritizations	Funding Change	(692)	0	692	
		6223.03	INVP 6223.03 DCT Mngd Svcs	YTD & Forecast	1,418	0	(1,418)			1,418	0	(1,418)	
		6223.04	INVP 6223.04 DCT Mainframe	YTD & Forecast	1,217	0	(1,217)			1,217	0	(1,217)	
		6223.05	INVP 6223.05 DCT App Migrat	YTD & Forecast	3,532	0	(3,532)			3,532	0	(3,532)	
		6489	I&O US Cloud & Hosting FY23	YTD & Forecast	0	8,129	8,129			7,623	16,622	8,999	Imprecise Estimates
		6489.01	INVP 6489 Public Cloud Platforms	YTD & Forecast	945	0	(945)			945	0	(945)	
		6489.02	INVP 6489 Cloud Containers	YTD & Forecast	511	0	(511)	Timing differences	Imprecise Estimates	511	0	(511)	
		6489.03	INVP 6489 Global Active Directory	YTD & Forecast	1,971	0	(1,971)			1,971	0	(1,971)	
		6489.04	INVP 6489 Hosted Solutions & Platf	YTD & Forecast	2,333	0	(2,333)			2,333	0	(2,333)	
	CNI	6199	I&O US CNI FY22	YTD & Forecast	344	0	(344)			240	0	(240)	
		6199.01.01	INVP 6199.01.01 Eol. Assets	YTD & Forecast	(174)	0	174			(174)	0	174	Accounting
		6199.01.02	INVP 6199.01.02 EOL Assets - Gas	YTD & Forecast	(144)	0	144		Accounting	(144)	0	144	
		6199.02.01	ITSM Implementation	YTD & Forecast	53	0	(53)			53	0	(53)	
		6488	I&O US CNI FY23	YTD & Forecast	0	3,016	3,016			2,347	4,500	2,153	Funding Bucket
		6488.01	INVP 6488.01 US CNI Networks	YTD & Forecast	184	0	(184)			184	0	(184)	
		6488.02	INVP 6488.02 Infrastructure Mgmt	YTD & Forecast	34	0	(34)	Timing differences	Other	34	0	(34)	
		6488.03	INVP 6488.03 CNI Tool Management	YTD & Forecast	100	0	(100)			100	0	(100)	
	Network	5774	WiFi Phase 2	No - Immaterial	(2)	0	2			(2)	0	2	
		6197	I&O US Networks FY22	YTD & Forecast	(77)	0	77			(186)	0	186	
		6197.01.01	INVP 6197.01.01 UPS Devices	YTD & Forecast	(709)	0	709			(709)	0	709	
		6197.01.02	INVP 6197.01.02 EOL Switch	YTD & Forecast	7	0	(7)			7	0	(7)	
		6197.01.03	INVP 6197.01.03 SD WAN	YTD & Forecast	155	0	(155)			155	0	(155)	
		6197.01.04	INVP 6197.01.04 NW Monitoring	YTD & Forecast	308	0	(308)			308	0	(308)	
		6197.01.05	INVP 6197.01.05 OOB Mgmt	YTD & Forecast	58	0	(58)			58	0	(58)	
		6197.01.06	INVP 6197.01.06 Storm Sites	YTD & Forecast	(49)	0	49		Imprecise Estimates	(49)	0	49	Imprecise Estimates
		6197.02.01	INVP 6197.02.01 WiFi	YTD & Forecast	159	0	(159)			159	0	(159)	
		6197.02.02	INVP 6197.02.02 WiFi Tooling	YTD & Forecast	(65)	0	65			(65)	0	65	
		6197.03.01	INVP 6197.03.01 Voice EOL	YTD & Forecast	71	0	(71)			71	0	(71)	
		6197.03.02	INVP 6197.03.02 MS Teams Intr	YTD & Forecast	(72)	0	72			(72)	0	72	
		6197.04.01	INVP 6197.04.01 vSlng Upgrade	YTD & Forecast	234	0	(234)			234	0	(234)	
		6197.05.01	INVP 6197.05.01 Data Ctr Transforma	YTD & Forecast	2	0	(2)			2	0	(2)	
		6450x	Accelerated Telecom Networks Audit (POTS and Circuits)	No - Opex Only	202	0	(202)			202	0	(202)	
		6486	I&O US Networks FY23	YTD & Forecast	148	9,028	8,880			10,841	19,300	8,457	
		6486.01	INVP 6486 LAN/Wireless	YTD & Forecast	1,744	0	(1,744)			1,744	0	(1,744)	
		6486.02	INVP 6486 WAN/SD-WAN	YTD & Forecast	1,099	0	(1,099)	Timing differences		1,099	0	(1,099)	
		6486.03	INVP 6486 Network Support Svcs	YTD & Forecast	2,746	0	(2,746)			2,746	0	(2,746)	
		6486.04	INVP 6486 Telephony	YTD & Forecast	2,829	0	(2,829)			2,829	0	(2,829)	
	Operations	5876	Pure Technical Debt Remediation	YTD & Forecast	(36)	0	36		Accounting	(36)	0	36	Accounting
		6114	Managed Workspace Services Transformation	No - Opex Only	(26)	0	26			(26)	0	26	
		6196	I&O US Operations FY22	YTD & Forecast	(277)	0	277			(277)	0	277	
		6196.01.01	INVP 6196.01.01 BAU Device Refresh	YTD & Forecast	12	0	(12)			12	0	(12)	Accounting
		6196.02.02	INVP 6196.02.02 CMDB Overhaul	YTD & Forecast	(61)	0	61		Accounting	(61)	0	61	
		6196.03.01	INVP 6196.03.01 Appl Perf Mgnt	YTD & Forecast	(611)	0	611			(611)	0	611	
		6196.03.02	INVP 6196.03.02 Auto Disaster Rec	YTD & Forecast	212	0	(212)			212	0	(212)	
		6468	GridStack - Infrastructure	YTD & Forecast	(1,217)	0	1,217		Timing	(1,217)	0	1,217	Timing
		6485	I&O US Digital Operations FY23	YTD Only	1,505	5,602	4,097			10,875	11,318	443	
		6485.01	INVP 6485 Service Aware CMDB	YTD Only	123	0	(123)			123	0	(123)	
		6485.02	INVP 6485 Observability	YTD Only	85	0	(85)	Timing differences		85	0	(85)	
		6485.03	INVP 6485 AI Operations	YTD Only	227	0	(227)			227	0	(227)	
		6485.04	INVP 6485.04 Automation	YTD Only	14	0	(14)			14	0	(14)	
		6485.05	INVP 6485.05 Unified Tech Ops Ctr	YTD Only	732	0	(732)			732	0	(732)	
	Other	6520U5	JIRA*Not DPIT	YTD & Forecast	0	600	600	Delays		0	900	900	Other
		Apptio	Apptio*Not DPIT	YTD & Forecast	0	100	100	Delays		0	600	600	Other
		EHUB	E-Hub*Moving to DPIT in Q3	YTD & Forecast	0	500	500	Delays		0	800	800	Other
	Digital Platforms	6519	DP GridStack	YTD & Forecast	0	7,962	7,962			6,084	15,943	9,859	
		6519.01	INVP 6519.01 GS Foundation GridInit	YTD & Forecast	1,752	0	(1,752)			1,752	0	(1,752)	
		6519.02	INVP 6519.02 GS Foundations GridCI	YTD & Forecast	1,554	0	(1,554)			1,554	0	(1,554)	
		6519.03	INVP 6519.03 GS Found. GridDataOps	YTD & Forecast	1,456	0	(1,456)	Timing differences		1,456	0	(1,456)	Other
		6519.04	INVP 6519.04 GS Found. GridIdentity	YTD & Forecast	153	0	(153)			153	0	(153)	
		6519.05	INVP 6519.05 GS Found. GridAccess	YTD & Forecast	38	0	(38)			38	0	(38)	
		6519.06	INVP 6519.06 GS Found. GridVault	YTD & Forecast	528	0	(528)			528	0	(528)	
		6519.07	INVP 6519.07 GridAPI/GridSystemInt	YTD & Forecast	1,938	0	(1,938)			1,938	0	(1,938)	
		6565	DP Secure & Compliant	YTD & Forecast	597	980	383	Timing differences	Other	2,972	2,657	(315)	Other
	Digital Workplace	4987	SOE (Windows) Upgrade and Device Refresh	No - Immaterial	(2)	0	2			(2)	0	2	
		6198	I&O US End User Experience FY22	YTD & Forecast	(57)	0	57			(57)	0	57	
		6198.01	INVP 6198.01.04 Mac OS / iOS Mgmt	YTD & Forecast	42	0	(42)			42	0	(42)	
		6198.01.01	INVP 6198.01.01 Real Time Collabora	YTD & Forecast	(152)	0	152		Accounting	(152)	0	152	Accounting
		6198.01.02	INVP 6198.01.02 Device and Data Acc	YTD & Forecast	45	0	(45)			45	0	(45)	
		6198.01.03	INVP 6198.01.03 Content Collaborati	YTD & Forecast	74	0	(74)			74	0	(74)	
		6487	I&O US End User Experience FY23	YTD Only	0	7,500	7,500			9,162	15,150	5,988	
		6487.01	INVP 6487 Device & Data Access	YTD Only	3,989	0	(3,989)			3,989	0	(3,989)	
		6487.02	INVP 6487 Real Time Collaboration	YTD Only	539	0	(539)			539	0	(539)	
		6487.03	INVP 6487 Content Collaboration	YTD Only	420	0	(420)	Timing differences		420	0	(420)	
		6487.04	INVP 6487.04 Support Assurance	YTD Only	372	0	(372)			372	0	(372)	
		6487.05	Digital Adopt & Change	YTD Only	709	0	(709)			709	0	(709)	
DPIT Total					**36,535**	**54,677**	**18,142**			**93,527**	**106,059**	**12,532**	

Impacts Year end Forecast

Category	Type of Variance	Explanation
Project Delivery - Scope change	Scope changes	Spend impacted by both reduced or increased scope
Project Delivery - Initial scope delivered	Imprecise Estimates	Cost and Quantity of resources were incorrectly assessed when creating the budget. (eg. Price Increases)
Project Delivery - Initial scope delivered	Complexity	Unforeseen challenges in delivering the solution drove costs up. No change in scope but amount of work to deliver increased or potentially decrease
Project Delivery - Initial scope delivered	Efficiencies	More cost effective implementation options - Found a better way to deliver the solution which reduced costs
Strategic Direction	Funding Bucket	Source of undefined funding where funds will be moved to other project
Strategic Direction	Funding Change	Move $ to or from another project (please specify INVP moving to or moving from
Annual Accounting	Accounting	Over or under accruals which reverse at year end leaving a balance on the INV
Annual Accounting	Timing	Project slipped and intended scope was not delivered this fiscal yea
Other	Other	Please explain

No impact to forecast

Category	Type of Variance	Explanation
Accounting Errors	Mischarges	Temporary due to unavailable accounting codes or charging errors which will be corrected
Accounting Errors	Accruals	which may have been missed / over accrue
Accounting Errors	Misclassifications	eg. Between opex and capex or expense types
Timing Changes	Timing differences	such as budget phasing, early / later agreed payments or delivery methodology impacting phasing of costs
Timing Changes	Prioritizations	Projects will begin earlier or later than originally planned - Management Determined
Timing Changes	Delays	Materials (chip shortage), Procurement issue (negotiations), resource issues (internal and external) - Beyond Management Contro
Other	Other	Please explain

Niagara Mohawk Power Corporation d/b/a National Grid

The Brooklyn Union Gas Company d/b/a National Grid NY

KeySpan Gas East Corporation d/b/a National Grid

Cases 20-E-0380, 20-G-0381, 19-G-0309 and 19-G-0310

INFORMATION TECHNOLOGY CAPITAL INVESTMENT REPORT FOR QUARTER ENDED DECEMBER 31, 2022

SUBMITTED TO:
NEW YORK STATE DEPARTMENT OF PUBLIC SERVICE STAFF

February 28, 2023

nationalgrid

Information Technology Capital Investment Report for Quarter Ended December 31, 2022

I. Introduction

Niagara Mohawk Power Corporation (Niagara Mohawk), The Brooklyn Union Gas Company (KEDNY), and KeySpan Gas East Corporation (KEDLI) (together "National Grid" or the "Companies") submit this Information Technology ("IT") Capital Investment Quarterly Report ("IT Quarterly Report") for the third quarter (Q3) of fiscal year 2023 ("FY23").[1] Pursuant to the Public Service Commission's ("Commission") January 20, 2022 Order in Cases 20-E-0380 and 20-G-0381, and August 12, 2021 Order in Cases 19-G-0309 and 19-G-0310, the Company is to provide quarterly variance reports to Department of Public Service Staff ("Staff") regarding IT capital projects and investments.[2] The IT Quarterly Report provides an explanation of the Company's overall IT investment plan approach, descriptions of program investment categories, and a comparison of budgeted to actual spending.

IT capital projects and investments that are shared investments across National Grid's operating companies are implemented and owned by National Grid USA Service Company, Inc. ("Service Company") and allocated to the benefiting US operating companies, including Niagara Mohawk, KEDNY, and KEDLI, in the form of Service Company rent expense once the investments are in-service. The Service Company rent expense includes a return on the capital investment (net of deferred taxes) plus booked depreciation expense.

Through Q3 FY23, National Grid spent $235.1 million compared to a year-to-date Service Company budget of $239.6 million, representing a $4.5 million underspend. Attachment 1 includes detail on budgets and actual spending for each of the portfolios and commentary regarding performance which has been presented to each of the Chief Information Officers responsible for the respective portfolios.

Attachment 2 includes a detailed breakdown of individual project spend by portfolio compared to the budgeted spend for each of those investments.

II. Program Categories/Spending Rationales

National Grid has organized the US Information Technology organization into six areas:

[1] The third quarter of FY23 is October 1, 2022 through December 31, 2022 ("Q3 FY23").

[2] *See* Cases 20-E-0380 and 20-G-0381, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service*, "Order Adopting Terms of Joint Proposal, Establishing Rate Plans and Reporting Requirements" (issued and effective January 20, 2022), Joint Proposal, Section IV.10.5; and Cases 19-G-0309 and 19-G-0310, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Energy East Corporation d/b/a National Grid for Gas Service*, "Order Approving Joint Proposal, as Modified, Imposing Additional Requirements" (issued and effective August 12, 2021), Joint Proposal, Section IV.6.5.

(1) Group Functions
(2) United States Electric
(3) United States Gas
(4) Customer
(5) Digital Platforms and Infrastructure Technology (Formerly Infrastructure and Operations)
(6) Security

IT capital investment is tracked based on these area designations. Descriptions of each portfolio category/spending rationale are provided below.

Group Functions

Group Functions executes National Grid's digital roadmaps, leveraging technology and reimagining processes in support of Finance, Enterprise Business Services, Human Resources, Legal, and Safety Health & Environmental, continuing to upgrade SOX related systems, and retiring outdated systems.

United States Electric

Provides IT Support to our US Electric Distribution and Transmission businesses. Modernizing our key legacy operational systems while building capabilities that will enable the energy transition and transformation to a Distribution System Operator (DSO). Through digital delivery, the Electric CIO organization provides greater efficiency and data driven insights into our workforce operations that provide:

- More automated and reliable grid controls systems
- Capital programs and projects that support the proliferation of Distributed Energy Resources (DERs)
- A growing market of prosumer electric customers

United States Gas

Provides IT Support to our US Gas Distribution and Transmission businesses by developing digital products based on the foundation of gas business enablement that deliver continuous improvement of our asset lifecycle management capabilities through:

- Risk-based asset investment planning
- Performance-based maintenance
- AI-based work optimization
- Mobile field products

Customer

Provides IT Support to our US Customer business, delivering best-in-class digital products that increase engagement and reduce costs to serve by:

- Modernizing our legacy "back-end" IT estate to support more digital products
- Liberating data and creating an abstracted and real-time customer data platform to assist with achieving energy efficiency targets

Digital Platforms and Infrastructure Technology (DPIT) (Formerly Infrastructure and Operations)
Supports our internal IT Architecture covering Cloud and Hosting, Critical Network Infrastructure, Networks, End User Experience and Operations. DPIT is focused on reducing National Grid's technical debt through cost effective, modernized IT infrastructure, enabling cloud capabilities that meet our risk profile. The organization enables a connected, seamless enterprise that is flexible, secure, reliable and scalable while maintaining acceptable levels of technical debt and risk.

Security
The Security function protects our people, premises, and digital systems with the objective of maintaining trust in National Grid services, focusing on:

- Reduction in likelihood and impact of security incidents
- Compliance with relevant laws, standards and regulations (*e.g.*, NERC, PCI DSS, GDPR, NIS-R)
- Improved user experience and standard of security practices across National Grid

The Security function aims to reduce the likelihood and impact of physical and cyber security incidents affecting National Grid. It does this by championing security best practices, technology, and promoting a security-awareness culture.

Other Category
For accounting purposes, there is an additional category listed in Attachment 2 to this report. The "Other" category has 3 main elements:

- The "IT4IT" sub-category acts as a funding bucket for NPV-positive projects to enable efficiency savings in our IT-run business operations.
- The "Other" sub-category is the allocation of operating expense overheads such as Enterprise Architecture and Solution Delivery charges to a centralized account. The capital charges for this overhead are made directly to individual projects.
- Finally, "Burdens" are the net removal of operating expense burdens to ensure our reporting agrees with our Corporate Controllable Cost reporting. Operating expense burdens are consolidated and reported separately under non-controllable costs in our corporate reporting packs.

For FY23 and FY24 there are Rhode Island separation costs flowing through this category. National Grid is currently providing IT services (applications, infrastructure, networks, etc..) to PPL as part of a 2-year Transition Services Agreement (TSA) following the sale of the RI Gas and Electric business. Significant effort is required to manage and track an orderly exit of all IT TSA services. National Grid must provide subject matter expertise, migrate critical application data, design and implement interim state processes and remove any customizations introduced to support delivery of the TSA. The costs for this are recovered from PPL and do not impact New York rate payers. The Group Functions portfolio is responsible for managing the expenditure although the costs role up through the "Other Category".

The summaries below explain the subjects presented in the attachments to the Q3 FY23 report.

A. Attachment 1

1. Summary (pages 2 – 3) – Overview of US IT as a whole.
    a. High-level executive summary, including work and challenges faced across the portfolios and a brief explanation around the forecast (p. 2).
    b. Actual expenditure versus budget by portfolio year to date, and full fiscal year forecast versus annual budget by portfolio (p. 2).
    c. Actual and forecasted spend by month for FY23 compared to budget, prior year and run rate. Each bar is color-coded to give an overview of the comparative spend per portfolio (p. 3).

2. Portfolio Sections – The remainder of Attachment 1 includes six sections, one for each of the portfolios: Group Functions; US Gas; US Electric; Customer; DPIT; and Security. Each portfolio consists of 3 pages:
    a. The first page provides a narrative financial and business overview. The three charts on the first page show: (1) operating expenses; (2) capital expenses; and, (3) total expenses. Each chart shows FY23 actuals and forecast by period versus budget, with trend lines showing prior year expenditure and run rate. This page also shows the full year forecast and year to date total and capital expenditure compared to the budget.
    b. The second page of the portfolio sections has two charts on the top left of the page: the top chart is actual expenditure versus budget by program year to date. The second chart compares full fiscal year forecast versus annual budget by program. There is also a key challenges section and the escalation point for that challenge. On the bottom half of this page is a chart showing actual and forecasted spend for FY23 compared to budget, prior year and run rate for the portfolio. Each bar is color-coded to give an overview of the comparative spend per program.
    c. The third page of each portfolio section provides a status update on the various projects. A priority investments list for the portfolio and their status appears at the top right hand of the page. Blue means complete, green means on track, amber means some risk, and red means focused intervention required.

    Attachment 1 is based on National Grid's monthly Strategic Portfolio Reviews (SPR) and reporting cadence to senior management of each portfolio, and ultimately to the IT leadership team.

B. Attachment 2

INVP Level Variances - This attachment provides a detailed listing of individual projects in each of the six portfolios explaining spend in FY23, including:

(1) Total actual spend compared to budget and variance;
(2) A variance reason for all capital investments which are outside the 10% tolerance to budget.

National Grid

Quarterly Report – Q3 FY23
Information Technology

28th February 2023

nationalgrid

US IT

Financial Performance

Portfolio View - Year to Date $M

Portfolio	Capex			Opex			Totex		
	Actuals YTD	Budget YTD	Variance	Actuals YTD	Budget YTD	Variance	Actuals YTD	Budget YTD	Variance
US Electric	25.4	42.4	17.0	2.7	6.0	3.3	28.1	48.4	20.3
US Gas	33.3	20.0	(13.3)	3.2	3.2	0.0	36.5	23.2	(13.3)
Customer	30.9	27.1	(3.8)	4.1	3.3	(0.8)	35.0	30.4	(4.6)
Group Functions	27.2	34.5	7.3	5.8	6.0	0.2	33.0	40.6	7.5
DPIT	61.3	70.1	8.8	4.6	9.7	5.1	65.9	79.8	13.9
Security	19.8	32.6	12.8	2.7	6.0	3.3	22.5	38.6	16.0
Other	0.6	(17.2)	(17.8)	13.5	(4.0)	(17.5)	14.1	(21.2)	(35.3)
Total	**198.5**	**209.4**	**11.0**	**36.6**	**30.2**	**(6.4)**	**235.1**	**239.6**	**4.5**

Portfolio View - Full Year $M

Portfolio	Capex			Opex			Totex		
	FY Forecast	FY Budget	Variance	FY Forecast	FY Budget	Variance	FY Forecast	FY Budget	Variance
US Electric	41.0	57.7	16.7	6.3	7.9	1.6	47.3	65.7	18.3
US Gas	50.8	28.0	(22.8)	4.5	4.5	(0.0)	55.3	32.5	(22.8)
Customer	43.3	33.5	(9.8)	5.5	4.0	(1.5)	48.8	37.5	(11.3)
Group Functions	40.6	41.9	1.3	8.4	8.3	(0.2)	49.0	50.2	1.2
DPIT	93.7	90.4	(3.3)	7.0	13.3	6.3	100.7	103.8	3.0
Security	33.2	42.1	9.0	3.1	7.1	4.0	36.3	49.2	13.0
Other	1.7	5.5	3.8	23.0	(3.4)	(26.4)	24.7	2.1	(22.6)
Total	**304.2**	**299.3**	**(5.0)**	**57.9**	**41.7**	**(16.1)**	**362.1**	**341.0**	**(21.1)**

IT Commentary

Delivery

Progress on delivery continues although slightly behind expectations. Significant releases this quarter include: Advanced Distribution Management Systems up feeders, Advanced Metering Infrastructure minimum viable product enabling customers to view energy usage and Procure to Pay data analytics tool to detect non-compliance.

Challenges

Commercial and contracting issues and chip shortages still exist in the post-COVID era which is slowing spend and creating challenges. Specialist IT resources and business resources are proving difficult to secure. However, capabilities are being delivered across the portfolios and management is actively re-prioritizing resources to deliver for customers.

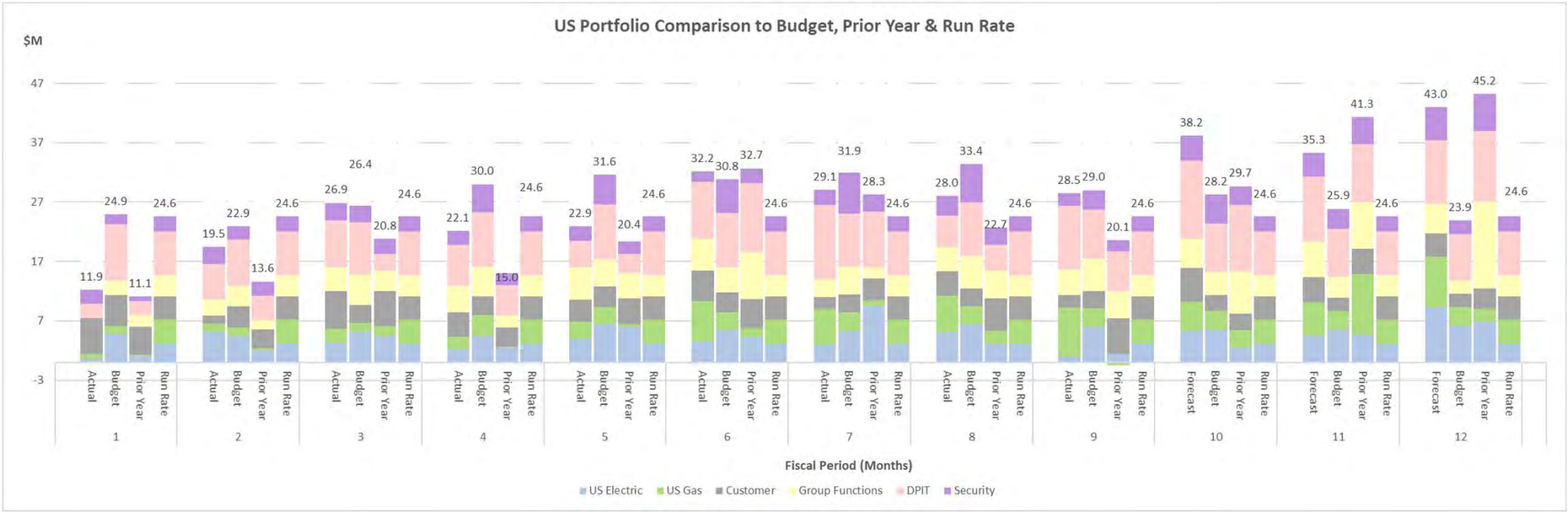
Financial Performance

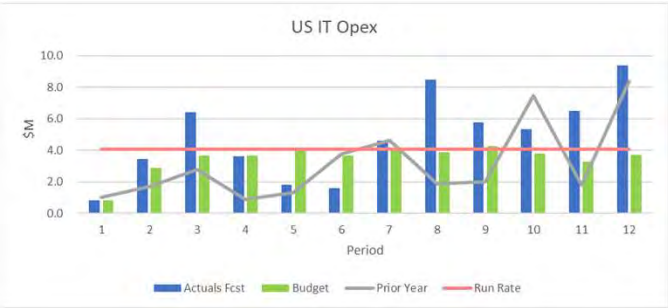
Financial performance is close to budget. The 'Other' category is offsetting the underspend in the Electric and Security portfolios, with a $23m budget phasing adjustment (following refinement of business plans) plus a reallocation of funds to Gas and Customer to meet user demands.

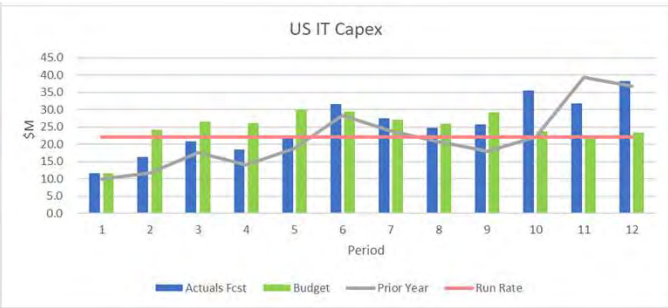
The remaining overspend in the "Other" category is the transition services agreement to PPL following the RI sale. The agreement ensures an orderly exit of all IT TSA services which National Grid is providing to cover subject matter expertise, migrate critical application data, design, implement interim state processes and remove customizations.

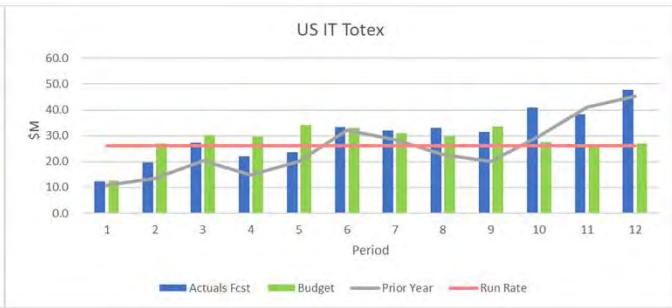
Although this is being managed through the Group Functions portfolio there is no impact to NY customers. Forecast for these costs is estimated at ~$22m which explains the forecasted overspend on budget across IT.

Portfolio Comparison & Cost Charts







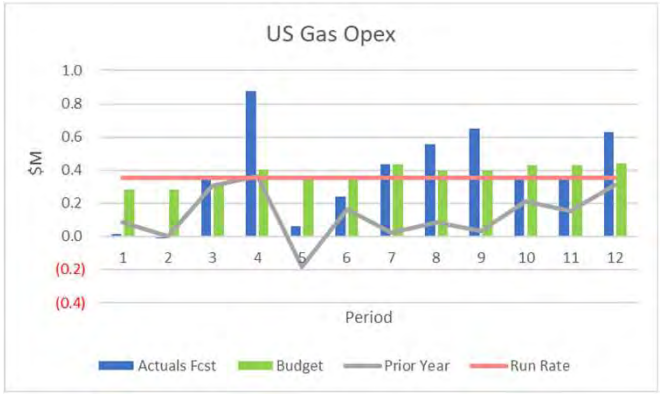


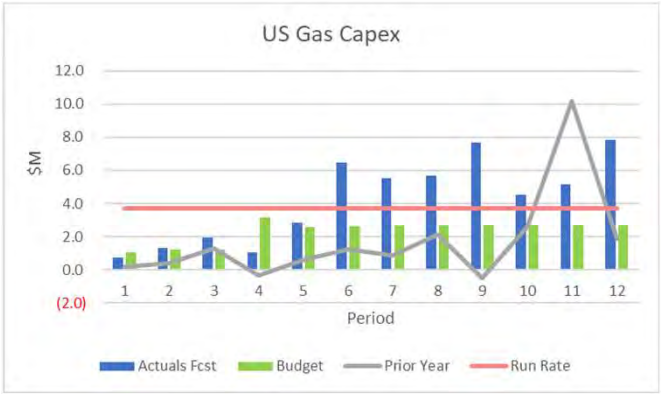
US Gas
Q3 FY23 Report

nationalgrid

US Gas

Financial Performance



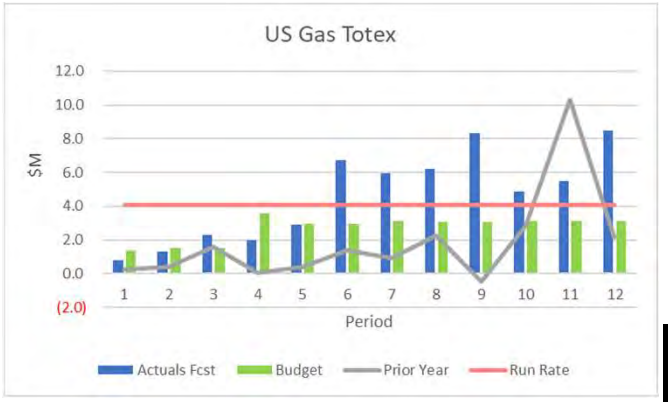


Totex

Budget ($M)	Forecast ($M)
32.5	55.3
YTD Budget($M)	YTD Actual ($M)
23.2	36.5

Capex

Budget ($M)	Forecast ($M)
28.0	50.8
YTD Budget($M)	YTD Actual ($M)
20.0	33.3



Portfolio Commentary

- U.S. Gas variance is largely due to the CS&E program, which is driving the forecast from the original plan and year to date variance to budget, with overruns on vendor labor and misalignment on where NG labor costs would fall. Re-sanction for the program is complete and we're on forecast with the revised sanction budget.
- Asset Management for LNG had a release in Q3 that stabilized the Platform and the integration between Maximo, Power plan and SAP. Improved User Interface validations were put in place to prevent any data quality issues with creation or processing of workorders
- Field Collection System(FCS) Upgrade project is to replace Automated and manual meter reading devices that are end of life with new devices, completed its analysis and design, proof of concept for application compatibility and expects to place the order of device in early Q4
- Gas Management System(SCADA) project will upgrade the hardware and software to the modern infrastructure, better security protocols and capabilities to handle outages. Team has completed business, technical and security requirements. Placed order for partial hardware related to security, software development and network
- Gas Transportation Information System (GTIS) consolidation program goal is to migrate UNY and MA regions to consolidated common GTIS platform and decommission of the several legacy applications. For part one, functional improvements made to the application customer data processing. Hardware was upgraded including servers and database to support the GTIS architecture. Improved application security by leveraging Single Sign-On (SSO) with Multi-Factored Authentication (MFA) for secured user login operation. Remediated all infrastructure vulnerabilities. For the consolidation program, high level requirements have been gathered

[redacted]

- New Storms Response Management System went into post go-live support and validation, implementing a new SaaS solution that is more reliable and automated, replacing an outdated and unsupported tool

US Gas Program View of Financials

Gas Program View Year to Date $M

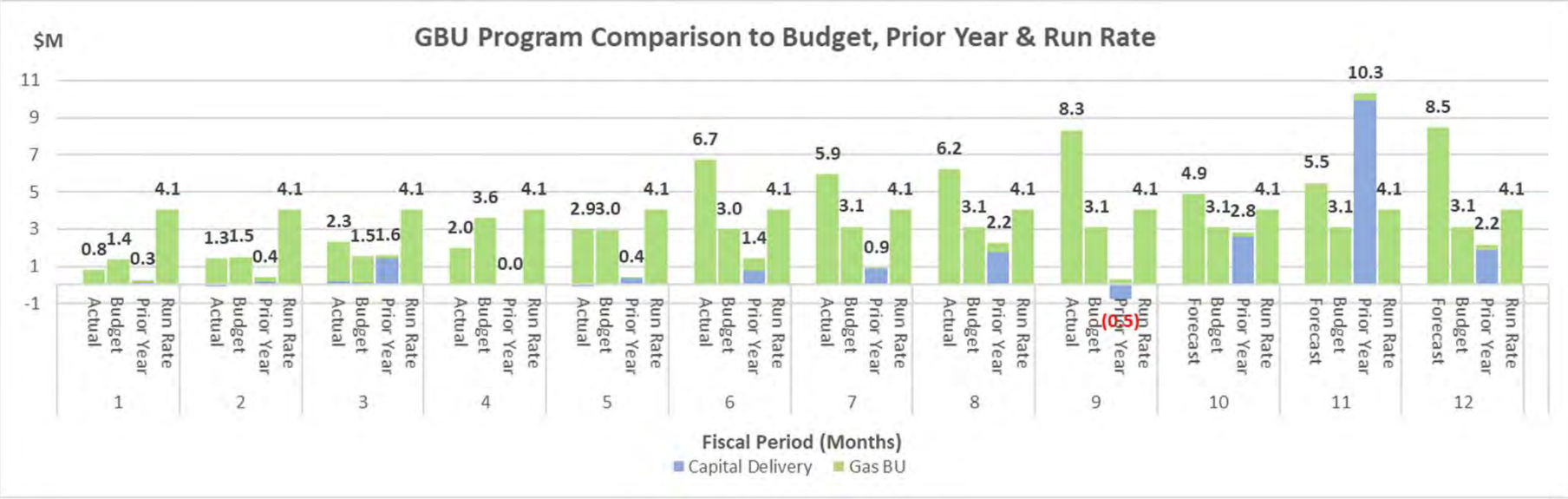
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
US Gas	Capital Delivery	0.1	0.3	0.2	(0.0)	0.0	0.0	0.1	0.3	0.2
	Gas BU	33.2	19.7	(13.5)	3.2	3.2	(0.0)	36.4	22.9	(13.5)
		33.3	**20.0**	**(13.3)**	**3.2**	**3.2**	**0.0**	**36.5**	**23.2**	**(13.3)**

Gas Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
US Gas	Capital Delivery	0.1	0.3	0.2	(0.0)	0.0	0.0	0.1	0.3	0.2
	Gas BU	50.7	27.8	(23.0)	4.5	4.5	(0.0)	55.2	32.2	(23.0)
		50.8	**28.0**	**(22.8)**	**4.5**	**4.5**	**(0.0)**	**55.3**	**32.5**	**(22.8)**

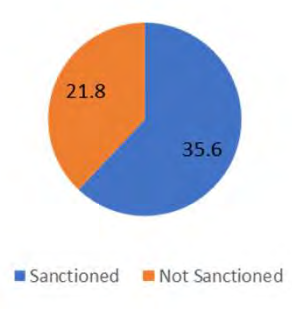
Key Challenges

Challenge	Escalation Point
Gas Storm Hardening project delays to network equipment backorder by Cisco	DPIT
GMS Upgrade (Scada)- Cisco network equipment delivery at risk for March delivery	DPIT



Mobilisation Summary

FY23 Sanction Status ($M Totex)



Status

- Based on 3+9 forecast, 6 projects left to sanction for FY23 have been either postponed or cancelled

Resources

- 3 new Project Managers have been hired, onboarded and are leading FY23 projects
- Overruns on vendor labor and misalignment on NG labor for CS&E Program, re-sanction is now completed

Scope

- Remaining pipeline projects need to complete scope assessment as part of business case analysis and sanctioning process.
- Additional Meter Shop project added to portfolio scope in FY23-Sanction completed and project in progress

National Grid

Priority Investments

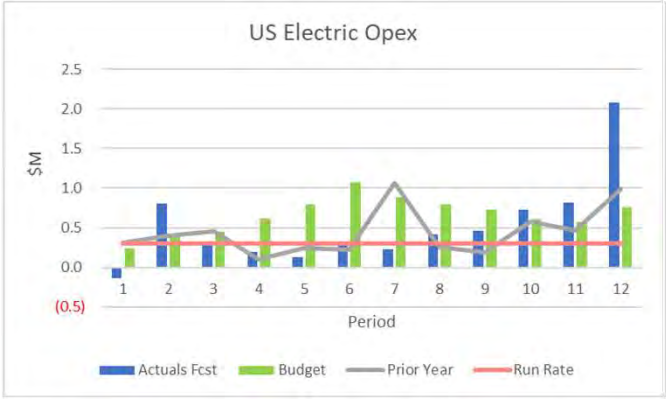
Program	Status	FY 23 Budget ($m)	RAG This Month	RAG Last Month
Gas Storm Hardening	Inflight	0.4		
LNG Asset & Work Mgmt.	Inflight	1.6		
Summons for Corrective Actions	PIS	0.1		
Enterprise Scalability and Resilience	Inflight	1.2		
Continuous Improvement for Gas Applications	Inflight	1.2		
PIF Replacement	PIS	0.0		
LNG IT support for OT	Start-up	0.9		
Gas I&R, Meter Ops, and Field Device Replacement	Inflight	0.1		
Enterprise Platform for Gas Retail Choice (GTIS)	Inflight	2.0		
Field Collection Systems (FCS) Upgrade	Prestart up	0.8		
Gas Management System (SCADA) Upgrade	Start-up	2.1		
New Gas DR Solution Evaluation	Inflight	0.9		
New Storms Response Management System Implementation	Inflight	0.3		
MOC Library & Action Item Automation	Inflight	0.2		
Continued Scaling & Evolution of Gas IT Transformation Program	Prestart up	20.1		
SOP Third-Party User Access for PE Review	Cancelled	0.1		
Enable Gas Curtailment Map in ESRI	Cancelled	0.04		
New System Operating Procedure Solution	Deferred/ FY24	0.1		
PIPES/Enterprise Maximo Integration	June decision	0.1		
Digital Asset & Work Management for CNG and I&R	Deferred/ FY25	0.3		
Enable Converted LI Gas Distribution Asset Data in ESRI	Deferred/ FY24	0.1		

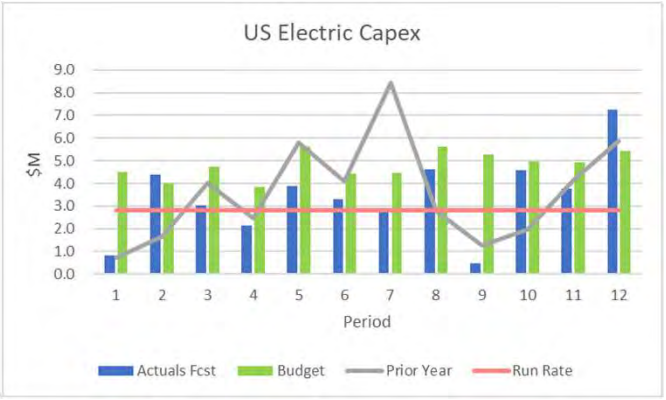
US Electric
Q3 FY23 Report

nationalgrid

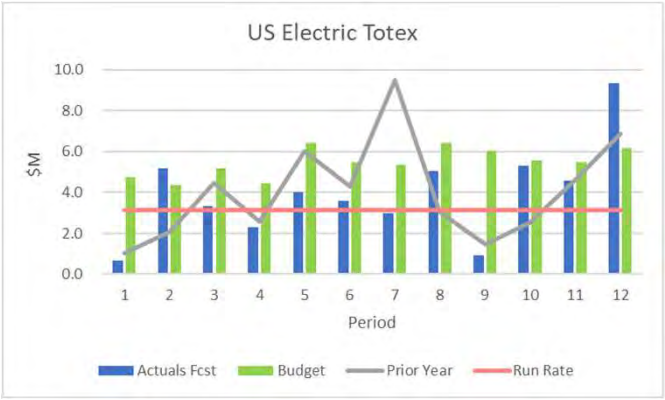
US Electric

Financial Performance





Totex	Budget ($M)	Forecast ($M)
	65.7	47.3
	YTD Budget($M)	YTD Actual ($M)
	48.4	28.1



Capex	Budget ($M)	Forecast ($M)
	57.7	41.0
	YTD Budget($M)	YTD Actual ($M)
	42.4	25.4

Portfolio Commentary

- Year to date under-spend of $17M CAPEX and $3M OPEX is being driven by supply chain delays, resource constraints, and various deliverability constraints that are delaying the start to planned products.
    - These delays are impacting the full year forecast and will reduce forecast v budget by $18.4M. This will, in part, be used to fund demand in other portfolios.
- **Field Collections System (FCS) Upgrade:** Remediated Tech Debt by upgrading our FCS system to its latest supportable version on October 28th. This system acts as our centralized billing hub for our Residential Customers by collecting, organizing, processing and distributing various data points to our critical billing systems (CSS, CRS)
- **MV90 Upgrade:** Successfully completed on November 11th, making our US Electric Meter Reading System more reliable and secure for our Commercial and Industrial Customers; this system supports 40% of US Electric Revenue
- **FutureNow:** Transformation Product released to the Transmission Business on November 7th, marking the completion of a key milestone in ensuring that the ~$22.6M in Forecasted US Type 1 Benefits are successfully delivered through FY24.
- **ADMS Program:** On track to reach target of 50% of feeders as “ADMS ready” by FYE; NY is currently at 644 of 1700 (38%).

US Electric Program View of Financials

Electric Program View Year to Date $M

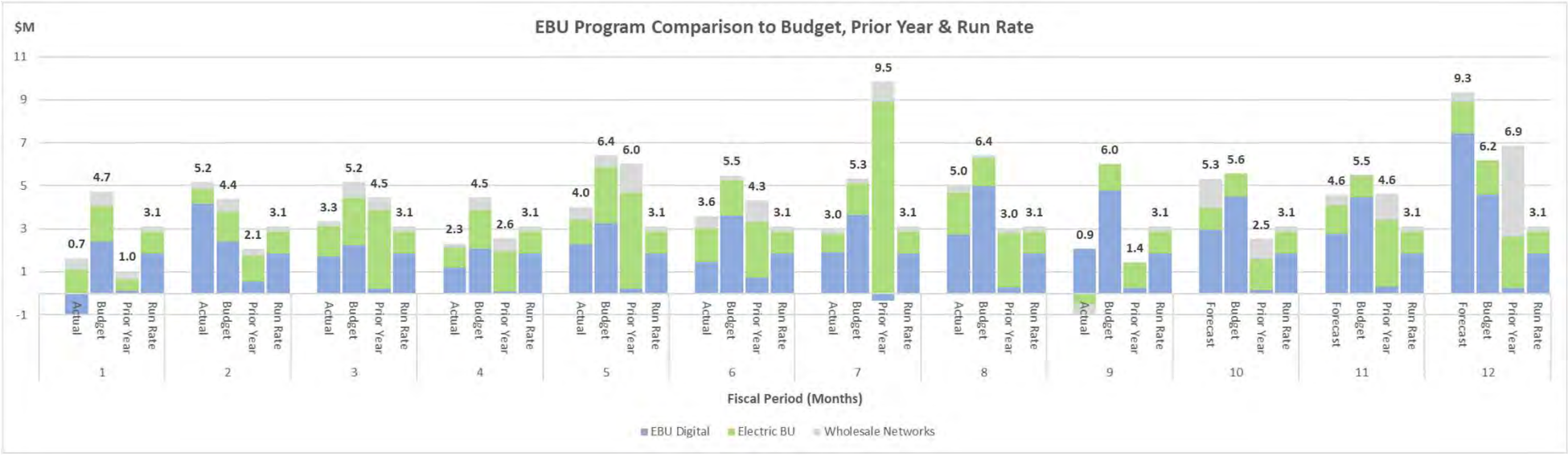
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
US Electric	EBU Digital	14.6	26.2	11.5	1.9	3.3	1.4	16.5	29.4	12.9
	Electric BU	8.4	12.6	4.2	0.7	2.6	1.9	9.1	15.2	6.1
	Wholesale Networks	2.3	3.6	1.3	0.1	0.1	0.0	2.4	3.8	1.3
	Totals	**25.4**	**42.4**	**17.0**	**2.7**	**6.0**	**3.3**	**28.1**	**48.4**	**20.3**

Electric Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
US Electric	EBU Digital	25.3	38.5	13.2	4.4	4.5	0.1	29.7	43.0	13.3
	Electric BU	11.1	15.6	4.5	1.8	3.3	1.5	12.9	18.9	6.0
	Wholesale Networks	4.6	3.6	(1.0)	0.1	0.1	0.0	4.8	3.8	(1.0)
	Totals	**41.0**	**57.7**	**16.7**	**6.3**	**7.9**	**1.6**	**47.3**	**65.7**	**18.3**

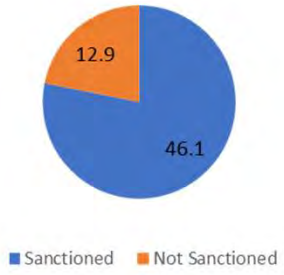
Key Challenges

Challenge	Escalation Point
1. Resource availability and resource quality challenges with ADAM Partners in the Core Portfolio; significant challenge within the Digital Lighthouse Portfolio.	CIO / HR / Commercial *(ADAM Partners)*
2. Global Supply Chain delays with critical hardware needed to deploy key programs/products*(i.e. ADMS, NYISO-PMU - resolved)*	Commercial, ITLT
3. Generation Asset Management Upgrade is experiencing schedule delays due to reprioritization and availability of integrated environments that are necessary for development and testing; *multiple business units are engaging the same pool of resources/environments*	CIO



Mobilisation Summary

FY23 Sanction Status ($M Totex)



Status

- Based on 9+3 forecast, ~78% of planned product spend has been Sanctioned.

Resources

- US Electric CIO Product Operating Model (POM) continue to make progress, with a goal to be rolled out by Fiscal Year End
- As part of the roll out of the new POM, interviews are ongoing for three (3) new Product Line Directors to help drive successful FY24 delivery.

Scope

- Remaining FY23 pipeline projects, including some FY24 projects are beginning to complete scope assessment as part of business case analysis and sanctioning process.

Priority Investments

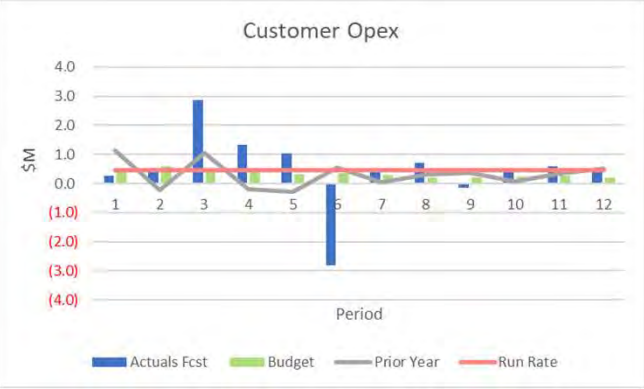
Program	Status	Budget ($m)	RAG This Month	RAG Last Month
GridMod	Inflight	45.0		
ADMS	Inflight	24.2		
FutureNow	Inflight	11.2		
VMO	Inflight	8.6		
OMW	Inflight	3.8		
Electric MV90 Upgrade	Complete	0.7		
FCS Upgrade	Complete	0.3		
Price Transmission	Complete	1.0		
Corporate PI	Inflight	1.6		
NYISO PMU – Large Generators	Inflight	0.7		
Content Management	Inflight	1.5		
Generation Asset Management Upgrade	Inflight	1.6		
Control Room EOL Server and Tool Replacement	Inflight	1.4		
Retail Settlement	Inflight	3.5		

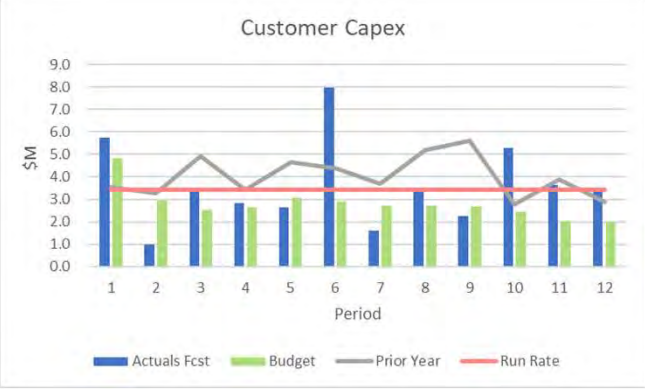
Customer
Q3 FY23 Report

nationalgrid

Customer

Financial Performance





Totex	Budget ($M)	Forecast ($M)
	37.5	48.8
	YTD Budget($M)	YTD Actual ($M)
	30.4	35.0

Capex	Budget ($M)	Forecast ($M)
	33.5	43.3
	YTD Budget($M)	YTD Actual ($M)
	27.1	30.9

Portfolio Commentary

- The Customer portfolio remains on track to meet additional demand ($11.3M) from the business over the initial plan.
  - Reallocation of funding across IT has been identified with demand for services remaining steady from Q2 to Q3;
  - Unified Web Portal 2.0 – significant driver of the year to date increase over budget. Additional spend will provide an improved user experience.
  - Technical Debt – Targeting technical debt remediations that support the Customer System and the Customer offices.

- Major releases in Q3 included:
  - Deployed Advanced Metering Infrastructure (AMI) Customer Portal update (My Energy Usage widget), allowing for MVP customers to see energy use within a specified period of time.
  - Implemented (5) Tactical ZBR Interactive Voice Response (IVR) improvements forecasting a savings of $225K/month. This brings the annualized total of IVR efficiency improvements to $10M.
  - Successfully implemented the 2nd release of “Traintrax” which provides system analysts improved system management of CSS batch process.
  - CSS Preferred Name, GBE Preferred Pronoun and GBE Preferred Name was successfully implemented into production to comply with the Preferred Name/Pronoun legislation passed in New York State.
  - Deployed (2) Bulk Pay change features into UWP production environment to improve Customer satisfaction.
  - Delivered critical Clean Energy update to Customer account validation service application, which is used by trade allies and vendors that provide services to NY customers.

Customer Program View of Financials

Customer Program View Year to Date $M

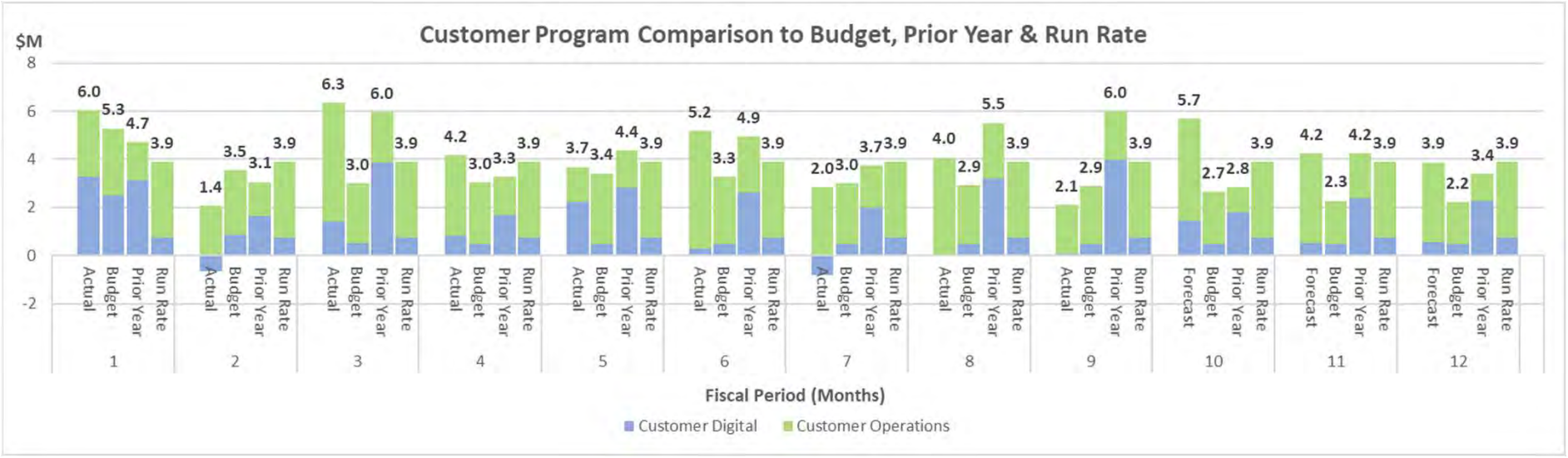
Portfolio / Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Customer Customer Digital	5.5	5.6	0.2	1.2	1.0	(0.2)	6.6	6.6	0.0
Customer Operations	25.4	21.4	(4.0)	3.0	2.3	(0.7)	28.4	23.7	(4.6)
Totals	**30.9**	**27.1**	**(3.8)**	**4.1**	**3.3**	**(0.8)**	**35.0**	**30.4**	**(4.6)**

Customer Program View Full Year $M

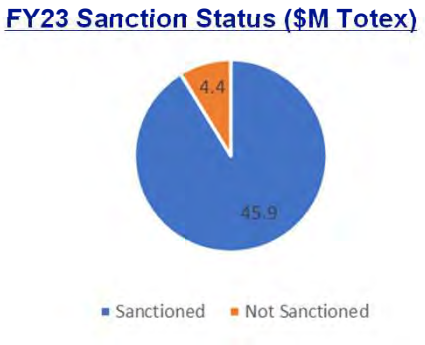
Portfolio / Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Customer Customer Digital	7.8	6.8	(1.0)	1.3	1.2	(0.2)	9.2	8.0	(1.2)
Customer Operations	35.5	26.7	(8.8)	4.2	2.8	(1.3)	39.6	29.5	(10.1)
Totals	**43.3**	**33.5**	**(9.8)**	**5.5**	**4.0**	**(1.5)**	**48.8**	**37.5**	**(11.3)**

Key Challenges

Challenge	Escalation Point
CRIS to CSS Timeline: *Development stream has fallen behind plan and additional quality testing has been added to the schedule. Additional resources being onboarded and increased governance implemented to reduce risk.*	Simon Baker
Virtual Assistant Timeline: *Implementation of a Virtual Assistant for US Customers has been delayed to July '23 due to challenges in deploying prerequisite infrastructure. Project team continues to develop longer term solution to deliver target customer benefits. (Part of the Zero-Based Review initiative)*	Simon Baker



Mobilisation Summary



Status

- A significant amount of the FY23 plan has been sanctioned

Resources

- All planned Q4 projects are active and resourced

Looking Forward

- Finalizing plan for FY24 investments in preparation to sanction on or before start of fiscal new year

Priority Investments

Program	Status	Budget ($m)	RAG	RAG Last Month
Zero Based Redesign (ZBR)	**Inflight**	**21.0**	Delays due to technical dependencies	
CRIS to CSS (C2C)	**Inflight**	**36.2**	Planned go-live at risk	
AMI	**Inflight**	**22.3**		
Clean Energy	**Inflight**	**9.3**		
Fix The Technology Foundations: UWP2.0	**Inflight**	**10.0**		
Customer Data Platform	**Inflight**	**7.0**		

Notes:

ZBR has delivered business benefits by seeing increases in Contact Center Interactive Voice Response (IVRs) containment, yet there continue to be delays on the Digital Assistant capability due to technical dependencies.

As noted in current challenges, C2C workstream has fallen behind plan and go-live is currently at risk.

AMI continues to make significant progress, having completed a major milestone by correctly billing MVP accounts for 6+ months.

US Security
Q3 FY23 Report

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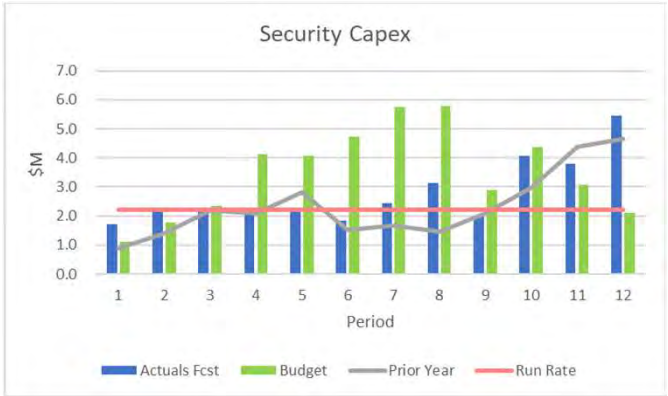
Security

Financial Performance



Totex

Budget ($M)	Forecast ($M)
49.2	36.3
YTD Budget($M)	YTD Actual ($M)
38.6	22.5

Capex

Budget ($M)	Forecast ($M)
42.1	33.2
YTD Budget($M)	YTD Actual ($M)
32.6	19.8

Portfolio Commentary

[redacted]

Security Program View of Financials

Security Program View Year to Date $M

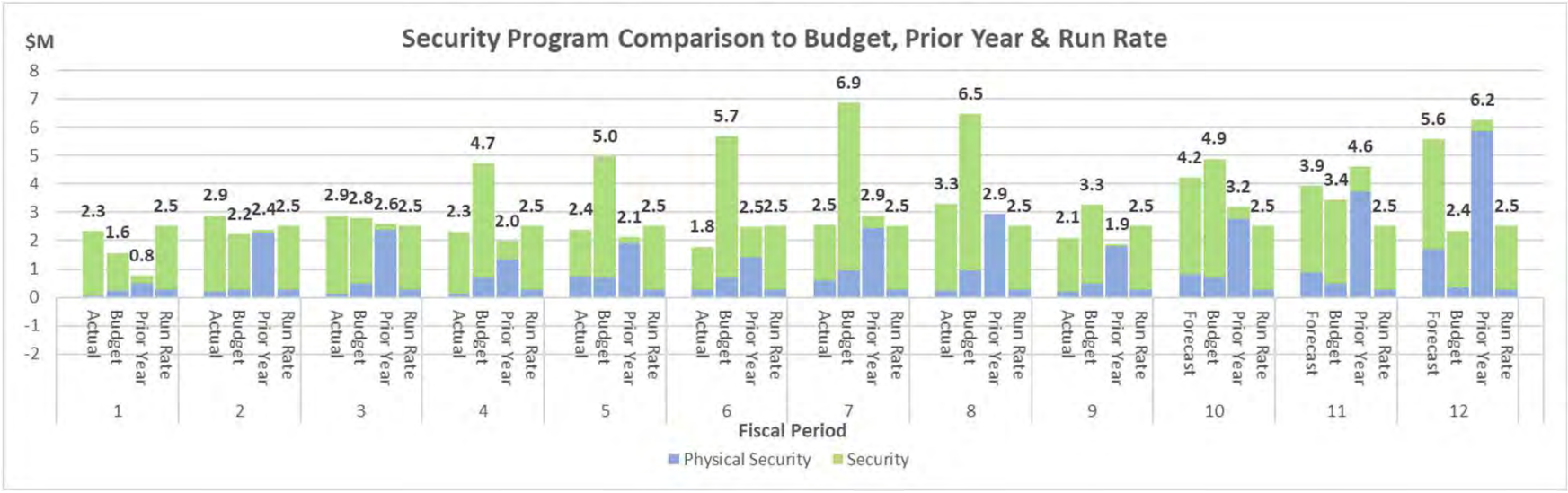
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Security	Physical Security	2.5	5.4	2.9	0.0	0.0	0.0	2.5	5.4	2.9
	Security	17.3	27.2	9.8	2.7	6.0	3.3	20.0	33.1	13.1
	Totals	**19.8**	**32.6**	**12.8**	**2.7**	**6.0**	**3.3**	**22.5**	**38.6**	**16.0**

Security Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Security	Physical Security	5.9	7.0	1.1	0.0	0.0	0.0	5.9	7.0	1.1
	Security	27.3	35.2	7.9	3.1	7.1	4.0	30.4	42.3	11.9
	Totals	**33.2**	**42.1**	**9.0**	**3.1**	**7.1**	**4.0**	**36.3**	**49.2**	**13.0**

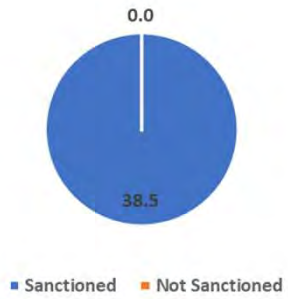
Key Challenges

Challenge	Escalation Point
OT Program - Earlier Supply Chain constraints on PAW and IDS components effectively mitigated through early procurement	Internal, Business and Supplier
Resource constraints are being addressed and future delivery plans will be adjusted with impacts to FY24.	Internal
Teams are working to identify pull-forward, advance purchasing and other opportunities to ensure we achieve Q3RF full year forecast in the event of additional challenges.	Internal



Mobilisation Summary

FY23 Sanction Status ($M Totex)



Status

- All projects have now been sanctioned.

Resources

- Resource constraints are being addressed with many key roles filled and new starters on-boarded. Process continues for vacancies.

Scope

[redacted]

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
[redacted]				

US Group Functions
Q3 FY23 Report

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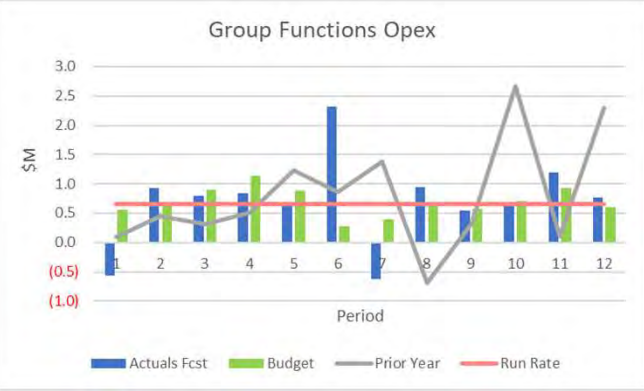
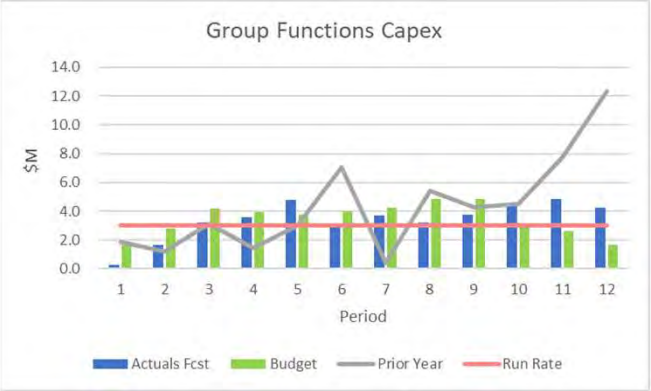
Group Functions

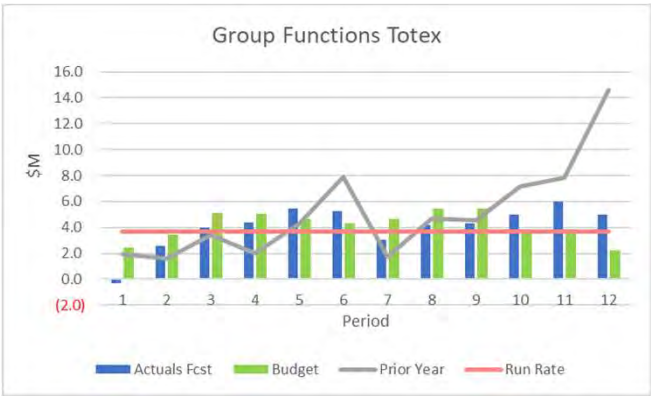
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	50.2	49.0
	YTD Budget($M)	YTD Actual ($M)
	40.6	33.0

Capex	Budget ($M)	Forecast ($M)
	41.9	40.6
	YTD Budget($M)	YTD Actual ($M)
	34.5	27.2



Portfolio Commentary

Group Functions is tracking to budget for full year forecast with accelerated run rate in Q4. YTD financial results are lagging to forecast due to delays in project start-ups.

Key Solutions Delivered in Q3:

Core PowerPlan Upgrade & Lease Consolidation

The upgrade allows us to decommission a standalone PowerPlan instance for Lease as well as to eliminate tech debt by bringing the application and underlying infrastructure to supported versions. Decreases the complexity of future upgrades reducing cost to maintain the application and ensures continued compliance of this critical SOX application.

Procure to Pay (P2P) Data Analysis Tool (Palantir)

Enhancements delivering advanced analytics on fraudulent activities by connecting to all the systems generating P2P transactions to enable 100% compliance. Enables auto alerts & outcome tracking on non-compliant activity, direct triage email capability from Foundry and trend reporting based on Detection Strategies.

HR Support Pack Upgrade (Q3 Release)

Delivers the Federal and State required updates for Payroll and Tax compliance and to comply with regulatory reporting.

Onboarding 2.0

Brings improved productivity for new hires by enhancing user interfaces and utilizing auto-population. Contributes to reducing days to onboard new resources for equipment and access provisioning.

Group Functions Program View of Financials

Group Functions Program View Year to Date $M

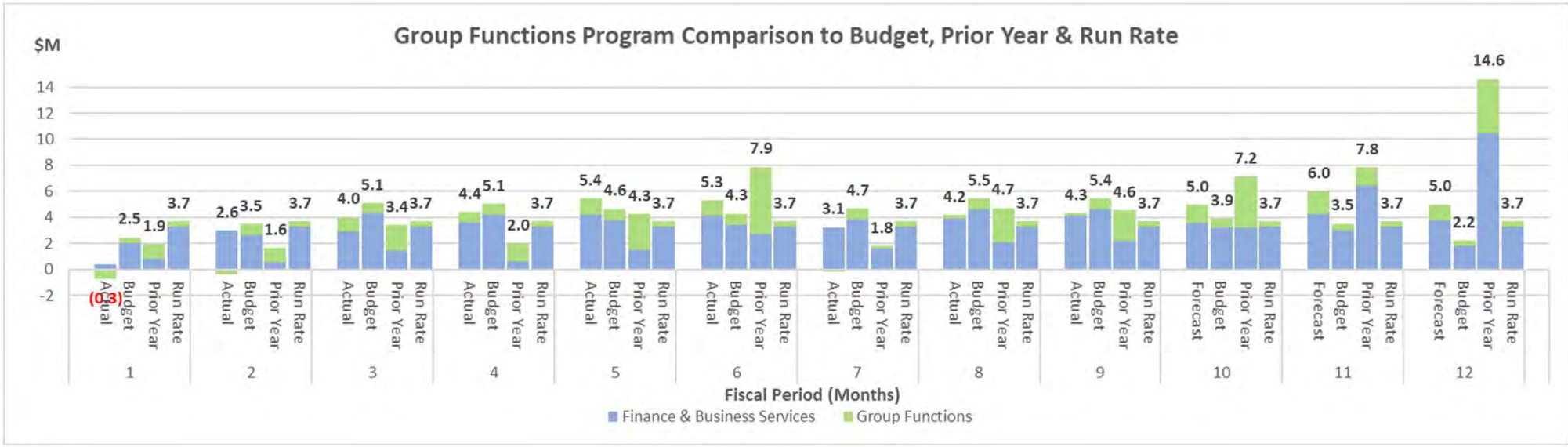
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Group Functions	Finance & Business Services	25.0	28.5	3.6	4.6	4.9	0.4	29.5	33.4	3.9
	Group Functions	2.2	6.0	3.8	1.3	1.1	(0.2)	3.5	7.1	3.6
	Totals	**27.2**	**34.5**	**7.3**	**5.8**	**6.0**	**0.2**	**33.0**	**40.6**	**7.5**

Group Functions Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Group Functions	Finance & Business Services	35.2	34.7	(0.5)	6.0	6.8	0.8	41.1	41.4	0.3
	Group Functions	5.4	7.3	1.8	2.5	1.5	(0.9)	7.9	8.8	0.9
	Totals	**40.6**	**41.9**	**1.3**	**8.4**	**8.3**	**(0.2)**	**49.0**	**50.2**	**1.2**

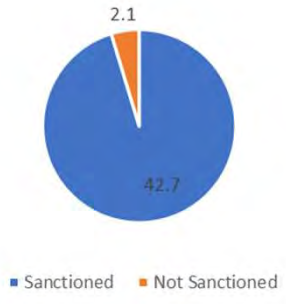
Key Challenges

Challenge	Escalation Point
• Resource constraints due to competing priorities and programs for both IT and Business resources	• Working with ADAM partners to backfill key roles • Recruitment ongoing to supplement with contractors
• Shortage of skilled/trained resources with Agile experience to help with transition to agile ways of working	• Upskill key resources in both IT and the business • Recruit and hire to fill open requisitions
• Risk – Capex moving to Opex due to new Finance policy on SaaS could impact ability to deliver financial commitments	• Working with finance to understand capitalization rules and ensure SaaS arrangements are GAAP compliant



Mobilisation Summary

FY23 Sanction Status ($M Totex)



Status

- The Ariba program required additional funding over the initial budget and has been re-sanctioned. The funding is required for additional resources to cover increased scope and an extended delivery timeline.
- Performing analysis on FY24 planned investments which can be moved to start-up in FY23

Resources

- Working with our vendor partners to staff for shortages so we can continue to deliver our business benefits planned for this fiscal year.

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
Blackline: Task Management	Complete	1.0		
Ariba Guided Buying	Inflight	1.4		
Core Power Plan Upgrade and Lease Consolidation	Complete	2.3		
Joiners, Movers, Leavers	Start-up	3.2		
Warehouse Investment Recovery Digitization	Inflight	1.0	Schedule	
R2R: Robotic Process Automation	Inflight	1.8	Schedule	
SAP Portals to Fiori	Complete	0.7		

Digital Platforms & Infrastructure Technology (DPIT)

Q3 FY23 Report

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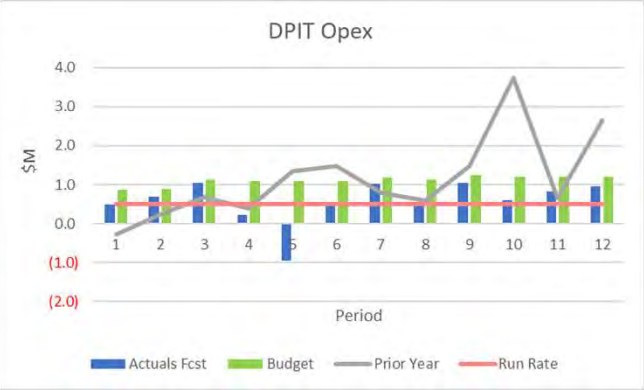
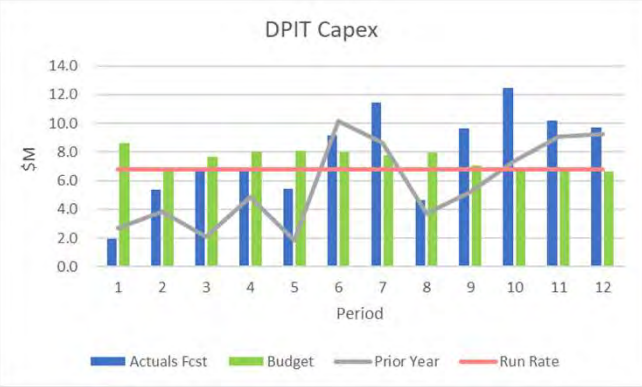
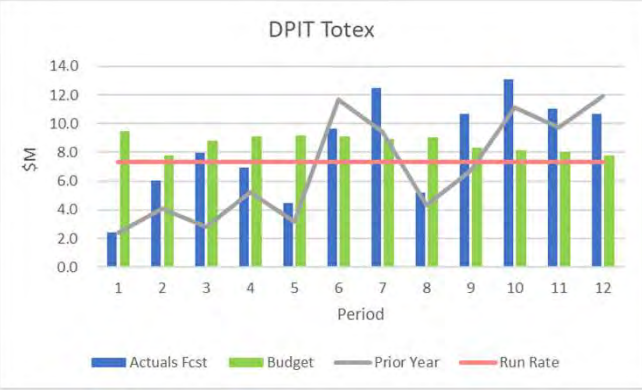
DPIT

Financial Performance







Totex	Budget ($M)	Forecast ($M)
	103.8	100.7
	YTD Budget($M)	YTD Actual ($M)
	79.8	65.9

Capex	Budget ($M)	Forecast ($M)
	90.4	93.7
	YTD Budget($M)	YTD Actual ($M)
	70.1	61.3

Portfolio Commentary

- **Financials:** Year to date ~$14M favorable to budget driven by budget phasing across the portfolio.
- The change in full year forecast since Q2 due to increased scope for Digital Workplace Services. DPIT is balancing spend within the portfolio and forecasting to FY23 budget, with:
  - Slight Capex overspend driven by Data Center Transformation due to increased upfront cost to drive additional downstream operational savings, and Supply Chain issues causing delays in the schedule
  - Opex underspend due to costs shifting out of DPIT to NG exceptional funding
- **Technical Debt:** Overall reduced to 16.9% (FY23 goal: 18%)
- **Benefit to Customers** delivered through Q3:
  - Unified Technology Operations Center(UTOC) Minimum Viable Product to provide overwatch on all technology and provide predictive analysis of asset health for proactive maintenance – ticketing solution is 100% live
  - Network and CNI upgrades to maintain reliability, availability and improve security
  - Windows as a Service deployed updates to >23K devices
  - Cloud & Hosting: migration of data ~ 98% complete to reduce risk of applications hosted on old and unsupported hardware, reduce Technical debt, and enable Data Center Transformation
  - Digital Platforms: several deliverables to provide secure /reliable/scalable/modernization improvements

DPIT Program View of Financials

DPIT Program View Year to Date $M

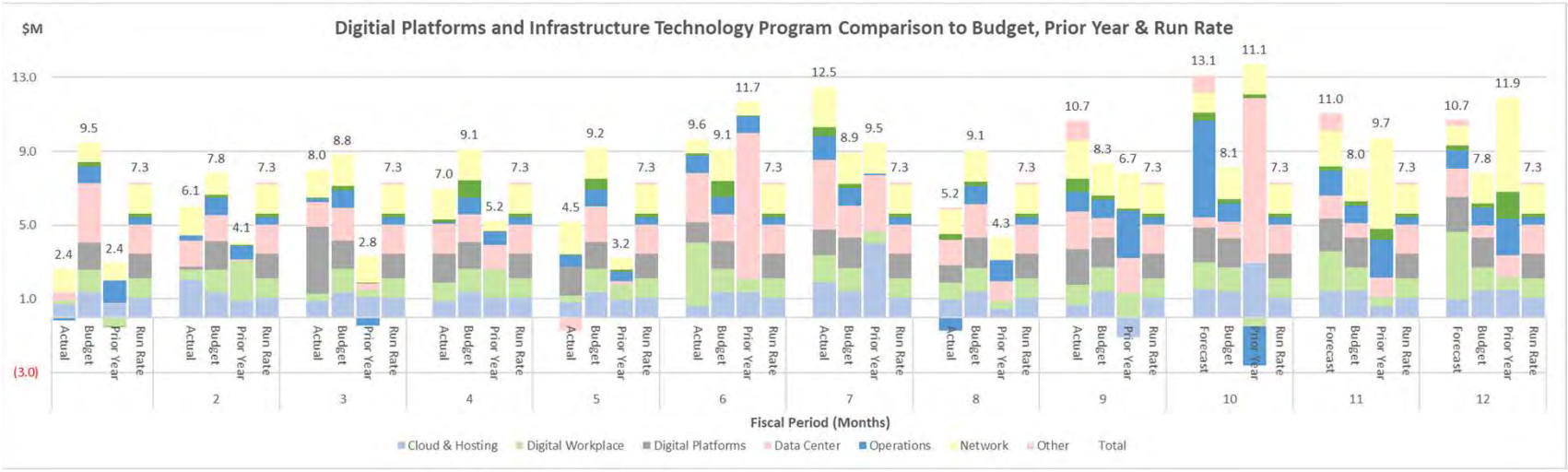
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
DPIT	Cloud & Hosting	8.3	11.2	2.9	1.0	1.2	0.1	9.4	12.4	3.0
	Digital Workplace	8.5	10.7	2.2	1.0	0.6	(0.4)	9.4	11.3	1.9
	Digital Platforms	11.1	12.1	1.0	1.1	1.7	0.6	12.3	13.8	1.6
	Data Center	13.6	11.1	(2.5)	0.4	4.8	4.4	14.0	15.9	1.9
	Operations	3.4	8.1	4.7	0.1	0.3	0.2	3.5	8.4	4.9
	CNI	1.8	3.6	1.8	0.1	0.1	(0.0)	1.9	3.8	1.8
	Network	13.4	13.2	(0.2)	0.8	1.0	0.2	14.2	14.2	0.0
	Other	1.1	0.0	(1.1)	0.0	0.0	(0.0)	1.2	0.0	(1.2)
	Totals	**61.3**	**70.1**	**8.8**	**4.6**	**9.7**	**5.1**	**65.9**	**79.8**	**13.9**

DPIT Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
DPIT	Cloud & Hosting	11.6	15.0	3.4	1.6	1.6	(0.0)	13.2	16.6	3.4
	Digital Workplace	15.5	14.3	(1.2)	1.2	0.8	(0.4)	16.7	15.2	(1.6)
	Digital Platforms	15.9	16.4	0.5	1.9	2.2	0.3	17.8	18.6	0.8
	Data Center	17.0	11.5	(5.5)	0.4	6.8	6.4	17.4	18.3	0.9
	Operations	10.8	10.9	0.1	0.3	0.4	0.1	11.1	11.3	0.2
	CNI	2.7	4.3	1.6	0.2	0.2	0.0	2.9	4.5	1.6
	Network	17.0	18.0	1.0	1.2	1.3	0.1	18.3	19.3	1.0
	Other	3.2	0.0	(3.2)	0.2	0.0	(0.2)	3.3	0.0	(3.3)
	Totals	**93.7**	**90.4**	**(3.3)**	**7.0**	**13.3**	**6.3**	**100.7**	**103.8**	**3.0**

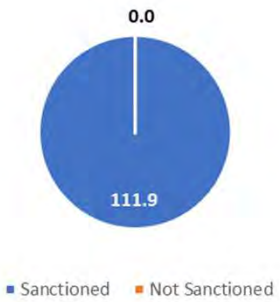
Key Challenges

Challenge	Escalation Point
Delays in US build, testing, and migration causing delays for US into FY24	• US Data Center is focused to complete in FY24
Higher spend rate in plan for second half FY23 requires close management of risks - driven by Ops Unified Technology Operations Centre (UTOC), Digital Workplace Services (HW, Atos), DCT (Mainframe SW, testing)	• Multiple monthly cost and forecast reviews • US portfolio forecast details provide breakout and risks
$9M HW/SW purchases to be completed in Q4	• Working closely with vendors, senior management, and commercial to mitigate risks associated with delivery



Mobilisation Summary

FY23 Sanction Status ($M Totex) *



*Total sanction includes exceptional costs as well as core portfolio costs

Status

- FY23 DPIT Product portfolios in the US are 100% sanctioned and inflight.
- **FY24 Investment Planning:** All FY24 product line epics are defined for FY24 by Product Managers
  - Product objectives align with Infrastructure and Business Unit objectives
  - Objectives and Epics will be prioritized to support FY24 final budget allocations
  - FY24 Sanctions are in process with a goal to have all approved before April 1 2023

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
Data Center Transformation (DCT) - *Modernization of Infrastructure & Networks*	Inflight	18.3	Schedule	
Cloud & Hosting - *Modernization of Infrastructure & Networks*	Inflight	16.6		
Digital Workplace Services - *Employee Satisfaction & Digital Product Delivery*	Inflight	15.2		
Networks - *Reduce Technical Debt*	Inflight	19.3		
Global Operations - *Employee Satisfaction & Digital Product Delivery*	Inflight	11.3		
Critical National Infrastructure (CNI) - *Reduce Technical Debt*	Inflight	4.5		
Digital Platforms - *Employee Satisfaction & Digital Product Delivery*	Inflight	18.6		

Resources

- No resource risks

Scope

- DPIT product portfolios on-track in FY23 to both Objectives/Key Results (OKRs) and financial forecast of $100.7m, except:
  - Data Center Transformation: External risks are being closely monitored

					Forecast to Budget Variance			Non Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it program	inv id	proj level descrip	Explain	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	9 + 3 Total	Budget Total	YTD Variance	Reason
Customer	Customer Digital	4750	Customer Experience Program	No - Opex Only	16,035	0	(16,035)			16,035	0	(16,035)	
		4750J	Web Self Service	YTD & Forecast	2,452,903	1,238,257	(1,214,646)	Mischarges	Complexity	3,385,772	1,238,257	(2,147,515)	Complexity
		4750M	ODS	YTD & Forecast	730,690	1,238,257	507,567		Funding Change	730,690	1,238,257	507,567	Funding Change
		5673	Mv Business Account (Nucleus Phase 2)	YTD & Forecast	3,000,759	4,145,250	1,144,491		Efficiencies	4,607,539	5,527,000	919,461	Efficiencies
		6206	Customer Smart Target	YTD & Forecast	413,875	0	(413,875)		Scope changes	413,875	0	(413,875)	Scope changes
	Customer Operations	3932	AIMS (TPV)	YTD & Forecast	(365,679)	0	365,679		Accounting	(365,679)	0	365,679	Accounting
		4904	Upgrade Servers for CCAE Reporting	No - Opex Only	(3,907)	0	3,907			(3,907)	0	3,907	
		5125	Customer Connection Program	Forecast Only	3,004,798	2,786,180	(218,618)		Scope changes	4,312,153	3,763,025	(549,128)	Scope changes
		5160	NY Value Stack Billing (Mandated)	YTD & Forecast	2,512,642	2,277,103	(235,539)		Complexity	3,361,919	2,295,586	(1,066,333)	Complexity
		5345B	Bill Print and Bill Composition Implementation	YTD & Forecast	1,469,436	1,055,130	(414,306)		Scope changes	2,373,421	1,055,130	(1,318,291)	Scope changes
		5345B	Bill Print and Bill Composition Implementation	See Above									Complexity
		5722	KEDU Rate Case (Mandated)	No - Opex Only	(1,640)	0	1,640			(1,640)	0	1,640	
		5723	KEDNY Rate Case (Mandated)	YTD & Forecast	288,874	380,331	91,457		Efficiencies	288,874	380,331	91,457	Efficiencies
		5778	Customer Ecosystem Stabilization	YTD & Forecast	928	991,583	990,655		Funding Bucket	311,462	991,583	680,121	Funding Bucket
		5795F	Community Distributed Generation (CDG) Consolidated Billing	YTD & Forecast	325,297	223,254	(102,043)		Other	325,297	223,254	(102,043)	Other
		5859A	Unified Web Portal	YTD & Forecast	1,848,581	2,100,000	251,419		Funding Change	1,848,581	2,800,000	951,419	Funding Change
		5863D	AIMS Stability	YTD & Forecast	(53,029)	0	53,029		Accounting	(53,029)	0	53,029	Accounting
		5926A	NiMO Rate Case	YTD & Forecast	(47,562)	0	47,562		Timing	(47,562)	0	47,562	Timing
		5987	Smart City Phase II	YTD & Forecast	38,519	0	(38,519)		Complexity	38,519	0	(38,519)	Complexity
		5996	Efficiency Target	Forecast Only	0	0	0			500,010	0	(500,010)	Other
		6007	Inside Agent	YTD & Forecast	(29,748)	0	29,748		Accounting	(29,748)	0	29,748	Accounting
		6032	Customer Billing COVID Transition (Minor Works)	YTD & Forecast	129,164	295,054	165,890		Scope chanaes	140,645	303,258	162,613	Scope chanaes
		6066	FCP Upgrade for CSS - Plan Forward	YTD & Forecast	586,008	372,526	(213,482)		Complexity	586,008	372,526	(213,482)	Complexity
		6112	WFH Hardware - Plan Forward	No Immaterial	(7,442)	0	7,442			(7,442)	0	7,442	
		6214	Customer Operational Enhancements FY22	No - Opex Only	7,896	0	(7,896)			7,896	0	(7,896)	
		6217	Customer Billing Minor Works FY22	No - Opex Only	(90,661)	0	90,661			(90,661)	0	90,661	
		6250	EV End-End Solution	YTD & Forecast	0	2,525,000	2,525,000		Scope changes	0	3,000,000	3,000,000	Scope changes
		6302	2022 AIMS Platform	YTD & Forecast	138,473	0	(138,473)		Timing	138,473	0	(138,473)	Timing
		6349	Customer Data Platform	YTD & Forecast	5,271,776	3,757,939	(1,513,837)		Complexity	6,938,427	5,010,585	(1,927,842)	Complexity
		6427	RI Purchase of Receivables (POR) Program	YTD & Forecast	54,496	87,197	32,701		Efficiencies	54,496	87,197	32,701	Efficiencies
		6435	Customer PCI Remediation	No - Within Limits	423,277	389,679	(33,598)			423,277	389,679	(33,598)	
		6492	NACHA Compliance- Security Encryption Effort	YTD & Forecast	885,373	474,850	(410,523)		Scope changes	1,517,950	474,850	(1,043,100)	Scope changes
		6496	NY Gender Pronoun and Name Mandate	YTD & Forecast	1,359,821	1,960,056	600,235		Efficiencies	1,696,674	1,960,056	263,382	Efficiencies
		6531	AIMS Upgrades and Enhancements	YTD & Forecast	664,363	483,750	(180,613)		Accounting	953,477	645,000	(308,477)	Accounting
		6538	Billing & Collections Mandates & Enhancements FY23	YTD & Forecast	0	3,240,596	3,240,596		Funding Bucket	625,009	5,294,426	4,669,417	Funding Bucket
		6543	Customer Minor Works	No - Opex Only	330,285	337,500	7,215			440,289	450,000	9,711	
		6584	UWP 2.0 Digital Re Architecture and Optimization	YTD & Forecast	8,064,535	0	(8,064,535)		Funding Change	11,330,275	0	(11,330,275)	Funding Change
		6451	One Step Auto Cashering	YTD & Forecast	57,113	0	(57,113)		Funding Change	314,040	0	(314,040)	Funding Change
		5778A	CBS Infrast MGKEDNY	YTD & Forecast	92,544	0	(92,544)		Funding Change	92,544	0	(92,544)	Funding Change
		5778B	CSS Infrast NGLeg+LI	YTD & Forecast	785,594	0	(785,594)		Funding Change	785,594	0	(785,594)	Funding Change
		5778C	Encryption & DR AR	No Immaterial	9,198	0	(9,198)			9,198	0	(9,198)	
		5778F	RightFax AR	YTD & Forecast	556,601	0	(556,601)		Funding Change	556,601	0	(556,601)	Funding Change
		5778D	Data Intee Svcs AR	No - Opex Only	5,882	0	(5,882)			5,882	0	(5,882)	
		6007X	Inside Agent	No - Opex Only	9,403	0	(9,403)			9,403	0	(9,403)	
		6654	Documentum Feasability & Analysis	No - Opex Only	64,467	0	(64,467)			247,239	0	(247,239)	
Customer Total					34,999,938	30,359,491	(4,640,447)			48,787,876	37,500,000	(11,287,876)	

					Forecast to Budget Variance			Non Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it_program	inv_id	prol_level_descrip	Explain	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	9 + 3 Total	Budget Total	YTD Variance	Reason
Group Functions	Finance & Business Services	3870A	Vision FM & Property Services Replacement	YTD & Forecast	75,421	0	(75,421)		Accounting	75,421	0	(75,421)	Accounting
		4418USB	RSA Archer Upgrade & Refresh	YTD & Forecast	71,980	0	(71,980)		Accounting	71,980	0	(71,980)	Accounting
		4562	US SAP: Business Warehouse (BW) Consolidation to HANA Enterprise Cloud (HEC)	YTD & Forecast	(58,530)	0	58,530		Accounting	(58,530)	0	58,530	Accounting
		5158	MyHub 2.0	YTD & Forecast	623,563	2,600,000	1,976,437		Funding Bucket	623,563	2,900,000	2,276,437	Funding Bucket
		5380A	Blackline: US Implementation	No - Opex Only	0	0	0			0	0	0	
		5466	AMAG HW and SW Upgrade	YTD & Forecast	1,191,338	800,000	(391,338)		Imprecise Estimates	1,585,073	800,000	(785,073)	Imprecise Estimates
		5497	Core Powerplan Upgrade and Lease Consolidation	YTD & Forecast	1,445,199	1,800,000	354,801		Efficiencies	1,572,355	2,300,000	727,645	Efficiencies
		5555	Fieldglass Replacement	YTD & Forecast	0	1,500,000	1,500,000		Funding Change	0	1,800,000	1,800,000	Funding Change
		5605	US SAP: Infrastructure Landscape FY21	No Immaterial	18,812	0	(18,812)			18,812	0	(18,812)	
		5606	US SAP: Infrastructure Landscape FY22	YTD & Forecast	(887,221)	0	887,221		Accounting	(887,221)	0	887,221	Accounting
		5610	Annual HR & Payroll Mandatory Service Pack Uperade (HRSP) FY22	No Immaterial	2.732	0	(2.732)			2.732	0	(2.732)	
		5611	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY23	YTD & Forecast	890,997	1,300,000	409,003		Imprecise Estimates	906,072	1,500,000	593,928	Imprecise Estimates
		5841A	Pension Administrator Replacement	No - Opex Only	0	0	0			0	0	0	
		5841C	401(k) Administration Changes	YTD & Forecast	123,090	0	(123,090)		Timing	123,090	0	(123,090)	Timing
		5842A	US SAP: GRC Upgrade	No - Opex Only	0	0	0			0	0	0	
		5842B	Mandatory FERC Form Changes	YTD & Forecast	(21,525)	0	21,525		Accounting	(21,525)	0	21,525	Accounting
		5843A	SAP Platform Upgrade: OS and Databases	No - Opex Only	0	0	0			0	0	0	
		5843B	US SAP Platform Uperade: Minor Releases	YTD & Forecast	(162,703)	0	162.703		Scope Chanees	(162,703)	0	162.703	Scope Chanees
		5843D	US SAP Healthcheck: Upgrade Critical Components - FY22	YTD & Forecast	869,528	700,000	(169,528)		Imprecise Estimates	869,528	700,000	(169,528)	Imprecise Estimates
		5895B	Everbridge / Critical Event Management	YTD & Forecast	(36,629)	0	36,629		Accounting	71,558	0	(71,558)	Complexity
		5896A	US Tax: SABRIX Upgrade	No - Opex Only	0	0	0			0	0	0	
		5896B	Warehouse SAP Scanner Replacement (Barcodine)	No Immaterial	(388)	0	388			(388)	0	388	
		5919	US SAP Minor Enhancements	No Immaterial	(5,246)	0	5,246			(5,246)	0	5,246	
		6002	Efficiency Target	No - Opex Only	0	0	0		Other	(600,000)	0	600,000	Other
		6137US	Enterprise PMO. Jira Alian release 2	YTD & Forecast	68,976	0	(68.976)		Fundine Chanee	68,976	0	(68.976)	Fundine Chanee
		6182US	Global Career Site	No - Opex Only	(48,999)	0	48,999		Accounting	(48,999)	0	48,999	Accounting
		6183US	HR Reporting Analytics	YTD & Forecast	56,699	1,000,000	943,301		Funding Change	56,699	1,300,000	1,243,301	Funding Change
		6185US	Onboarding 2.0	YTD & Forecast	208,660	0	(208.660)		Funding Change	208,660	0	(208.660)	Funding Change
		6219	US F&BS Minor Enhancements (Non-SAP)	No - Opex Only	14.542	0	(14.542)			14.542	0	(14.542)	
		6220	US F&BS Minor Enhancements (SAP)	YTD & Forecast	69,823	0	(69.823)		Timing	69,823	0	(69.823)	
		6221	Tech Debt: RiskMaster Replacement	YTD & Forecast	626,037	0	(626.037)		Funding Change	632,838	0	(632.838)	Funding Change
		6231	US Treasury: Allegro Upgrade Phase 2	YTD & Forecast	63,755	0	(63,755)		Timing	63,755	0	(63,755)	Timing
		6242US	ARIBA System Enhancements	YTD & Forecast	165,816	0	(165,816)		Timing	165,816	0	(165,816)	Timing
		6265	HR Policy Compliance	No - Opex Only	33,783	0	(33,783)			33,783	0	(33,783)	
		6270US	Digitising Learning	YTD & Forecast	449,798	0	(449,798)		Funding Change	423,887	0	(423,887)	Funding Change
		6273US	Workforce Data Domain. HR data. governance. Tools. Processes. EDM	YTD & Forecast	1,440,487	700,000	(740,487)		Timine	2,414,870	700,000	(1,714,870)	Timine
		6289B	FTOM: Record to Report	No Immaterial	(5,455)	0	5,455			(5,455)	0	5,455	
		6289D	FTOM: Record to Report – Task Management with Blackline	YTD & Forecast	1,180,296	1,000,000	(180,296)		Imprecise Estimates	1,180,296	1,000,000	(180,296)	Imprecise Estimates
		6289E	FERC Form Integration with eDesk	YTD & Forecast	501,771	1,100,000	598,229		Imprecise Estimates	501,771	1,100,000	598,229	Imprecise Estimates
		6290A	Source To Pay: Release 1 – P2P Roll Out Program	YTD & Forecast	6,326,801	1,433,000	(4,893,801)		Imprecise Estimates	9,592,532	1,433,000	(8,159,532)	Imprecise Estimates
		6290A	Source To Pay: Release 1 – P2P Roll Out Program	See Above					Complexity				Complexity
		6290B	Source To Pay: Release 2 - Supplier Onboarding	YTD & Forecast	103,189	1,300,000	1,196,811		Timing	103,189	1,300,000	1,196,811	Other
		6346US	CA: Grouo We Dieital Evolution Phase 1 - FY22	YTD & Forecast	(87.667)	0	87.667		Accountine	(87.667)	0	87.667	Accountine
		6346USC	Domain Management Refresh	No - Opex Only	(466,838)	0	466,838			(466,838)	0	466,838	
		6346USE	Brand Central Design and Implementation	No - Opex Only	(10,015)	0	10,015			(10,015)	0	10,015	
		6346USF	Brand Central Design and Implementation	No - Opex Only	(67,193)	0	67,193			(67,193)	0	67,193	
		6346USI	Video Strateaic Manaaement and Archivine (Enterprise)	No - Opex Only	(53,405)	0	53.405			(53,405)	0	53.405	
		6377	US Group Function Operational Requests	YTD & Forecast	(353,562)	0	353.562		Accounting	(353,562)	0	353,562	Accounting
		6421	P2P Data Analysis Tool	YTD & Forecast	1,647,222	900,000	(747,222)		Scope changes	1,664,990	1,200,000	(464,990)	Scope changes
		6426	Investment Recoverv Digitization	YTD & Forecast	891.257	1.000.000	108.743	Delavs		1.108.813	1.000.000	(108.813)	Timine
		6450x	Accelerated Telecom Networks Audit (POTS and Circuits)	No - Opex Only	843,891	0	(843,891)			843,891	0	(843,891)	
		6520US	Jira Align Phase 3	YTD & Forecast	1,144,600	0	(1,144,600)		Imprecise Estimates	1,946,172	0	(1,946,172)	Imprecise Estimates
		5607	US SAP: Infrastructure Landscape FY23	YTD & Forecast	2,057,584	3,000,000	942,416		Funding Bucket	2,804,671	4,000,000	1,195,329	Funding Bucket
		5843E	US SAP: BW/BI Uperade	YTD & Forecast	101.215	600.000	498.785		Timine	149.740	700.000	550.260	Timine
		5843F	US SAP PI/PO Upgrade	YTD & Forecast	1,025,557	800,000	(225,557)		Scope changes	2,377,390	900,000	(1,477,390)	Scope changes
		6529	US Group Function Minor Enhancements (SAP)	YTD & Forecast	723,799	900,000	176,201		Other	925,429	1,100,000	174,571	Other
		6530	US Group Function Operational Requests - FY23	YTD & Forecast	1,724,347	1,000,000	(724,347)		Imprecise Estimates	2,321,030	1,300,000	(1,021,030)	Imprecise Estimates
		4418USC	GRC Archer FY23	YTD & Forecast	86,084	1,125,000	1,038,916	Timing differences		1,160,214	1,500,000	339,786	Scope changes
		6514US	NG.com Refresh	YTD & Forecast	0	1,645,000	1,645,000		Funding Bucket	0	1,900,000	1,900,000	Funding Bucket
		6533US	SHE FY23: Group SHE Digital Evolution	YTD & Forecast	0	1,299,921	1,299,921		Timing	0	1,500,000	1,500,000	Timing
		6534US	Leeal FY23: Grouo Leeal Dieital Evolution	YTD & Forecast	0	568.713	568.713		Timine	0	708.800	708.800	Timine
		Corp23/6596	US Placeholder - Corp Tech Debt Remediation	YTD & Forecast	0	1,100,000	1,100,000		Funding Bucket	0	1,800,000	1,800,000	Funding Bucket
		6270	Digitising Learning	YTD & Forecast	0	400,000	400,000		Funding Bucket	0	600,000	600,000	Funding Bucket
		HR23	Tactical HR	YTD & Forecast	0	1,100,000	1,100,000		Funding Bucket	0	1,800,000	1,800,000	Funding Bucket
		OPS23	Placeholder (Tech Debt & App Rat)	YTD & Forecast	0	480,000	480,000		Funding Bucket	0	680,000	680,000	Funding Bucket
		6225US	R2R Reporting Use Cases	YTD & Forecast	0	580,000	580,000		Funding Bucket	0	1,300,000	1,300,000	Funding Bucket
		6289C	FTOM Record to Report: Process Mining with Celonis	No - Opex Only	0	0	0			0	0	0	
		6289G	FTOM R2R: Robotic Process Automation of Manual Journal Entries	YTD & Forecast	1.170.341	1.300.000	129.659		Scope chanees	1.265.666	1.800.000	534.334	Scope chanees
		R2R23	Finance + placeholder (Regulatory Compliance)	YTD & Forecast	0	400,000	400,000		Funding Bucket	0	800,000	800,000	Funding Bucket
		6467	Union Bidding	YTD & Forecast	374,781	0	(374,781)		Funding Change	428,125	0	(428,125)	Funding Change
		6566US	HR Policy Compliance FY23	No - Opex Only	131,911	0	(131,911)			239,565	0	(239,565)	
		6180US	CV Talent Screenine / Blind CVs	YTD & Forecast	264.158	0	(264.158)		Funding Chanee	350.593	0	(350.593)	Funding Chanee
		6424US	Infonet decommissioning	No - Opex Only	49,866	0	(49,866)			49,866	0	(49,866)	
		3870B	Vision FM replacement, Phase 2	YTD & Forecast	1,231,833	0	(1,231,833)		Funding Change	1,390,634	0	(1,390,634)	Funding Change
		6574	US Grouo Function: Tech Debt and Application Rationalization	No - Opex Only	0	0	0			200,000	0	(200,000)	
		6289F	US GAAP Statutory reporting with WDesk	YTD & Forecast	328,584	0	(328,584)		Funding Change	456,120	0	(456,120)	Funding Change
		6621US	My Hub	YTD & Forecast	659,082	0	(659,082)		Funding Change	2,132,065	0	(2,132,065)	Funding Change
		6618	US S4 Discovery Refresh	No - Opex Only	696,143	0	(696,143)			696,143	0	(696,143)	
	Grouo Functions	5899A	Cloud API Miarations Phase 2	YTD & Forecast	112.485	0	(112.485)		Accountine	56.964	0	(56.964)	Accountine
		5917A	Data Mgmt Tooling	No - Opex Only	(2)	0	2			(2)	0	2	
		5983	DB Landscape Optimization	YTD & Forecast	1,360,025	0	(1,360,025)		Funding Change	1,928,806	0	(1,928,806)	Funding Change
		6137	Enterprise PMO	No - Opex Only	3	0	(3)			3	0	(3)	
		6252US	Data Mgmt Tools and Initiatives	YTD & Forecast	(179,821)	0	179,821		Accounting	(179,821)	0	179,821	Accounting
		6253US	Cloud API Migration - Legacy Integration Tools	YTD & Forecast	692,322	0	(692,322)		Funding Change	692,322	0	(692,322)	Funding Change
		6254US	Platform as a Service (PaaS) on Azure	No Immaterial	11,909	0	(11,909)			11,909	0	(11,909)	
		6255US	Dialtal Foundation Tooline and DevSecOps	YTD & Forecast	(82.383)	0	82.383		Accountine	(82.383)	0	82.383	Accountine
		6256US	Platform Upgrades: Salesforce/Maximo/Muleosft	YTD & Forecast	(28,251)	0	28,251		Accounting	(28,251)	0	28,251	Accounting
		6258USA	Data Archival Platform Implementation	YTD & Forecast	74,806	0	(74,806)		Funding Change	74,806	0	(74,806)	Funding Change
		6258USB	Data Archival: US Group Function Portfolio	YTD & Forecast	1,625,832	900,000	(725,832)		Scope changes	2,327,220	1,100,000	(1,227,220)	Scope changes
		6260US	EA Foundation: Increased Capability	YTD & Forecast	102,990	0	(102,990)		Accounting	100,122	0	(100,122)	Accounting
		6261US	ServiceNow Upgrade: Stay Current FY22	YTD & Forecast	(182,874)	0	182,874		Accounting	(182,874)	0	182,874	Misclassifications
		6266US	EA Strategic Portfolio Planning & Delivery / Additional Capability	YTD & Forecast	0	800,000	800,000		Efficiencies	0	800,000	800,000	Efficiencies
		6313US	GDPR Purge Compliance	No - Opex Only	853	0	(853)			853	0	(853)	
		6461USA	Business Planning F&A (FP&A Enablement & Modelling Tool)	YTD & Forecast	(146,049)	1,978,182	2,124,231		Timing	1,489,176	2,500,000	1,010,824	Timing
		6468	GridStack - Infrastructure	YTD & Forecast	(1,219,690)	1,993,499	3,213,189		Accounting	(1,219,690)	2,657,999	3,877,689	Accounting
		6520US	Jira Align Phase 3	No - Opex Only	0	900,000	900,000		Other	0	900,000	900,000	Other
		6513US	CA: Grouo Web Dieital Evolution - FY23	YTD & Forecast	116.678	553.333	436.655		Efficiencies	225.926	800.000	574.074	Efficiencies
		6587US	E-Hub	YTD & Forecast	818,202	0	(818,202)		Funding Change	1,280,006	0	(1,280,006)	Funding Change
		6522US	SHE: Digitisation of the CMT Process	No - Opex Only	180,064	0	(180,064)			180,064	0	(180,064)	
		6549US	SHE: Resilience Dieital Evolution	YTD & Forecast	36.513	0	(36.513)		Funding Chanee	324.518	0	(324.518)	Funding Chanee
		6391US	SHE: Consolidated Incident Management and Reporting	No - Opex Only	0	0	0			333,112	0	(333,112)	
		6402US	Legal Operations	No - Opex Only	123,305	0	(123,305)			238,309	0	(238,309)	
		6563US	CA:NG.com Design	No - Opex Only	69,761	0	(69,761)			269,761	0	(269,761)	
		6458	OES IT Tool Sunset	No - Opex Only	20.088	0	(20.088)			20.088	0	(20.088)	
		6660	Prog Repository F&A	No - Opex Only	5,832	0	(5,832)			5,832	0	(5,832)	
Group Functions Total					33,022,570	40,556,649	7,534,079			49,000,767	50,179,799	1,179,032	

					Forecast to Budget Variance			Non Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it_program	inv_id	proi_level_descrip	Explain	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	9 + 3 Total	Budget Total	YTD Variance	Reason
Security	Physical Security	3617A	Physical Security - G20	YTD & Forecast	51,560	0	(51,560)		Funding Change	401,560	0	(401,560)	Funding Change
		3617D	Physical Security - NIMOE	YTD & Forecast	1,327,794	0	(1,327,794)		Funding Change	1,327,794	0	(1,327,794)	Funding Change
		3617F	PS - NIMOE	Forecast Only	0	0	0		Funding Change	1,030,000	0	(1,030,000)	Funding Change
		3617G	Physical Security - KEDNY - Equip Reolacement	YTD & Forecast	127,712	0	(127,712)		Funding Change	347,712	0	(347,712)	Funding Change
		3617H	Physical Security - KEDLI - Equip Replacement	YTD & Forecast	281,900	0	(281,900)		Funding Change	501,900	0	(501,900)	Funding Change
		3617I	Physical Security - MASS Electric - Equip Replacement	YTD & Forecast	136,267	0	(136,267)		Funding Change	506,267	0	(506,267)	Funding Change
		3617J	Physical Security - Boston Gas	YTD & Forecast	256,726	0	(256,726)		Funding Change	1,006,726	0	(1,006,726)	Funding Change
		3617K	Physical Security - Colonial Gas	YTD & Forecast	44,074	0	(44,074)		Funding Change	164,074	0	(164,074)	Funding Change
		3617L	Physical Security - RI Electric	Forecast Only	18,773	0	(18,773)			248,773	0	(248,773)	Funding Change
		3617M	Physical Security - RI Gas	YTD & Forecast	256,934	0	(256,934)		Funding Change	326,934	0	(326,934)	Funding Change
		3617	Physical Security	YTD & Forecast	0	5,436,667	5,436,667		Imprecise Estimates	0	6,976,667	6,976,667	Imprecise Estimates
	Security	3683USF	Enterprise Centralized Patch Management	YTD & Forecast	(161,401)	0	161,401		Annual Accounting	(161,401)	0	161,401	Funding Change
		3683USH	Network Security Improvements	YTD & Forecast	468,747	0	(468,747)		Funding Change	468,747	0	(468,747)	Funding Change
		3683USN	SOAR - Security Orchestration Automation and Response	YTD & Forecast	208,111	0	(208,111)		Funding Change	208,111	0	(208,111)	Funding Change
		5702USA	Malware Defense - Endpoint Protection	No Immaterial	1,777	0	(1,777)		Funding Change	1,777	0	(1,777)	Funding Change
		5820	Data Protection	No - Opex Only	259,967	0	(259,967)		Funding Change	259,967	0	(259,967)	Funding Change
		5848	Vulnerability Management	Forecast Only	0	0	0			93,205	0	(93,205)	Funding Change
		5850	Network Security	No - Opex Only	(78,825)	0	78,825		Funding Change	(78,825)	0	78,825	Funding Change
		5850USA	SSN Optimization US	No Immaterial	(8,932)	0	8,932		Funding Change	(8,932)	0	8,932	Funding Change
		5850USB	vSTIG Firewall Upgrade US	YTD & Forecast	69,689	0	(69,689)		Funding Change	69,689	0	(69,689)	Funding Change
		5850USD	Intrusion Detection System and Intrusion Prevention System US	YTD & Forecast	259,930	0	(259,930)		Funding Change	259,930	0	(259,930)	Funding Change
		5870	Identity and Access Manaement	YTD & Forecast	2,412,474	3,602,000	1,189,526		Fundina Bucket	2,412,474	3,602,000	1,189,526	Fundina Bucket
		5933	Operational Technology Intrusion Detection	YTD & Forecast	4,284,626	17,904,000	13,619,374		Complexity	8,826,574	23,250,000	14,423,426	Funding Bucket
		5940	Global VM Enterprise	YTD & Forecast	887,782	0	(887,782)		Funding Change	887,782	0	(887,782)	Funding Change
		6210US	FY22 Platform Tower	YTD & Forecast	761,526	0	(761,526)		Fundina Change	761,526	0	(761,526)	Fundina Change
		6211US	FY22 Network Tower	YTD & Forecast	471,536	0	(471,536)		Funding Change	471,536	0	(471,536)	Funding Change
		6245US	Security Orchestration Automation & Response	YTD & Forecast	1,467,139	0	(1,467,139)		Funding Change	1,619,323	0	(1,619,323)	Funding Change
		6416	Operational Technology Privileged Access Workstation	No Immaterial	(8,147)	0	8,147			(8,147)	0	8,147	Funding Change
		6417US	Vulnerability Manaement T2 Project	YTD & Forecast	615,871	3,269,767	2,653,896		Scope Chanees	896,897	4,195,967	3,299,070	Funding Bucket
		6480us	FY23 Platform Tower Project	YTD & Forecast	2,802,358	4,784,267	1,891,909		Funding Bucket	4,271,617	6,139,467	1,867,850	Funding Bucket
		6526us	FY23- Network Security Program	YTD Only	1,701,942	2,015,000	313,058		Timing	3,022,140	3,100,000	77,860	
		6540us	FY23_24 SOAR	YTD & Forecast	575,095	1,550,067	974,972		Impercise Estimates	802,124	1,981,267	1,179,143	Imprecise Estimates
		6510	OT PAW FY23	YTD & Forecast	1,404,154	0	(1,404,154)		Funding Change	2,235,778	0	(2,235,778)	Funding Change
		6613US	FY23 Identity and Access Management	YTD & Forecast	1,545,666	0	(1,545,666)		Funding Change	3,080,569	0	(3,080,569)	Funding Change
Security Total					22,532,825	38,561,767	16,028,942			36,254,201	49,245,367	12,991,166	

					Forecast to Budget Variance			Non Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it program	inv id	erol level descrip	Explain	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	9 + 3 Total	Budget Total	YTD Variance	Reason
US Electric	EBU Digital	5054	On My Way (EBU Digital Solution) Phase 2	YTD & Forecast	4,126,217	2,935,548	(1,190,669)		Imprecise Estimates	5,359,283	3,761,961	(1,597,322)	Imprecise Estimates
		5702	GridStack	YTD & Forecast	26,971	0	(26,971)	Mischarges		26,971	0	(26,971)	Mischarges
		5890C	GridInspector F&A	No - Opex Only	10,680	0	(10,680)		Other	337,406	0	(337,406)	Timing
		5912	On My Way (EBU Digital Solution)	YTD & Forecast	42,203	0	(42,203)		Funding Change	42,203	0	(42,203)	Funding Change
		6073	Veg Management Optimization	YTD & Forecast	3,393,804	6,208,364	2,814,560		Scope changes	4,854,309	8,566,395	3,712,086	Scope changes
		6073	Veg Management Optimization	See Above					Imprecise Estimates				Imprecise Estimates
		6076	Future Now - validation phase	YTD & Forecast	6,723,707	9,039,425	2,315,718		Efficiencies	8,896,581	11,186,890	2,290,309	Efficiencies
		6291	ICU-Out (Electric Digital)	YTD & Forecast	757,714	3,983,897	3,226,183		Timing	1,329,515	5,647,785	4,318,270	Timing
		6425	DER Active Resource Integration NY	YTD & Forecast	552,096	0	(552,096)		Funding Change	441,561	0	(441,561)	Funding Change
		6437	Grid MD	YTD & Forecast	4,687	1,020,103	1,015,416		Timing	4,687	1,020,103	1,015,416	Timing
		6388	FY22 - EBU Daital Placeholder	YTD & Forecast	0	6.245.000	6.245.000		Fundina Bucket	2.575.219	12.800.000	10.224.781	Fundina Bucket
		6437A	Grid MD	YTD & Forecast	824,334	0	(824,334)		Funding Change	1,249,171	0	(1,249,171)	Funding Change
		6662	Proj Costing Cont Mgmt	YTD & Forecast	75,183	0	(75,183)		Funding Change	4,295,525	0	(4,295,525)	Funding Change
		6661	Demand Planning and Material Track & Trace	No - Opex Only	0	0	0			263,939	0	(263,939)	
	Electric BU	4481	Energy Accounting System Migration to Wholesale Settlement Application - NY	YTD & Forecast	(94,129)	0	94,129		Accounting	(94,129)	0	94,129	Accounting
		4483	US Control Transmission	YTD & Forecast	49,547	0	(49,547)		Timing	49,547	0	(49,547)	Timing
		5241	East Pulaski Energy Storage System IS Network	YTD & Forecast	27,928	0	(27,928)		Funding Change	56,928	0	(56,928)	Funding Change
		5242	Troy Energy Storage System IS Network	No Immaterial	(1,241)	0	1,241			(1,241)	0	1,241	
		5773B	Visitor Logging for NERC-CIP Reliability Compliance	YTD & Forecast	527,260	439,714	(87,546)		Scope changes	531,198	439,714	(91,484)	Scope changes
		5773E	NERC-CIP Reliability Compliance	YTD & Forecast	234,055	0	(234,055)		Funding Change	234,055	0	(234,055)	Other
		5871A	NY Battery Energy Storage Systems	YTD & Forecast	0	1,825,000	1,825,000		Other	0	2,675,000	2,675,000	Other
		5871D	Clean Energy Initiatives	YTD & Forecast	0	380.000	380.000		Timing	331.623	625.000	293.377	Timing
		5871E	Power Marketer Pilot	No - Opex Only	158,503	0	(158,503)			165,939	0	(165,939)	
		5871F	Battery Storage Communications Upgrade	YTD & Forecast	0	1,030,000	1,020,000		Timing	550,000	1,500,000	950,000	Timing
		5887A	ARCOS Reporting Replacement	YTD & Forecast	377,492	780.689	403.197		Efficiencies	404.756	780.689	375.933	Efficiencies
		5888D	Fluke & WECO Calibration licenses, hardware updates and data migration	YTD & Forecast	516,636	118,000	(398,636)		Scope changes	526,866	118,000	(408,866)	Scope changes
		5889C	Electric FY22 Regulatory Mandates and Enhancements	YTD & Forecast	(13,812)	155,000	168,812		Other	(13,812)	225,000	238,812	Other
		5890A	Corporate PI	YTD & Forecast	1,243,210	1,568,907	325,697	Timing differences		1,751,274	1,588,789	(162,485)	Imprecise Estimates
		5890C	GridInsoector F&A	No - Opex Only	0	0	0			0	0	0	
		5891A	Electric MV90 Upgrade	YTD & Forecast	1,557,042	741,520	(815,522)		Timing	1,582,752	741,520	(841,232)	Complexity
		5891C	Field Collection System (FCS) Upgrade	YTD & Forecast	339,521	294,558	(44,963)		Imprecise Estimates	437,789	294,558	(143,231)	Imprecise Estimates
		5891D	IEE Integration/Upgrades	YTD & Forecast	0	1,065,000	1,065,000		Efficiencies	0	1,150,000	1,150,000	Efficiencies
		5891E	MTS Retail Settlement (MDS)	YTD & Forecast	2,115,228	2,788,654	673,426		Timing	2,714,786	3,530,991	816,205	Timing
		5891F	Meter Data Services Consolidation Phase 1	YTD & Forecast	0	920,000	920,000		Timing	10,000	1,725,000	1,715,000	Timing
		5891I	Price Transmission (Retail Pricing) Replacement	YTD & Forecast	1,303,112	1,005,377	(297,735)		Imprecise Estimates	1,304,535	1,005,377	(299,158)	Imprecise Estimates
		5892A	Control Room EOL Server and Tool Reolacement	YTD & Forecast	222,829	0	(222,829)		Imprecise Estimates	222,829	0	(222,829)	Imprecise Estimates
		5892B	Control Room Servers and Tools Upgrade/Replacement	YTD & Forecast	245,808	960,000	714,192		Timing	951,811	1,380,000	428,189	Funding Bucket
		5892C	IOS-DVD 811 Dig Safe DVD Replacement	YTD & Forecast	84,811	1,127,890	1,043,079		Efficiencies	190,613	1,127,890	937,277	Efficiencies
		5893A	Digitizing the Electric Business - Mobility (Mwork Replacement) *A	No - Opex Only	14,137	0	(14,137)			14,137	0	(14,137)	
		6043	Plan Forward	No - Opex Only	3,298	0	(3,298)			3,298	0	(3,298)	
		6320	Telecom & Network Deep Dive	No - Opex Only	75,005	0	(75,005)			75,005	0	(75,005)	
		6571	GridAPI Electric	Forecast Only	0	0	0			500,004	0	(500,004)	Funding Change
		5889D	Electric FY23 Reeulatory Mandates and Minor Enhancements	No - Opex Only	76.570	0	(76.570)			146.570	0	(146.570)	
		5871DA	SD-WAN for Distributed Communications	Forecast Only	19,879	0	(19,879)		Funding Bucket	155,002	0	(155,002)	Funding Change
		6631	US Electric Rapid Rebuild	No - Opex Only	15,365	0	(15,365)			100,273	0	(100,273)	
	Wholesale Networks	5773D	NYISO PMU Requirement for Large Generators	YTD & Forecast	996,681	736,119	(260,562)		Complexity	1,329,962	736,119	(593,843)	Complexity
		5822C	Generation Asset Management Upgrade	YTD & Forecast	786,990	1.547.623	760.633	Accruals		2.502.660	1.547.623	(955.037)	Scope changes
		6043G1	Content Management for Engineering	YTD & Forecast	663,965	1,493,248	829,283		Scope changes	919,905	1,493,248	573,343	Scope changes
US Electric Total					28,083,286	48,399,636	20,316,350			47,331,305	65,667,653	18,336,348	

CIO Group	it_program	inv_id	proj_level_descrip	Explain
US Gas	Capital Delivery	5274A	New Storms Response Management System Implementation	YTD & Forecast
		6222	PIF Replacement	YTD & Forecast
	Gas BU	4769	Gas Storm Hardening Using Remote Service Valves (RSV's)	YTD & Forecast
		4898	Asset and Work Manaement Sxstem for LNG	YTD & Forecast
		5785I	NRG GIS Replacement Project	No Immaterial
		5787D	Gas Transportation System (GTS) Product	Forecast Only
		5787G	F&A Study for Gas Demand Response (DR)	No - Opex Only
		5788D	Interface MyHub To OnBoard LMS OQ External Site	No Immaterial
		5788G	FY22 Gas Regulatory and Operational Enhancements (Minor Works)	No - Opex Only
		5789E	Summons and Corrective Actions Response (CAR) Work Tracker	YTD & Forecast
		6015	Continuous Imorovement for Gas Aoolications	YTD & Forecast
		6353A	FY22 Gas Application Rationalization	No - Opex Only
		6392	Contractor Ipad for Gas Business	YTD & Forecast
		6441	Scalability & Resilience for Enterprise Gas Platforms	YTD & Forecast
		6479	LNG IT Support for OT Systems	YTD & Forecast
		6481	Pathfinder for Leak Survey for MA	YTD & Forecast
		5785Q	Field Collection Systems (FCS) Upgrade (Gas Only)	YTD & Forecast
		6472	MOC Library & Action Item Loa Automation	No - Opex Only
		6509	Gas I&R, Gas Meter Ops, and Electric Complex Construction Field Device Replacement	YTD & Forecast
		6539	Continued Scaling and Evolution of Gas IT Transformation Program	YTD & Forecast
		6542	New Gas DR Solution Evaluation	No - Opex Only
		6550	Gas Manaaement Svstem (SCADA) Uoerade'	YTD & Forecast
		6552	System Operating Procedure Third-Party User Access for PE Review	No - Opex Only
		6553	Enable Gas Curtailment Map in ESRI	No - Opex Only
		6554	New Svstem Ooeratine Procedure (SOP) Solution	No - Opex Only
		6555	PIPES/Enterprise Maximo Integration	No - Opex Only
		6556	Digital Asset & Work Management for CNG and I&R?	YTD & Forecast
		6557	Enable Converted LI Gas Distribution Asset Data in ESRI	No - Opex Only
		6559	FY23 Gas Reaulatorv and Ooerational Enhancements (Minor Works)?	No - Opex Only
		6582	Gas Application Continuous Improvement	YTD & Forecast
		6573	FY23 Gas Regulatory and Operational Enhancements (Minor Works)	No - Opex Only
		6610	Gas App Rationalization	YTD & Forecast
		6642	Tesco	YTD & Forecast
		6671D	US Gas Data Modern-E	No - Opex Only
US Gas Total				

					Forecast to Budget Variance			Non Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Explain	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	9 + 3 Total	Budget Total	YTD Variance	Reason
DPIT	Cloud & Hosting	6200	I&O US Cloud and Hosting FY22	YTD & Forecast	26,304	0	(26,304)		Accounting	26,304	0	(26,304)	Accounting
		6200.01.01	INVP 6200.01.01 HCI / V-Block Migra	See Above	97,236	0	(97,236)			97,236	0	(97,236)	
		6200.01.02	INVP 6200.01.02 DPS Enablement (Mig	See Above	(86,615)	0	86,615			(86,615)	0	86,615	
		6200.02.01	INVP 6200.02.01 Storage Maration C	See Above	105,171	0	(105,171)			105,171	0	(105,171)	
		6200.02.02	INVP 6200.02.02 Data Management / A	See Above	(106,698)	0	106,698			(106,698)	0	106,698	
		6200.03.01	INVP 6200.03.01 Public Cloud (Azure	See Above	184,973	0	(184,973)			184,973	0	(184,973)	
		6200.04.01	INVP 6200.04.01 Azure and GBE Palo	See Above	(165,117)	0	165,117			(165,117)	0	165,117	
		6200.04.02	INVP 6200.04.02 Horizon on Azure Ce	See Above	(47,110)	0	47,110			(47,110)	0	47,110	
		6200.05.01	INVP 6200.05.01 New Global Active D	See Above	(15,669)	0	15,669			(15,669)	0	15,669	
		6489	I&O US Cloud & Hosting FY23	YTD & Forecast	3,855,943	12,355,520	8,500,177		Imprecise Estimates	7,723,701	16,622,203	8,898,502	Complexity
		6489.01	INVP 6489 Public Cloud Platforms	See Above	1,294,810	0	(1,294,810)			1,294,810	0	(1,294,810)	
		6489.02	INVP 6489 Cloud Containers	See Above	1,092,426	0	(1,092,426)			1,092,426	0	(1,092,426)	
		6489.03	INVP 6489 Global Active Directory	See Above	2,471,958	0	(2,471,958)			2,471,958	0	(2,471,958)	
		6489.04	INVP 6489 Hosted Solutions & Platf	See Above	659,296	0	(659,296)			659,296	0	(659,296)	
	CNI	6199	I&O US CNI FY22	YTD & Forecast	341,779	0	(341,779)		Accounting	341,779	0	(341,779)	Accounting
		6199.01.01	INVP 6199.01.01 EoL Assets	See Above	(174,429)	0	174,429			(174,429)	0	174,429	
		6199.01.02	INVP 6199.01.02 EOL Assets - Gas	See Above	(149,863)	0	149,863			(149,863)	0	149,863	
		6199.02.01	ITSM Implementation	See Above	(51,578)	0	51,578			(51,578)	0	51,578	
		6488	I&O US CNI FY23	YTD & Forecast	0	3,754,370	3,754,370	Timing differences	Other	917,032	4,499,596	3,582,564	Efficiencies
		6488.01	INVP 6488.01 US CNI Networks	See Above	1,249,400	0	(1,249,400)			1,249,400	0	(1,249,400)	
		6488.02	INVP 6488.02 Infrastructure Mgmt	See Above	303,134	0	(303,134)			303,134	0	(303,134)	
		6488.03	INVP 6488.03 CNI Tool Manaaement	See Above	420,201	0	(420,201)			420,201	0	(420,201)	
	Network	5774	WiFi Phase 2	No Immaterial	(1,763)	0	1,763			(1,763)	0	1,763	
		6197	I&O US Networks FY22	YTD & Forecast	(76,428)	0	76,428		Imprecise Estimates	(76,428)	0	76,428	Imprecise Estimates
		6197.01.01	INVP 6197.01.01 UPS Devices	See Above	(484,206)	0	484,206			(484,206)	0	484,206	
		6197.01.02	INVP 6197.01.02 EOL Switch	See Above	6,730	0	(6,730)			6,730	0	(6,730)	
		6197.01.03	INVP 6197.01.03 SD WAN	See Above	34,732	0	(34,732)			34,732	0	(34,732)	
		6197.01.04	INVP 6197.01.04 NW Monitoring	See Above	331,278	0	(331,278)			331,278	0	(331,278)	
		6197.01.05	INVP 6197.01.05 OOB Mamt	See Above	58,750	0	(58,750)			58,750	0	(58,750)	
		6197.01.06	INVP 6197.01.06 Storm Sites	See Above	(49,046)	0	49,046			(49,046)	0	49,046	
		6197.02.01	INVP 6197.02.01 WiFi	See Above	50,124	0	(50,124)			50,124	0	(50,124)	
		6197.02.02	INVP 6197.02.02 WiFi Tooling	See Above	(65,393)	0	65,393			(65,393)	0	65,393	
		6197.03.01	INVP 6197.03.01 Voice EOL	See Above	55,197	0	(55,197)			55,197	0	(55,197)	
		6197.03.02	INVP 6197.03.02 MS Teams Intr	See Above	(134,380)	0	134,380			(134,380)	0	134,380	
		6197.04.01	INVP 6197.04.01 vStig Upgrads	See Above	234,349	0	(234,349)			234,349	0	(234,349)	
		6197.05.01	INVP 6197.05.01 Data Ctr Transforma	See Above	2,340	0	(2,340)			2,340	0	(2,340)	
		6486	I&O US Networks FY23	YTD & Forecast	10,830,650	14,189,291	3,358,641	Timing differences	Imprecise Estimates	14,956,101	19,300,124	4,344,023	Efficiencies
		6486.01	INVP 6486 LAN/Wireless	See Above	317,816	0	(317,816)			317,816	0	(317,816)	
		6486.02	INVP 6486 WAN/SD-WAN	See Above	163,562	0	(163,562)			163,562	0	(163,562)	
		6486.03	INVP 6486 Network Support Svcs	See Above	199,687	0	(199,687)			199,687	0	(199,687)	
		6486.04	INVP 6486 Telephony	See Above	2,683,659	0	(2,683,659)			2,683,659	0	(2,683,659)	
	Operations	5876	Pure Technical Debt Remediation	YTD & Forecast	(35,775)	0	35,775		Accounting	(35,775)	0	35,775	Accounting
		6114	Manaaed Worksoace Services Transformation	YTD & Forecast	(38,108)	0	38,108		Accounting	(38,108)	0	38,108	Accounting
		6196	I&O US Operations FY22	YTD & Forecast	(275,043)	0	275,043		Accounting	(275,043)	0	275,043	Accounting
		6196.01.01	INVP 6196.01.01 BAU Device Refresh	See Above	12,392	0	(12,392)			12,392	0	(12,392)	
		6196.01.02	INVP 6196.01.02 Application Mainten	See Above	351	0	(351)			351	0	(351)	
		6196.02.02	INVP 6196.02.02 CMDB Overhaul	See Above	(54,690)	0	54,690			(54,690)	0	54,690	
		6196.03.01	INVP 6196.03.01 Appl Perf Mgmt	See Above	(609,848)	0	609,848			(609,848)	0	609,848	
		6196.03.02	INVP 6196.03.02 Auto Disaster Rec	See Above	208,986	0	(208,986)			208,986	0	(208,986)	
		6485	I&O US Operations FY23	YTD & Forecast	2,164,359	8,445,551	6,281,192	Timing differences		9,741,034	11,318,284	1,577,250	
		6485.01	INVP 6485 Service Aware CMDB	See Above	285,195	0	(285,195)			285,195	0	(285,195)	
		6485.02	INVP 6485 Observability	See Above	248,667	0	(248,667)			248,667	0	(248,667)	
		6485.03	INVP 6485 AI Operations	See Above	583,426	0	(583,426)			583,426	0	(583,426)	
		6485.04	INVP 6485.04 Automation	See Above	45,940	0	(45,940)			45,940	0	(45,940)	
		6485.05	INVP 6485.05 Unified Tech Ops Ctr	See Above	962,671	0	(962,671)			962,671	0	(962,671)	
	Other	6641	FY23 Service Now Upgrade - US	YTD & Forecast	342,316	0	(342,316)		Funding Change	1,689,737	0	(1,689,737)	
		6652	Microstrategy Sunset	YTD & Forecast	826,124	0	(826,124)		Funding Change	1,643,042	0	(1,643,042)	
	Data Center	6223	I&O US Data Center Transformation	YTD & Forecast	429,903	15,913,725	15,483,822	Timing differences	Funding Change	3,755,219	18,268,518	14,513,299	Funding Change
		6223	I&O US Data Center Transformation	See Above									Scope changes
		6223.01	INVP 6223.01 DCT Other Costs	See Above	1,399,994	0	(1,399,994)			1,399,994	0	(1,399,994)	
		6223.02	INVP 6223.02 DCT Co-Location	See Above	(515,981)	0	515,981			(515,981)	0	515,981	
		6223.03	INVP 6223.03 DCT Mngd Svcs	See Above	2,949,692	0	(2,949,692)			2,949,692	0	(2,949,692)	
		6223.04	INVP 6223.04 DCT MainFrame	See Above	3,801,061	0	(3,801,061)			3,801,061	0	(3,801,061)	
		6223.05	INVP 6223.05 DCT App Migrat	See Above	5,977,088	0	(5,977,088)			5,977,088	0	(5,977,088)	
	Digital Platforms	6565	Secure & Compliant FY23	Forecast Only	1,665,030	1,818,515	153,485	Timing differences		3,180,061	2,656,677	(523,384)	
		6519	GridStack Foundations	YTD & Forecast	437,943	12,003,916	11,565,973		Funding Change	4,480,417	15,943,364	11,462,947	Imprecise Estimates
		6519.01	INVP 6519.01 G5 Foundation GridInit	See Above	2,196,711	0	(2,196,711)			2,196,711	0	(2,196,711)	
		6519.02	INVP 6519.02 G5 Foundations GridCl	See Above	1,978,348	0	(1,978,348)			1,978,348	0	(1,978,348)	
		6519.03	INVP 6519.03 G5 Found. GridDataOps	See Above	2,306,376	0	(2,306,376)			2,306,376	0	(2,306,376)	
		6519.04	INVP 6519.04 G5 Found. GridIdentity	See Above	253,996	0	(253,996)			253,996	0	(253,996)	
		6519.05	INVP 6519.05 G5 Found. GridAccess	See Above	98,411	0	(98,411)			98,411	0	(98,411)	
		6519.06	INVP 6519.06 G5 Found. GridVault	See Above	491,202	0	(491,202)			491,202	0	(491,202)	
		6519.07	INVP 6519.07 GridAPI/GridSystemInt	See Above	2,842,706	0	(2,842,706)			2,842,706	0	(2,842,706)	
	Digital Workplace	4987	SOE (Windows) Upgrade and Device Refresh	No Immaterial	(614)	0	614			(614)	0	614	
		6198	I&O US End User Experience FY22	YTD & Forecast	(57,138)	0	57,138		Accountine	(57,138)	0	57,138	Accountine
		6198.01	INVP 6198.01.04 Mac OS / iOS Mgmt	See Above	42,028	0	(42,028)			42,028	0	(42,028)	
		6198.01.01	INVP 6198.01.01 Real Time Collabora	See Above	(151,810)	0	151,810			(151,810)	0	151,810	
		6198.01.02	INVP 6198.01.02 Device and Data Acc	See Above	44,954	0	(44,954)			44,954	0	(44,954)	
		6198.01.03	INVP 6198.01.03 Content Collaborati	See Above	74,929	0	(74,929)			74,929	0	(74,929)	
		6487	I&O US Digital Workplace Services FY23	YTD & Forecast	897,430	11,305,912	10,408,482	Timing differences		8,201,000	15,150,061	6,949,061	Imprecise Estimates
		6487.01	INVP 6487 Device & Data Access	See Above	4,707,647	0	(4,707,647)			4,707,647	0	(4,707,647)	
		6487.02	INVP 6487 Real Time Collaboration	See Above	1,150,029	0	(1,150,029)			1,150,029	0	(1,150,029)	
		6487.03	INVP 6487 Content Collaboration	See Above	936,726	0	(936,726)			936,726	0	(936,726)	
		6487.04	INVP 6487.04 Support Assurance	See Above	406,905	0	(406,905)			406,905	0	(406,905)	
		6487.05	Digital Adopt & Change	See Above	1,061,107	0	(1,061,107)			1,061,107	0	(1,061,107)	
		6602	Microstrategy Mod	No - Opex Only	325,566	0	(325,566)			325,566	0	(325,566)	
DPIT Total					65,879,812	79,786,801	13,906,989			100,718,058	103,758,827	3,040,769	

Impacts Year end Forecast

Category	Type of Variance	Explanation
Project Delivery - Scope change	Scope changes	Spend impacted by both reduced or increased scope
Project Delivery - Initial scope delivered	Imprecise Estimates	Cost and Quantity of resources were incorrectly assessed when creating the budget. (eg. Price Increases)
Project Delivery - Initial scope delivered	Complexity	Unforeseen challenges in delivering the solution drove costs up. No change in scope but amount of work to deliver increased or potentially decreased
Project Delivery - Initial scope delivered	Efficiencies	More cost effective implementation options - Found a better way to deliver the solution which reduced costs
Strategic Direction	Funding Bucket	Source of undefined funding where funds will be moved to other projects
Strategic Direction	Funding Change	Move $ to or from another project (please specify INVP moving to or moving from)
Annual Accounting	Accounting	Over or under accruals which reverse at year end leaving a balance on the INVP
Annual Accounting	Timing	Project slipped and intended scope was not delivered this fiscal year
Other	Other	Please explain

No impact to forecast

Category	Type of Variance	Explanation
Accounting Errors	Mischarges	Temporary due to unavailable accounting codes or charging errors which will be corrected
Accounting Errors	Accruals	which may have been missed / over accrued
Accounting Errors	Misclassifications	eg. Between opex and capex or expense types.
Timing Changes	Timing differences	such as budget phasing, early / later agreed payments or delivery methodology impacting phasing of costs
Timing Changes	Prioritizations	Projects will begin earlier or later than originally planned - Management Determined
Timing Changes	Delays	Materials (chip shortage), Procurement issue (negotiations), resource issues (internal and external) - Beyond Management Control
Other	Other	Please explain

Niagara Mohawk Power Corporation d/b/a National Grid

The Brooklyn Union Gas Company d/b/a National Grid NY

KeySpan Gas East Corporation d/b/a National Grid

Cases 20-E-0380, 20-G-0381, 19-G-0309, and 19-G-0310

INFORMATION TECHNOLOGY CAPITAL INVESTMENT REPORT FOR QUARTER ENDED MARCH 31, 2023

SUBMITTED TO:
NEW YORK STATE DEPARTMENT OF PUBLIC SERVICE STAFF

May 31, 2023

nationalgrid

Information Technology Capital Investment Report for Quarter Ended March 31, 2023

I. Introduction

Niagara Mohawk Power Corporation (Niagara Mohawk), The Brooklyn Union Gas Company (KEDNY), and KeySpan Gas East Corporation (KEDLI) (together "National Grid" or the "Companies") submit this Information Technology ("IT") Capital Investment Quarterly Report ("IT Quarterly Report") for the fourth quarter (Q4) of fiscal year 2023 ("FY23").[1] Pursuant to the Public Service Commission's ("Commission") January 20, 2022 Order in Cases 20-E-0380 and 20-G-0381, and August 12, 2021 Order in Cases 19-G-0309 and 19-G-0310, the Company is to provide quarterly variance reports to Department of Public Service Staff ("Staff") regarding IT capital projects and investments.[2] The IT Quarterly Report provides an explanation of the Company's overall IT investment plan approach, descriptions of program investment categories, and a comparison of budgeted to actual spending.

IT capital projects and investments that are shared investments across National Grid's operating companies are implemented and owned by National Grid USA Service Company, Inc. ("Service Company") and allocated to the benefiting US operating companies, including Niagara Mohawk, KEDNY, and KEDLI, in the form of Service Company rent expense once the investments are in-service. The Service Company rent expense includes a return on the capital investment (net of deferred taxes) plus booked depreciation expense.

Through Q4 FY23, National Grid spent $367.1 million compared to a year-to-date Service Company budget of $341.0 million, representing a $26.1 million overspend. Attachment 1 includes detail on budgets and actual spending for each of the portfolios and commentary regarding performance which has been presented to each of the Chief Information Officers responsible for the respective portfolios.

Attachment 2 includes a detailed breakdown of individual project spend by portfolio compared to the budgeted spend for each of those investments.

II. Program Categories/Spending Rationales

National Grid has organized the US Information Technology organization into six areas:

[1] The fourth quarter of FY23 is January 1, 2023 through March 31, 2023 ("Q4 FY23").

[2] *See* Cases 20-E-0380 and 20-G-0381, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service*, "Order Adopting Terms of Joint Proposal, Establishing Rate Plans and Reporting Requirements" (issued and effective January 20, 2022), Joint Proposal, Section IV.10.5; and Cases 19-G-0309 and 19-G-0310, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Energy East Corporation d/b/a National Grid for Gas Service*, "Order Approving Joint Proposal, as Modified, Imposing Additional Requirements" (issued and effective August 12, 2021), Joint Proposal, Section IV.6.5.

(1) Group Functions
(2) United States Electric
(3) United States Gas
(4) Customer
(5) Digital Platforms and Infrastructure Technology (Formerly Infrastructure and Operations)
(6) Security

IT capital investment is tracked based on these area designations. Descriptions of each portfolio category/spending rationale are provided below.

Group Functions

Group Functions executes National Grid's digital roadmaps, leveraging technology and reimagining processes in support of Finance, Enterprise Business Services, Human Resources, Legal, and Safety Health & Environmental, continuing to upgrade SOX related systems, and retiring outdated systems.

United States Electric

Provides IT Support to our US Electric Distribution and Transmission businesses. The group's primary focus is modernizing our key legacy operational systems while building capabilities that will enable the energy transition and transformation to a Distribution System Operator (DSO). Through digital delivery, the United State Electric organization provides greater efficiency and data driven insights into our workforce operations that provide:

- More automated and reliable grid controls systems
- Capital programs and projects that support the proliferation of Distributed Energy Resources (DERs)
- A growing market of prosumer electric customers

United States Gas

Provides IT Support to our US Gas Distribution and Transmission businesses by developing digital products that deliver continuous improvement of our asset lifecycle management capabilities through:

- Risk-based asset investment planning
- Performance-based maintenance
- AI-based work optimization
- Mobile field products

Customer

Provides IT Support to our US Customer business, delivering best-in-class digital products that increase engagement and reduce costs to serve by:

- Modernizing our legacy "back-end" IT estate to support more digital products
- Liberating data and creating an abstracted and real-time customer data platform to assist with achieving energy efficiency targets

Digital Platforms and Infrastructure Technology (DPIT) (Formerly Infrastructure and Operations)
Supports our internal IT Architecture covering Cloud and Hosting, Critical Network Infrastructure, Networks, End User Experience and Operations. DPIT is focused on reducing National Grid's technical debt through cost effective, modernized IT infrastructure, enabling cloud capabilities that meet our risk profile. The organization enables a connected, seamless enterprise that is flexible, secure, reliable and scalable while maintaining acceptable levels of technical debt and risk.

Security
The Security function protects our people, premises, and digital systems with the objective of maintaining trust in National Grid services, focusing on:

- Reduction in likelihood and impact of security incidents
- Compliance with relevant laws, standards and regulations (*e.g.*, NERC, PCI DSS, GDPR, NIS-R)
- Improved user experience and standard of security practices across National Grid

The Security function aims to reduce the likelihood and impact of physical and cyber security incidents affecting National Grid. It does this by championing security best practices, adopting technology advancements, and promoting a security-awareness culture.

Other Category
For accounting purposes, there is an additional category listed in Attachment 2 to this report. The "Other" category has 3 main elements:

- The "IT4IT" sub-category acts as a funding bucket for NPV-positive projects to enable efficiency savings in our IT-run business operations.
- The "Other" sub-category is the allocation of operating expense overheads such as Enterprise Architecture and Solution Delivery charges to a centralized account. The capital charges for this overhead are made directly to individual projects.
- "Burdens" are the net removal of operating expense burdens to ensure our reporting agrees with our Corporate Controllable Cost reporting. Operating expense burdens are consolidated and reported separately under non-controllable costs in our corporate reporting packs.

Following the sale of the Rhode Island Gas and Electric business to PPL, associated separation costs are and will flow through this category in FY23 and FY24. National Grid is currently providing IT services (applications, infrastructure, networks, etc..) to PPL as part of a 2-year Transition Services Agreement (TSA) following the sale. Significant effort is required to manage and track an orderly exit of all IT TSA services. National Grid must provide subject matter expertise, migrate critical application data, design and implement interim state processes, and remove any customizations introduced to support delivery of the TSA. The costs for this are recovered from PPL and do not impact New York rate payers. The Group Functions portfolio is responsible for managing the expenditure but the costs role up through the "Other Category".

The summaries below explain the subjects presented in the attachments to the Q4 FY23 report.

A. Attachment 1

1. Summary (pages 2-3) – Overview of US IT .
    a. High-level executive summary, including work and challenges faced across the portfolios and a brief explanation around the forecast (p. 2).
    b. Actual expenditure versus budget by portfolio year to date, and full fiscal year forecast versus annual budget by portfolio (p. 2).
    c. Actual and forecasted spend by month for FY23 compared to budget, prior year and run rate. Each bar is color-coded to give an overview of the comparative spend per portfolio (p. 3).

2. Portfolio Sections – The remainder of Attachment 1 includes six sections, one for each of the IT portfolios: Group Functions; US Gas; US Electric; Customer; DPIT; and Security. Each portfolio consists of 3 pages:
    a. The first page provides a narrative financial and business overview. The three charts on the first page show: (1) operating expenses; (2) capital expenses; and, (3) total expenses. Each chart shows FY23 actuals versus budget, with trend lines showing prior year expenditure and run rate. This page also shows the full year total expenditure and capital expenditure compared to the budget.
    b. The second page of the portfolio sections has a chart on the top left of the page: The chart compares full fiscal year expenditure versus annual budget by program. There is also a key challenges section and the escalation point for that challenge. On the bottom half of this page is a chart showing actual spend for FY23 compared to budget, prior year, and run rate for the portfolio. Each bar is color-coded to give an overview of the comparative spend per program.
    c. The third page of each portfolio section provides a status update on the various projects. A priority investments list for the portfolio and their status appears at the top right hand of the page. Blue means complete, green means on track, amber means some risk, and red means focused intervention required.

Attachment 1 is based on National Grid's quarterly Strategic Portfolio Reviews (SPR) and reporting cadence to senior management of each portfolio, and ultimately to the IT leadership team.

B. Attachment 2

INVP Level Variances - This attachment provides a detailed listing of individual projects in each of the six portfolios explaining spend in FY23, including:

(1) Total actual spend compared to budget and variance;
(2) A variance reason for all capital investments which are outside the 10% tolerance to budget.

C. Attachment 3

FY 23 Closure Summary – This attachment provides a detailed listing of individual projects in each of the six IT portfolios that were closed during the fiscal year. The attachment shows:

(1) The list of closure papers written and approved in FY23
(2) Total costs incurred in the closure papers
(3) Explanation of the benefits derived from the closure papers
(4) Summary of important/relevant lessons learned

In total, 105 projects were closed during FY23 with a total cost of $268.582 million.

National Grid

Quarterly Report – Q4 FY23
Information Technology

31st May 2023

nationalgrid

US IT

Financial Performance

Portfolio View - Full Year $M

Portfolio	Capex			Opex			Totex		
	FY Actual	FY Budget	Variance	FY Actual	FY Budget	Variance	FY Actual	FY Budget	Variance
US Electric	35.7	57.7	22.0	6.4	7.9	1.6	42.1	65.7	23.6
US Gas	44.4	28.0	(16.4)	4.3	4.5	0.2	48.8	32.5	(16.3)
Customer	38.5	33.5	(5.0)	5.6	4.0	(1.6)	44.2	37.5	(6.7)
Group Functions	48.2	41.9	(6.3)	8.6	8.3	(0.3)	56.8	50.2	(6.7)
DPIT	105.7	90.4	(15.3)	8.1	13.3	5.3	113.8	103.8	(10.0)
Security	34.1	42.1	8.1	3.2	7.1	3.9	37.2	49.2	12.0
Other	1.2	5.5	4.3	23.0	(3.4)	(26.4)	24.2	2.1	(22.1)
Total	**307.9**	**299.3**	**(8.6)**	**59.2**	**41.7**	**(17.4)**	**367.1**	**341.0**	**(26.1)**

Financial Performance

With the exception of the variance in the "Other" category, portfolio performance is very close to budget. Throughout the year there has been reprioritization of expenditure, including in the Gas portfolio to deliver Continued Scaling and Evolution, to shift focus to projects as needed. Group Functions required additional resources to integrate applications in Procure to Pay, Customer continues to develop the unified web portal, and DPIT capitalized software following a review of accounting policies, all of which drove budget variances. These were offset by the Electric and Security portfolios, each of which reprioritized work based on demand and resources.

The 'Other' category includes the transition services agreement to PPL following the RI sale. The agreement ensures an orderly exit of all IT TSA services which National Grid is providing to cover subject matter expertise, migrate critical application data, design, implement interim state processes and remove customizations. Although this is being managed through the Group Functions portfolio, there is no impact to NY customers and they will not be charged for ~$20m of RI spend that is the main driver of the overall budget variance.

IT Commentary

Delivery

The IT team delivered a number of key investments over the past quarter to enhance both user and customer experiences while maintaining a focus on safety and security. Key deliverables include:

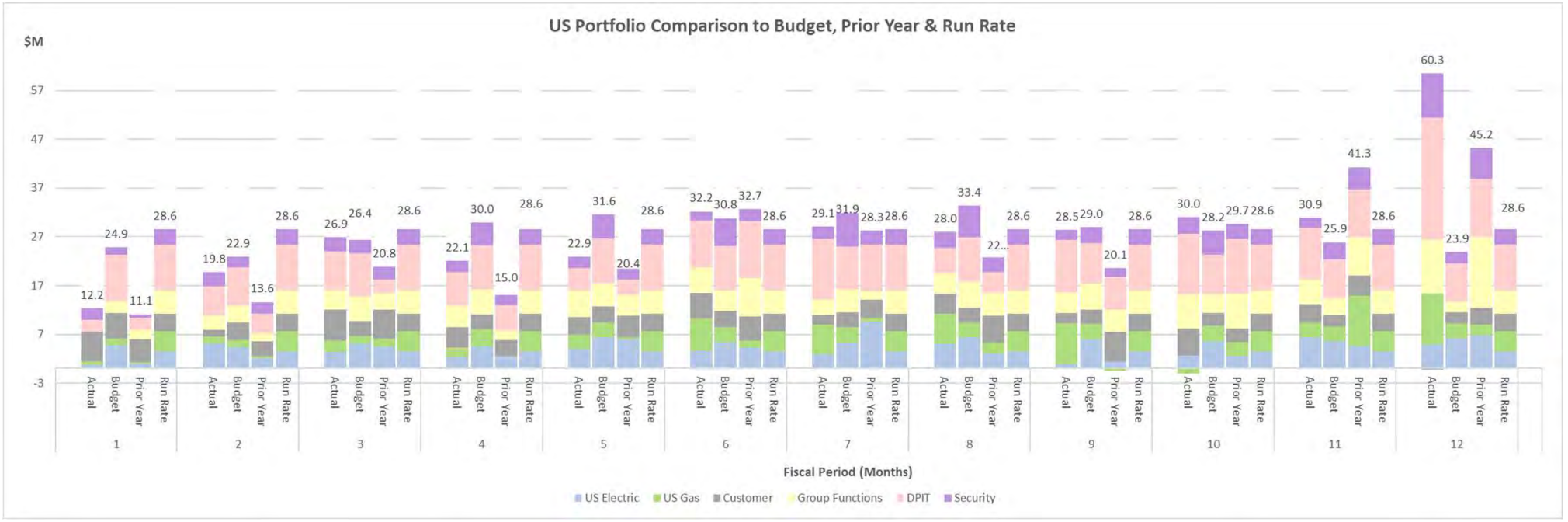
- Gas Document Management System
- Call Center Technology
- Unified Web Portal
- My Business Account
- Future Now – Work Planning for the Transmission Business
- MV 90 Meter Reading Upgrade
- Unified Technology Operations Center

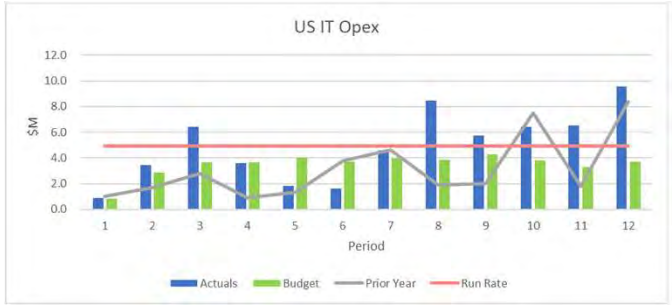
Challenges

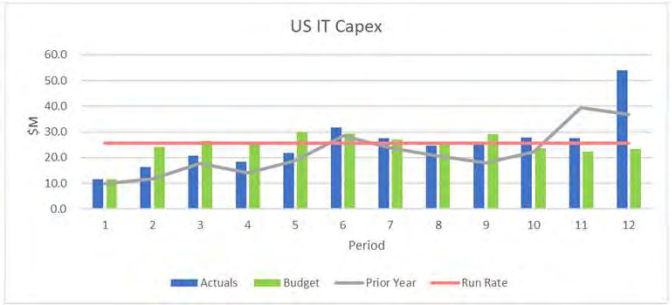
Commercial and contracting issues and chip shortages still exist in the post-COVID era which slowed spend and created challenges.

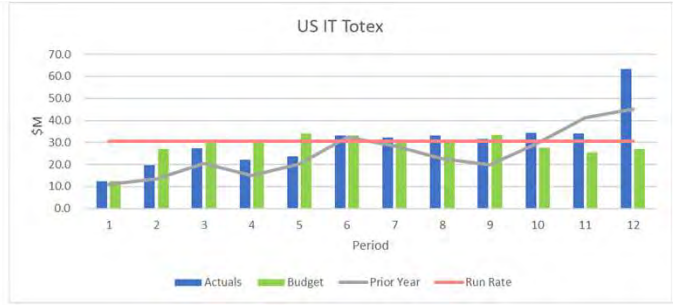
Specialist IT resources and business resources proved difficult to secure. However, capabilities are being delivered across the portfolios and management actively re-prioritized resources to deliver for customers.

Portfolio Comparison & Cost Charts









US Gas
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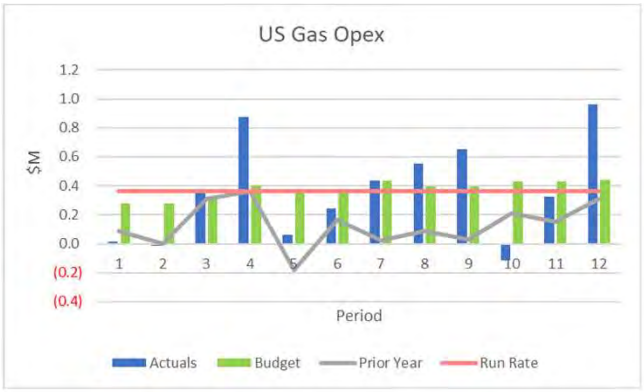
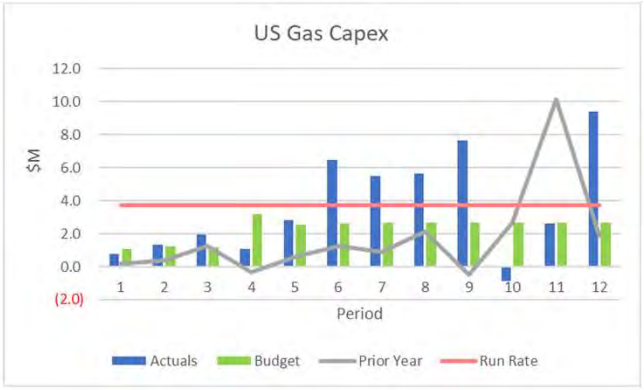
US Gas

Financial Performance



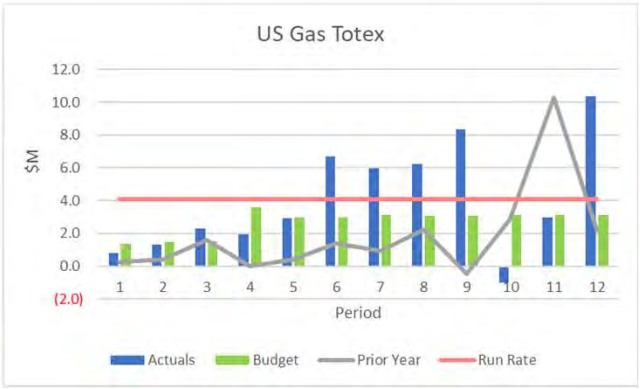


Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
32.5	48.8	55.3

Capex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
28.0	44.4	50.8



Portfolio Commentary

Variance Drivers

- U.S. Gas variance is largely due to the CS&E program ($12m), attributable to vendor and NG labor cost overrun. Re-sanction for the program is complete and Docuware document management has been deployed to Long Island (~1000 users) and NYC (~50 users), decreasing system costs while improving user experience.
- Tesco Meter Management Project was reprioritized in FY23 ($2.5m) by the Business as the current Meter shop solution deployed in UNY and LI didn't meet business needs. System final testing is underway.

Major Releases

- Asset Management for LNG successfully completed final go-live which better tracks LNG facility assets through their lifecycle;
- Gas Transportation Information System (GTIS) consolidation program goal is to migrate UNY and MA regions to consolidated common GTIS platform and decommission of the several legacy applications. Design work for cloud migration was completed. Phase-1 development that included foundational features was completed. Scoping for MA and UNY started.
- Gas Application Continuous Improvement - Safety Tagging System upgraded from on-premises SQL server to cloud based solution.
- Upgraded legacy Gas Work Methods Website to SharePoint, enabling IT Support and functionality.

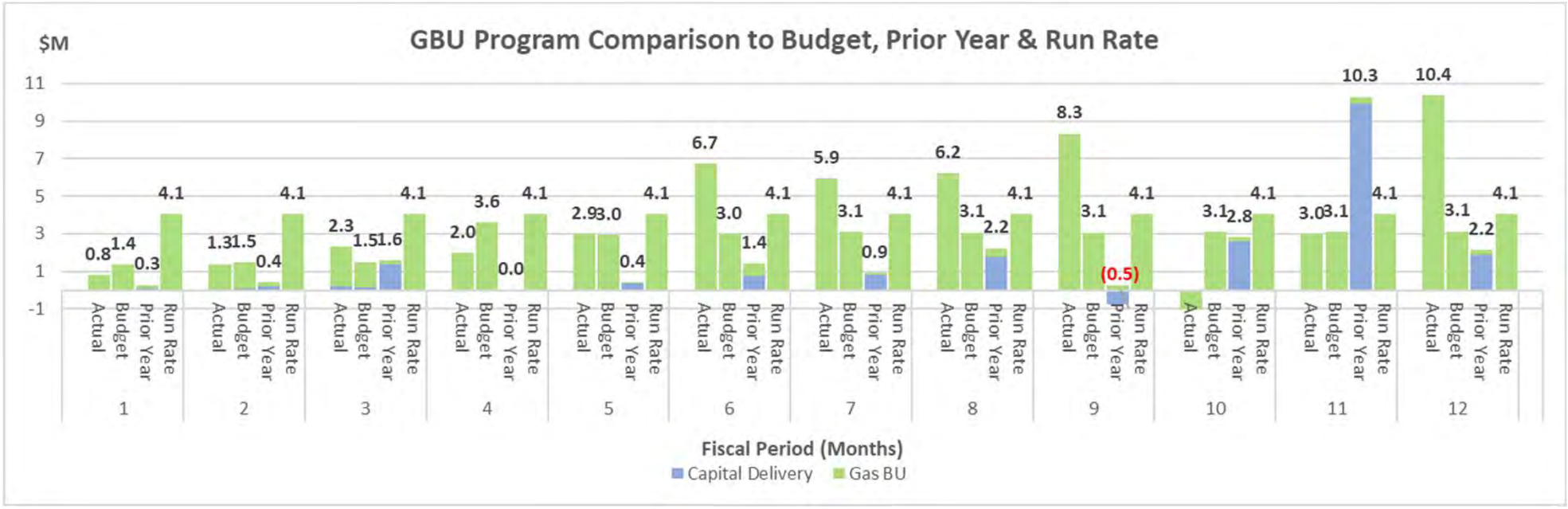
US Gas Program View of Financials

Gas Program View Full Year $M

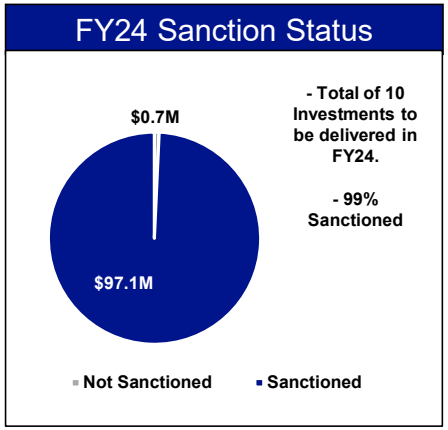
Portfolio	Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
US Gas	Capital Delivery	0.1	0.3	0.2	(0.0)	0.0	0.0	0.1	0.3	0.2
	Gas BU	44.4	27.8	(16.6)	4.4	4.5	0.1	48.7	32.2	(16.5)
		44.4	**28.0**	**(16.4)**	**4.3**	**4.5**	**0.2**	**48.8**	**32.5**	**(16.3)**

Key Challenges

Challenge	Escalation Point
GMS Upgrade (Scada)- Cisco network equipment delivery at risk for March delivery	DPIT



Mobilisation Summary



Status

- Based on 1+2 forecast, 9 projects are sanctioned and 1 left to sanction for FY24

Resources

- 1 new Business Analyst has been hired, onboarded for the FY24 projects

Scope

- Remaining projects need to complete scope assessment as part of business case analysis and sanctioning process.

Priority Investments

Program	Status	FY 23 Budget ($m)	RAG This Month	RAG Last Month
Gas Storm Hardening	Inflight	0.4		
LNG Asset & Work Mgmt.	Inflight	1.6		
Summons for Corrective Actions	PIS	0.1		
Enterprise Scalability and Resilience	Inflight	1.2		
Continuous Improvement for Gas Applications	Inflight	1.2		
PIF Replacement	PIS	0.0		
LNG IT support for OT	Start-up	0.9		
Gas I&R, Meter Ops, and Field Device Replacement	Inflight	0.1		
Enterprise Platform for Gas Retail Choice (GTIS)	Inflight	2.0		
Field Collection Systems (FCS) Upgrade	Prestart up	0.8		
Gas Management System (SCADA) Upgrade	Start-up	2.1		
New Gas DR Solution Evaluation	Inflight	0.9		
New Storms Response Management System Implementation	Inflight	0.3		
MOC Library & Action Item Automation	Inflight	0.2		
Continued Scaling & Evolution of Gas IT Transformation Program	Prestart up	20.1		
SOP Third-Party User Access for PE Review	Cancelled	0.1		
Enable Gas Curtailment Map in ESRI	Cancelled	0.04		
New System Operating Procedure Solution	Deferred/ FY24	0.1		
PIPES/Enterprise Maximo Integration	June decision	0.1		
Digital Asset & Work Management for CNG and I&R	Deferred/ FY25	0.3		
Enable Converted LI Gas Distribution Asset Data in ESRI	Deferred/ FY24	0.1		

US Electric
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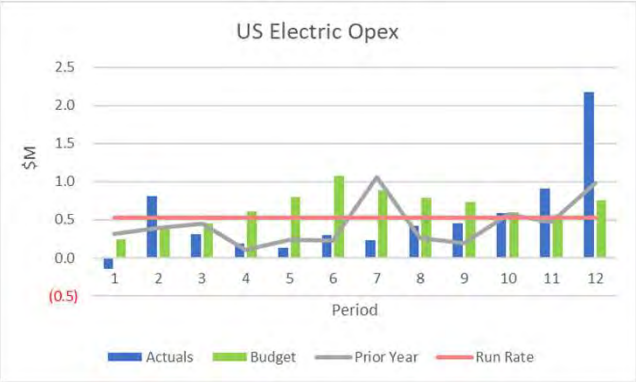
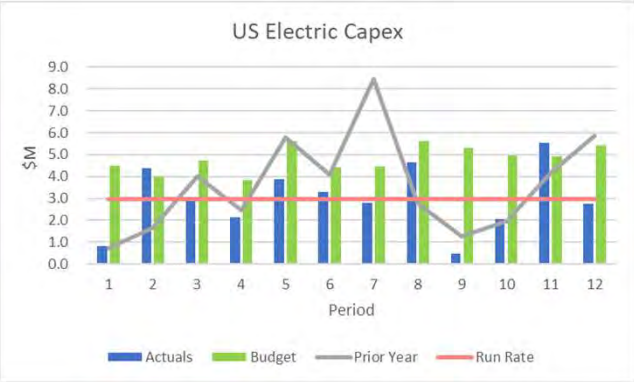
US Electric

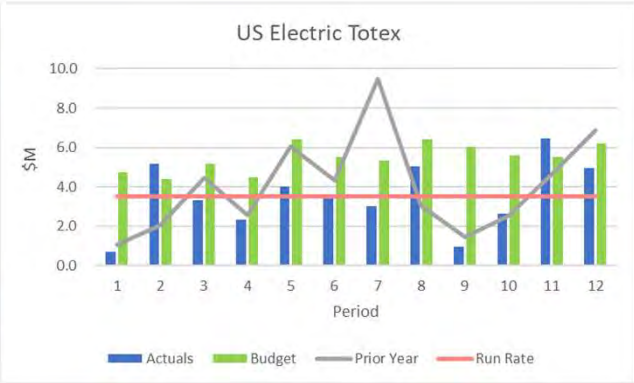
Financial Performance





Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
65.7	42.1	47.3



Capex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
57.7	35.7	41.0

Portfolio Commentary

Variance Drivers

- Full year under-spend of $22M CAPEX and $1.6M OPEX is being driven by supply chain delays, resource constraints, and various deliverability constraints that delayed the start of planned products. There was also a $4.8m underspend due to scope reduction for Vegetation Management Optimization as it was determined this scope would not provide sufficient customer value.

Major Releases

- **Field Collections System (FCS) Upgrade:** Remediated Tech Debt by upgrading our FCS system to its latest supportable version on October 28th. This system acts as our centralized billing hub for our Residential Customers by collecting, organizing, processing and distributing various data points to our critical billing systems (CSS, CRS)
- **MV90 Upgrade:** Successfully completed on November 11th, making our US Electric Meter Reading System more reliable and secure for our Commercial and Industrial Customers; this system supports 40% of US Electric Revenue
- **FutureNow:** Transformation Product released to the Transmission Business on November 7th, marking the completion of a key milestone in ensuring that the ~$22.6M in Forecasted US Type 1 Benefits are successfully delivered through FY24.
- **Corporate PI (Process Intelligence):** Went live on 3/28, allowing quicker and more reliable access to SCADA Data for end-users outside CNI to facilitate planning and analytics of load demand and supply

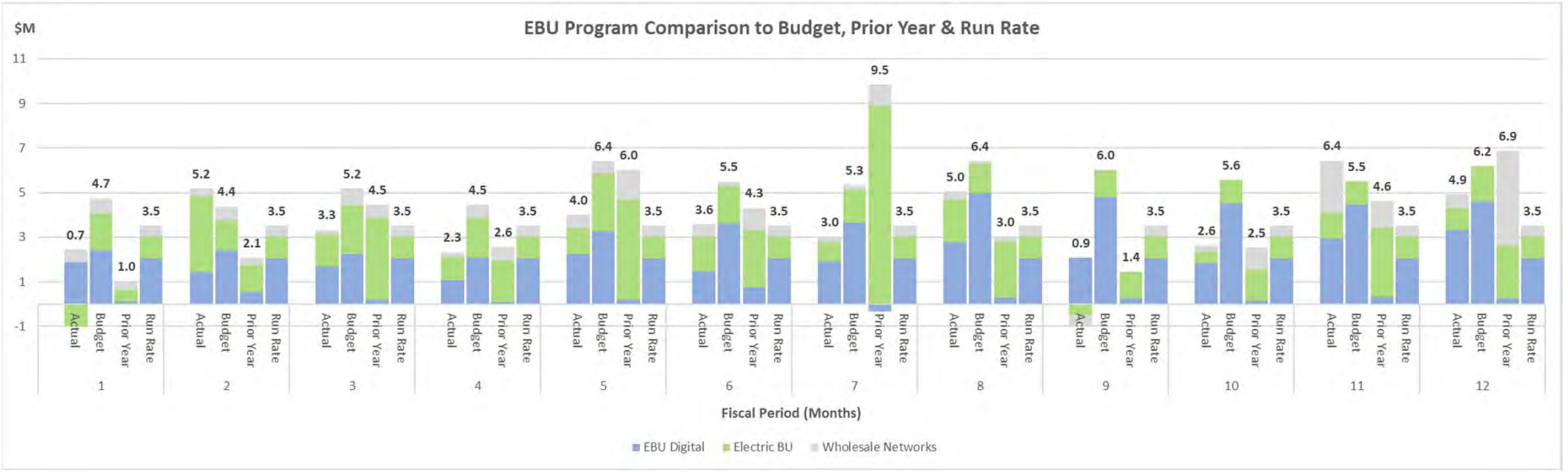
US Electric Program View of Financials

Electric Program View Full Year $M

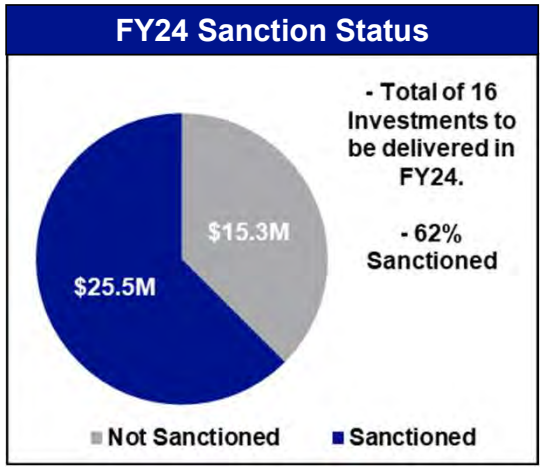
Portfolio	Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
US Electric	EBU Digital	20.2	38.5	18.3	4.4	4.5	0.1	24.6	43.0	18.4
	Electric BU	9.9	15.6	5.7	1.8	3.3	1.4	11.8	18.9	7.1
	Wholesale Networks	5.6	3.6	(2.0)	0.1	0.1	0.1	5.7	3.8	(1.9)
	Totals	**35.7**	**57.7**	**22.0**	**6.4**	**7.9**	**1.6**	**42.1**	**65.7**	**23.6**

Key Challenges

Challenge	Escalation Point
1. Resource availability and resource quality challenges with ADAM Partners in the Core Portfolio; significant challenge within the Digital Lighthouse Portfolio.	CIO / HR / Commercial *(ADAM Partners)*
2. Global Supply Chain delays with critical hardware needed to deploy key programs/products*(i.e. ADMS, NYISO-PMU - resolved)*	Commercial, ITLT
3. Generation Asset Management Upgrade is experiencing schedule delays due to reprioritization and availability of integrated environments that are necessary for development and testing; *multiple business units are engaging the same pool of resources/environments*	CIO



Mobilisation Summary



Status

- Based on 0+12 forecast, ~62% of planned product spend has been Sanctioned.

Scope:

- Initial assessment for 3 new pipeline initiatives currently underway.

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
GridMod	Inflight	26.8		
ADMS	Inflight	24.3		
DSO	Inflight	15.8		
FutureNow	Inflight	5.5		
VMO	Inflight	2.0		
OMW	Inflight	5.1		
GridInspector (Computapole Replacement)	Inflight	4.2		
Project Costing & Contract Management	Inflight	2.5		
SD-WAN Distributed Communications	Inflight	0.3		
NYISO PMU – Large Generators	Inflight	0.1		
Content Management	Inflight	0.3		
Generation Asset Management Upgrade	Inflight	0.2		
Retail Settlement	Inflight	3.5		
US Rapid Rebuild	Inflight	2.5		

Customer
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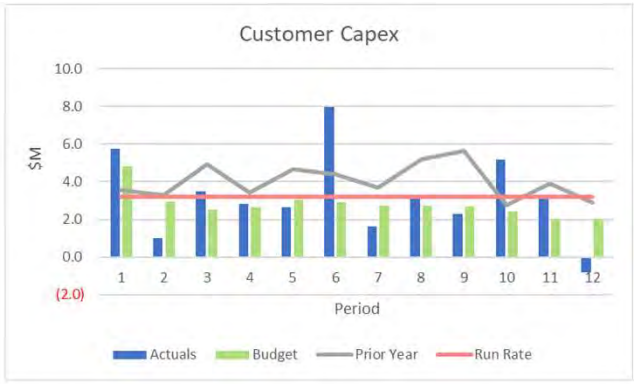
Customer

Financial Performance



Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
37.5	44.2	48.8

Capex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
33.5	38.5	43.3

Portfolio Commentary

Variance Drivers

- Reallocation of funding across IT to meet customer demand with Unified Web Portal 2.0 – a significant driver of the increase over budget. The additional spend will provide an improved user experience.
- Technical Debt – Targeting technical debt remediations that support the Customer System and the Customer offices contributed to the reallocation of costs over initial budget.

- **Major Releases:**
  - **Advanced Metering Infrastructure (AMI).** Released several features to support rapid installation of additional AMI meters and handling billing exceptions
  - **My Business Account (MBA).** Achieved 36% FY23 Product Adoption, delivering $2M in FYTD cost savings, 66% above target. Releases include: (1) CRIS Paperless Billing capability, enabling customers to easily bulk enroll/de-enroll in Paperless Billing; (2) Ability to view Account Contact Information to more easily identify accounts; (3) MuleSoft Platform email capability to proactively alert any API issues before widespread customer impact.
  - **Call Center Technologies.** completed Startek (3rd party vendor managing customer calls) onboarding, which will manage a portion of call center activity (50% anticipated in future state); benefits include increased customer satisfaction and cost savings due to competitive bid process. Completed consolidation of data centers, which reduced RTB costs by $72k annually

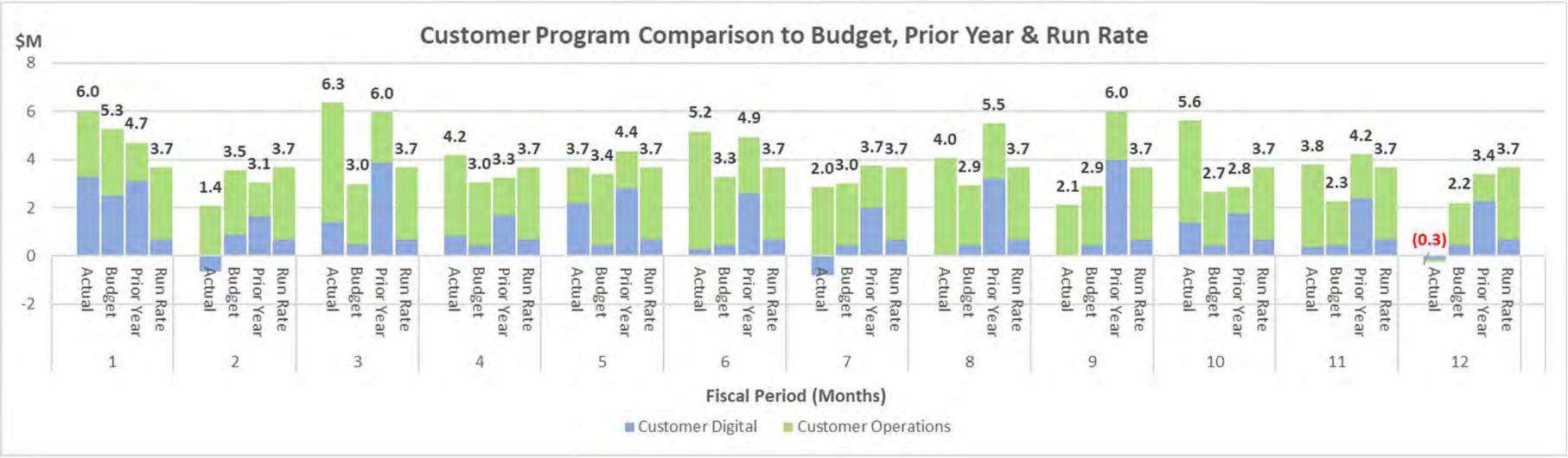
Customer Program View of Financials

Customer Program View Full Year $M

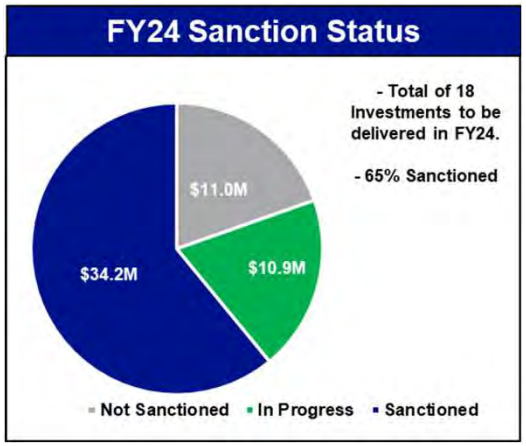
Portfolio / Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
Customer Customer Digital	6.8	6.8	0.1	1.4	1.2	(0.3)	8.2	8.0	(0.2)
Customer Operations	31.7	26.7	(5.1)	4.2	2.8	(1.4)	35.9	29.5	(6.4)
Totals	**38.5**	**33.5**	**(5.0)**	**5.6**	**4.0**	**(1.6)**	**44.2**	**37.5**	**(6.7)**

Key Challenges

Challenge	Escalation Point
***CRIS to CSS* Timeline**: *official go-live date has shifted to May 2024 due to program delays.*	Simon Baker
Advanced Metering Infrastructure (AMI): *Continue dialog with DPS staff to address concerns with Gas meter module, which has not yet been approved for deployment.*	Simon Baker



Mobilisation Summary



Status

- A significant amount of the FY24 plan has been sanctioned

Resources

- Resources are ramping up in support of the FY24 plan and currently there are no constraints identified

Looking Forward

- The FY24 investment plan continues to focus on improving and enabling the customer experience, remediating technical debt, increasing our position on use and quality of secured Customer data and implementing/resolving mandate and compliance related requirements

Priority Investments

Program	Status	Budget ($m)	RAG	RAG Last Month
Zero Based Redesign (ZBR)	Inflight	21.0		
CRIS to CSS (C2C)	Inflight	36.2		
AMI	Inflight	22.3		
Clean Energy	Inflight	9.3		
Fix The Technology Foundations: UWP2.0	Inflight	10.0		
Customer Data Platform	Inflight	7.0		

Notes:

ZBR has delivered business benefits by seeing increases in Contact Center Interactive Voice Response (IVRs) containment, yet there continue to be delays on the Digital Assistant capability due to technical dependencies.

As noted in current challenges, C2C workstream has been re-baselined for a May 2024 delivery.

AMI continues to make significant progress ahead of deployment

US Security
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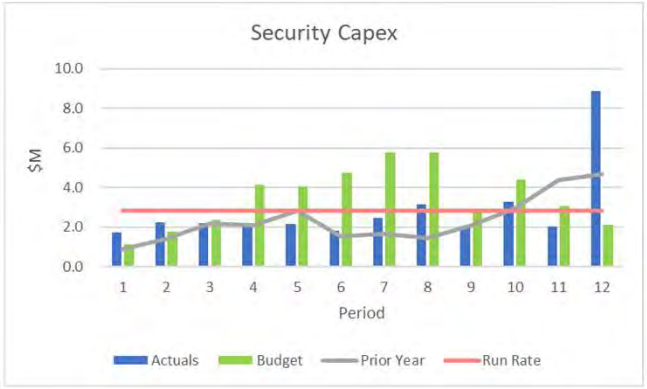
Security

Financial Performance

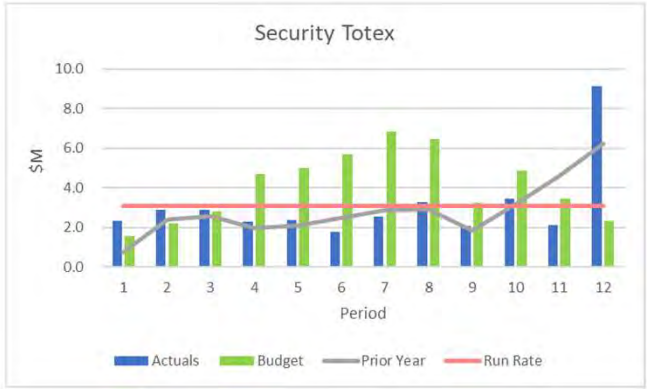


Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
49.2	37.2	36.3

Capex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
42.1	34.1	33.2



Portfolio Commentary

Variance Drivers

- The main driver of underspend to budget is a reduction in OT Security Point of Presence and Intrusion Detection System site deployments due to resource constraints related to Cloud Security and Vulnerability Management.
- These reductions have freed up budgets and have been repurposed for initiatives outside of Security.

Major Releases

[redacted]

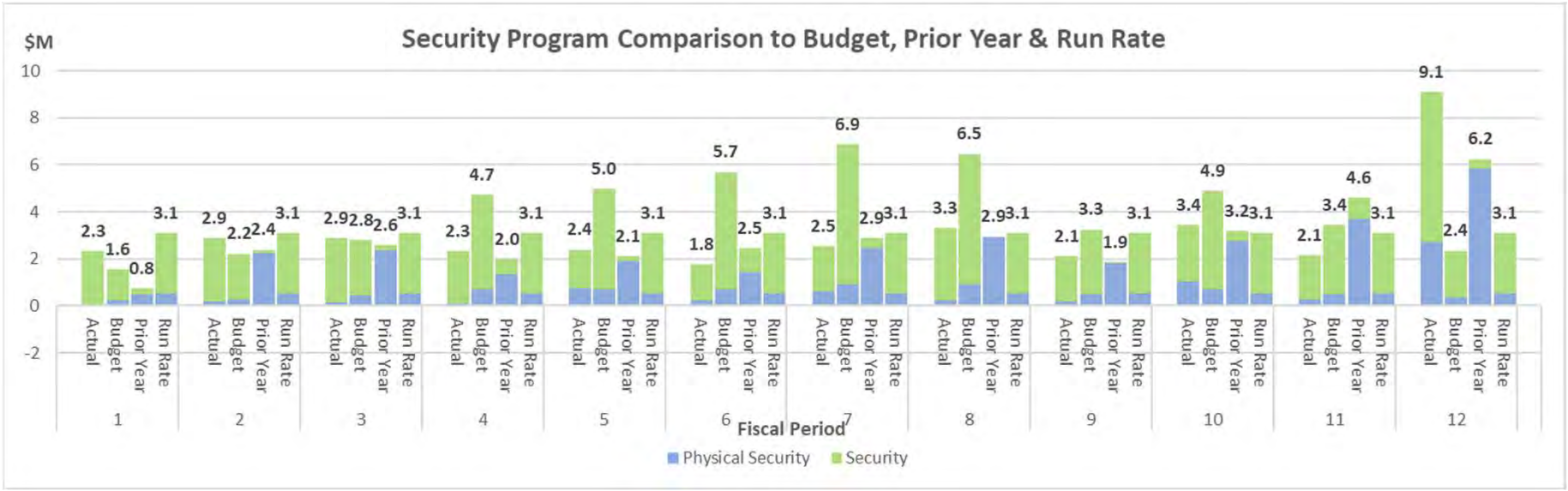
Security Program View of Financials

Security Program View Full Year $M

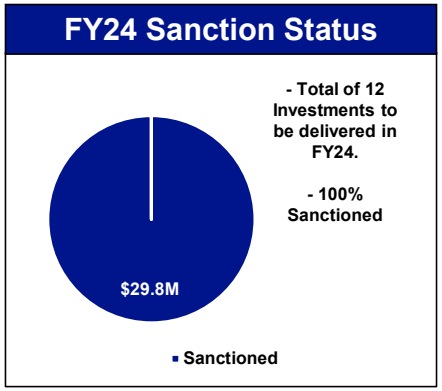
Portfolio	Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
Security	Physical Security	6.5	7.0	0.4	0.0	0.0	0.0	6.5	7.0	0.4
	Security	27.5	35.2	7.6	3.2	7.1	3.9	30.7	42.3	11.6
	Totals	**34.1**	**42.1**	**8.1**	**3.2**	**7.1**	**3.9**	**37.2**	**49.2**	**12.0**

Key Challenges

Challenge	Escalation Point
OT Program - Earlier Supply Chain constraints on PAW and IDS components effectively mitigated through early procurement	Internal, Business and Supplier
Resource constraints are being addressed and future delivery plans will be adjusted with impacts to FY24.	Internal



Mobilisation Summary



Status

- FY24 Sanctions completed for each Security Tower ensuring continuity of capital expenditure.

Resources

- Resource constraints are being managed with many key roles filled and new starters on-boarded. Process continues for vacancies.

Scope

- Projects have been sanctioned. Q2RF forecast reflective of FY24 Budget, no reductions.
- Investment prioritization completed and budget set at $26.4m Capex for FY24.

Priority Investments

Program	Status	Capex Budget ($m)	RAG This Month	RAG Last Month
[redacted]				

US Group Functions
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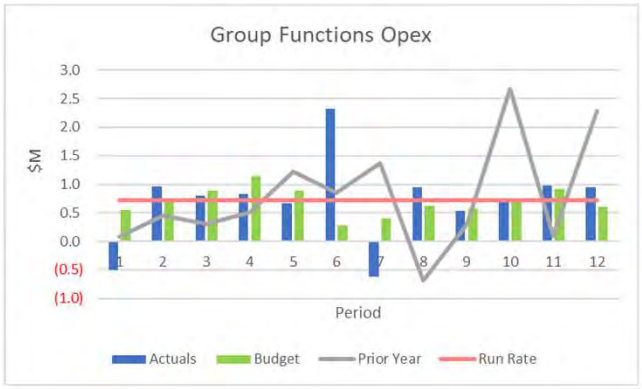
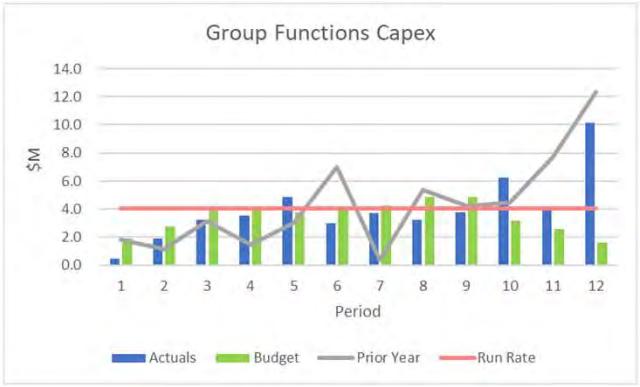
Group Functions

Financial Performance



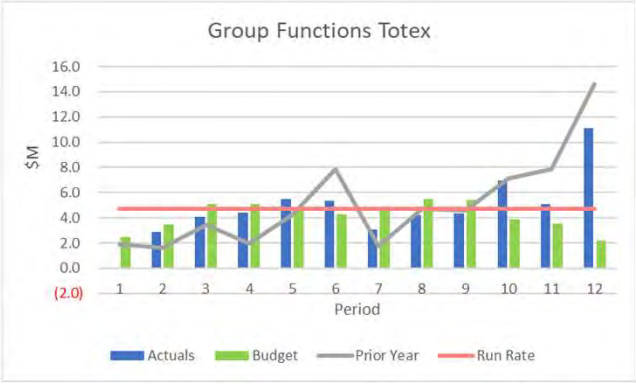


Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
50.2	56.8	49.0

Capex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
41.9	48.2	40.6



Portfolio Commentary

Variance Drivers:

Increase in Q4 spend was mostly driven by the Source to Pay Ariba program. Program delays associated with resource constraints and additional scope adds for Palantir and Pathlock required an extended timeline and drove the variance.

Major Releases:

Maximo Real Estate/Right of Way – Release 3

Final release delivers enhanced report and record management capabilities for easier and faster access to system data and tools to better manage user requests.

Workforce Data Domain

Delivers single source of truth (golden record) for employee data drives accuracy and compliance. Minimizes cost and friction by reducing data defects, fewer manual interventions for data quality checks and maintenance of integrations.

SAP PI to PO Upgrade – Release 2

Migrated 36 interfaces to the new PO platform to ensure we are meeting the latest encryption security standard.

RPA Manual Journal Entries

Reduces risk and streamlines process in reduction of manual journal entry tracking and consolidation of journal data sources while improving the employee experience.

Digitizing Learning Global Support Model

Drives streamlining & operational excellence towards Global Learning Management Services.

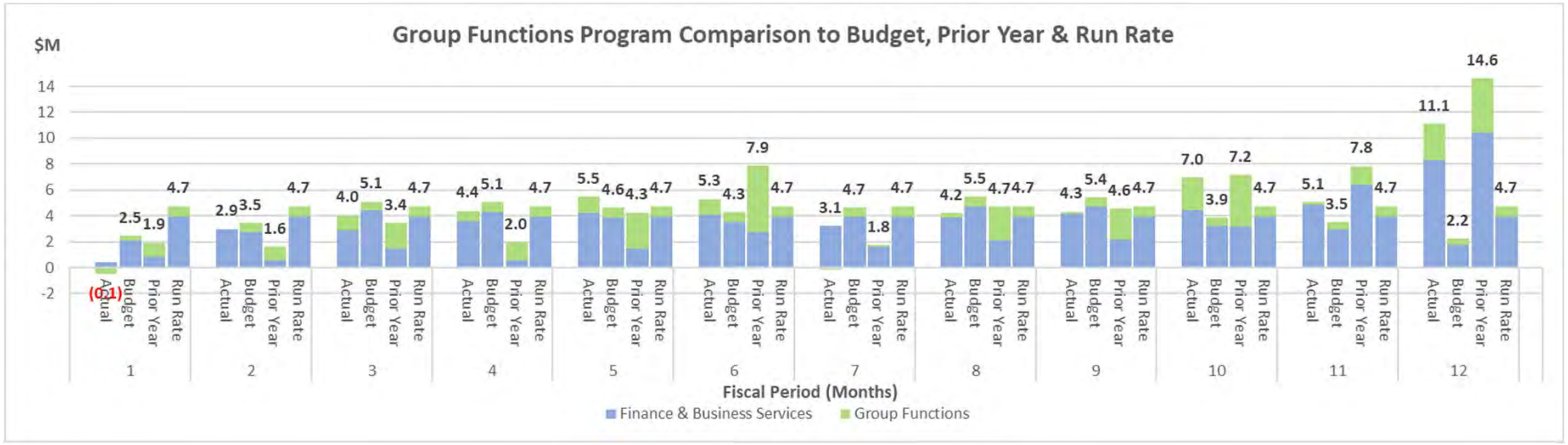
Group Functions Program View of Financials

Group Functions Program View Full Year $M

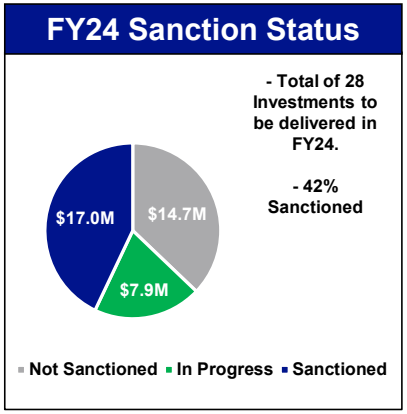
Portfolio	Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
Group Functions	Finance & Business Services	40.7	35.6	(5.2)	6.4	6.8	0.4	47.1	42.3	(4.8)
	Group Functions	7.5	6.4	(1.1)	2.3	1.5	(0.7)	9.7	7.9	(1.9)
	Totals	**48.2**	**41.9**	**(6.3)**	**8.6**	**8.3**	**(0.3)**	**56.8**	**50.2**	**(6.7)**

Key Challenges

Challenge	Escalation Point
• Resource constraints due to competing priorities and programs for both IT and Business resources	• Working with ADAM partners to backfill key roles • Recruitment ongoing to supplement with contractors
• Shortage of skilled/trained resources with Agile experience to help with transition to agile ways of working	• Upskill key resources in both IT and the business • Recruit and hire to fill open requisitions
• Risk – Capex moving to Opex due to new Finance policy on SaaS could impact ability to deliver financial commitments	• Working with finance to understand capitalization rules and ensure SaaS arrangements are GAAP compliant



Mobilization Summary - FY24



Key Outcomes to be Achieved in FY24

- **Asset Health** - Migrate 101 interfaces from PI to PO to mitigate security risks and ensure latest encryption standards are met.
- **Join to Leave** - People Digital Transformation - Focus on Core, Data and Processes, revert MyHub to standard, enabling People Function Target Operating Model. This is intended to drive Operational Efficiency, User Experience focussed on Self Service, leading to improved Data Quality and Compliance.
- **Record to Report** - Increased standardization and automation in reporting. Automate data feeds and leverage Wdesk to reduce use of offline files. Continue to reduce manual effort associated with journal entries via RPA and Blackline integration.
- **Source to Pay** - Implement Ariba Guided Buying across US (NY and NE) enabling intelligent cataloge search resulting in increased electronic purchasing, reducing free text po's, and increasing electronic invoicing. Implement next phase of Ariba and upon completion enable retiring SAP SRM system.
- **Plan to Perform** - Implement three separate product releases to lessen the dependency on offline financial models by automating calculations. Will enable collaborative working in connected data environments which will reduce the planning cycle from 15 to 3 weeks, thereby releasing resource effort to focus on other priorities.

Priority Investments

FY	Program	Status	Totex Budget ($M)	RAG This Qtr.	RAG Last Qtr.
'23	Blackline: Task Management	Complete	1.0		
	Core Power Plan Upgrade and Lease Consolidation	Complete	2.3		
	SAP Portals to Fiori	Complete	0.7		
	Warehouse Investment Recovery Digitization	Complete	1.0		
'24	Ariba	Inflight	6.6		
	Joiners, Movers, Leavers	Inflight	3.4		
	Robotic Process Automation of Manual Journal Entries	Inflight	1.1		
	HR Service Pack	Inflight	1.4		
	Business Planning Modeling	Inflight	1.8		
	SAP Cloud Analytics	Pipeline	0.7		

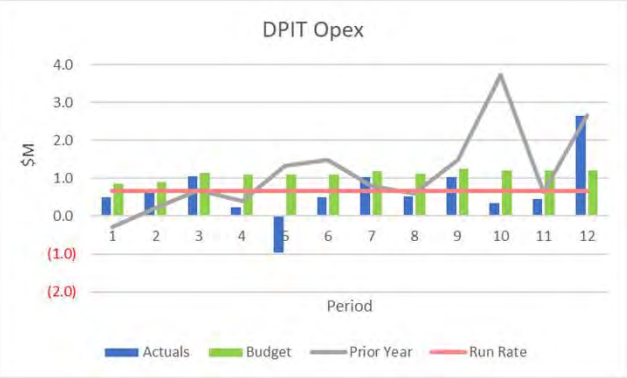
Digital Platforms & Infrastructure Technology (DPIT)

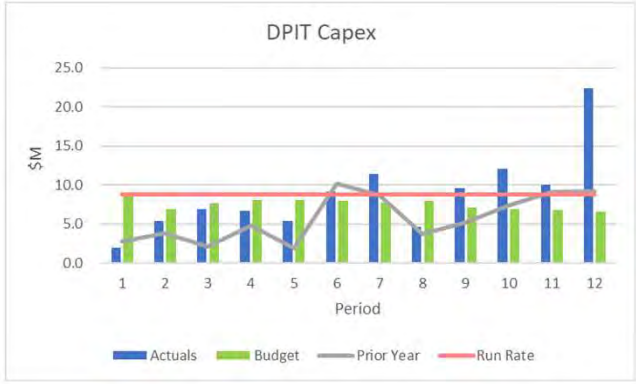
Q4 FY23 Report

nationalgrid

DPIT

Financial Performance



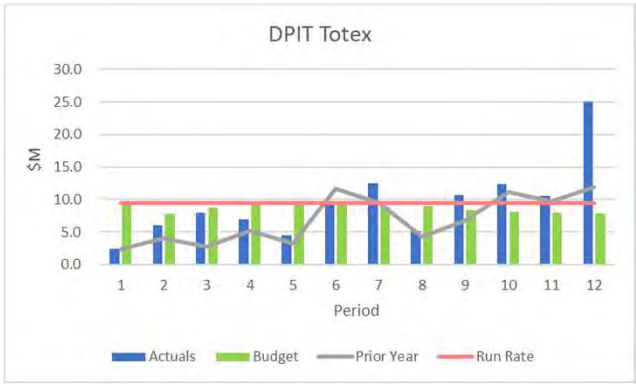


Totex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
103.8	113.8	100.7

Capex

Budget ($M)	Actual ($M)	Q3 Forecast ($M)
90.4	105.7	93.7



Portfolio Commentary

Variance Drivers

- **Financials:** Delivered within 3% of TotEx Investment Plan
  - Capex overspend driven by planned scope increases in the Digital Workplace Services and Digital Platform areas. There was also a reclassification of licence costs to Capex which drove the overall variance.
  - Opex underspend is due to costs shifting out of DPIT to NG exceptional funding.
- **Technical Debt:** Exceeded FY23 stretch goal (15%). Efforts now focused on maintenance in FY24.

Major Releases:

- Unified Technology Operations Center(UTOC) live 1 month early and already reduced truck rolls and pole climbs by 30%. 17% reduction in unplanned outages year to date, due to reactive problem management and new methods of fixing underlying issues.
- Networks delivering increased efficiency, network speed, and security to voice & data applications for a better user experience.
- CNI developed a Rapid Recovery plan for critical systems.
- Windows as a Service deployed updates to >23K devices.
- Cloud & Hosting: migration of data ~ Complete to reduce risk of applications hosted on old and unsupported hardware, reduce Technical debt, and enable Data Center Transformation.
- Digital Platforms: several deliverables to provide secure /reliable/scalable/modernization improvements.

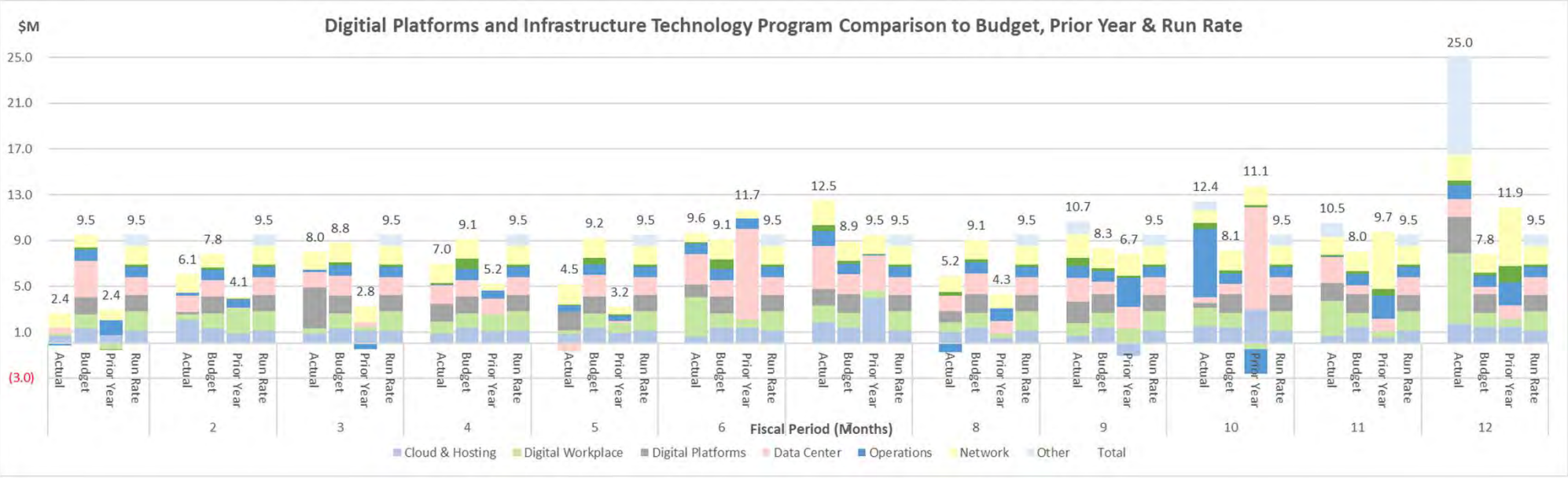
DPIT Program View of Financials

DPIT Program View Full Year $M

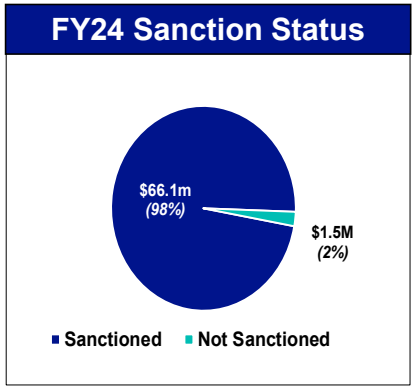
Portfolio	Program	Capex FY Actual	Capex FY Budget	Capex Variance	Opex FY Actual	Opex FY Budget	Opex Variance	Totex FY Actual	Totex FY Budget	Totex Variance
DPIT	Cloud & Hosting	11.2	15.0	3.8	2.0	1.6	(0.4)	13.3	16.6	3.4
	Digital Workplace	18.9	14.3	(4.6)	1.3	0.8	(0.5)	20.3	15.2	(5.1)
	Digital Platforms	15.6	16.4	0.8	1.9	2.2	0.3	17.5	18.6	1.1
	Data Center	17.9	11.5	(6.5)	0.4	6.8	6.4	18.3	18.3	(0.0)
	Operations	10.0	10.9	0.9	0.8	0.4	(0.4)	10.8	11.3	0.5
	CNI	2.8	4.3	1.5	0.2	0.2	0.0	3.0	4.5	1.5
	Network	17.8	18.0	0.2	1.3	1.3	0.0	19.1	19.3	0.2
	Other	11.4	0.0	(11.4)	0.2	0.0	(0.2)	11.6	0.0	(11.6)
	Totals	**105.7**	**90.4**	**(15.3)**	**8.1**	**13.3**	**5.3**	**113.8**	**103.8**	**(10.0)**

Key Challenges

Challenge	Escalation Point
Delays in US build, testing, and migration causing delays for US into FY24	• US Data Center is focused to complete in FY24



Mobilisation Summary



*Total sanction includes exceptional costs as well as core portfolio costs

Status

- FY23 DPIT Product portfolios in the US are complete.
- **FY24 Sanction Status:**
  - 98% of the portfolio is sanctioned, Digital Workplace Services has a partial sanction
  - All FY24 Sanctions approved by April 1, 2023, except Digital Platforms, which was approved April 18, 2023
  - Objectives and Epics were prioritized to support FY24 final budget allocations
  - Product objectives align with Infrastructure and Business Unit objectives

Priority Investments FY24

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
Data Center Transformation (DCT) - *Modernization of Infrastructure & Networks*	Complete	10.7	Schedule	
Cloud & Hosting - *Modernization of Infrastructure & Networks*	Complete	4.4		
Digital Workplace Services - *Employee Satisfaction & Digital Product Delivery*	Complete	16.3		
Networks - *Reduce Technical Debt*	Complete	12.0		
Global Operations - *Employee Satisfaction & Digital Product Delivery*	Complete	3.6		
Critical National Infrastructure (CNI) - *Reduce Technical Debt*	Complete	3.6		
Digital Platforms - *Employee Satisfaction & Digital Product Delivery*	Complete	10.5		

Resources

- No resource risks

Scope

- DPIT product portfolios on-track in FY24 to achieve both Objectives/Key Results (OKRs) and financial forecast of $67.7m.

				Actual to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason
Customer	Customer Digital	4750	Customer Experience Program	16,035		(16,035)	Opex Only Investment
Customer	Customer Digital	4750J	Web Self Service	3,465,139	1,238,257	(2,226,882)	Complexity
Customer	Customer Digital	4750M	ODS	833,343	1,238,257	404,914	Funding Change
Customer	Customer Digital	5673	My Business Account (Nucleus Phase 2)	2,609,271	5,527,000	1,446,342	Efficiencies
Customer	Customer Digital	5673	My Business Account (Nucleus Phase 2)			1,471,387	Accounting
Customer	Customer Digital	6206	Customer Smart Target	404,586		(404,586)	Scope changes
Customer	Customer Digital	6713	Customer-Facing Mobile Application	877,233		(877,233)	Funding Change
Customer	Customer Digital	3697	Computing platform for Gas Load Forecasting	2,903		(2,903)	Opex Only Investment
Customer	Customer Operations	3932	AIMS (TPV)	(365,679)		365,679	Accounting
Customer	Customer Operations	4904	Upgrade Servers for CCAE Reporting	(3,907)		3,907	Opex Only Investment
Customer	Customer Operations	5125	Customer Connection Program	3,385,810	3,763,025	377,215	Scope changes
Customer	Customer Operations	5160	NY Stack VDER and Portal Integration	3,395,045	2,295,586	(1,099,459)	Complexity
Customer	Customer Operations	5345B	Bill Print and Bill Composition Implementation	2,381,011	1,055,130	(1,325,881)	Scope changes
Customer	Customer Operations	5722	KEDLI Rate Case (Mandated)	(1,640)		1,640	Opex Only Investment
Customer	Customer Operations	5723	KEDNY Rate Case	288,874	380,331	91,457	Efficiencies
Customer	Customer Operations	5778	Customer Billing Improvement Program - Phase 1	928	991,583	990,655	Funding Bucket
Customer	Customer Operations	5795F	Community Distributed Generation (CDG) Consolidated Billing	275,567	223,254	(24,707)	Imprecise Estimates
Customer	Customer Operations	5795F	Community Distributed Generation (CDG) Consolidated Billing			(27,606)	Timing
Customer	Customer Operations	5859A	Customer and Agent Experience - IVR Baseline	1,851,194	2,800,000	948,806	Funding Change
Customer	Customer Operations	5863D	AIMS Stability	(53,029)		53,029	Accounting
Customer	Customer Operations	5926A	NiMO Rate Case	(47,562)		47,562	Accounting
Customer	Customer Operations	5987	Smart City Phase II	39,363		(39,363)	Complexity
Customer	Customer Operations	6007	Inside Agent	(29,748)		29,748	Accounting
Customer	Customer Operations	6032	Customer Billing COVID Transition	194,523	303,258	108,735	Scope changes
Customer	Customer Operations	6066	FCP Upgrade for CSS - Plan Forward	606,232	372,526	(233,706)	Complexity
Customer	Customer Operations	6112	WFH Hardware - Plan Forward	(7,442)		7,442	Immaterial
Customer	Customer Operations	6214	Customer Operational Enhancements FY22	7,896		(7,896)	Opex Only Investment
Customer	Customer Operations	6217	Customer Billing Minor Works FY22	(90,564)		90,564	Opex Only Investment
Customer	Customer Operations	6250	EV End-End Solution		3,000,000	3,000,000	Funding Change
Customer	Customer Operations	6302	2022 AIMS Platform	179,250		(179,250)	Funding Change
Customer	Customer Operations	6349	Customer Data Platform	4,965,608	5,010,585	44,977	Accounting
Customer	Customer Operations	6427	RI Purchase of Receivables (POR) Program	54,496	87,197	32,701	Complexity
Customer	Customer Operations	6434	CRIS to CSS Migration	2,081		(2,081)	Opex Only Investment
Customer	Customer Operations	6435	Customer PCI Remediation	420,896	389,679	(31,217)	Variance less than 10%
Customer	Customer Operations	6492	NACHA Compliance- Security Encryption Effort	1,654,235	474,850	(1,179,385)	Complexity
Customer	Customer Operations	6496	NY Gender Pronoun and Name Mandate	1,635,826	1,960,056	324,230	Complexity
Customer	Customer Operations	6531	AIMS Upgrades and Enhancements	565,636	645,000	79,364	Accounting
Customer	Customer Operations	6538	Billing & Collections Mandates & Enhancements		5,294,426	5,294,426	Funding Bucket
Customer	Customer Operations	6543	Customer Minor Works	463,690	450,000	(13,690)	Variance less than 10%
Customer	Customer Operations	6584	UWP 2.0 Digital Re Architecture and Optimization	11,987,457		(11,357,457)	Funding Change
Customer	Customer Operations	6584	UWP 2.0 Digital Re Architecture and Optimization			(345,000)	Scope changes
Customer	Customer Operations	6584	UWP 2.0 Digital Re Architecture and Optimization			(285,000)	Accounting
Customer	Customer Operations	6451	One Step Auto Cashering	168,942		(168,942)	Timing
Customer	Customer Operations	5778A	CRIS Infrast MGKEDNY	92,544		(92,544)	Funding Change
Customer	Customer Operations	5778B	CSS Infrast NGLeg+LI	806,907		(806,907)	Scope changes
Customer	Customer Operations	5778C	Encryption & DR AR	10,104		(10,104)	Immaterial
Customer	Customer Operations	5778F	RightFax AR	751,956		(751,956)	Complexity
Customer	Customer Operations	5778D	Data Integ Svcs AR	5,881		(5,881)	Immaterial
Customer	Customer Operations	6007X	Inside Agent	9,403		(9,403)	Opex Only Investment
Customer	Customer Operations	6654	Documentum Feasability & Analysis	254,267		(254,267)	Opex Only Investment
Customer	Customer Operations	5345BX	Bill Print and Bill Composition Implementation	14,246		(14,246)	Opex Only Investment
Customer	Customer Operations	6686	Documentum Application Modifications	73,650		(73,650)	Opex Only Investment
Customer Total				44,152,457	37,500,000	(6,652,457)	

				Actual to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason
Group Functions	Finance & Business Service	3870A	Vision FM & Property Services Replacement	75,421		(75,421)	Complexity
Group Functions	Finance & Business Service	4418USB	RSA Archer Upgrade & Refresh	71,980		(71,980)	Accounting
Group Functions	Finance & Business Service	4562	US SAP: Business Warehouse (BW) Consolidation to HANA Enterprise Cloud (HEC)	(58,530)		58,530	Accounting
Group Functions	Finance & Business Service	5158	MyHub Compliance & Enhancements	623,563	2,900,000	2,276,437	Scope changes
Group Functions	Finance & Business Service	5380A	Blackline: US Implementation	0		0	Immaterial
Group Functions	Finance & Business Service	5466	AMAG HW and SW Upgrade	1,516,710	800,000	(716,710)	Imprecise Estimates
Group Functions	Finance & Business Service	5497	Core Powerplan Upgrade and Lease Consolidation	1,176,659	2,300,000	1,123,341	Efficiencies
Group Functions	Finance & Business Service	5555	Fieldglass Replacement		1,800,000	1,800,000	Funding Change
Group Functions	Finance & Business Service	5605	US SAP: Infrastructure Landscape FY21	18,812		(18,812)	Immaterial
Group Functions	Finance & Business Service	5606	US SAP: Infrastructure Landscape FY22	(909,261)		909,261	Accounting
Group Functions	Finance & Business Service	5610	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY22	2,732		(2,732)	Immaterial
Group Functions	Finance & Business Service	5611	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY23	897,447	1,500,000	602,553	Imprecise Estimates
Group Functions	Finance & Business Service	5841A	Pension Administrator Replacement	0		0	Immaterial
Group Functions	Finance & Business Service	5841C	401(k) Administration Changes	123,090		(123,090)	Timing
Group Functions	Finance & Business Service	5842A	US SAP: GRC Upgrade	0		0	Immaterial
Group Functions	Finance & Business Service	5842B	Mandatory FERC Form Changes	(21,525)		21,525	Accounting
Group Functions	Finance & Business Service	5843A	SAP Platform Upgrade: OS and Databases	0		0	Immaterial
Group Functions	Finance & Business Service	5843B	US SAP Platform Upgrade: Minor Releases	(162,703)		162,703	Accounting
Group Functions	Finance & Business Service	5843D	US SAP Healthcheck: Upgrade Critical Components - FY22	869,524	700,000	(169,524)	Imprecise Estimates
Group Functions	Finance & Business Service	5895B	Everbridge / Critical Event Management	6,967		(6,967)	Immaterial
Group Functions	Finance & Business Service	5896A	US Tax: SABRIX Upgrade	0		0	Immaterial
Group Functions	Finance & Business Service	5896B	Warehouse SAP Scanner Replacement (Barcoding)	(388)		388	Immaterial
Group Functions	Finance & Business Service	5919	US SAP Minor Enhancements	(5,246)		5,246	Immaterial
Group Functions	Finance & Business Service	6137US	Enterprise PMO, Jira Align release 2	(39,062)		39,062	Accounting
Group Functions	Finance & Business Service	6182US	Global Career Site	(48,999)		48,999	Opex Only Investment
Group Functions	Finance & Business Service	6183US	HR Reporting Analytics	59,864	1,300,000	1,240,136	Funding Change
Group Functions	Finance & Business Service	6185US	Onboarding 2.0	428,376		(428,376)	Funding Change
Group Functions	Finance & Business Service	6219	US F&BS Minor Enhancements (Non-SAP)	14,542		(14,542)	Opex Only Investment
Group Functions	Finance & Business Service	6220	US F&BS Minor Enhancements (SAP)	75,777		(75,777)	Funding Change
Group Functions	Finance & Business Service	6221	Tech Debt: RiskMaster Replacement	645,277		(645,277)	Funding Change
Group Functions	Finance & Business Service	6231	US Treasury: Allegro Upgrade Phase 2	63,755		(63,755)	Timing
Group Functions	Finance & Business Service	6242US	ARIBA System Enhancements	165,816		(165,816)	Timing
Group Functions	Finance & Business Service	6265	HR Policy Compliance	33,783		(33,783)	Opex Only Investment
Group Functions	Finance & Business Service	6270US	Digitising Learning	440,595		(440,595)	Funding Change
Group Functions	Finance & Business Service	6273US	Workforce Data Domain, HR data, governance, Tools, Processes, EDM	2,687,096	700,000	(1,987,096)	Imprecise Estimates
Group Functions	Finance & Business Service	6289B	FTOM: Record to Report	(5,455)		5,455	Immaterial
Group Functions	Finance & Business Service	6289D	FTOM: Record to Report – Task Management with Blackline	1,180,296	1,000,000	(180,296)	Imprecise Estimates
Group Functions	Finance & Business Service	6289E	FERC Form Integration with wDesk	525,400	1,100,000	574,600	Imprecise Estimates
Group Functions	Finance & Business Service	6290A	Source To Pay: Release 1 – P2P Roll Out Program	12,028,955	1,433,000	(10,595,955)	Imprecise Estimates Complexity
Group Functions	Finance & Business Service	6290B	Source To Pay: Release 2 - Supplier Onboarding	109,471	1,300,000	1,190,529	Timing
Group Functions	Finance & Business Service	6346USC	Domain Management Refresh	(466,838)		466,838	Opex Only Investment
Group Functions	Finance & Business Service	6346USE	Brand Central Design and Implementation	(10,015)		10,015	Opex Only Investment
Group Functions	Finance & Business Service	6346USF	Brand Central Design and Implementation	(67,193)		67,193	Opex Only Investment
Group Functions	Finance & Business Service	6346USI	Video Strategic Management and Archiving (Enterprise)	(53,405)		53,405	Opex Only Investment
Group Functions	Finance & Business Service	6346US	CA: Group We Digital Evolution Phase 1 - FY22	(87,667)		87,667	Accounting
Group Functions	Finance & Business Service	6377	US Group Function Operational Requests	(353,562)		353,562	Accounting
Group Functions	Finance & Business Service	6421	P2P Data Analysis Tool	1,690,327	1,200,000	(490,327)	Scope changes
Group Functions	Finance & Business Service	6426	Investment Recovery Digitization	1,155,399	1,000,000	(155,399)	Complexity
Group Functions	Finance & Business Service	6450x	Accelerated Telecom Networks Audit (POTS and Circuits)	1,324,040		(1,324,040)	Opex Only Investment
Group Functions	Finance & Business Service	6520US	Jira Align Phase 3	1,735,172	900,000	(835,172)	Other
Group Functions	Finance & Business Service	5607	US SAP: Infrastructure Landscape FY23	3,138,031	4,000,000	861,969	Other
Group Functions	Finance & Business Service	5843E	SAP BW/BI Upgrade	167,279	700,000	532,721	Timing
Group Functions	Finance & Business Service	5843F	SAP PI NW 7.40 (PI -> PO upgrade) to NW 750	3,016,247	900,000	(2,116,247)	Scope changes
Group Functions	Finance & Business Service	6529	US Group Function Minor Enhancements (SAP)	1,090,806	1,100,000	9,194	Variance less than 10%
Group Functions	Finance & Business Service	6530	US Group Function Operational Requests - FY23	3,193,797	1,300,000	(1,893,797)	Scope changes
Group Functions	Finance & Business Service	4418USC	CRO FY23: GRC Archer	1,113,271	1,500,000	386,729	Imprecise Estimates
Group Functions	Finance & Business Service	6514US	NG.com Refresh		1,900,000	1,900,000	Funding Bucket
Group Functions	Finance & Business Service	6533US	SHE FY23: Group SHE Digital Evolution		1,500,000	1,500,000	Timing
Group Functions	Finance & Business Service	6534US	Legal FY23: Group Legal Digital Evolution		708,800	708,800	Timing
Group Functions	Finance & Business Service	Corp23	US Placeholder - Corp Tech Debt Remediation		1,800,000	1,800,000	Funding Bucket
Group Functions	Finance & Business Service	6270	Digitising Learning		600,000	600,000	Funding Bucket
Group Functions	Finance & Business Service	HR23	Tactical HR		1,800,000	1,800,000	Funding Bucket
Group Functions	Finance & Business Service	OP523	Placeholder (Tech Debt & App Rat)		680,000	680,000	Funding Bucket
Group Functions	Finance & Business Service	6225US	R2R Reporting Use Cases		1,300,000	1,300,000	Funding Bucket
Group Functions	Finance & Business Service	6289G	FTOM R2R: Robotic Process Automation of Manual Journal Entries	1,719,399	1,800,000	80,601	Variance less than 10%
Group Functions	Finance & Business Service	R2R23	Finance + placeholder (Regulatory Compliance)		800,000	800,000	Funding Bucket
Group Functions	Finance & Business Service	6467	Union Bidding	472,806		(472,806)	Funding Change
Group Functions	Finance & Business Service	6566US	HR Policy Compliance FY23	212,123		(212,123)	Opex Only Investment
Group Functions	Finance & Business Service	6180US	CV Talent Screening / Blind CVs	287,313		(287,313)	Funding Change
Group Functions	Finance & Business Service	6424US	Infonet decommissioning	49,866		(49,866)	Opex Only Investment
Group Functions	Finance & Business Service	3870B	Vision FM replacement, Phase 2	1,448,602		(1,448,602)	Funding Change
Group Functions	Finance & Business Service	6289F	US GAAP Statutory reporting with WDesk	597,682		(597,682)	Funding Change
Group Functions	Finance & Business Service	6621US	My Hub	2,434,606		(2,434,606)	Funding Change
Group Functions	Finance & Business Service	6618	US S4 Discovery Refresh	696,143		(696,143)	Opex Only Investment
Group Functions	Group Functions	5899A	Cloud API Migrations Phase 2	33,198		(33,198)	Timing
Group Functions	Group Functions	5917A	Data Mgmt Tooling	(2)		2	Opex Only Investment
Group Functions	Group Functions	5983	DB Landscape Optimization	1,984,377		(1,984,377)	Funding Change
Group Functions	Group Functions	6137	Enterprise PMO	3		(3)	Opex Only Investment
Group Functions	Group Functions	6252US	Data Mgmt Tools and Initiatives	7,476		(7,476)	Immaterial
Group Functions	Group Functions	6254US	Platform as a Service (PaaS) on Azure	11,909		(11,909)	Immaterial
Group Functions	Group Functions	6253US	Cloud API Migration - Legacy Integration Tools	760,143		(760,143)	Funding Change
Group Functions	Group Functions	6255US	Digital Foundation Tooling and DevSecOps	(82,383)		82,383	Accounting
Group Functions	Group Functions	6256US	Platform Upgrades: Salesforce/Maximo/Mulesoft	(28,251)		28,251	Accounting
Group Functions	Group Functions	6258USA	Data Archival Platform Implementation	74,806		(74,806)	Complexity
Group Functions	Group Functions	6258USB	Data Archival: US Group Function Portfolio	2,439,161	1,100,000	(1,339,161)	Scope changes
Group Functions	Group Functions	6260US	EA Foundation: Increased Capability	181,288		(181,288)	Timing
Group Functions	Group Functions	6261US	ServiceNow Upgrade: Stay Current FY22	(182,874)		182,874	Accounting
Group Functions	Group Functions	6266US	E-Hub	499,724	800,000	300,276	Imprecise Estimates
Group Functions	Group Functions	6313US	GDPR Purge Compliance	853		(853)	Opex Only Investment
Group Functions	Group Functions	6360US	IT Planning: Enterprise PMO	132,501		(132,501)	Funding Change
Group Functions	Group Functions	6461USA	Business Planning F&A (FP&A Enablement & Modelling Tool)	2,261,115	2,500,000	238,885	Timing
Group Functions	Group Functions	6468	GridStack - Infrastructure	(1,219,471)	2,657,999	3,877,470	Accounting
Group Functions	Group Functions	6513US	CA: Group Web Digital Evolution - FY23	142,676	800,000	657,324	Timing
Group Functions	Group Functions	6587US	E-Hub	1,037,335		(1,037,335)	Funding Change
Group Functions	Group Functions	6522US	SHE: Digitisation of the CMT Process	352,541		(352,541)	Funding Change
Group Functions	Group Functions	6549US	SHE: Resilience Digital Evolution	410,752		(410,752)	Other
Group Functions	Group Functions	6391US	SHE: Consolidated Incident Management and Reporting	342,949		(342,949)	Opex Only Investment
Group Functions	Group Functions	6402US	Legal Operations	226,945		(226,945)	Opex Only Investment
Group Functions	Group Functions	6563US	CA:NG.com Design	100,780		(100,780)	Opex Only Investment
Group Functions	Group Functions	6458	QES IT Tool Sunset	20,377		(20,377)	Opex Only Investment
Group Functions	Group Functions	6660	Ping Repository F&A	22,356		(22,356)	Opex Only Investment
Group Functions	Group Functions	6690	Cost center and Time improvements	59,340		(59,340)	Opex Only Investment
Group Functions	Group Functions	6714	Join to Leave	150,000		(150,000)	Opex Only Investment
Group Functions Total				56,834,592	50,179,799	(6,654,793)	

CIO Group	It_program	[redacted]	Actual to Budget Variance			Permanent
			Actual YTD	YTD Budget	YTD Variance	Reason
Security	Physical Security					
Security	Physical Security					
Security	Physical Security					
Security	Physical Security					
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Security	Security					
Security Total						

				Actual to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason
US Electric	EBU Digital	5054	On My Way (EBU Digital Solution) Phase 2	7,247,592	3,761,961	(3,485,631)	Imprecise Estimates
US Electric	EBU Digital	5890C	GridInspector F&A	233,908		(233,908)	Timing
US Electric	EBU Digital	5912	On My Way (EBU Digital Solution)	42,203		(42,203)	Timing
US Electric	EBU Digital	6073	Veg Management Optimization	3,774,332	8,566,395	4,792,063	Scope changes
US Electric	EBU Digital	6076	FutureNow Validation Phase	9,017,589	11,186,890	2,169,301	Efficiencies
US Electric	EBU Digital	6291	ICU-Out (Electric Digital)	1,115,767	5,647,785	4,532,018	Timing
US Electric	EBU Digital	6425	DER Active Resource Integration NY	(352,953)		352,953	Funding Change
US Electric	EBU Digital	6437	GridMD	4,687	1,020,103	1,015,416	Timing
US Electric	EBU Digital	6388	EBU Placeholder		12,800,000	12,800,000	Funding Bucket
US Electric	EBU Digital	6437A	Grid MD	1,326,728		(1,326,728)	Timing
US Electric	EBU Digital	6631	US Electric Rapid Rebuild	51,699		(51,699)	Opex Only Investment
US Electric	EBU Digital	6662	Proj Costing Cont Mgmt	1,071,213		(1,071,213)	Timing
US Electric	EBU Digital	6661	Demand Planning and Material Track & Trace	897,596		(897,596)	Funding Change
US Electric	EBU Digital	6672	DER Scenario Forecast & Planning	197,807		(197,807)	Opex Only Investment
US Electric	EBU Digital	6673	DER Connections	203,713		(203,713)	Opex Only Investment
US Electric	Electric BU	4481	Energy Accounting System Migration to Wholesale Settlement Application - NY	(94,129)		94,129	Efficiencies
US Electric	Electric BU	4483	US Control Transmission	49,547		(49,547)	Timing
US Electric	Electric BU	5241	East Pulaski Energy Storage System IS Network	65,053		(65,053)	Scope changes
US Electric	Electric BU	5242	Troy Energy Storage System IS Network	(1,241)		1,241	Immaterial
US Electric	Electric BU	5702	GridStack	27,088		(27,088)	Accounting
US Electric	Electric BU	5773B	Visitor Logging for NERC-CIP Reliability Compliance	336,933	439,714	102,781	Other
US Electric	Electric BU	5773E	NERC-CIP Reliability Compliance	235,843		(235,843)	Timing
US Electric	Electric BU	5871A	NY Battery Energy Storage Systems		2,675,000	2,675,000	Timing
US Electric	Electric BU	5871D	Clean Energy Initiatives		625,000	625,000	Timing
US Electric	Electric BU	5871E	Power Marketer Pilot	183,003	0	(183,003)	Timing
US Electric	Electric BU	5871F	Battery Storage Communication Upgrades	4,882	1,500,000	1,495,118	Timing
US Electric	Electric BU	5887A	ARCOS - Reporting Replacement	544,006	780,689	236,683	Efficiencies
US Electric	Electric BU	5888D	Fluke & WECO Calibration licenses, hardware updates and data migration	371,287	118,000	(253,287)	Scope changes
US Electric	Electric BU	5889C	Electric Regulatory Enhancements	(13,812)	225,000	238,812	Funding Bucket
US Electric	Electric BU	5890A	Corporate PI	1,816,739	1,588,789	(227,950)	Imprecise Estimates
US Electric	Electric BU	5891A	Electric MV90 Upgrade	1,628,607	741,520	(887,087)	Scope changes
US Electric	Electric BU	5891C	Field Collection System (FCS) Upgrade	146,712	294,558	147,846	Scope changes
US Electric	Electric BU	5891D	IEE Integration/Upgrades		1,150,000	1,150,000	Funding Change
US Electric	Electric BU	5891E	MTS Retail Settlement (MDS)	2,691,238	3,530,991	839,753	Timing
US Electric	Electric BU	5891F	MDS Consolidation Phase 1		1,725,000	1,725,000	Timing
US Electric	Electric BU	5891I	Price Transmission (Retail Price) System Replacement	1,324,831	1,005,377	(319,454)	Timing
US Electric	Electric BU	5892A	Control Room EOL Server and Tool Replacement	222,829		(222,829)	Timing
US Electric	Electric BU	5892B	Control Room Servers and Tools Upgrade/Replacement	1,358,491	1,380,000	21,509	Variance less than 10%
US Electric	Electric BU	5892C	IDS-DVD 811 Dig Safe DVD Replacement	156,506	1,127,890	971,384	Efficiencies
US Electric	Electric BU	5893A	Digitizing the Electric Business - Mobility (Mwork Replacement) *A	14,137		(14,137)	Opex Only Investment
US Electric	Electric BU	6043	Plan Forward	3,298		(3,298)	Opex Only Investment
US Electric	Electric BU	6320	Telecom & Network Deep Dive	75,005		(75,005)	Opex Only Investment
US Electric	Electric BU	6571	GridAPI Electric	147,349		(147,349)	Opex Only Investment
US Electric	Electric BU	5889D	Electric FY23 Regulatory Mandates and Minor Enhancements	154,482		(154,482)	Opex Only Investment
US Electric	Electric BU	5871DA	SD-WAN for Distributed Communications	88,308		(88,308)	Funding Change
US Electric	Wholesale Networks	5773D	NYISO PMU Requirement for Large Generators	1,179,108	736,119	(442,989)	Complexity
US Electric	Wholesale Networks	5822C	Generation Asset Management Upgrade	3,868,721	1,547,623	(2,321,098)	Complexity
US Electric	Wholesale Networks	6043G1	Content Management for Engineering	653,704	1,493,248	839,544	Complexity
US Electric Total				42,070,406	65,667,653	23,597,247	

				Actual to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason
US Gas	Capital Delivery	5274A	New Storms Response Management System Implementation	320,969	277,032	(43,937)	Timing
US Gas	Capital Delivery	6222	PIF Replacement	(270,663)		270,663	Accounting
US Gas	Gas BU	4769	Gas Storm Hardening Using Remote Service Valves (RSV's)	513,974	382,812	(131,162)	Timing
US Gas	Gas BU	4898	Asset Management System for LNG	3,217,488	1,607,000	(1,610,488)	Complexity
US Gas	Gas BU	578SI	NRG GIS Replacement Project	672		(672)	Immaterial
US Gas	Gas BU	5787D	Gas Transportation System (GTS) Product	3,159,834	1,980,000	(1,179,834)	Imprecise Estimates
US Gas	Gas BU	5787G	F&A Study for Gas Demand Response (DR)	13,914		(13,914)	Other
US Gas	Gas BU	5788D	Interface MyHub To OnBoard LMS OQ External Site	898		(898)	Immaterial
US Gas	Gas BU	5788G	FY22 Gas Regulatory and Operational Enhancements (Minor Works)	(7,959)		7,959	Opex Only Investment
US Gas	Gas BU	5789E	Summons and Corrective Actions Response (CAR) Work Tracker	97,195		(97,195)	Other
US Gas	Gas BU	6015	Continuous Improvement for Gas Applications	(226,010)	1,210,000	1,436,010	Funding Bucket
US Gas	Gas BU	6353A	FY22 Gas Application Rationalization	13,902		(13,902)	Opex Only Investment
US Gas	Gas BU	6392	Contractor Ipad for Gas Business	(129,892)		129,892	Accounting
US Gas	Gas BU	6441	Scalability & Resilience for Enterprise Gas Platforms	1,387,115	1,190,000	(197,115)	Accounting
US Gas	Gas BU	6479	LNG IT Support for OT Systems	296,051	850,000	553,949	Timing
US Gas	Gas BU	6481	Pathfinder for Leak Survey for MA	52,903		(52,903)	Other
US Gas	Gas BU	5785Q	Field Collection Systems (FCS) Upgrade (Gas Only)	581,187	840,000	258,813	Timing
US Gas	Gas BU	6472	MOC Library & Action Item Log Automation	188,372	150,000	(38,372)	Accounting
US Gas	Gas BU	6509	Gas I&R, Gas Meter Ops, and Electric Complex Construction Field Device Replacement	497,024	134,000	(363,024)	Imprecise Estimates
US Gas	Gas BU	6539	Continued Scaling and Evolution of Gas IT Transformation Program	32,110,044	20,070,000	(12,040,044)	Imprecise Estimates
US Gas	Gas BU	6542	New Gas Demand Response Solution POC and Pilot?	76,309	863,083	786,774	Scope changes
US Gas	Gas BU	6550	Gas Management System (SCADA) Upgrade'	2,950,911	2,098,333	(852,578)	Scope changes
US Gas	Gas BU	6552	System Operating Procedure Third-Party User Access for PE Review		100,000	100,000	Funding Change
US Gas	Gas BU	6553	Enable Gas Curtailment Map in ESRI		40,000	40,000	Funding Change
US Gas	Gas BU	6554	New System Operating Procedure (SOP) Solution		100,000	100,000	Timing
US Gas	Gas BU	6555	PIPES/Enterprise Maximo Integration		50,000	50,000	Funding Change
US Gas	Gas BU	6556	Digital Asset & Work Management for CNG and I&R?		290,000	290,000	Timing
US Gas	Gas BU	6557	Enable Converted LI Gas Distribution Asset Data in ESRI		140,000	140,000	Timing
US Gas	Gas BU	6559	FY23 Gas Regulatory and Operational Enhancements (Minor Works)?		150,000	150,000	Funding Change
US Gas	Gas BU	6582	Gas Application Continuous Improvement	849,757		(849,757)	Scope changes
US Gas	Gas BU	6573	FY23 Gas Regulatory and Operational Enhancements (Minor Works)	200,476		(200,476)	Funding Change
US Gas	Gas BU	6610	Gas App Rationalization	356,378		(356,378)	Funding Change
US Gas	Gas BU	6642	Tesco	2,452,241		(2,452,241)	Funding Change
US Gas	Gas BU	6671D	US Gas Data Modern-E	88,505		(88,505)	Accounting
US Gas Total				48,791,595	32,522,261	(16,269,334)	

				Actual to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason
DPIT	Data Center	6223	I&O US Data Center Transformation FY22	372,226	18,268,518	(9,679)	Funding Change
DPIT	Data Center	6223.01	INVP 6223.01 DCT Other Costs	1,776,003		0	See total 6223 variance
DPIT	Data Center	6223.02	INVP 6223.02 DCT Co-Location	(286,911)		0	See total 6223 variance
DPIT	Data Center	6223.03	INVP 6223.03 DCT Mngd Svcs	3,086,597		0	See total 6223 variance
DPIT	Data Center	6223.04	INVP 6223.04 DCT Mainframe	3,059,856		0	See total 6223 variance
DPIT	Data Center	6223.05	INVP 6223.05 DCT App Migrat	10,270,426		0	See total 6223 variance
DPIT	Operations	6196	I&O US Operations FY22	(275,043)		717,852	Accounting
DPIT	Operations	6196.01.01	INVP 6196.01.01 BAU Device Refresh	12,392		0	See total 6196 variance
DPIT	Operations	6196.01.02	INVP 6196.01.02 Application Mainten	351		0	See total 6196 variance
DPIT	Operations	6196.02.02	INVP 6196.02.02 CMDB Overhaul	(54,690)		0	See total 6196 variance
DPIT	Operations	6196.03.01	INVP 6196.03.01 Appl Perf Mgnt	(609,848)		0	See total 6196 variance
DPIT	Operations	6196.03.02	INVP 6196.03.02 Auto Disaster Rec	208,986		0	See total 6196 variance
DPIT	Operations	6485	I&O US Digital Operations FY23	5,489,792	11,318,284	(267,601)	Funding Change
DPIT	Operations	6485.01	INVP 6485 Service Aware CMDB	1,667,972		0	See total 6485 variance
DPIT	Operations	6485.02	INVP 6485 Observability	1,008,486		0	See total 6485 variance
DPIT	Operations	6485.03	INVP 6485 AI Operations	2,294,350		0	See total 6485 variance
DPIT	Operations	6485.04	INVP 6485.04 Automation	84,984		0	See total 6485 variance
DPIT	Operations	6485.05	INVP 6485.05 Unified Tech Ops Ctr	1,040,301		0	See total 6485 variance
DPIT	Network	6197	I&O US Networks FY22	(76,428)		(13,479)	Accounting
DPIT	Network	6197.01.01	INVP 6197.01.01 UPS Devices	(484,206)		0	See total 6197 variance
DPIT	Network	6197.01.02	INVP 6197.01.02 EOL Switch	6,730		0	See total 6197 variance
DPIT	Network	6197.01.03	INVP 6197.01.03 SD WAN	34,732		0	See total 6197 variance
DPIT	Network	6197.01.04	INVP 6197.01.04 NW Monitoring	331,278		0	See total 6197 variance
DPIT	Network	6197.01.05	INVP 6197.01.05 OOB Mgmt	58,750		0	See total 6197 variance
DPIT	Network	6197.01.06	INVP 6197.01.06 Storm Sites	(49,046)		0	See total 6197 variance
DPIT	Network	6197.02.01	INVP 6197.02.01 WiFi	50,124		0	See total 6197 variance
DPIT	Network	6197.02.02	INVP 6197.02.02 WiFi Tooling	(15,679)		0	See total 6197 variance
DPIT	Network	6197.03.01	INVP 6197.03.01 Voice EOL	55,197		0	See total 6197 variance
DPIT	Network	6197.03.02	INVP 6197.03.02 MS Teams Intr	(8,505)		0	See total 6197 variance
DPIT	Network	6197.04.01	INVP 6197.04.01 vStig Upgrads	234,349		0	See total 6197 variance
DPIT	Network	6197.05.01	INVP 6197.05.01 Data Ctr Transforma	(123,817)		0	See total 6197 variance
DPIT	Network	6486	I&O US Networks FY23	16,228,659	19,300,124	188,394	Imprecise Estimates
DPIT	Network	6486.01	INVP 6486 LAN/Wireless	418,464		0	See total 6486 variance
DPIT	Network	6486.02	INVP 6486 WAN/SD-WAN	943,660		0	See total 6486 variance
DPIT	Network	6486.03	INVP 6486 Network Support Svcs	(1,030,792)		0	See total 6486 variance
DPIT	Network	6486.04	INVP 6486 Telephony	2,551,739		0	See total 6486 variance
DPIT	Digital Workplace	6198	I&O US End User Experience FY22	(57,103)		46,773	Accounting
DPIT	Digital Workplace	6198.01	INVP 6198.01.04 Mac OS / iOS Mgmt	42,028		0	See total 6198 variance
DPIT	Digital Workplace	6198.01.01	INVP 6198.01.01 Real Time Collabora	(151,810)		0	See total 6198 variance
DPIT	Digital Workplace	6198.01.02	INVP 6198.01.02 Device and Data Acc	44,954		0	See total 6198 variance
DPIT	Digital Workplace	6198.01.03	INVP 6198.01.03 Content Collaborati	74,929		0	See total 6198 variance
DPIT	Digital Workplace	6198.01.04	Mac OS/iOS Mgmt AC	229		0	See total 6198 variance
DPIT	Digital Workplace	6487	I&O US Digital Workplace Services FY23	10,172,699	15,150,061	(3,845,005)	Scope changes
DPIT	Digital Workplace	6487	I&O US Digital Workplace Services FY23			(948,830)	Accounting
DPIT	Digital Workplace	6487.01	INVP 6487 Device & Data Access	5,225,865		0	See total 6487 variance
DPIT	Digital Workplace	6487.02	INVP 6487 Real Time Collaboration	1,477,832		0	See total 6487 variance
DPIT	Digital Workplace	6487.03	INVP 6487 Content Collaboration	1,019,420		0	See total 6487 variance
DPIT	Digital Workplace	6487.04	INVP 6487.04 Support Assurance	336,725		0	See total 6487 variance
DPIT	Digital Workplace	6487.05	Digital Adopt & Change	1,711,355		0	See total 6487 variance
DPIT	CNI	6199	I&O US CNI FY22	341,779		34,091	Accounting
DPIT	CNI	6199.01.01	INVP 6199.01.01 EoL Assets	(174,429)		0	See total 6199 variance
DPIT	CNI	6199.01.02	INVP 6199.01.02 EOL Assets - Gas	(149,863)		0	See total 6199 variance
DPIT	CNI	6199.02.01	ITSM Implementation	(51,578)		0	See total 6199 variance
DPIT	CNI	6488	I&O US CNI FY23	2,432,899	4,499,596	1,512,983	Funding Change
DPIT	CNI	6488.01	INVP 6488.01 US CNI Networks	244,236		0	See total 6488 variance
DPIT	CNI	6488.02	INVP 6488.02 Infrastructure Mgmt	249,492		0	See total 6488 variance
DPIT	CNI	6488.03	INVP 6488.03 CNI Tool Management	59,986		0	See total 6488 variance
DPIT	Cloud & Hosting	6200	I&O US Cloud and Hosting FY22	26,304		(2,400)	Accounting
DPIT	Cloud & Hosting	6200.01.01	INVP 6200.01.01 HCI / V-Block Migra	97,236		0	See total 6200 variance
DPIT	Cloud & Hosting	6200.01.02	INVP 6200.01.02 DPS Enablement (Mig	(86,615)		0	See total 6200 variance
DPIT	Cloud & Hosting	6200.02.01	INVP 6200.02.01 Storage Migration C	105,171		0	See total 6200 variance
DPIT	Cloud & Hosting	6200.02.02	INVP 6200.02.02 Data Management / A	(96,773)		0	See total 6200 variance
DPIT	Cloud & Hosting	6200.03.01	INVP 6200.03.01 Public Cloud (Azure	184,973		0	See total 6200 variance
DPIT	Cloud & Hosting	6200.04.01	INVP 6200.04.01 Azure and GBE Palo	(165,117)		0	See total 6200 variance
DPIT	Cloud & Hosting	6200.04.02	INVP 6200.04.02 Horizon on Azure Ce	(47,110)		0	See total 6200 variance
DPIT	Cloud & Hosting	6200.05.01	INVP 6200.05.01 New Global Active D	(15,669)		0	See total 6200 variance
DPIT	Cloud & Hosting	6489	I&O US Cloud & Hosting FY23	6,289,037	16,622,203	3,369,718	Imprecise Estimates
DPIT	Cloud & Hosting	6489.01	INVP 6489 Public Cloud Platforms	1,404,346		0	See total 6489 variance
DPIT	Cloud & Hosting	6489.02	INVP 6489 Cloud Containers	976,951		0	See total 6489 variance
DPIT	Cloud & Hosting	6489.03	INVP 6489 Global Active Directory	2,821,357		0	See total 6489 variance
DPIT	Cloud & Hosting	6489.04	INVP 6489 Hosted Solutions & Platf	1,760,794		0	See total 6489 variance
DPIT	Digital Platforms	6519	GridStack Foundations	1,757,733	15,943,364	1,140,364	Imprecise Estimates
DPIT	Digital Platforms	6519.01	INVP 6519.01 GS Foundation GridInit	2,976,527		0	See total 6519 variance
DPIT	Digital Platforms	6519.02	INVP 6519.02 GS Foundations GridCl	2,151,558		0	See total 6519 variance
DPIT	Digital Platforms	6519.03	INVP 6519.03 GS Found. GridDataOps	2,706,919		0	See total 6519 variance
DPIT	Digital Platforms	6519.04	INVP 6519.04 GS Found. GridIdentity	354,456		0	See total 6519 variance
DPIT	Digital Platforms	6519.05	INVP 6519.05 GS Found. GridAccess	251,856		0	See total 6519 variance
DPIT	Digital Platforms	6519.06	INVP 6519.06 GS Found. GridVault	628,016		0	See total 6519 variance
DPIT	Digital Platforms	6519.07	INVP 6519.07 GridAPI/GridSystemInt	3,975,935		0	See total 6519 variance
DPIT	Digital Platforms	6565	DP Secure & Compliant	2,704,251	2,656,677	(47,574)	Imprecise Estimates
DPIT	Digital Workplace	4987	SOE (Windows) Upgrade and Device Refresh	(614)		614	Immaterial
DPIT	Network	5774	WiFi Phase 2	(1,763)		1,763	Immaterial
DPIT	Operations	5876	Pure Technical Debt Remediation	(35,775)		35,775	Accounting
DPIT	Operations	6114	Managed Workspace Services Transformation	(38,108)		38,108	Accounting
DPIT	Digital Workplace	6602	Microstrategy Mod	356,249		(356,249)	Opex Only Investment
DPIT	Other	6641	FY23 Service Now Upgrade - US	2,143,845		(2,143,845)	Funding Change
DPIT	Other	6652	MicroStrategy Sunset	2,202,433		(2,202,433)	Funding Change
DPIT	Other	6708	Microsoft E3/E5 Licenses	7,280,243		(7,280,243)	Accounting
DPIT Total				113,789,730	103,758,827	(10,030,903)	
Grand Total				342,877,796	338,873,907	(4,003,889)	

Impacts Year end Forecast

Driver	Variance Category	Explanation
Project Delivery - Scope change	Scope changes	Spend impacted by both reduced or increased scope
Project Delivery - Initial scope delivered	Imprecise Estimates	Cost and Quantity of resources were incorrectly assessed when creating the budget. (eg. Price Increases)
Project Delivery - Initial scope delivered	Complexity	Unforeseen challenges in delivering the solution drove costs up. No change in scope but amount of work to deliver increased or potentially decreased
Project Delivery - Initial scope delivered	Efficiencies	More cost effective implementation options - Found a better way to deliver the solution which reduced costs
Strategic Direction	Funding Bucket	Source of undefined funding where funds will be moved to other projects
Strategic Direction	Funding Change	Move $ to or from another project (please specify INVP moving to or moving from)
Annual Accounting	Accounting	Over or under accruals which reverse at year end leaving a balance on the INVP
Annual Accounting	Timing	Project slipped and intended scope was not delivered this fiscal year
Other	Other	Please explain

No impact to forecast

Driver	Variance Category	Explanation
Accounting Errors	Mischarges	Temporary due to unavailable accounting codes or charging errors which will be corrected
Accounting Errors	Accruals	which may have been missed / over accrued
Accounting Errors	Misclassifications	eg. Between opex and capex or expense types.
Timing Changes	Timing differences	such as budget phasing, early / later agreed payments or delivery methodology impacting phasing of costs
Timing Changes	Prioritizations	Projects will begin earlier or later than originally planned - Management Determined
Timing Changes	Delays	Materials (chip shortage), Procurement issue (negotiations), resource issues (internal and external) - Beyond Management Control
Other	Other	Please explain

INVID	Inv Name	Actual Capex	Actual Opex	Actual TotEx	Lessons Learned?	Benefits	Allocation Code
4481	US MDS-Wholesale Settlement System Replacement	4,805,900	316,300	5,122,200	Using a fix price contract where feasible and practical should be used as appropriate to reduce risks to the project. "Collaboratively developed software" as in this case, requires Senior Subject Matter Expert (SME) Interface resources during coding and implementation due to the maiden development of the IEE Settlement application. Given this software was not "an off the shelf solution", agile delivery over the applied waterfall delivery would have identified the significance of interface design errors causing delivery delays.	This investment will have a positive impact on the ability to complete the daily settlement and reconciliation process, as the automation of the load data will mitigate the risk of manual error and shorten the time required to complete the tasks associated with the process. l By consolidating the two platforms into a vender supported application we will insure a disaster recovery plan that is site tested for the NY Wholesale Energy Settlement Market and improve performance. l Stability and availability of a key business critical application. l Elimination of manual data transfers from the MV-90 xi system. l Eliminate manual comparisons and aggregation calculations between the respective ISO's data and National Grids data.	541T
4483	US Control-Transmission Outage Application (iTOA) Upgrade	4,158,000	550,000	4,708,000	The new and improved Integrated Outage Application delivered a new web application platform, improved user access, and increased application stability in a new and supported environment. From a business perspective, it was unclear what the full breadth of expectations were. What were the responsibilities of the business roles - It took a long time for the business to catch up with what was going on within IT Tools and delivery methodologies (Sprints and Iterations).	This investment will ensure the continued ability to track electric transmission and distribution outages, both planned and unplanned, allowing the control rooms to share information with the field and other required third parties. This will assist in running the electric network safely and meeting National Grid's regulatory requirements. The new web-based technology and components of the application will allow for easier sharing of information and ensure continued support, reliability, and security. This will provide National Grid with the future ability to connect directly with other Local Control Centres in the region who also use or will use the iTOA application. The new application will be housed in National Grid's Azure cloud that will provide increased redundancy, resiliency, support, and serviceability.	G185
4562	US SAP: Business Warehouse (BW) Consolidation	454,700	287,700	742,400	1. Scope, technical capabilities and compatibility matrix should be clarified before sanctioning. 2. Dependencies on other major programs should be reviewed and agreed upon with all stakeholders as early as possible.	This investment will allow the Finance Data Reporting Centre of Excellence to: Reduce Service cost for compliance support Enhance reporting capabilities and provide better support Avoid loss of productivity by consolidating support to one platform In addition, National Grid will: Reduce the cost of support for IT application maintenance team (Wipro) Reduce the number of servers used for ongoing services (running the business) Improve the efficiency for setting up new environments, by decreasing the number of servers used for future projects and, as a result, avoiding the related cost for Maintenance needs (patch, upgrade, restart) Monitoring needs Testing effort	G020
4987	SOE Windows Upgrade and Device Refresh	22,077,070	176,007	22,253,077	Work with budget holders and partners to ensure funding and resources are available for deployment and smooth execution. Engage Vendors to analyse full NG application estate. This helped in identifying incompatible applications and to recommend remediation as well as determine what activities were outside the scope of the project. Taking an more Agile approach to the project allowed the project to shift gears on how the devices would be delivered to the end user due to Covid constraints. Clearing and ensuring applications were Win10 compatible Making the business more accountable on business deployed applications	Provide a fully supported operating system for end user devices with functional benefits improving employee's effectiveness Mitigate cyber security risk associated with running un-patchable software Increase security from newer applications, as all the latest critical security patches and upgrades should be utilized in full across all devices in the network Refresh all laptops and desktops that are more than 3 years old with higher spec and more secure Dell devices	G020
5375	SAP/PowerPlan/Front Office Maintenance of Business (MOB) - FY20	376,000	77,000	453,000	This project managed enhancements for SAP platform to support various business teams. There is a need to improve a delivery time line make this delivery more efficient.	The benefits are expected to contribute to improved system reliability and business functionality, fulfil the organization's operating requirements and comply with regulatory mandates.	G020
5430	NY Green Button Connect	2,963,000	218,000	3,181,000	None listed	Key BusinessBenefits: 1. Regulatory Compliance –Solution build needs to be compliant with GBC standards and requirements per PSC mandate. 2. There are regulatory incentives for meeting the timelines for providing GBC compliant data –Benefit for the Customer rate decrease. 3. It is envisioned that GBC will also support the proposed National Grid Customer Energy Management Platform (CEMP) initiatives using AMI/smart meter data once AMI approval is secured. Key Customer Benefits: l Saving Money: understanding home energy usage and finding ways to reduce consumption and reduce bills l Insight: access to web portals that can analyse energy usage and provide actionable tips. l Heating and Cooling: customized heating and cooling activities for savings and comfort l Education: community and energy-efficiency competitions. l Retrofits: improved decision-support tools to facilitate energy-efficiency retrofits l Verification: measurement of energy-efficiency investments l Real Estate: provision of energy costs for tenants and/or new home purchasers l Solar: optimize the size and cost-effectiveness of rooftop solar panels	C113
5472	INVP 5472 GRC Archer Risk and Migration	2,629,000	676,000	3,305,000	Purchase orders raised in the US in particular, caused issues for business program team who were all based in the UK operating global roles – all costs need to be pulled together to get the true cost of the project. Should have had an understanding of all Statements of work and what is being agreed so these can be fed into costs and managed appropriately. Long running programs can experience changes impacting the design from outside the control of the program. This can be from changing strategic architecture, security standards, regulatory demands or even because new teams or individuals decide on differing solutions. Once the system is released it is important that users commence adoption as soon as possible so that training and decisions are fresh in everyone's mind. This is more efficient and can lead to less changes or challenges later	Embracing an integrated GRC process approach will provide a centralized method for gathering important risk and compliance data, conducting assessments and tests and most importantly, reporting to management the findings and overall risk and compliance position the company is currently facing in the most efficient way, empowering effective decision making. By replacing the outdated and manual processes with a central integrated approach, National Grid will significantly improve its ability to manage its risk and compliance obligations.	G020
5487	Collaboration & Unified Communications	1,379,000	844,000	2,223,000	Engagement: Customer engagement earlier on in the project would have made planning for Implementation easier. Daily stand up meetings with the vendor worked well for the team once in Implementation stage. Ensure full understanding of security requirements as agreed upon by IT Security across the Program. Understanding the impacts of SharePoint to SharePoint Online and Intelligent Communications from the onset was critical in aligning the financial forecasts and spend.	5487C: Collaboration Migration - Box& Huddle to SharePoint Online l Increase security l Single, unified strategic platform l Reduction of Box licenses l Potential to reduce operational costs l Increased collaboration and efficiency as SharePoint online allows for End Users to access their documents on any device on any network. l Improved End-User experience 5487E: Collaboration Migration - ICE Public Folders - Migration l Exchange 2010 platform is out of support and hence no security updates available l Public Folders migration is an enabler to decommission the ICE Platform as all other components migrated 5487F: Collaboration Migration - Legacy Device Refresh l Improvement of performance of the laptops enabled users to work more efficiently l Improvement of employee morale. l The user's frustration was reduced by the higher performance of the new laptops. 5487G: Azure Cloud Mail Routing & Exchange 2010 Decom l Upgrading the email environment to newer, supportable technology l Improves reliability and productivity l Enables a better Customer Experience l Reduction of RTB costs due to the decom of ICE	N012
5523	Voice Infrastructure Program	1,439,000	242,000	1,681,000	1. The AS-IS documentation needs to be maintained by National Grid and not by third-party vendors. There was no documentation for the control room phones within National Grid. It is recommended that new documentation should have all published listed for each control room. 2. Existing control rooms with Cisco phones must be treated like new control rooms which led to performing development and full user acceptance testing at locations. 3. Resiliency plans for control rooms using a call from home can have issues if jabber fails and the cell phone takes too long to ring in some areas. The recommended action is; for smaller control rooms we should implement Normal Call Flow and emergency call flow, Emergency Call Flow rings all jabber phones, desk phones, and cell phones immediately, and larger control rooms may need to incorporate cell phones into a normal call flow.	•Support more efficient use of National Grid Facilities •More effective collaboration •Increase end user experience •This program of work is aligned to the strategies and other efforts within National Grid •IPT Systems will afford greater potential to support integration of voice, data services and applications	G020

5524	WAN Infrastructure Program	1,410,000	130,000	1,540,000	1. As a collaboration with the facilities team was critical when scheduling on-site work in order to provide the survey team with site access. We learned how to and delivered a detailed schedule for on-site work while contacting the facilities team in advance to get approval to access the site. 2. During the Covid-19 Pandemic, HSOA's and operational activities access to many sites were restricted. When site access was not available and the team was put on hold, we discovered the criticality of developing a detailed plan to quickly get the team ramped up when restrictions were lifted. 3. When Verizon notified National Grid a site was completed, it was unclear as to whether the work being invoiced had been carried out in a manner benefitting the project scope. To resolve this issue, we required Verizon or any other third-party vendor to provide photographic and detailed evidence that the work was actually completed to its specifications.	•Support more efficient use of National Grid Facilities •More effective collaboration •Support of more device types in the environment •Increase of end user experience	G020
5605	US SAP: Infrastructure Landscape - FY21	2,158,000	48,000	2,206,000	The new process introduced in FY21 was successful in supporting all projects. The team should plan to use the same approach in FY23.	This project is intended to support projects by: I Reducing the lead time to start projects and initiatives within the portfolio I Increasing accuracy of cost estimates I Alleviating project startup bottlenecks I Increasing reliability for SAP related project delivery I Reducing one-time capex startup costs associated with standing up new environments for each project	G020
5610	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY22	959,000	72,000	1,031,000	HR Service Pack(HRSP) used automated Topology and Orchestration Specification for Cloud Applications (TOSCA) Health Scripts during cutover as standard health checks. To improve the benefits for the HRSP delivery, the testing team should connect with functional teams to define scripts in scope for the project instead of running all standard scripts. Improve coordination between SAP PI and the DXC teams to ensure that all load balancing servers used by SAP PI are up and running; and receive notifications of any FTP load balancer server issues to be able to correct them immediately	The project is intended to implement and comply with mandatory federal and state regulatory and legal changes. For example, new tax tables and any new changes to employer tax reporting are achieved through applying these HR support packs. The anticipated benefits of upgrading from current patch level to the new patch level or applying the HR support pack are listed below. Produce weekly, monthly and special payroll runs Ensure correct federal and state withholdings and legal reporting requirements Provide a more stable and reliable core SAP solution Reduce need for incident resolution and associated patches Provide an opportunity to eliminate and reduce custom code for changes Allow for faster SAP vendor resolution times for production incidents/issues	G020
5611	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY23	662,000	235,000	897,000	There were multiple projects sharing the development/testing environment which led to project delays and testing issues. When possible projects should use a non shared environment. Older/existing devices that will be used after an upgrade need to be evaluated to ensure that they meet the systems minimum requirements. Use of mock cutover plans should help define the actual length of time needed for production cutover.	The project is intended to implement and comply with mandatory federal and state regulatory and legal changes. For example, new tax tables and any new changes to employer tax reporting are achieved through applying these HR support packs. The anticipated benefits of upgrading from current patch level to the new patch level or applying the HR support pack are listed below. Produce weekly, monthly and special payroll runs Ensure correct federal and state withholdings and legal reporting requirements Provide a more stable and reliable core SAP solution Reduce need for incident resolution and associated patches Provide an opportunity to eliminate and reduce custom code for changes Allow for faster SAP vendor resolution times for production incidents/issues	G020
5631	Global Booking Mgmt (GBM): US implementation	-	202,000	202,000	Complete onboarding of 3rd party support team and deliver training before closing the project	The main benefits of this project are: Physical resource utilization: improved management of office space and increase resource utilization. Global Support: full IT support and incident management. Employee Enablement: providing greater booking management functionality, enhanced visibility of meeting room & car parking availability. Stakeholder Experience: enabling greater flexibility for users booking needs to allow them to selfserve and track their bookings;also introduces digital signage (floor plans) to help navigation and improve the flexibility of workspaces. COVID Compliance: I The solution supports National Grid stay compliant with COVID regulations (State and Federal levels).	G020
5722	KEDLI Rate Case (Mandated)	657,000	115,000	772,000	The team has implemented practices to re-enforce good coding practices and peer review checklists to eliminate errors such as the EAP rate error introduced into production over a year earlier and not caught due to lack of peer review of the code.	This project allows National Grid to comply with the Keyspan Energy Delivery Long Island (KEDLI) Rate Case which entails rate structure changes and billing process modifications. National Grid's billing system will automate billing calculations in compliance with the rate case.	5230
5723	KEDNY Rate Case (CRIS)	2,396,000	157,000	2,553,000	Due to changing business Priorities the project received multiple change requests which impact project scope, direction, and support. Technical design was impacted as the business requirements were not well defined and clarification was needed from the business on multiple occasion. Current performance issues required multiple data repairs as the business was not aware there were data issues until the project started analysis and design.	This project allows National Grid IT to comply with the Keyspan Energy Delivery New York (KEDNY) Rate Case which entails rate structure changes and billing process modifications. National Grid's billing system will automate billing calculations in compliance with the rate case.	5220
5774	Wifi Phase 2 - (Cancelled investment)	-	-	-	Projects will be more thoroughly planned to ensure sanction commitments are met	The key business benefits are to have: • Reduce risk of End of Life Equipment • Improved Performance • Improve end user experience through the delivery of Wi-Fi services • Supports more efficient use of the National Grid facilities • Rightsizing the LAN to support wireless and wired • Support of more device types in the environment • Increase mobility • Simplify User access management • Reduce risk & improve service performance • The pre-emptive fixes to field devices • Technology that allows IT to engage employees ahead of problems to repair issues • Ability to identify threat vectors ahead of vendor notification	G020
5820	US Personal Identifiable Information	-	276,856	276,856	Loss of the Project Manager towards the end of the project lead to a delay in the closure activities of this project. Improvement/Lesson Learned: Ensure ongoing actions are actively reassigned and actioned going forward.	The benefits of this investment are: I Reduction of the risk of a data breach and resultant notification costs I Implement any additional security controls required to mitigate identified data privacy related risks I Provide customer data protection through data lineage and data classification I Ensure adequate organizational competencies and capabilities to support current and targeted operational models I Improve our compliance posture with mandatory regulatory requirements I Remediate data design flaws to improve any ineffective business processes In addition, National Grid will: I Establish and design the processes and technologies necessary to nurture a culture of enablement and transparency through data governance and privacy by design I Standardize data intake, data flows, data dictionaries, and metadata management processes to enable more consistent usage and interpretation of its data assets I Manage a governance committee and data management team to drive key compliance and data privacy initiatives	G020
5836	Lockbox Payments and Credit Payments Implementation	941,000	107,000	1,048,000	Customer payment processing time will be improved. Vendor's service is also projected to encounter less exceptions and quicker processing of errors when they arise, improving customer experience. Savings will be determined when the requirements and design (R&D) phase is complete.	Stay engaged with the Business team as their priorities can change at any time and impact project scope, direction, and support. MuleSoft has a known limitation of 1GB for file transmission. This limitation should be communicated early in the process to the project team. Which would result in last minute design changes for the project. Volume testing should be conducted as early as possible during the testing phases thus avoiding potential abends in the final cutover processes	C175
[redacted]							
5874	Cloud Centre of Excellence	2,117,000	1,587,000	3,704,000	1. Lesson: Need to be more pro-active in scheduling Pen Testing, because late timing caused delays on deliverables. 2. Lesson: Work closer with the security team on deliverables and issues.	Key business benefits of this are : Reduction in time needed to deploy services National Grid will have cloud native and DevOps capabilities in house Will provide cloud-native security and service management framework to assist National Grid stay in control of its services Faster time to implement applications Increased options available to support new services Improved reliability of services enabled through Microsoft assisting National Grid with the implementation of the Azure DevOps process and tooling capabilities needed to achieve the Cloud Centre of Excellence engagement scope	G020

5876	US Applications – Infrastructure & Framework Upgrade	1,674,000	46,000	1,720,000	In case of application rewrite, perform one round of code scan on the current application code to find the list of open vulnerabilities to compare against the re-written code. ⍰ Should be done either prior to the start the project or at an early stage of the project Capacity planning for storage, software licenses should be considered by evaluating both current and future needs. Perform an additional integrated round of testing during the development phase itself, to minimize the System Integration testing/User Acceptance testing (SIT/UAT) defects.	These critical applications will be under vendor support for the software, database and hardware components for better supportability ⍰ Flagged security vulnerabilities will be remediated by being able to apply the latest security patches ⍰ Provide better turn around times to recover from incidents in case of issues and reduce business downtime impact ⍰ Implement automated monitoring and alerting using the latest tools	G020
5905	Nucleus (Customer Digital Solution)	10,542,000	-	10,542,000	Proactively identifying product dependencies, contingency plans and short term solutions helped for on-time monthly production deployments. Product roll out plan was successful with improved communications, on-time delivery of core features and strong support from change champions. The highest quality product feedback came when customers used their real data in the product. Learnt to build in more time in between code freeze and security testing to avoid last minute scrambling before releases. Increased transparency into 3rd party developer resourcing and work assignments could have led to rework	Capacity generated savings: I Capacity creation for Account Managers and Energy Efficiency sales reps. Direct cost savings: I Reduced call centre volume, postage costs, lockbox fees, and collection activity Additional benefits: ⍰ Customer satisfaction – increased transparency, reduced time spent on billing and payment process, digital adoption will increase customer ease and satisfaction ⍰ Improved data quality- collect data to build an 360 degree view of the customer to: ⍰ Deliver personalized solutions at the right time ⍰ Provide deep insights on customer business ⍰ Increase awareness of value added services. ⍰ Technology platform – basis for future digital initiatives	C175
5912	On My Way	11,782,000	326,000	12,108,000	Engagement and alignment with systems that we depend on (STORMS system etc) even earlier on. We tried to engage with this team early on, but the timing did not work out: they were experiences transitions on their team and did not have the capacity to support our needs. Looking at broader roadmaps and more explicit alignment will benefit future additions to OMW. Documentation – budgeting more of my time towards documentation instead of doing it in between other responsibilities. I know that some documentation lingered because once the roll out began it was difficult for me to find time. Shift security assessment early in process	Capacity generated savings: ⍰ Expected savings will be generated from electric line workers, trouble workers, supervisors, and clerks having more capacity to increase workload throughput and potentially avoid contractor spend Direct cost savings: ⍰ Savings from IT (replacing obsolete applications), paper, and vehicle fuel and maintenance expenditures Additional benefits: ⍰ Customer satisfaction – transparency of work progress will increase customer ease ⍰ New technology platform – enable future digital initiatives to be implemented to further improve business processes ⍰ Improved data quality – leveraged across the EBU for improve analytics, performance mgmt., and planning	G198
[redacted]							020
5999	Worcester AMI SaaS Upgrade	-	853,000	853,000	National Grid changed direction to using a new interface tool MuleSoft, while the project was starting up, the project needed to redesign the integration modules to accommodate this new tool.	This upgrade will ensure the Worcester AMI meters continue to communicate effectively for meter reading and customer billing. In the event of any issues, this upgrade will ensure we are still under supported vendor products and associated service agreement(s). This upgrade will also align both the Worcester and the Clifton Park SaaS software platforms allowing for more standardized employee training and knowledge sharing	5310
6007	Inside Agent Remediation	884,000	133,000	1,017,000	The project didn't properly identify the amount of work and time it would take to decommission the old servers and application. Also, good succession planning for key project team members could help reduce the impact of productivity.	This project will address the non-compliant application and ultimately it will be retired. I Replacing Inside Agent with MuleSoft/GoAnywhere will reduce risk associated with transferring data. I MuleSoft/GoAnywhere will provide a single platform for security, full-lifecycle API management integration and end-to-end monitoring.	C165
6015	Gas Application Continuous Improvements	2,833,000	1,000	2,834,000	Follow up on the invoices with the vendor as the delay in getting them can delay the project closure and increase the risk for charges to come into the investment incorrectly	Improvements to usability, scalability, data integrity, and application business logic for gas applications.	C210
6046	INVP 6046: SDWAN Expansion	4,655,000	2,000	4,657,000	1. Communicate more frequently with vendors to stay on top of invoicing. 2. Extensive pre-installation site readiness survey. 3. Post implementation photos to confirm site is left in acceptable operational conditions to eliminate additional site visits to resolve any issues such as open cabinets.	Centralized orchestration and policy control (leading to faster provisioning of network policy changes). Better performance for cloud-based applications. Better visibility into applications traversing the network (SD-WAN recognizes over 3000 applications by name through Deep Packet Inspection (DPI)) SD-WAN continuously monitors the performance of each WAN transport (Multi-Protocol Label Switching (MPLS) or internet) and routes traffic over the appropriately performing transport, based on application policy (latency, jitter, packet loss).	G020
6051	STORMS Extended Performance	1,174,000	37,000	1,211,000	The level of detail in the project plan included comprehensive steps for the Engineers and Data Base Administration teams to complete activities accurately and on time. Recommendation: All project plans should get to this level of detail. Project schedule had to be adjusted multiple times due to other project teams using the same test environment. Recommendation: Stand up more than one end-to-end test environment to ensure that one project does not impact another project.	Business will have improved performance and stabilized applications by enhancing the performance of the applications and middleware for STORMS and ARMScheduler to ensure benefits from the systems are realized	G160
6114	Managed Workspace Services Transformation	-	2,178,000	2,178,000	•Improve scope definition for future End User Experience (EUX) transitions (split transition and transformation more clearly) •Take more time planning the program timescales	This infrastructure implementation reduces technical risk and increases uptime capabilities for end-users to ensure end-customers receive the level of service expected.	G020
6196	I&O US Operations FY22	7,595,000	636,000	8,231,000	A minimum cadence of monthly cross charging between regions is important to maintain accurate positions of forecast against budget I The difference in tracking methods between the two regions makes it challenging for the budget owner to switch between formats I Weekly review of the budget lines keeps the forecast in order	⍰ Improve End User Experience ⍰ Improve equipment performance ⍰ Improves reliability and productivity ⍰ Better customer experience ⍰ Less Support request ⍰ Automated maintenance ⍰ Centralized monitoring ⍰ Single delivery toolset ⍰ Disaster Recover	G020
6197	I&O US Networks FY22	20,778,000	1,186,000	21,964,000	The age of documentation and estate mentioned was considered for future investment planning. Issues with lead times on delivery of equipment led to more tactical procurement and adjustment of priorities within this year, and accounting for planning for subsequent years.	The work completed during the release train will enable future requirements for Corporate network modernization Improvement of the end-user experience by utilizing subject matter experts to deliver an updated design for Verizon services that improve the operation and delivery of the Verizon services at a reduced cost This work will be a driving force to ensure we are meeting our IT Focus Areas and other efforts within IT National Grid Supports more efficient use of National Grid facilities by improving delivery and resilience of communications across sites	G020
6199	I&O US CNI FY22	2,069,000	493,000	2,562,000	1. Scope changes are a factor in Agile team delivery. As we sanction for a full year of work, we have learned to be better prepared through intricate planning to avoid re-sanctions which is what happened in this investment. 2. Initial team forming took longer then expected. In turn, this better prepared for kicking off FY23 delivery under the new Agile delivery framework.	Reduction of assets that are end of service life within the National Grid CNI estate Improved vulnerability management resulting in reduced security issues Enhanced regulatory compliance reporting Enhanced system security by swapping legacy devices out for new kit Increased operational availability and vendor support Enable abilities and increase capacity to meet demand	G186

6200	I&O US Cloud and Hosting FY22	6,694,000	5,900,000	12,594,000	1. Require better planning for data transformation effort (Hosted Estate Modernization), which work was not completed in FY22, requiring this to extend to FY23 scope. Improvement to effectively plan fo+K39r road bumps and coordinate closer with Service Owners. 2.Worked directly with Service Transition to adjust to Agile Delivery, from change management to release management	Key business benefits of these efforts are: Cloud Modernization - Ability to leverage market investment in commoditized cloud services - Improved resilience from public cloud through enhanced deployment models. Standardized technologies and processes for private cloud - Creation of secure public cloud service and development of in house capability to efficiently manage and operate the public cloud infrastructure - Automation enabling National Grid to scale and absorb new workloads and contain costs in provisioning and decommissioning Hosted Estate Modernization - Migration of applications to appropriate internal/external hosting environment - Enhanced agility and flexibility for workload deployment supporting digital initiatives - Upgrade of internal infrastructure including server infrastructure, private cloud and associated storage to ensure that it is secure and fully vendor supported - Improve service availability through implementing new a back-up service based on disk rather than tape-based storage Active Directory Modernization - Delivery of up-to-date Active Directory infrastructure using latest software version and security standards - Increased security - Increased reliability with long term support by vendors - Delivery of stop-gap solutions to Legacy Active Directory to secure NG while implementation of Global Active Directory takes place	G020
6219	US F&BS Minor Enhancements (Non-SAP)	-	157,000	157,000	None listed in Closure Sanction Paper	The requests worked under this project are expected to contribute to improved system reliability and business functionality, fulfil the organization's operating requirements, and comply with regulatory mandates.	G020
6222	PIF Replacement	609,000	49,000	658,000	Lower WIPRO MuleSoft resources costs/time utilized than earlier estimated. Recommendation: PM to work with MuleSoft development lead to agree on more accurate resources cost/allocation prior to the project initiation.	The MuleSoft Runtime Fabric on Azure managed by National Grid solution doesn't represent a Minimum Cost Alternative solution, however it is still in the best interest of National Grid as it: Reduces security risk Reduces complexity for maintenance and support	X148
6231	US Treasury: Allegro Upgrade Phase 2	268,000	26,000	294,000	1. Azure scale-sets product (VMSS) was not production ready while this project was in-flight. The project team had to request several enhancements / new code to support Azure DevOps requirements. In the future, the non-business requirements, such as infrastructure details, should be gathered in the startup phase. 2. Global AD (GAD) was standing up a new domain architecture while this project was in-flight. This created a stronger dependency on the GAD team than anticipated and led to some delays and issues finding the proper resources for specific requirements. While it added complexity to this project, Active Directory team gained expertise will be benefiting the future projects. 3. Good collaboration with the various teams, especially with CCoE, Global AD and IT security, allowed team to deliver on time and under the budget.	•Cost Reduction: Implementing Azure scale-sets in the Allegro Horizon system will have a positive impact by effectively reducing costs in a pay-as-you-go model. System resources will become available based on a predefined usage forecast or as needed, per user demand. •Maintenance Simplification and Efficiency: For optimal performance, Allegro architecture employs the use of multiple VM application servers to process load from user requests and backend jobs. The transition to Azure scale-sets will enable the use of server images as templates that can be applied to VMs. Application testing, configuration changes, OS patch application, etc., can be done once instead of multiple times	G175
6320	Telecom & Network Deep Dive	-	180,000	180,000	Project Management: Deliverable framework, including read-out status, should have been reviewed with Steerco early on. Team Work: Business and IT partnered and worked well.	This investment will provide the detail to pursue network transformation and optimization opportunities, including potential strategic sourcing events. Also work to reduce and/or eliminate underutilized or unnecessary services.	G020
6377	US Group Functions Operational Req	908,000	336,000	1,244,000	By consolidating multiple small investments, National Grid was able to reduce the number of sanction papers, forecasts, and stage gates which saved on labor costs. The consolidation of these projects also allowed National Grid to capitalize build and test activities and manage delivery in efficient manner	The requests worked under this project are expected to contribute to improved system reliability and business functionality, fulfil the organization's operating requirements, and comply with regulatory mandates.	G020
6407	Online Marketplace Replacement	-	51,000	51,000	Make sure Requirements are clearly documented. Was given this project with no clearly defined Requirements and it was a challenge.	1.Enhanced *eCommerce platform* – provides customers an enhanced shopping experience as well as the ability to cross promote other relevant energy efficiency programs 2.Personalized experience – uses customer data (through single sign on) to customize the user experience with data-driven recommendations. 3.Engaged customer relationship – makes it easier to do business with us by providing a seamless digital experience (single sign-on) and builds trust as their energy partner by providing valuable information and options. 4.New revenue streams and savings – increases earning opportunities through smart bundling, partnerships, cross promotion and advertising, while increasing adoption of services to drive efficiencies and savings.	C170
[redacted]							
6427	RI Purchase of Receivables (POR) Program	727,000	96,000	823,000	Engagement with the business team on their multiple priorities and required support is imperative to avoid negative impact to the project scope, direction and support. Confirmation that the business has access to designated testing environments earlier in the project timeline would eliminate delays. Confirmation that the required EDI batch jobs and transactions should be available in selected testing environments at the onset of testing activities would avoid negative impact to timelines.	- Compliance with RI PUC - Ability to process POR payments accurately and efficiently for ~58,000 customer enrolled with 32 suppliers - Number of ESCO customers expected to increase to over 100,000 with the implementation of aggregations in RI - Will prevent the billing business from making a large number of manual calculations and payments required for monthly payments to suppliers	S36E
6542	New Gas DR Solution Evaluation	-	76,000	76,000	Create clear expectations with team members, specially when they are not involved daily in the project, so there is accountability on the deliverable and the timeline.	This investment mitigates the risk that NG cannot continue offering its DNY GDR program and would be unable to balance gas supply and demand on peak days by providing a thorough assessment and recommendation(s) for a new GDR hardware and software solution to potentially replace Enel in 2024 or 2025.	C22S
6618	Backoffice Discovery Refresh	-	696,000	696,000	There are no Lessons Learned for this project	As a result of this effort, the following benefits should be achieved: - The Discovery Roadmap Refresh will provide timing, sequencing, staffing considerations, costs and benefits associated with the US S/4 business transformation - The Project team will leverage the lessons learned from the National Grid UK S/4 implementation and incorporate these themes into the project deliverables - The Discovery Roadmap Refresh will consider early value opportunities throughout the US S/4 business transformation - Provide the path for future S/4 HANA transformation implementation The work will conclude with an Executive Alignment session to review outcomes from Discovery Roadmap Refresh and seek approval to move forward with the next phase - Enterprise Data Model design.	GT53
1002X	In Demand Energy Efficiency Enhancements CY2020 Closure	-	2,862,000	2,862,000	Review estimation / work load, Plan for separate BO scrum team. Holistically plan for the remaining work on LCI merger. A template for business users to help better defined the requirements. For SBS - engage business Program Managers on time to have acceptance criteria defined	The Indemand system supports implementation, audit and reporting of Energy Efficiency (EE) provided by National Grid to its customers as mandated by state regulators. The system spans commercial, residential, gas and electricbusiness segments to allow support of EE programs in the National Grid regions.	C17S
1002Y	InDemand Energy Efficiency CY2021 Release Enhance Closure	-	2,931,000	2,931,000	Need planning for addressing the parallel initiative of LCI merger in order for resources to be available for BAU work. Need to have good visibility of prioritized work to avoid overtime, need planning for addressing the parallel initiative of LCI merger in order for resources to be available for BAU work. Review other estimation techniques.	The InDemand system enables implementation, audit and reporting of Energy Efficiency (EE) programs provided by National Grid to its customers as mandated by state regulators. The system spans commercial, residential, gas and electric business segments to allow support of Energy Efficiency programs in all the National Grid regions	C17S
3683USF	Enterprise Centralized Patch Management	210,000	126,000	336,000	Opportunities for Improvement:- Project costs could have been transferred to the follow on project earlier allowing for the earlier closure of accounting codes and hence overall project closure to be undertaken earlier.	None documented	G020
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3870A	Vision FM & Property services replacement	3,757,000	226,000	3,983,000	An Agile project requires that the Product Owner is defined by the business prior to the project starting to make overall decisions. Avoid scope creep by ensuring that the Minimal Value Product (MVP) is clearly defined as to what is and is not included Gain more exposure to the various processes used by other departments by levering knowledge through the Product Owner	1. Introduce one common application for Facilities Management and Real Estate departments 2. Vastly improve Facility Management environments in terms of reliability and efficiency 3. Streamline property records management with an operationally efficient workflow and improve process accuracies 4. Modernize GIS platform 5. Improve ease of access through consolidation of systems	G020
4418USA	Archer Optimization	-	22,000	22,000	None listed in Closure Sanction Paper	This investment will have a positive impact on the ability to complete the daily settlement and reconciliation process, as the automation of the load data will mitigate the risk of manual error and shorten the time required to complete the tasks associated with the process. By consolidating the two platforms into a vender supported application we will insure a disaster recovery plan that is site tested for the NY Wholesale Energy Settlement Market and improve performance. Stability and availability of a key business critical application. Elimination of manual data transfers from the MV-90 xi system. Eliminate manual comparisons and aggregation calculations between the respective ISO's data and National Grids data.	G020
4704R	NY REV Street Lighting Delivery	1,676,000	232,000	1,908,000	Delivered alignment to regulatory requirement of the NIMO rate case and building existing technology used in the current OMS application with the ability to visually monitor the operational status of the respective locational-based street light inventory. NYSL introduced a quick and easy method for customers to identify and report outage/maintenance issues while making requests for new or other related lighting services. The direct and automatic creation of outage investigation orders improved responsiveness without incurring new labor cost. Enhanced the customer's ability to obtain, monitor and review billing inventory information timely and on-demand. Eliminating the need to maintain tariff-mandated, paper-based schedules and other manually generated reports.	Benefits to National Grid and the customer: Deliver alignment to regulatory requirement of the NiMO rate case; Building on similar technology used in the current OMS application, the ability to visually monitor the operational status of the respective locational-based street light inventory; The introduction of a quick and easy method for customers to identify and report outage/maintenance issues while making requests for new or other related lighting services; The direct and automatic creation of outage investigation orders, improving responsiveness without a need to incur additional labor cost; Enhancing the customer's ability to obtain, monitor and review billing inventory information in a timely and OnDemand basis, eliminating the need to maintain tariff-mandated, paper-based schedules and other manually generated reports;	S21E
5471A	ADMS Grid Modernization	26,988,385	6,615,721	33,604,106	GM-LL0078: Without access to the vendor's system (ABB/Hitachi ADMS system) it was hard to know what data elements that was needed to make it work. The documentation from the vendor was good, but actual hands-on would have been better for data requirements. The team should have planned time for the vendor to complete the build of their system, then allow ADMS team to access it for data load analysis. NOTE: Getting a test system ahead of time is helpful and holds true for any major systems. GM-LL0077: Test Procedures took longer to execute because the team did not have access to the system when the procedures were created and reviewed. Lesson is to insist on access to a working system during the Test Procedure creation and validation, well in advance of the Testing phase beginning GM-LL0075: During initial communications to the Control Center(CC) operators should make sure the level of information is clear. Reach out to the CC Directors to let them know when high level communication is sent, and when more detailed communication will be sent out. Manage the recipient's expectations on what level of detail is in each communication. GM-LL0030: Introduce change impact assessment approach earlier in the effort to reduce risks at the end and avoid re-work. And socialize overall change management validation approach earlier. GM-LL0026: In person engagement from all 4 control rooms during Business Process Design (BPD) workshops was extremely crucial to the success of our work products and the in person visits/tours of the 4 control rooms benefited not only the BPD team, but also the change management and training team.	Strategic Objectives: Affordability • ADMS will add a level of efficiency to planned and emergency feeder reconfigurations by the ability to simulate conditions days ahead. Optimizing switching solutions, as well as optimizing asset utilization. The complexity of the grid is expected to significantly grow as the volume of DER resources rise to meet the needs of our state renewable policies, and these efficiencies will help offset increases in resources need to maintain the safe and reliable operation of the grid. Clean Energy • Monitoring, analyzing and incorporating operational data from distribution devices into centralized advanced applications such as operations load flow, the ADMS will support the improved management of the distribution system which will support greater operational integration of DERs Operational Objectives: Reliability • ADMS Restoration Switching Analysis will support quicker restoration by offering expedited solutions for alternate switching configurations for use by system operators. • ADMS will be capable of performing advanced Fault Location Analysis thereby reducing restoration response time and improving overall reliability Safety • Enhanced monitoring capabilities can improve safety to employees and the public through faster location and response to abnormal conditions and short circuit analysis Regulatory Alignment • ADMS strongly aligns with the regulatory goals across all three US jurisdiction.	T198
5471D	Grid Modernization Data Management	3,291,000	1,082,000	4,373,000	GM-LL0160: Engage a security consultant during the start of the project and secure an exception if the project team needs unmasked, production data for development/testing work in lower environments. GM-LL0177: Maintain a Source Reference sheet including Source Type (Oracle, SQL, File), Frequency (weekly, monthly, on-demand), and Source Refresh/Load schedule. GM-LL0179: To avoid delays due to the non-availability of key project resources, ensure backups for each key role. GM-LL0180: Submit a "Use of Live Data Request Form" for Security approval when its necessary to use Production data in non-Prod environments.	This project will enable the following benefits: 1. Improve data quality for identified critical data elements after data quality profiling and source system remediation as applicable. 2. Provide business support for better alignment with National Grid's Data Management, Business Management System standards 3. Heightened business partnership and awareness around Data Management practices needed for increasing digitization across National Grid 4. Reduced SME time to consolidate data, validate, combine and analyze data for business processes 5. Enable the business Asset Management Maturity ambitions through improved understanding of data quality, challenges and remediation progress	T198
5471E	Grid Modernization Enterprise Integration Platform	3,609,000	1,009,000	4,618,000	GM-LL0148: Currently GIS and ADMS teams are setting up a new environment while Mule is existing platform. This is not allowing end systems to carry out test cases of TO-BE production with Mule Prod. This is mitigated by advancing the Mule interfaces Go Live accordingly GM-LL0147: By bringing critical approvers to a single call, will help save the time and effort of multiple members and improves productivity through effective review. Lesson learnt from SVD and CSA preparation. GM-LL0146 : By involving decision makers in discussions, rework can be reduced resulting in more productivity and to bring everyone on to the same page by sharing first hand information. GM-LL0145 : To enable platform side activities to start early to provide required customized features as per project requirement. Results in reducing last minute dependency.	Apart from delivering a core enabling infrastructure and services across the Grid Modernization landscape, this project will provide the following benefits towards ongoing integrated, secure, scalable operation of Grid Modernization business capabilities: · Provide agile business process & information orchestration across systems covering variety of scenarios such as Application to Application (A2A), Business to Business (B2B) & Business to Consumer (B2C) integrations. · Enable Model Driven Service based approach focused on reusability and high availability. · Enable a Secure Gateway for integrating external services rendered over Cloud or by third parties which are outside National Grid network. · Improve user productivity by leveraging and extending services appropriately utilizing right protocol and integration standards (e.g. lightweight APIs for Mobile and RICH Web consumption) · Provide End to End traceability and audit capability and alerting for operational efficiency.	T198
5471F	Grid Modernization GIS - Phase 1	4,412,000	849,000	5,261,000	GM-LL0122 : Lack of repository for Small World application - this is an existing application however there was lack of documentation i.e. test scripts, list of interfaces and details thereof GM-LL0141:Business Process documentations like Workflow, Job aid, Best Practices etc. were not available on the business side. GM-LL0172 : SOW with GE, there are no penalty clauses. For example, in case defect is not fixed within "x" duration there is no bearing on the vendor. This was evident in resolution for multi merge issue which could have been resolved faster. GM-LL0173: In the SOW with GE, Acceptance criteria for Milestones is not spelled out in detail, which leads to issues when milestone is completed and misinterpretation of "done" between vendor and NG. GM-LL0174: With offshore teams, need to have a onshore representation to mitigate the risk like COVID which can severely impact many team members at once. GM-LL0187: Vendor resources often relied on NG team to have all the knowledge of the product and customizations whereas the benefits of vendor services was assumed that they will bring the knowledge and guide the project team. This resulted in many defects and also longer time taken to make the necessary changes. Also, the resources were continually changing during project execution leaving NG to bring each resource up to speed and investing more time/ effort. Resolution: Ensure vendor resources understand expectations and are prepared to be aware of the NG customizations and guide the project team on how some changes can impact the functionality. GM-LL0188: Sprint planning as done without participation from NG. The priorities set forth by NG were not considered during Sprint planning. Involvement from NG is important to ensure project delivery (by the vendor) meets the customer (NG) expectation.	• Implementing the GIS data enhancement and Optimization project will enable network models to be developed consistently and efficiently for distribution system planning, hosting capacity analysis and operations utilizing the ADMS. • By providing accurate data with the appropriate level of granularity, the maintenance of system modelscan be further refreshed more frequently to provide more timely and accurate system assessments. These efforts are foundational to enable the desired granular management of the grid envisioned in the Grid Modernization Program.	C198
5471XA	DERMS Implementation Roadmap	-	569,000	569,000	Preparatory sessions for stakeholder workshops and presentations were useful in conducting efficient and quick reviews with business stakeholders.	The DERMS Implementation Roadmap shall provide the direction and timeline for implementing the required DERMS capabilities over the course of multiple years. This is a significant planning exercise that will guide the setup and sanctioning of future projects within the DERMS program. The roadmap will help direct the funding, resourcing, delivery management and risk management actions across the DERMS program and help manage interdependencies with other major programs.	G198
5471Z	Short Term Forecasting POC	-	312,000	312,000	There was no NG reference documentation available on the current Short Term Forecasting process, describing the details needed. In the future, having groups provide this information up front will speed learning and time spent understanding the process and calculations needed.	☑ Help National Grid resources to understand the complexity to train forecasting models. ☑ Ability to improve forecast model accuracy through continuous AI (Artificial Intelligence) based training. ☑ Help to identify data accuracy and dependency with W either and Scada data. ☑ Ability to identify security risks and define mitigation plans for an enterprise scale Forecasting solution implementation.	S21E
5487G	Azure Cloud Mail Routing & Exchange 2010 Decom	-	74,000	74,000	Testing of applications by Service/Application Owner was completed in a prior project 4491, 12 - 18 months prior. When it was time to implement the cutover from legacy platform to the cloud, the service/application owners requested another testing effort. The 3rd party service/application owners were able to confirm the testing within a 2 month timeframe but internal Service/application owners needed more time to complete their testing. ☑ Review previous project work to determine if extensive testing or regression testing is most applicable. Security sign off on the project was required for closure. The remaining task was to conduct an external pen test of our production environment for email. ☑ Ensure required resources are available to complete the work within the required timeframe.	☑ Upgrading the email environment to newer, supportable technology ☑ Improves reliability and productivity, removing technical risk from the estate ☑ Enables a better Customer Experience ☑ Reduction of RTB costs due to the decom of ICE	G020

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5785I	NRG GIS Replacement Project	3,457,000	40,000	3,497,000	Performance baseline between the new and old platform should have been done in a better way. Team encountered quite a few issues during User Access Testing. Ticketing and issue tracking from business could have been managed better.	This is a tactical investment to bridge the gap for an unstable platform until full implementation and roll out of GBE. At that time this solution will be decommissioned. The investment will convert and move data to an enterprise strategic format and platform and discontinue use of an unsupported platform. There will be savings as the infrastructure for NRG is decommissioned and the business	G225
5787G	Gas Demand Response F&A	-	141,000	141,000	1. Develop and document a standardized RFP process for IT projects. This will help train IT staff on the RFP process and help them more effectively plan RFP projects and more accurately estimate IT resources, time, and cost to conduct RFPs. 2. Engage NG's Procurement team earlier in the RFP process to secure Procurement resources for IT project RFPs and get their input on RFP plans and timelines.	Gas supply and regulatory constraints have increased the importance of GDR programs to meet peak customer demand without expanding gas infrastructure. This F&A project will identify, evaluate, and recommend 1-3 solutions to be piloted in a future project that can improve the Company's end-to-end GDR processes while supporting its current and future GDR programs to safely and reliably operate the gas network while meeting customer demand	G210
5788D	Interface MyHub To OnBoard LMS OQ External Site	571,000	123,000	694,000	The design option to implement MyHub to Onboard LMS were not thoroughly vetted by MyHub/SAP team. Recommendation: MyHub/SAP team should have performed enough internal due diligence before presenting solution options to Domain Architect for approval. Security requirements changed midway into the project and that required rework. Recommendation: As National Grid gets more and more security conscious, these requirements will get tighter and, therefore, prior to security rules/requirements being updated for National Grid systems, the full impact of such changes to in-flight projects needs to be evaluated.	Key business benefits are: Digitization of Business Processes: ⍰ Builds employee information interface between SAP/MyHub and Gas Operational Qualifications (OQ) certification provider (ITS) Accurate and Consistent Data Stream: ⍰ Eliminates errors due to manual data processing and transfer Secure Data Transmission: ⍰ Migrates data transfer from unencrypted email-based Excel transfers to a secure and encrypted mechanism Reduced Business Cost: ⍰ Eliminates the need for managers and business analysts to manually process data, when they could be performing more productive work Better Regulatory Compliance: ⍰ Reduces fines and re-work due to work performed by Field Technicians with expired certifications.	G210
5841c	401(k) Administration Changes	836,000	80,000	916,000	Lesson 1: Scope Management – What we could do better: The connectivity testing with the external vendor did not consider exact steps during Testing as it would be in Production. The Test setup was different than in Production and the Testing team was not aware of this. As a result, the connectivity failed during go-live. We had taken next 30 days to set it up correctly and test again. All 30 days the files were exchanged manually. Recommendation: It is important to ensure that Test environment is setup exactly as in Production and THEN verify it. Lesson 2: Business Case & Benefits Management – What we did right The team discovered a complex defect during user acceptance testing. The defect, which was not related to the changes made by the project, was subsequently fixed by the team. By fixing the issue we made sure the promised business benefits were not compromised. The team was able to discover a complex issue outside of the scope and resolve it as part of the delivery. Recommendation: Always look at a long term benefit for the overall product, not just what is in scope for an individual project.	Benefits the Company and participants will reap by the implementation of the approved plan design changes: 1. Plans Consolidation: Merging the two 401(k) Plans into one will simplify and provide efficiencies in plan administration for both National Grid and Vanguard, and in return, Vanguard has offered to pass those savings onto participants. This equates to participant savings of 30% year over year. ⍰ Streamlines participant data transfer process with employee job transfers (union to management and vice versa) and eliminates manual process by Vanguard and improves efficiency through automation. ⍰ The Summary Plan Description is combined to one master document using appendices for match formulas for each employee grouping. ⍰ The weekly inbound participant election file will be combined to one file. ⍰ Eliminates the need for a Master Trust on the stable value fund and required filings for the Master Trust. ⍰ Labor efforts for Compliance and mandatory filings will reduce costs. Elimination of one Plan audit, two form 5500s (one plan and the stable value Master Trust) and one financial statement. Thereby reduces auditing, financial statement compilation and tax preparation costs by approximately $35,000 per year. ⍰ Non-Discrimination Testing is conducted on one Plan and reducing the chance of plan failure. ⍰ Vanguard adhoc reporting becomes more efficient and timelier. 2. New Features: The Roth 401(k) is a feature that has several potential benefits. The Roth 401 (k) was first introduced in 2006 and has grown in popularity since then. Currently, 74% of Vanguard clients offer the Roth 401(k) feature. Adding this feature provides employees with another way to save for retirement within the 401(k) plans in a tax-friendly way, allowing participants to prepare better for retirement and taxes they will have to pay in retirement. It also provides new hires into the Company who have a Roth 401(k) at another employer the ability to move their Roth 401(k) savings into the National Grid 401(k) plans. This feature is not currently an option, causing some concern among new hires. The Services Delivery Center consistently receives inquiries from new hires if a Roth 401(k) is available allowing them to rollover money from previous employers. Adding the Roth 401(k) feature to the National Grid 401(k) plans is another step in meeting the needs of our multi generational workforce and keeping	G020
5842B	Mandatory FERC Form Changes	431,000	186,000	617,000	Engagement with National Grid End Users, relevant IT vendors, Operational Finance and Controllers during requirements ensured the requirements were captured correctly. ⍰ Representation from stakeholders across all departments/jurisdictions during the testing phase allowed for extensive feedback, resulting in minimal issues during service transition. ⍰ An effective training plan that was established and signed off prior to development ensured great end user participation	1. Avoiding Risk of Non-Compliance I Inability to meet reporting requirements can cause daily fines of up to $1 million 2. Following BMS Standards: IT Support I Implementation can help ensure appropriate internal and external support. I The Company can provide the appropriate resources and support to internal users including troubleshooting, training and access verification (downloading software for end-user). I Reporting teams will no longer need to allocate time and resources to the maintenance and up keep of the current filing format, Visual FoxPro, which is no longer supported by the original manufacturer. 3. Increasing Value: I XBRL provides consistent structure of data ideal for financial information by making it more easily consumed by both the FERC and data users. I Currently, to analyze FERC data, consumers must access one report or download a series of files. Structured data will allow greater comparison and reporting. I XBRL Tagging allows for data to be validated before it is file, improving quality, reducing error and driving consistency. I XBRL format can decrease costs and improve efficiency by following a uniform standard. I XBRL reporting aligns with international and domestic adoption of structured data by both industry and regulators, such as the SEC and EU Inline XBRL mandates. I Enable future improvement and automation of the reporting process. I Can be integrated, or even drive, alignment with the expected future state of the reporting process.	G020
5843B	US SAP Platform Upgrade: Minor Releases	368,000	95,000	463,000	Engagement with National Grid End Users, relevant IT vendors, Operational Finance and Controllers during requirements ensured the requirements were captured correctly. ⍰ Representation from stakeholders across all departments/jurisdictions during the testing phase allowed for extensive feedback, resulting in minimal issues during service transition. ⍰ An effective training plan that was established and signed off prior to development ensured great end user participation.	Requests worked under this project are expected to contribute to: ⍰ Improved system reliability and business functionality ⍰ Fulfill the organization's operating requirements ⍰ Comply with regulatory mandates.	G020
5843D	US SAP Healthcheck: Upgrade Critical Components - FY22	1,404,000	514,000	1,918,000	Develop a standard set of tests that contain data, insight tasks and recurrence schedules that will serve as benchmarks to test the stability of the system. Completely resolve pending issues before moving on to cutover. Ensure that the vendor provides adequate support for all issues encountered, as tickets were hung for days without response, requiring escalations and additional time to resolve.	Staying complaint with National Grid Policies, Digital Risk and Security Policies and Procedures. ⍰ Adhere to the defined asset management lifecycle policy and usage policies for IT assets to avoid jeopardizing the platform which falls under SOX compliance regulations. ⍰ Apply the required changes in accordance with National Grid IT guidelines to ensure ongoing successful operation. ⍰ Avoiding increased vulnerability caused by expired hardware, software or applications which are not receiving patches and/or vendor support.	G020
5850USA	Secure Services Network (SSN) optimization	448,546	25,057	473,603	The procurement and installation of physical data center racks was subject to multiple delays - special heavy delivery, cabling PWOs, etc. and it is recommended a plan for extended time window for this activity to accommodate potential delays be included in future	Improved network security: Upgrading end of life network devices and applications will enhance National Grid's ability to safeguard digital assets from cyber-attacks by proactively identifying threats and responding in a timely manner and minimizing the impact.	G020

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5863D	AIMS Stability	609,000	84,000	693,000	Ensure Vendors are responsible for 'proactive monitoring' within the solution they are providing, and or ensure there are means within National Grid.	Following benefits are associated with this initiative: I Increased capacity and resiliency will result in steady state operation of the platform including reduction in platform instability instances due to capacity issues. This will consequently reduce the number of priority incidents attributed to platform performance to achieve operational business benefits and platform availability as expected.	C175
5877A	Intelligent Substation Design	303,000	408,000	711,000	1. Stronger vendor management is required in dealings with Bentley, to ensure prompt resolution on product issues/escalations, to avoid any compromises to the project schedule. 2. Deeper Bentley product architecture with integrations into National Grid's infrastructure environments, to be well understood, which will enable future Bentley product roadmap opportunities.	Deployment of the SaaS "Bentley Connect Edition Suite" CAD application will provide the following benefits: Deliver current and supported applications, driving modernization of toolsets used by engineering teams with new capabilities in support of efficient business processes; Support standardization of technology resources and processes delivering repeatability and error reductions; Reduce time to deliver substation designs including efficiencies due to reduced error rates; Reduced substation operating costs due to improved, enhanced, and more accurate substation design data; FERC Order 1000 is presently necessitating the transition of utilities to more automated and standardized workflows, facilitating a shift to Intelligent 3D Design. National Grid will be better poised to compete in FERC 1000 projects. This investment will directly support the Digital Substation effort to deploy IEC61850 technology and substation monitoring, estimated at 150 substations over the next 10 years.	G188
5889A	Electric Incident Reporting System (EIRS)	547,000	55,000	602,000	Working with external entities, such as the DPS Staff, at project start to define a clear cadence of communication to be established with the DPS's technical team and agree on deliverables through project life cycle.	Benefits of the project include: I. The project will ensure National Grid remains compliant in meeting its regulatory obligation to provide the NY State real time notification of major events and statuses of the Electric System. II. Provide National Grid access and control over its own incident data while providing entry, dashboard and reporting features not available today. III. Simplify the recording and reporting of the events allowing for a more efficient capture of the information.	G181
5895A	Investment Recovery IT Strategy & Roadmap	-	99,000	99,000	The inclusion of subject matter expert engagement from many business areas in the project plan was valuable, as it helped to ensure the final proposal was feasible, properly estimated, and met all business and IT requirements. Subject matter engagement should be estimated, confirmed, and built into future project plans.	• Elimination of manual processes related to Warehouse inventory Record keeping and sales order processing. • Small to Moderate Increase in sales. • Reduction in Costs. • Increase Efficiency and Effectiveness of Internal Processes. • Aligns to US Business strategy through inventory optimization and process improvements	T362
5896B	Warehouse SAP Scanner Replacement (Barcoding)	382,000	217,000	599,000	Confirm hardware availability as early in the project as possible, so that contingencies can be built into the project plan. In this project the delivery of new scanners was impacted by a Blue Chip shortage due to the COVID pandemic. The project team identified the minimum number of scanners required to continue development and testing, and the deployment was phased based on availability of these scanners. The custom patch to fix the NG-M network connectivity issue was delayed by several months. The project team utilized the existing NG-P network until the custom patch was available. This avoided further delays in implementation of new scanners. It was beneficial to consider alternate solutions, even ones requiring re-work, in order to reduce project delays.	The warehouse handheld devices currently used in the 5 main Distribution Centers are critical to the efficiencies of Receiving, Putaway, Counting, Picking, and Delivering material to the Field for Operations. Should they fail, the resulting manual workaround activity will be prohibitively costly and impact the ability to deliver material to the field in a timely manner. The new devices will provide improved scanning abilities and extended battery life.	G020
5896D	WEM Waste Environmental Management	312,000	36,000	348,000	Previous version of WEM System was working only on Windows XP machines. As cost and risk to maintain Windows XP machines is high, wherever possible, the project team overlapped project phases which helped to reduce the project timeline. Testing of the new WEM SaaS System with Chrome and Edge browsers uncovered issues with the Edge browser. It is recommended to test the web based applications with various browsers. Engage the Cyber security team early in the project. During this project, the NG Cyber Security team recommended engaging an external vendor to perform a Penetration Test. During this test, two vulnerabilities were identified and fixed by Chemical Safety (WEM vendor).	The complexity of the Federal, State and Local regulations and their interrelationships require a software solution to support proper management of waste activities, recordkeeping and reporting. This upgrade will reduce the Company risk for compliance violations by removing the reliance on manual processes which are extremely cumbersome and prone to error.	G020
5917a	Enterprise Data Management	1,709,000	271,000	1,980,000	After evaluating the project, the following improvements / lessons learned have been identified: Security BSR item actions and potential complexities of those items should be identified at the beginning of the project. During this project, additional analysis and configuration was required as a result of not having identified all BSR item and actions and potential complexities. Onboarding of security team members at the start of the project. By onboarding these resources, the process for satisfying individual BSR items may have been known earlier, thus allowing us to avoid a re-sanction. External team resources required to support resolution of BSR items should be identified during project startup.	I EDM Capabilities: New and in-flight projects require the data capabilities this fiscal year (Customer Transformation Blueprint, GridMod). Successful delivery of these programs and sustained health of the associated data calls for enablement of data management tools. I IT Principles: Capabilities are required to support IT Principles like Data is an Asset; Data is published; Data is accessible, Information is an asset which is fundamental to the efficient and effective delivery of IT services. I Efficiency: Disparate solutions exist today, within both IT and shadow IT, causing inefficiencies in support and annual RTB (Ex – GBE, ADA). I Cost Reduction: Projects that utilize these enterprise capabilities will see reduce one time and operational costs due to utilizing a centralized and standardized management of data. I Regulatory/Compliance: Capabilities are required to manage long term support of compliance with Business Management Standards and Data Management Policies. I Mastering Data: There is consensus across National Grid that there is a significant problem with data quality, with negative impact on efficiency/productivity, quality of decision making, and level of risk.	G020
5926A	NiMO Rate Case	1,157,000	115,000	1,272,000	Flexibility was required due to changing requirements, which resulted in code and table changes and reworking of testing. These changes also resulted in extended timelines for testing and implementation. Engagement with the Business team was a priority, as many resources were not available due to production issues and other project initiatives and deadlines. This resulted in delays and testers working overtime to meet timelines	This project allows National Grid to comply with the NMPC Rate Case which entails rate structure changes and billing process modifications.	C113

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6043B6	Mobile devices rollout for Plan Forward	-	241,000	241,000	None	This effort will enable workers to remain remote and not have to congregate at common sites. This will ensure that all levels of leadership will be able to communicate with all of their employees. Employees will have access to applications such as MS Teams and Outlook, SAP time entry, and company infonet. This will also enable remote training.	G210

[redacted]

6254US	Platform as a Service (PaaS) on Azure	769,000	117,000	886,000	During the project delivery, the team encountered defects in the pipelines provided by the Cloud Center of Excellence (CCOE) team. These issues caused a bottleneck for team resource availability. In the future the DevOps team will better plan for any unforeseen issues with the CCOE pipeline team, to avoid any project issues moving forward.	Reduction in time to provision infrastructure and increase in productivity due to infrastructure availability. - Reduction in configuration build errors due to infrastructure properties (server names, connection strings, etc) being hard-coded. - Improved adoption rate of automation across the projects for infrastructure provisioning. - Improved adoption rate for new Azure Resource types and capabilities. - Reduction of cost due to integration with log analytics/alerting.	G020
6255US	Digital Foundation Tooling and DevSecOps	531,000	125,000	656,000	1. During the implementation of this investment, the project team could not deliver a few of the originally planned components due to the addition of higher priority tasks and some resource constraints. In the future the team should analyze all components of the project scope to ensure high priority scope items are not omitted and what is delivered better aligns with the original scope. 2. There were a few unknowns that forced the team to change the project implementation plan in the middle of development. In the future, the team should plan for some buffer in case of any unforeseen issues to allow for minimal project impact.	This project will deliver: - Greater efficiencies and productivity within the application life cycle. - Increased operational stability, and improved software quality. - Increased agility with the end to end automation. Additional benefits will result in standardized Agile and DevOps tool offerings and include; - Reduced operational support costs, - Reduced operational complexity, and - Efficient talent development and ramp up on standard tool offerings. This investment will provide: 1. Dashboard Capability. Project will implement Hygieia, which is an Open Source configurable DevOps dashboard and provides: (1) a visual display for developers to track progress; (2) a near real-time consolidated view of the entire end to end lifecycle delivery; (3) a faster failure detection and shortened Root Cause Analysis timeframe; (4) improved end to end performance by eliminating bottlenecks and providing real time notifications of failures and outages. 2. Migration of tools (Jira, Confluence, Bitbucket, and SonarQube) from program specific (GBE) to corporate (National Grid) Azure environment will provide alignment with corporate National Grid standards; allow usage of Azure services for site disaster recovery not supported in GBE Azure; eliminate the use of redundant legacy implementations of various tools (Jenkins, SAAS Artifactory, and Ansible Tower). 3. Internet Accessibility for planning / collaboration tools (Jira and Confluence) to allow secure remote access to the tools and remove need of Virtual Desktop (VDIs) for remote users. Elimination of VDIs will results in cost savings through cost avoidance for future projects. 4. Defining CI/CD best practices and patterns will drive improved reusability and reduce effort needed to onboard new projects; will provide consistent metrics and measurements across projects and improve maintainability of CI/CD pipelines. 5. Incorporating security tools to improve product security and potential threat detection. Static Application Security Testing (SAST) and Dynamic Application Security Testing (DAST) tools will help to identify security flaws hidden inside code before they get to a production environment which will	G020
6256US	Platform Upgrades- Mulesoft, Informatica and DevOps	703,000	123,000	826,000	Thoroughly evaluate project scope during the requirements phase of the project. This project had a scope increase as a result of changing requirements. On top of the initially planned scope, the project upgraded 21 additional DevOps platforms. This involved multiple version and functionality upgrades. In the future better analyze and refine scope to avoid project re-sanctioning.	Benefits of the upgrading these foundational platforms include: Mulesoft - Improved Security - Improved Compatibility - Improved platform Efficiency at lowered costs - Improved Scalability - Ensure platform Compliance Informatica - Improved Security - Improved Compatibility - Improved platform Efficiency - Improved Data management capabilities - Ensure platform Compliance Devops: - Improved Security - Improved Stability - Reduced operation costs and complexities - Improved efficiencies of development - Ensure platform Compliance - Future proof the investment made supporting up to 2000 users.	G020
6258USA	Data Archival - Platform Implementation	1,641,536	314,230	1,955,766	Improvement - Project Management: Level of complexity of business requirements for two of the applications in scope, PowerPlants, were not fully understood. Team had to reach a common understanding of these types of requirements to proactively mitigate the delays.	Implementation of Enterprise Archival Platform will: Enable the decommissioning of legacy systems, leading to optimized technology landscape. Enable the maintenance of long-term storage and access to application data to meet retention and knowledge needs of the business. Enable the business to cost-effectively store data over time and to align with retention, auditability and accessibility requirements Reduce the time and cost associated with storing/consuming historical data Provide the ability to search, preserve and maintain chain of custody of data over the long term for investigative and litigation needs. Improve the ability to produce electronic information requested during legal proceedings. Adhere to the internal records management policies including Improved end users access to application data for ongoing knowledge management and analytics purposes Ability to archive data in line with technology trends and demands cloud computing, big data, proliferation of apps, platforms and networ	G020
6261US	ServiceNow Upgrade & Additional Capabilities	549,000	78,000	627,000	Sharing resources across different initiatives within the same department impacts delivery plan. A dedicated team must be assigned for each initiative/project. - Frequent resource changes impacted the delivery as not all analysis was transferred with resource/assignment changes. Thorough documentation is recommended. Frequent switching of resources across assignments should be avoided.	The following benefits are expected by upgrading the Service Now application: . 1. Improved response times due to new search features 2. Improved mobile device accessibility and use 3. Improved employee experience 4. Increase in Key Performance Indicator (KPI) metrics 5. Improved employee experience 6. Reduce issue resolution	G020
6266US	EA Strategic Portfolio Planning & Delivery / Additional Capability	1,574,000	112,000	1,686,000	Better identify and engage appropriate resources: - The project was set to be a global project with impact to both the UK and the US. There was a knowledge gap when it came to the UK sanctioning process resulting in delays. In the future we'll want to better coordinate and align with UK PMO team members to ensure the UK and US project timelines are in sync to avoid any dependency issues and/or constraints. - Meet with vendors early to set invoicing timeline expectations - During this project there were invoicing issues, resulting in delayed vendor payments. In the future we need to ensure that invoices are provided monthly to allow the team to reduce the number of accruals that carry over, resulting in improved project financial management.	The consolidation and maintenance of Business and IT Strategies in this platform and the elimination of strategies maintained in Excel and PowerPoint. - The ability to create an integrated plan prioritizing investments across the Business and IT. - The decommissioning of the Focal Point system.	G020
6289B	Record to Report – Close and Reporting simplification	1,149,000	744,000	1,893,000	Effective project management: All relevant parties (National Grid and vendor) were consulted during requirements gathering ensuring that the requirements captured covered IT, Operational Finance and the Controllership. - Tools limitation: The Jira testing tool was limited and unable to meet SDF required testing reporting requirements. - Effective planning allowed for success in training, along with great end user participation. For example, recording the sessions allowed users to access and reference the course material after training was complete.	This project will deliver such benefits as: - Revised and improved Account Reconciliation scoping process - Provisioning of reporting dashboards and maintenance documentation and threshold sensitivity analysis - Provisioning of Master data recommendations - Cash reconciliation automation evaluation - Bank feed requirements, design and implementation for 3-4 bank feeds - Better Blackline License rationalization - Establishing efficiencies for the month end close process, including an improved month end close task list - Updated close procedural documents, Close metrics and KPIs - Identified Automation opportunity (for future improvements) and recommended solution (tool) for month end close process - Delivered current state evaluation and future state assessment for Record To Report Reporting Strategy	G020

6289D	FTOM: Record to Report - Task Management with Blackline	1,268,000	439,000	1,707,000	Having a Project Manager from National Grid and an experienced BlackLine implementation team from PricewaterhouseCoopers (PwC) allowed for effective collaboration and the development of realistic timelines for milestones. The National Grid Enterprise Project Management Office (EPMO) generated a project plan plus PwC developed a task Management-specific plan that was embedded in the National Grid version. The combination of the two offered a higher level of completeness and understanding about the work being performed. Access to the project plan was limited to only specific stakeholders as the Microsoft (MS) Project technology used to store the project plan was not widely accessible.	This project will deliver such benefits as: ⍰ Enhance traditionally manual process by using automated notifications, workflows and automations where applicable ⍰ Milestone tasks can be used to track activity and automatically certify when child tasks are executed ⍰ Management-level reports give visibility into the global close calendar ⍰ Tasks will be configured for future linkage with reconciliations and journals, guaranteeing the correct sequence, flow, and rollup of related activities ⍰ Optimization of the close process by removing remaining manual component	G020
6313US	GDPR Purge Compliance	-	116,000	116,000	1. Requirements - Positive - Good teamwork and engagement with Data Stewards and Data Owner to clarify Retention Policies. 2. Requirements - Negative - Retention Policy Master record should have been aligned to the published retention and kept separate from editable version. 3. Test - Negative - Test Strategy Doc not created prior to testing which led to so some ambiguity on testing scope. 4. Test - Negative - Live Data Request form to be completed prior to testing.	Legal risk – we are able to deliver our legal obligations to set retention periods and implement them against the personal data of our workers. These obligations arise from GDPR in the UK and multiple US laws	GT63
6346US	US CA Group Web Digital Evolution	-	705,000	705,000	Resources to be identified and on-boarded prior to commencing the initiative. Definition and alignment on Objectives & Key Results (OKRs) from the outset so we know what good looks like. Review Sanctioning process to simplify overall governance to start up and close down project.	Increase external stakeholder, regulator and partner satisfaction. Reduce Opex spend on Domain Management Services (DMS) through vendor review and simplification of key domain management capabilities ensuring they are fit for purpose, establishing ownership clarity and cost control. Protection of our Brand through use of compliant materials and making communications more professional. Stronger National Grid identity through a consistent and personalized brand experience for stakeholders. Improve user experience, so that the NG site and search results are more discoverable by search algorithms, ensuring that our campaigns and communications are among the top results. Measure and analyze the reputational impact of our campaigns across different audiences e.g. policy makers, stakeholders, employees and the general public. Better understanding our target audience, content consumed, time spent on, their point of view, feeding into our Content and Search Engine Optimization (SEO) strategy. Reduce proliferation of websites across group and bring them under a central control or purview. Reduction in Shadow IT, simplification of our application landscape. Replacement of existing Shadow IT external email campaign tools with single compliant solution.	G020
6424US	Infonet Decommissioning	-	49,000	49,000	Resource Management – ensuring the required skills are identified during start up phase of the project in both regions. This will support the project team onboarding ensuring the right skill sets are within the team. Scope Management – ensure the business provide the correct level of details as to what sharepoint sites are in scope and what are deemed out of scope	The cost savings of $130K on annual basis will be made when the platform is eventually shut down.	G020
6440US	MA EV Off Peak Charging Program	-	49,000	49,000	⍰ No IT ownership at the start of this project ⍰ IT Communication was very poor at the start of this project- Need more program level solution architects who understand scope instead of relying on single contact ⍰ Full documentation should be prepared and never sent in piece meal as this causes confusion to stakeholders Closeout Activities	Program seeks to help National Grid better understand peak and off peak charging behavior of EV drivers to develop future rate structures that encourage efficient charging behavior.	C277

Niagara Mohawk Power Corporation d/b/a National Grid

The Brooklyn Union Gas Company d/b/a National Grid NY

KeySpan Gas East Corporation d/b/a National Grid

Cases 20-E-0380, 20-G-0381, 19-G-0309, and 19-G-0310

INFORMATION TECHNOLOGY & DIGITAL CAPITAL INVESTMENT REPORT FOR QUARTER ENDED JUNE 30, 2023

SUBMITTED TO:
NEW YORK STATE DEPARTMENT OF PUBLIC SERVICE

August 29, 2023

nationalgrid

Information Technology & Digital Capital Investment Report for Quarter Ended June 30, 2023

I. Introduction

Niagara Mohawk Power Corporation (Niagara Mohawk), The Brooklyn Union Gas Company (KEDNY), and KeySpan Gas East Corporation (KEDLI) (together "National Grid" or the "Companies") submit this Information Technology & Digital ("IT&D") Capital Investment Quarterly Report ("IT&D Quarterly Report") for the first quarter (Q1) of fiscal year 2024 ("FY24").[1] Pursuant to the Public Service Commission's ("Commission") January 20, 2022 Order in Cases 20-E-0380 and 20-G-0381, and August 12, 2021 Order in Cases 19-G-0309 and 19-G-0310, the Company is to provide quarterly variance reports to Department of Public Service Staff ("Staff") regarding IT&D capital projects and investments.[2] The IT&D Quarterly Report provides an explanation of the Company's overall IT&D investment plan approach, descriptions of program investment categories, and a comparison of budgeted to actual spending.

IT&D capital projects and investments that are shared investments across National Grid's operating companies are implemented and owned by National Grid USA Service Company, Inc. ("Service Company") and allocated to the benefiting US operating companies, including Niagara Mohawk, KEDNY, and KEDLI, in the form of Service Company rent expense once the investments are in-service. The Service Company rent expense includes a return on the capital investment (net of deferred taxes) plus booked depreciation expense.

Through Q1 FY24, National Grid spent $75.7 million compared to a year-to-date Service Company budget of $67.3 million, representing an $8.4 million overspend. Attachment 1 includes detail on budgets and actual spending for each of the portfolios and commentary regarding performance which has been presented to each of the Chief Information and Digital Officers ("CIDOs") responsible for their respective portfolios.

Attachment 2 includes a detailed breakdown of individual project spend by portfolio compared to the budgeted spend for each of those investments.

II. Program Categories/Spending Rationales

National Grid has organized the US Information & Digital Technology organization into six areas:

[1] The first quarter of FY24 is April 1, 2023 through June 30, 2023 ("Q1 FY24").

[2] *See* Cases 20-E-0380 and 20-G-0381, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service*, "Order Adopting Terms of Joint Proposal, Establishing Rate Plans and Reporting Requirements" (issued and effective January 20, 2022), Joint Proposal, Section IV.10.5; and Cases 19-G-0309 and 19-G-0310, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Energy East Corporation d/b/a National Grid for Gas Service*, "Order Approving Joint Proposal, as Modified, Imposing Additional Requirements" (issued and effective August 12, 2021), Joint Proposal, Section IV.6.5.

(1) Group Functions
(2) United States Electric
(3) United States Gas
(4) Customer
(5) Digital Platforms and Infrastructure Technology
(6) Security

IT&D capital investment is tracked based on these area designations. Descriptions of each portfolio category/spending rationale are provided below.

Group Functions

Group Functions executes National Grid's digital roadmaps, leveraging technology and reimagining processes in support of Finance, Enterprise Business Services, Human Resources, Legal, and Safety Health & Environmental, continuing to upgrade SOX related systems, and retiring outdated systems.

United States Electric

Provides IT&D support to our US Electric Distribution and Transmission businesses. The group's primary focus is modernizing our key legacy operational systems while building capabilities that will enable the energy transition and transformation to a Distribution System Operator (DSO). Through digital delivery, the United State Electric organization provides greater efficiency and data driven insights into our workforce operations that provide:

- More automated and reliable grid controls systems
- Capital programs and projects that support the proliferation of Distributed Energy Resources (DERs)
- A growing market of prosumer electric customers

United States Gas

Provides IT&D Support to our US Gas Distribution and Transmission businesses by developing digital products that deliver continuous improvement of our asset lifecycle management capabilities through:

- Risk-based asset investment planning
- Performance-based maintenance
- AI-based work optimization
- Mobile field products

Customer

Provides IT&D Support to our US Customer business, delivering best-in-class digital products that increase engagement and reduce costs to serve by:

- Modernizing our legacy "back-end" IT&D systems to support more digital products
- Liberating data and creating an abstracted and real-time customer data platform to assist with achieving energy efficiency targets

Digital Platforms and Infrastructure Technology (DPIT)

Supports our internal IT&D Architecture covering Cloud and Hosting, Critical Network Infrastructure, Networks, End User Experience and Operations. DPIT is focused on reducing National Grid's technical debt through cost effective, modernized IT&D infrastructure, enabling cloud capabilities that meet our risk profile. The organization enables a connected, seamless enterprise that is flexible, secure, reliable and scalable while maintaining acceptable levels of technical debt and risk.

Security

The Security function protects our people, premises, and digital systems with the objective of maintaining trust in National Grid services, focusing on:

- Reduction in likelihood and impact of security incidents
- Compliance with relevant laws, standards and regulations (*e.g.*, NERC, PCI DSS, GDPR, NIS-R)
- Improved user experience and standard of security practices across National Grid

The Security function aims to reduce the likelihood and impact of physical and cyber security incidents affecting National Grid. It does this by championing security best practices, adopting technology advancements, and promoting a security-awareness culture.

Other Category

For accounting purposes, there is an additional category listed in Attachment 2 to this report. The "Other" category has 4 main elements:

- The "IT4IT" sub-category acts as a funding bucket for NPV-positive projects to enable efficiency savings in our IT&D-run business operations.
- The "Other" sub-category is the allocation of operating expense overheads such as Enterprise Architecture and Solution Delivery charges to a centralized account. The capital charges for this overhead are made directly to individual projects.
- "Burdens" are the net removal of operating expense burdens to ensure our reporting agrees with our Corporate Controllable Cost reporting. Operating expense burdens are consolidated and reported separately under non-controllable costs in our corporate reporting packs.
- "RI Sale"; following the sale of the Rhode Island Gas and Electric business to PPL, associated separation costs flow through this category. National Grid is currently providing IT&D services (applications, infrastructure, networks, etc.) to PPL as part of a 2-year Transition Services Agreement (TSA) following the sale. Significant effort is required to manage and track an orderly exit of all IT&D TSA services. National Grid must provide subject matter expertise, migrate critical application data, design and implement interim state processes, and remove any customizations introduced to support delivery of the TSA. The costs for this are recovered from PPL and do not impact New York rate payers. The Group Functions portfolio is responsible for managing the expenditure but the costs role up through the "Other Category".

The summaries below explain the subjects presented in the attachments to the Q1 FY24 report.

A. Attachment 1

1. Summary (pages 2-3) – Overview of US IT&D.
    a. High-level executive summary, including work and challenges faced across the portfolios and a brief explanation around the forecast (p. 2).
    b. Actual expenditure versus budget by portfolio year to date, and full fiscal year forecast versus annual budget by portfolio (p. 2).
    c. Actual and forecasted spend by month for FY24 compared to budget, prior year and run rate. Each bar is color-coded to give an overview of the comparative spend per portfolio (p. 3).

2. Portfolio Sections – The remainder of Attachment 1 includes six sections, one for each of the IT&D portfolios: Group Functions; US Gas; US Electric; Customer; DPIT; and Security. Each portfolio consists of 3 pages:
    a. The first page provides a narrative financial and business overview. The three charts on the first page show: (1) operating expenses; (2) capital expenses; and, (3) total expenses. Each chart shows FY24 actuals and forecast versus budget, with trend lines showing prior year expenditure and run rate. This page also shows the year to date actual and full year forecast for capital, and total expenditure compared to budget.
    b. The second page of the portfolio sections has two charts on the top left of the page: the top chart is actual expenditure versus budget by program year to date. The second chart compares full fiscal year forecast versus annual budget by program. There is also a key challenges section and the escalation point for that challenge. On the bottom half of this page is a chart showing actual and forecasted spend for FY24 compared to budget, prior year and run rate for the portfolio. Each bar is color-coded to give an overview of the comparative spend per program.
    c. The third page of each portfolio section provides a status update on the various projects. A priority investments list for the portfolio and their status appears at the top right hand of the page. Blue means complete, green means on track, amber means some risk, and red means focused intervention required.

    Attachment 1 is based on National Grid's quarterly Portfolio Performance Reviews (PPR) and reporting cadence to senior management of each portfolio, and ultimately to the IT&D leadership team.

B. Attachment 2

INVP Level Variances - This attachment provides a detailed listing of individual projects in each of the six portfolios and the "Other" category explaining spend in Q1 FY24, including:

(1) Total actual spend compared to budget and variance;
(2) A variance reason for all capital investments which are outside the 10% tolerance to budget.

National Grid

Quarterly Report – Q1 FY24 Information Technology & Digital

29th August 2023

nationalgrid

US IT&D

Financial Performance

Portfolio View - Year to Date $M

Portfolio	Capex			Opex			Totex		
	Actuals YTD	Budget YTD	Variance	Actuals YTD	Budget YTD	Variance	Actuals YTD	Budget YTD	Variance
US Electric	7.0	5.9	(1.1)	1.8	0.7	(1.1)	8.8	6.6	(2.2)
US Gas	20.6	14.9	(5.7)	2.6	1.4	(1.2)	23.2	16.3	(6.9)
Customer	12.7	11.0	(1.7)	1.9	1.0	(0.9)	14.6	12.0	(2.6)
Group Functions	0.5	6.4	5.9	1.2	0.7	(0.5)	1.7	7.1	5.4
DPIT	15.8	15.6	(0.2)	1.4	1.6	0.2	17.2	17.2	0.0
Security	4.6	6.5	1.9	0.5	0.6	0.0	5.1	7.1	2.0
Other	0.5	0.6	0.1	4.6	0.5	(4.1)	5.2	1.1	(4.1)
Total	**61.7**	**60.9**	**(0.8)**	**14.0**	**6.4**	**(7.6)**	**75.7**	**67.3**	**(8.4)**

Portfolio View - Full Year $M

Portfolio	Capex			Opex			Totex		
	FY Forecast	FY Budget	Variance	FY Forecast	FY Budget	Variance	FY Forecast	FY Budget	Variance
US Electric	39.0	36.8	(2.2)	3.9	4.0	0.2	42.9	40.8	(2.0)
US Gas	83.5	89.6	6.2	9.3	8.2	(1.0)	92.7	97.8	5.1
Customer	52.1	47.8	(4.3)	4.4	4.7	0.3	56.6	52.5	(4.1)
Group Functions	31.1	34.6	3.5	5.5	5.1	(0.4)	36.6	39.7	3.2
DPIT	61.0	61.2	0.2	6.9	6.5	(0.4)	67.9	67.7	(0.2)
Security	26.4	26.4	0.0	1.7	3.4	1.7	28.1	29.8	1.7
Other	0.3	4.2	3.9	27.1	0.5	(26.6)	27.3	4.7	(22.7)
Total	**293.3**	**300.6**	**7.3**	**58.7**	**32.5**	**(26.2)**	**352.0**	**333.1**	**(18.9)**

IT&D Commentary

Q1 Releases

- By transitioning to an agile approach, the company has made some impactful releases across the IT&D portfolio this quarter:
  - Continued Scaling & Evolution (CS&E) has released a document management system to over 1,000 Downstate NY users.
  - Customer has improved contact center technology to be compliant with FCC mandates and improving functionality and resiliency to around 2,000 users.
  - Ariba release in May impacts 1,400 suppliers enhancing e-invoicing capabilities and catalog coverage.

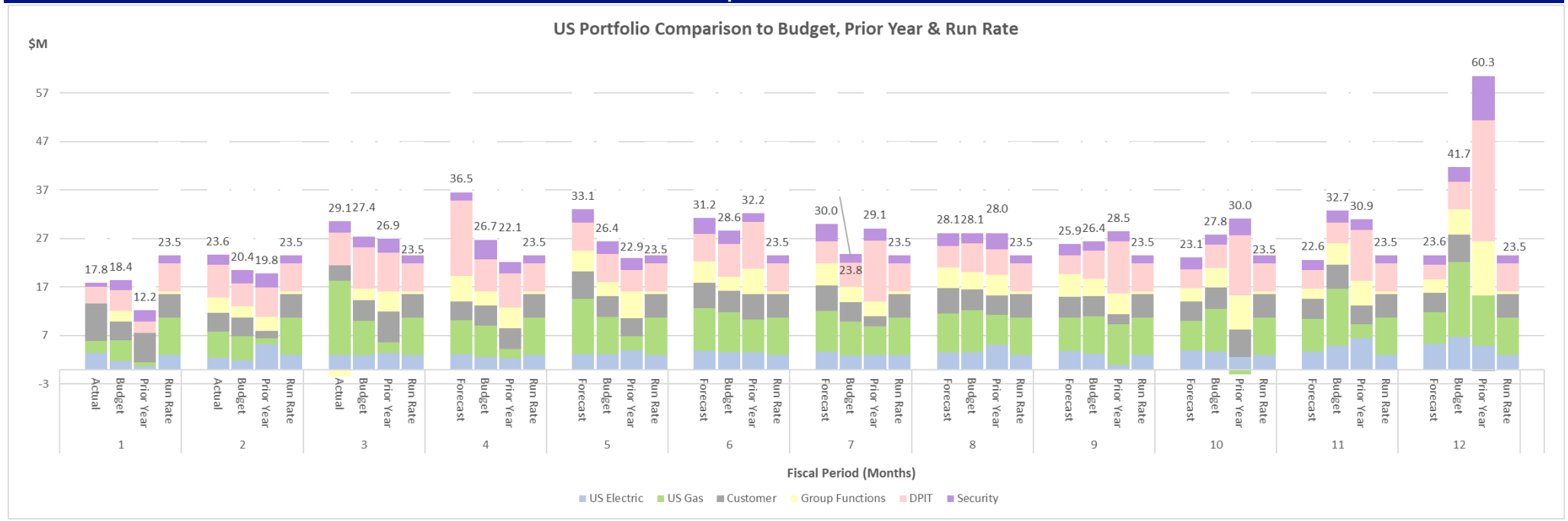
Challenges

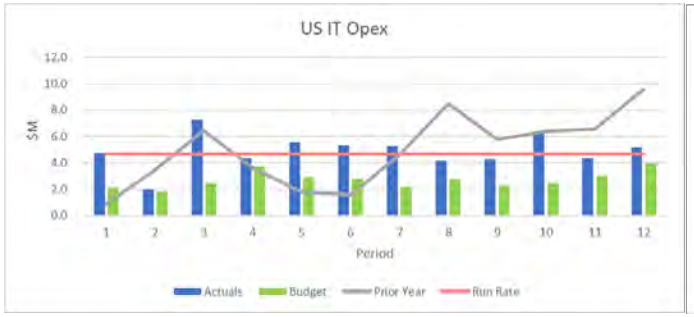
- Business engagement and customer input is undergoing a learning experience as we transition to agile delivery.
- Cost pressures and resource challenges are being managed across the portfolio.

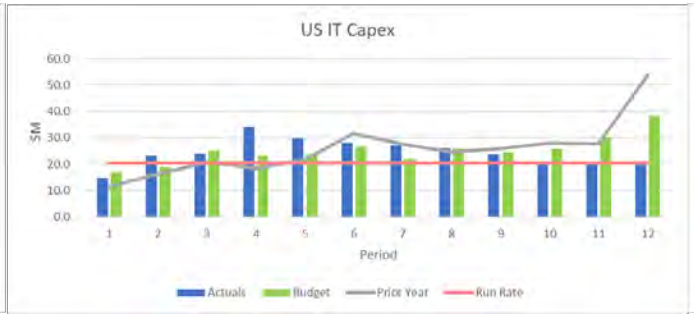
Financial Performance

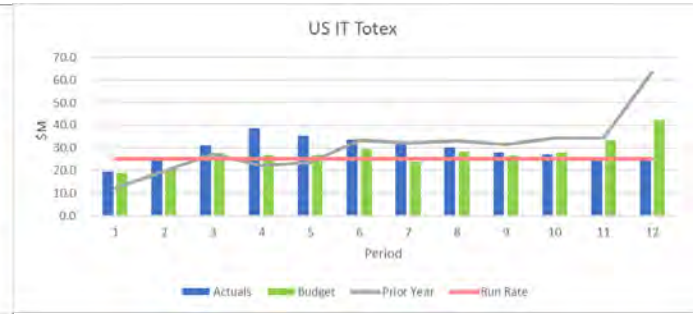
- Year to Date financial performance is tracking close to budget once accounting adjustments are corrected and RI costs excluded. There was an accrual duplication of $7.1m in CS&E which is partially offset by intercompany credits of $2.6m for eHub and $1.1m for MyHub.
- The full year forecast overage is driven by separation activities for RI which will not impact NY customers.
- Note: a Construction Overhead Accounting adjustment has been made to correct an FY23 miss-allocation to individual projects. For FY24 this means individual portfolios received unexpected credits or debits. The net impact is zero across IT&D but it does present challenges in individual portfolios. This is though being managed across the organization.

Portfolio Comparison & Cost Charts









US Gas
Q1 FY24 Report

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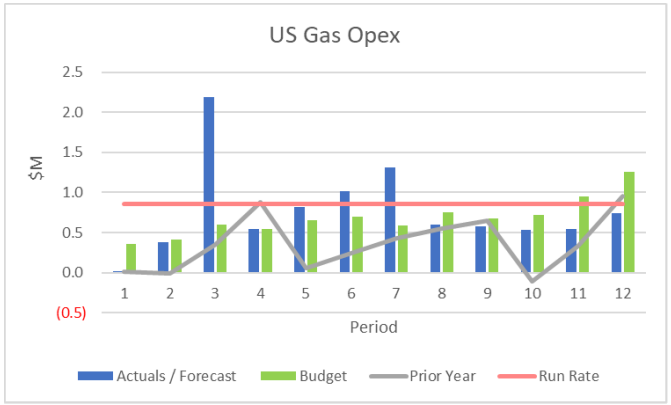
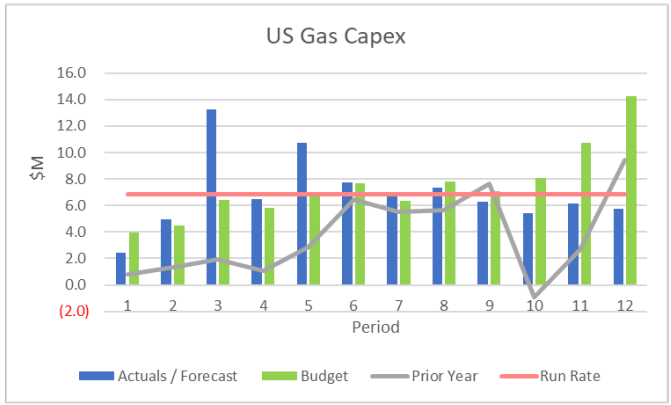
US Gas

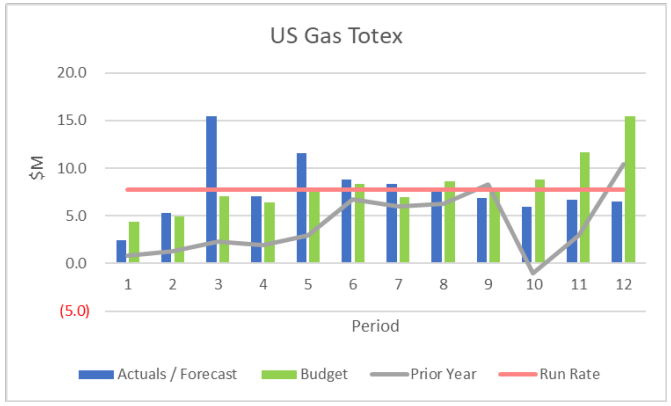
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	92.7	97.8
	YTD Budget($M)	YTD Actual ($M)
	16.3	23.2

Capex	Budget ($M)	Forecast ($M)
	89.6	83.5
	YTD Budget($M)	YTD Actual ($M)
	14.9	20.6



Portfolio Commentary

Financials

- Year to date the Gas business is on budget. The variance is primarily attributable to a double accrual for CS&E of $7.1m which reversed in July.
- Full year forecast is favorable due to a re-forecast of Gas Transportation System costs and bringing more work in house ($3.9m).

Q1 Releases

- CS&E – Tesco Meter Management system went live on 5/8 enabling the MA Metershop to have online traceability of the status of the meters from the time of receipt at the shop to the time they are either retired or installed at customer's site. Deployed Docuware document management to Long Island (~1000 users) and NYC (~50 users) last week of April, decreasing system costs while improving user experience
- Digital and IT Foundations - Leak Mgmt. System (LMS) LI archive went live in Operational Data Store (ODS) and will enable reporting and analysis of legacy closed leaks for Compliance and Asset Management, that were not migrated to GBE Maximo. LMS LI decommission will allow us to eliminate security risks and discontinue application support, which will save $64k annual costs
- Data Center Transformation (DCT) – Gas Transportation Information System (GTIS) migrated in April, part of NG's continuing commitment to consolidate 7 US legacy data centers down to 2,FIS, Maximo-NE, Maximo T&D, and Gas T migrated in May

US Gas Program View of Financials

Gas Program View Year to Date $M

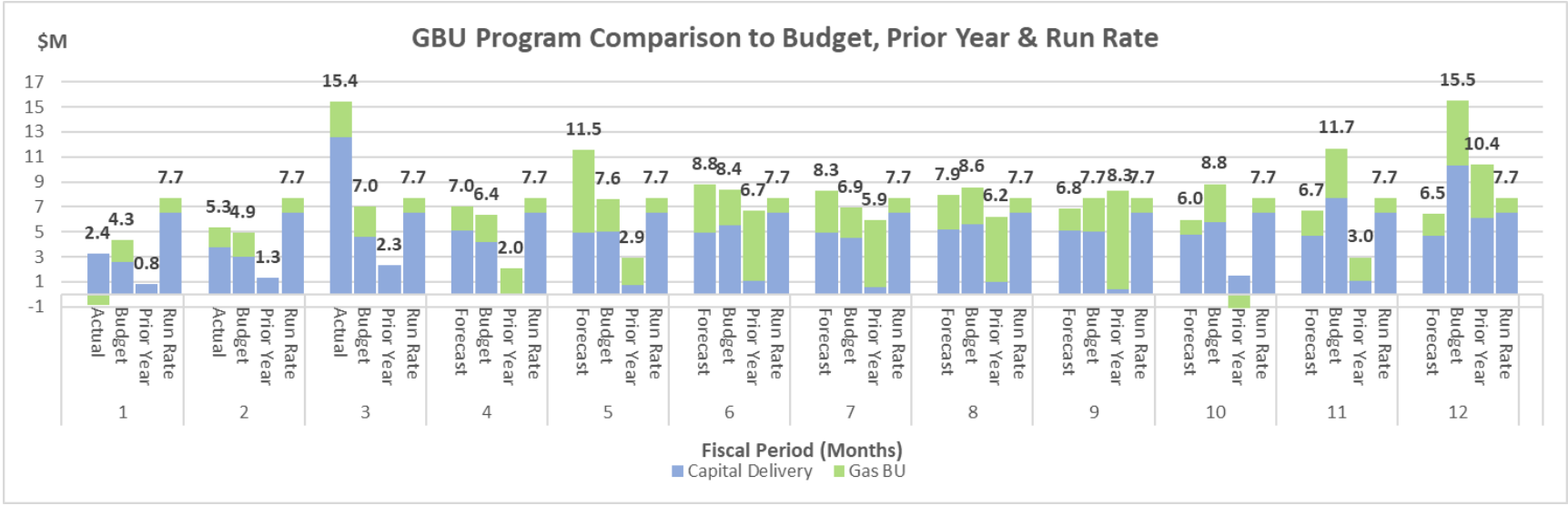
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
US Gas	CS&E	17.0	9.4	(7.6)	2.7	0.8	(1.8)	19.6	10.2	(9.4)
	Gas BU	3.6	5.5	1.9	(0.1)	0.5	0.6	3.5	6.0	2.5
		20.6	**14.9**	**(5.7)**	**2.6**	**1.4**	**(1.2)**	**23.2**	**16.3**	**(6.9)**

Gas Program View Full Year $M

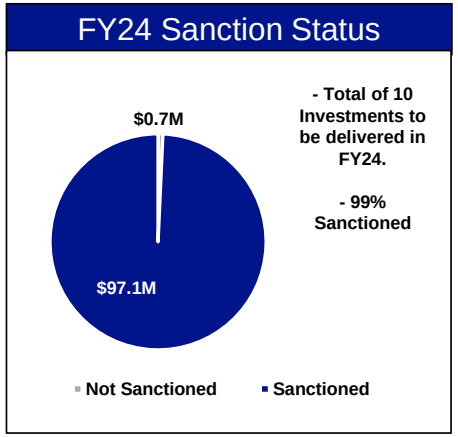
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecas	Totex FY Budget	Totex Variance
US Gas	CS&E	57.0	58.8	1.7	7.0	5.1	(1.9)	64.0	63.9	(0.2)
	Gas BU	26.5	30.9	4.4	2.2	3.1	0.9	28.7	34.0	5.3
		83.5	**89.6**	**6.2**	**9.3**	**8.2**	**(1.0)**	**92.7**	**97.8**	**5.1**

Key Challenges

Challenge	Escalation Point
• **CS&E**- GBE capabilities have customer information dependencies, the additional C2C program delay will lead to higher CS&E program costs as well as an extended timeline. The CS&E program conducted a detailed analysis of cost impacts and shared with the executive committee in June	NY CIO
• **Enterprise Retail Choice Retail Program(GTIS)-** IBM costs came back significantly higher than expected, contract is delayed while negotiations are ongoing and we explore insourcing the work as an alternative. RFP initiated to engage a vendor for re-evaluation for final delivery. Will know the final cost by end of September	GTIS/NG Product Team(NYCIO)
• **SCADA(GMS)-** Supply chain issues can delay the delivery of Cisco, Dell servers, and cabinets that can impact the project timelines. Team is working with CNI to order hardware in Q1 ahead of time and monitor.	DPIT



Mobilisation Summary



Status

- All projects are sanctioned for FY24 and started

Resources

- Overruns on vendor labor and misalignment on NG labor for CS&E Program, re-balancing of resource allocations underway

Scope

- All investments are active

Priority Investments

Program	Status	FY 24 Budget ($m)	RAG This Month	RAG Last Month
CS&E	Inflight	63.9		
Work Management	Inflight	1.1		
Gas Network Operations	Inflight	23.5		
Network and Asset Management	Inflight	0.5		
Digital & IT Foundations	Inflight	5.6		

US Electric
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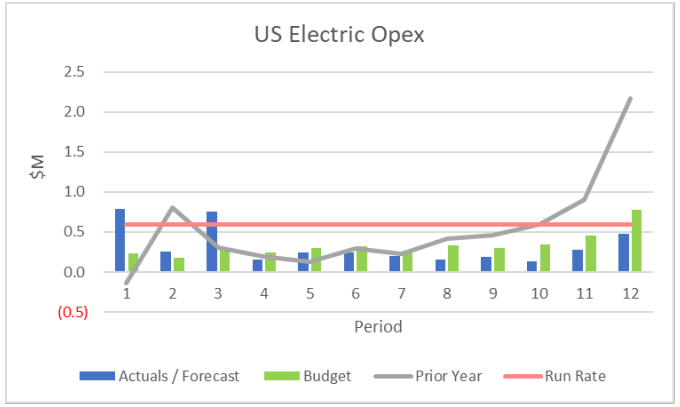
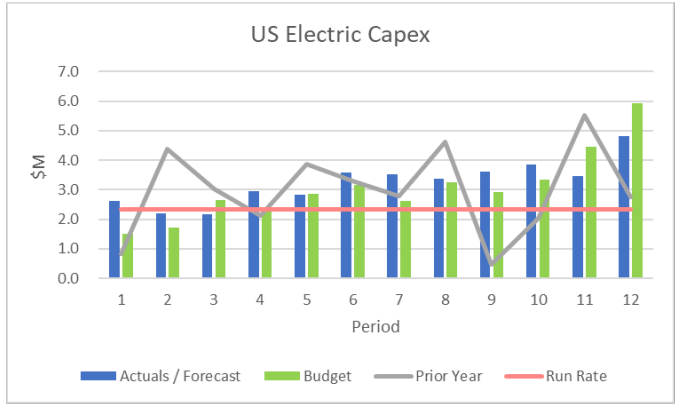
US Electric

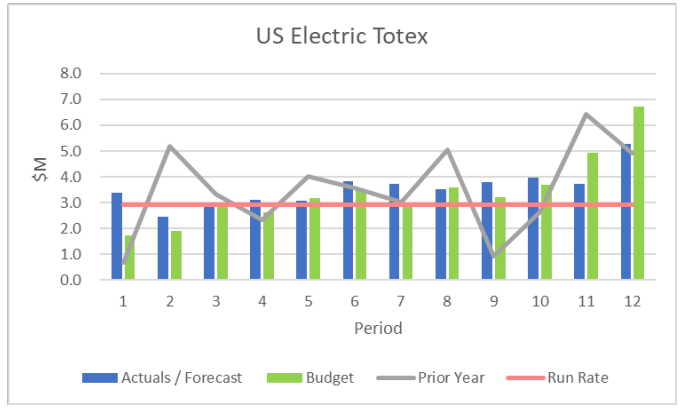
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	40.8	42.9
	YTD Budget($M)	YTD Actual ($M)
	6.6	8.8

Capex	Budget ($M)	Forecast ($M)
	36.8	39.0
	YTD Budget($M)	YTD Actual ($M)
	5.9	7.0



Portfolio Commentary

Financials

YTD Totex actuals vs budget variance ($2.2M) mainly driven by

- Construction Overhead charges of ~1M Capex to multiple initiatives, creating an unplanned spend spike
- Generation Asset Management – walked in at the reduction of existing initiatives without re-forecasting to include $1.03M within Q1
- FutureNow – overrun of $800K Capex
- Shift to right for IDS Replacement and Demand Planning & Track and Trace initiatives, causing underspends to date

Q1 Releases

Core / Digital:

- **FutureNow:** Forecasting 2.0 MVP (financial forecast entry tool to flow project financials into Copperleaf) implemented. Forecasting now lives within One Integrated Plan and replaces the PowerBI forecast file. In the first 3 weeks about 134 Forecasters updated 1211 projects using this tool that improves user experience and reduces time required to complete Forecasting.
- **Veg+:** recently deployed ViperPro Workflow for Electric Transmission, building on the previous release for Distribution, which enables optimization of vegetation management through the advanced digital tool (ViperPro) providing whole cost vegetation management control and optimization
- **NYISO PMU Large Generators** initiative was completed, bringing NG in compliance with the NYISO mandate to support 3rd party Phasor Measurement Unit (PMU) data transmission to NYISO, and addressing technical debt.

Strategic:

- **DERA+:** successfully completed a production release which includes our initial commitment of Operate "Most Valuable Features" - Making operating DER aggregations and their associated DERs on our networks safe and reliable - Achieve 80% coverage of key operational features required by the control center(s) for DER aggregation operations. 2. Delivered initial features to handle resubmitted applications (earlier than expected)
- **GridMod TOMS:** Integrated system and user functionality to support GIS/Snowflake/Mulesoft integration into TOMS on weekly basis including circuit information data updates as well as 4RF NMS interface integration and data loading
- **GridMod Cyber Security:** Strategic Privileged Access Management (PAM) tool delivered to manage, track and enhance the security of privileged accounts across the US CNI environment for ADMS
- **GridMod Data Management:** Loaded PI Tag data from Corporate PI into Snowflake to enable users to search PI data based on assets, circuit names, substation names and geographical locations. Migrated P6 historical data from PostgreSQL legacy database into Snowflake and built generic views for project, activity, and resource, so that the Capital Delivery Team can meet all their analytical needs. Continue to build "One system One Model" to improve data access and analytics capabilities

US Electric Program View of Financials

Electric Program View Year to Date $M

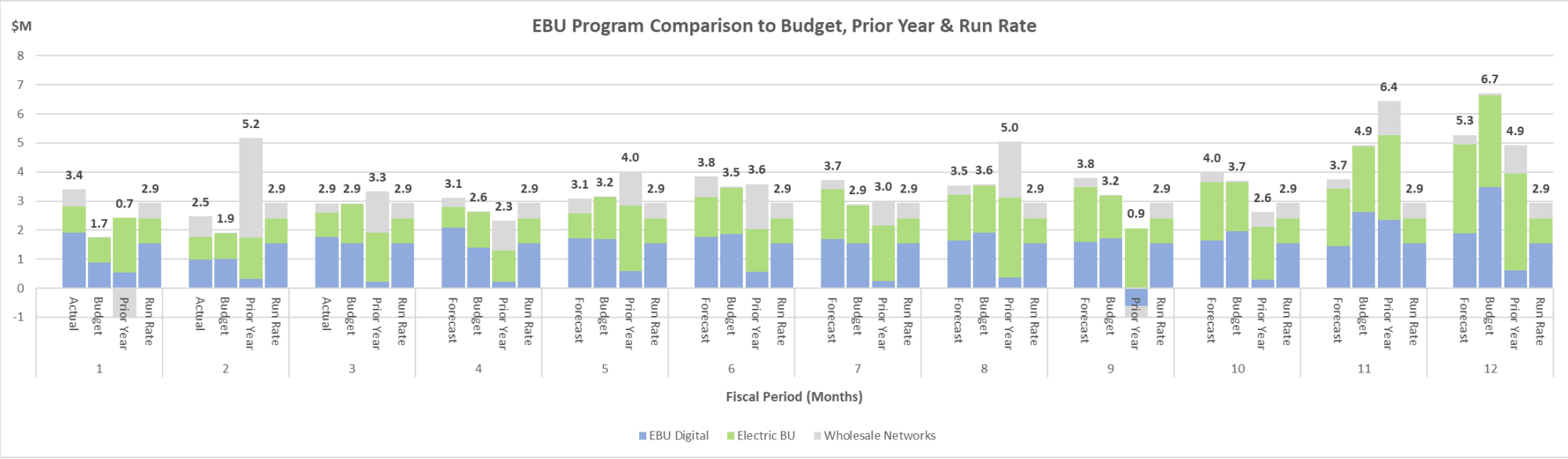
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
US Electric	EBU Digital	3.6	3.2	(0.4)	1.0	0.2	(0.7)	4.6	3.5	(1.2)
	Electric BU	1.7	2.6	0.9	0.8	0.4	(0.4)	2.5	3.1	0.5
	Wholesale Networks	1.6	0.1	(1.6)	0.0	0.0	0.0	1.6	0.1	(1.6)
	Totals	**7.0**	**5.9**	**(1.1)**	**1.8**	**0.7**	**(1.1)**	**8.8**	**6.6**	**(2.2)**

Electric Program View Full Year $M

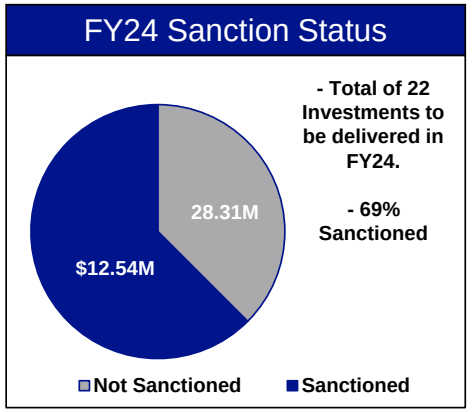
Portfolio	Program	Capex FY Forecas	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
US Electric	EBU Digital	18.4	20.1	1.7	1.8	1.5	(0.3)	20.1	21.6	1.5
	Electric BU	15.6	16.4	0.8	2.1	2.5	0.4	17.7	18.8	1.2
	Wholesale Networks	5.0	0.3	(4.7)	0.1	0.1	0.0	5.1	0.4	(4.7)
	Totals	**39.0**	**36.8**	**(2.2)**	**3.9**	**4.0**	**0.2**	**42.9**	**40.8**	**(2.0)**

Key Challenges

Challenge	Escalation Point
• **FutureNow:** Optimization for Distribution is forecasting to deliver less value than the anticipated 8% CapEx efficiency due to low business trust in optimized planning results. Plan in place for further product feature release to increase confidence and focus on business owner role in implementing solution benefit for FY24 and beyond	***Product Line Director***
• **Core:** Generation Asset Management has encountered many E2E testing blocker issues which are being resolved. The blocker issues have elongated the E2E testing cycle. The release schedule is being replanned to reduce costs and duration. Several project closures have been impacted by the SAP/Ariba conversion and are slowly being resolved. The review and approval of sanction papers continues to be a challenge to project closure.	***IT Director***
• **ADMS OMS:** The RI divestiture will require new MA data center build-out; Also, the legacy OMS system must operate in parallel until PPL has migrated customers to their outage management and customer systems	***Product Line Director***
• **ADMS OMS:** NY WAN circuit deployment is being delayed due to Verizon resource challenges and permit issues. If WAN circuits are delayed beyond August, then NY Go Live will be pushed out	***Product Line Director***



Mobilisation Summary



US Electric IT Core & Digital:

Status

- Based on 3+9 forecast, ~69% of planned product spend has been sanctioned with credible plan in place to effectively sanction and deliver throughout FY24

Resources

- Core & Digital IT – continues to be a heavy reliance on contracted resources, which requires resolution to move to an effective long-term product model
- Strategic program – CNI resource onboarding challenge to ensure continued operation and performance of solutions post implementation.

Scope

- 100% Investments are Active, with $22.9M worth of investments in ‘Deliver’ phase, and $18.5M in ‘Define’ Phase

Priority Investments

Program	Status	Budget ($m)	RAG This Month (June)	RAG Last Month (May)
Core *(Fix the Foundation & Regulatory Compliance)*	Inflight	19.3		
Digital Lighthouse *(Transformational Products)*	Inflight	21.6		
GridMod	Inflight	29.5		
ADMS / FLISR Acceleration	Inflight	33.4		
DSO	Inflight	19.5		

Customer
Q1 FY24 Report

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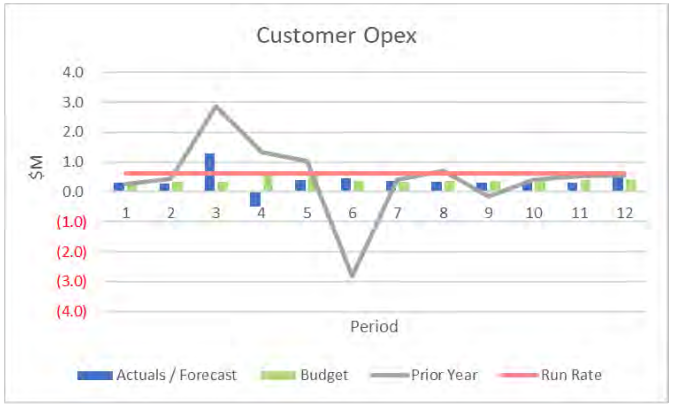
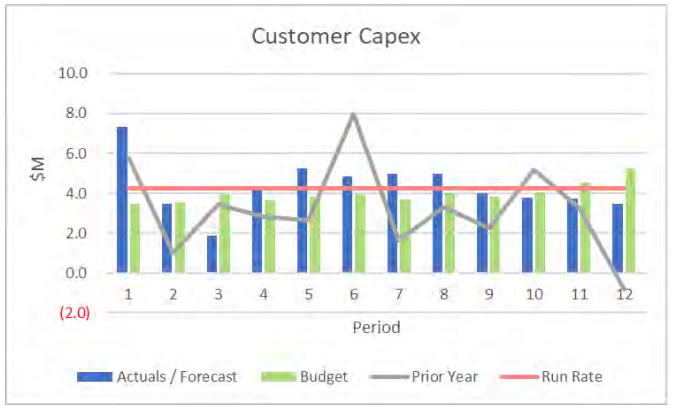
Customer

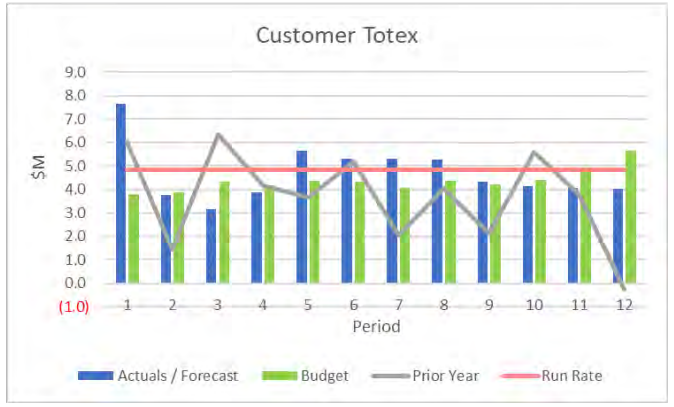
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	52.5	56.6
	YTD Budget($M)	YTD Actual ($M)
	12.0	14.6

Capex	Budget ($M)	Forecast ($M)
	47.8	52.1
	YTD Budget($M)	YTD Actual ($M)
	11.0	12.7



Portfolio Commentary

Financials

- YTD actuals and forecast are running slightly higher than planned due to Construction Overhead charges that were applied in P1 FY24 as a correction from P12 FY23. The current forecast is impacted by this adjustment but the portfolio is assessing how to bring costs back to budget.

Q1 Releases

- Advanced Metering Infrastructure (AMI) – NY scaled deployment continues with 300+ electric meters and 230+ field network devices installed by the end of June. The measured ramp-up has enabled learning to address challenges before full-scale deployment which begins in September FY24.
- CRIS to CSS (C2C) – Testing the solution continues to progress to plan in line with the May FY25 delivery.
- Contact Center Technologies – Significant upgrade delivered enhanced operational resiliency, improved business functionality, and brought us into compliance with the latest FCC regulations. Approximately 2,000 Call Center Agents will benefit from these improvements.
- My Business Account (MBA) – Upgraded technology stack to latest version of code base and released features such as personalization enhancements response to customer feedback. Adoption rates for MBA are on target to reach 43% by end of fiscal year 2024.

Customer Program View of Financials

Customer Program View Year to Date $M

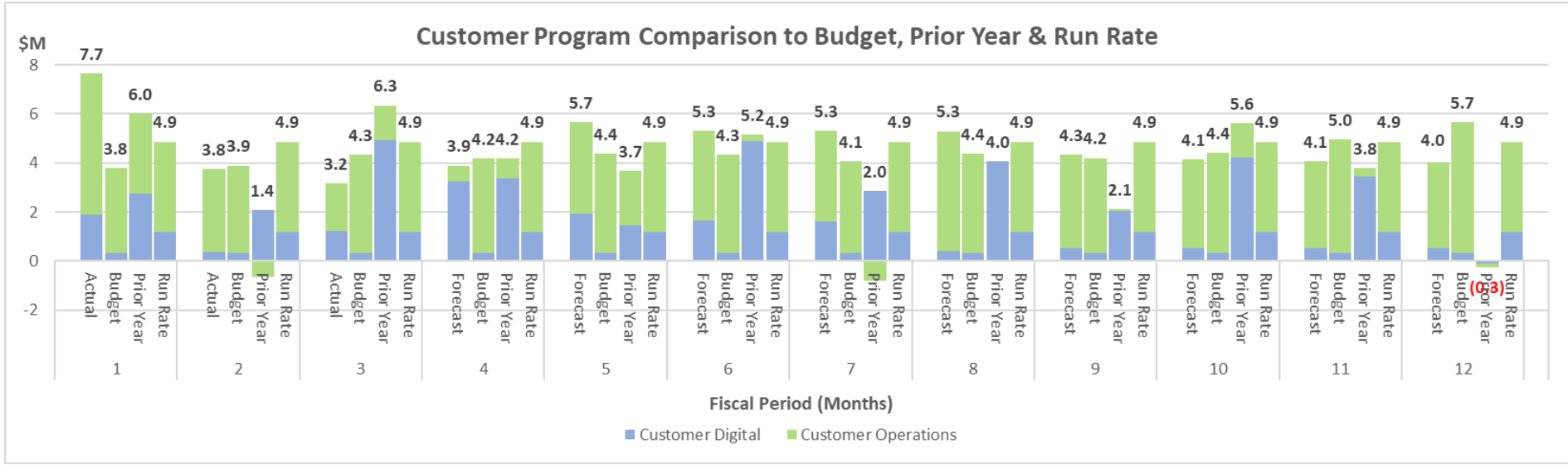
Portfolio / Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Customer Customer Digital	2.3	0.8	(1.6)	1.2	0.2	(1.0)	3.5	1.0	(2.5)
Customer Operations	10.4	10.2	(0.1)	0.7	0.8	0.1	11.1	11.0	(0.1)
Totals	**12.7**	**11.0**	**(1.7)**	**1.9**	**1.0**	**(0.9)**	**14.6**	**12.0**	**(2.6)**

Customer Program View Full Year $M

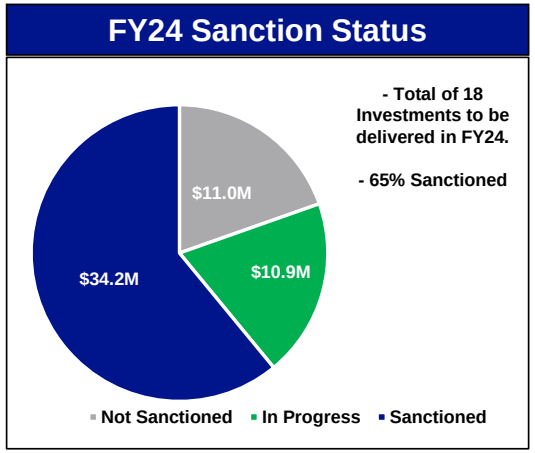
Portfolio / Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Customer Customer Digital	13.2	3.1	(10.2)	1.1	0.9	(0.2)	14.3	4.0	(10.4)
Customer Operations	38.9	44.7	5.8	3.3	3.8	0.5	42.2	48.6	6.3
Totals	**52.1**	**47.8**	**(4.3)**	**4.4**	**4.7**	**0.3**	**56.6**	**52.5**	**(4.1)**

Key Challenges

Challenge	Escalation Point
AMI – Continue work with NY PSC staff to address concerns with Gas meter module, coating was approved 8/2, working through final approval. Currently investigating alternative options.	Simon Baker



Mobilisation Summary



Status

- The team continues to work on sanctioning the remaining funds for investments planned to be delivered in FY24

Resources

- Most of the planned investments to be delivered in FY24 are active and resourced, with the exception of three that are currently undergoing requirements analysis.

Looking Forward

- Continue to sanction and resource investments for FY24 delivery.

Priority Investments

	Program	Status	Totex Budget ($m)	RAG	RAG Last Month
Core IT Investments	CIS Platform	Inflight	18.8		
	Contact Center	Inflight	10.5		
	Customer Data Platform	Inflight	6.0		
	Digital Experience	Inflight	13.4		
	Energy Services	Inflight	3.8		
Strategic IT-Enabled Investments for the Customer Business (tracked outside of Core IT Portfolio)	Zero Based Redesign (ZBR)	Inflight	7.0		
	Advanced Metering Infrastructure (AMI) – NY	Inflight	7.1		
	Advanced Metering Infrastructure (AMI) – MA	Inflight	1.7		
	CRIS to CSS (C2C)	Inflight	9.4		
	Energy Efficiency (Clean Energy 2.0)	Inflight	1.8		

US Security
Q1 FY24 Report

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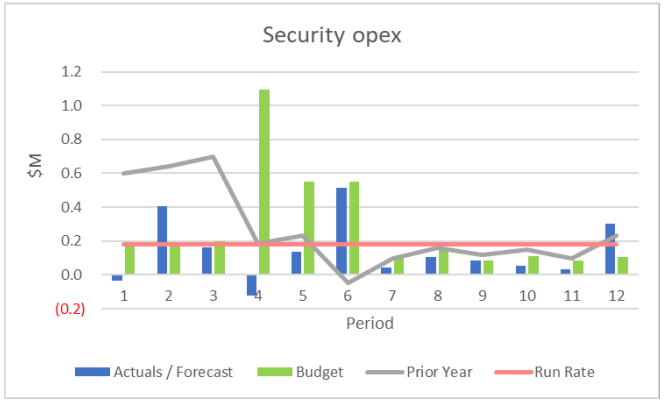
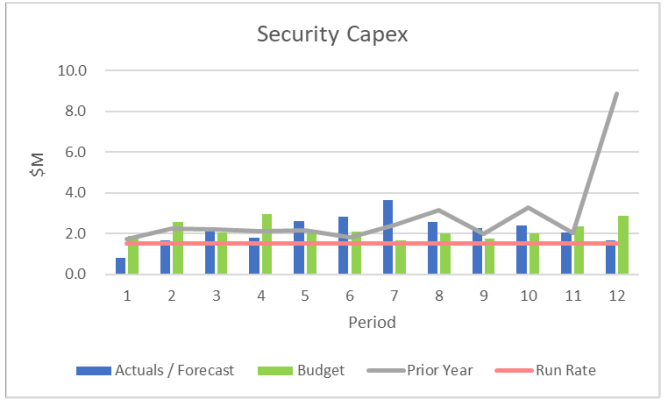
Security

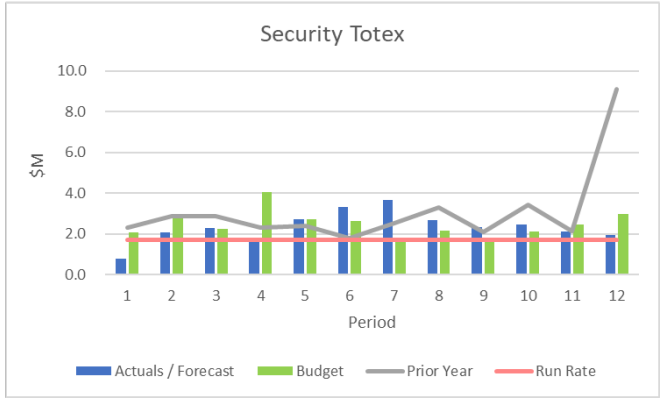
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	29.8	28.1
	YTD Budget($M)	YTD Actual ($M)
	7.1	5.1

Capex	Budget ($M)	Forecast ($M)
	26.4	26.4
	YTD Budget($M)	YTD Actual ($M)
	6.5	4.6



Portfolio Commentary

Financials

- YTD performance below budget due to adjustments to project phasing. Expected to ramp up to budget targets as sanctions have been completed.
- Physical Security break-fix lower than forecasted.
- Prior Year Construction Overhead adjustment (~500k) resulted in project credits.
- Lower priority initiatives have been reprioritized to fund and resource walk-in initiatives, including Data Privacy Automation.

Q1 Releases

- [redacted]

Security Program View of Financials

Security Program View Year to Date $M

Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Security	Physical Security	0.3	1.0	0.8	0.0	0.0	0.0	0.3	1.0	0.8
	Cyber Operational Technology	2.2	3.0	0.8	0.0	0.0	(0.0)	2.2	3.0	0.8
	Security	2.1	2.5	0.4	0.5	0.6	0.0	2.6	3.0	0.4
	Totals	**4.6**	**6.5**	**1.9**	**0.5**	**0.6**	**0.0**	**5.1**	**7.1**	**2.0**

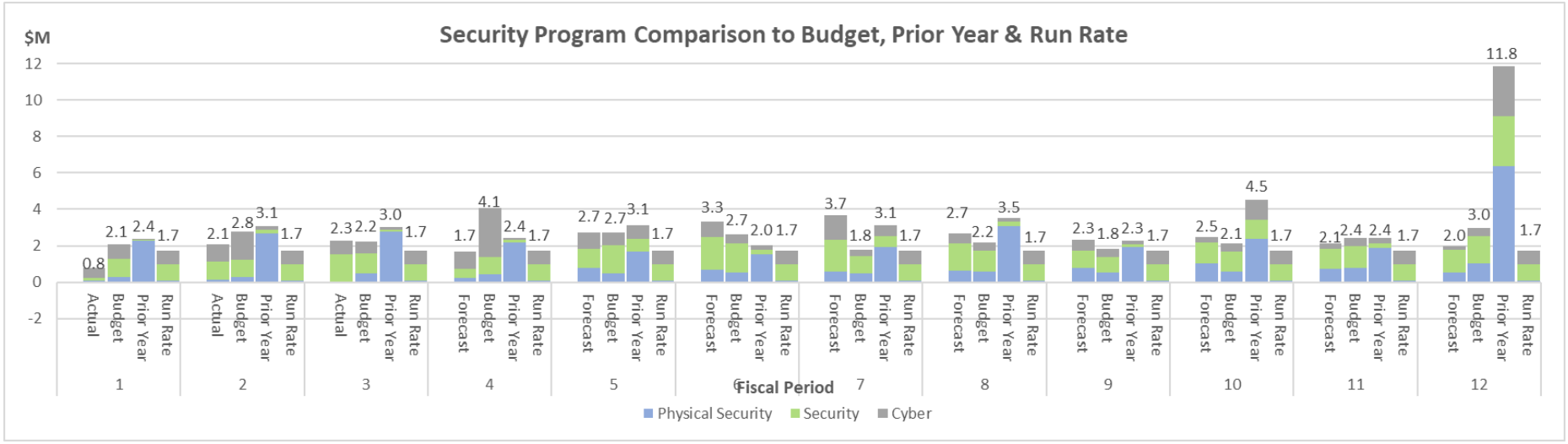
Security Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Security	Physical Security	6.3	6.4	0.1	0.0	0.0	0.0	6.3	6.4	0.1
	Cyber Operational Technology	8.1	8.5	0.4	0.0	1.0	1.0	8.2	9.5	1.3
	Security	12.0	11.5	(0.5)	1.7	2.4	0.7	13.6	13.9	0.3
	Totals	**26.4**	**26.4**	**0.0**	**1.7**	**3.4**	**1.7**	**28.1**	**29.8**	**1.7**

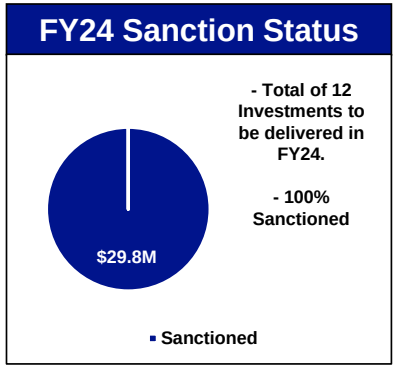
Key Challenges

Challenge	Escalation Point
Final budget of $26.4m below projected FY23 out-turn of ~$34m Capex. Deferred spend continues to be an opportunity if need to over-spend is required.	Not required

[redacted]



Mobilisation Summary



Status

- All planned projects have been sanctioned. Walk-In's are in review process

Resources

- Projects are resourced appropriately.

Scope

- Project scope is continuously assessed to ensure alignment to strategic objectives are responsive to real-time threat intelligence.

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
[redacted]				

US Group Functions
Q1 FY24 Report

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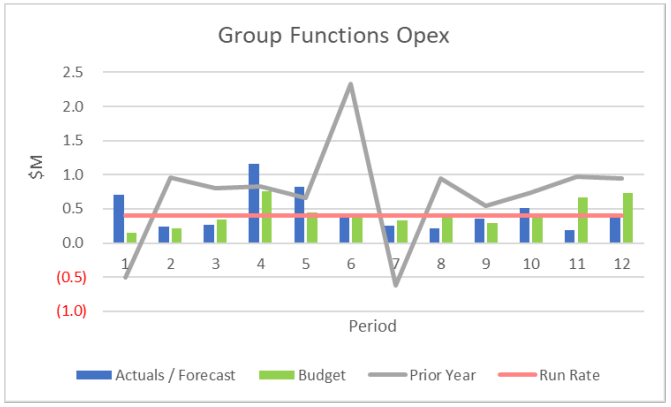
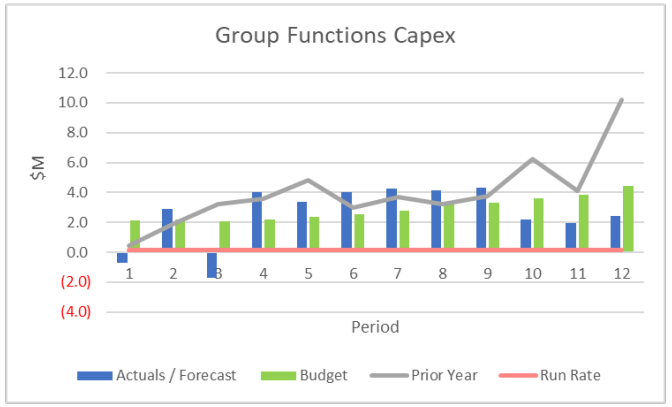
Group Functions

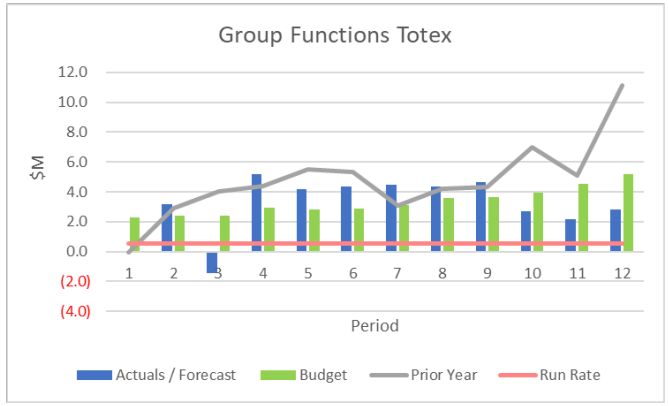
Financial Performance





Totex	Budget ($M)	Forecast ($M)
	39.7	36.6
	YTD Budget($M)	YTD Actual ($M)
	7.1	1.7

Capex	Budget ($M)	Forecast ($M)
	34.6	31.1
	YTD Budget($M)	YTD Actual ($M)
	6.4	0.5



Portfolio Commentary

Financials

- The majority of the year-to-date variance is made of accounting adjustments - $2.6m credit for E-hub and $1.1m credit for MyHub for a missed accrual, both related to inter-company adjustments. An additional $1.1m was the COH adjustment credit.
- Full year forecast variance is due to scope and cost pressures across the portfolio.

Q1 – Successes and Go-Lives

- **R2R:** Reduction of manual journals through business adoption of RPA for prep, park and post of regulatory deferrals and pension journals (3%). Deployed 12 US GAAP templates for statutory reporting within Wdesk. Use of standardized templates will reduce cycle times and minimize manual preparation and risk of error.
- **S2P - Ariba:** Release 1 Go-Live in May 2023, enabled 1,400 suppliers, 11 buying channels, and 48 catalogs to continue to increase e-invoicing and catalog coverage. Ariba Release 2 sanction paper to be submitted imminently and will automate accruals in Ariba to further reduce manual journal entries and complete the exit from SRM for all procurement activities. Completed F&A for Planning Repository and are recommending replacing Quickbase with Planview, the team is working to build out the business case and sanction paper with a target kickoff in Q3 for Quickbase implementation.
- **J2L:** Sodales Union Job Bidding solution digitizing paper-based process for union job bidding successfully rolled out to all US jurisdiction recruiters and hiring managers (~950). Payroll - Cost Center & Time Reporting data quality improvement of ~30% by remediating data sync issues (R1 May 2023).
- **PLTP:** Implementation has commenced. First phase of go-live for Group functions Opex in October. Architectural agreement on the use of Snowflake has been achieved.
- **SEA**: Launched initiative for a global Incident Management and reporting product/platform (Cority). Commencing execution in August and once fully launched, will reduce NG RTB by ~$200k.

Group Functions Program View of Financials

Group Functions Program View Year to Date $M

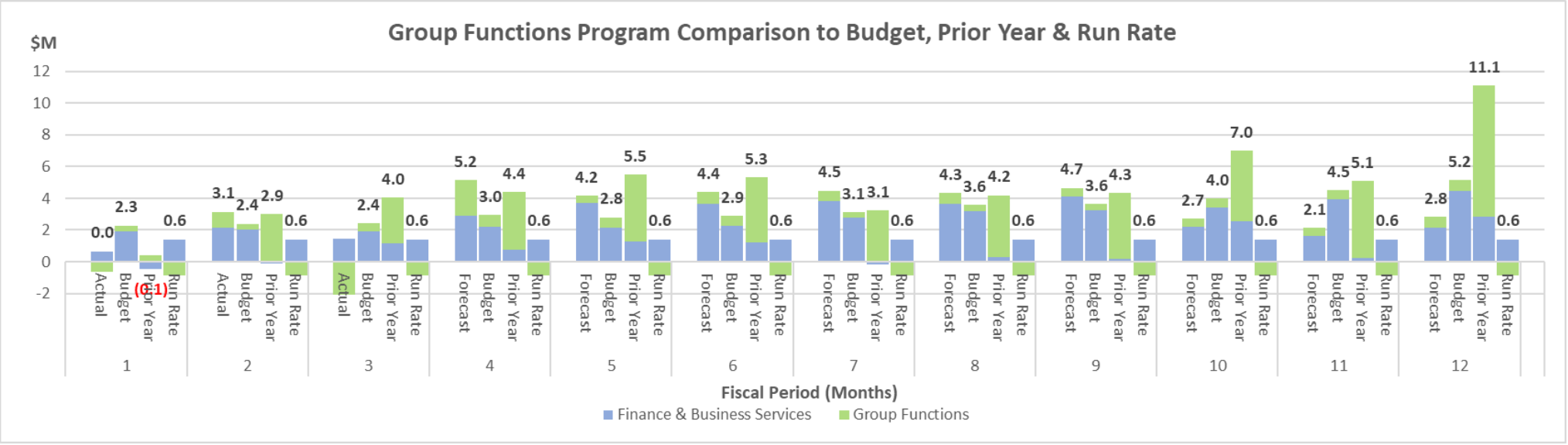
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
Group Functions	Finance & Business Services	3.3	5.2	2.0	1.0	0.7	(0.3)	4.2	5.9	1.7
	Group Functions	(2.8)	1.1	4.0	0.3	0.0	(0.2)	(2.6)	1.2	3.7
	Totals	**0.5**	**6.4**	**5.9**	**1.2**	**0.7**	**(0.5)**	**1.7**	**7.1**	**5.4**

Group Functions Program View Full Year $M

Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
Group Functions	Finance & Business Services	28.0	29.4	1.3	4.1	4.3	0.1	32.2	33.6	1.5
	Group Functions	3.1	5.3	2.2	1.4	0.8	(0.5)	4.4	6.1	1.7
	Totals	**31.1**	**34.6**	**3.5**	**5.5**	**5.1**	**(0.4)**	**36.6**	**39.7**	**3.2**

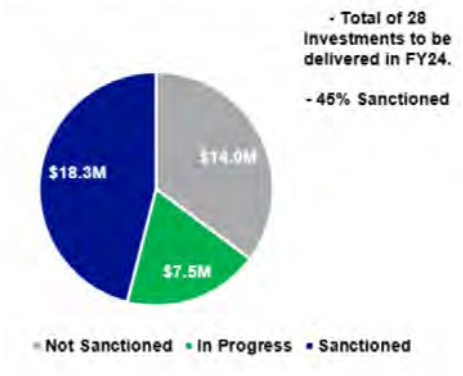
Key Challenges

Challenge	Escalation Point
R2R - Risk of full business engagement for R2R initiatives delivered near fiscal year end freeze.	Engaging business early to agree release schedule and delivery plans.
J2L - Re-assessing business requirements and developing an HR Digital roadmap along with business case	Align to investment strategy and funding needs. Contingent workforce tracking and funding discussions are underway with stakeholders.
S2P - Palantir support structure vendor agreement ends. (Palantir supports the Fraud Analytics in P2P)	Negotiations with WIPRO to identify long term application support
PLTP - Work on controls and security approvals for use of Snowflake in progress but need to be in place to satisfy readiness for Go live criteria.	Key dependency identified for data quality workstream.



Mobilisation Summary

FY24 Sanction Status



Key Outcomes to be Achieved in FY24

- **Asset Health** - Migrate 101 interfaces from PI to PO to mitigate security risks and ensure latest encryption standards are met.
- **Join to Leave** - People Digital Transformation - Focus on Core, Data and Processes, revert MyHub to standard, enabling People Function Target Operating Model. This is intended to drive Operational Efficiency, User Experience focussed on Self Service, leading to improved Data Quality and Compliance.
- **Record to Report** - Increased standardization and automation in reporting. Automate data feeds and leverage Wdesk to reduce use of offline files. Continue to reduce manual effort associated with journal entries via RPA and Blackline integration.
- **Source to Pay** - Implement Ariba Guided Buying across US (NY and NE) enabling intelligent catalog search resulting in increased electronic purchasing, reducing free text po's, and increasing electronic invoicing. Implement next phase of Ariba and upon completion enable retiring SAP SRM system.
- **Plan to Perform** - Implement three separate product releases to lessen the dependency on offline financial models by automating calculations. Will enable collaborative working in connected data environments which will reduce the planning cycle from 15 to 3 weeks, thereby releasing resource effort to focus on other priorities.

Priority Investments

Program	Status	Budget ($M)	RAG This Qtr.	RAG Last Qtr.
Record to Report (R2R)	Inflight	2.3		
Plan to Perform (PLTP)	Inflight	1.8		
Join To Leave (J2L)	Inflight	6.6		
Strategic External Affairs (SEA) / General Counsel (GC)	Inflight	7.7		
Asset Health (AH)	Inflight	13.2		
Source to Pay (S2P) / Operations (OPS)	Inflight	8.1		

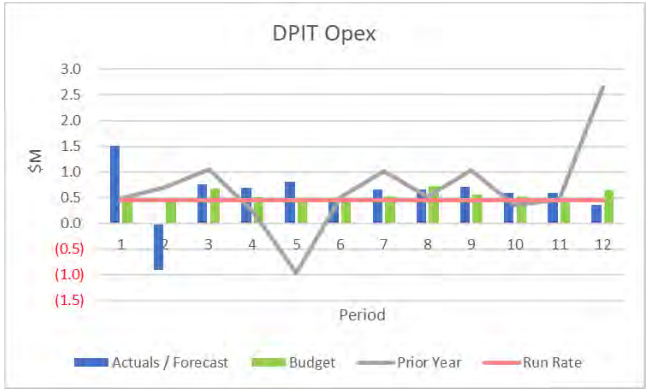
Digital Platforms & Infrastructure Technology (DPIT)

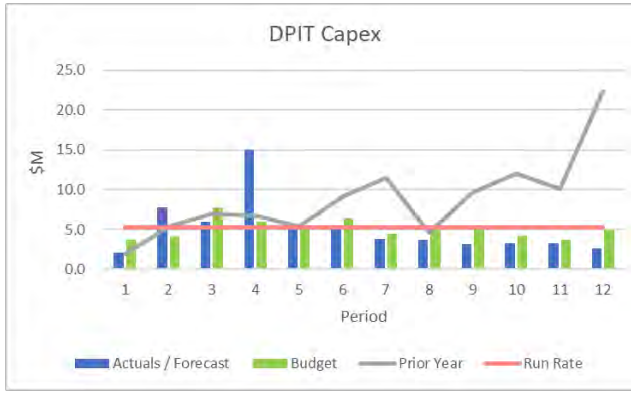
Q1 FY24 Report

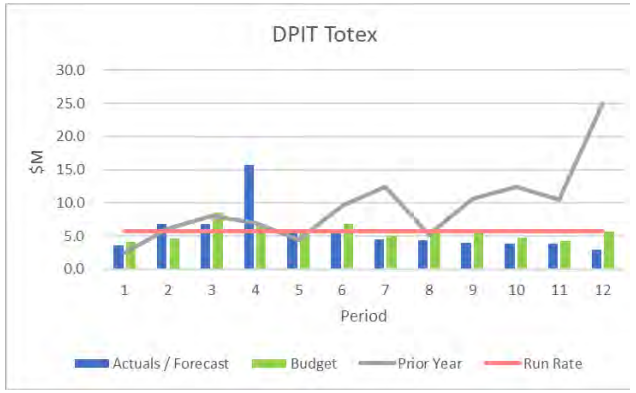
nationalgrid

DPIT

Financial Performance







Totex	Budget ($M)	Forecast ($M)
	67.7	67.9
	YTD Budget($M)	YTD Actual ($M)
	17.2	17.2

Capex	Budget ($M)	Forecast ($M)
	61.2	61.0
	YTD Budget($M)	YTD Actual ($M)
	15.6	15.8

Portfolio Commentary

Financials

- YTD and forecast are on track.

Q1 Releases

- US Data Center Transformation (DCT) on track for migrations and decommissions to exit DXC Datacenter by August 31. 744 of 1010 servers in scope for migrations complete (74%), 533 servers decommissioned.
- SharePoint 2007 Migrations completed reducing Technical Debt enabling US DCT.
- Microsoft Teams improvements include – user naming conventions, external tools to improve collaboration and productivity external Webex and Zoom users.
- NOW Mobile – application implemented to provide fast access to IT support services on all corporate mobile devices
- Network updates:
- 14 sites (512 users) - Teams Voice rollout to replace EOL solutions.
- 9 of 13 Syracuse voice gateways were consolidated and decommissioned
- 2 sites (Ogdensburg and Albany State Street) successfully converted to SD WAN.
- CNI – Replaced 28 EOL Wireless Routers in NY, to improve security, reduce technical debt, reduce operational risks, and support Optel circuit replacements
- SolarWinds successfully integrated with ServiceNow, allowing thresholds and Alert levels from SolarWinds to automatically generate incidents in ServiceNow for Unified Technology Operations Center (UTOC) to receive.
- Configuration Management Database (CMDB) – auto-discovery of 120 undiscovered devices bringing total auto-discovered devices to 97%
- Work Management Product – improvements to enable users to access work request data providing insight into work performed on our assets

DPIT Program View of Financials

DPIT Program View Year to Date $M

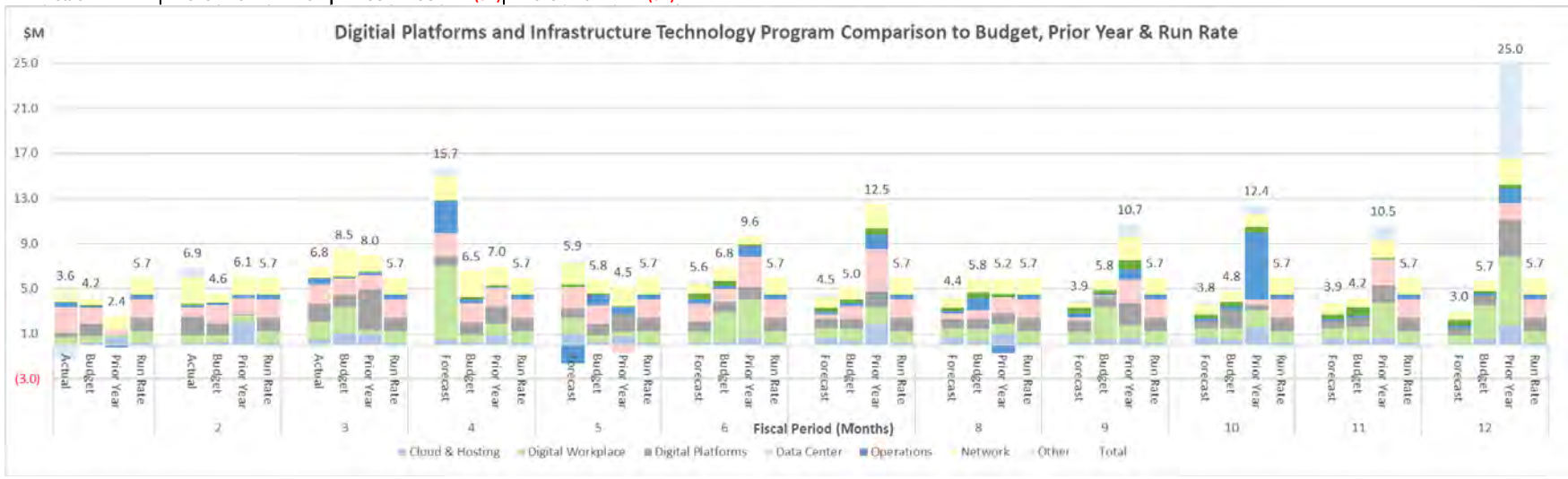
Portfolio	Program	Capex Actuals YTD	Capex Budget YTD	Capex Variance	Opex Actuals YTD	Opex Budget YTD	Opex Variance	Totex Actuals YTD	Totex Budget YTD	Totex Variance
DPIT	Cloud & Hosting	0.6	1.3	0.8	0.0	0.2	0.2	0.6	1.6	0.9
	Digital Workplace	2.8	3.2	0.4	0.2	0.2	(0.1)	3.0	3.4	0.3
	Digital Platforms	3.3	2.7	(0.6)	0.2	0.6	0.3	3.5	3.3	(0.2)
	Data Center	4.9	4.5	(0.5)	0.0	0.0	(0.0)	5.0	4.5	(0.5)
	Operations	0.9	0.4	(0.5)	0.2	0.0	(0.1)	1.0	0.4	(0.6)
	CNI	0.2	0.3	0.1	0.0	0.0	0.0	0.2	0.3	0.1
	Network	4.0	3.3	(0.7)	0.5	0.5	0.0	4.4	3.8	(0.6)
	Other	(0.8)	0.0	0.8	0.2	(0.0)	(0.2)	(0.6)	(0.0)	0.6
	Totals	**15.8**	**15.6**	**(0.2)**	**1.4**	**1.6**	**0.2**	**17.2**	**17.2**	**0.0**

DPIT Program View Full Year $M

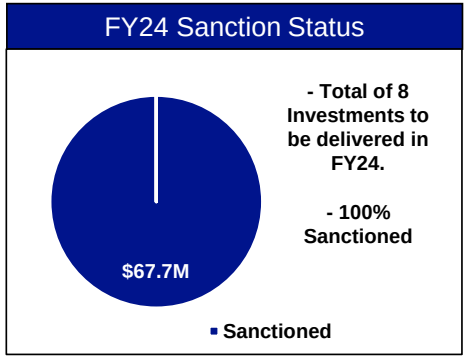
Portfolio	Program	Capex FY Forecast	Capex FY Budget	Capex Variance	Opex FY Forecast	Opex FY Budget	Opex Variance	Totex FY Forecast	Totex FY Budget	Totex Variance
DPIT	Cloud & Hosting	4.3	4.4	0.1	0.8	0.8	(0.0)	5.1	5.2	0.1
	Digital Workplace	16.2	16.3	0.1	0.8	0.9	0.0	17.0	17.2	0.1
	Digital Platforms	8.9	10.5	1.6	1.8	2.0	0.2	10.7	12.5	1.8
	Data Center	11.4	10.7	(0.7)	0.3	0.3	(0.0)	11.7	11.1	(0.7)
	Operations	3.5	3.6	0.0	0.5	0.4	(0.0)	4.0	4.0	(0.0)
	CNI	3.6	3.6	0.0	0.1	0.1	0.0	3.7	3.7	0.0
	Network	12.2	12.0	(0.1)	2.1	2.0	(0.1)	14.2	14.0	(0.2)
	Other	0.9	0.0	(0.9)	0.5	(0.0)	(0.5)	1.4	(0.0)	(1.4)
	Totals	**61.0**	**61.2**	**0.2**	**6.9**	**6.5**	**(0.4)**	**67.9**	**67.7**	**(0.2)**

Key Challenges

Challenge	Escalation Point
DCT Exceptional OpEx funding – still pending approval. Get to Green: Continuing to work with Finance/Strategy for Approval of these funds. Temporary solution in place.	Finance & Strategy Teams
DCT Challenge - program will remain Amber due to tight timescales and amount of remaining servers to migrate.	Ongoing monitoring
US Device refresh on hold due to spike in break/fix and new hires	Ongoing monitoring



Mobilisation Summary



Status
- All FY24 investments have been fully sanctioned, except for DWS (partial), Ehub walk-in (in progress)

Resources
- No resource risks

Scope
- DPIT product portfolios on-track in FY24 to both Objectives/Key Results (OKRs) and financial forecast of $67.7m, except:
  - Data Center Transformation: Internal and external risks are being closely monitored

Priority Investments

Program	Status	Budget ($m)	RAG This Month	RAG Last Month
Data Center Transformation (DCT)	Inflight	11.06		
Cloud & Hosting	Inflight	5.18		
Digital Platforms Includes $1.62M for eHub	Inflight	14.14		
Digital Workplace Services	Inflight	17.15		
Networks	Inflight	14.05		
Operations	Inflight	3.99		
CNI	Inflight	3.73		

				YTD					Full Year Forecast			
				YTD Budget Variance			Non_Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	3 + 9 Total	Budget Total	YTD Variance	Reason
US Gas	Gas BU	4769	Gas Storm Hardening Using Remote Service Valves (RSV's)	60,849		(60,849)		Timing	60,849		(60,849)	Timing
US Gas	Gas BU	4898	Asset and Work Management System for LNG	4,827		(4,827)		Immaterial	4,827		(4,827)	Immaterial
US Gas	Gas BU	5274A	New Storms Response Management System Implementation	961		(961)		Immaterial	961		(961)	Immaterial
US Gas	Gas BU	5785Q	Field Collection Systems (FCS) Upgrade (Gas Only)	88,579	600,800	512,221		Timing	388,937	1,130,000	741,063	Imprecise Estimates
US Gas	Gas BU	5787D	Gas Transportation System (GTS) Product	1,115,003	784,000	(331,003)		Imprecise Estimates	1,115,003	4,999,998	3,884,995	Imprecise Estimates
US Gas	Gas BU	6392	Contractor Ipad for Gas Business	14,529		(14,529)		Accounting	14,529		(14,529)	Accounting
US Gas	Gas BU	6441	Scalability & Resilience for Enterprise Gas Platforms	3,276		(3,276)		Immaterial	3,276		(3,276)	Immaterial
US Gas	Gas BU	6472	MOC Library & Action Item Log Automation	(35,298)		35,298		Accounting	(35,298)		35,298	Accounting
US Gas	Gas BU	6479	LNG IT Support for OT Systems	255,388	229,600	(25,788)		Scope changes	1,615,830	1,435,000	(180,830)	Scope changes
US Gas	Gas BU	6550	Gas Management System (SCADA) Upgrade'	868,380	2,781,840	1,913,460	Delays		17,059,294	17,100,002	40,708	Variance less than 10%
US Gas	Gas BU	6573	FY23 Gas Regulatory and Operational Enhancements (Minor Works)	(24,364)		24,364		Accounting	(24,364)		24,364	Accounting
US Gas	Gas BU	6582	Gas Application Continuous Improvement	(187,714)		187,714	Mischarges		(187,714)		187,714	Accounting
US Gas	Gas BU	6610	Gas App Rationalization	19,929		(19,929)		Accounting	19,929		(19,929)	Accounting
US Gas	Gas BU	6642	Tesco	663,316	513,600	(149,716)	Timing differences		2,858,387	3,210,000	351,613	Accounting
US Gas	Gas BU	6664A	Enable Converted LI Gas Distribution Asset Data in ESRI		80,000	80,000		Funding Change		500,000	500,000	Funding Change
US Gas	Gas BU	6664F	Standalone RMD Solution	36,865		(36,865)		Opex Only Investment	369,868		(369,868)	Opex Only Investment
US Gas	Gas BU	6671	US Gas Data Modern-E	208,447		(208,447)	Timing differences	Funding Change	3,974,315		(3,974,315)	Funding Change
US Gas	Gas BU	6671A	Gas Application Lifecycle Management (ALM) Application Rationalization		164,800	164,800		Funding Change		1,030,000	1,030,000	Funding Change
US Gas	Gas BU	6671B	GACI (Gas Application Continuous Improvements)		219,200	219,200		Funding Change		1,370,000	1,370,000	Funding Change
US Gas	Gas BU	6671C	ES&R (Enterprise Scalability & Resilience)		232,000	232,000		Funding Change		1,450,000	1,450,000	Funding Change
US Gas	Gas BU	6671D	US Gas Data Modern-E	430,254	399,999	(30,255)		Variance less than 10%	1,438,799	1,599,996	161,197	Imprecise Estimates
US Gas	Gas BU	6671E	FY24 Gas Regulatory and Operational Enhancements (Minor Works)		36,655	36,655		Funding Change		150,001	150,001	Funding Change
US Gas	Gas BU	PZG24G	Opco Adj NY & US Gas		3	3		Immaterial		12	12	Immaterial
US Gas	CS&E	6539	Continued Scaling and Evolution of Gas IT Transformation Program	19,637,712	10,217,600	(7,100,000)	Accruals		64,046,952	63,860,000	(186,952)	Variance less than 10%
US Gas	CS&E	6539	Continued Scaling and Evolution of Gas IT Transformation Program			(2,320,112)	Timing differences					See total 6539 variance
US Gas Total				**23,160,939**	**16,260,097**	**(6,900,842)**			**92,724,380**	**97,835,009**	**5,110,629**	

				YTD					Full Year Forecast			
				YTD Budget Variance			Non_Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	3 + 9 Total	Budget Total	YTD Variance	Reason
US Electric	EBU Digital	5054	On My Way (EBU Digital Solution) Phase 2	485,754	856,000	370,246	Timing differences	Accounting	4,944,897	5,350,000	405,103	Variance less than 10%
US Electric	EBU Digital	6073	Veg Management Optimization	1,432,293	344,000	(1,088,293)	Timing differences	Accounting	3,168,456	2,150,000	(1,018,456)	Accounting
US Electric	EBU Digital	6076	Future Now - validation phase	1,605,917	944,000	(661,917)	Other	Funding Change	6,779,193	5,900,000	(879,193)	Funding Change
US Electric	EBU Digital	6291	ICU-Out (Electric Digital)	260,115		(260,115)		Accounting	260,115		(260,115)	Accounting
US Electric	EBU Digital	6425	DER Active Resource Integration NY	442		(442)		Immaterial	442		(442)	Immaterial
US Electric	EBU Digital	6437A	Grid MD	271,154		(271,154)	Prioritizations		358,053		(358,053)	Funding Change
US Electric	EBU Digital	6661	Demand Planning and Material Track & Trace	105,958	880,000	774,042	Timing differences	Accounting	105,958	5,500,000	5,394,042	Funding Bucket
US Electric	EBU Digital	6662	Proj Costing Cont Mgmt	448,621	432,000	(16,621)		Variance less than 10%	4,470,865	2,700,000	(1,770,865)	Accounting
US Electric	EBU Digital	6672	DER Scenario Forecast & Planning	10,725		(10,725)		Accounting	10,725		(10,725)	Accounting
US Electric	EBU Digital	6673	DER Connections	11,031		(11,031)		Accounting	11,031		(11,031)	Accounting
US Electric	Electric BU	5773B	Visitor Logging for NERC-CIP Reliability Compliance	58,446		(58,446)	Other	Accounting	58,446		(58,446)	Accounting
US Electric	Electric BU	5773E	NERC-CIP Reliability Compliance	(69,277)		69,277	Accruals		(69,277)		69,277	Accounting
US Electric	Electric BU	5871DA	SD-WAN for Distributed Communications	111,625	64,000	(47,625)	Other		367,851	400,000	32,149	Variance less than 10%
US Electric	Electric BU	5871E	Power Marketer Pilot	4,404		(4,404)		Immaterial	4,404		(4,404)	Immaterial
US Electric	Electric BU	5871F	Battery Storage Communications Upgrade	64,208		(64,208)		Opex Only Investment	64,208		(64,208)	Opex Only Investment
US Electric	Electric BU	5887A	ARCOS Reporting Replacement	16,158		(16,158)	Accruals	Accounting	16,158		(16,158)	Accounting
US Electric	Electric BU	5889D	Electric FY23 Regulatory Mandates and Minor Enhancements	320		(320)		Immaterial	320		(320)	Immaterial
US Electric	Electric BU	5889E	Regulatory Enhancements (Minor Works)	12,808	16,000	3,192		Imprecise Estimates	148,808	100,000	(48,808)	Imprecise Estimates
US Electric	Electric BU	5890A	Corporate PI	66,533	3,200	(63,333)	Delays	Other	70,712	20,000	(50,712)	Accounting
US Electric	Electric BU	5890C	GridInspector F&A	1,070,251	757,445	(312,806)		Timing	4,643,846	4,480,000	(163,846)	Variance less than 10%
US Electric	Electric BU	5890D	IDS Replacement	6,794	440,000	433,206	Timing differences	Imprecise Estimates	2,756,794	2,750,000	(6,794)	Variance less than 10%
US Electric	Electric BU	5891A	Electric MV90 Upgrade	181,757		(181,757)		Scope changes	263,591		(263,591)	Scope changes
US Electric	Electric BU	5891C	Field Collection System (FCS) Upgrade	647,797		(647,797)		Scope changes	621,101		(621,101)	Scope changes
US Electric	Electric BU	5891E	MTS Retail Settlement (MDS)	430,303	646,200	215,897	Delays	Accounting	3,750,014	4,038,751	288,737	Variance less than 10%
US Electric	Electric BU	5891F	Meter Data Services Consolidation Phase 1		273,600	273,600	Prioritizations		1,700,000	1,710,000	10,000	Variance less than 10%
US Electric	Electric BU	5891I	Price Transmission (Retail Pricing) Replacement	1,160		(1,160)		Opex Only Investment	1,160		(1,160)	Opex Only Investment
US Electric	Electric BU	5892B	Control Room Servers and Tools Upgrade/Replacement	(209,001)		209,001		Scope changes	(199,021)		199,021	Other
US Electric	Electric BU	5892C	IDS-DVD 811 Dig Safe DVD Replacement	48,454		(48,454)		Accounting	48,454		(48,454)	Accounting
US Electric	Electric BU	6571	GridAPI Electric	28,374		(28,374)		Accounting	28,374		(28,374)	Accounting
US Electric	Electric BU	6631	US Electric Rapid Rebuild	50,029	464,000	413,971	Timing differences		2,607,836	2,900,000	292,164	Other
US Electric	Electric BU	6736	m360	22,000		(22,000)			22,000		(22,000)	Funding Change
US Electric	Electric BU	6741B	Ambient Adjusted Line Ratings Management			0		Variance less than 10%	774,175		(774,175)	Funding Change
US Electric	Electric BU	NTWKEQUIP	Network Equipment End of Life Replacement - LAN		392,000	392,000		Funding Change		2,450,000	2,450,000	Funding Change
US Electric	Electric BU	PZG24E	Opco Adj NE & US Electric		12	12		Immaterial		48	48	Immaterial
US Electric	Wholesale Networks	5773D	NYISO PMU Requirement for Large Generators	232,809	24,000	(208,809)		Complexity	191,092	150,000	(41,092)	Complexity
US Electric	Wholesale Networks	5822C	Generation Asset Management Upgrade	912,334	40,000	(872,334)		Complexity	4,211,530	250,000	(3,961,530)	Complexity
US Electric	Wholesale Networks	6043G1	Content Management for Engineering	472,836		(472,836)		Scope changes	672,692		(672,692)	Scope changes
US Electric Total				**8,793,135**	**6,576,457**	**(2,216,678)**			**42,865,005**	**40,848,799**	**(2,016,207)**	

				YTD					Full Year Forecast			
				YTD Budget Variance			Non_Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	3 + 9 Total	Budget Total	YTD Variance	Reason
Customer	Customer Digital	4750J	Web Self Service	4,219		(4,219)		Immaterial	4,219		(4,219)	Immaterial
Customer	Customer Digital	4750M	ODS	(12,711)		12,711		Accounting	(12,711)		12,711	Accounting
Customer	Customer Digital	5673	My Business Account (Nucleus Phase 2)	2,575,998	519,102	(1,474,369)		Accounting	5,413,963	2,099,997	(1,474,369)	Accounting
Customer	Customer Digital	5673	My Business Account (Nucleus Phase 2)			(582,527)		Scope changes			(1,839,597)	Scope changes
Customer	Customer Digital	5859A	Unified Web Portal	(90)		90		Immaterial	(90)		90	Immaterial
Customer	Customer Digital	6706	UWP Enhancements		465,000	465,000	Timing differences		1,233,993	1,860,000	626,007	Funding Bucket
Customer	Customer Digital	6715	UWP 2.0 MyAccount Implementation and Delivery	932,120		(932,120)		Funding Change	7,703,906		(7,703,906)	Funding Change
Customer	Customer Operations	5125	Customer Connection Program	2,045,310	996,208	(778,733)		Accounting	4,988,621	3,749,995	(778,733)	Accounting
Customer	Customer Operations	5125	Customer Connection Program			(270,369)		Complexity			(459,893)	Complexity
Customer	Customer Operations	5160	NY Value Stack Billing (Mandated)	1,016,145	690,001	(326,144)		Complexity	1,846,672	1,070,001	(776,671)	Complexity
Customer	Customer Operations	5345B	Bill Print and Bill Composition Implementation	35,206		(35,206)		Timing	35,206		(35,206)	Timing
Customer	Customer Operations	5778	Customer Ecosystem Stabilization		1,056,000	(1,409)		Accounting	5,851,929	6,600,000	1,409	Accounting
Customer	Customer Operations	5778	Customer Ecosystem Stabilization			1,057,409		Funding Bucket			746,662	Funding Bucket
Customer	Customer Operations	5778B	CSS Infrast NGLeg+LI	666		(666)		Immaterial	666		(666)	Immaterial
Customer	Customer Operations	5778C	Encryption & DR AR	1,140		(1,140)		Immaterial	1,140		(1,140)	Immaterial
Customer	Customer Operations	5778F	RightFax AR	218,864		(218,864)		Other	218,864		(218,864)	Funding Change
Customer	Customer Operations	5996	Simon Baker's Efficiency Target			0		Variance less than 10%	(9,646,676)		9,646,676	Other
Customer	Customer Operations	6032	Customer Billing COVID Transition (Minor Works)	(1,599)		1,599		Immaterial	(1,599)		1,599	Immaterial
Customer	Customer Operations	6066	FCP Upgrade for CSS - Plan Forward	(1,026)		1,026		Immaterial	(1,026)		1,026	Immaterial
Customer	Customer Operations	6302	2022 AIMS Platform	68,341	2,374,998	68,341		Accounting	68,341	9,499,992	(68,341)	Accounting
Customer	Customer Operations	6302	2022 AIMS Platform			2,238,316		Accounting			9,499,992	Funding Change
Customer	Customer Operations	6349	Customer Data Platform	3,865,317	1,500,000	(2,365,317)		Accounting	3,865,317	6,000,000	2,134,683	Accounting
Customer	Customer Operations	6451	One Step Auto Cashering	196,515		8,892		Accounting	1,122,680		8,892	Accounting
Customer	Customer Operations	6451	One Step Auto Cashering			(205,407)		Funding Change			(1,131,572)	Funding Change
Customer	Customer Operations	6492	NACHA Compliance- Security Encryption Effort	382,260		33,869		Accounting	1,464,164		33,869	Accounting
Customer	Customer Operations	6492	NACHA Compliance- Security Encryption Effort			(416,129)		Complexity			(1,498,033)	Complexity
Customer	Customer Operations	6496	NY Gender Pronoun and Name Mandate	(21,675)		21,675		Accounting	(21,675)		21,675	Accounting
Customer	Customer Operations	6517	Digital Gas Connetion Experience		17,600	17,600		Mischarges		110,000	110,000	Mischarges
Customer	Customer Operations	6531	AIMS Upgrades and Enhancements	290,256	255,000	(68,714)		Accounting	108,639	1,020,000	(68,714)	Accounting
Customer	Customer Operations	6531	AIMS Upgrades and Enhancements			33,458		Timing			980,075	Funding Change
Customer	Customer Operations	6538	Billing & Collections Mandates & Enhancements FY23		1,782,400	1,782,400		Funding Bucket	13,274,000	11,140,000	(2,134,000)	Funding Bucket
Customer	Customer Operations	6543	Customer Minor Works	5,999		(5,999)		Accounting	5,999		(5,999)	Accounting
Customer	Customer Operations	6584	UWP 2.0 Digital Re Architecture and Optimization	2,532,343	2,345,001	289,421		Accounting	1,123,510	9,380,004	289,421	Accounting
Customer	Customer Operations	6584	UWP 2.0 Digital Re Architecture and Optimization			(476,763)		Timing			7,967,073	Timing
Customer	Customer Operations	6592	Transmission Services Agreement/Transmission Services Contract (TSA/TSC) Application Repla	306,717		(306,717)		Timing	316,906		(316,906)	Timing
Customer	Customer Operations	6654	Documentum Feasability & Analysis	115,550		(115,550)		Accounting	115,550		(115,550)	Timing differences
Customer	Customer Operations	6686	Documentum Application Modifications	(65,253)		65,253		Accounting	234,747		(234,747)	Funding Change
Customer	Customer Operations	6687	FY 24 Customer Minor Works	208,599		(208,599)		Funding Change	412,394		(412,394)	Other
Customer	Customer Operations	6713	Customer-Facing Mobile Application	(138,251)		138,251		Accounting	2,661,745		148,807	Accounting
Customer	Customer Operations	6713	Customer-Facing Mobile Application					See total 6713 variance			(2,810,552)	Funding Change
Customer	Customer Operations	6727	AIMS FY24 Product Team			0		Variance less than 10%	1,185,455		(1,185,455)	Other
Customer	Customer Operations	6728	AIMS Replatform	10,967		(10,967)		Other	8,865,637		(8,865,637)	Funding Change
Customer	Customer Operations	6740	FY 24 Customer Data Platform			0		Variance less than 10%	4,144,884		(4,144,884)	Funding Bucket
Customer	Customer Operations	PZG24C	Opco Adj Customer		1	1		Immaterial		3	3	Immaterial
Customer Total				**14,571,927**	**12,001,311**	**(2,570,616)**			**56,585,371**	**52,529,992**	**(4,055,379)**	

CIO Group	it_program
Security	Physical Security
Security	Physical Security
Security	Physical Security
Security	Physical Security
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Security	Security
Security	Security
Security	Cyber Operational Technology
Security	Cyber Operational Technology
Security Total	

[redacted]

				YTD					Full Year Forecast			
				YTD Budget Variance			Non_Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	3 + 9 Total	Budget Total	YTD Variance	Reason
Group Functions	Finance & Business Services	3870B	Vision FM replacement, Phase 2	(13,848)		13,848		Accounting	(13,848)		13,848	Accounting
Group Functions	Finance & Business Services	4418USC	GRC Archer FY23	(99,466)		99,466		Accounting	(99,466)		99,466	Accounting
Group Functions	Finance & Business Services	4418USD	CRO FY24: GRC Archer US	6,037		(6,037)		Funding Change	1,149,037		(1,149,037)	Funding Change
Group Functions	Finance & Business Services	5466	AMAG HW and SW Upgrade	(28,218)		28,218		Accounting	(28,218)		28,218	Accounting
Group Functions	Finance & Business Services	5497	Core Powerplan Upgrade and Lease Consolidation	33,773		(33,773)		Accounting	33,773		(33,773)	Accounting
Group Functions	Finance & Business Services	5606	US SAP: Infrastructure Landscape FY22	33,883		(33,883)		Accounting	33,883		(33,883)	Accounting
Group Functions	Finance & Business Services	5607	US SAP: Infrastructure Landscape FY23	(115,533)		115,533		Accounting	(115,533)		115,533	Accounting
Group Functions	Finance & Business Services	5608	US SAP: Infrastructure Landscape FY24	847,118	700,820	(146,298)	Accruals		3,742,006	4,050,820	308,814	Variance less than 10%
Group Functions	Finance & Business Services	5612	Annual HR & Payroll Mandatory Service Pack Upgrade (HRSP) FY24	327,102	224,000	(103,102)	Timing differences		1,208,330	1,400,000	191,670	Imprecise Estimates
Group Functions	Finance & Business Services	5842H	SAP Cloud Analytics		112,000	112,000	Delays		700,000	700,000	0	Variance less than 10%
Group Functions	Finance & Business Services	5843	SAP and BackOffice Health Program		288,000	288,000		Funding Bucket	1,640,000	1,800,000	160,000	Variance less than 10%
Group Functions	Finance & Business Services	5843E	US SAP: BW/BI Upgrade	370,672	164,765	(205,907)		Imprecise Estimates	1,984,946	1,600,000	(384,946)	Imprecise Estimates
Group Functions	Finance & Business Services	5843F	US SAP PI/PO Upgrade	573,881	720,000	146,119		Accounting	2,806,270	2,773,000	(33,270)	Variance less than 10%
Group Functions	Finance & Business Services	5895	US Group Functions: Operational Excellence		118,000	118,000		Funding Bucket		580,000	580,000	Funding Bucket
Group Functions	Finance & Business Services	5895B	Everbridge / Critical Event Management	(24,738)		24,738		Accounting	(24,738)		24,738	Accounting
Group Functions	Finance & Business Services	5895C	US Group Function: Tools and Technologies	209,267		(209,267)		Funding Change	487,013		(487,013)	Funding Change
Group Functions	Finance & Business Services	5902	Applications Health Check (Non-SAP) Program		192,000	192,000		Funding Bucket		1,227,730	1,227,730	Funding Bucket
Group Functions	Finance & Business Services	5902A	US Group Functions Operational requests	196,150		(196,150)		Funding Change	1,816,365		(1,816,365)	Funding Change
Group Functions	Finance & Business Services	5919	US SAP Minor Enhancements	(25,065)		25,065		Accounting	(25,065)		25,065	Accounting
Group Functions	Finance & Business Services	6137US	Enterprise PMO, Jira Align release 2	(5)		5		Immaterial	(5)		5	Immaterial
Group Functions	Finance & Business Services	6180US	CV Talent Screening / Blind CVs	58,137		(58,137)		Other	58,137		(58,137)	Other
Group Functions	Finance & Business Services	6183US	HR Reporting Analytics	(8,370)		8,370		Accounting	(8,370)		8,370	Accounting
Group Functions	Finance & Business Services	6185US	Onboarding 2.0	(18,603)		18,603		Accounting	(18,603)		18,603	Accounting
Group Functions	Finance & Business Services	6220	US F&BS Minor Enhancements (SAP)	9,064		(9,064)		Accounting	9,064		(9,064)	Accounting
Group Functions	Finance & Business Services	6221	Tech Debt: RiskMaster Replacement	(1,522)		1,522		Immaterial	(1,522)		1,522	Immaterial
Group Functions	Finance & Business Services	6270US	Digitising Learning	7,788		(7,788)		Accounting	7,788		(7,788)	Accounting
Group Functions	Finance & Business Services	6273US	Workforce Data Domain, HR data, governance, Tools, Processes, EDM	9,838		(9,838)		Accounting	9,838		(9,838)	Accounting
Group Functions	Finance & Business Services	6289E	FERC Form Integration with wDesk	(30,142)		30,142		Accounting	(30,142)		30,142	Imprecise Estimates
Group Functions	Finance & Business Services	6289F	US GAAP Statutory reporting with WDesk	81,587		(81,587)		Scope changes	81,587		(81,587)	Scope changes
Group Functions	Finance & Business Services	6289G	FTOM R2R: Robotic Process Automation of Manual Journal Entries	99,006		(99,006)		Imprecise Estimates	99,006		(99,006)	Imprecise Estimates
Group Functions	Finance & Business Services	6289H	RPA Strategic / Robotic Process Automation of Manual Journal Entries - Phase 2	13,839	174,400	160,561	Prioritizations		1,318,662	1,090,000	(228,662)	Imprecise Estimates
Group Functions	Finance & Business Services	6290A	Source To Pay: Release 1 – P2P Roll Out Program	1,967,993	2,100,000	132,007	Timing differences		2,877,123	2,202,678	(674,445)	Imprecise Estimates
Group Functions	Finance & Business Services	6290B	Source To Pay: Release 2 - Supplier Onboarding			0		Variance less than 10%	5,489,615	4,400,000	(1,089,615)	Scope changes
Group Functions	Finance & Business Services	6421	P2P Data Analysis Tool	877		(877)		Immaterial	877		(877)	Immaterial
Group Functions	Finance & Business Services	6426	Investment Recovery Digitization	97,230		(97,230)		Imprecise Estimates	103,096		(103,096)	Imprecise Estimates
Group Functions	Finance & Business Services	6467	Union Bidding	(105,630)		105,630	Mischarges		105,293		(105,293)	Timing
Group Functions	Finance & Business Services	6529	US Group Function Minor Enhancements (SAP)	(103,232)		103,232		Accounting	(103,232)		103,232	Accounting
Group Functions	Finance & Business Services	6530	US Group Function Operational Requests - FY23	137,778		(137,778)		Imprecise Estimates	137,778		(137,778)	Imprecise Estimates
Group Functions	Finance & Business Services	6533US	Group SHE Digitial Evolution - FY23		48,000	48,000		Funding Change		1,600,000	1,600,000	Funding Bucket
Group Functions	Finance & Business Services	6566US	HR Policy Compliance FY23	16,918		(16,918)		Accounting	16,918		(16,918)	Accounting
Group Functions	Finance & Business Services	6575A	R2R: Regulatory and Compliance - FY24	14,185		(14,185)		Funding Change	14,185		(14,185)	Funding Change
Group Functions	Finance & Business Services	6597	US Group Function: R2R regulatory and Compliance		86,400	86,400		Funding Bucket		540,000	540,000	Funding Bucket
Group Functions	Finance & Business Services	6621	JML MyHub - US		470,000	470,000		Funding Bucket		3,350,000	3,350,000	Funding Bucket
Group Functions	Finance & Business Services	6621US	My Hub	(1,137,070)		1,137,070	Accruals		(1,137,070)		1,137,070	Accounting
Group Functions	Finance & Business Services	6653	US Group Function Minor Enhancements (SAP) - Minor Works FY24-27		188,800	188,800		Funding Bucket		1,180,000	1,180,000	Funding Bucket
Group Functions	Finance & Business Services	6653USA	US Group Function Minor Enhancements (SAP) - Minor Works FY24-27	274,293		(274,293)		Funding Change	1,196,048		(1,196,048)	Funding Change
Group Functions	Finance & Business Services	6657US	Sustainability Apps and Tools FY24-27			0		Variance less than 10%	1,950,000	2,100,000	150,000	Variance less than 10%
Group Functions	Finance & Business Services	6690	Cost center and Time improvements	355,200		(355,200)		Funding Change	548,485		(548,485)	Funding Change
Group Functions	Finance & Business Services	6695US	CRO Funding Line - RSA Archer Upgrade & Refresh US		16,000	16,000		Funding Bucket		1,150,000	1,150,000	Funding Bucket
Group Functions	Finance & Business Services	6714	Join to Leave	211,137		(211,137)		Funding Change	211,137		(211,137)	Funding Change
Group Functions	Finance & Business Services	6716US	Business Services Upgrades and Enhancements			0		Variance less than 10%	1,109,702		(1,109,702)	Funding Change
Group Functions	Finance & Business Services	6716USA	Business Services Upgrades and Enhancements		80,000	80,000		Funding Change		500,000	500,000	Funding Change
Group Functions	Finance & Business Services	6716USB	WDD FY24+ US		64,000	64,000		Funding Change		400,000	400,000	Funding Change
Group Functions	Finance & Business Services	6716USC	RPA J2L US		53,600	53,600		Funding Change		335,000	335,000	Funding Change
Group Functions	Finance & Business Services	6717US	Tactical HR FY24+ US	8,570	107,200	98,630	Delays		2,820,118	670,000	(2,150,118)	Funding Change
Group Functions	Group Functions	5983	DB Landscape Optimization	(179,689)		179,689		Accounting	(179,689)		179,689	Accounting
Group Functions	Group Functions	6252US	Data Mgmt Tools and Initiatives	1,897		(1,897)		Immaterial	1,897		(1,897)	Immaterial
Group Functions	Group Functions	6253US	Cloud API Migration - Legacy Integration Tools	(10,220)		10,220		Accounting	(10,220)		10,220	Accounting
Group Functions	Group Functions	6258USB	Data Archival: US Group Function Portfolio	171,317		(171,317)		Scope changes	347,064		(347,064)	Scope changes
Group Functions	Group Functions	6360US	IT Planning: Enterprise PMO	5,379		(5,379)		Accounting	5,379		(5,379)	Accounting
Group Functions	Group Functions	6391US	SHE: Consolidated Incident Management and Reporting	0		0		Variance less than 10%	1,358,000		(1,358,000)	Funding Change
Group Functions	Group Functions	6402US	Legal Operations			0		Variance less than 10%	734,000		(734,000)	Funding Change
Group Functions	Group Functions	6461US	Financial Transformation: Plan to Perform		750,000	750,000		Funding Bucket	0	1,800,000	1,800,000	Funding Bucket
Group Functions	Group Functions	6461USA	Business Planning F&A (FP&A Enablement & Modelling Tool)	(49,143)		49,143		Accounting	1,750,860		(1,750,860)	Funding Change
Group Functions	Group Functions	6522US	SHE: Digitisation of the CMT Process	(81,391)		81,391		Accounting	(81,391)		81,391	Accounting
Group Functions	Group Functions	6549US	SHE: Resilience Digital Evolution	8,377		(8,377)		Accounting	8,377		(8,377)	Accounting
Group Functions	Group Functions	6563US	CA:NG.com Design	53,520		(53,520)		Funding Change	1,138,520		(1,138,520)	Funding Change
Group Functions	Group Functions	6587US	E-Hub	(2,591,783)		2,591,783		Accounting	(2,591,783)		2,591,783	Accounting
Group Functions	Group Functions	6643US	CA: Group Web Digital Evolution FY24-FY27		100,000	100,000		Funding Bucket		1,430,000	1,430,000	Funding Bucket
Group Functions	Group Functions	6660	Plng Repository F&A	6,566		(6,566)		Funding Change	695,855		(695,855)	Funding Change
Group Functions	Group Functions	6693US	Procurement Systems Digital Enablement (Global US)		112,000	112,000		Funding Bucket		700,000	700,000	Funding Bucket
Group Functions	Group Functions	6696US	Property Services/ Real Estate/ Fleets / Meter Shop/IMWM - IT Sys Modernization	111,935		(111,935)		Timing	1,237,178	770,000	(467,178)	Scope Changes
Group Functions	Group Functions	6719US	Legal FY24: Group Legal Digital Evolution		224,000	224,000		Funding Bucket		1,400,000	1,400,000	Funding Bucket
Group Functions	Group Functions	PZG24U	Opco Adj US Group Functions		(4)	(4)		Immaterial		(17)	(17)	Immaterial
Group Functions Total				**1,696,643**	**7,093,981**	**5,397,337**			**36,574,314**	**39,749,211**	**3,174,897**	

				YTD					Full Year Forecast			
				YTD Budget Variance			Non_Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	3 + 9 Total	Budget Total	YTD Variance	Reason
DPIT	Cloud & Hosting	6489	I&O US Cloud & Hosting FY23	(100,553)		139,102		Accounting	(100,553)		139,102	Accounting
DPIT	Cloud & Hosting	6489.01	INVP 6489 Public Cloud Platforms	(49,369)				See total 6489 variance	(49,369)			See total 6489 variance
DPIT	Cloud & Hosting	6489.02	INVP 6489 Cloud Containers	(4,839)				See total 6489 variance	(4,839)			See total 6489 variance
DPIT	Cloud & Hosting	6489.03	INVP 6489 Global Active Directory	(5,280)				See total 6489 variance	(5,280)			See total 6489 variance
DPIT	Cloud & Hosting	6489.04	INVP 6489 Hosted Solutions & Platf	20,939				See total 6489 variance	20,939			See total 6489 variance
DPIT	Cloud & Hosting	6679US	DPIT US Cloud & Hosting FY24	662,054	1,568,000	810,600	Timing differences		5,116,150	5,183,700	(27,796)	Variance less than 10%
DPIT	Cloud & Hosting	6679US.01	Platform Ops	36,579			See total 6679US variance		36,579			See total 6679US variance
DPIT	Cloud & Hosting	6679US.02	Cloud Platforms	13,841			See total 6679US variance		13,841			See total 6679US variance
DPIT	Cloud & Hosting	6679US.03	Devel Auto	44,926			See total 6679US variance		44,926			See total 6679US variance
DPIT	CNI	6680US	US CNI FY24	211,265	302,900	91,635	Timing differences		3,682,558	3,728,040	45,482	Variance less than 10%
DPIT	Network	6486	I&O US Networks FY23	156,676		(54,222)		Accounting	156,676		54,222	Accounting
DPIT	Network	6486	I&O US Networks FY23			(140,034)		Imprecise Estimates			(248,478)	Imprecise Estimates
DPIT	Network	6486.01	INVP 6486 LAN/Wireless	(48,599)				See total 6486US variance	(48,599)			See total 6486US variance
DPIT	Network	6486.02	INVP 6486 WAN/SD-WAN	(27,491)				See total 6486US variance	(27,491)			See total 6486US variance
DPIT	Network	6486.03	INVP 6486 Network Support Svcs	50,740				See total 6486US variance	50,740			See total 6486US variance
DPIT	Network	6486.04	INVP 6486 Telephony	62,929				See total 6486US variance	62,929			See total 6486US variance
DPIT	Network	6488.01	INVP 6488.01 US CNI Networks	80,681		(34,704)		Accounting	80,681		(34,704)	Accounting
DPIT	Network	6488.02	INVP 6488.02 Infrastructure Mgmt	(229,069)				See total 6488.01 variance	(229,069)			See total 6488.01 variance
DPIT	Network	6488.03	INVP 6488.03 CNI Tool Management	183,092				See total 6488.01 variance	183,092			See total 6488.01 variance
DPIT	Network	6682US	Networks FY24	3,751,903	3,780,700	(413,381)	Timing differences		13,564,846	14,043,700	36,675	Variance less than 10%
DPIT	Network	6682US.02	Network Services	63,070			See total 6682US variance		63,070			See total 6682US variance
DPIT	Network	6682US.03	Telephone	124,308			See total 6682US variance		124,308			See total 6682US variance
DPIT	Network	6682US.04	WAN	81,219			See total 6682US variance		81,219			See total 6682US variance
DPIT	Network	6682US.05	Wi-Fi	173,582			See total 6682US variance		173,582			See total 6682US variance
DPIT	Operations	6485	I&O US Operations FY23	(77,024)		(194,597)		Accounting	(77,024)		(194,597)	Accounting
DPIT	Operations	6485.01	INVP 6485 Service Aware CMDB	(14,545)				See total 6485US variance	(14,545)			See total 6485US variance
DPIT	Operations	6485.02	INVP 6485 Observability	29,936				See total 6485US variance	29,936			See total 6485US variance
DPIT	Operations	6485.03	INVP 6485 AI Operations	(52,338)				See total 6485US variance	(52,338)			See total 6485US variance
DPIT	Operations	6485.04	INVP 6485.04 Automation	264,178				See total 6485US variance	264,178			See total 6485US variance
DPIT	Operations	6485.05	INVP 6485.05 Unified Tech Ops Ctr	44,390				See total 6485US variance	44,390			See total 6485US variance
DPIT	Operations	6683US	Global Technology Operations	769,617	419,700	(427,692)	Timing differences		3,745,036	3,993,200	170,389	Variance less than 10%
DPIT	Operations	6683US.01	Service Aware CMDB	5,890			See total 6683US variance		5,890			See total 6683US variance
DPIT	Operations	6683US.05	UTOC	71,885			See total 6683US variance		71,885			See total 6683US variance
DPIT	Other	6641	FY23 Service Now Upgrade - US	(301,846)		98,144		Accounting	(301,846)		98,144	Accounting
DPIT	Other	6641	FY23 Service Now Upgrade - US			203,702		Funding Bucket			203,702	Funding Bucket
DPIT	Other	6652	MicroStrategy Sunset	800,461		88,195		Accounting	1,352,060		88,195	Accounting
DPIT	Other	6652	MicroStrategy Sunset			(888,656)		Funding Bucket			(1,440,255)	Funding Bucket
DPIT	Other	6698US	FY24 eHub	125,855		(125,855)		Funding Change	1,595,009		(1,595,009)	Funding Change
DPIT	Other	6708	Microsoft E3/E5 Licenses	(1,238,031)		1,238,031		Accounting	(1,238,031)		1,238,031	Accounting
DPIT	Other	P2G24D	Opco Adj Instrastructure		(6)	(6)		Immaterial		(24)	(24)	Immaterial
DPIT	Data Center	6223	I&O US Data Center Transformation	4,287	4,490,000	295,071		Accounting	6,778,896	11,057,000	295,071	Accounting
DPIT	Data Center	6223	I&O US Data Center Transformation			(755,451)		Imprecise Estimates			(963,060)	Imprecise Estimates
DPIT	Data Center	6223.01	INVP 6223.01 DCT Other Costs	613,318				See total 6223 variance	613,318			See total 6223 variance
DPIT	Data Center	6223.02	INVP 6223.02 DCT Co-Location	(70,562)				See total 6223 variance	(70,562)			See total 6223 variance
DPIT	Data Center	6223.03	INVP 6223.03 DCT Mngd Svcs	232,696				See total 6223 variance	232,696			See total 6223 variance
DPIT	Data Center	6223.04	INVP 6223.04 DCT Mainframe	31,408				See total 6223 variance	31,408			See total 6223 variance
DPIT	Data Center	6223.05	INVP 6223.05 DCT App Migrat	4,139,233				See total 6223 variance	4,139,233			See total 6223 variance
DPIT	Digital Platforms	6519	GridStack Foundations	328,745		223,454		Accounting	328,745		(456,268)	Accounting
DPIT	Digital Platforms	6519.01	INVP 6519.01 GS Foundation GridInit	122,677		(679,722)	Mischarges		122,677			See total 6519 variance
DPIT	Digital Platforms	6519.02	INVP 6519.02 GS Foundations GridCl	88,522				See total 6519 variance	88,522			See total 6519 variance
DPIT	Digital Platforms	6519.03	INVP 6519.03 GS Found. GrdiDataOps	103,765				See total 6519 variance	103,765			See total 6519 variance
DPIT	Digital Platforms	6519.04	INVP 6519.04 GS Found. GridIdentity	(67,078)				See total 6519 variance	(67,078)			See total 6519 variance
DPIT	Digital Platforms	6519.05	INVP 6519.05 GS Found. GridAccess	(7,576)				See total 6519 variance	(7,576)			See total 6519 variance
DPIT	Digital Platforms	6519.06	INVP 6519.06 GS Found. GridVault	(60,831)				See total 6519 variance	(60,831)			See total 6519 variance
DPIT	Digital Platforms	6519.07	INVP 6519.07 GridAPI/GridSystemInt	(51,956)				See total 6519 variance	(51,956)			See total 6519 variance
DPIT	Digital Platforms	6565	Secure & Compliant FY23	(341,005)		341,005		Accounting	(341,005)		341,005	Accounting
DPIT	Digital Platforms	6678US	DPIT US Digital Platforms FY24	3,366,638	3,298,900	99,987		Variance less than 10%	10,543,231	12,514,000	1,922,574	Funding Bucket
DPIT	Digital Platforms	6678US.0.1	Sec&Compl-DevOps	3,750		(215,920)		See total 6678 variance	3,750			See total 6678 variance
DPIT	Digital Platforms	6678US.07	PowerApps & Automate	16,437				See total 6678 variance	16,437			See total 6678 variance
DPIT	Digital Platforms	6678US.08	ServiceNow	23,347				See total 6678 variance	23,347			See total 6678 variance
DPIT	Digital Platforms	6678US.09	(blank)	4,661				See total 6678 variance	4,661			See total 6678 variance
DPIT	Digital Workplace	6198.01.04	Mac OS/iOS Mgmt AC	(35)		35		Immaterial	(35)		35	Immaterial
DPIT	Digital Workplace	6487	I&O US Digital Workplace Services FY23	(834,119)		506,746		Accounting	(834,119)		506,746	Accounting
DPIT	Digital Workplace	6487.01	INVP 6487 Device & Data Access	(46,742)				See total 6487 variance	(46,742)			See total 6487 variance
DPIT	Digital Workplace	6487.02	INVP 6487 Real Time Collaboration	(27,414)				See total 6487 variance	(27,414)			See total 6487 variance
DPIT	Digital Workplace	6487.03	INVP 6487 Content Collaboration	57,197				See total 6487 variance	57,197			See total 6487 variance
DPIT	Digital Workplace	6487.04	INVP 6487.04 Support Assurance	414,150				See total 6487 variance	414,150			See total 6487 variance
DPIT	Digital Workplace	6487.05	Digital Adopt & Change	(69,818)				See total 6487 variance	(69,818)			See total 6487 variance
DPIT	Digital Workplace	6602	Microstrategy Mod	(25,880)		25,880		Accounting	(25,880)		25,880	Accounting
DPIT	Digital Workplace	6687US	Secure&Compl Data	188,448		(188,448)		Funding Change	188,448		(188,448)	Funding Change
DPIT	Digital Workplace	6681US	DPIT US Digital Workplace Services	3,169,822	3,351,000	(24,306)		Variance less than 10%	17,153,494	17,151,000	(207,978)	Variance less than 10%
DPIT	Digital Workplace	6681US.01	Devices	97,096				See total 6681US variance	97,096			See total 6681US variance
DPIT	Digital Workplace	6681US.02	Conferencing	15,704				See total 6681US variance	15,704			See total 6681US variance
DPIT	Digital Workplace	6681US.03	Collaboration	39,056				See total 6681US variance	39,056			See total 6681US variance
DPIT	Digital Workplace	6681US.05	Access Management	13,150				See total 6681US variance	13,150			See total 6681US variance
DPIT	Digital Workplace	6681US.06	DA&CM	40,477				See total 6681US variance	40,477			See total 6681US variance
DPIT Total				**17,192,600**	**17,211,194**	**18,594**			**67,861,979**	**67,670,616**	**(191,363)**	

				YTD					Full Year Forecast			
				YTD Budget Variance			Non_Permanant	Permanent	Forecast to Budget Variance			Permanent
CIO Group	it_program	inv_id	proj_level_descrip	Actual YTD	YTD Budget	YTD Variance	Reason	Reason	3 + 9 Total	Budget Total	YTD Variance	Reason
Other	IT4IT		IT Investments	655,113	379,840	(275,273)		Scope Changes	366,368	2,794,000	2,427,632	Scope Changes
Other	COH		Construction Overhead	1,168,494	724,003	(444,491)		Opex Only	1,168,494	1,900,012	731,518	Opex Only
Other	Burdens		Burdens	(1,548,901)	0	1,548,901		Opex Only	(1,548,901)	0	1,548,901	Opex Only
Other	RI Separation		RI Separation	4,901,290	0	(4,901,290)		RI Pass Through	27,362,592	0	(27,362,592)	RI Pass Through
Other Total				**5,175,996**	**1,103,843**	**(4,072,153)**			**27,348,553**	**4,694,012**	**(22,654,540)**	

Exhibit ___ (ITDP-3R)

Information Technology Capital Investment Plans for fiscal year ("FY") 2023 and FY 2024

Niagara Mohawk Power Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY
KeySpan Gas East Corporation d/b/a National Grid

Information Technology Capital Investment Plan

Cases 20-E-0380, 20-G-0381, 19-G-0309 and 19-G-0310

FISCAL YEAR 2023

PREPARED FOR:
DEPARTMENT OF PUBLIC SERVICE STAFF
3 EMPIRE STATE PLAZA
ALBANY, NY 12223

March 31, 2022

nationalgrid

Fiscal Year 2022 Information Technology Capital Investment Plan

I. Introduction

Niagara Mohawk Power Corporation (Niagara Mohawk), The Brooklyn Union Gas Company (KEDNY), and KeySpan Gas East Corporation (KEDLI) (together "National Grid" or the "Companies") submits its Information Technology ("IT") Capital Investment Plan (the "Investment Plan") for fiscal year 2023 ("FY23 Plan").[1] This submission is in compliance with Section IV.6.5 of the Joint Proposal adopted by the New York Public Service Commission ("PSC" or the "Commission") January 20, 2022 in Cases 20-E-0380 and 20-G-0381, and Section IV.10.5 of the Joint Proposal adopted by the Commission August 12, 2021 in Cases 19-G-0309 and 19-G-0310.[2]

IT capital projects and investments that are shared investments across National Grid's operating companies are implemented and owned by National Grid USA Service Company, Inc. ("Service Company") and allocated to the benefiting US operating companies, including Niagara Mohawk, KEDNY and KEDLI in the form of an annual rent expense once the investments are in-service. The rent expense includes a return on the capital investment plus depreciation expense.

The FY23 Plan is made up of in-flight projects that will require capital spending to complete as well as projects that will be sanctioned and begin to incur capital charges during the fiscal year. In total, the FY23 Plan reflects Service Company capital expenditures of $299.257 million across the US operating companies. Of this amount, approximately $288.894 million is for projects that benefit and, therefore, are chargeable to Niagara Mohawk, $228.342 million to KEDNY and $227.365 million to KEDLI.[3] The FY23 Plan provided in Attachment 1 includes a list of projects, forecast in-service dates, and the allocators used to determine the rent expense for the operating companies that will benefit from each project, along with FY23 Service Company IT capital costs and the full Service Company project spend expected for each investment.

National Grid's IT organization continuously reviews the Plan relative to current business requirements, risks and information. The FY23 Plan assumes that the impacts of the novel coronavirus (COVID–19) pandemic or any impacts from the conflict in Ukraine will not

[1] FY23 is from April 1, 2022 through March 31, 2023.
[2] Cases 20-E-0380 and 20-G-0381, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service*, "Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans" (issued and effective January 20, 2022) and Cases 19-G-0309 and 19-G-0310, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Energy East Corporation d/b/a National Grid for Gas Service*, "Order Adopting Joint Proposal, and Establishing Gas Rate Plans" (issued and effective August 12, 2021).
[3] The Joint Proposal adopted in Cases 20-E-0380 and 20-G-0381 reflected a Service Company investment level for FY23 of $252 million for projects that benefit and, therefore, are chargeable to Niagara Mohawk, and the Joint Proposal adopted in Cases 19-G-0309 and 19-G-0310 reflected a Service Company investment level for FY23 of $150 million for projects that benefit and therefore, are chargeable to KEDNY and KEDLI.

significantly impact the Plan or National Grid's ability to deliver it. However, these are unprecedented circumstances, and National Grid will continue to review its system and customer needs and will revise the Plan as necessary to meet those needs.

II. Development of the Capital Plan

To determine projects for inclusion in the FY23 Plan, National Grid engaged in an iterative process that involved collaboration with Jurisdictional representatives and the functional business areas (*e.g.,* Operations, Customer and Digital, Network Strategy, and Finance and Shared Services) to identify the investments of highest priority for inclusion in the Plan. This process was aligned to the corporate budgeting process and strategic portfolio reviews.

The development of the FY23 Plan (portfolio reviews) was based on an assessment and analysis of current and proposed programs aligned to the Company's IT strategic focus - build and maintain the foundation, drive new capabilities, and enable cost savings. We reviewed each program and balanced funding across focus areas to optimize investments and drive value. The FY23 portfolio is focused on the following:

- Fulfilling regulatory and legal mandates
- Reducing cyber and operational risk
- Driving standardization and modernization
- Building tools and platforms to empower our people
- Safeguarding critical business applications
- Building new capabilities for improved business performance
- Implementing cost efficiencies and saving programs to achieve year-over-year run-the-business ("RTB") cost reductions

In building the FY23 Plan, National Grid first identified the amount of work remaining and the delivery timeline for in-flight, multi-year investments. Next, new investments to address compliance with a legal or regulatory requirement/mandate or an open corporate risk were added to the Plan. Third, investments that focused on IT infrastructure – primarily investments to upgrade/replace/add functionality that would create reliability in systems and fix foundational capabilities necessary for operations and/or future aspirations of the business (including ensuring digital security) were included. Finally, the IT team worked closely with the Jurisdiction and business executive sponsors to focus on their top project requests.

After this analysis, IT distributed a draft investment plan to business functions, Jurisdictional representatives, and other key stakeholders for feedback and held review sessions on the proposed plan. A final Investment Plan review occurs each fiscal year with the Jurisdictional Presidents, Jurisdictional delegates, and functional executives. Upon the review and approval of the Investment Plan by the Jurisdictional Presidents and functional executives, the consolidated financial information from the Plan, by jurisdiction and company, is incorporated with the IT O&M requirements into the IT budget. The IT budget is then reviewed and approved by the New York Chief Information Officer ("CIO"), Global CIO, and Corporate Finance, and is incorporated into a consolidated National Grid USA budget for approval by the National Grid plc Board.

III. Formal Sanctioning Support and Oversight of Projects Included in the Investment Plan

Throughout the year, IT projects are subject to a formal governance process that includes a review and approval for all projects greater than $0.030 million. Within IT, there is a team that reviews all IT projects before forwarding the project to the appropriate sanctioning authority for approval. Business Sponsored IT Investments up to $50 million that are classified as "Grow" will require both IT & Business approval.

- Projects less than $10 million are approved by the US CIOs for their portfolios through a workflow approval process and under their delegations of authority.
- Projects from $10 million to $25 million are approved by the Senior IT Leadership through a workflow approval process and under their delegations of authority.
- Projects from $25 million to $50 million are approved by the Global CIO through a workflow approval process and under their delegation of authority.
- Projects from $50 million to $203 million are approved by the New York and New England Executive Sanctioning Committees ("NYESC" and "NEESC"), under delegation of authority by the President, New York Jurisdiction and President, New England Jurisdiction. In addition, these committees will also approve any projects less than this amount which are IT projects classified as Transform and any other Projects at its discretion without regard to whether such Projects exceed the $50 million threshold.
- Projects which exceed the $203 million threshold must go to the Group Investment Committee for approval.

IT projects are typically initially sanctioned for Requirements and Design work (also referred to as "Partial Sanction") and then will return for Development and Implementation sanction (also referred to as "Sanction" or "Full Sanction"). If project scope and costs are well understood (*e.g.,* upgrades to hardware that do not necessitate requirements and design work or whose cost estimate can be reliably informed by similar previous installations), a project may proceed directly to "Sanction." In the event a project is forecast to exceed its sanction amount, a project must return to the appropriate Governance Committee for "Re-Sanction" consideration. A "Closure" paper is prepared at the completion of a project detailing the financial and objective outcomes of the project.

The Investment Plan includes all IT projects that progress from initial Feasibility & Analysis phases to ultimate closure to plant in service, as defined below, and supported by the formal sanctioning and governance process discussed above.

- **Feasibility & Analysis** – A study conducted in cases where a significant level of analysis is required to better determine the direction or approach for a project. Because such a study is likely to have a significant impact on IT resources and spend, the effort is treated as a project to be included in the Investment Plan.

- **Requirements & Design** – This phase includes requirements definition and completion of analysis, confirmed project scope, agreed functional design, and technology solution to meet business and technical requirements, confirmed plan for Development and Implementation, and confirmed production handover, support strategy, and sourcing approach.
- **Development & Implementation** – This phase includes development to the specified designs, testing, preparation for training, deployment of the solution to the business, and post-implementation support.
- **Project Closure** – This phase closes the project, including capturing ongoing RTB costs, completing post implementation review and lessons learned, closing open actions and change requests, implementing a decommissioning plan for replaced technologies, and submitting a closure paper to the appropriate sanctioning committee.

Cost estimates for projects that have completed the Requirements & Design stage with full scope and solution design generally have +/- 10% cost estimates. Other projects at earlier stages, and the budgets for those projects, are less refined and more susceptible to changes in scope and budget. The projects in the Investment Plan are continuously reviewed for changes in assumptions, constraints, project delays, accelerations, and system operations, performance, safety, regulatory/compliance and customer driven needs that arise. The sanctioning process allows a further evaluation of the total expected costs of a project at key points in the project's lifecycle, including a view of how a given fiscal year's spend on the project relates to the costs included in the Investment Plan as incorporated in the corporate budget.

IV. Capital Plan Investments by Spending Categories

The Company's FY23 Plan is presented by portfolio category/spending rationale as shown below.

A. Group Functions

Group Functions executes National Grid's digital roadmaps, leveraging technology and reimagining processes in support of Finance, Enterprise Business Services, Human Resources, Legal, and Safety Health & Environmental, continuing to upgrade SOX related systems, and retiring outdated systems. The 5 largest Investments within this portfolio are as follows:

1. NG.com Refresh (INVP 6514US) – This investment is to refresh our externally facing web platform NG.com. NG.com provides the public face to National Grid's customers, communities and stakeholders. It offers the starting point for engagement with our organisation and key information to wider businesses who work with National Grid.

2. SAP non-production landscapes (INVP 5607) - Procuring, setting up and configuring project environments is a critical path task for SAP related projects and initiatives. These activities can be time consuming and costly when done in isolation. To improve IT project implementation schedules, National Grid IT introduced a process to provide annual funding at the beginning of each fiscal year to procure and configure a set of project environments in support of all initiatives within the SAP portfolio for that fiscal year. Appropriate costs will be associated with individual projects by issuing separate

work orders for each project. This method enables accurate cost treatment of the environments.

3. My Hub 2.0 (INVP 5158) – The investment will move myHub to a standard configuration and reinforce it as a single source of workforce data. The control environment will be strengthened through simplifying integrations and ensuring workforce data is aligned across the Company's systems. MyHub will improve efficiency in supporting our Operating Model by removing Joiners to Leavers complexity, improving user experience and process efficiency. It will unlock standard reporting and full workforce analytics capability and define the workforce master data model, ownership and governance.

4. Lease Accounting Update (PowerPlan) (INVP 5497) - Powerplan is National Grid's mission critical financial asset, project and tax system. This project will upgrade the application and all its underlying architecture to the latest version to bring National Grid back into support. The standalone Lease module also will be consolidated into the upgraded core Powerplan solution.

5. FP&A Business Planning Modelling and Use cases (Global US) (INVP 6461USA) - This program of work will support overall Plan to Perform end to end process to deliver the Group Finance ambition to ensure flawless service and proactive, quality insights that drives performance and enhance forecasting and budgeting process. The investment will provide planning and modelling tooling for group planning and budgeting, including financial and non-financial key drivers, automation of cost allocations and variance analysis reporting. Data flows will be streamlined from source systems.

B. United States Electric Business Unit

IT supports the US Electric Distribution and Transmission businesses in modernizing key legacy operational systems while building capabilities that will effectively enable the energy transition. Through digital delivery, the Electric CIO organization provides greater efficiency and data driven insights into our workforce operations:

- More automated and reliable grid controls systems
- Capital programs and projects that support the growth of Distributed Energy Resources (DERs)
- A growing market of prosumer electric customers

Key projects within the Electric portfolio in FY23 include:

1. Digital Warehouse (Track Me) (INVP 6388) - Validation and development work with a suite of Digital Products (*i.e.*, ICU-Out, TrackMe, Digital Supervisor, *etc*.). Digital Warehouse will provide automation of inventory tracking, sophisticated demand forecasting and cross-shop visibility of warehousing. The outcomes of this product will be reductions in obsolete materials, decreasing frequency of manual inventory inspections, avoiding double ordering, reduction in overall inventory size and reduction in schedule delays.

2. Digital Supervisor (SuperView) (INVP 6388) - Digital Supervisor Assistant will streamline and automate the processing of daily paperwork. The system will work alongside On My Way (OMW). SuperView will further enable barns and yards to go paperless and reduce time spent manually processing paper forms, manual input errors and improve compliance activities.

3. ICU - Out (INVP 6291) - This product is part of the electric digital portfolio that will accelerate benefits to the electric business through digital transformation. The product will provide an advanced customer outage solution that will analyze and integrate outage information from multiple systems and sources to generate a holistic view of an outage event. This is expected to reduce time and costs in restoring power outages through better data analytics that will enable us to better understand customer outage status / location, grid damage, along with having more advanced prediction of storm impacts.

   Currently we rely primarily on customers calling to report outages to determine where outages are. This causes inefficiencies because not all customers call within a timely period of the outage occurring.

   Opportunities available through digitization include:
   - Improved data quality
   - Improved outage area predictions
   - Improved or eliminated manual processes
   - Improved visibility into field conditions
   - Improved customer communications

4. Future Now (INVP 6076) – Work planning in the Electric Business Unit (EBU) is a manual process involving multiple systems which are not fully integrated. This investment will complete the development and roll out of FutureNow optimization capabilities to the Transmission Business enabling the EBU to maximize strategic objectives against budget and resources.

5. Vegetation. Management Optimization (VMO) (INVP 6073) - This digital project seeks to maximize the value of National Grid's vegetation management programs. The product aims to provide the capability to plan tree pruning activities based on condition, using advanced analytics and other available technologies, instead of pruning on fixed time cycles. It will assess technologies to maximize value during the execution of the pruning activities. The product also seeks opportunities to drive value in the hazard tree program management, in terms of both planning and execution. Features will include Sub-Circuit pruning and improved reliability.

6. Retail Settlement (INVP 5891E) - Planning and development work to implement a new Retail Settlement system for FY24. The company has recently implemented a new Wholesale Settlement solution, replacing two end of life systems which were no longer supportable due to out dated components, infrastructure and application code. This investment will allow for consolidation with the new Wholesale Settlement system released in FY22.

C. United States Gas Business Unit

IT supports the US Gas Distribution and Transmission businesses by developing digital products based on the foundation of gas business enablement that deliver continuous improvement of our asset lifecycle management capabilities through:

- Risk based asset investment planning
- Performance based maintenance
- AI based work optimization
- Mobile field products

Key Gas Investments for FY23 include:

1. Continued Scaling and Evolution of GBE Transformation (INVP 6539) - Customer initiated work-types adding additional features and capabilities to the core GBE products. The investment will unlock expanded value by evolving and adding to our products lines. Areas of focus will include:
    - Data transparency for assets
    - Asset risk modelling and prioritization for remaining work types
    - Portfolio/project management
    - Work planning and dispatching with scheduling optimization for Maintenance and Construction and Customer Meter Services
    - Work Floor Software enhancements
    - Operation Qualifications for employee calendar and time off management
    - Customer and Dispute Management enhancements
    - Bundling for asset order initiation & queue management
    - Replacement of Legacy E-Permits system, for Permit & Property Rights Management
    - Pathfinder Enhancements
    - Material traceability for Meter Shops

2. Enterprise Platform for Gas Retail Choice Programs (INVP 5787D) – This investment will create a roadmap and begin the migration of legacy market nomination and confirmation systems to the enterprise GTS platform. It will address security gaps and risks with other legacy systems and implement corrections to current code in GTIS to resolve errors in customer file processing from CRIS.

3. Gas Management System (Supervisory Control and Data Acquisition SCADA) Upgrade (INVP 6550) – This project will design and provide the estimation of the deployment planning of the Gas Management System (GMS). Outputs will include:
    - Define hardware and software (MS servers, network devices, OSs, vendor software, and security and monitoring tools) requirements and specifications for the GMS upgrade to comply with National Grid IT zero based risk Security requirements, TSA regulations, and NERC-CIP requirements.
    - Complete high-level system design.
    - Create multi-year GMS Upgrade contract(s) negotiations plan.

- Finalize the solution implementation plan along with the estimated cost effort.

4. GACI (Gas Application Continuous Improvements) (INVP 6015) – The investment will complete the migration of gas applications to a new data center. It will address high and medium severity defect tickets for multiple applications in the Gas Business portfolio. These defects will address functionality, usability and data integrity issues.

5. Asset and Work Management System for LNG (INVP 4898) - The LNG Department does not have an automated Maintenance Management System (MMS). This project will complete the deployment of the solution to the remaining 7 plants in scope for FY23. It will provide automated integrations to SAP and Power Plan for financial tracking. Implement HSE (Health, Safety & Environment) functionality that include lockout / tagout and permitting

   The solution is specifically designed for both the gas and oil industries for deficiency and audit tracking. The system will consolidate information into a single system with standard formats to ensure the Company is compliant with applicable mandates and legislation.

D. Customer

IT provides support to our US Customer business, delivering best-in-class digital products that increase engagement and reduce cost to serve. Areas of focus will include:

- Modernizing our legacy "back-end" IT estate to support more digital products
- Liberating data and creating an abstracted and real-time customer data platform to assist with achieving energy efficiency ambitions

Key projects in the Customer Area include:

1. Billing & Collections Mandates & Enhancements (INVP 6538) - This is a funding source for new Billing & Collections projects in FY23 which will arise due to future mandates.

2. Customer Data Platform (INVP 6349) - Customer Data Platform - A comprehensive data assessment highlighted the need for improvement in basic master data capabilities, governance and the necessary tools to support the highly complex demands coming from the business. There are ~300 business capabilities at National Grid that make up the delivery of customer service and operations. This investment will ensure data is accessed consistently and accurately from a single source. It will enable the Company to speak to customers with one voice across the customer lifecycle, and the ability for the customer to contact the Company in the manner of their choice.

3. My Business Account (Nucleus Phase 2) (INVP 5673) - This project's focus is on re-engineering the National Grid commercial and industrial (C&I) customers' digital interactions to create a streamlined and seamless customer experience throughout the customer journey. The project will enhance existing transaction capabilities and account management tools to support the unique C&I needs for multi-account management. It will continue to build in the National Grid managed Azure Cloud Instance using the Enterprise Solution Platform Mulesoft as the middleware to connect to existing core system AGS,

.Net, CSS, CRIS and upcoming Salesforce applications. The driver of this project is to reduce challenges our C&I customers face surrounding their National Grid experience, address key customer pain points and enable core capabilities.

4. Customer Connection Program (INVP 5125) – The Customer Connection Program covers mandated, operational and system reliability enhancements for the GridForce system and related interfaces. This program is needed to assure compliance with tariffs and regulations for Electric and Gas customer connections as well as Distributed Generation tariffs.

5. Electric Vehicle (EV) End-End Solution (INVP 6250) - This project will create the End to End solution blueprint for Electric Transportation Connection Programs. It will include the following areas of focus:
    - EV Customer Portal (Application automation)
    - Centralized Project Management and Tracking system
    - Reporting and Forecasting visualization capability
    - Storage of Project Documentation and charging station data.

E. Infrastructure and Operations ("I&O")

The I&O area supports National Grid's internal IT Architecture covering Cloud and Hosting, Critical Network Infrastructure, Networks, End User Experience and Operations. I&O is focused on reducing National Grid's technical debt through cost effective, modernized IT infrastructure enabling cloud capabilities that meet our risk profile. The organization enables a connected, seamless enterprise that is flexible, secure, reliable and scalable while maintaining acceptable levels of technical debt and risk. Key I&O programs include:

1. I&O US End User Experience FY23 (INVP 6487) - End users need to be more engaged, agile, secure, and therefore productive and efficient through the technologies that they interact with daily. The investment will focus on improving the technology experience for our end users, employees, contractors and partners. Initiatives include:
    - Enhanced developer capabilities with fully managed MAC solution. Devices (laptops/desktops) refreshed as per policy.
    - Perpetual Windows, Browser & Office Currency (Windows as a Service). Remote PC/MAC provisioning and configuration.
    - Expanded replacement of aging Video Conferencing equipment with MS Team compatible equipment providing frictionless, simplified and improved end user experience.
    - Device & Data Access (DDA) - Provide our colleagues with appropriate devices which enable them to securely access applications and data.
    - Real Time Collaboration (RTC) - Enable high quality synchronous digital interactions between people with very low latency.
    - Content Collaboration (CC) - Deliver asynchronous, highly available applications which allow colleagues to co-create business workflows and documents.
    - Support Assurance - Deliver a multi-channel, cost-effective and digital suite of provisioning, support and knowledge experiences.
    - Digital Adoption and Change (DA&CM) - Prepare and enable the business to efficiently adopt our digital productivity and collaboration experiences.

2. I&O US Data Center Transformation (INVP 6223) - Currently our Cloud and Hosting environment and partnership are failing to keep pace with strategic requirements. A wholesale re-establishment of our data center and computer foundation is required in order to remove inefficiencies and complexities which have developed over a number of years. Four new Data Centers will be created with reduced operating costs, modernized infrastructure, increased security and a reduction in carbon footprint (~40% reduction of CO2 from current DXC locations).

3. I&O US CNI FY23 (INVP 6488) – This investment will reduce technical debt and improve availability while mitigating security risk due to replacement of End of Service/Support Life assets (EoSL). Areas of focus will be:
    - CNI Networks - Network support and lifecycle management for CNI environment.
    - CNI Infrastructure Management - Lifecycle management of all server assets within CNI environment.
    - CNI Tool Management - Procuring and managing tooling to support CNI.

4. I&O US Cloud & Hosting FY23 (INVP 6489) - High levels of aged technology have resulted in increased cyber and operational risks. This investment will mitigate risk and performance issues by modernizing assets and processes. It will provide faster application delivery enabled through simplified, platform agnostic provisioning and increased automation of workloads.
    - Application portability for cost effective and robust application availability options (Container as a Service CaaS).
    - Technical debt reduction – decommission of EoSL devices and continued migration of applications to new Active Directory.
    - Hosted Solutions & Platforms - Continue the efforts to reduce non-virtualized infrastructure and technical debt while modernizing management systems and standardizing technology.
    - Private Cloud Platforms - An automated, secure and standardized application platform similar to Public Cloud but within the dedicated hosting environment for National Grid.
    - Public Cloud Platforms - Public Cloud enabled for National Grid specific business/security requirements.
    - Cloud Containers - Optimize application utilization and enable workload mobility.
    - Global Access Management - Global AD for secure access to entire corporate estate.

5. I&O US Digital Operations FY23(INVP 6485) - Improving IT Operations in supporting the Digital Agenda by using advances in technology to reduce waste and enable faster development cycles. This program will deliver:
    - Improved end to end monitoring with layered monitoring tools across full stack Minimum Viable Product (MVP) of Artificial Intelligence (AI) Operations.
    - Improved stability and reduced Mean Time to Resolve for eligible applications.
    - Foundational build of Technology Operations Center – improving efficiency and reducing complexity through centralization.

- Service Aware Configuration Management Database (CMDB) - Automated estate scanning with mapping from component to business service.
- Observability - Layered monitoring tools for full stack visibility of transactions.
- AI Operations - Aggregation of diverse telemetry with applied AI for insight and predictive intelligence.
- Automation - Automation of action spanning Disaster Recovery (DR), self-healing, provisioning and ITSM administration.
- Unified Technology Ops Centre (UTOC) - A single Operations Centre spanning Enterprise, CNI and Operational Technology.

6. I&O US Networks FY23 (INVP 6486) - The US Networks Agile Release Train (ART) consists of four major initiatives to mitigate end of service life hardware/software and high demand to meet our network service, customer, and regulatory requirements. These four initiatives are; WAN/SD-WAN, LAN/Wireless, Telephony, and Network Support Services.
    - Cisco IP Telephony - Upgrading, replacing, or new Telephony infrastructure.
    - WAN/SD-WAN - Provide a secure and reliable Wide Area Network to enable employees to be productive when working in National Grid offices.
    - Network Support Services - Provide improved and robust network services and monitoring.
    - LAN/Wireless - Provides a reliable and functional wired and wireless network ensuring proper coverage, uptime, and performance for employees and guests of National Grid.

F. Security

The Security function protects our people, premises, and digital systems with the objective of maintaining trust in National Grid services, focusing on:

- Reduction in likelihood and impact of security incidents
- Compliance with relevant laws, standards and regulations (*e.g.*, NERC, PCI DSS, GDPR, NIS-R)
- Improved user experience and standard of security practices across National Grid

The Security function aims to reduce the likelihood and impact of physical and cyber security incidents affecting National Grid. It does this by championing security best practices, technology and promoting a security-awareness culture.

1. Physical Security (INVP 3617) – This project will deliver improvements to security at physical locations by upgrading monitoring systems and supporting infrastructure to maintain regulatory compliance for monitoring and safety of our people and premises. The project will replace out of support and warranty field units where failure may result in extended replacement lead times or inability to obtain like-for-like replacements. It will also improve the quality of monitoring with high quality video capture and detection systems.

2. Cyber Operational Technology (INVP 5933) – This project will deliver improvements to National Grid's network resilience and defense in depth capabilities to facilitate our ability

to identify cyber risks and threats, protect and detect against malicious cyber activity and effectively respond to and recover from cyber incidents at operational sites.

3. Vulnerability Management (INVP 6417US) – This project will deliver improvements to National Grid's penetration testing, vulnerability scanning and web application scanning capabilities that identify vulnerabilities and facilitate prioritization for remediation based on risk.

4. Network Security (INVP 6526US) This project will deliver improvements that reduce the likelihood of cyber incidents related to distributed denial of service attacks, and protect against and enable detection of network traffic related to malware, which can propagate through the network if traffic is not monitored or alerted upon.

5. Platform Security (INVP 6480US) – This project will deliver improvements that reduce the likelihood of cyber incidents related to malware and exploitation of insecure system configurations, identify high value hardening activities, assess impact on operational stability and environments, test and implement system hardening and improve mobile threat defense.

6. Security Orchestration Automation and Response (INVP 6540US) - This project will deliver improvements to response and recovery capability, reduce the overall impact duration of an incident and implement advanced threat detection capabilities that factor changes to the threat landscape.

7. Identity and Access Management (INVP 5870) - This project will deliver improvements that reduce the likelihood of cyber incidents relating to unauthorized access or misuse of access privileges, and privileges on critical applications.

V. Conclusion

Consistent with Section IV.10.5 of the Joint Proposal for cases 20-E-0380 and 20-G-0381 and Section IV.6.5 for Cases 19-G-0309 and 19-G-0310, the Company will provide quarterly variance reports within 60 days after the end of each Fiscal Year quarter. The reports will address the status of IT programs and budgets. On a semi-annual basis, the Company will also meet with Department of Public Service Staff to discuss budget and actual spending to date and discuss project status.

Niagara Mohawk Power Corp., The Brooklyn Union Gas Co., KeySpan Gas East Corp. d/b/a National Grid
Cases 20-E-0380 20-G-0381, 19-G-0309 and 19-G-0310
FY23 IT Capital Investment Plan
Attachment 1

NMPC/KEDNY/KEDLI
In-Flight and Forecasted Information Technology (IT) Capital Projects

Line	Investment Name	INVP #	Bill Pool	In-Service Date	Inception To 2/28/22	FY23 CAPEX	FY23 Spend for Projects Of Any NY Allocation	Category
	Customer Business Unit							
1	Billing & Collections Mandates & Enhancements	6538	C175	3/31/2023		$ 4,642,783	$ 4,642,783	Upgrade/Mod
2	NY Value Stack Billing	5160	521E	2/15/2023		$ 2,271,130	$ 2,271,130	Upgrade/Mod
3	UWP (Launch)	4750J	C175	6/30/2022	37,690,201	$ 984,841	$ 984,841	Grow
4	ODS (Launch)	4750M	C343	6/30/2022	7,141,394	$ 984,841	$ 984,841	Grow
5	KEDNY Rate Case (CRIS)	5723	5220	6/30/2022		$ 300,672	$ 300,672	Upgrade/Mod
6	Customer PCI Remediation	6435	C175	6/30/2022		$ 378,100	$ 378,100	Upgrade/Mod
7	Customer Data Platform	6349	G148	3/31/2023		$ 5,010,585	$ 5,010,585	Grow
8	Preferred Pronouns	6496	C170	10/30/2022		$ 1,645,758	$ 1,645,758	Upgrade/Mod
10	AIMS FY23	6531	C175	3/31/2023		$ 545,000	$ 545,000	Upgrade/Mod
11	NACHA	6492	C175	8/30/2022		$ 424,749	$ 424,749	Upgrade/Mod
12	Customer Billing COVID Transition	6032	C175	2/28/2023		$ 271,649	$ 271,649	Upgrade/Mod
13	FCP Upgrade for CSS - Plan Forward	6066	C434	8/30/2022		$ 308,165	$ 308,165	Upgrade/Mod
14	Customer Billing Improvement Program - Phase 1	5778	C175	8/14/2022		$ 790,835	$ 790,835	Upgrade/Mod
15	My Business Account (Nucleus Phase 2)	5673	C175	3/31/2023		$ 4,874,000	$ 4,874,000	Grow
16	Customer Connection Program	5125	C175	3/31/2023		$ 3,540,476	$ 3,540,476	Upgrade/Mod
18	CDG Net Metering	5795F	521E	8/9/2022	1,253,146	$ 215,330	$ 215,330	Upgrade/Mod
19	EV End-End Solution	6250	521E	2/28/2023		$ 2,850,000	$ 2,850,000	Grow
20	Bill Print and Bill Composition Implementation	5345B	C175	9/30/2022	2,290,216	$ 997,086	$ 997,086	Upgrade/Mod
21	UWP FY23 Product	5859A	C175	3/31/2023		$ 2,464,000	$ 2,464,000	Upgrade/Mod
	United States Electric Business Unit							
22	GridStack Funding	6468	C198	3/31/2022		$ 1,751,617	$ 1,751,617	Transform
23	Digital Warehouse (Track Me)	6388	C902	3/31/2022		$ 5,000,000	$ 5,000,000	Transform
24	Digital Supervisor (SuperView)	6388	C902	3/31/2022		$ 6,000,000	$ 6,000,000	Transform
25	Battery Storage Communications Upgrade	5871F	C902	3/31/2022		$ 300,000	$ 300,000	Upgrade/Mod
26	IEE Pilot Upgrades	5891D	C902	12/1/2022		$ 1,000,000	$ 1,000,000	Upgrade/Mod
27	Meter Data Services Consolidation Phase I	5891F	C902	6/30/2023		$ 1,500,000	$ 1,500,000	Upgrade/Mod
28	Price Transmission (Retail Pricing) Replacement	5891I	521E	10/30/2022	82,271	$ 936,656	$ 936,656	Upgrade/Mod
29	Control Room EOL Server and Tool Replacement	5892B	C902	3/31/2023		$ 1,200,000	$ 1,200,000	Upgrade/Mod
30	IDS 811 DVD Replacement	5892C	C198	9/28/2022		$ 907,428	$ 907,428	Upgrade/Mod
32	Regulatory Enhancements (Minor Works)	5889C	C198	3/31/2023		$ 200,000	$ 200,000	Upgrade/Mod
33	Generation Asset Management Upgrade	5822C	G324	10/30/2022	2,415,522	$ 1,480,877	$ -	Grow
34	On My Way (OMW)	5054	C902	3/31/2025		$ 3,612,426	$ 3,612,426	Transform
35	ICU - Out	6291	C198	3/31/2023		$ 5,177,780	$ 5,177,780	Transform
36	Future Now	6076	G187	12/31/2024		$ 10,051,337	$ 10,051,337	Transform
37	GridMD / Predictament	6437	G187	4/1/2024		$ 895,635	$ 895,635	Transform
38	Clean Energy Initiatives	5871D	C902	3/31/2022		$ 550,000	$ 550,000	Transform
39	Veg. Management Optimization (VMO)	6073	G187	12/31/2022		$ 7,746,691	$ 7,746,691	Transform
40	WECO / Fluke	5888D	C198	3/31/2023		$ 118,000	$ 118,000	Upgrade/Mod
42	3rd Party Battery Storage & Green Initiatives	5871A	521E	3/31/2022		$ 2,350,000	$ 2,350,000	Transform
43	NY ISO PMU	5773D	521T	7/30/2022	142,274	$ 705,973	$ 705,973	Upgrade/Mod
44	Visitor Log for NERC-CIP	5773B	G186	7/30/2022	35,513	$ 414,062	$ 414,062	Upgrade/Mod
45	ARCOS Reporting Tool Upgrade/Replacement	5887A	G187	9/30/2022		$ 617,971	$ 617,971	Upgrade/Mod
46	Retail Settlement	5891E	G370	10/9/2023	40,397	$ 3,281,841	$ 3,281,841	Upgrade/Mod
47	Corporate PI	5890A	G186	11/30/2022	60,655	$ 1,482,625	$ 1,482,625	Grow
48	Content Management	6043G1	T186	6/30/2022	260,257	$ 1,441,718	$ 1,441,718	Grow
49	Electric MV90 Upgrade	5891A	G186	4/1/2022	1,039,000	$ 567,693	$ 567,693	Upgrade/Mod
50	Field Collections System Upgrade	5891C	C175	4/1/2022	821,100	$ 209,669	$ 209,669	Upgrade/Mod
	United States Gas Business Unit							
52	Continued Scaling and Evolution of GBE Transformation	6539	C210	3/30/2026		$ 18,970,000	$ 450,000	Grow
53	Digital Asset & Work Management for CNG and I&R	6556	C210	3/31/2022		$ 140,000	$ 140,000	Upgrade/Mod
54	I&R and Meter Ops Field Device Improvements	6509	G444	3/31/2022		$ 124,000	$ 124,000	Upgrade/Mod
55	Field Collection Systems (FCS) Upgrade (Gas Only	5785Q	C209	11/1/2023		$ 670,000	$ 670,000	Upgrade/Mod
56	ES&R (Enterprise Scalability & Resilience)	6441	C210	3/30/2026		$ 1,070,000	$ 1,070,000	Upgrade/Mod
57	GACI (Gas Application Continuos Improvements)	6015	C210	3/30/2026		$ 1,090,000	$ 1,090,000	Upgrade/Mod
58	Enterprise Platform for Gas Retail Choice Programs	5787D	C210	3/31/2022	288,996	$ 1,580,000	$ 1,580,000	Grow
59	Gas Storm Hardening Using Remote Service Valves (RSV's)	4769	C225	6/30/2022		$ 302,812	$ 302,812	Upgrade/Mod
60	Asset and Work Management System for LNG	4898	G383	5/27/2022		$ 1,547,000	$ 1,547,000	Grow
61	Gas Management System (SCADA) Upgrade	6550	C239	3/31/2022		$ 1,758,333	$ 1,758,333	Upgrade/Mod
62	LNG IT Support for OT Systems	6479	C239	6/30/2022		$ 500,000	$ 500,000	Upgrade/Mod
63	New Storms Response Management System Implementation (electric)	5274A	G148	5/30/2022	1,233,722	$ 267,032	$ 267,032	Grow
	Group Functions							
64	Digitising Learning	6270	G020	12/31/2022		$ 400,000	$ 400,000	Upgrade/Mod
65	US Placeholder - Corp Tech Debt Remediation	Corp23	G020	10/15/2022		$ 1,600,000	$ 1,600,000	Grow
66	Tactical HR	HR23	G020	3/31/2022		$ 1,600,000	$ 1,600,000	Upgrade/Mod
67	Placeholder (Tech Debt & App Rat)	OPS23	G020	3/31/2023		$ 480,000	$ 480,000	Upgrade/Mod
68	Finance + placeholder (Regulatory Compliance)	R2R23	G020	3/31/2022		$ 600,000	$ 600,000	Grow
69	Workforce Data Domain	6273US	G020	6/30/2022	635,196	$ 400,000	$ 400,000	Upgrade/Mod
70	CRO FY23: GRC Archer	4418USC	G020	3/31/2023		$ 1,400,000	$ 1,400,000	Upgrade/Mod
71	SAP BW/BI Upgrade	5843E	G020	1/31/2023		$ 500,000	$ 500,000	Grow
72	SAP PI NW 7.40 (PI -> PO upgrade) to NW 750	5843F	G020	1/31/2023		$ 700,000	$ 700,000	Grow
73	R2R Reporting Use Cases	6225US	G020	3/31/2023		$ 1,000,000	$ 1,000,000	Transform
74	FTOM Record to Report: R2R Automation	6289G	G020	3/31/2022		$ 1,500,000	$ 1,500,000	Transform
75	Group Web Digital Evolution FY23	6513US	G020	3/31/2023		$ 500,000	$ 500,000	Grow
76	NG.com Refresh	6514US	G020	3/31/2023		$ 1,700,000	$ 1,700,000	Grow
77	SHE FY23: Group SHE Digital Evolution	6533US	G020	3/31/2023		$ 1,400,000	$ 1,400,000	Upgrade/Mod
78	Legal FY23: Group Legal Digital Evolution	6534US	G020	3/1/2023		$ 408,800	$ 408,800	Upgrade/Mod
79	AMAG	5466	G020	7/29/2022		$ 800,000	$ 800,000	Upgrade/Mod
80	Global Workforce Tracking	5555	G020	3/31/2022		$ 1,500,000	$ 1,500,000	Upgrade/Mod
81	SAP Healthcheck	5843D	G020	6/3/2022	553,757	$ 600,000	$ 600,000	Upgrade/Mod
82	HR Analytics	6183US	G020	5/31/2022	13,478	$ 1,000,000	$ 1,000,000	Upgrade/Mod
83	SAP non-production landscapes	5607	G020	3/31/2023		$ 4,000,000	$ 4,000,000	Upgrade/Mod
84	HRSP FY23	5611	G020	1/31/2023		$ 1,200,000	$ 1,200,000	Upgrade/Mod
85	My Hub 2.0	5158	G020	3/30/2022		$ 2,600,000	$ 2,600,000	Grow
86	Procurement Systems Digital Enablement (Global US)	6421	G020	9/30/2022		$ 900,000	$ 900,000	Transform
87	Non-SAP Minor Works	6530	G020	3/31/2023		$ 1,000,000	$ 1,000,000	Upgrade/Mod
88	FTOM Record to Report: WDesk Phase 2	6289E	G020	11/30/2022		$ 800,000	$ 800,000	Transform
89	Investment Recovery	6426	GT53	1/31/2023		$ 870,000	$ 870,000	Transform
90	S2P Transformation release 2	6290B	GT53	1/31/2023		$ 1,100,000	$ 1,100,000	Transform

NMPC/KEDNY/KEDLI
In-Flight and Forecasted Information Technology (IT) Capital Projects

Line	Investment Name	INVP #	Bill Pool	In-Service Date	Inception To 2/28/22	FY23 CAPEX	FY23 Spend for Projects Of Any NY Allocation	Category
91	Lease Accounting Update (PowerPlan)	5497	GT53	10/13/2022		$ 2,000,000	$ 2,000,000	Upgrade/Mod
92	Data Archival	6258USB	GT53	6/9/2022	546,870	$ 1,100,000	$ 1,100,000	Upgrade/Mod
93	S2P Transformation release 1	6290A	GT53	10/14/2022	347,582	$ 1,100,000	$ 1,100,000	Transform
94	SAP Minor Works	6529	GT53	3/31/2023		$ 800,000	$ 800,000	Upgrade/Mod
95	FP&A Business Planning Modelling and Use cases (Global US)	6461USA	GT53	3/31/2023		$ 2,200,000	$ 2,200,000	Transform
96	FTOM Record to Report: Task Management	6289D	GT53	6/30/2022	223,774	$ 700,000	$ 700,000	Transform
	Infrastructure & Operations							
97	I&O US End User Experience FY23	6487	G020	7/31/2023		$ 14,346,647	$ 14,346,647	Grow
98	I&O US Data Center Transformation	6223	G020	3/31/2023		$ 11,455,413	$ 11,455,413	Transform
99	I&O US CNI FY23	6488	G020	7/31/2023		$ 4,299,602	$ 4,299,602	Grow
100	I&O US Cloud & Hosting FY23	6489	G020	7/31/2023		$ 15,016,209	$ 15,016,209	Grow
101	I&O US Digital Operations FY23	6485	G020	7/31/2023		$ 10,917,055	$ 10,917,055	Grow
102	I&O US Networks FY23	6486	G020	7/31/2023		$ 17,999,792	$ 17,999,792	Upgrade/Mod
	Other Investments							
103	Evolution Process Enhancements & Improvements	6297	G020	3/31/2023		$ 4,900,000	$ 4,900,000	Grow
104	Apptio	6264US	G020	3/31/2023	351,155	$ 600,000	$ 600,000	Grow
105	E-Hub	6266US	G020	3/31/2023	878,346	$ 800,000	$ 800,000	Grow
106	DP Secure & Compliant	6565	G020	3/31/2023		$ 1,874,631	$ 1,874,631	Grow
107	JIRA	6520US	GT53	3/31/2023		$ 900,000	$ 900,000	Grow
108	DP GridStack	6519	GT53	3/31/2023		$ 14,525,369	$ 14,525,369	Grow
	Security							
109	Physical Security Child Project	3617A - G20	G020	3/31/2022		$ 498,333	$ 498,333	Upgrade/Mod
110	Physical Security Child Project	3617F - NIMO	G114	3/31/2022		$ 2,990,000	$ 2,990,000	Upgrade/Mod
111	Physical Security Child Project	3617G - KEDNY	5220	3/31/2022		$ 498,333	$ 498,333	Upgrade/Mod
112	Physical Security Child Project	3617H- KEDLI	5230	3/31/2022		$ 498,333	$ 498,333	Upgrade/Mod
113	Physical Security Child Project	3617I -Mass Electric	5310	3/31/2022		$ 498,333	$ -	Upgrade/Mod
114	Physical Security Child Project	3617J - Boston Gas	5330	3/31/2022		$ 996,667	$ -	Upgrade/Mod
115	Physical Security Child Project	3617K - Colonial Gas	5340	3/31/2022		$ 249,167	$ -	Upgrade/Mod
116	Physical Security Child Project	3617L - RI Electric	536E	3/31/2022		$ 498,333	$ -	Upgrade/Mod
117	Physical Security Child Project	3617M - RI Gas	536G	3/31/2022		$ 249,167	$ -	Upgrade/Mod
118	Cyber Operational Technology	5933	G020	3/31/2022		$ 21,400,000	$ 21,400,000	Upgrade/Mod
119	Vulnerability Management	6417US	G020	3/31/2022		$ 2,491,667	$ 2,491,667	Upgrade/Mod
120	Network Security	6526US	GT53	3/31/2022		$ 2,800,000	$ 2,800,000	Upgrade/Mod
121	Platform Security	6480US	GT53	3/31/2022		$ 5,581,333	$ 5,581,333	Upgrade/Mod
122	Security Orchestration Automation and Response	6540US	GT53	3/31/2022		$ 1,495,000	$ 1,495,000	Upgrade/Mod
123	Identity and Access Management	5870	G020	6/30/2022		$ 1,400,000	$ 1,400,000	Upgrade/Mod
	FY23 Investment Plan					**$ 299,257,364**	**$ 276,764,820**	

Niagara Mohawk Power Corporation d/b/a National Grid, The Brooklyn Union Gas Company d/b/a National Grid, NYKeySpan Gas East Corporation d/b/a National Grid
Allocations used for IT Projects

SAP Alloc. Code	Description	Allocation Basis	5210E NMPC - ELEC	5210G NMPC - GAS	5210T NMPC - TRANS	NMPC Total	5220G KEDNY	5230G KEDLI	ALL OTHER CO's	TOTAL
5220	KEDNY	Direct Charge	0.00%	0.00%	0.00%	0.00%	100.00%	0.00%	0.00%	100.00%
5230	KEDLI	Direct Charge	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	0.00%	100.00%
5310	Massachusetts Electric Company	Direct Charge	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	100.00%
5330	Boston Gas Company	Direct Charge	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	100.00%
5340	Colonial Gas Company	Direct Charge	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	100.00%
521E	Niagara Mohawk Electric Distribution	Direct Charge	100.00%	0.00%	0.00%	100.00%	0.00%	0.00%	0.00%	100.00%
521T	Niagara Mohawk Electric Transmission	Direct Charge	0.00%	0.00%	100.00%	100.00%	0.00%	0.00%	0.00%	100.00%
536E	Narragansett Electric Company	Direct Charge	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	100.00%
536G	Narragansett Gas Company	Direct Charge	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	100.00%
C170	NMPC-E&G, KEDNY, KEDLI	C - Number of Customers via Count of Meters Installed	39.82%	15.05%	0.00%	54.87%	30.57%	14.56%	0.00%	100.00%
C175	All Retail Companies	C - Number of Customers via Count of Meters Installed	22.84%	8.63%	0.00%	31.47%	17.53%	8.35%	42.65%	100.00%
C198	All Electric Retails	C - Number of Customers via Count of Meters Installed	46.82%	0.00%	0.00%	46.82%	0.00%	0.00%	53.18%	100.00%
C209	All Gas Retails not including RI	C - Number of Customers via Count of Meters Installed	0.00%	18.19%	0.00%	18.19%	36.96%	17.60%	27.25%	100.00%
C210	All Gas Retails	C - Number of Customers via Count of Meters Installed	0.00%	16.85%	0.00%	16.85%	34.23%	16.31%	32.61%	100.00%
C225	KEDNY and KEDLI	C - Number of Customers via Count of Meters Installed	0.00%	0.00%	0.00%	0.00%	67.73%	32.27%	0.00%	100.00%
C239	KEDNY, KEDLI, Boston Gas and Colonial Gas	C - Number of Customers via Count of Meters Installed	0.00%	0.00%	0.00%	0.00%	45.18%	21.52%	33.30%	100.00%
C343	KEDNY, Boston Gas and Colonial Gas	C - Number of Customers via Count of Meters Installed	0.00%	0.00%	0.00%	0.00%	57.56%	0.00%	42.44%	100.00%
C434	Legacy NG Retails and KEDLI	C - Number of Customers via Count of Meters Installed	32.84%	12.41%	0.00%	45.25%	0.00%	12.01%	42.74%	100.00%
C902	All Electric Distribution not including RI	C - Number of Customers via Count of Meters Installed	54.64%	0.00%	0.00%	54.64%	0.00%	0.00%	45.36%	100.00%
G114	Niagara Mohawk Power Corporation	G - General Allocator, 3-Point Formula	63.29%	16.68%	20.03%	100.00%	0.00%	0.00%	0.00%	100.00%
G020	Parents, Retail, TRAN, Hydros, NG LNG Reg, KS GEN, GW, PJ, Metrowest, Wayfinder, Transgas, KEDC, KSI	G - General Allocator, 3-Point Formula	16.42%	4.36%	5.28%	26.06%	13.69%	8.10%	52.15%	100.00%
G148	Retails, Transission	G - General Allocator, 3-Point Formula	17.58%	4.63%	5.56%	27.77%	14.49%	8.56%	49.18%	100.00%
G186	NMPC-E&T, MECO-E&T, Nantucket, NECO-E&T and NEP-T	G - General Allocator, 3-Point Formula	30.40%	0.00%	10.51%	40.91%	0.00%	0.00%	59.09%	100.00%
G187	NMPC-E&T, MECO-E&T, Nantucket and NEP-T	G - General Allocator, 3-Point Formula	35.91%	0.00%	12.27%	48.18%	0.00%	0.00%	51.82%	100.00%
G324	KS Generation, Glenwood and Port Jefferson	G - General Allocator, 3-Point Formula	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	100.00%
G370	Legacy NG Elec Retails, NMPC-T and NEP-T	G - General Allocator, 3-Point Formula	31.54%	0.00%	11.04%	42.58%	0.00%	0.00%	57.42%	100.00%
G383	Gas Retails excluding Niagara Mohawk	G - General Allocator, 3-Point Formula	0.00%	0.00%	0.00%	0.00%	38.44%	21.89%	39.67%	100.00%
G444	NIMO-EGT,KEDNY, KEDLI, MECO-ET, Boston, Colonial Gas, NECO-G, NEP	G - General Allocator, 3-Point Formula	19.37%	5.08%	6.04%	30.49%	15.86%	9.33%	44.32%	100.00%
GT53	G-020 plus LIPA TSA Residual entity	LIPA TSA G - General Allocator, 3-Point Formula	18.53%	4.90%	5.91%	29.34%	15.36%	9.09%	46.21%	100.00%
T186	NMPC-E&T, MECO-E&T, Nantucket, NECO-E&T, and NEP-T	T - Total T&D Expenditures	31.10%	0.00%	9.84%	40.94%	0.00%	0.00%	59.06%	100.00%

Niagara Mohawk Power Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY
KeySpan Gas East Corporation d/b/a National Grid

Information Technology Capital Investment Plan

Cases 20-E-0380, 20-G-0381, 19-G-0309 and 19-G-0310

FISCAL YEAR 2024

PREPARED FOR:
DEPARTMENT OF PUBLIC SERVICE STAFF
3 EMPIRE STATE PLAZA
ALBANY, NY 12223

March 31, 2023

Fiscal Year 2024 Information Technology Capital Investment Plan

I. Introduction

Niagara Mohawk Power Corporation (Niagara Mohawk), The Brooklyn Union Gas Company (KEDNY), and KeySpan Gas East Corporation (KEDLI) (together "National Grid" or the "Companies") submits its Information Technology ("IT") Capital Investment Plan (the "Investment Plan") for fiscal year 2024 ("FY24 Plan").[1] This submission is in compliance with Section IV.6.5 of the Joint Proposal adopted by the New York Public Service Commission ("PSC" or the "Commission") January 20, 2022 in Cases 20-E-0380 and 20-G-0381, and Section IV.10.5 of the Joint Proposal adopted by the Commission August 12, 2021 in Cases 19-G-0309 and 19-G-0310.[2]

IT capital projects and investments that are shared investments across National Grid's operating companies are implemented and owned by National Grid USA Service Company, Inc. ("Service Company") and allocated to the benefiting US operating companies, including Niagara Mohawk, KEDNY and KEDLI in the form of an annual rent expense once the investments are in-service. The rent expense includes a return on the capital investment plus depreciation expense.

The FY24 Plan is made up of in-flight projects that will require capital spending to complete as well as projects that will be sanctioned and begin to incur capital charges during the fiscal year. In total, the FY24 Plan reflects Service Company capital expenditures of $295.625 million across the US operating companies. Of this amount, approximately $290.563 million is for projects that benefit and, therefore, are chargeable to Niagara Mohawk, $225.530 million to KEDNY and $261.730 million to KEDLI.[3] The FY24 Plan provided in Attachment 1 includes a list of projects, forecast in-service dates, and the allocators used to determine the rent expense for the operating companies that will benefit from each project, along with FY24 Service Company IT capital costs and the full Service Company project spend expected for each investment.

National Grid's IT organization continuously reviews the Investment Plan relative to current business requirements, risks and information. The FY24 Plan assumes that large-scale economic, political, and environmental events will not significantly impact the Capital Investment Plan or

[1] FY24 is from April 1, 2023 through March 31, 2024.

[2] Cases 20-E-0380 and 20-G-0381, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Niagara Mohawk Power Corporation d/b/a National Grid for Electric and Gas Service*, "Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans" (issued and effective January 20, 2022) and Cases 19-G-0309 and 19-G-0310, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Energy East Corporation d/b/a National Grid for Gas Service*, "Order Adopting Joint Proposal, and Establishing Gas Rate Plans" (issued and effective August 12, 2021).

[3] The Joint Proposal adopted in Cases 20-E-0380 and 20-G-0381 reflected a Service Company investment level for FY24 of $252 million for projects that benefit and, therefore, are chargeable to Niagara Mohawk, and the Joint Proposal adopted in Cases 19-G-0309 and 19-G-0310 reflected a Service Company investment level for FY24 of $150 million for projects that benefit and therefore, are chargeable to KEDNY and KEDLI.

National Grid's ability to deliver it. However, National Grid will continue to review its technology and customer needs and revise the Investment Plan as necessary.

II. Development of the Capital Plan

National Grid engages in an iterative process that involves collaboration with Jurisdictional representatives and the functional business areas (*e.g.,* Operations, Customer and Digital, Network Strategy, and Finance and Shared Services) to identify the investments of highest priority for inclusion in the Capital Investment Plan. This process aligns with the corporate budgeting process and strategic portfolio reviews.

The development of the IT FY24 Plan was based on an assessment and analysis of current and proposed programs aligned to the Company's IT strategic focus - build and maintain the foundation, drive new capabilities, and enable cost savings. We reviewed each program and balanced funding across focus areas to optimize investments and drive value. The FY24 Plan focuses on the following:

- Fulfilling regulatory and legal mandates
- Reducing cyber and operational risk
- Driving standardization and modernization
- Building tools and platforms to empower our people
- Safeguarding critical business applications
- Building new capabilities for improved business performance
- Implementing cost efficiencies and saving programs to achieve year-over-year run-the-business ("RTB") cost reductions

In building the IT FY24 Plan, National Grid first identified the amount of work remaining and the delivery timeline for in-flight, multi-year investments. Next, new investments to address compliance with a legal or regulatory requirement/mandate or an open corporate risk were added to the Plan. Third, investments that focused on IT infrastructure – primarily investments to upgrade/replace/add functionality that would create reliability in systems and fix foundational capabilities necessary for operations and/or future aspirations of the business (including ensuring digital security) were included. Finally, the IT team worked closely with the Jurisdiction and business executive sponsors to focus on their top project requests.

After this analysis, IT distributed a draft Investment Plan to business functions, Jurisdictional representatives, and other key stakeholders for feedback and held review sessions on the proposed plan. A final Investment Plan review occurs each fiscal year with the Jurisdictional Presidents, Jurisdictional delegates, and functional executives. Upon the review and approval of the Investment Plan by the Jurisdictional Presidents and functional executives, the consolidated financial information from the Plan, by jurisdiction and company, is incorporated with the IT O&M requirements into the IT budget. The IT budget is then reviewed and approved by the New York Chief Information Officer ("CIO"), Global CIO, and Corporate Finance, and is incorporated into a consolidated National Grid USA budget for approval by the National Grid plc Board.

III. Formal Sanctioning Support and Oversight of Projects Included in the Investment Plan

Throughout the year, IT projects are subject to a formal governance process that includes a review and approval for all projects greater than $0.030 million. Within IT, there is a team that reviews all IT projects before forwarding the project to the appropriate sanctioning authority for approval. Business Sponsored IT Investments up to $50 million that are classified as "Grow" will require both IT & Business approval.

- Projects less than $10 million are approved by the CIOs for their portfolios through a workflow approval process and under their delegations of authority.
- Projects from $10 million to $25 million are approved by the Senior IT Leadership through a workflow approval process and under their delegations of authority.
- Projects from $25 million to $50 million are approved by the Global CIO through a workflow approval process and under their delegation of authority.
- Projects from $50 million to $203 million are approved by the New York and New England Executive Sanctioning Committees ("NYESC" and "NEESC"), under delegation of authority by the President, New York Jurisdiction and President, New England Jurisdiction. In addition, these committees will also approve any projects less than this amount which are IT projects classified as Transform and any other Projects at its discretion without regard to whether such Projects exceed the $50 million threshold.
- Projects which exceed the $203 million threshold must go to the Group Investment Committee for approval.

IT projects are typically initially sanctioned for Requirements and Design work (also referred to as "Partial Sanction") and then will return for Development and Implementation sanction (also referred to as "Sanction" or "Full Sanction"). If project scope and costs are well understood (*e.g.*, upgrades to hardware that do not necessitate requirements and design work or whose cost estimate can be reliably informed by similar previous installations), a project may proceed directly to "Sanction." In the event a project is forecast to exceed its sanction amount, a project must return to the appropriate Governance Committee for "Re-Sanction" consideration. A "Closure" paper is prepared at the completion of a project detailing the financial and objective outcomes of the project.

The Investment Plan includes all IT projects that progress from initial Feasibility & Analysis phases to ultimate closure to plant in service, as defined below, and supported by the formal sanctioning and governance process discussed above.

- **Feasibility & Analysis** – A study conducted in cases where a significant level of analysis is required to better determine the direction or approach for a project. Because such a study is likely to have a significant impact on IT resources and spend, the effort is treated as a project to be included in the Investment Plan.

- **Requirements & Design** – This phase includes requirements definition and completion of analysis, confirmed project scope, agreed functional design, and technology solution to meet business and technical requirements, confirmed plan for Development and Implementation, and confirmed production handover, support strategy, and sourcing approach.
- **Development & Implementation** – This phase includes development to the specified designs, testing, preparation for training, deployment of the solution to the business, and post-implementation support.
- **Project Closure** – This phase closes the project, including capturing ongoing RTB costs, completing post implementation review and lessons learned, closing open actions and change requests, implementing a decommissioning plan for replaced technologies, and submitting a closure paper to the appropriate sanctioning committee.

Cost estimates for projects that have completed the Requirements & Design stage with full scope and solution design generally have +/- 10% cost estimates. Other projects at earlier stages, and the budgets for those projects, are less refined and more susceptible to changes in scope and budget. The projects in the Investment Plan are continuously reviewed for changes in assumptions, constraints, project delays, accelerations, and system operations, performance, safety, regulatory/compliance and customer driven needs that arise. The sanctioning process allows a further evaluation of the total expected costs of a project at key points in the project's lifecycle, including a view of how a given fiscal year's spend on the project relates to the costs included in the Investment Plan as incorporated in the corporate budget.

IV. Capital Plan Investments by Spending Categories

The Company's FY24 Plan is presented by major program spending category/spending rationale. Descriptions of each portfolio category/spending rationale are provided below.

A. Group Functions

Group Functions executes National Grid's digital roadmaps, leveraging technology and reimagining processes in support of Finance, Enterprise Business Services, Human Resources, Legal, and Safety Health & Environmental, continuing to upgrade SOX related systems, and retiring outdated systems. The 5 largest Investments within this portfolio are as follows:

1. SAP Infrastructure Landscapes FY24 (INVP 5608) – Procuring, setting up and configuring project environments is a critical path task for SAP related projects and initiatives. These activities can be time consuming and costly when done in isolation. To improve IT project implementation schedules, National Grid IT introduced a process to provide annual funding at the beginning of each fiscal year to procure and configure a set of project environments in support of all initiatives within the SAP portfolio for that fiscal year. Appropriate costs will be associated with individual projects by issuing separate work orders for each project. This method enables accurate cost treatment of the environments.

2. S2P Transformation R1: Ariba Guided Buying ~~(US)~~ (INVP 6290A) – This investment will implement and extend the SAP Ariba platform and retire the current Supplier Relationship

Management (SRM) module of SAP. This project forms part of a larger group of investments which will redesign the Source to Pay process and implement SAP Ariba as part of the Finance Transformation program. The goal of this initiative is to improve the buying experience, streamline the process including removal of manual efforts and workarounds, and simplify the technical landscape.

3. S2P Transformation R2: Supplier Onboarding ~~(Global - US)~~ (INVP 6290B) – This investment will deliver Smart Supplier On-Boarding processes, replacing Manual vendor on boarding work with Data Governance and integration with the SAP Landscape. This project forms part of a larger group of investments which will redesign the Source to Pay process and implement SAP Ariba as part of the Finance Transformation program. The goal of this initiative is to improve the buying experience, streamline the process, including removal of manual efforts and workarounds, and simplify the technical landscape.

4. JML (Joiners, Movers & Leavers) MyHub ~~US~~ (INVP 6621) –
   The J2L transformation aims to simplify and digitally enable core People & Culture processes and systems to deliver a consistent, end-user focused experience that delivers three key outcomes:
   - User experience – reduce process errors, improve data quality and analytics to better inform decision making, build colleague capabilities to ensure processes are executed as designed.
   - Efficiency – automation, simplification, and standardization to reduce time to execute and transact end-to-end processes.
   - Compliance – ensure processes are executed as designed in line with company policies, reducing risk.

5. SAP PI NW 7.40 (PI -> PO upgrade) to NW 750 (INVP 5843F) - National Grid uses the SAP platform for back-office operations in support of various processes, including Record to Report (R2R), Join to Leave (J2L) and Supply Chain Management (SCM) among others. This project will upgrade SAP Process Integration (PI) to SAP Process Orchestration (PO) which is a more current and advanced SAP integration module. SAP PI went out of support on December 31, 2020.

B. United States Electric Business Unit

IT supports the US Electric Distribution and Transmission businesses in modernizing key legacy operational systems while building capabilities that will effectively enable the energy transition. Through digital delivery, the Electric CIO organization provides greater efficiency and data driven insights into our workforce operations:

- Modernization of our Foundational Infrastructure
- Delivery of new Business Capabilities and Ways of Working for our End Users
- More automated and reliable grid controls systems
- Capital programs and projects that support the growth of Distributed Energy Resources (DERs)
- A growing market of prosumer electric customers

Key projects within the Electric portfolio in FY24 include:

1. Future Now (INVP 6076) – Work planning in the Electric Business Unit (EBU) is a manual process involving multiple systems which are not fully integrated. This investment will complete the development and roll out of FutureNow optimization capabilities to the Transmission Business, enabling the EBU to further maximize strategic objectives against budget and resources in both the Distribution and Transmission businesses.

2. On My Way (OMW) (Excluding Mobile Outage Dispatch) (INVP 5054) – This investment will continue to release new Digital capabilities that result in improved reliability and/or efficiencies for the Electric Business. In FY24, the team's top priorities are the deployment of digitized trouble work/CWR, sketch markups to improve job closeout, digitized dispatch maintenance work, along with getting more job types (e.g. LED conversions, UG, etc.) into OMW.

3. Demand Planning & Material Track and Trace (INVP 6661) – Pending Validation Completion, delivery of the Demand Planning tool is expected to better predict demand for goods at distribution centers and automatically generate orders with recommended supplier and bundling options, while delivery of the Material Track & Trace tool is expected to enable our field workers to view and locate materials from nearby barn or distribution center, then easily check out material with mobile device to improve inventory level accuracy and complete construction jobs more timely/efficiently.

4. GridInspector (Computapole Replacement) (INVP 5890C) – Replacement of legacy Inspection and Maintenance T&D System that is nearing the end of its life. This investment will deliver a supported I&M system that's Windows 10 compliant, can be available/used on mobile devices and most importantly, ensures National Grid can effectively and efficiently run our mandated inspection programs.

5. Retail Settlement (INVP 5891E) –Implementation of a new Retail Settlement system expected to be completed by the end of FY24. The company implemented a new Wholesale Settlement solution in FY22, replacing two end of life systems which were no longer supportable due to outdated components, infrastructure, and application code. This investment allows for consolidation of our legacy Retail Settlement system, into the new Wholesale Settlement system.

C. United States Gas Business Unit

IT supports the US Gas Distribution and Transmission businesses by developing digital products based on the foundation of gas business enablement that deliver continuous improvement of our asset lifecycle management capabilities through:

- Risk based asset investment planning
- Performance based maintenance
- AI based work optimization
- Mobile field products

Key Gas Investments for FY24 include:

1. Continued Scaling and Evolution of GBE Transformation (INVP 6539) – Development and deployment work in support of the GBE program as it evolves, as well as additional features and capabilities beyond the core suite of GBE products that enhance GBE's value and impact. Areas of focus for FY24 will be separated between core functionality and enhancements, with the former focused on a range of services including asset and resource management, asset investment planning, construction cycles, and contractor invoicing, and the latter seeking to enhance workflow, data consistency, and collections processes, among others.

2. Gas Management System (Supervisory Control and Data Acquisition SCADA) Upgrade (INVP 6550) – This multiyear project will upgrade the operationally critical Gas Management System (GMS) Supervisory Control and Data Acquisition (SCADA) for Gas businesses in New York and Massachusetts. FY24 outcome will complete the detailed solution design, and implementation of the SCADA development environment in preparation for production implementation and deployment in FY25.

3. Enterprise Platform for Gas Retail Choice Programs (INVP 5787D) – The Gas Transportation Information System (GTIS), currently used by US Gas supply and retail choice programs in DNY, and Long Island. GTIS consolidation program goal is to migrate UNY and MA regions to consolidated common GTIS platform and decommission of the several legacy applications characterized as being at security risk (multiple critical vulnerabilities which expose NG to external threats) due to old unsupported technologies.

4. Gas Meter Management System Deployment and Integration (INVP 6642) – This project is to enhance the gas meter lifecycle management solution deployed in Maximo through improvements to the meter test/inventory process. The Maximo asset management solution works well when meters have been installed but is underperforming as an asset inventory system. This is resulting in a bottleneck during the receipt, test and shipping phase of meter shop operations. The TESCO Asset Management Solution provides this functionality out-of-the box, with some minor configuration. This project encompasses the phased deployment of the solution followed by integration to the Customer Service System (CSS) and Maximo Work & Asset Management solution.

5. Data Modernization (INVP 6671D) – Implementation of the Gas Data Modernization effort is intended to develop a modern and centralized data platform that streamlines data retention, integration, integrity, transformation, archiving, analytics, compliance, reporting, quality, governance, and decision making required to meet regulatory mandates. The solution will be implemented for New York State Energy Research and Development Authority (NYSERDA) Integrated Energy Data Resource (IEDR) and prioritized use cases.

[redacted] LNG OT Cybersecurity Remediation (INVP 6479) – [redacted]

D. Customer

IT provides support to our US Customer business, delivering best-in-class digital products that increase engagement and reduce cost to serve. Areas of focus will include:

- Modernizing our legacy "back-end" IT estate to support more digital products.
- Liberating data and creating an abstracted and real-time customer data platform to assist with achieving energy efficiency targets.

Key projects in the Customer Area include:

1. Billing & Collections Mandates & Enhancements (INVP 6538) - This is a funding source for new billing and collections projects in FY24 which will arise due to future mandates.

2. UWP 2.0 Re-Platform (INVP 6584) – The Unified Web Portal enables customers to interact with National Grid online, providing services such as bill pay and energy insights, which are key enablers for customer satisfaction. Key strategic programs, such as AMI and ZBR are critically dependent on UWP for customers to utilize core functionality. The UWP 2.0 Re-Platform will build upon the Unified Web Portal (UWP) Minimum Viable Product, further establishing a platform that provides a highly reliable, seamless experience, delivering on customers' expectations.

3. 2022 AIMS Platform (INVP 6302) – This investment is the continued expansion and technology refresh of the Contact Center technologies including telephony, self-service Interactive Voice Response and Work Management and Call Recording systems. Continued updates and enhancements to the existing the Contact Center technologies will enable increased customer self-service and operational efficiencies.

4. Customer Ecosystem Stabilization (IT Project) (INVP 5787) – This investment seeks to stabilize our customer ecosystem (including remediation of technical debt), which focuses on the Customer Related Information System (CRIS), Customer Service System (CSS), as well as the applications around our core billing systems. Much of current infrastructure within Customer is out of support, and the outcome of this investment is to address the most pressing, highest risk areas of technical debt to ensure that the customer ecosystem can continue to operate in its full capacity. Benefits include reducing frequent outages to peripheral applications, bringing into support applications which have fallen out of support, and remediating applications which are out of support and are critical to the business.

5. Customer Data Platform (INVP 6349) – A comprehensive data assessment highlighted the need for improvement in basic master data capabilities, governance, and the necessary tools to support the highly complex demands coming from the business. There are ~300 business capabilities at National Grid that make up the delivery of customer service and operations. This investment will ensure data is accessed consistently and accurately from a single source. It will enable the Company to speak to customers with one voice across the customer lifecycle, and the ability for the customer to contact the Company in the manner of their choice.

6. Customer Connection Program (INVP 5125) – The Customer Connection Program covers mandated, operational and system reliability enhancements for the GridForce system and related interfaces. This program is needed to assure regulatory compliance for Electric and Gas customer connections, as well as Distributed Generation.

7. My Business Account (Nucleus Phase 2) (INVP 5673) - This project's focus is on re-engineering the National Grid commercial and industrial (C&I) customers' digital interactions to create a streamlined and seamless customer experience throughout the customer journey. The project will enhance existing transaction capabilities and account management tools to support the unique C&I needs for multi-account management. It will continue to build in the National Grid managed Azure Cloud Instance using the Enterprise Solution Platform Mulesoft as the middleware to connect to existing core system AGS, .Net, CSS, CRIS and upcoming Salesforce applications. The driver of this project is to reduce challenges our C&I customers face surrounding their National Grid experience, addressing key customer pain points and enabling core capabilities.

E. Digital Platforms and Infrastructure Technology ("DPIT")

DPIT supports National Grid's internal IT Architecture covering Cloud and Hosting, Critical Network Infrastructure, Networks, End User Experience and Operations. DPIT is focused on reducing National Grid's technical debt through cost effective, modernized IT infrastructure, enabling cloud capabilities that meet our risk profile. The organization enables a connected, seamless enterprise that is flexible, secure, reliable and scalable while maintaining acceptable levels of technical debt and risk. Key DPIT programs include:

1. DPIT Digital Workplace Services FY24 (INVP 6681US) - End users need to be more engaged, agile, secure, and therefore productive and efficient through the technologies that they interact with daily. The investment will focus on improving the technology experience for our end users, employees, contractors and partners. Initiatives include:
    - Devices - Provide our colleagues with appropriate devices which enable them to securely access applications and data.
    - Conferencing - Enable high quality synchronous digital interactions between people with very low latency.
    - Collaboration - Deliver asynchronous, highly available applications which allow colleagues to co-create business workflows and documents.
    - Access Management - Simplification of Active Directory Domain Structure to provide stronger security controls in a more easily maintained manner

- Support Assurance - Deliver a multi-channel, cost-effective and digital suite of provisioning, support and knowledge experiences.
- Digital Adoption and Change (DA&CM) - Prepare and enable the business to efficiently adopt our digital productivity and collaboration experiences.

2. Networks FY24 (INVP 6682) - The US Networks Agile Release Train (ART) consists of the following initiatives to mitigate end of service life hardware/software and high demand to meet our network service, customer, and regulatory requirements.
    - Cisco IP Telephony - Upgrade voice infrastructure design improving resiliency for Voice and deliver carrier diversity to improve service availability
    - WAN/SD-WAN - Transforming the legacy WAN environment to SD-WAN across National Grid US sites to modernize infrastructure and increase capacity driven by demand from digitalization and the continued demand to support our employees.
    - Delivering new unified communications capabilities within the US regional estate and decommissioning legacy infrastructure
    - Network Support Services - Provide improved and robust network services and monitoring.
    - LAN/Wireless - Provides a reliable and functional wired and wireless network ensuring proper coverage, uptime, and performance for employees and guests of National Grid.
    - Fully migrating all connected devices from End-of-Life Secure Telecommunications Internet Gateway (STIG) core routing and switching infrastructure (edge routers and access and distribution switches) to a new modern switch infrastructure within each of the STIG locations

3. I&O US Data Center Transformation (INVP 6223) – Currently, our Cloud and Hosting environment and partnership are failing to keep pace with strategic requirements. A wholesale re-establishment of our data center and computer foundation is required in order to remove inefficiencies and complexities which have developed over a number of years. Four new Data Centers will be created with reduced operating costs, modernized infrastructure, increased security and a reduction in carbon footprint (~40% reduction of CO2 from current DXC locations).

4. DPIT Digital Platforms (INVP 6678US) – Digital Platforms is responsible for providing a perennial secure and reliable digital arena to enable the enterprise's clean energy transition for all. Elements of this investment are:
    - GridStack is the suite of reusable, enterprise platforms that software and data engineers use to develop, cost-effective, secure, compliant and well-governed software and data insights at National Grid.
    - PowerApps & Power Automate - to complete the build-out of the Power Platform COE, including developing governance and control over use of the Power Apps and Power Automate capabilities
    - Secure & Compliant - to maintain/optimize our Trusted Platforms to an optimal level, enable adherence to contractual support agreements, enact continued availability through standard procedures such as DR testing and embed the

necessary security rigor and practices to mitigate service disruptions due to bad actors, malicious actions and product deficiencies
- ServiceNow - to improve platform health, deliver a ServiceNow platform that is secure and compliant to National Grid requirements and provide the foundations for further digital transformation

5. I&O US Cloud & Hosting FY24 (INVP 6679US) - Cloud & Hosting is on a multi-year journey to transform the core computing infrastructure from traditional physical distributed IT assets to centralized Cloud Platforms (both public and private). The overarching goals are to deliver the necessary computing power to support our business applications in the most sustainable and cost-effective manner, while mitigating risk and performance issues by modernizing assets and processes.

6. US Critical National Infrastructure (CNI) FY24 (INVP 6680US) – This investment ensures continued reliable delivery of energy to customers by ensuring that the CNI assets remain secure, supported and reliable. Areas of focus will be:
   - CNI Networks - Network support and lifecycle management for CNI environment.
   - Reducing overall technical debt
   - Replacing unsupported Operational Telemetry (OpTel) circuits
   - Configuring CNI Operational Telemetry networks to encrypt data

7. Global Technology Operations FY24 (INVP 6683US) - Digital Operations transforms technology operations from manual and reactive to proactive and automated. The net effect is to increase business agility by enabling National Grid to exploit dynamic technologies such as hyper converged infrastructure and containerisation. The functionality delivers near real-time status and performance data with automated anomaly detection to pin point the root cause of faults in seconds. Digitising operations supports National Grid. This investment will deliver:
   - Unified Technology Ops Centre (UTOC) - A single Operations Centre spanning Enterprise, CNI and Operational Technology.
   - Service Aware Configuration Management Database (CMDB) - Near real-time status and relationship tracking of technology components and mapping to business services, which enables data driven decision making.

F. Security

The Security function protects our people, premises, and digital systems with the objective of maintaining trust in National Grid services, focusing on:
- Reduction in likelihood and impact of security incidents
- Compliance with relevant laws, standards and regulations (*e.g.*, NERC, PCI DSS, GDPR, NIS-R)
- Improved user experience and standard of security practices across National Grid

The Security function aims to reduce the likelihood and impact of physical and cyber security incidents affecting National Grid. It does this by championing security best practices, technology and promoting a security-awareness culture.

1. Cyber Operational Technology (INVP 5933) – This project will deliver improvements to National Grid's network resilience and defense in depth capabilities to facilitate our ability to identify cyber risks and threats, protect and detect against malicious cyber activity and effectively respond to and recover from cyber incidents at operational sites.

2. Physical Security (INVP 3617) – This project will deliver improvements to security at physical locations by upgrading monitoring systems and supporting infrastructure to maintain regulatory compliance for monitoring and safety of our people and premises. The project will replace out of support and warranty field units where failure may result in extended replacement lead times or inability to obtain like-for-like replacements. It will also improve the quality of monitoring with high quality video capture and detection systems.

3. Platform Security (INVP 6699US) – This project will deliver improvements that reduce the likelihood of cyber incidents related to malware and exploitation of insecure system configurations, identify high value hardening activities, assess impact on operational stability and environments, test and implement system hardening and improve mobile threat defence.

4. Network Security (INVP 6700US) This project will deliver improvements that reduce the likelihood of cyber incidents related to distributed denial of service attacks and protect against and enable detection of network traffic related to malware, which can propagate through the network if traffic is not monitored or alerted upon.

5. Identity and Access Management (INVP 6613) - This project will deliver improvements that reduce the likelihood of cyber incidents relating to unauthorized access or misuse of access privileges, and privileges on critical applications.

6. Security Orchestration Automation and Response (INVP 6540US) - This project will deliver improvements to response and recovery capability, reduce the overall impact duration of an incident and implement advanced threat detection capabilities that factor changes to the threat landscape.

7. Vulnerability Management (INVP 6417US) – This project will deliver improvements to National Grid's penetration testing, vulnerability scanning and web application scanning capabilities that identify vulnerabilities and facilitate prioritization for remediation based on risk.

V. Conclusion

Consistent with Section IV.10.5 of the Joint Proposal for cases 20-E-0380 and 20-G-0381 and Section IV.6.5 for Cases 19-G-0309 and 19-G-0310, the Company will provide quarterly variance reports within 60 days after the end of each Fiscal Year quarter. The reports will address the status of IT programs and budgets. On a semi-annual basis, the Company will also meet with Department of Public Service Staff to discuss budget and actual spending to date and discuss project status.

Niagara Mohawk Power Corp., The Brooklyn Union Gas Co., KeySpan Gas East Corp. d/b/a
National Grid
Cases 20-E-0380 20-G-0381, 19-G-0309 and 19-G-0310
FY24 IT Capital Investment Plan
Attachment 1
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NMPC/KEDNY/KEDLI
In-Flight and Forecasted Information Technology (IT) Capital Projects

Line	Investment Name	INVP #	PRJ#	Bill Pool	In-Service Date	Inception To 2/28/23	FY24 CAPEX	FY24 Spend for Projects Of Any NY Allocation	Category
	Customer Business Unit								
1	Billing & Collections Mandates & Enhancement	6538	PRJ-3756	C175	2/1/2024	$ -	$ 10,640,000	$ 10,640,000	Run
2	UWP 2.0 Re-platform	6584	PRJ-3828	C175	12/30/2023	$ 9,107,372	$ 9,000,000	$ 9,000,000	Run
3	AIMS Replatform	6302	PRJ-4867	C175	1/8/2024	$ 707,927	$ 8,500,000	$ 8,500,000	Grow
4	Customer Ecosystem Stabilization (IT Project	5778	PRJ-4865	C434	2/1/2024	$ 1,199,000	$ 6,200,000	$ 6,200,000	Run
5	Customer Data Platform Project	6349	PRJ-2083	G148	3/1/2024	$ 10,835,976	$ 5,000,000	$ 5,000,000	Grow
6	Customer Connection Program	5125	PRJ-3633	C175	1/8/2024	$ -	$ 3,560,000	$ 3,560,000	Run
7	My Business Account	5673	PRJ-587	C175	12/1/2024	$ 6,192,000	$ 1,800,000	$ 1,800,000	Grow
8	FY24 UWP Enhancements	6706	PRJ-2104	C175	12/15/2023	$ -	$ 1,290,000	$ 1,290,000	Run
10	AIMS Sustain Product Team	6531	PRJ-2102	C175	1/8/2024	$ 630,000	$ 1,020,000	$ 1,020,000	Run
11	NY Value Stack Billing	5160	PRJ-2094	521E	6/30/2023	$ 7,628,515	$ 690,000	$ 690,000	Run
12	Connections Portal	6517	PRJ-4593	C210	1/8/2024	$ -	$ 110,000	$ 110,000	Grow
	United States Electric Business Unit								
13	Future Now	6076	PRJ-2106	C198	3/31/2024	$ 5,669,749	$ 5,500,000	$ 5,500,000	Transform
14	On My Way (OMW) (Excluding Mobile Outage Dispatch)	5054	PRJ-2043	C198	3/29/2024	$ 10,123,000	$ 5,100,000	$ 5,100,000	Transform
15	Demand Planning & Material Track and Trace	6661	PRJ-4510	C902	3/31/2024	$ -	$ 5,000,000	$ 5,000,000	Transform
16	Computapole Replacement	5890C	PRJ-4524	G187	3/31/2025	$ -	$ 4,230,000	$ 4,230,000	Run
17	Retail Settlement	5891E	PRJ-659	G370	10/9/2023	$ 2,669,369	$ 3,590,000	$ 3,590,000	Run
18	IDS Replacement	5890D	PRJ-4546	G187	12/31/2024	$ -	$ 2,500,000	$ 2,500,000	Run
19	Project Costing / Contract Management	6662	PRJ-4559	C902	3/31/2024	$ -	$ 2,500,000	$ 2,500,000	Transform
20	US Rapid Rebuild	6631	PRJ-4874	G187	12/31/2023	$ -	$ 2,500,000	$ 2,500,000	Run
21	Veg. Management Optimization (VMO)	6073	PRJ-2105	C198	3/1/2024	$ 4,881,863	$ 2,000,000	$ 2,000,000	Transform
22	Deploy New Routers/Switches for NY and MA Non-Transformed Substation	TBD	PRJ-5145	G901	3/31/2024	$ -	$ 1,750,000	$ 1,750,000	Grow
23	MDS Consolidation Phase 1	5891F	PRJ-5149	G187	3/31/2024	$ -	$ 1,500,000	$ 1,500,000	Run
24	SD-WAN Distributed Communications	5871DA	PRJ-5150	521E	12/31/2023	$ -	$ 300,000	$ 300,000	Grow
25	Generation Asset Management Upgrade	5822C	PRJ-3640	G324	6/30/2023	$ 5,700,436	$ 200,000	$ -	Run
26	NY ISO PMU	5773D	PRJ-2722	521T	4/27/2023	$ 1,177,845	$ 130,000	$ 130,000	Run
	United States Gas Business Unit								
27	CS&E	6539	PRJ-3651	C903	3/31/2024	$ 25,965,653	$ 58,750,000	$ 58,750,000	Grow
28	Gas Management System (SCADA) Upgrade	6550	PRJ-3657	C209	10/31/2026	$ 756,548	$ 16,800,000	$ 16,800,000	Run
29	Enterprise Platform for Gas Retail Choice Programs	5787D	PRJ-1571	C210	3/31/2025	$ 2,786,000	$ 4,900,000	$ 4,900,000	Run
30	Gas Meter Management System Deployment and Integratio	6642	PRJ-4639	C209	3/31/2024	$ 351,000	$ 2,950,000	$ 2,950,000	Run
31	Data Modernization	6671D	PRJ-4862	G911	3/31/2024	$ -	$ 1,500,000	$ 1,500,000	Run
32	ES&R (Enterprise Scalability & Resilience)	6671C	PRJ-4943	C210	3/31/2024	$ -	$ 1,320,000	$ 1,320,000	Run
33	LNG IT Support for OT Systems - IT Funded	6479	PRJ-3659	G443	3/31/2024	$ 63,000	$ 1,277,000	$ 1,277,000	Run
34	GACI (Gas Application Continuous Improvements	6671B	PRJ-4942	C209	3/31/2024	$ -	$ 1,230,000	$ 1,230,000	Run
35	Field Collection Systems (FCS) Upgrade (Gas Only	5785Q	PRJ-1575	C210	3/15/2024	$ 60,000	$ 500,000	$ 500,000	Run
36	Gas Application Lifecycle Management (ALM) Application Rationalizatio	6671A	PRJ-4162	C210	3/31/2024	$ -	$ 400,000	$ 400,000	Run
	Group Functions								
37	SAP Infrastructure Landscapes FY24	5608	PRJ-4570	G020	3/31/2024		$ 3,099,920	$ 3,099,920	Run
38	S2P Transformation R2: Supplier Onboarding (Global - US)	6290B	PRJ-3700	G020	1/31/2024		$ 3,000,000	$ 3,000,000	Transform
39	JML MyHub - US	6621	PRJ-5006	G020	3/31/2024	$ 1,290,000	$ 3,000,000	$ 3,000,000	Run
40	SAP PI NW 7.40 (PI -> PO upgrade) to NW 750	5843F	PRJ-3679	G020	3/31/2024	$ 2,061,000	$ 2,443,000	$ 2,443,000	Grow
41	S2P Transformation R1: Ariba Guided Buying (US	6290A	PRJ-1904	G020	5/31/2023	$ 9,263,000	$ 2,000,000	$ 2,000,000	Transform
42	SAP and BackOffice Health Program (Upgrades	5843	PRJ-4567	G020	3/31/2024		$ 1,530,000	$ 1,530,000	Run
43	FP&A Business Planning Modelling and Use cases (Global US	6461USA / 6461USB	PRJ-3689	G020	6/30/2023	$ 1,642,000	$ 1,500,000	$ 1,500,000	Transform
44	SHE Funding Line: Group SHE Digital Evolution US	6533US	PRJ-3698	G020	3/31/2024		$ 1,300,000	$ 1,300,000	Run
45	SAP BW/BI Upgrade	5843E	PRJ-3680	G020	11/30/2023		$ 1,200,000	$ 1,200,000	Grow
46	Legal FY24: Group Legal Digital Evolution	6534US	PRJ-4600	C175	3/31/2024		$ 1,200,000	$ 1,200,000	Run
47	HRSP FY24	5612	PRJ-4543	G020	6/30/2023		$ 1,130,000	$ 1,130,000	Run
48	US Group Functions Operational Requests	5902	PRJ-4581	G020	3/31/2024		$ 1,097,730	$ 1,097,730	Run
49	CRO Funding Line - RSA Archer Upgrade & Refresh US	6695US	PRJ-4529	GT53	3/31/2024		$ 1,050,000	$ 1,050,000	Run
50	US Group Function Minor Enhancements (SAP) - Minor Works FY24-27	6653	PRJ-4995	G020	3/31/2024		$ 1,010,000	$ 1,010,000	Run
51	RPA Strategic / Robotic Process Automation of Manual Journal Entries - Phase	6289H	PRJ-4565	G020	3/31/2024		$ 960,000	$ 960,000	Grow
52	Sustainability Apps and Tools FY24-27	6657US	PRJ-4576	GT53	3/31/2024		$ 900,000	$ 900,000	Grow
53	Procurement Systems Digital Enablement (Global US	6693US	PRJ-4631	G020	3/31/2024		$ 600,000	$ 600,000	Transform
54	Tactical HR FY24+ US	6717US	PRJ-5017	G020	3/31/2024		$ 550,000	$ 550,000	Run
55	US Group Functions: Tools and Technologies	5895	PRJ-4582	G020	3/31/2024		$ 550,000	$ 550,000	Grow
56	SAP Cloud Analytics	5842H	PRJ-4568	G020	3/31/2024		$ 500,000	$ 500,000	Grow
57	US Group Function: R2R Regulatory and Compliance	6597	PRJ-4947	G020	3/31/2024		$ 490,000	$ 490,000	Run
58	Business Services Upgrades and Enhancements	6716US	PRJ-5014	G020	3/31/2024		$ 400,000	$ 400,000	Transform
59	WDD FY24+ US	6716US	PRJ-5019	C175	11/30/2023		$ 320,000	$ 320,000	Grow
60	RPA J2L US	6716US	PRJ-5013	C175	9/30/2023		$ 300,000	$ 300,000	Grow
	Digital Technology & Infrastructure Platforms								
61	DPIT Digital Workplace Services FY24	6681US	PRJ-5066	G020	3/31/2024	$ 6,635,000	$ 16,288,000	$ 16,288,000	Grow
62	Networks FY24	6682	PRJ-5067	G020	3/31/2024	$ 8,150,000	$ 12,024,000	$ 12,024,000	Run
63	I&O US Data Center Transformation	6223	PRJ-3665	G020	12/31/2023	$ 40,028,000	$ 10,740,000	$ 10,740,000	Transform
64	DPIT Digital Platforms	6678US	PRJ-5142	G020	3/31/2024		$ 10,520,000	$ 10,520,000	Grow/Transform
65	DPIT US Cloud & Hosting FY24	6679US	PRJ-5065	G020	3/31/2024	$ 3,480,000	$ 4,430,000	$ 4,430,000	Grow
66	US CNI FY24	6680US	PRJ-5062	G020	3/31/2024	$ 2,474,000	$ 3,630,000	$ 3,630,000	Grow
67	Global Technology Operations	6683US	PRJ-5068	G020	3/31/2024	$ 7,476,000	$ 3,550,000	$ 3,550,000	Grow
	Other Investments								
68	Jira Align US	6520US	PRJ-4624	GT53	3/29/2024	$ 1,423,000	$ 1,960,000	$ 1,960,000	Grow
69	IT Strategic and Investment Planning Tooling	6767	PRJ-4587	GT53	3/29/2024		$ 500,000	$ 500,000	Grow
70	WorkBoard	6615	PRJ-4171	GT53	3/29/2024		$ 420,000	$ 420,000	Grow
71	Apptio Development	6264US	PRJ-1899	G020	3/29/2024	$ 988,000	$ 400,000	$ 400,000	Grow
72	Software Asset Management	6568X	PRJ-5003	G020	3/29/2024		$ 400,000	$ 400,000	Grow
	Security								
73	Cyber Operational Technology	5933	PRJ-3675	G020	3/31/2024	$ 7,742,000	$ 8,500,000	$ 8,500,000	Run
74	Physical Security (US)	3617	PRJ-3676	G020	3/31/2024		$ 6,400,000	$ 6,400,000	Run
75	Platform Security	6699US	PRJ-5333	G020	3/31/2024		$ 3,300,000	$ 3,300,000	Run
76	US Network Security	6700US	PRJ-5334	G020	3/31/2024		$ 3,100,000	$ 3,100,000	Run
77	Identity and Access Management	6613	PRJ-3741	G020	3/31/2024		$ 3,100,000	$ 3,100,000	Run
78	Security Orchestration Automation and Response	6540US	PRJ-3673	G020	3/31/2024	$ 654,000	$ 1,000,000	$ 1,000,000	Run
79	Vulnerability Management	6417US	PRJ-3674	G020	3/31/2024	$ 647,000	$ 1,000,000	$ 1,000,000	Run
	FY24 Investment Plan						**$ 295,629,650**	**$ 295,429,650**	

Niagara Mohawk Power Corporation d/b/a National Grid, The Brooklyn Union Gas Company d/b/a National Grid, NYKeySpan Gas East Corporation d/b/a National Grid
Allocations used for IT Projects

SAP Alloc. Code	Description	Allocation Basis	5210E NMPC - ELEC	5210G NMPC - GAS	5210T NMPC - TRANS	5220G NMPC Total	5230G KEDNY	KEDLI	ALL OTHER CO's	TOTAL
521E	Niagara Mohawk Electric Distribution	Direct Charge	100.00%	0.00%	0.00%	100.00%	0.00%	0.00%	0.00%	100.00%
521T	Niagara Mohawk Electric Transmission	Direct Charge	0.00%	0.00%	100.00%	100.00%	0.00%	0.00%	0.00%	100.00%
C175	All Retail Companies	C - Number of Customers via Count of Meters Installed	22.82%	8.62%	0.00%	31.44%	17.45%	8.38%	42.73%	100.00%
C198	All Electric Retails	C - Number of Customers via Count of Meters Installed	46.74%	0.00%	0.00%	46.74%	0.00%	0.00%	53.26%	100.00%
C209	All Gas Retails excluding NECO	C - Number of Customers via Count of Meters Installed	0.00%	18.19%	0.00%	18.19%	36.81%	17.68%	27.32%	100.00%
C210	All Gas Retails	C - Number of Customers via Count of Meters Installed	0.00%	16.85%	0.00%	16.85%	34.09%	16.37%	32.69%	100.00%
C434	Legacy NG Retails and KEDLI	C - Number of Customers via Count of Meters Installed	32.79%	12.39%	0.00%	45.18%	0.00%	12.04%	42.78%	100.00%
C902	All Electric Distribution excluding NECO	C - Number of Customers via Count of Meters Installed	54.55%	0.00%	0.00%	54.55%	0.00%	0.00%	45.45%	100.00%
C903	All Retails excluding NECO	C - Number of Customers via Count of Meters Installed	25.58%	9.66%	0.00%	35.24%	19.55%	9.39%	35.82%	100.00%
G020	Parents, Retail, TRAN, Hydros, NG LNG Reg, KS GEN, GW, PJ, Metrowest, Wayfinder, Transgas, KEDC, KSI	G - General Allocator, 3-Point Formula	17.83%	4.43%	5.55%	27.82%	13.75%	8.27%	50.16%	100.00%
G148	Retails, Transission	G - General Allocator, 3-Point Formula	19.06%	4.70%	5.83%	29.59%	14.53%	8.71%	47.16%	100.00%
G187	NMPC-E&T, MECO-E&T, Nantucket and NEP-T	G - General Allocator, 3-Point Formula	39.18%	0.00%	12.93%	52.11%	0.00%	0.00%	47.89%	100.00%
G324	KS Generation, Glenwood and Port Jefferson	G - General Allocator, 3-Point Formula	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%	100.00%
G370	Legacy NG Elec Retails, NMPC-T and NEP-T	G - General Allocator, 3-Point Formula	0.00%	0.00%	0.00%	0.00%	34.38%	11.67%	53.95%	100.00%
G443	KEDNY, KEDLI, Boston Colonial Gas, NG LNG	G - General Allocator, 3-Point Formula	0.00%	0.00%	0.00%	0.00%	39.78%	23.36%	36.86%	100.00%
G901	Transmission, excluding Hydro & NECO	G - General Allocator, 3-Point Formula	0.00%	0.00%	53.47%	53.47%	0.00%	0.00%	46.53%	100.00%
G911	Retails, Transmission, excluding NECO	G - General Allocator, 3-Point Formula	21.59%	5.30%	6.52%	33.41%	16.31%	9.76%	40.52%	100.00%
GT53	G-020 plus LIPA TSA Residual entity, excluding NECO	LIPA TSA G - General Allocator, 3-Point Formula	20.00%	4.95%	6.17%	31.12%	15.33%	9.20%	44.35%	100.00%

Exhibit ___ (ITDP-4R)

Excerpt from the Companies' Implementation Plan Update

Comprehensive Management and Operations Audit of National Grid USA's New York Electric and Gas Utilities

Case No. 18-M-0185

IMPLEMENTATION PLAN UPDATE

PREPARED FOR:

THE STATE OF NEW YORK PUBLIC SERVICE COMMISSION
THREE EMPIRE STATE PLAZA
ALBANY, NEW YORK 12223

October 31, 2022

Chapter 2: Information Systems

Recommendation: 12 (page 40)

The Utilities should improve quarterly variance reports for Information Systems projects.

Implementation Leads

Executive Sponsor:	**Erik Barthel** – NY CIO
Team Lead(s):	Daniel J. DeMauro, Jr. – Director, IT Regulatory Jim Reynolds – Director, IT Finance Caitlin Davidson – Director, US IT PMO

Implementation Priority: High

Recommendation Accepted: Yes (with modification)

Current Status: Implemented

Background Information:

As part of its rate plan in Cases 17-E-0238 and 17-G-0239, NMPC provides information to Staff and the Commission about IT project selection and budgeting. Prior to each of its rate years, NMPC files with the Commission its IT Capital Investment Plan for the upcoming rate year, including the overall IT investment plan for that year, the process for developing the plan, and identification of the proposed IT projects and their estimated costs. NMPC also files quarterly variance reports to address the status of the Service Company IT programs/projects and budgets. The variance reports include (i) a narrative explaining overall IT investment plan approach; (ii) a description of defined program categories (*e.g.,* Regulatory, Legal and Compliance Mandates; Business Programs and Projects; Technology/Infrastructure Foundation and Reliability) that form the spending rationale for IT projects; (iii) detail on budgets and actual spending for each of the program categories, and for the top ten projects by budget within the program categories; (iv) explanations of variances between budgets and actual spending; (v) identification of allocations to NMPC; and (vi) a report on budget exceptions by program category and top ten projects based on allocations to NMPC. In the report for the last quarter of each Rate Year, NMPC also includes (i) an explanation of any cost or timeline differences exceeding ten percent for projects completed; (ii) the latest partial, full, re-sanction or closure paper for any projects exceeding $1 million that were authorized during the previous rate year; and (iii) performance on the IT capital cost incentive. NMPC also meets with Staff on a semi-annual basis to (i) discuss proposed projects for the next six months; (ii) provide an update on the status of in-flight projects; (iii) discuss proposed projects for the next six months; and (iv) review quarterly filings. The Audit Report states (at 34) that the information currently provided by NMPC is generally adequate but could benefit from additional detail.

At present, KEDNY and KEDLI do not have an IT reporting requirement in their current rate plan, but Staff filed testimony in their ongoing rate proceedings (Cases 19-G-0309 and 19-G-0310) (the "2019 KEDNY and KEDLI Rate Cases") recommending that KEDNY and KEDLI adopt similar reporting requirements as NMPC and meet with Staff on a semi-annual basis.

Most recently, on November 25, 2020, Staff filed testimony in NMPC's current rate proceeding (Cases 20-E-0380 and 20-G-0381) (the "2020 NMPC Rate Case") stating that the IT reporting requirements currently in effect for NMPC (as described above) require significant efforts from both NMPC to develop the reports and from Staff to review the reports. At the same time, Staff testified that the IT reporting requirements do not necessarily provide the information in a format that permits easy monitoring and understanding of any individual IT project's progress or enables full comprehension of the entire IT portfolio. Therefore, Staff recommends collaborating with NMPC to develop streamlined and improved IT reporting requirements to enable easier monitoring of project progress in terms of schedule, scope, and budget.[2]

Proposal to Implement Recommendation:

KEDNY and KEDLI will implement IT reporting requirements that are ultimately adopted by the Commission in the 2019 KEDNY and KEDLI Rate Cases. At the same time, the NY Operating Companies believe that collaboration with Staff is necessary to streamline the IT reporting process to provide benefits to National Grid and Staff, as recognized by the Staff Information Technology Panel in the 2020 NMPC Rate Case. Therefore, the NY Operating Companies propose to meet with Staff within 120 days of the Commission's order in the 2019 KEDNY and KEDLI Rate Cases to adopt an IT reporting framework that provides Staff with greater transparency into National Grid's IT investment, sanctioning, budgeting, and approval process without creating an undue burden on the NY Operating Companies to compile the information and on Staff to review the information.

Schedule:

KEDNY and KEDLI will implement the new reporting consistent with the timelines adopted by the Commission in the 2019 KEDNY and KEDLI Rate Cases to be followed by a collaborative meeting with Staff within 120 days of that order. At that meeting, the NY Operating Companies and Staff will develop a framework for IT reporting that can be utilized by the NY Operating Companies to address both the Audit Report recommendation as well as any obligations established in rate plans in a manner that aligns reporting across the NY Operating Companies.

Summary of Cost/Benefit and Risk Analysis:

Extending the current NMPC reporting requirements to KEDNY and KEDLI will require a significant amount of incremental information and documentation across the NY Operating Companies. An incremental full time equivalent will be needed to compile this reporting on an ongoing basis. The expectation is that National Grid and Staff can meet to discuss an enhanced

[2] *See* Cases 20-E-0380 and 20-E-3081, Staff Information Technology Panel's testimony at 73-74.

process that streamlines reporting while at the same time providing better transparency to Staff. Benefits are qualitative in nature and include improved visibility to variances and further transparency to Staff on IT investments, sanctioning, budgeting, and approval.

Measures of Success:

The deliverable to demonstrate that the recommendation has been implemented is submission to Staff of the enhanced IT reporting for the NY Operating Companies.

Implementation Plan Update (January 31, 2022):

As discussed in the October 31, 2021 Implementation Plan Update, the Joint Proposal in the 2019 KEDNY and KEDLI Rate Cases required KEDNY and KEDLI to collaborate with Staff to develop modified and improved IT reporting. On December 22, 2021, KEDNY and KEDLI filed the agreed upon Information Technology Reporting Format that will govern future IT reporting. A copy of that filing is included in Attachment 3.

The Joint Proposal in the recent Niagara Mohawk rate cases (Cases 20-E-0380 and 20-G-0381) requires that the reporting modifications implemented as part of the KEDNY/KEDLI collaborative be applied the same to Niagara Mohawk. Accordingly, the Company submits that this recommendation has been implemented.

Exhibit ___ (ITDP-5R)

Responses to information requests ("IR") provided in these proceedings and cited in this rebuttal testimony

Date of Request: June 12, 2023 Request No. DPS-459
Due Date: June 22, 2023 NG Request No. NG-487

KeySpan Gas East Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY

KEDNY KEDLI Case No. 23-G-0225/0226 Data Requests

Request for Information

FROM: Sandra Sweet

TO: National Grid

SUBJECT: Digital Platform Information Technology (DPIT)

Request:

In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

Exhibit___(ITDP-4) shows capital expenditures for the "Digital Platform Information Technology" category in the amounts of $75.2 million for Fiscal Year (FY) 2025, $71.4 million for FY 2026, $74.9 million for FY 2027, and $54.1 for FY 2028.

1. Describe the approval process for this project category?

2. Provide a breakdown of the costs incurred to date related to this project.

3. Explain how the Companies arrived at the budgeted amount for both capital and operating expenses for each of the years listed. Include all supporting workpapers and calculations.

4. Did either of the Companies forecast any savings associated within any of these budgets? If so, how much savings and where is it included in the Rate Year (RY) forecast? If not, why do each of the Companies believe that no savings will be realized?

5. Explain whether implementation beginning in FY 2025 is on track.

Response:

1. National Grid's jurisdictions identify project priorities with technology components. As part of that prioritization effort, business needs, infrastructure requirements, sequencing, resource availability and capacity, size and impact of investment requests, and the overall benefit of the investment compared to other proposals is assessed.

   Once engaged, IT&D considers existing multi-year investments and identifies the amount of work remaining, and the delivery timeline for those projects. IT&D will also support

new investments needed to comply with legal or regulatory requirements and mandates, or other critical corporate objectives. Lastly, projects to address IT&D infrastructure are targeted, including investments to upgrade, replace, and add functionality to ensure system reliability and to establish foundational capabilities required for future operational needs of the business.

Working as part of the senior leadership team in each jurisdiction, IT&D collaborates on creating an investment plan in line with jurisdictional priorities. A final investment plan review occurs each fiscal year with the Jurisdictional Presidents, jurisdictional delegates, and functional executives. Upon the review and approval of the investment plan by the Jurisdictional Presidents and functional executives, consolidated financial information from the investment plan, by jurisdiction and company, is incorporated with IT&D operations and maintenance ("O&M") requirements into the IT&D budget. The IT&D budget is then reviewed and approved by the New York CIDO, the New England CIDO, the Global CIDO, and Corporate Finance. Finally, the plan is incorporated into a consolidated National Grid USA budget for approval by the National Grid plc board.

2. The capital costs incurred to date for each investment are listed below. Digital Platforms and Information Technology (DPIT) transitioned to a product operating model during FY22. In this model, DPIT investments and services are now grouped into "investment towers" to align with business value.

Investment Name	FY21 CAPEX	FY22 CAPEX	FY23 CAPEX
Networks	0	20,688,086	17,836,753
Operations	0	8,038,079	10,048,754
US CNI	0	2,103,497	2,782,371
GridStack & Digital Platforms	0	1,983,350	13,242,472
US Cloud & Hosting	0	6,718,360	11,248,832
US Digital Workplace Services	0	2,910,986	18,924,045
US E-hub	1,155,103	1,025,906	1,244,086

3. Please see the response to question 4 in DPS-457.

4. Please see the response to question 5 in DPS-457.

5. Please see the response to question 6 in DPS-457.

Name of Respondent:
Jeffrey Knighton

Date of Reply:
June 22, 2023

Date of Request: July 21, 2023 Request No. DPS-850
Due Date: July 31, 2023 NG Request No. NG-1057

KeySpan Gas East Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY

KEDNY KEDLI Case 23-G-0225/0226 Data Requests

Request for Information

FROM: Sandra Sweet

TO: National Grid

SUBJECT: DPIT

Request:

In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

Page 14 of the Information Technology and Digital Panel's initial testimony discusses the Network Infrastructure project under Digital Platform & Information Technology category.

1. Describe the Companies' approach(es) for developing the cost estimates for each of these projects, including any assumptions and methodologies.

2. What is the dollar amount (or percentage) of these cost estimates representing contingency, and what estimation approach was used to develop the costs?

3. What amount of the dollars estimated for this project represent insurance against faulty estimating?

4. If there are anticipated savings to result from these technology investments, where in the Companies' cost structure will the savings manifest?

5. Are the savings identified in question 4, above, one-time or ongoing savings?

    a. When will those savings occur?

    b. Are there conditions that are required for the cost savings to be realized (milestones met, dates met, other systems work completed, etc.)?

6. What form will the cost savings take (reduction in number, & dollar amounts, of support contracts for maintaining legacy systems, fewer FTEs, lower software costs, etc.)?

7. What personnel or team at the Companies is implementing this initiative?

    a. Explain how the team will be organized and where in the organization will the team be situated?

8. Explain whether there is organizational capacity to perform this work?

9. Who is the executive sponsor?

10. What is the governance structure for this project?

11. What are the major risks for this project and what is being done to manage and/or mitigate those risks?

12. What are the long-term implications for maintenance, staffing, costs, obsolescence risk associated with this project?

Response:

1. As provided in the Company's response to DPS-297, Question 1.c., future IT&D projects that have not gone through the sanctioning review and approval process are estimated and described using an Investment Estimation Template ("IET"). The IET captures key attributes of proposed future investments, including the investment description, cost estimates, scope, timing, and the expected benefiting operating companies, among other details. Leveraging standard internal and external labor rate information, project manager experience, and historical cost trends, the template provides a detailed estimate of labor, hardware, software, and other expenses. In addition to the individual project estimates, the Company prepares a high-level Strategic Business Plan ("SBP") outlining the IT&D investment horizon for the next 10 years. Future investments proposed in the Rate Year and Data Years broadly align with the SBP outlook. A component of the overall Network Infrastructure project has been through sanctioning review and approval for a portion of the overall investment in FY23 and FY24. The remaining components of the project beyond FY24 have an IET with supporting details and cost-estimating information. Please see Attachment 8 and Attachment 10 to the Company's response to DPS-457 for the associated sanction papers and IET, respectively.

2. The Network Infrastructure project under the Digital Platform & Information Technology ("DPIT") category incorporated a 10 percent risk margin in its forecast.

3. This investment does not include any costs representing insurance against faulty estimating.

4. Savings anticipated from the Network Infrastructure investment will offset IT&D RTB costs. They are included in Exhibit ____(ITDP-6) and are also shown on Attachment 10 to the Company's response to DPS-457. Please see the response to DPS-516 for an explanation for how those savings are reflected in the revenue requirements.

5. a. The investment in Network Infrastructure is expected to yield $2.5M of cost benefits in both FY25 and FY26, separately.

   b. The cost benefits associated with the Network Infrastructure investment require that it be placed in service and meet its anticipated milestones outlined in the IET.

6. The savings will be realized by reduction of non-managed network spend, and by removing devices that are out of service and not currently managed within the network.

7. The resources required to deliver this investment are detailed in the IET included in Attachment 10 to the Company's response to DPS-457. The team is in the IT&D Digital Platforms and Infrastructure Technology ("DPIT") organization. For reference, a visual of the IT&D organization can be found in Exhibit_(ITDP-1). Specifically, the Network Infrastructure team within the DPIT organization will deliver this investment.

8. Please see the Company's response to DPS-541. The resources, both internal and external, for FY23 and FY24 have been sourced, identified, and are currently progressing the investment deliverables. The Network Infrastructure team will remain intact and continue to follow the Network Infrastructure milestone schedule through the Rate Year and Data Years. If needed, the Company will augment the project team from within the DPIT organization or with third party provides to ensure it remains on schedule.

9. Shannon Soland, Chief Information Digital Officer ("CIDO") of IT&D DPIT, is the executive sponsor for the Network Infrastructure investment.

10. Please see Exhibit_(ITDP-1) included in the IT&D Panel's submission in the Company's rate filing in these proceedings for a graphical depiction of the IT&D management organization. The delivery of this investment will be managed by a project team within National Grid's DPIT function, led by the DPIT CIDO. The project team will be subject to direction and oversight by the New York CIDO and National Grid's US sanctioning procedures and associated committees which govern, review and approve all capital investments, including IT&Ds, that are greater than or equal to $1 million dollars.

11. The risks associated with the Network Infrastructure investment are detailed in the project sanction paper and IET, included in Attachments 8 and 10 to the Company's response to DPS-457, respectively. The project team is responsible for managing risks and for escalating them to senior leadership as they arise.

12. The long-term maintenance, staffing, costs, and obsolescence risks and requirements associated with the Network Infrastructure investment are evaluated during the sanctioning review and approval process at routine intervals to ensure the project continues to meet its investment objectives.

Name of Respondent: Jeffrey Knighton

Date of Reply: July 31, 2023

Date of Request: July 24, 2023 Request No. DPS-879
Due Date: August 3, 2023 NG Request No. NG-1086

KeySpan Gas East Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY

KEDNY KEDLI Case 23-G-0225/0226 Data Requests

Request for Information

FROM: Philip Tabor

TO: National Grid

SUBJECT: Physical Security

Request:

In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

Referring to Exhibit___(ITDP-4) IT Service Company Capital Forecast line items "Physical Security".

1. Provide a detailed description beyond what was included in the Companies' response to DPS-457 Supplemental Attachment 69 CONFIDENTIAL, and provide:

    a. How priority is determined for which sites are worked on through this program.

    b. The methodology and information used to determine which physical threats and vulnerabilities are most important and addressed by this program.

    c. How the program is split between critical national infrastructure, critical facilities, and non-critical facilities.

    d. A timeline and expected "in-service" date for each significant part of the program.

    e. Any cost savings associated with this line item.

    f. Any improvements to existing capabilities.

    g. Any additional capabilities added by this line item.

2. Provide proposed budgets and actual spend for this line item by year for 2019 through 2022 for Cases 19-G-0309 & 19-G-0310.

3. Provide a detailed explanation of how this program does not overlap with the LNG Security System Upgrade in KEDNY Exhibit___(GIOP-5) pp. 143-145.

Response:

1. a. The IT&D Security function assesses risks that each physical security investment seeks to address and prioritizes them accordingly. This risk-based approach is used to determine which sites are worked on and considers the following factors:

- **Break-Fix** – When an asset used to secure a physical location fails and the Company's ability to secure a site is compromised, break-fix work provides temporary compensating controls to be implemented until longer-term solutions can be completed to replace the failed asset with a like-for-like asset. Break-Fix is completed on an on-going and as-needed basis and takes funding priority.

- **The Company's knowledge and management of assets** – The Company manages the lifecycle of assets used to secure sites and maintains an inventory of asset age, vendor, make, model, warranty, and support. A campaign of replacements based on asset lifecycle proactively reduces break-fix work and predicts which assets should be replaced ahead of failure. It is based on whether an asset is no longer supported, is discontinued, or is no longer as effective as current solutions are.

- **Changes to physical site classification and physical security requirements –** The Company reviews physical sites on an on-going basis to assess site criticality and vulnerability. Changes to site criticality also change physical security needs based on updated site classifications of low, medium, or high.

- **Site specific reports and required improvements –** The Company assesses sites on a proactive basis to identify gaps in physical security. When gaps are identified, they are prioritized for investment. On a reactive basis, after a physical security breach has occurred, a post-incident assessment is completed to understand what controls are required to mitigate future events from occurring. Those controls are prioritized for investment.

b. The method and information used to determine which physical security threats and vulnerabilities are prioritized and addressed is driven by the risk-based approach described in the response to Question 1, as well as:

- Site Criticality
- Annual vulnerability assessments conducted at all critical facilities
- Dedicated threat intelligence teams continuously monitor and assess the threat landscape. (tactics, techniques, procedures ("TTPs"))
- Collaboration and participation in a broad range of information sharing groups, including with other utilities. Information sharing groups include:
  - Electrical Information Sharing and Analysis Center ("E-ISAC")
  - Downstream Natural Gas Information Sharing and Analysis Center ("DNG-ISAC")
  - Homeland Security Information Network ("HSIN")
  - Energy Sector Analyst Roundtable ("ESAR")
  - International Association of Law Enforcement Intelligence Analysts ("IALEIA")

- Edison Electric Institute ("EEI")
- American Gas Association ("AGA")
- Interstate Natural Gas Association of America ("INGAA")
- Oil and Natural Gas Sector Coordinating Council ("ONGSCC")
- Pipeline Sector Coordinating Council ("PSCC")

- Law enforcement guidance
- Regulations (*e.g.,* Transportation Security Administration ("TSA"), Maritime Transportation Security Act ("MTSA"), Chemical Facility Anti-Terrorism Standards ("CFATS"))
- The Company's analysis of historic internal and external trends

c. Physical Security investments provided in the Service Company Capital Forecast line items in Exhibit___(ITDP-4) prioritize projects at sites based on overall criticality and risk, including consideration for sites deemed critical national infrastructure and, more generally, critical facilities essential to the Company's ability to operate and serve customers.

d. Physical Security projects with a forecast above $1m are sanctioned annually as part of the IT&D sanctioning governance procedures. Generally, in-service dates for each Physical Security program investment occur at the end of a given fiscal year.

e. There are no any anticipated costs savings associated with Physical Security investments presented by the Company's IT&D Panel in these proceedings. Benefits associated with Physical Security investments include:

- Long-term reduction of break-fix work by proactively replacing and upgrading equipment before failure or the end of useful life.
- Leveraging internal resources when possible.
- Securing and protecting the assets that are essential to the Company's ability to provide safe and reliable service to customers.

f. – g. Improvements and additional capabilities for Physical Security assets proposed by IT&D include:

- increasing compatibility and adopting updated firmware and operating systems supported by the applicable vendor.
- investments required to upgrade and maintain the supporting IT infrastructure that enables physical site monitoring.

2. The table below provides the total budget and actual spend for IT&D Physical Security investments for FY19 through FY22:

Category	**Budget/Actuals**	**FY19**	**FY20**	**FY21**	**FY22**
Physical Security	Budget	3.3m	3.8m	7.0m	6.5m
	Actuals	7.2m	7.8m	9.1m	4.7m

3. The proposed investment for the Greenpoint Liquefied Natural Gas ("LNG") Facility Security System Upgrade included in Exhibit___(GIOP-5) at pages 143-145 is being supported by the Company's Gas function and is separate from the IT&D Security portfolio submission. Investments in site-specific assets such as card readers, cameras, physical protections, and intrusion detection systems that are physically on site, like the proposed investment at the Greenpoint LNG facility, are generally supported by IT&D but owned and funded by the applicable operating company and are not included in the IT&D Service Company capital forecasts.

Name of Respondent:
Jeffrey Knighton
Dennis Leung

Date of Reply:
August 2, 2023

Date of Request: July 31, 2023 Request No. DPS-942
Due Date: August 10, 2023 NG Request No. NG-1234

KeySpan Gas East Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY

KEDNY KEDLI Case 23-G-0225/0226 Data Requests

Request for Information

FROM: Paul Hammond

TO: National Grid

SUBJECT: Exhibit___(IDTP-4) IT Service Company Capital Forecast

Request:

In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

Referring to Exhibit___(IDTP-4) Lines 440-444, Physical Security:

1. Provide Sanctions Papers for the Programs included in Physical Security Lines 440-444.

2. Provide a list of alternatives for the proposed Programs included in Physical Security Lines 440-444.

Response:

1. Sanction papers for the Physical Security programs listed in lines 440-444 in Exhibit ___ (ITDP-4) have not been presented or approved by the IT&D sanctioning review committee yet. Similar to all Physical Security programs in excess of $1 million that go through annual review, these investments are expected to begin annual sanctioning review in FY24 and to commence in FY25.

2. The following alternatives have been considered for the physical security program:

    a. **Do Nothing and replace through break-fix only – Not Accepted** - While the existing physical security capabilities are functional, they do not feature capabilities on the market today, constraining the effectiveness of capabilities. A break-fix approach does not bring capabilities to current standards because it is like-for-like to ensure compatibility.
    b. **Accelerate proactive replacement program – Not Accepted** – The current proactive replacement program has considered the useful life of existing assets, the useful life of replaced assets, internal resource availability for IT and Business

stakeholders to support on-going break-fix requirements, upgrades required due to site classification changes, regulatory changes, and site availability when determining the budget and in-scope sites. Acceleration of the proactive replacement campaign would require increased resources to deliver and increased funding to achieve.

c. **Prioritize site-specific break-fix, proactive replacement of end-of-life assets, and site by site physical security requirements – Accepted** – The Company's proposal for funding considers the different aspects of the physical security program and the Company's ability to deliver it, while appropriately managing risks and any changes to the regulatory requirements. This proposal enables the Company to ramp up or down the different aspects of the program to address any immediate needs and balance future needs for remediation or upgrade.

Name of Respondent:
Jeffrey Knighton
Dennis Leung

Date of Reply:
August 10, 2023